

Technical Reference Manual for Castalia

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1 System Configuration

The microcontroller (MCU) described in this document is a five-hart (five-core) multiprocessor built from five identical VestaRV CPU cores, a custom 32-bit RISC-V processor. The microcontroller is composed of the five VestaRV harts, a ROM that contains boot software and a Forth interpreter, private RAM memory cells per hart, an arbitrated shared memory window for inter-hart communication, and several peripherals. The multi-core architecture is described in Section 4. The modular architecture enables customization of the peripheral set, memory configuration, and system features to match specific application requirements. Figure 1 shows the top-level organization of this configuration.

The following list details the MCU configuration:

- Chip name: Castalia
- Compatible with RISC-V compiler RV32I[M][A][C][Zba][Zbb][Zbs][Zbc]
- 32-bit memory bus
- Contains 32 general purpose dual-port CPU registers
- Supports the RISC-V compressed instruction set and allows the use of the [C] compiler extension. Allowing the compiler to use the [C] extension reduces executable code size, but also takes the processor longer to fetch 32-bit long instructions that are aligned on a 2 but not 4 byte boundary from memory.
- Includes a multi-stage bit shifter in the ALU to save power and chip area for bit shift operations at the expense of speed
- Includes a fast 32-bit signed integer hardware multiplier in the ALU and allows the use of the [M] compiler extension
- Includes a medium speed 32-bit signed integer hardware divider and remainder calculator in the ALU
- Supports the RISC-V atomic instruction set (LR/SC and AMOs) and allows the use of the [A] compiler extension; on multi-hart configurations atomics are globally coherent across the shared window
- Supports the RISC-V bit-manipulation extensions Zba (address generation), Zbb (basic bit manipulation), Zbs (single-bit operations), and Zbc (carry-less multiplication)
- The read-only misa CSR (0x301) advertises the enabled instruction-set extensions at run time
- Supports maskable hardware interrupts generated by peripheral signals
- 5× VestaRV harts (cores), each with private RAM; hart ID readable via the `mhartid` CSR
- An arbitrated shared memory window with 80 KiB of shared RAM, a CLINT (inter-processor and per-hart timer interrupts), 16 hardware mutexes, and a per-hart peripheral interrupt router
- 6× 8-pin general purpose I/O (GPIO) ports with edge-triggered interrupts and per-pin multiplexed alternate functions (GPIO + up to 8 alternate functions per pin)
- 2× SPI interfaces (SPI0 provides memory-mapped access to external flash memory)
- 2× UART interfaces with hardware parity support

- 2× 32-bit timers with pulse-width modulation outputs and input capture units
- A CRC calculation engine (CRC16_CDMA2000)
- 2× internal digitally controllable oscillators
- 2× external clock pins for clock generation and accurate timing
- A windowed watchdog timer
- A neural processing unit (NPU) co-processor for hardware acceleration of machine learning tasks
- Per-tile MTCMOS power gating with hardware gate/wake sequencing (cold-boot wake)
- 2× I²C interfaces (both master and slave mode)
- 1× NFC ISO 14443A tag / card-emulation engine (Miller/Manchester codec, CRC-A, anticollision, digital AFE boundary)
- Claim/complete peripheral interrupt delivery (PLIC-style) with any-vector-to-any-hart routing

Figure 1 is the view to read first if the question is *what is on this chip*: the masters in a band across the top, the arbiter drawn as the single bus bar it behaves like directly under them, every peripheral this configuration instantiates hanging off it, the shared memories, and the package boundary in red with the outside world crossing it. Figure 2 is that same generated content drawn *flat*, for the whole chip in one landscape view: one bus bar, the masters above it, and the peripherals cut to a name and a line in one rank below it. Both figures are generated from the same configuration, so the peripheral set, the instance counts and the addresses in them are this build's and no other's; the memory system's own view of the chip, the one address column all five harts share, is Figure 4 in Section 3, and the per-pin bonding is Section 2. Both draw the analog front end the way Figure 18 does — one site per hart, in the column of the hart that owns it: a row under the bus bar in the first, a hat on each channel hart in the second — because that alignment *is* the claim: there is no per-hart wire to an analog site, only the gate printed in each one and hart 0's reach over all of them (Section 7.2).

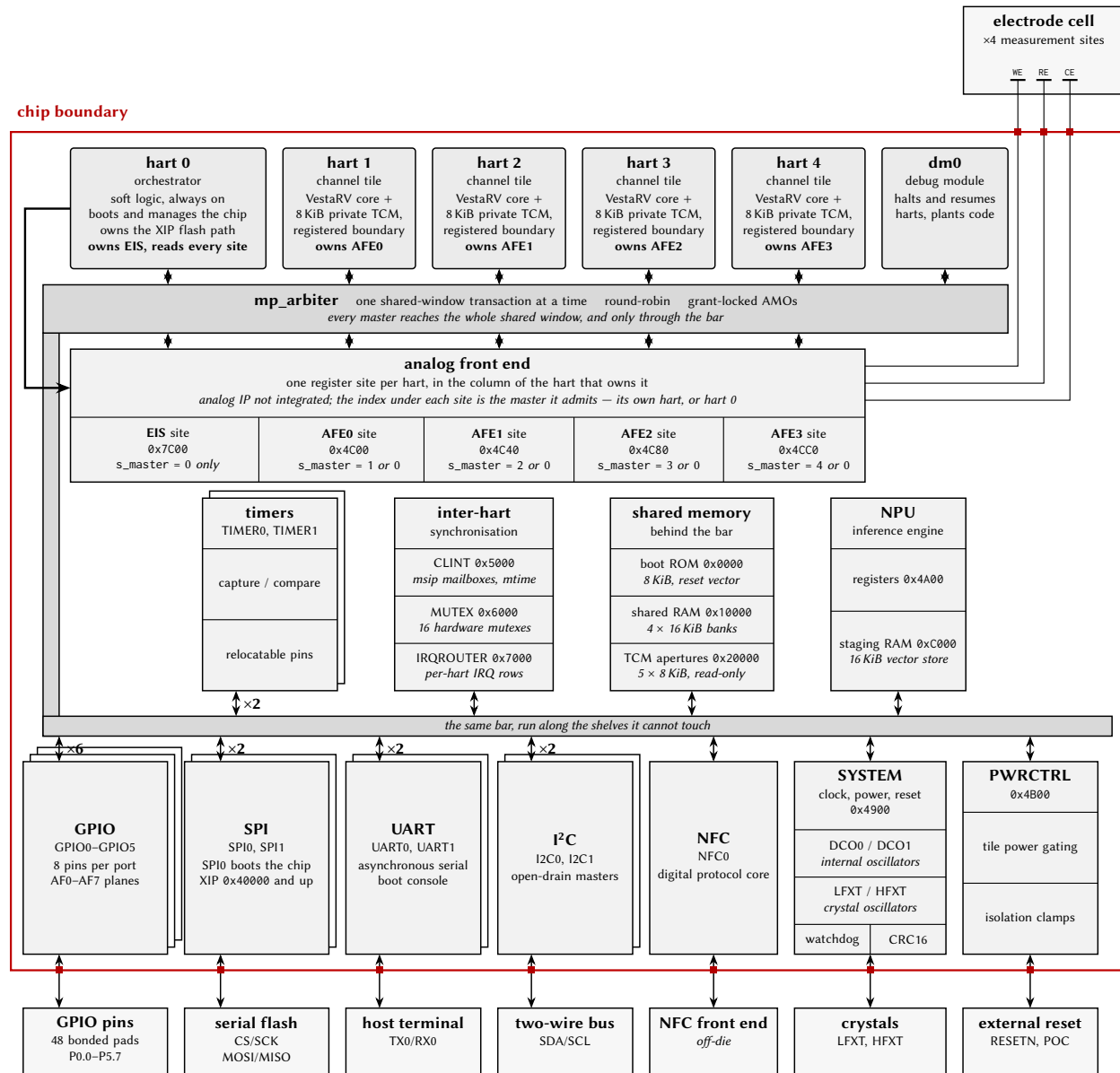


Figure 1: Castalia whole-chip diagram: the bus and the peripherals. One serializing shared-window bus reaches everything on the die, so the `mp_arbiter` is drawn as the grey bar every block taps — run down the margin and along the lower shelves, because one bar cannot touch three shelves without a tap crossing a box. Masters on top; the pin-facing shelf against the red package boundary, its outside world beyond it; an $\times N$ badge is N instances of one peripheral, not one block. In the analog row the *column is the ownership*: a site sits under the hart that owns it, and what enforces that is the gate printed inside it — a comparison against the arbiter’s granted-master index, not a wire; hart 0’s margin arrow is the same comparison from the other side. Everything drawn is generated from this configuration.

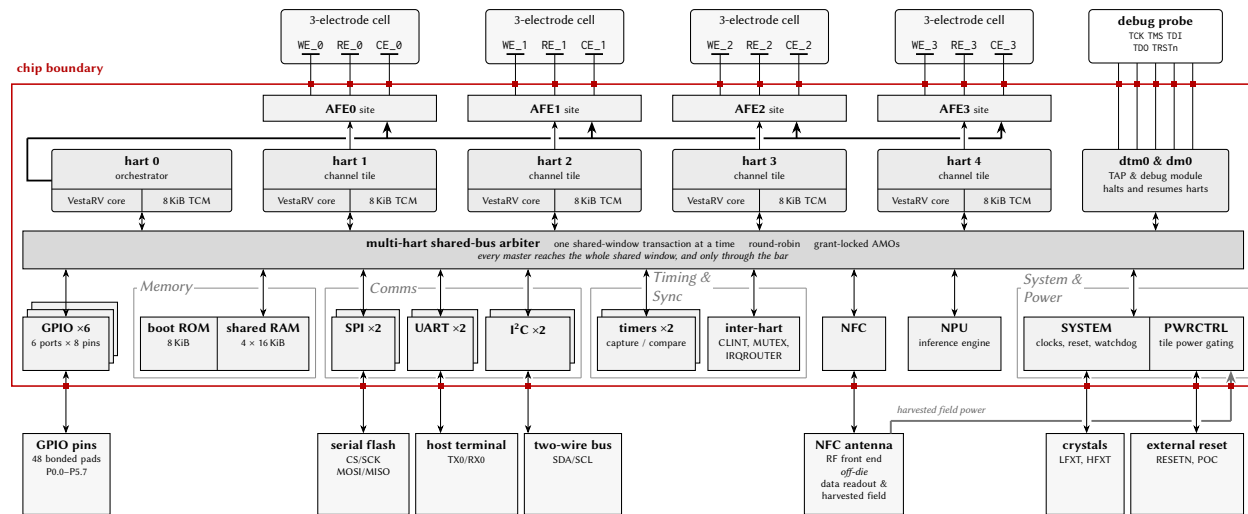


Figure 2: Castalia whole-chip diagram, flat companion to Figure 1: the same generated content on *one* bus bar and *one* rank. The bar is the `mp_arbiter` entity of the RTL. Every master taps it from above and every peripheral taps it from below, straight up, so there is one bus in this drawing and no second segment of it. A hart box is split into the two things the tile is built from — the VestaRV core and its private TCM. Each of those TCMs is also windowed read-only into the shared map (Section 3), which is why the rank carries no aperture box of its own. Peripheral boxes carry a name and one line only; the addresses are in Section 3, and a $\times N$ in a title — with the offset squares stacked behind the box — is N instances of one peripheral, not one block. The rank is grouped by type: the grey headings, and the thin outlines round every type that has more than one box or compartment, are organisation only — every block under them is an arbiter slave in its own right, tapping the bar on its own wire. Every off-chip partner hangs on a straight vertical wire dropped from the tap above it, entering the partner's top edge off-centre where the partner had to be nudged aside. The JTAG/debug column and the NFC block are drawn *solid* in every build, whether or not this configuration contains them: what a given build actually has is the `debug.enable` and `peripherals.nfc` knobs of Table 1, and the JTAG column prints *debug builds only* in the box where this build has no debug module in it. The one stroke in the drawing that still changes with the configuration is the harvested-supply rail, drawn *dashed* where this build does not run the chip off the field. The Debug Module is one more master on the bar, reached over five JTAG pins from an off-chip probe, and is present in debug-enabled builds only. NFC has two roles and this view draws both: its digital core reads the tag through the bar like any peripheral — the RF front end is off-die — and on a field-powered board the harvested field is the chip's *supply*, which is the heavy grey rail running along the off-chip band into PWRCTRL's PGOOD and boot-strap taps, whose boot gate holds every hart in reset until that supply is good. This view is the *channels*: each channel's analog site is drawn as a hat on the hart that owns it, with a thin arrow between the two — that arrow is the ownership gate the site enforces (a comparison against the arbiter's granted-master index), not a wire of its own. The analog IP is not integrated. The sites are ordinary arbiter slaves and no site is reached except through the bar, like everything else in this drawing; what the row draws is the narrower fact, which is *who may reach one*: each site's own arrow down from the hart it admits, and hart 0's heavy rail under the row, which is the same comparison from the other side, reaching every site at once. The register that enforces it is Section 7.2's: the row carries the reach and leaves the register name to the text. Each channel brings its own three-electrode cell straight up across the boundary on unbroken wires. The shared EIS sweep engine has no channel of its own and is omitted here; its register site is in Section 7 and in Figure 1. Everything drawn is generated from this configuration.

1.1 Chip Configuration

This document, the memory-map headers, the linker scripts, and the drop-in RTL are all generated from one chip configuration by `make chip`. The configuration is a small JSON document (every key optional; missing keys keep the Castalia defaults) passed as `make chip CONFIG=config.json`; an interactive front-end for producing it ships with the generator (`docs/chip_configurator.html`). Table 1 lists every configuration knob together with the value used to build *this* document, and Table 2 lists the geometry derived from those knobs. The resolved configuration is also written to `config/ChipConfig.resolved.json` at every build, and `make verify [CONFIG=config.json]` *proves* a configuration: it stages the generated RTL into a behavioral simulation environment and runs the multi-core boot/ISA smoke suite against it. The core peripheral set and the pad ring are not configuration knobs; they are fixed template content, and the pad ring (Section 2) is derived from the generator’s package model. The second-instance peripherals (UART1, SPI1, TIMER1, I2C1) and the NPU, however, are individually droppable through the peripherals section: a dropped instance’s register window reads zero, its interrupt vectors are reserved (the numbering is frozen), and its pins revert to plain GPIO.

Table 1: Chip configuration knobs (`make chip CONFIG=config.json`) and the values of this build

Configuration key	This build	Meaning / valid values
<code>chipName</code>	Castalia	non-empty string: renames the chip in the TRM and headers (CHIP_NAME env wins)
<code>numHarts</code>	5	int 1..32: hart/tile count (5 = Castalia default, 18 = Argus)
<code>orchestrator</code>	true	bool: hart 0 is the always-on orchestrator; harts 1..N-1 are gateable tiles
<code>numMutexes</code>	16	int 1..1024: HW mutex bank size (16 = Castalia, 32 = Argus)
<code>registerFileDualPort</code>	true	bool: docs-only, the register file is dual-port in the RTL either way
<code>core.fetchAhead</code>	true	bool: C-extension fetch-ahead (one flip-flop; removes most of the straddling-fetch stall)
<code>isa.mul</code>	true	bool: M multiply
<code>isa.fastMul</code>	true	bool: docs-only, the multiplier is already single-cycle
<code>isa.div</code>	true	bool: M divide
<code>isa atomics</code>	true	bool: A extension (LR/SC + AMO)
<code>isa.compressed</code>	true	bool: C extension
<code>isa.bitmanip</code>	true	bool: Zba/Zbb/Zbs/Zbc
<code>isa.minimalTiles</code>	true	bool: harts 1..N-1 drop M and B (rv32iac); hart 0 keeps the full ISA
<code>isa.counters</code>	false	bool: Zicntr march suffix only (cycle/instreet always exist, time/timeh never do)
<code>isa.counters64</code>	false	bool: docs-only high counter halves (always present); needs isa.counters
<code>isa.zicond</code>	false	bool: Zicond conditional-zero ops (czero.eqz/czero.nez)
<code>isa.zcb</code>	false	bool: Zcb extra compressed loads/stores and ALU ops; needs isa.compressed

Table 1 continued on next page

Table 1 continued from previous page

Configuration key	This build	Meaning / valid values
isa.zimop	false	bool: Zimop+Zcmop may-be-ops (mop.r/mop.rr rd<-0, c.mop.n nops)
isa.zihint	false	bool: Zihintpause+Zihintntl (PAUSE = 16-cycle arbiter-yield window; ntl.* nops)
isa.zihpm	false	bool: Zihpm performance counters 3 and 4 (four chip events)
isa.zawrs	false	bool: Zawrs wrs.nto/wrs.sto wait-on-reservation-set (needs isa.atomics)
isa.zabha	false	bool: Zabha byte/halfword AMOs; needs isa.atomics
isa.zacas	false	bool: Zacas compare-and-swap (amo-cas.w/.b/.h); needs isa.atomics
isa.zicboz	false	bool: Zicboz cbo.zero cache-block zero (64-byte block)
isa.zcmp	false	bool: Zcmp compressed push/pop and register moves; needs isa.compressed
isa.zcmt	false	bool: Zcmt compressed table jump and the jvt CSR; needs isa.compressed
isa.zbkb	false	bool: Zbkb crypto bit-manipulation (pack/brev8/zip/unzip)
isa.zbkc	false	bool: Zbkc carry-less multiply (clmul/clmulh)
isa.zbkx	false	bool: Zbkx crossbar permute (xperm8/xperm4)
isa.zkn	false	bool: Zkn AES+SHA (Zknd+Zkne+Zknh)
isa.zfinx	false	bool: Zfinx single-precision floating point in the x registers
priv.trapCsr	true	bool: standard M-mode trap CSRs and delivery; firmware opts in via mtrapctl
priv.umode	false	bool: user mode (privilege register, MPP stack, mcounteren); needs priv.trapCsr
priv.pmp	false	bool: PMP (Smpmp) CSR bank and match unit; needs priv.umode
priv.pmpEntries	16	int, 8 or 16: PMP entry count when priv.pmp is true (default 16)
debug.enable	true	bool: debug mode, the Debug Module and the JTAG DTM; needs priv.trapCsr
memory.romSize	8192 (8 KiB)	int bytes, 1 KiB multiple <= 0x4000: boot ROM (0x0-0x1FFF at the shipped 8 KiB; the page runs to 0x3FFF and its tail is unmapped)
memory.tcmSizePerHart	8192 (8 KiB)	int bytes, 1 KiB multiple <= 0x4000: per-hart TCM based at 0x8000 (top = 0x8000 + size - 1; 0x9FFF at the shipped 8 KiB)
memory.sharedBulkRamSize	65536 (64 KiB)	int bytes, multiple of 0x4000: shared bulk RAM from 0x10000, one bank per 16 KiB

Table 1 continued on next page

Table 1 continued from previous page

Configuration key	This build	Meaning / valid values
memory.npuStagingRamSize	16384 (16 KiB)	int bytes, 1 KiB multiple <= 0x4000: NPU staging RAM (0xC000-0xFFFF)
peripherals.npu	true	bool: false drops the NPU (slot 10 and the 0xC000 staging window read zero)
peripherals.i2c1	true	bool: false drops I2C1 (slot 15 reads zero; SDA1/SCL1 revert to plain GPIO)
peripherals.uart1	true	bool: false drops UART1 (slot 5 reads zero; TX1/RX1 revert to plain GPIO)
peripherals.spi1	true	bool: false drops SPI1 (slot 3 reads zero; its four pins revert to plain GPIO)
peripherals.timer1	true	bool: false drops TIMER1 (slot 7 reads zero; the T1 pins revert to plain GPIO)
peripherals.cqAfeStubs	true	bool: the four AFE register stubs and the EIS engine stub (slot 12, 0x4C00)
peripherals.qspi	false	bool: QSPI0 controller in slot 12 (0x4C00); needs cqAfeStubs false
peripherals.i3c	false	bool: I3C0 controller (MVP + DAA + IBI) at 0x6100
peripherals.nfc	true	bool: NFC0 ISO 14443A tag engine at 0x6200 (the RF front end is off-die)
peripherals.rtc	false	bool: RTC0 32.768 kHz wall clock with alarm and tick at 0x6500
peripherals.pwm	false	bool: PWM0 two-channel buffered PWM at 0x6600 (outputs on P2.2/P2.3)
peripherals.onewire	false	bool: OW0 1-Wire master at 0x6700, open-drain DQ on P4.7
peripherals.fieldPower	true	bool: wires the field-power pins (PGOOD, harvest strap) into PWRCTRL
peripherals.dma	false	bool: DMA0 multi-channel DMA at 0x6800; adds one more arbiter master
peripherals.dmaChannels	4	int, 2 or 4: DMA0 channel count when peripherals.dma is true (default 4)
peripherals.i2ctarget	false	bool: I2CT0 hardware I2C target at 0x6A00, sharing the I2C0 pads
peripherals.trng	false	bool: TRNG0 ring-oscillator entropy source at 0x6900 (firmware must DRBG it)
peripherals.trngRings	8	int, 4 or 8: TRNG0 ring count when peripherals.trng is true (default 8)
peripherals.eventFabric	false	bool: EVFAB0 event/trigger crossbar at 0x6B00 (8 channels, no IRQ vector)
package.model	castalia-lqfp100	string: package pinout model, one of myshkin-qfn44, castalia-quad-qfn64 or castalia-lqfp100
package.preliminary	true	bool: prints the Preliminary note in the TRM package section

Table 2: Derived geometry of this configuration (computed, not configurable)

Derived value	This build	Notes
ISA string (march)	rv32imac_zba_zbb_zbs_zbc	Advertised in the read-only misa CSR
Shared-window address width	16 bits (words)	Arbiter/tile word-address width; the window is $0x0 - 2^{w+2} - 1$
Shared RAM banks	4×16 KiB	One SRAM macro per bank behind the arbiter
Extended flash base	0x40000	First address decoded to the SPI-flash XIP path (hart 0 only); strictly the complement of the shared window
Interrupt vectors	121	Vectors 83/84 are the CLINT software/timer interrupts
CLINT MTIME	0x5020	Layout is hart-count-derived: MSIPx at $0x5000 + 4h$, MTIMECMPx from 0x5030
Boot-loader mailbox rows	$0x10500 + 0x10 \cdot \text{hartid}$	Tile-loading rows {SRC, LEN, ENTRY} consumed by the boot ROM
Stack pointer at reset	0xA000	Top of each hart's private TCM, growing down
Peripheral count	21	Instantiated peripherals (including CLINT/MUTEX/IRQROUTER)

2 Packaging and Pins Configuration

Preliminary: the package model for this configuration (LQFP-100) has not been finalized for Castalia; the bonding shown here is the planned assignment, not a confirmed bond-out. Pin assignments, package type, and pin count are subject to change.

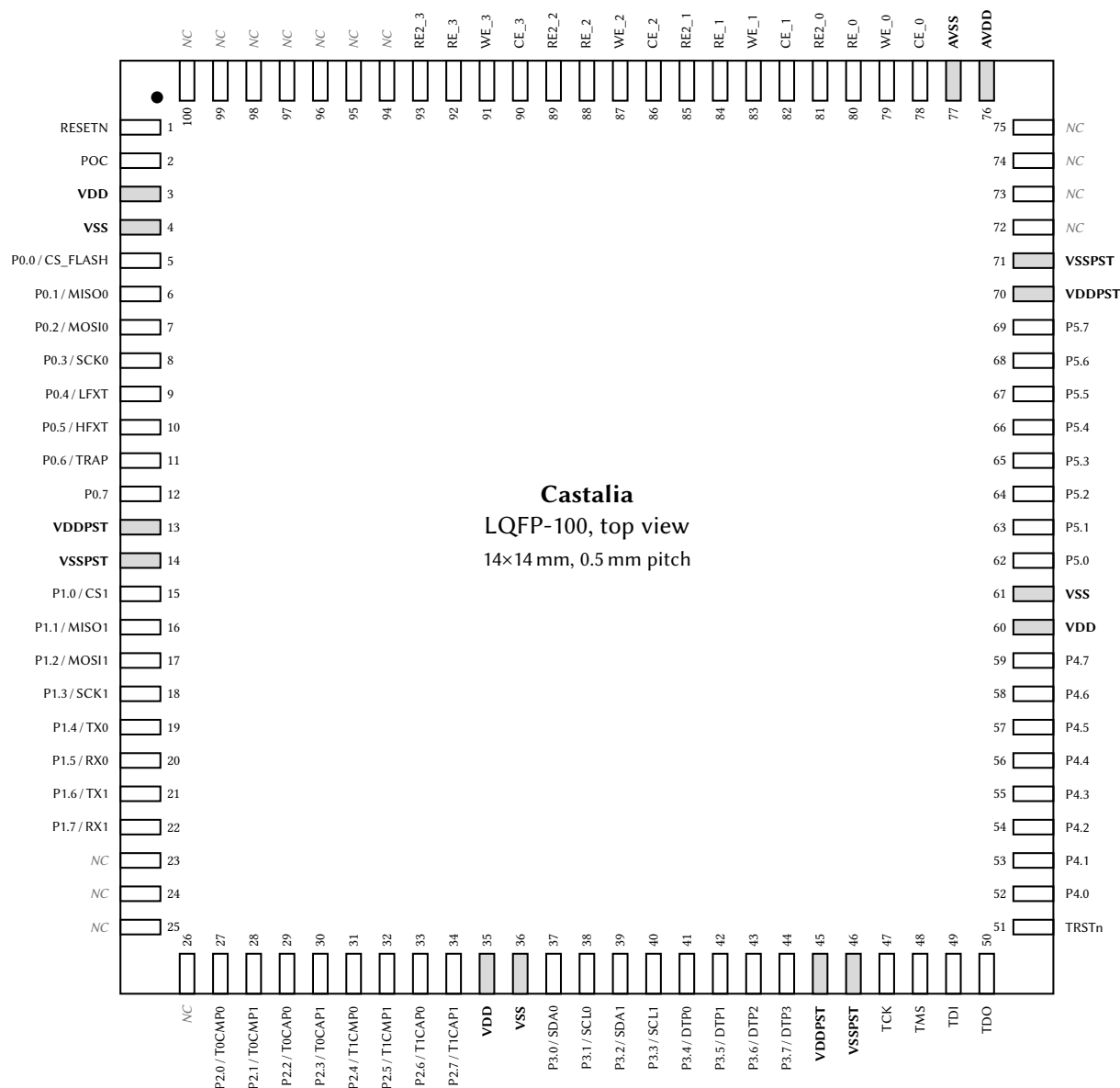


Figure 3: LQFP-100 pinout, top view (14 × 14 mm, 0.5 mm pitch). Each pin is labeled with its pad name and, for GPIO pins, its AF0 (reset-selected) alternate function; power-rail pins are shaded and bold. The full per-pin alternate-function matrix is in Table 5.

The chip package is a LQFP-100 with dimensions 14 × 14 mm and pitch 0.5 mm. The package footprint (as seen from the top down) is shown in Figure 3. The pins on the package are listed in Table 3. The pin ordering, side assignment, and power domains in both the figure and the table are derived from the generator's package model and are also exported machine-readably to config/PadRing.json at every make chip build.

Table 3: Package Pins

#	Pin	I/O	Power Domain	Power Pins
1	RESETN	Input	Digital I/O (3.3 V)	VDDPST/VSSPST
2	POC	Input	Digital I/O (3.3 V)	VDDPST/VSSPST
3	VDD	Power Input	Digital Core (1.0 V)	VDD/VSS
4	VSS	Power Input	Digital Core (1.0 V)	VDD/VSS
5	P0.0	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
6	P0.1	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
7	P0.2	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
8	P0.3	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
9	P0.4	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
10	P0.5	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
11	P0.6	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
12	P0.7	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
13	VDDPST	Power Input	Digital I/O (3.3 V)	VDDPST/VSSPST
14	VSSPST	Power Input	Digital I/O (3.3 V)	VDDPST/VSSPST
15	P1.0	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
16	P1.1	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
17	P1.2	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
18	P1.3	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
19	P1.4	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
20	P1.5	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
21	P1.6	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
22	P1.7	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
23	NC	–	–	–
24	NC	–	–	–
25	NC	–	–	–
26	NC	–	–	–
27	P2.0	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
28	P2.1	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
29	P2.2	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
30	P2.3	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
31	P2.4	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
32	P2.5	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
33	P2.6	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
34	P2.7	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
35	VDD	Power Input	Digital Core (1.0 V)	VDD/VSS
36	VSS	Power Input	Digital Core (1.0 V)	VDD/VSS
37	P3.0	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
38	P3.1	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
39	P3.2	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
40	P3.3	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
41	P3.4	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
42	P3.5	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
43	P3.6	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
44	P3.7	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST

Table 3 continued on next page

Table 3 continued from previous page

#	Pin	I/O	Power Domain	Power Pins
45	VDDPST	Power Input	Digital I/O (3.3 V)	VDDPST/VSSPST
46	VSSPST	Power Input	Digital I/O (3.3 V)	VDDPST/VSSPST
47	TCK	Input	Digital I/O (3.3 V)	VDDPST/VSSPST
48	TMS	Input	Digital I/O (3.3 V)	VDDPST/VSSPST
49	TDI	Input	Digital I/O (3.3 V)	VDDPST/VSSPST
50	TDO	Output	Digital I/O (3.3 V)	VDDPST/VSSPST
51	TRSTn	Input	Digital I/O (3.3 V)	VDDPST/VSSPST
52	P4.0	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
53	P4.1	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
54	P4.2	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
55	P4.3	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
56	P4.4	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
57	P4.5	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
58	P4.6	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
59	P4.7	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
60	VDD	Power Input	Digital Core (1.0 V)	VDD/VSS
61	VSS	Power Input	Digital Core (1.0 V)	VDD/VSS
62	P5.0	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
63	P5.1	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
64	P5.2	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
65	P5.3	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
66	P5.4	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
67	P5.5	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
68	P5.6	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
69	P5.7	Input/Output	Digital I/O (3.3 V)	VDDPST/VSSPST
70	VDDPST	Power Input	Digital I/O (3.3 V)	VDDPST/VSSPST
71	VSSPST	Power Input	Digital I/O (3.3 V)	VDDPST/VSSPST
72	NC	–	–	–
73	NC	–	–	–
74	NC	–	–	–
75	NC	–	–	–
76	AVDD	Power Input	Analog (2.5 V)	AVDD/AVSS
77	AVSS	Power Input	Analog (2.5 V)	AVDD/AVSS
78	CE_0	Input/Output	Analog (2.5 V)	AVDD/AVSS
79	WE_0	Input/Output	Analog (2.5 V)	AVDD/AVSS
80	RE_0	Input/Output	Analog (2.5 V)	AVDD/AVSS
81	RE2_0	Input/Output	Analog (2.5 V)	AVDD/AVSS
82	CE_1	Input/Output	Analog (2.5 V)	AVDD/AVSS
83	WE_1	Input/Output	Analog (2.5 V)	AVDD/AVSS
84	RE_1	Input/Output	Analog (2.5 V)	AVDD/AVSS
85	RE2_1	Input/Output	Analog (2.5 V)	AVDD/AVSS
86	CE_2	Input/Output	Analog (2.5 V)	AVDD/AVSS
87	WE_2	Input/Output	Analog (2.5 V)	AVDD/AVSS
88	RE_2	Input/Output	Analog (2.5 V)	AVDD/AVSS
89	RE2_2	Input/Output	Analog (2.5 V)	AVDD/AVSS

Table 3 continued on next page

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#	Pin	I/O	Power Domain	Power Pins
90	CE_3	Input/Output	Analog (2.5 V)	AVDD/AVSS
91	WE_3	Input/Output	Analog (2.5 V)	AVDD/AVSS
92	RE_3	Input/Output	Analog (2.5 V)	AVDD/AVSS
93	RE2_3	Input/Output	Analog (2.5 V)	AVDD/AVSS
94	NC	–	–	–
95	NC	–	–	–
96	NC	–	–	–
97	NC	–	–	–
98	NC	–	–	–
99	NC	–	–	–
100	NC	–	–	–

Many of the digital I/O pins are GPIO pins. These pins can be in either primary/GPIO mode, or secondary/peripheral mode (see Section 16 on the GPIO peripherals for more details). A list of all the GPIO pins, along with their primary function, their reset-selected alternate function (AF0), and the reset state of their PxSEL, PxOUT, PxDIR, and PxREN bits, is shown in Table 4. The remaining alternate functions each pin can select (AF1–AF7) are listed separately in the alternate-function matrix, Table 5.

Table 4: GPIO Pin Functions and Reset Values

Pin	Primary Function	Alternate Function (AF0)	Reset Values			
			SEL	OUT	DIR	REN
P0.0	GPIO0	CS_FLASH	0 (Primary)	1 (High)	1 (Output)	0 (Disabled)
P0.1	GPIO1	MISO0	1 (Alternate)	0 (Low)	0 (Input)	0 (Disabled)
P0.2	GPIO2	MOSI0	1 (Alternate)	0 (Low)	0 (Input)	0 (Disabled)
P0.3	GPIO3	SCK0	1 (Alternate)	0 (Low)	0 (Input)	0 (Disabled)
P0.4	GPIO4	LFXT	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P0.5	GPIO5	HFXT	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P0.6	GPIO6	TRAP	1 (Alternate)	0 (Low)	1 (Output)	0 (Disabled)
P0.7	BOOT	–	0 (Primary)	0 (Low)	0 (Input)	1 (Enabled)
P1.0	GPIO8	CS1	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P1.1	GPIO9	MISO1	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P1.2	GPIO10	MOSI1	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P1.3	GPIO11	SCK1	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P1.4	GPIO12	TX0	1 (Alternate)	0 (Low)	1 (Output)	0 (Disabled)
P1.5	GPIO13	RX0	1 (Alternate)	0 (Low)	0 (Input)	0 (Disabled)
P1.6	GPIO14	TX1	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P1.7	GPIO15	RX1	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P2.0	GPIO16	T0CMP0	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P2.1	GPIO17	T0CMP1	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P2.2	GPIO18	T0CAP0	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P2.3	GPIO19	T0CAP1	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P2.4	GPIO20	T1CMP0	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P2.5	GPIO21	T1CMP1	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P2.6	GPIO22	T1CAP0	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)

Table 4 continued on next page

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Pin	Primary Function	Alternate Function (AF0)	Reset Values			
			SEL	OUT	DIR	REN
P2.7	GPIO23	T1CAP1	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P3.0	GPIO24	SDA0	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P3.1	GPIO25	SCL0	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P3.2	GPIO26	SDA1	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P3.3	GPIO27	SCL1	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P3.4	GPIO28	DTP0	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P3.5	GPIO29	DTP1	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P3.6	GPIO30	DTP2	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P3.7	GPIO31	DTP3	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P4.0	GPIO32	–	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P4.1	GPIO33	–	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P4.2	GPIO34	–	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P4.3	GPIO35	–	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P4.4	GPIO36	–	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P4.5	GPIO37	–	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P4.6	GPIO38	–	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P4.7	GPIO39	–	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P5.0	GPIO40	–	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P5.1	GPIO41	–	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P5.2	GPIO42	–	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P5.3	GPIO43	–	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P5.4	GPIO44	–	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P5.5	GPIO45	–	0 (Primary)	0 (Low)	0 (Input)	0 (Disabled)
P5.6	GPIO46	–	0 (Primary)	0 (Low)	0 (Input)	1 (Enabled)
P5.7	GPIO47	–	0 (Primary)	0 (Low)	0 (Input)	1 (Enabled)

When a pin is in alternate function (peripheral) mode (its PxSEL bit = 1), its 3-bit PxAFS field selects which of up to eight alternate functions (AF0–AF7) drives the pad. Table 5 lists, for every GPIO pin, the exact function selected by each PxAFS value. Plane 0 is the pin's AF0 (legacy secondary) function and is the reset selection; planes marked **Hi-Z** have no function assigned for that pin and configure the pad as a high-impedance input. The superscript on each function denotes its direction (ⁱⁿ input, ^{out} output, ^{io} bidirectional), and a [†] marks each pin's reset plane (its PxAFS reset value). While a pin is in GPIO (primary) mode the selected plane has no effect on the pad, but it still routes relocatable peripheral *inputs* (see the GPIOx chapter, Section 16).

Table 5: GPIO Alternate Function Selection (PxAFS)

Pin	PxAFS field value (selected AF plane)							
	0 (AF0)	1	2	3	4	5	6	7
P0.0	CS_FLASH ^{out†}	TX0 ^{out}	TX1 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}
P0.1	MISO0 ^{io†}	TX1 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}
P0.2	MOSI0 ^{out†}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}	TX0 ^{out}
P0.3	SCK0 ^{out†}	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}	TX0 ^{out}	TX1 ^{out}
P0.4	LFXT ^{io†}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}	TX0 ^{out}	TX1 ^{out}	T0CMP0 ^{out}
P0.5	HFXT ^{io†}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}	TX0 ^{out}	TX1 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}
P0.6	TRAP ^{out†}	SCK1 ^{out}	MOSI1 ^{out}	TX0 ^{out}	TX1 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}

Table 5 continued on next page

Table 5 continued from previous page

Pin	PxAFS field value (selected AF plane)							
	0 (AF0)	1	2	3	4	5	6	7
P0.7	Hi-Z [†]	MOSI1 ^{out}	TX0 ^{out}	TX1 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}
P1.0	CS1 ^{in†}	T0CMP0 ^{out}	TX1 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}
P1.1	MISO1 ^{io†}	T0CMP1 ^{out}	T0CMP0 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}	TX0 ^{out}
P1.2	MOSI1 ^{io†}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	TX0 ^{out}	TX1 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}
P1.3	SCK1 ^{io†}	T1CMP1 ^{out}	MOSI1 ^{out}	TX0 ^{out}	TX1 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}
P1.4	TX0 ^{out†}	SDA1 ^{io}	SCK1 ^{out}	MOSI1 ^{out}	TX1 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}
P1.5	RX0 ^{io†}	SCL1 ^{io}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}	TX0 ^{out}	TX1 ^{out}	T0CMP0 ^{out}
P1.6	TX1 ^{out†}	SDA0 ^{io}	TX0 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}
P1.7	RX1 ^{io†}	SCL0 ^{io}	MOSI1 ^{out}	TX0 ^{out}	TX1 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}
P2.0	T0CMP0 ^{out†}	TX1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}	TX0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}
P2.1	T0CMP1 ^{out†}	RX1 ^{io}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}	TX0 ^{out}	TX1 ^{out}
P2.2	T0CAP0 ^{in†}	SDA1 ^{io}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}
P2.3	T0CAP1 ^{in†}	SCL1 ^{io}	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}	TX0 ^{out}
P2.4	T1CMP0 ^{out†}	TX0 ^{out}	T0CMP1 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}	TX1 ^{out}	T0CMP0 ^{out}
P2.5	T1CMP1 ^{out†}	RX0 ^{io}	TX0 ^{out}	TX1 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}	SCK1 ^{out}
P2.6	T1CAP0 ^{in†}	SDA0 ^{io}	SCK1 ^{out}	MOSI1 ^{out}	TX0 ^{out}	TX1 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}
P2.7	T1CAP1 ^{in†}	SCL0 ^{io}	MOSI1 ^{out}	TX0 ^{out}	TX1 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}
P3.0	SDA0 ^{io†}	T0CAP0 ⁱⁿ	TX0 ^{out}	TX1 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}
P3.1	SCL0 ^{io†}	T0CAP1 ⁱⁿ	TX1 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}
P3.2	SDA1 ^{io†}	T1CAP0 ⁱⁿ	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}
P3.3	SCL1 ^{io†}	T1CAP1 ⁱⁿ	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}	TX0 ^{out}
P3.4	DTP0 ^{io†}	T0CMP0 ^{out}	TX0 ^{out}	TX1 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}
P3.5	DTP1 ^{io†}	T0CMP1 ^{out}	RX0 ^{io}	T0CMP0 ^{out}	T1CMP0 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}
P3.6	DTP2 ^{io†}	T1CMP0 ^{out}	T0CMP0 ^{out}	T0CMP1 ^{out}	T1CMP1 ^{out}	SCK1 ^{out}	MOSI1 ^{out}	MISO1 ^{io}
P3.7	DTP3 ^{io†}	T1CMP1 ^{out}	T0CMP1 ^{out}	T1CMP0 ^{out}	SCK1 ^{out}	MOSI1 ^{out}	TX0 ^{out}	TX1 ^{out}
P4.0	Hi-Z [†]	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P4.1	Hi-Z [†]	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P4.2	Hi-Z [†]	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P4.3	Hi-Z [†]	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P4.4	Hi-Z [†]	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P4.5	Hi-Z [†]	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P4.6	Hi-Z [†]	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P4.7	Hi-Z [†]	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P5.0	Hi-Z [†]	NFC_RF_CLK ⁱⁿ	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P5.1	Hi-Z [†]	NFC_RF_RX ⁱⁿ	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P5.2	Hi-Z [†]	NFC_FIELD_DETECT ⁱⁿ	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P5.3	Hi-Z [†]	NFC_RF_TXMOD ^{out}	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P5.4	Hi-Z [†]	NFC_RF_TX_EN ^{out}	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P5.5	Hi-Z [†]	NFC_AFE_EN ^{out}	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P5.6	Hi-Z [†]	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
P5.7	Hi-Z [†]	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z

3 Address Space

0x00000	Boot ROM Size = 8 KiB	Shared window (all harts, arbitrated)
0x01FFF 0x02000	<i>Unmapped (reads zero)</i>	
0x03FFF 0x04000	Peripheral registers Size = 4 KiB	Shared window (all harts, arbitrated)
0x04FFF 0x05000	CLINT Size = 4 KiB	
0x05FFF 0x06000	Mutex bank Size = 4 KiB	
0x06FFF 0x07000	IRQ router Size = 4 KiB	
0x07FFF 0x08000	Private TCM Size = 8 KiB per hart	Private to each hart (not arbitrated)
0x09FFF 0x0A000	<i>Unmapped (reads zero)</i>	
0x0BFFF 0x0C000	NPU staging RAM Size = 16 KiB	Shared window (all harts, arbitrated)
0x0FFFF 0x10000	Shared RAM Size = 64 KiB	
0x1FFFF 0x20000	TCM aperture 0 Size = 16 KiB	
0x23FFF 0x24000	TCM aperture 1 Size = 16 KiB	Shared window (hart 0's read-only view of each hart's TCM)
0x27FFF 0x28000	TCM aperture 2 Size = 16 KiB	
0x2BFFF 0x2C000	TCM aperture 3 Size = 16 KiB	
0x2FFFF 0x30000	TCM aperture 4 Size = 16 KiB	
0x33FFF 0x34000	<i>Unmapped (reads zero)</i>	
0x3FFFF 0x40000	SPI flash (XIP) read only $(addr + SPI0FOS) \bmod 2^{24}$	External SPI flash (hart 0 only)
0xFFFFFFF		

Figure 4: Castalia address space: one 32-bit column, drawn once for all five harts. Every row is a decoded region and the column is gapless; the grey rows are unmapped and read zero. The braces down the right name the *access class* of each band rather than the device in it, because two views overlap in one address space: the shared window, which every hart reaches through the arbiter, and the private band carved out of it, where the identical address selects a different 8 KiB of tightly-coupled RAM in each hart. The aperture brace is the one sanctioned crossing between the two, and it is read-only and hart 0's alone. Every region, size and address here is derived from this build's memory map, the same map the headers and linker scripts are generated from; the text below walks the column.

The 32-bit memory bus on the MCU is in the von Neumann architecture. All addressable memory elements (i.e. the ROM, RAM, and memory mapped registers) are all located on a single bus. A pointer can access any of these locations in memory.

The ROM cell contains boot code, which is read-only. Each SRAM cell can be read or written to, and may contain both program and variable information. The chip carries no on-die non-volatile memory: it boots from an *external* SPI flash device, and hart 0, and only hart 0, can additionally read that flash directly, and execute in place from it, through the extended-flash window at `0x40000` and above (the last region in Figure 4; see Section 17.10). At reset every hart fetches from the boot ROM at `0x00000` and dispatches on its hart ID: hart 0 copies the application image out of the flash into memory and enters it, and the remaining harts park until hart 0 loads and launches them (Section 4.3). Each SRAM and ROM cell can be individually powered off or on using the SYSTEM peripheral. When powered off, the cells use less power and cannot be read or written to. SRAM cells that are powered off do not retain their previous data, and the data becomes undefined.

The interrupt vector table (IVT) begins at address `0x8000` and contains jump instructions to the interrupt service routines (ISRs). When an interrupt occurs, the CPU directly vectors to the corresponding entry in the IVT, which then jumps to the appropriate ISR. Each IVT entry is 4 bytes and typically contains a RISC-V jump instruction (e.g., `j handler`). The IVT can be initialized and configured in software, allowing flexible interrupt routing and handler assignment at runtime.

Figure 4 is a single address column drawn for all five harts, and its braces name who reaches each region. Everything below `0x40000` other than the private band is the *shared window*: every hart reaches it through the arbiter, and all five harts see the same memory there (Section 4). The private band is the exception: at those addresses a hart reaches its own tightly-coupled memory (TCM) instead of the shared bus, so the same address names a different 8 KiB of RAM in each hart, and no hart can reach another's directly. The TCM apertures are the one sanctioned way across that line: hart 0, the orchestrator, reads hart h 's private TCM through the aperture at $0x20000 + h \times 8 \text{ KiB}$. The apertures are read-only, and any other hart reading them gets zeros (Section 4). The extended-flash window is likewise hart 0's alone; the other harts' accesses in that range read zeros without stalling.

4 Multi-Core Architecture

Castalia is a five-hart (five-core) multiprocessor built from five identical VestaRV cores. Each hart is a complete, single-issue, in-order RV32IMAC processor with its own private *tightly-coupled memory* (TCM), so the common case (instruction fetch and private data access) never contends with the other harts and runs at full core clock. Everything else in the low address space is *shared*: a single boot ROM, all peripherals, the inter-processor interrupt hardware (CLINT), a hardware mutex bank, an interrupt router, the NPU staging RAM, and 64 KiB of bulk RAM, all reached through one serializing round-robin arbiter. The main features of the multi-core architecture are listed below:

- 5 identical VestaRV harts, each identified by the read-only `mhartid` CSR (address `0xF14`)
- One shared boot ROM at `0x0`: all five harts reset to PC `0x0` and dispatch on `mhartid` (see Section 4.3)
- 16 KiB of private TCM per hart at `0x8000–0xBFFF` (the same local address in every hart), holding each hart’s vector table, code, data, and stack
- All peripherals shared by all five harts at their legacy `0x4000`-page addresses, with the arbiter guaranteeing register-access atomicity
- 64 KiB of shared bulk RAM at `0x10000–0x1FFFF` for locks, mailboxes, staged tile programs, and inter-hart data
- CLINT with per-hart software interrupts (inter-processor interrupts) and a shared 64-bit `mtime` with per-hart compare registers
- 16 single-instruction hardware mutexes (MUTEX bank)
- Claim/complete peripheral interrupt delivery (IRQROUTER): any peripheral interrupt can be routed to any hart over one external-interrupt wire per hart, with exactly-once claiming
- LR/SC and AMO atomic instructions supported to shared RAM, including sub-word shared stores (byte lanes)
- Instruction fetch from shared memory is supported (the boot ROM is fetched this way by construction); parked harts sleep via the custom `sleep` instruction and are woken by a software interrupt; no busy-wait power burn

4.1 Harts and Roles

All five harts are instances of the same *hart tile*: an unmodified VestaRV core plus its own private TCM, replicating the proven single-core memory path, with every connection to the rest of the chip registered at the tile boundary. Since every hart sees the same address map (shared boot ROM, shared peripherals, private TCM at the same local address), the five harts are architecturally identical and code is hart-portable by construction.

Hart 0 is additionally the *management hart*. It is the only hart attached to the SPI-flash boot path and the extended flash region ($\geq 0x40000$), and by convention it owns system management: the SYSTEM controller’s clock configuration registers reprogram the master clock that all five harts run on, so only the management hart may touch them, and only after quiescing the other harts (see the SYSTEM chapter). Peripheral interrupts are uniform across harts: every hart, hart 0 included, routes and receives them through its IRQROUTER row and the claim/complete mechanism (see the IRQROUTER chapter).

Hart 0 (mhartid zero, the management and boot hart) is additionally the *orchestrator*, and it is the one hart that is not a corner tile. The channel harts, mhartid 1 through 4, are hardened *hart tile* macros at the corners of the die; the orchestrator is soft logic placed in the centre band alongside the shared fabric, with its own 16 KiB TCM at the same private 0x8000–0xBFFF window. It is architecturally identical to the tiles (same ISA, same private memory map, same mhartid discovery) and differs in exactly three ways that software can observe:

- **It is always on, and every other hart is not.** The orchestrator sits outside the switched-rail power fabric entirely: there are no header switches for the centre band, so PWRCCR’s bit 0 reads 0 and ignores writes, as it always has. What changes in this configuration is the other side of that sentence: *every* channel hart 1–4 now has a gate bit, including the corner tile that a four-hart chip would have called hart 0 and kept always-on. A blanket PWRCCR write therefore gates all 4 channel tiles and leaves the orchestrator running (see the PWRCTRL chapter).
- **It boots the chip and starts the others.** Boot mastership is the orchestrator’s, because the orchestrator is hart 0: it is the hart attached to the SPI-flash boot path and the extended flash region, it stages each channel hart’s image into that hart’s loader mailbox row and ignites it with a CLINT software interrupt (Section 4.3), and it holds the management privilege from reset rather than receiving it in a hand-over.
- **It can read every hart’s private TCM.** Each hart’s TCM is also visible read-only in the shared window, hart h at $0x20000 + 0x4000 \times h$ (the orchestrator’s own included, at 0x20000), so the indexing is uniform. Only the management hart may read them; any other hart reads zero and has its write dropped, and writes are dropped for everyone (the windows are read-only by design: a write path into a running core’s memory is a coherence hazard). A window belonging to a power-gated tile completes normally and returns zeros, so software checks the PWRCTRL sequencer state before trusting what it reads; zero is a legal TCM value.

The intended firmware model follows: the orchestrator boots the chip, starts and coordinates the channel harts, owns the front-end and its interrupts, inspects any channel hart’s working memory through its TCM window, and stays up across any power-gating of the tiles.

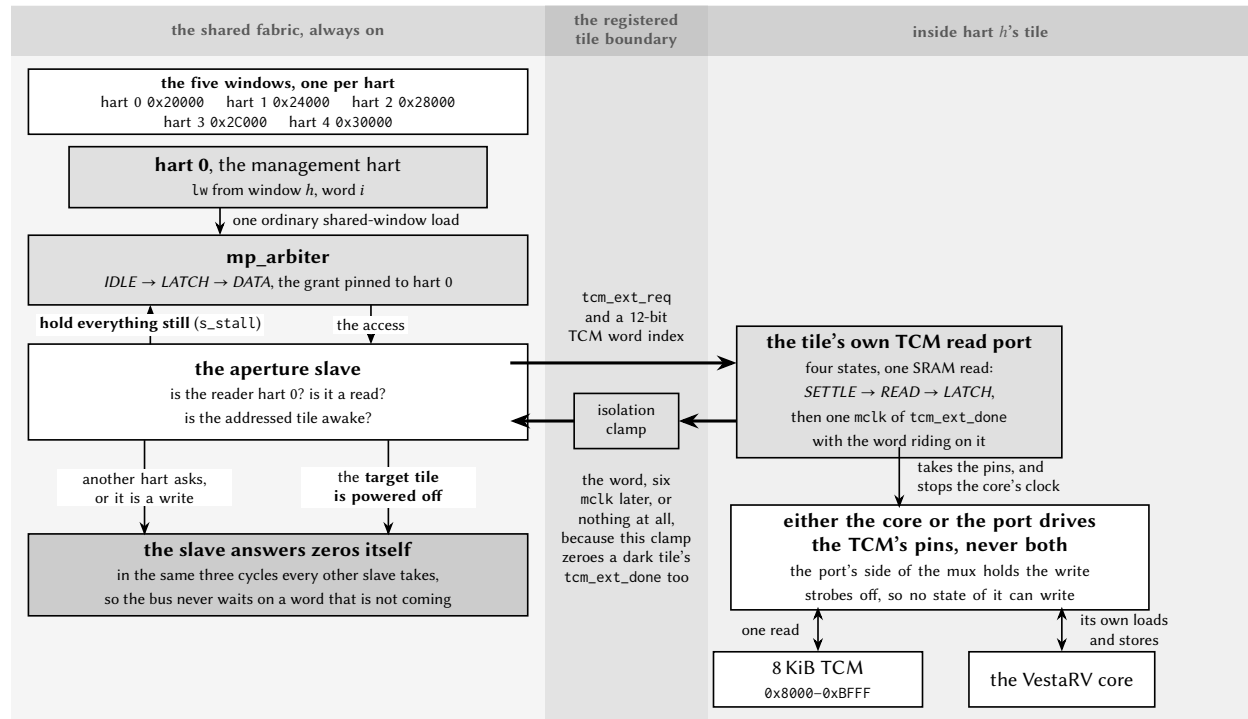
Figure 5 follows one aperture read through the hardware, because the window is not a second port onto a shared memory: it is a request that leaves the shared fabric, borrows the target tile’s own memory for four cycles, and comes back.

Software discovers which hart it is running on by reading the mhartid CSR:

```
csrr    t0, mhartid      # t0 = 0 .. (number of harts - 1)
```

4.2 The Shared Window

Everything below the extended flash region except each hart’s private TCM is behind the arbiter. Every hart sees the same physical devices at the same addresses:



While the port has those pins the core's clock is gated off for seven edges, and the Q shadow keeps showing it the word it last read, so the frozen core can never mistake the aperture's word for its own.

Figure 5: One read through a TCM aperture. The management hart issues an ordinary shared-window load; the aperture slave decides, in the cycle the arbiter presents it, which of three answers this access gets. Two of them the slave produces itself, in the three cycles every other shared-window slave takes: a read by any other hart, and any write, are answered with zeros, and so is a read of a tile that is powered off: the isolation clamp that zeroes a dark tile's data also zeroes its tcm_ext_done , so a slave that simply waited for the tile would park the management hart on the bus forever. The third answer is a real transaction: the slave holds the arbiter still with s_stall (it is the only slave that does, because a tile read takes six mclk and the arbiter's walk is built for a slave that answers in one) and drives the addressed tile's read port. Inside the tile that port takes the TCM's pins for four cycles, with the core's clock gated off and its write strobes held inactive, which is what makes the window read-only in the hardware rather than by convention. Software still checks the PWRCTRL state before trusting a word: zero is a legal TCM value.

Address range	Device	Notes
0x00000–0x01FFF	Boot ROM (8 KiB)	Shared, read-only; all harts reset here
0x02000–0x03FFF	Unmapped	The ROM is smaller than its page; reads return zero
0x04000–0x04FFF	Peripheral slots	16 slots of 256 bytes, legacy numbering
0x05000–0x05FFF	CLINT	IPIs (MSIPx), MTIME/MTIMECMPx
0x06000–0x06FFF	MUTEX bank	16 hardware mutexes
0x07000–0x07FFF	IRQROUTER	Per-hart routing rows + CLAIM/COMPLETE
0x0C000–0x0FFFF	NPU staging RAM (16 KiB)	Vector storage; NPU-owned during a run
0x10000–0x1FFFF	Shared bulk RAM (64 KiB)	Locks, mailboxes, inter-hart data

(0x08000–0x0BFFF is each hart’s **private** TCM and never reaches the arbiter.) The sixteen peripheral slots at 0x4000 keep the original single-core VestaRV slot numbering (GPIO0 at 0x4000, UART0 at 0x4400, SYSTEM at 0x4900, and so on), so single-core driver code runs unchanged; the difference is that every hart can now reach every peripheral.

Instruction fetch from the shared window is supported: the boot flow depends on it, and programs may execute directly from shared RAM. One caveat: compressed (RVC) instructions in shared-window code are not currently validated; code intended to execute from shared memory should be built without the C extension. Code in the private TCM has no such restriction.

4.2.1 Arbitration and Stalling

A round-robin arbiter serializes all shared-window transactions. The arbiter grants one hart at a time and holds the grant until that hart’s whole transaction completes, then advances the round-robin pointer, so no hart can be starved. While a hart waits for its grant, its clock is stalled transparently; software never sees a partial transaction and needs no special code for shared accesses. Because a stalled hart contributes nothing until its turn comes, frequently-pollled data structures (spin locks in particular) should use a backoff strategy (see Section 4.4).

An AMO (atomic read-modify-write) instruction holds the arbiter grant across both halves of its read-modify-write pair, so AMOs to shared RAM are atomic with respect to all other harts. A shared-window access costs a few master-clock cycles uncontended (versus a single cycle to the private TCM), which is why per-hart code and stacks belong in the TCM. Figure 6 shows one uncontended read at the arbiter’s own pins.

One slave does not fit that walk. A read through a TCM aperture has to leave the shared fabric, borrow the target tile’s own memory and come back, which takes six mclk, so the aperture slave holds the arbiter in *LATCH* with *s_stall* until the word is actually in its hand. It is the only slave in the design that does this, and the only one that can: nothing else could have been granted meanwhile, because the arbiter had already pinned the bus to this transaction. Figure 7 is the same set of pins as Figure 6, on that read.

4.3 Boot and Tile Loading

All five harts come out of reset at PC 0x0 and fetch the shared boot ROM through the arbiter. The ROM immediately dispatches on *mhartid*; the whole flow is summarized in Figure 8:

Hart 0 runs the full boot sequence, exactly as on the single-core chip: it configures GPIO0/SPI0, copies the program from SPI flash into the load window 0x8000–0xFFFFC (its TCM plus the NPU staging RAM), and jumps to the program at 0x8200. Before any tile can be released, the boot ROM zeroes the shared-RAM mailbox region 0x10000–0x107FF: shared RAM powers up with *unknown* contents on silicon, and this write-before-read guarantee is what makes “mailbox reads as zero” checks safe.

Harts 1–4 set up a minimal interrupt context (a stack at the top of their TCM and a software-interrupt vector) and park in low-power sleep. A parked tile wakes only when hart 0 (or any running program)

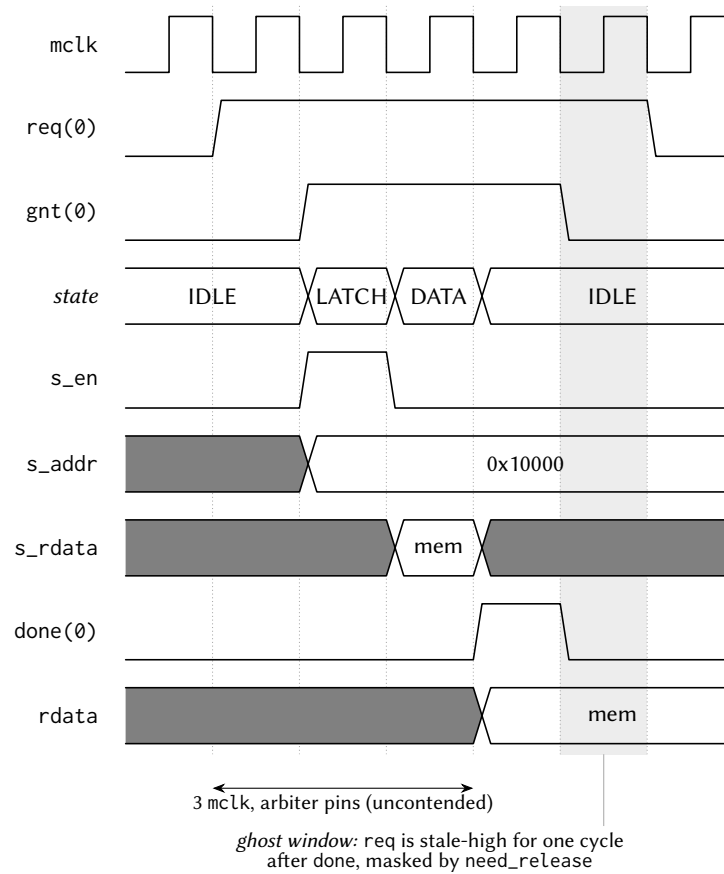


Figure 6: One uncontended shared-window read, at the `mp_arbiter` pins. The arbiter walks `IDLE` → `LATCH` → `DATA`: the transaction is presented to the slave for one cycle, the synchronous slave returns data the following cycle, and `done` and `rdata` are registered together at the edge leaving `DATA`, three `mclk` from an observed `req`. The hart itself sees about two cycles more, one in each direction across the registered tile boundary. In the shaded cycle the master's `req` is still high although its transaction is over; the arbiter masks that stale request until it is observed low, which is what stops a completed access from being served twice.

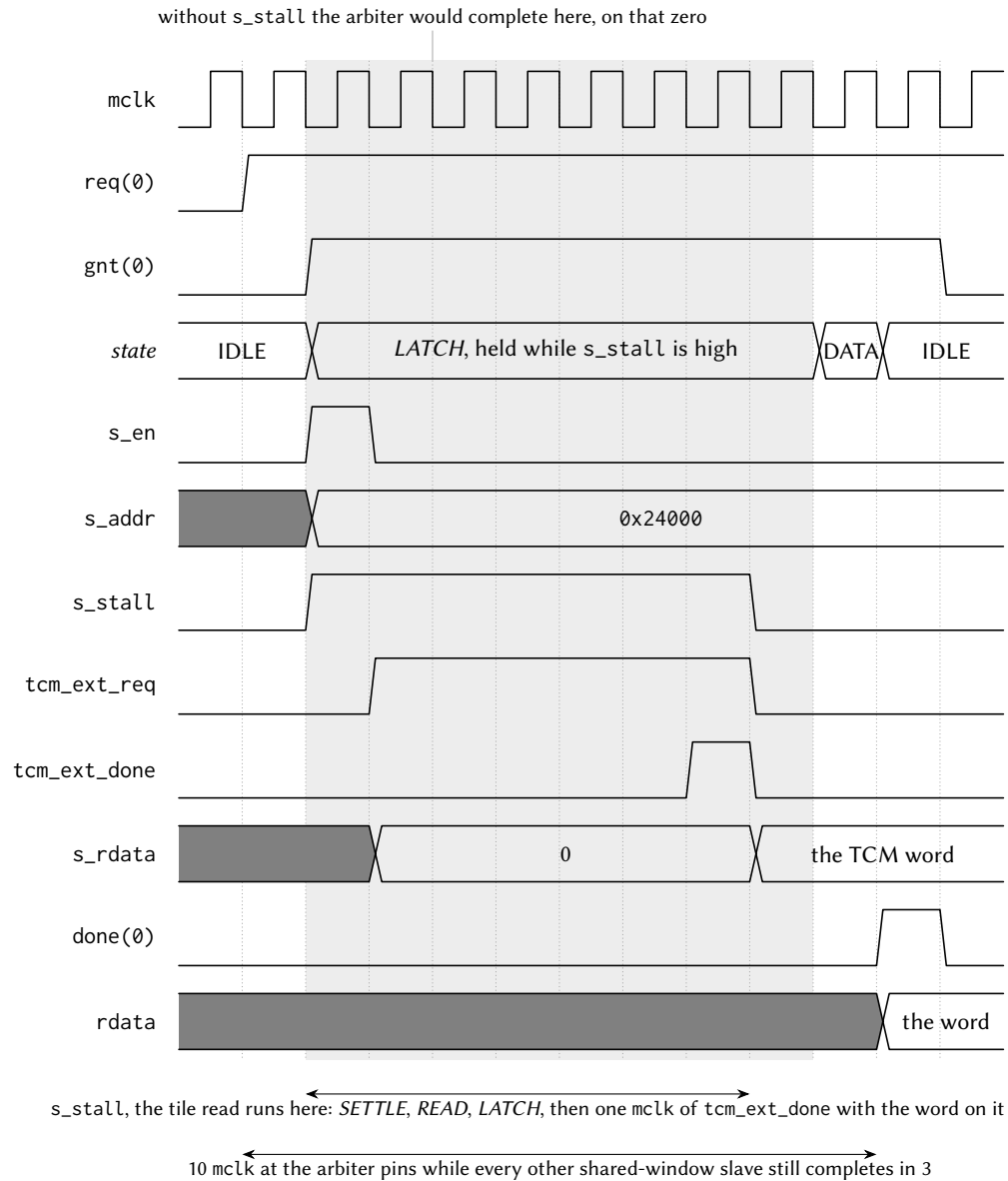


Figure 7: One stalled shared-window read: the management hart through a TCM aperture, at the same `mp_arbiter` pins as Figure 6. Up to the shaded region it is the ordinary walk: the grant, the one-cycle `s_en` strobe, the address presented. The aperture slave raises `s_stall` in that same cycle, and the arbiter stays in *LATCH*: grant pinned, address still presented, only the completion moves. Underneath, the tile's read port takes one SRAM read and returns the word on a single `mclk` of `tcm_ext_done`; the slave captures it, drops `s_stall`, and the arbiter finishes the transaction it was holding. Without the stall it would have completed on the marked cycle, handing back the zero the slave had not filled in yet.

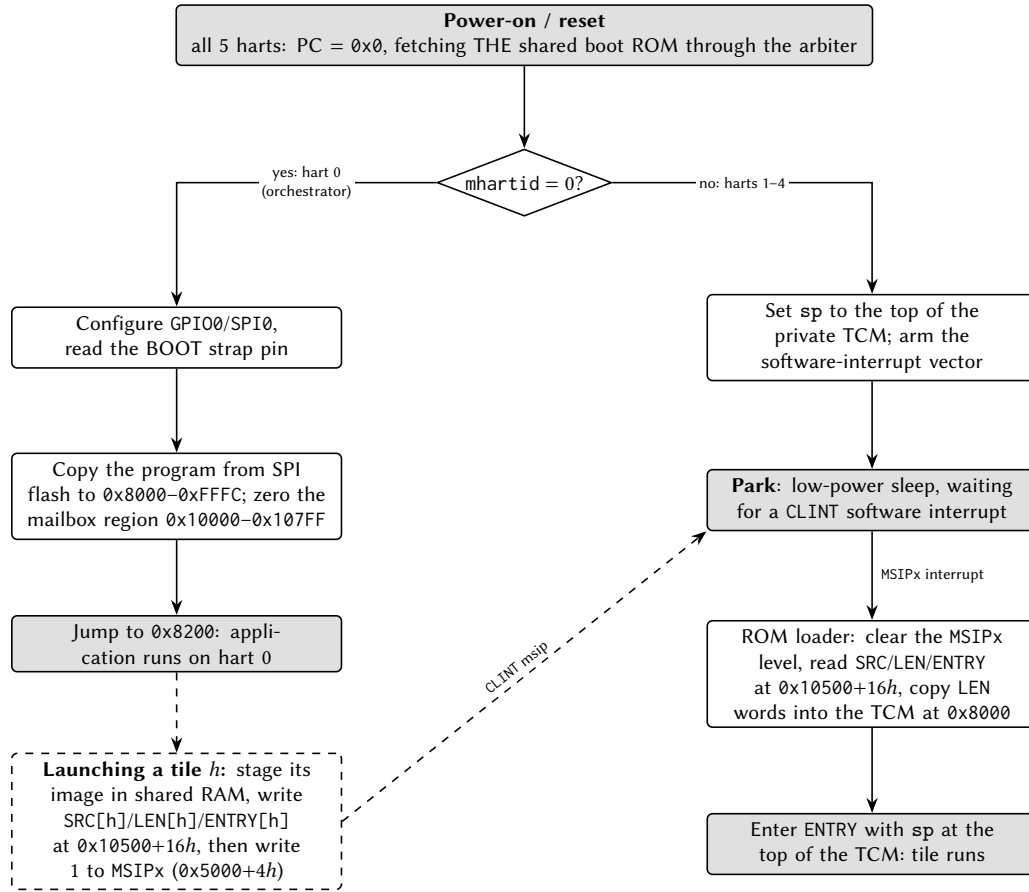


Figure 8: Boot and tile-loading flow. All harts execute the same shared boot ROM; hart 0 runs the legacy SPI-flash boot while every other hart parks in low-power sleep until it is loaded and released through its loader mailbox row and a CLINT software interrupt.

sends it a CLINT software interrupt. On wake, the ROM-resident loader reads the tile’s *loader mailbox row* in shared RAM:

Mailbox	Address	Meaning
SRC[h]	$0x10500 + 16h + 0$	Word-aligned source address in shared space
LEN[h]	$0x10500 + 16h + 4$	Words to copy to the tile’s TCM at $0x8000$ (0 = none)
ENTRY[h]	$0x10500 + 16h + 8$	PC the tile enters with

The loader clears the tile’s own MSIP level, copies LEN[h] words from SRC[h] into the tile’s private TCM at $0x8000$ (a LEN of 2048 loads the full 8 KiB image: vector table, code, and interrupt handlers included), and enters ENTRY[h] with the stack pointer set to the top of the tile’s TCM and all other registers undefined.

The canonical release sequence for hart h is therefore: stage the tile’s program somewhere in shared RAM, write SRC[h] and LEN[h] and ENTRY[h], then write 1 to the hart’s MSIP register ($0x5000 + 4h$). Because the mailbox row is consumed by the ROM only at wake time, the row must be complete *before* the MSIP write. Software interrupts sent to a tile *after* it has been loaded are handled by the loaded program’s own vector table, so the same CLINT mechanism serves both boot-time loading and run-time inter-processor interrupts.

Stacks: each hart needs its own stack in its own private TCM. When an interrupt is taken, the hardware pushes the return program counter at $sp - 4$ (see Section 10), so every hart, including a freshly loaded tile

hart, must have a valid stack pointer before enabling interrupts or sleeping. The boot ROM guarantees this for parked and freshly-entered tiles; application code keeps the convention of placing each hart's stack at the top of its TCM, growing downward.

4.3.1 Harvested (field-powered) boot

In configurations wired for field power, hart 0's boot path has a second entry, selected by a hardware strap sampled at reset and reported by the PWRCTRL controller. Before any SPI or flash activity, hart 0 polls the strap-valid flag and, if the harvested-boot strap is set, branches to a ROM-resident service path instead of the SPI-flash download described above. This path is built for a chip running on harvested field energy, where the multi-kilobyte flash copy is the single most power-hungry cold-start step and is therefore skipped entirely.

The harvested path keeps both MCLK and SMCLK on HFXT (the SPI path's mid-boot SMCLK-to-LFXT switch does not run, so the clock configuration is written explicitly rather than inherited), then power-gates every tile hart through PWRCTRL (the tiles are already ROM-parked in WFI, the sanctioned quiesced state, so gating them is safe at any hart count). It configures the port-6 GPIO alternate-function plane for the six off-die NFC analog-front-end wires, provisions the NFC tag identity (a ROM-default UID, ATQA and SAK) and a short status payload, and arms the NFC block in LISTEN mode with auto-read enabled, so the hardware answers Type-2 read commands with no firmware in the loop. Finally it arms its own software-interrupt vector with a harmless clear-and-return handler (a stray MSIP must not disturb the parked hart), divides MCLK down (the parked service loop needs no speed, and dynamic power scales with the divide), and sleeps.

On a configuration built without the NFC block, the NFC register writes land in an unused alias of the MUTEX page and are harmless; the path is written to never *read* an NFC register, because a read of that alias would claim a mutex.

The exact sequence, in the order the ROM executes it (register addresses in the peripheral chapters):

1. **Common entry (shared with the SPI path):** interrupts off, all memory blocks powered, stack pointer set, and the shared-RAM mailbox region 0x10000–0x107FC zeroed (the write-before-read contract). At a 24 MHz MCLK this zeroing, 512 word writes through the shared-bus arbiter, dominates the harvested cold-start (about 625 μ s of the roughly 690 μ s from gate release to sleep).
2. **Strap branch:** poll PWRSTS until PWSTRAPVLD, then branch on PWSTRAP. (The sample completed a handful of cycles after reset release, long before this code runs; the poll is cheap insurance.)
3. **Clock story, explicitly:** write SYSCLOCKR = 0; both MCLK and SMCLK stay on HFXT. The SPI path's mid-boot switch of SMCLK to LFXT never runs on this path, so the choice is written rather than inherited.
4. **Gate the tiles:** write all-ones to PWRCTRL PWRCR. The register masks the write to the tile bits that exist, so the same ROM gates every tile hart at any hart count; the tiles are parked in their boot-ROM WFI, the sanctioned quiesced state.
5. **Pin mux:** select the alternate-function planes for the six off-die NFC AFE wires on port 6 pins 0–5 (PxSEL, then the AF-select fields). Pins 6 and 7 (the strap and PGOOD) are untouched and stay plain inputs.
6. **Provision the tag:** write the ROM-default identity (UID 0xC0FFEE01, ATQA 0x0044, SAK 0x00), point the payload index register at byte 0 with auto-increment, and write a three-byte status payload: 'H', 'V', then the live PWRSTS snapshot, the honest “sensor record” of the base chip. A fielded product replaces all of this by relaunching with real firmware.

7. **Arm LISTEN:** a *byte* store of NFCEN|LISTEN to the NFC control register's low lane. A full-word store would also write the second byte lane and clear the auto-read enable, whose reset default is set; auto-read is what lets the hardware answer Type-2 READs with the processor asleep.
8. **Stray-interrupt guard:** arm interrupt-vector slot 83 (the software interrupt) in hart 0's own TCM with a return stub, so an unexpected msip is cleared and returns the hart to sleep instead of vectoring into uninitialized memory.
9. **Slow down and sleep:** divide MCLK by eight (CLKDIVCR), then sleep. The clock reconfiguration is legal here because every other bus master is gated. From this point the NFC hardware serves readers autonomously; a field arrival can be taken as an interrupt only if firmware later opts in.

4.4 Synchronization

Castalia offers three synchronization mechanisms, from cheapest to most flexible. Figure 9 summarizes how to choose between them:

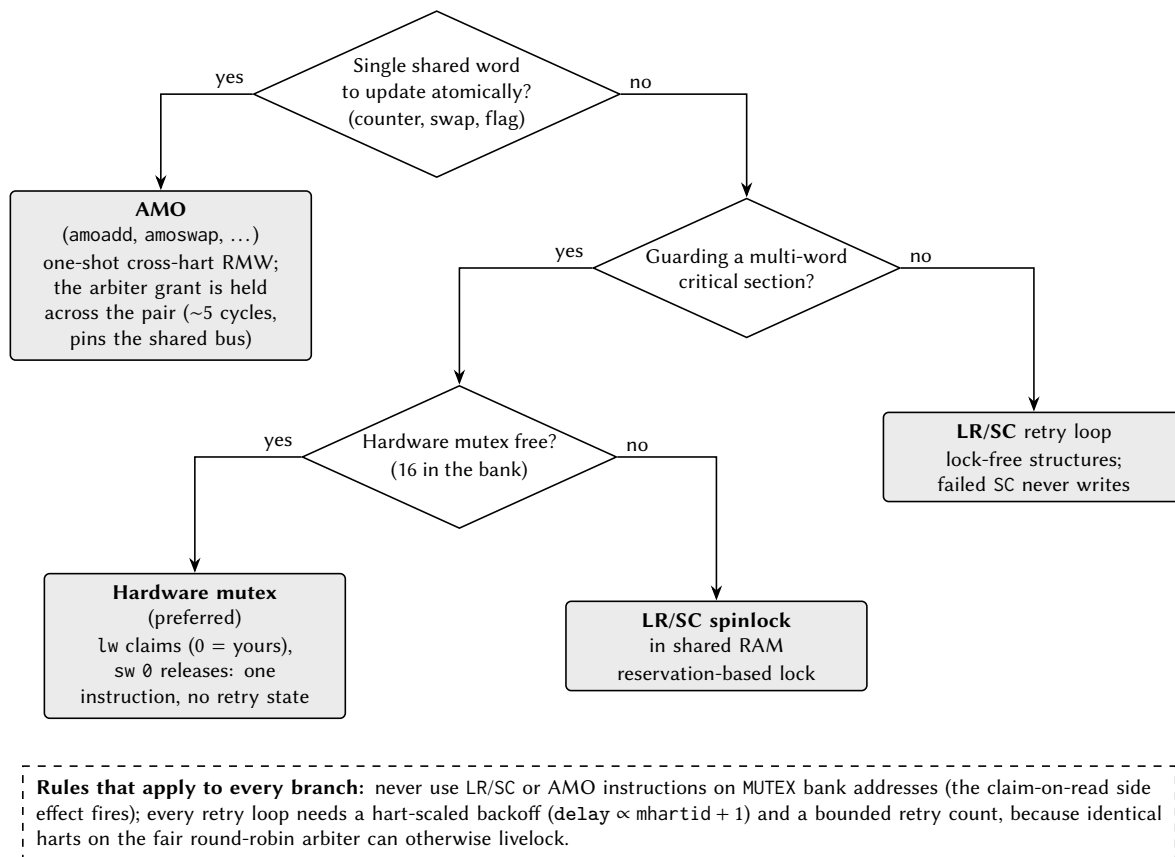


Figure 9: Choosing a synchronization primitive. The dashed box lists the two rules that apply to every branch.

1. **Hardware mutexes** (MUTEX bank): a single load claims a free mutex; a single store releases it. No retry loop hardware state, no reservation. This is the preferred lock for most uses. See the MUTEX chapter.

2. **LR/SC spinlocks in shared RAM:** standard RISC-V reservation-based locks. A failed SC has its write suppressed by the reservation unit, so it is safe under contention.
3. **AMO instructions to shared RAM:** atomic add, swap, and friends; the arbiter locks the grant across the read-modify-write pair.

Two rules apply to all of them:

- **Never** use LR/SC or AMO instructions on MUTEX bank addresses; hardware mutexes are claimed with plain loads and released with plain stores only.
- Under contention, spin with a *hart-scaled backoff*: delay between retries by an amount proportional to `mhartid` (for example `delay = k · (mhartid + 1)`) and bound the retry count. Multiple harts hammering the arbiter in lockstep can otherwise livelock a naive retry loop.
- Additionally, do not access the NPU staging RAM (`0xC000–0xFFFF`) while an NPU run may be active; poll the NPU's busy flag first (see the NPU chapter).

There are no data caches, so there is no cache-coherence protocol to consider: a shared-RAM write is globally visible as soon as the arbiter completes the transaction.

5 Privileged Architecture

Every VestaRV hart executes at the RISC-V *machine* (M) privilege level, the level that owns the whole address space and every control register. What differs between configurations is how a hart *takes a trap* (an interrupt or an exception) and whether it can drop to the *user* (U) privilege level and be fenced off from memory it does not own.

Two trap architectures exist in this family. The **legacy** architecture is the one described in Section 10: a table of jump instructions at `0x8000`, a hardware push of the return address, and the `retirq` instruction to return. It is inherited from the single-core VestaRV chip, it is what the boot ROM, the interrupt dispatcher and every shipped example use, and it is active out of reset in *every* configuration. The **standard** architecture is the one written in the RISC-V privileged specification: a single trap entry point in `mtvec`, the cause and value registers `mcause/mtval`, a saved program counter in `mepc`, an interrupt-enable stack in `mstatus`, and the `MRET` instruction to return. It is a configuration option (`priv.trapCsr` in Table 1), and when it is built the two architectures *coexist* in the same hart, selected at run time by one bit.

This configuration includes the standard trap architecture; the sections below document it.

5.1 Machine Information Registers

Five read-only registers identify the hart and the implementation. They are present in *every* configuration (they are not part of the trap architecture and are not affected by `priv.trapCsr`), and reading any of them is always legal, at machine level, in either delivery mode.

Address	Register	Value
0xF11	<code>mvendorid</code>	0x00000000: no commercially registered JEDEC vendor.
0xF12	<code>marchid</code>	0x00000000: no architecture ID assigned by the RISC-V registry.
0xF13	<code>mimpid</code>	0x00000000: no implementation revision encoded.
0xF14	<code>mhartid</code>	The hart's own index, 0 to 4 (Section 4). The only one of the five that differs between harts, and the only one with a non-zero value.
0xF15	<code>mconfigptr</code>	0x00000000: no configuration data structure in memory.

Zero is a defined answer, not a missing one. The privileged specification requires all five registers to exist and gives zero a specific meaning in each: it says “*not implemented*” or “*not assigned*”, which is the truthful answer for a non-commercial teaching chip. Discovery software must therefore read them and interpret zero, rather than treat a zero as evidence that the register is absent; there is no way to tell the two apart by reading, which is why the specification requires the registers rather than the values.

Writing one is an illegal instruction. All five sit in the read-only quadrant of the register address space (`csr[11:10] = 11`), and any CSR instruction in a *write* form aimed at that quadrant traps, in every configuration, including the read-modify-write forms that would write back an unchanged value. Reading with `csrr` (or any form whose source operand is `x0`, which the hardware recognizes as a pure read) is the only legal access.

5.2 The Legacy Trap Mechanism

The legacy mechanism is a pure *interrupt* architecture: it delivers the hardware interrupt sources of Section 10 and nothing else.

- **Delivery** is vectored in hardware. Each of the 121 interrupt sources owns one 4-byte slot of the interrupt vector table at 0x8000, and the hardware jumps directly to the slot for the source it is taking. The slot holds a jump instruction to the service routine.
- **Entry** pushes the return program counter, and only that, to the stack at `sp - 4` and decrements `sp` by four. The stack pointer must therefore be valid before any interrupt can arrive.
- **Return** is the custom `ret_irq` instruction, which pops the saved program counter, increments `sp` by four and resumes. The custom sleep and wake instructions park and un-park the hart (the generated header spells the three of them `irq_return()`, `cpu_sleep()` and `cpu_wake()`; the bare-metal assembly environment spells them `iret`, `extinguish` and `ignite`).
- **Exceptions do not exist.** There is no recoverable exception in the legacy architecture. An illegal instruction, including an access to an unimplemented control register, puts the hart into a terminal trap state: the program counter stops advancing and the hart never retires another instruction. This is a deliberate, deterministic failure mode for a teaching chip, not an error state software can catch.
- **Handling is non-recursive** and the enable state of the three per-hart interrupt lines is fixed in hardware; there is no interrupt-enable control register in the legacy path.

Because the legacy mechanism pushes to the stack on entry, it assumes the stack pointer belongs to trusted code. That assumption is exactly what the standard architecture removes, which is why a chip that wants user mode needs the standard architecture underneath it.

5.3 Selecting a Delivery Mode: `mtrapctl`

The two architectures are selected by bit 0 (LEGACY) of the custom machine register `mtrapctl` at address 0x7C0, which lives in the architecturally reserved custom machine read/write range. All other bits read zero and ignore writes.

<code>mtrapctl.LEGACY</code>	Mode	Behavior
1 (reset)	Legacy	Exactly the mechanism of Section 5.2: vector table delivery, the stack push on entry, <code>ret_irq</code> to return, and a terminal halt on any illegal instruction.
0	Standard	Interrupts and exceptions vector through <code>mtvec</code> with the full <code>mepc/mcause/mtval/mstatus</code> sequence, and <code>MRET</code> returns. The vector table is inert, and nothing is ever pushed to the stack by trap entry.

LEGACY resets to 1, so a chip built with the standard architecture still *boots* on the legacy path: the boot ROM, the tile loader, the interrupt dispatcher and existing application code run unmodified, and a program that never writes `mtrapctl` cannot tell the difference. Selecting the standard mode is an explicit software act.

Software contract: change `mtrapctl` only while no trap is in flight; that is, from ordinary program flow with interrupts quiesced, never from inside an interrupt service routine and never between a trap entry and its matching return. The rule is the same class as the clock-reconfiguration contract in the SYSTEM chapter: the state of a trap that was taken under one delivery mode cannot be unwound by the other. Switching back to legacy is legal and is the recovery path.

The two custom instruction groups keep their encodings in both modes. `ret_irq` is only *defined* in legacy mode, where an interrupt sequence is in progress for it to acknowledge; executing it in standard mode is a software error whose only consequence is the stack-pointer damage the instruction itself does.

5.4 The Machine Trap Registers

Table 6 lists every control register the standard trap architecture adds, with the behavior of each field. Fields not listed read as zero and ignore writes; every field is WARL (write any value, read a legal value), so software must read back what it wrote rather than assume a value took.

Table 6: Machine trap control and status registers

Address	Register	Fields and behavior
0x300	mstatus	MIE (bit 3) global machine interrupt enable, and MPIE (bit 7) its saved copy. MPP (bits 12:11) is the saved previous privilege and is pinned to 11 (machine). MPRV (bit 17) makes <i>data</i> accesses use the privilege in MPP. TW (bit 21) traps WFI (no effect without user mode).
0x310	mstatush	Implemented as a legal register that reads zero and ignores writes (the high half of mstatus holds no field this chip implements).
0x305	mtvec	BASE (bits 31:2) is the trap entry address, read/write. MODE (bits 1:0) is pinned to 00, <i>direct</i> mode: every trap enters at BASE, and the handler dispatches on mcause.
0x304	mie	Per-source interrupt enables: MSIE (bit 3) software, MTIE (bit 7) timer, MEIE (bit 11) external. Resets to zero: standard mode wakes up fully masked, unlike the legacy path whose three lines are hard-wired enabled.
0x344	mip	Read-only mirror of the three live interrupt lines: MSIP (bit 3), MTIP (bit 7), MEIP (bit 11). Writes are ignored: every source is level-driven and is cleared at the source (a CLINT MSIPh write, an MTIMECMPx write, or the IRQROUTER claim/complete sequence).
0x340	mscratch	32 bits of read/write scratch storage reserved for the trap handler. In standard mode the hardware saves nothing to memory, so this is where a handler keeps the pointer it needs before it owns a stack.
0x341	mepc	The trapped program counter. Bit 0 always reads zero; bit 1 is writable on a chip built with the compressed instruction set and reads zero otherwise.
0x342	mcause	The trap cause: bit 31 (the interrupt flag) distinguishes an interrupt from an exception, bits 3:0 carry the code. Bits 30:4 read zero. Hardware writes it on every trap entry per Table 7; software may write it.
0x343	mtval	The trap value, written by hardware per Table 7 and freely writable by software.
0x306	mcounteren	A legal register pinned to zero (its enables only have meaning with user mode built).
0x7C0	mtrapctl	Custom. LEGACY (bit 0) selects the delivery mode and resets to 1; bits 31:1 are pinned to zero. See Section 5.3.

5.5 Trap Entry, Trap Return, and Causes

In standard mode a trap entry performs **no memory transaction at all**. In one atomic step the hardware writes mepc, mcause and mtval per Table 7, copies MIE into MPIE and clears MIE (so the handler starts

with interrupts disabled), records the interrupted privilege level in MPP, enters machine mode, and sets the program counter to `mtvec`'s BASE. The stack pointer is not read, not written and not adjusted, which is the entire point, because the interrupted stack pointer may belong to untrusted code.

MRET reverses it: the program counter is loaded from `mepc`, MIE is restored from `MPIE`, `MPIE` is set to 1, and the privilege level returns to MPP. ECALL and EBREAK are decoded as the environment call and breakpoint exceptions; both are illegal instructions on a chip without the standard trap architecture.

A minimal handler installation looks like this (the assembler needs the Zicsr extension enabled for the control-register instructions):

```
.option push
.option arch, +zicsr
la      t0, trap_handler
csrw    mtvec, t0          # direct-mode trap entry point
csrw    mie, zero          # start fully masked
csrwi   mtrapctl, 0        # leave legacy delivery: standard mode from here
.option pop
```

Table 7: Trap causes, values, and saved program counters

Condition	mcause	mtval	mepc
Instruction address misaligned	0	The misaligned address	The faulting instruction
Instruction access fault (memory protection)	1	The faulting fetch address	The faulting instruction
Illegal instruction (including an access to an unimplemented or privileged control register, or a <i>write</i> to a read-only one)	2	The faulting instruction encoding	The faulting instruction
Breakpoint (EBREAK)	3	0	The EBREAK
Load access fault (memory protection)	5	The faulting data address	The faulting instruction
Store or atomic access fault (memory protection)	7	The faulting data address	The faulting instruction
Environment call from machine mode	11	0	The ECALL
Machine software interrupt (CLINT MSIPh)	0x80000003	0	The resume address
Machine timer interrupt (CLINT MTIMECMP0L)	0x80000007	0	The resume address
Machine external interrupt (IRQROUTER)	0x8000000B	0	The resume address

Three details of Table 7 are worth stating explicitly.

- **The illegal-instruction value for a compressed instruction is the expanded encoding, not the raw halfword.** When a 16-bit compressed instruction traps, `mtval` carries the 32-bit instruction the decompressor produced from it, which is what the core actually rejected. The specification permits either the faulting encoding or zero here; reporting the expansion is more informative than

zero and costs nothing, but a debugger that expects the raw halfword will not find it. This is a deliberate, documented deviation.

- **Interrupt priority is fixed:** external, then software, then timer. An interrupt is taken only when MIE is set and the same bit is set in both mip and mie.
- **Interrupts are sampled at instruction boundaries only**, and the multi-cycle instruction sequences (the compressed push/pop sequences, the cache-block zero instruction, and the table-jump sequence) run to completion uninterrupted, exactly as in the legacy mechanism.

5.6 Waiting for an Interrupt

Two instructions park a hart, and they are not interchangeable.

- The standard WFI instruction is available whenever the standard trap registers are built, in either delivery mode. It stops the hart until $(mip \wedge mie) \neq 0$, *regardless* of MIE. If MIE is set the hart then takes the interrupt; if it is clear the hart simply resumes at the following instruction, which is what makes the standard “sleep until an event, handle it inline” idiom work.
- The custom sleep instruction keeps its legacy meaning: it parks the hart until an interrupt is actually *taken*, and the service routine must execute wake before returning or the hart goes straight back to sleep. This is the instruction the boot ROM parks tile harts with.

Because WFI waits on $mip \wedge mie$, a hart that executes it with mie still zero waits forever. That is the specified behavior, not a defect; TW exists so that machine-mode software can stop less privileged code from doing it.

6 Debug Support

Every program eventually needs a way to ask the hardware what it is actually doing. The chapters so far have described two: the Forth interpreter over UART0 (Section 13), which lets you poke at a running system from a terminal, and the disassembly listings the compiler emits with `-g`, which let you read what the machine was told to do. Both are *cooperative*: they need the chip to still be executing your code, correctly enough to talk back.

Hardware debug is the non-cooperative kind. It reaches into the chip over dedicated wires that do not depend on any software running, stops a processor between instructions, reads and writes its registers and memory, and starts it again. It works on a hart that is stuck in an infinite loop, on a hart that has taken an illegal instruction and stopped retiring, and, with one configuration bit set, on a hart before it has executed its first instruction. That is what this chapter is about.

This configuration includes the debug system; the sections below document it.

6.1 What JTAG Is

JTAG is a serial protocol standardised as IEEE 1149.1 (four mandatory wires plus an optional fifth) that was invented to test the solder joints on a circuit board and has since become the way almost every processor in the world is debugged. Understanding it is worth twenty minutes, because everything above it in the stack is built on one very simple idea.

The wires are:

Wire	Direction	Purpose
TCK	in	Test clock. Everything on the port is synchronous to it, and it is supplied by the debug probe, not by the chip. It has no relationship to any clock inside the chip.
TMS	in	Test mode select. Sampled on every rising TCK edge; this one wire drives the entire state machine.
TDI	in	Test data in: the serial input to whichever shift register is currently selected.
TDO	out	Test data out: the serial output of that register.
TRSTn	in	The optional fifth wire: an asynchronous reset for the port. Not required by the standard, which can always reset the port with TMS alone, but present here.

Inside the chip these wires drive a **Test Access Port**, or TAP: a sixteen-state finite state machine whose only input is TMS. The states are fixed by the standard and every JTAG device in the world has the same sixteen. They form two nearly identical lobes, one for scanning an *instruction* and one for scanning *data*, joined at a common idle state:

- **Test-Logic-Reset** is the top of the graph and the reset state. **Holding TMS high for five TCK clocks reaches it from anywhere in the graph**, which is the standard's guaranteed recovery: a debugger that has lost track of where the state machine is does not need to know, it just clocks five ones.
- **Run-Test/Idle** is the resting state between operations.
- The **IR lobe** (Select-IR, Capture-IR, Shift-IR, Exit1-IR, Pause-IR, Exit2-IR, Update-IR) shifts a new value into the *instruction register*.
- The **DR lobe** (the same seven states with DR in place of IR) shifts a value through whichever *data register* the current instruction has selected.

Figure 10 is the whole machine, with every transition this chip implements.

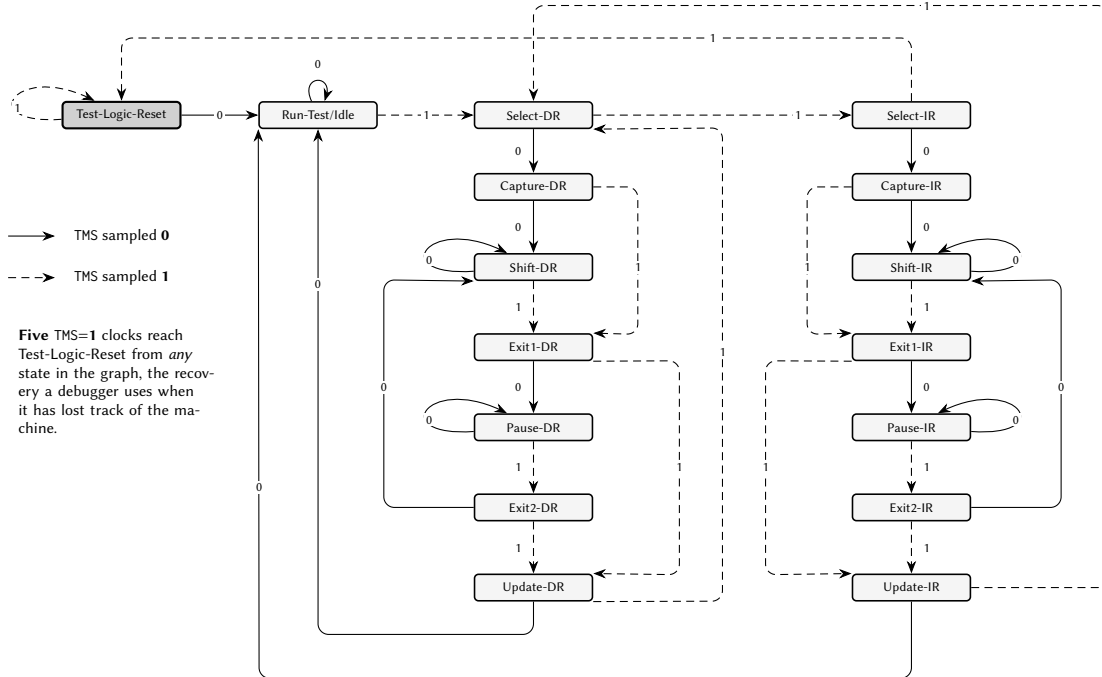


Figure 10: The sixteen-state JTAG test access port. Solid arrows are taken when TMS is sampled low, dashed when it is sampled high. The two lobes hanging below Select-DR and Select-IR are the same seven states drawn as mirror images of one another, one scanning a data register and one the instruction register. Note that five consecutive high samples reach Test-Logic-Reset from every state in the graph, which is the recovery a debugger uses when it does not know where the machine is.

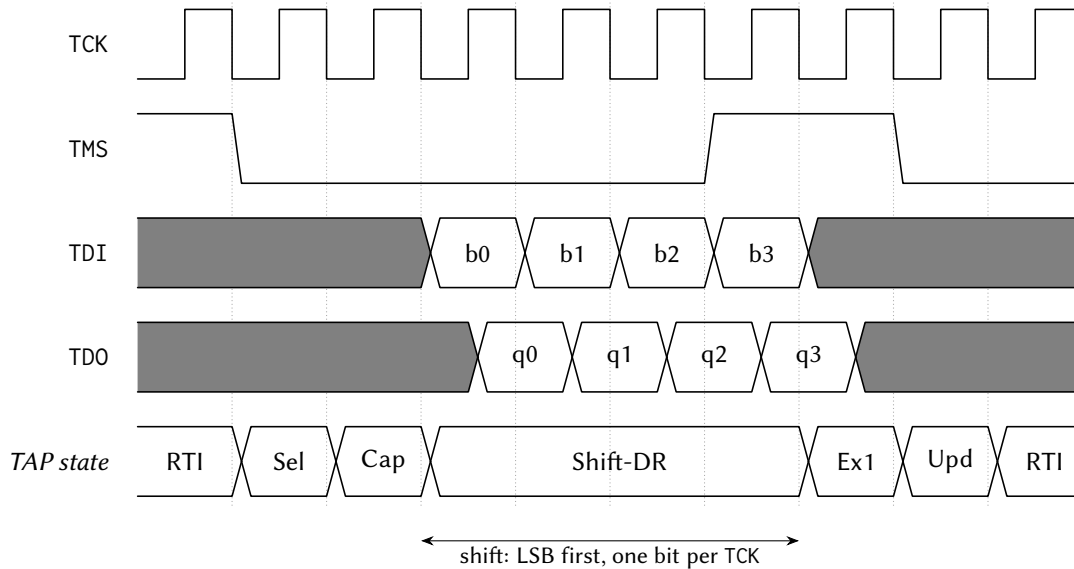
The operation you actually perform is called a **scan**, and it has three phases that matter. In **Capture**, the selected register is loaded in parallel from whatever it is shadowing. In **Shift**, the register behaves as a shift chain: on each TCK it takes one bit from TDI and presents one bit on TDO, least-significant bit first. In **Update**, the value that has just been shifted in is committed. **So one scan simultaneously reads the old contents and writes the new ones**, which is why a debugger's traffic always looks like a stream of fixed-length shifts rather than the read-then-write pairs a parallel bus would use. Figure 11 follows one four-bit scan through those phases.

The instruction register selects *which* data register a subsequent DR scan will reach. Every JTAG device is therefore fully described by three things: the width of its instruction register, the set of instruction values it recognises, and the width and meaning of the data register each one selects. For this chip, that is Table 8.

Two consequences are worth stating because they surprise people. First, the port is entirely passive with respect to the chip's own clocks: it will answer while the processor is running, while it is stopped, and while it is held in reset. Second, because the standard defines a **BYPASS** instruction that selects a single flip-flop, several devices can be wired in a chain (TDO of one to TDI of the next), and a debugger can talk to any one of them while the others contribute one bit each of padding.

6.2 The RISC-V Debug Stack

JTAG by itself only gives you a way to shift bits in and out. The RISC-V debug specification defines what those bits *mean*, and it does so in four layers. Each layer knows nothing about the internals of the one



Capture loads the register in parallel from what it shadows; **Update** commits what was shifted in. One scan therefore reads the old contents and writes the new ones at the same time. TMS is sampled on the RISING edge; TDO changes on the FALLING edge, which is why its transitions sit half a cycle after the others.

Figure 11: One four-bit data-register scan. The state row is the sequence the port walks under the TMS pattern above it. Data is shifted least-significant bit first, one bit per TCK; TDI is sampled on the rising edge and TDO changes on the falling one, which is why the two rows are offset by half a cycle. A real scan is 32 or 41 bits wide rather than four, but the shape is exactly this.

below it, which is what allows the same debugger software to drive very different chips.

Layer	What it is
DTM	The <i>Debug Transport Module</i> . It is the JTAG-facing half: the TAP itself, plus the logic that turns one DR scan into one transaction on the interface below. Its job is transport and nothing else; it has no idea what a hart is.
DMI	The <i>Debug Module Interface</i> . A tiny address/data bus, here 7 address bits and 32 data bits, with three operations: no-op, read, write. This is the seam. Anything that can drive a DMI can debug the chip; JTAG is merely the transport this chip provides.
DM	The <i>Debug Module</i> . A block of registers sitting on the DMI that implements <i>run control</i> : select a hart, halt it, ask what state it is in, run a small program on it, resume it. There is one DM for the whole chip and it can reach every hart.
Debug mode	A state of the <i>hart itself</i> , effectively a privilege level above machine mode. A hart in debug mode has stopped executing your program, is executing the debugger's code instead, and has a small set of control registers that only exist while it is there.

The layer that most often confuses newcomers is the last one. **A halted hart is not a frozen hart.** On this architecture, halting a processor does not stop its clock and freeze it mid-instruction; it *diverts* it, between instructions, to a fixed address where the debugger's own code is waiting. The hart then sits in a

loop at that address, still fetching and executing, waiting to be told what to do. When the debugger wants to read a register, it does not reach into the register file with wires; it writes a two- or three-instruction program into memory, tells the hart to run it, and reads back where the program put the answer.

This is the single most important thing to understand about RISC-V debug, and almost every property in the rest of this chapter follows from it: that the debugger needs somewhere in memory to put code, that reading a register can report an *exception*, that the hart's own `s0` and `s1` registers need saving before anything else can happen, and that memory accesses on behalf of the debugger go through the ordinary bus, with ordinary arbitration, exactly as the program's own would.

Figure 12 shows those layers as they are actually built, in a configuration that includes the debug system: which clock each part runs on, what crosses between them, and the three ways the Debug Module reaches the rest of the chip.

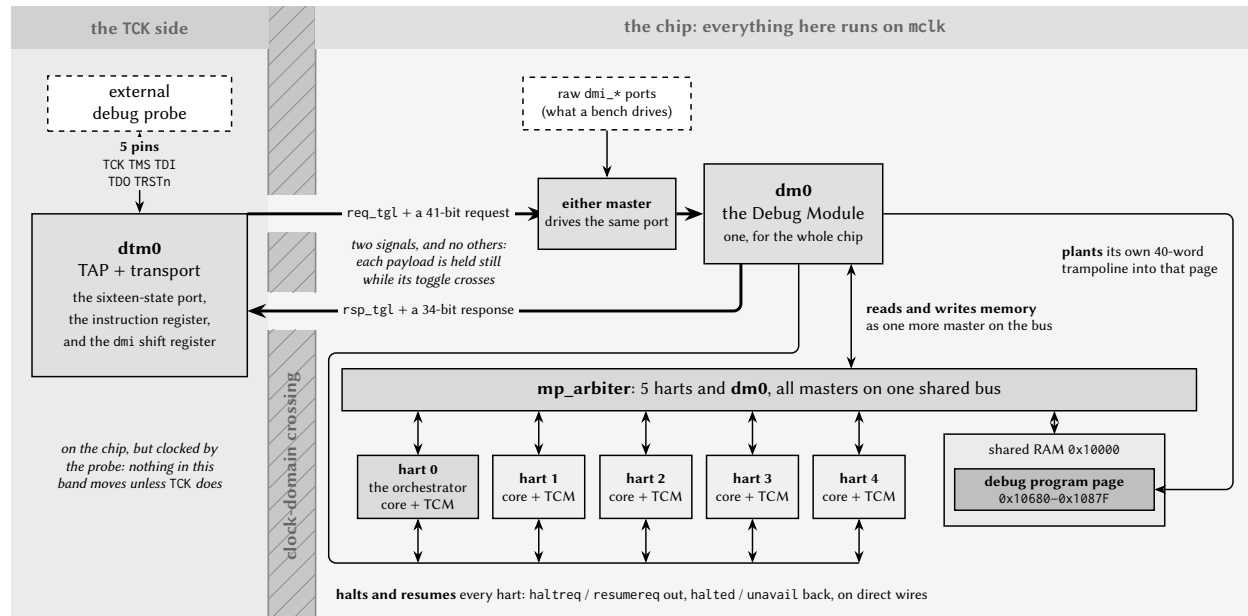


Figure 12: The debug path, from the probe to a hart, in a configuration built with debug support. The transport and the Debug Module sit in *different clock domains*, and the wall between them is crossed by exactly two signals: one toggle each way, each carrying a payload that is held still while its toggle is in flight. Everything to the right of that wall is ordinary chip fabric, and the one Debug Module reaches it in three ways: it halts and resumes the harts over direct wires, it reads and writes memory as one more master on the same arbiter the harts use, and it plants its own trampoline into the debug program page. The count of hart tiles follows this configuration.

6.3 The Debug Port

The debug port occupies five pins, which are present only on the LQFP-100 package; a configuration built for a smaller package has the debug system in silicon but nowhere to bond it.

Pin	Signal	I/O	Idle state when unconnected
47	TCK	Input	Pulled low . TCK idles low.
48	TMS	Input	Pulled high . This is what IEEE 1149.1 requires, and it is what allows five clocks from a disconnected probe to still return the port to Test-Logic-Reset.
49	TDI	Input	Pulled high .
50	TDO	Output	Driven. Held quiet between shift states rather than toggling, so the pin is well behaved in a multi-device chain.
51	TRSTn	Input	Pulled low ; see below.

The pull on TRSTn is a deliberate, documented deviation from IEEE 1149.1. The standard wants TRSTn to idle high, so that a port with a floating reset pin is available to a probe that turns up later. This chip pulls it *low*, which holds the TAP in reset. The reasoning is that the debug transport here is optional: a board that does not use it leaves all five pins unconnected, and in that case the desirable behaviour is a port that is completely inert (issuing nothing, clocking nothing, and unable to reach the Debug Module at all) rather than a port that is quietly attachable by anyone who touches the pins. **Fail-safe-inert was chosen over fail-safe-attachable.** A board that wants to use the port must therefore drive TRSTn high, and cannot rely on the pull to do it.

The five pins are ordinary digital I/O on the 3.3 V VDDPST/VSSPST domain and may be left unconnected; the chip is unaffected. They are deliberately placed on the south and east edges, away from the analog band on the north edge.

Identification. A DR scan of the IDCODE register returns a 32-bit value that identifies the part before the debugger knows anything else about it:

Chip	IDCODE	Decomposition
Castalia	0x1CA57EEF	version 1, part number 0xCA57, manufacturer 0x777, mandatory LSB 1
Argus	0x1A265EEF	version 1, part number 0xA265, manufacturer 0x777, mandatory LSB 1

The manufacturer field is 0x777, which is *not* a valid JEDEC JEP106 code and is not registered to anyone. That is intentional and it is the honest answer for a non-commercial teaching chip: rather than borrow an identifier belonging to a real company, the field carries a value that is recognisably invalid. Discovery software should identify this part by its full IDCODE, not by its manufacturer field.

6.4 The Debug Transport Module

The DTM implements the TAP described in Section 6.1 and four data registers. Its instruction register is **five bits** wide and resets to the IDCODE instruction.

Test-Logic-Reset sets the instruction register to IDCODE and does nothing else. In particular it does *not* clear the sticky status described below, and does not discard a result waiting to be read. That is what the dmireset strobe is for, and it is why a debugger that has just recovered a lost TAP still has to clear the status explicitly. TRSTn, by contrast, resets the entire transport.

DTMCS reports what the transport can do and carries the two recovery strobes:

IR value	Instruction	DR width	Selects
0x01	IDCODE	32	The identification value above. This is the reset instruction, so the first DR scan after a port reset always returns the ID-CODE.
0x10	DTMCS	32	The transport's own control and status register.
0x11	DMI	41	One transaction on the Debug Module Interface.
0x1F	BYPASS	1	A single flip-flop, for use in a multi-device chain.
<i>anything else</i>	–	1	BYPASS. Every unrecognised instruction selects the one-bit chain, so the length of the selected register never depends on what was scanned before.

Table 8: Debug Transport Module instruction register values

Bits	Field	Meaning
[3:0]	version	1: this transport satisfies both the 0.13 and the 1.0 debug specifications.
[9:4]	abits	7: the DMI address is seven bits wide, so the DMI data register is $7 + 32 + 2 = 41$ bits.
[11:10]	dmistat	Sticky transaction status; see below.
[14:12]	idle	7: the number of Run-Test/Idle clocks the debugger should insert between a DMI scan and the scan that collects its result.
[16]	dmireset	Write 1 to clear dmistat and abandon any outstanding transaction. Reads as 0.
[17]	dmihardreset	Write 1 to do the above <i>and</i> clear the stored result. Reads as 0.

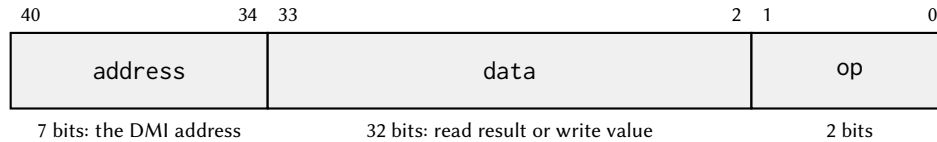
The idle value is truthful, and that is a deviation worth knowing about. Many implementations advertise 0 here while in fact requiring idle clocks, leaving the debugger to discover the requirement by failing. The value 7 reported here is derived from the real latency of the slowest transaction class. A DMI access to one of the Debug Module's own registers completes in eight to ten cycles of the internal main clock; an access to data0 or a program-buffer word is *proxied* into shared memory through the arbiter and takes tens of cycles. Against an assumed TCK period of at least 140 ns (that is, TCK at or below 7.14 MHz) against a 24 MHz main clock, the worst class needs six idle cycles, and 7 gives one cycle of margin. It is also the largest value a three-bit field can express.

idle is a recommendation, not a guarantee. At a faster TCK, or while the shared bus is heavily contended by the harts, a transaction can still be outstanding when the debugger comes back for it. That is not an error; it is reported as *busy* and the correct response is to clear the status and reissue.

The DMI data register is 41 bits, laid out as in Figure 13: op[1:0], data[33:2], address[40:34].

On the way *in*, op is 00 for no-operation, 01 for read and 10 for write. On the way *out*, it is 00 for success, 10 for failed and 11 for busy. **A no-operation scan never starts anything**, which is what makes a capture-only scan (shifting in zeros purely to read out the previous result) safe.

dmistat is sticky, and it has two flavours. It takes the value 3 (busy) if a scan is captured while a



Shifted LSB first, so op goes in and comes out first. On the way *in*: 00 no-op, 01 read, 10 write.
On the way *out*: 00 success, 10 failed, 11 busy. Field widths here are for legibility; the bit numbers are exact.

Figure 13: The 41-bit DMI data register. One scan of it is one transaction on the Debug Module Interface.

transaction is still in flight, or if a new transaction is scanned in while one is outstanding; and 2 (failed) if a transaction completes with a failure, which happens, for instance, on a DMI access to an address the Debug Module does not implement. **While `dmistat` is nonzero, in either flavour, further DMI scans are dropped.** This is deliberate: it stops a debugger from issuing a long sequence of writes into the void after the first one failed. The recovery is always the same (write `dmireset`, then reissue), and the status is cleared by `dmireset` or `dmihardreset` and by nothing else. Neither Test-Logic-Reset nor a system reset clears it.

Figure 14 follows a single transaction across the clock boundary, and is where the idle value above comes from.

6.5 The Debug Module

There is one Debug Module for the whole chip. It sits on the always-running main clock, it is reachable by the DTM through the DMI, and, because it must be able to place instructions where a hart will fetch them, it is also a *master* on the shared-memory arbiter, alongside the harts (Section 4). Table 9 is its DMI address map.

Table 9: Debug Module registers, at their DMI addresses

DMI	Register	Purpose
0x04	data0	The single abstract-command data word. Backed by shared memory, not by a flip-flop.
0x10	dmcontrol	Run control: enable the module, select a hart, halt it, resume it.
0x11	dmstatus	The selected hart’s state, and the module’s capabilities. Read-only.
0x12	hartinfo	Reads 0x00000000: no scratch registers, no debugger-visible data0 address. Read-only, and the zero is deliberate; see below.
0x13	haltsum1	Reads zero. Present, and answering successfully, because debuggers probe it.
0x16	abstractcs	Abstract-command status: busy, error code, and the module’s sizes.
0x17	command	Write to start an abstract command. Reads zero.
0x18	abstractauto	Reads zero; automatic command re-execution is not implemented.
0x20	progbuf0	Program-buffer word 0. Backed by shared memory.

Table 9 continued on next page

Table 9 continued from previous page

DMI	Register	Purpose
0x21	progbuf1	Program-buffer word 1. Backed by shared memory.
0x32	dmcs2	Halt-group membership of the selected hart.
0x40	haltsum0	One bit per hart: halted and reachable. The only status register that ignores the hart selection.

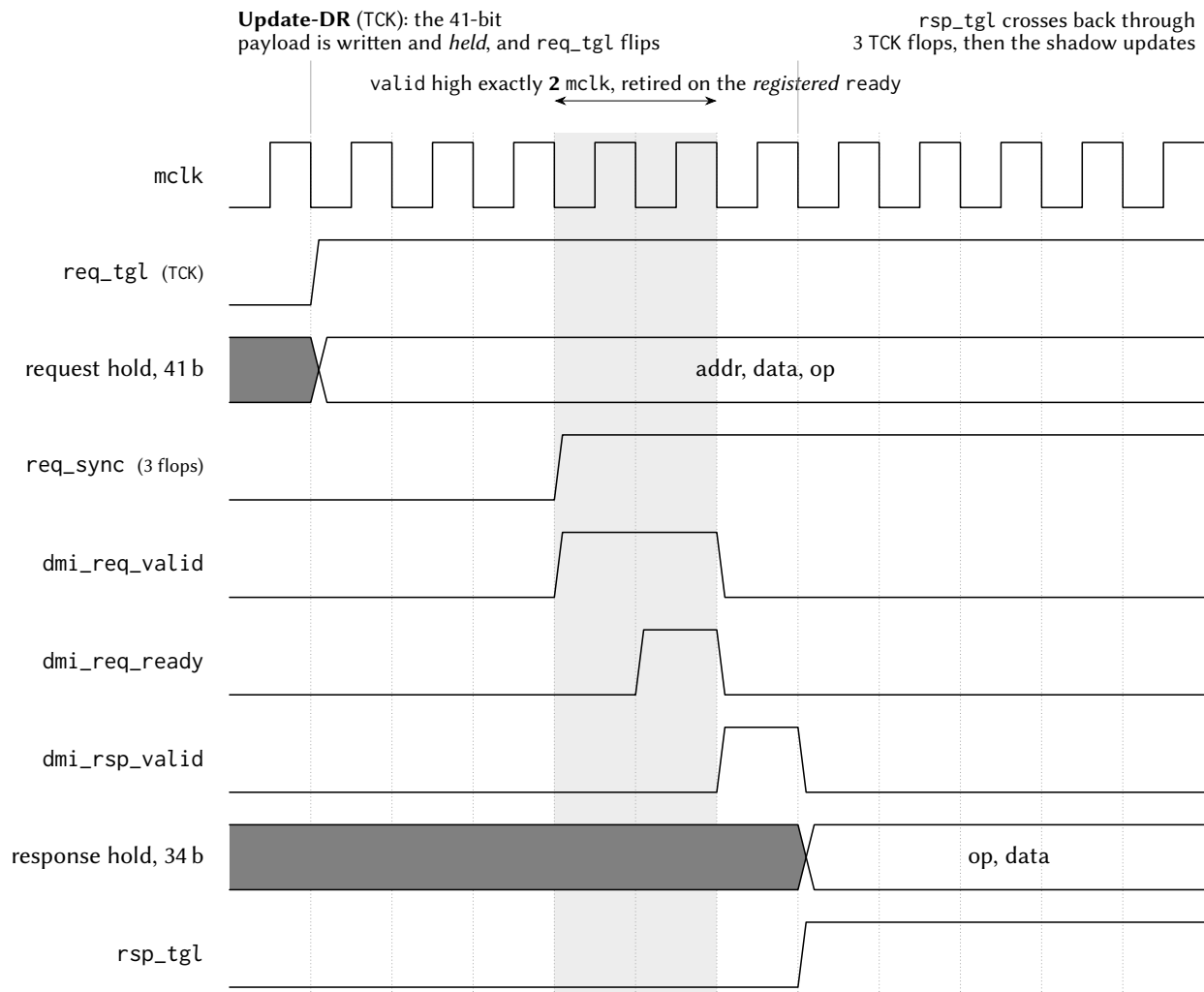
hartinfo reads zero on purpose, and a debugger should want it to. The register exists to advertise per-hart scratch registers and the address of data0, so that a debugger can use them directly. This implementation advertises neither: its nscratch, dataaccess, datasize and dataaddr fields are all zero. With dataaccess = 0 a conforming debugger does not consult dataaddr at all and reaches memory through the program buffer and its own work area instead, which is the only path this chip supports anyway (there is no System Bus Access; see Section 6.11). Advertising an address here would also have been actively wrong: the field is twelve bits wide, and the real data0 address does not fit in it. A truthful zero is better than a truncated pointer into somebody else's memory.

Any other DMI address returns *failed*, which, because dmistat is sticky, will also stall the transport until the debugger clears it. Probing for unimplemented registers is therefore not free, and this is why haltsum1 and abstractauto answer with a successful zero instead of a failure: they are addresses debuggers routinely read during discovery, and failing them would punish a debugger for being thorough.

dmcontrol is the register everything starts with. dmactive (bit 0) must be written 1 before the module will do anything; writing it 0 resets the module's own state, but *not* the harts. hartsel (bits [25:16]) selects which hart the status and command registers refer to; all ten bits are implemented, and selecting a hart index that does not exist is reported honestly as *nonexistent* rather than silently clamped to the highest real hart, which is what makes hart-count discovery work. haltreq (bit 31) is a level: while it is set, the selected hart is asked to halt. resumereq (bit 30) is a write-one request to resume; it is queued rather than refused, and it is ignored for a hart that is not halted or is not reachable. setresethaltreq and clrresethaltreq (bits 3 and 2) arm and disarm halt-on-reset for the selected hart. ndmreset, hartreset and hasel are not implemented; they read zero and ignore writes.

dmstatus reports version = 3 (debug specification 1.0), authenticated = 1 (there is no authentication on this chip), hasresethaltreq = 1 and impebreak = 1. Its state is reported through the usual paired bits: allhalted/anyhalted, allrunning/anyrunning, allunavail/anyunavail, allnonexistent/anynonexistent, allresumeack/anyresumeack, allhavereset/anyhavereset. Because only one hart is ever selected, both bits of every pair always carry the same value.

abstractcs reports progbufsize = 2 and datacount = 1: two program-buffer words and one data word. busy (bit 12) is set while a command is running, and the module remains readable while it is, so the debugger can poll. cmderr (bits [10:8]) is the error code, and it is **write-one-to-clear**:



The Debug Module's re-accept lockout is a **timer**, not a handshake: it reopens 9 mclk after a capture. A master that held valid until its response came back would still be asserting it then and would earn a **second, duplicate accept**: two responses for one request, sliding every later pair by one. The symptom is not a wrong answer; it is the *previous* answer. Two cycles leaves seven inside the window.

Where idle = 7 comes from. A debugger sees the result no earlier than 3 TCK (this response synchroniser) + $\lceil t_{DM}/T_{TCK} \rceil$. Here t_{DM} is 8–10 mclk for a Debug Module register, and *tens* for data0 or a program-buffer word, which are proxied into shared RAM through the arbiter. At $TCK \leq 7.14$ MHz against a 24 MHz mclk that is 6 cycles for the worst class; 7 adds one of margin and is the largest value the 3-bit field can hold. Faster TCK, or a contended bus, can still report busy.

Figure 14: One DMI transaction crossing between the two clock domains, drawn on the internal clock. The two events in the debug probe's own clock domain are marked above the waveforms rather than drawn, because the ratio between the two clocks is a board-level choice. The shaded cycles are the request handshake: the request is presented for exactly two cycles and retired as soon as the acceptance is observed, and the note below explains why holding it any longer would make the same request be answered twice.

cmderr	Meaning	Raised when
0	none	–
1	busy	A command was written while one was running, or data0/a program-buffer word was accessed while the module's bus engine was busy.
2	not supported	The command type is not access-register, or the transfer size is not 32-bit.
3	exception	The hart took an exception while executing the command.
4	halt/resume	The selected hart was not halted when the command was written.
7	other	A command did not complete within the module's internal timeout.

A command written while cmderr is already nonzero is ignored entirely: it does not run, and it does not clear the error on its way past. Software must clear cmderr before issuing the next command.

command implements one command type: **access register**. It transfers one 32-bit value between data0 and a register of the halted hart, optionally executing the program buffer afterwards. The fields are aarsize ([22:20], which must be 2 for 32-bit), postexec (bit 18), transfer (bit 17), write (bit 16) and regno ([15:0]). Register numbers 0x1000–0x101F are the general-purpose registers x0–x31; any other number is treated as a control and status register address. **Access memory is deliberately not implemented** and returns *not supported*. Memory is reached instead by writing an lw or sw into the program buffer and executing it on the halted hart; see Section 6.11.

data0 and the program-buffer words are memory, not registers. A DMI access to 0x04, 0x20 or 0x21 becomes a read or write of a word in shared memory, issued by the Debug Module through the ordinary arbiter. This is why those three addresses are the slow class that sets idle to 7, and why an access to them while the module's bus engine is busy answers successfully with zero data *and* sets cmderr to *busy*: the answer arrives, but it is not the answer you wanted, and the error code is how you find out.

The program buffer is two DMI-visible words plus an **implicit third word**, which is why impebreak reads 1. Software may therefore place up to three instructions' worth of work in the buffer, with the third slot supplied by hardware.

6.6 Debug Mode in the Hart

A hart enters debug mode for one of four reasons, recorded in dcsr.cause:

cause	Reason	Notes
1	EBREAK	A software breakpoint, taken only when dcsr.ebreakm is set (or when the hart is already in debug mode).
3	Halt request	The Debug Module asserted halreq.
4	Single step	The one instruction permitted by dcsr.step has retired.
5	Halt on reset	setresethalreq was armed when this hart came out of reset.

On entry, the hart saves its program counter to dpc, records the cause and its previous privilege level in dcsr, sets debug mode, and jumps to a fixed address, **0x00010780**, in the shared memory window. It is now executing the debugger's code.

Four control registers exist while it is there, and **only** while it is there:

Address	Register	Contents
0x7B0	dcsr	Debug control and status: xdebugver [31:28] reads 4; ebreakm (15) makes EBREAK enter debug mode; ebreaku (12) does the same for user mode, where user mode is configured; cause [8:6] is read-only and set by hardware; step (2) enables single-stepping; prv [1:0] is the privilege level to return to.
0x7B1	dpc	The saved program counter. Bit 0 is hardwired zero; bit 1 is writable only where the compressed extension is configured.
0x7B2	dscratch0	A 32-bit scratch word for the debugger's own use.
0x7B3	dscratch1	A second scratch word.

Accessing any of them from machine or user mode is an illegal instruction, and the denial is wider than the four addresses. Every CSR instruction naming an address in 0x7B0–0x7BF is rejected outside debug mode. This is not an optimisation: it means a program cannot accidentally, or deliberately, read the debugger's saved state or forge an entry.

DRET (0x7B200073) leaves debug mode, restoring the program counter from dpc and the privilege level from dcsr.prv. Outside debug mode it is an illegal instruction.

Five behaviours are worth calling out because they are the ones that decide whether a debug session works:

- **The halt request is unmaskable.** It is recognised at fourteen points in the processor's instruction sequencer: the thirteen at which an interrupt is recognised, *plus the terminal trap state*. A hart that has taken an illegal instruction and stopped retiring, which is otherwise unrecoverable short of reset (Section 5.2), can therefore be halted and inspected. Neither mstatus.MIE nor mtrapctl.LEGACY has any effect on it. Note that this rescues a wedged hart only if something is there to request the halt: with no debugger attached the wedge is exactly as terminal as it always was.
- **An EBREAK records its own address in dpc**, not the address after it. A debugger resuming past a software breakpoint must add the instruction's length itself. Every other cause is taken between instructions, so dpc is already the resume address.
- **No interrupt is taken in debug mode.** Interrupt sources remain pending as levels and are delivered after the DRET, so a debug session neither loses interrupts nor lets the application's handlers run underneath it.
- **An exception taken in debug mode re-enters debug mode** at the same fixed address, leaving dpc and dcsr untouched, and without disturbing mepc, mcause or mtval. This is what makes a faulting abstract command (reading a CSR the configuration does not implement, say) report cmderr = 3 instead of destroying the session.
- **Halt-on-reset is sampled once, at reset release.** Arming setresethaltreq affects the *next* time that hart comes out of reset; it will not halt a hart that is already running. This is what lets a debugger catch a hart before its first instruction, and it is why arming it does not disturb a running system.

Finally: **single-step diverts at the stepped instruction's own dispatch**, so exactly one instruction retires per step. Stepping is inactive while in debug mode, so the debugger's own stub does not step itself.

Figure 15 collects the whole of the above into the two states a hart can be in.

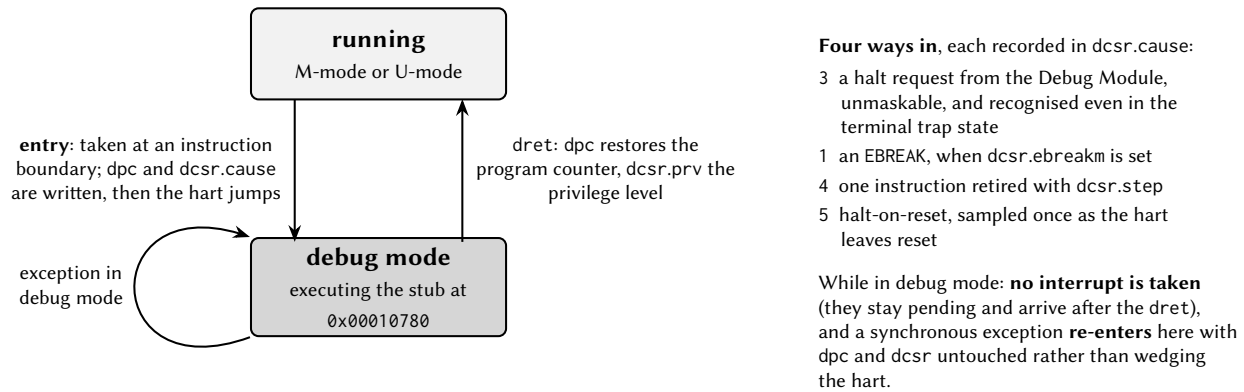


Figure 15: Debug mode, from the hart’s point of view. Debug mode behaves as a privilege level above machine mode: the hart has not stopped, it has been diverted to code the debugger controls, and it returns by executing dret. The self-loop is the property that makes a faulting abstract command recoverable rather than fatal.

6.7 Halt Groups and Unavailable Harts

Two facilities exist because this is a multi-hart chip.

Halt groups solve the problem that stopping one hart of a cooperating set, while the others run on, changes the very behaviour you were trying to observe. dmcs2 assigns the selected hart to one of eight groups; group 0, the reset value, means “no group”. **When any member of a group halts, for any reason, including its own EBREAK, the Debug Module requests a halt of every other member.** Setting a breakpoint in one hart therefore stops the whole group at very nearly the same instant. Note that this is halt-only: resuming a group is done by resuming its members individually.

Unavailable is a third state, and it is not the same as halted. This chip power-gates the harts (Section 4); a hart whose power domain is off cannot answer the Debug Module at all, and the signals that would report its state are clamped to zero at the domain boundary. A clamped “not halted” is indistinguishable from “running”, so the Debug Module does not rely on it: it takes a separate, deliberate *unavailable* signal from the power controller, and consults it **before** halted or running. The reported state is therefore resolved in strict order (**nonexistent, then unavailable, then halted, then running**), and a gated hart reports allunavail/anyunavail with both halted and running clear. It is never misreported as running.

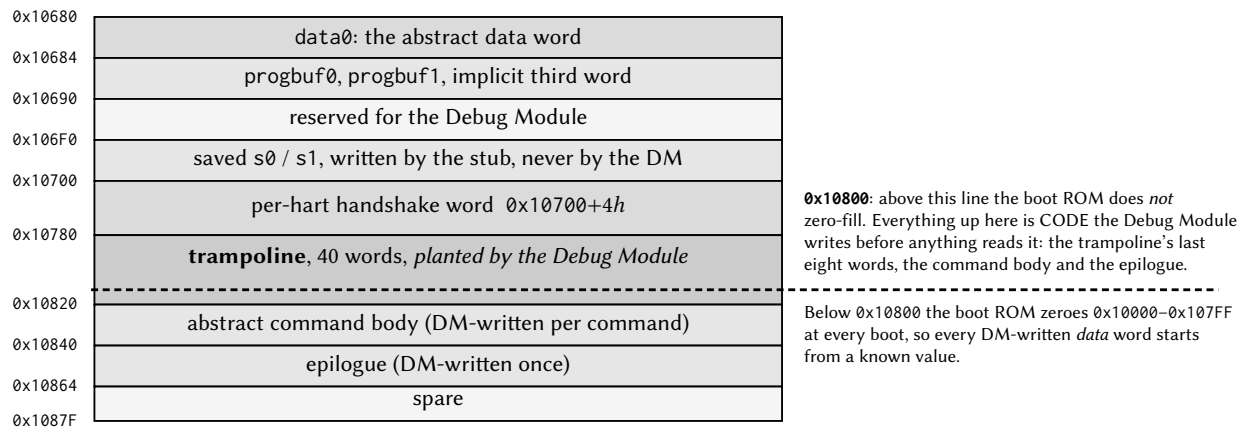
Two consequences follow. First, a halt request aimed at a gated hart does nothing; the debugger must power the hart up first, through the power controller, exactly as software would. Second, **a hart returning from power-down cold-boots the boot ROM and parks; it does not resume where it left off.** If you want to catch it as it comes back, arm halt-on-reset before powering it up. Note that hart 0 is not exempt from any of this.

6.8 The Debug Program Page

Because a halted hart executes code, the debug system needs memory it owns. It uses one 512-byte region of the shared window, listed here so that application software does not collide with it.

Address	Contents
0x10680	data0, the abstract-command data word.
0x10684	progbuf0
0x10688	progbuf1
0x1068C	The implicit third program-buffer word, written by the Debug Module.
0x10690–0x106EC	Reserved for the Debug Module.
0x106F0, 0x106F4	The entry stub’s private repair copies of s0 and s1. They are not where a debugger reads those registers from; see below.
0x10700 + 4 <i>h</i>	The per-hart handshake word between the Debug Module and hart <i>h</i> .
0x10780–0x1087F	The debug entry page, 64 words: the entry stub, the area into which the Debug Module writes each abstract command, and the common exit sequence.

Figure 16 draws the same region to scale, with the one boundary inside it that software has to know about.



The whole span 0x10680–0x1087F is reserved. It is ordinary shared RAM, and it cannot be read-only, because the Debug Module rewrites the command body at every abstract command.

Figure 16: The debug program page. The boot ROM zero-fills shared memory only up to 0x107FF, so the dashed line marks where that guarantee stops: everything above it is code the Debug Module writes before anything can read it, which is why the page needs no staging and cannot be read-only memory.

The whole region 0x10680–0x1087F is reserved. Application software, including software on harts that are not being debugged, must not write to it in a configuration built with debug support. A stray store into the entry page does not merely corrupt data; it corrupts the code a halted hart is executing.

Two details matter to anyone writing code that runs *in* that page. It is assembled without compressed instructions, because execution of compressed or misaligned instructions from the shared window is outside the tested envelope (Section 4).

And the debug system needs two working registers of its own, which is why s0 and s1 are handled differently from every other register. While a hart is halted, **the single home of the interrupted program’s s0 and s1 is the pair of CSRs dscratch0 and dscratch1**. A debugger that reads s0 gets the value out of dscratch0; one that writes s0 writes dscratch0; and code in the program buffer that changes s0 has that change survive to the debugger’s next read, because the exit sequence adopts the architectural pair into the dscratch0/dscratch1 pair before it does anything else. The two words at 0x106F0 and 0x106F4

are the entry stub's own repair copies, used to rebuild that pair if a hart re-enters debug mode from inside the page; the Debug Module writes them only when it is itself the author of a new value, and nothing else should read them at all. The visible consequence is the one that matters: **you see the interrupted program's values, not the stub's working values**, and a plain halt-and-resume with no debugger register writes leaves the program's `s0` and `s1` exactly as it found them.

6.9 A Worked Halt, Read and Resume

The following is a complete session at the DMI level: attach, halt hart 2, read its `a1` register, and resume it. Each numbered step is one or more JTAG scans. A "DMI read" is two scans (one to issue the read, and one, after the recommended idle clocks, to collect the result), because the transport returns the *previous* transaction's result on every capture.

1. **Reset the port and identify the chip.** Drive `TRSTn` high, then clock `TMS` high for five `TCK` cycles to reach Test-Logic-Reset. The instruction register is now `IDCODE`, so a 32-bit DR scan returns `0x1CA57EEF`.
2. **Read DTMCS.** Scan `0x10` into the instruction register, then a 32-bit DR scan. Confirm `version = 1` and `abits = 7`, and note `idle = 7`: insert seven Run-Test/Idle clocks after every DMI scan from here on.
3. **Select the DMI register.** Scan `0x11` into the instruction register. Every scan below is a 41-bit DR scan carrying {address, data, op}.
4. **Enable the Debug Module.** Write `0x00000001` to `dmcontrol` (`0x10`): {`0x10`, `0x00000001`, write}.
5. **Select hart 2 and request a halt.** Write `0x80020001` to `dmcontrol`: `haltreq` set (bit 31), `hartsel = 2` (bits [25:16]), `dmactive` set.
6. **Wait for the halt.** Read `dmstatus` (`0x11`) repeatedly until `allhalted` (bit 9) is set. If `allunavail` is set instead, hart 2 is powered down; power it up before continuing. If `allnonexistent` is set, this configuration has fewer than three harts.
7. **Drop the halt request.** Write `0x00020001` to `dmcontrol`, leaving hart 2 selected. The hart stays halted; halting is a state of the hart, not a level the Debug Module holds.
8. **Clear any stale error.** Write `0x00000700` to `abstractcs` (`0x16`) to clear `cmderr`. A command written while an error stands is ignored.
9. **Read a1.** `a1` is `x11`, so `regno = 0x100B`. Write `0x0022100B` to `command` (`0x17`): `aarsize = 2` (`0x00200000`) + `transfer` (`0x00020000`) + `regno`.
10. **Wait for the command.** Read `abstractcs` until `busy` (bit 12) is clear, then check `cmderr` [10:8]. Zero means the value is ready.
11. **Collect the value.** Read `data0` (`0x04`). This one is proxied through shared memory, so it is the transaction the seven idle clocks were sized for.
12. **Resume.** Write `0x40020001` to `dmcontrol`: `resumereq` (bit 30), hart 2 still selected. Poll `dmstatus` until `allresumeack` (bit 17) is set.

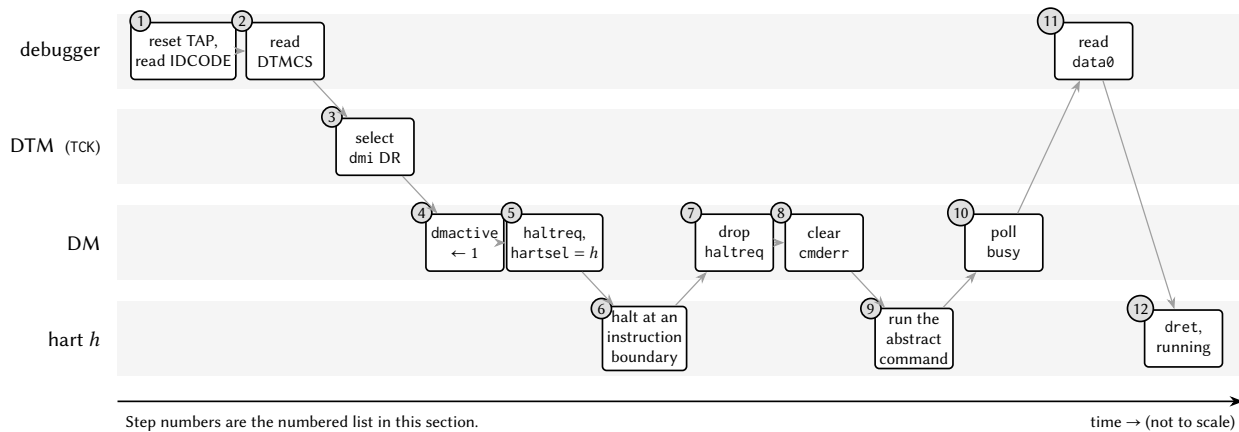


Figure 17: The twelve steps above, across the four agents that perform them. The numbers are the list items. Note how little of the sequence is the debugger talking to the hart directly: most of it is the debugger arranging for the *hart* to do the work, which is the consequence of there being no independent path from the debug port to memory.

To write a register instead of reading one, put the value in `data0` first and set `write` (bit 16) in the command. **To read or write memory**, put an `lw` or `sw` in the program buffer, put the address in a register of the halted hart, and issue a command with `postexec` set; the hart executes your instruction with its own privilege and its own view of the address space.

If a scan ever returns `op = 10` or `11`, the transport has latched a sticky status and is dropping further scans. Read `DTMCS`, write `dmireset`, and reissue.

6.10 Connecting a Debugger

The sequences above are at the DMI level. In practice a debugger generates them, and this part has been brought up end to end against stock, unmodified tools: **OpenOCD 0.12.0** and **riscv-none-elf-gdb 13.2**. Nothing in either was patched. Four facts about that combination are worth knowing before the first session, because each of them is invisible until it costs an afternoon.

Clock the adapter at 3 MHz or below. The Debug Transport Module reports `idle = 7`, meaning it needs seven Run-Test/Idle cycles after a DMI scan before the result is ready. A debugger that honours that field may clock TCK at up to the port's rated 7.14 MHz. **Stock OpenOCD does not honour it:** it learns the required delay by observing failures, starting from zero. What sets the safe speed is then the requirement that three TCK periods cover one DMI transaction's time in the module's own clock domain, $3 T_{\text{TCK}} \geq t_{\text{DM}}$, which gives

Traffic	Minimum T_{TCK}	Maximum f_{TCK}
Register-class (<code>dmcontrol</code> , <code>dmstatus</code> , ...)	139 ns	7.19 MHz
Program-buffer class (all memory access)	278 ns	3.60 MHz

The program-buffer figure is the one that binds, because every memory access and every software breakpoint goes through the program buffer, and it shrinks further when other harts are contending for the shared bus. **Set the adapter to 3 MHz or below;** 3.60 MHz is a hard bound, not a target. The 7.14 MHz rating applies only to a master that honours `idle`.

reset halt succeeds and resets nothing. The Debug Module has no system reset (`ndmreset` is unimplemented), so the command degrades: it asserts halt on every hart, re-reads the ID code, and **reports success**. Nothing is reset: memory keeps its contents, harts keep their execution state, and a running hart's counters carry on from where they were rather than restarting. The failure is silent, so do not infer a reset from the absence of an error. Where a genuine halt-on-reset is needed, arm `setresethaltreq` for the hart and power-cycle its tile through the power controller (Section 4).

That recovery works for harts 1 to $N - 1$ only. The per-tile power controller is the lever behind every recovery procedure here: the wedged-first-word case of Section 6.11, and halt-on-reset above. **Hart 0 has no such lever:** it is always on, its power-controller bit ignores writes, and `ndmreset` and `hartreset` read zero. The only thing that resets hart 0 is the `resetn` pin, which also resets the Debug Module and therefore ends the debug session. Plan accordingly when choosing which hart to experiment on.

An EBREAK written into the program buffer becomes a jump. Debuggers terminate short program-buffer sequences with an explicit EBREAK. On this Debug Module a program must instead leave through the common exit sequence, which is what signals completion and restores `dscratch0` and `dscratch1`; an EBREAK that bypasses it would be reported as an exception even though the work had been done. So a DMI write of exactly the 32-bit EBREAK encoding into `progbuf0` or `progbuf1` stores a jump to that exit sequence instead. Reading the word back returns the jump, not the EBREAK; the substitution is not hidden. Only that exact encoding in those two registers is affected: `data0` stores whatever it is given, and a word one bit away from EBREAK is stored verbatim. A program that genuinely traps still never reaches the exit sequence and is still reported as an exception, so this does not blunt error reporting.

6.11 Limitations and Status

This section states plainly what the debug system does not do, and what is not yet finished.

- **There is no System Bus Access, and there never will be.** Many RISC-V Debug Modules include a second bus master that reads and writes memory directly, independently of any hart. This one does not, by design. Memory is reached only by executing `lw` and `sw` on a halted hart through the program buffer. The consequences are worth understanding: memory accesses on behalf of the debugger go through the same arbiter, with the same permissions and the same view of the address space, as the hart's own (which is a feature, not a workaround), but they require a hart that is halted and healthy, so a chip with no hart in a debuggable state cannot have its memory inspected.
- **There are no hardware triggers.** Address and data watchpoints, and hardware breakpoints on instruction fetch, are not implemented; the trigger registers are absent. Breakpoints are software breakpoints: replace the instruction with EBREAK and set `dcsr.ebreakm`. This means breakpoints cannot be set in the boot ROM or in any read-only region.
- **The debug system requires the standard trap architecture.** `debug.enable` cannot be configured without `priv.trapCsr` (Section 5); the generator rejects the combination. EBREAK is decoded on the standard privileged instruction path, so a chip without it could not recognise a software breakpoint at all.
- **The debug system is a configuration option and is off by default.** A build without `debug.enable` contains none of this hardware (no debug port, no Debug Module, no debug mode), and its processor is bit-identical to one built before the debug system existed. See Table 1 for the setting in this build.

- **ndmreset and hartreset are not implemented.** A debugger cannot reset the system or an individual hart through the Debug Module; they read zero and ignore writes. Resetting the chip is done with the resetn pin (Section 9).
- **Resume groups are not implemented.** Halt groups stop a set of harts together; resuming them is done one at a time.
- **The entry page is volatile, and a wrong first word wedges the hart that halts into it.** The code a hart runs when it enters debug mode occupies 0x10780 onwards, and the Debug Module places it there itself: once when a debugger raises dmactive, and again before it acts on any hart it has just seen halt. A production part therefore needs nothing staged in advance, and there is no debug ROM: that page *cannot* be read-only memory, because the Debug Module rewrites part of it at every abstract command. What it is instead is ordinary shared RAM, so any master on the chip can overwrite it. Most such damage repairs itself, because the next halt rewrites the page before the Debug Module depends on it, but not damage to the **first** word. A hart that halts into a wrong first word takes an exception, re-enters at that same address, and asks the bus for the same word again; the tile answers from the copy it is already holding, so it never observes the repair and spins there, retiring nothing. Recovering that hart requires a power cycle of its tile through the power controller (Section 4); ndmreset, being unimplemented, cannot do it.
- **Debugger support exists, and its rough edges are known.** Stock OpenOCD 0.12.0 and GDB 13.2 attach, enumerate every hart, halt and resume individually, read and write registers and memory, set software breakpoints and single-step (Section 6.10). Two limits are worth stating with it. The adapter must be clocked at 3 MHz or below, for the reason given in that section. And **bulk memory writes are not yet usable**: abstractauto is unimplemented, and a debugger that assumes it, as OpenOCD's burst-write path does, writes only the first word of a block and silently drops the rest, which is why loading a program image over the debug port does not currently work. One word at a time is correct; a block is not. Single-stepping is also slow enough that data watchpoints implemented by stepping are impractical.
- **The debug port is not yet part of a signed-off physical implementation.** The five pins exist in the chip-level netlist and in the LQFP-100 package definition, but the current signed-off layout predates them.

One further note for anyone comparing this implementation against the RISC-V debug specification or against a reference simulator. Several behaviours here differ deliberately from the most widely used reference model, and in every case this implementation follows the specification's text where the reference does not: an unrecognised JTAG instruction selects BYPASS rather than leaving the previous register selected; dmistat is actually driven, and reports sticky failure as well as sticky busy; idle reports the real requirement instead of zero; an out-of-range hart selection reports *nonexistent* rather than being clamped to the highest hart; and dmstatus.version reports 3 for specification 1.0. A debugger written against the specification will find this part conformant; one written against a particular simulator's quirks may not.

7 Analog Front-End Subsystem

This chip carries a bipolar-potentiostat analog front-end (AFE) for electrochemical impedance measurement, organised as four per-quadrant measurement *sites* plus one shared electrochemical-impedance-spectroscopy (EIS) sweep engine. Each site drives its own electrode group (counter/working/reference/RE2) brought out on the package (Section 2). The analog blocks themselves (the potentiostat, the transimpedance ADC path, and the shared EIS engine + analog multiplexer) are analog IP that is not yet integrated; what *is* present is the complete *digital access path*: a register stub for each site and for the EIS engine, each a fully-functional shared-window arbiter slave with the ownership gate and interrupt path described below. Software (and the verification suite) programs and reads these exactly as it will the final analog blocks; the stubs hold placeholder storage and drive their interrupt from a software-settable flag until the analog IP replaces them.

Figure 18 is the arrangement: the same five harts and the same arbiter as Figure 1, with the analog register sites in the row underneath and the electrodes leaving the die at the bottom.

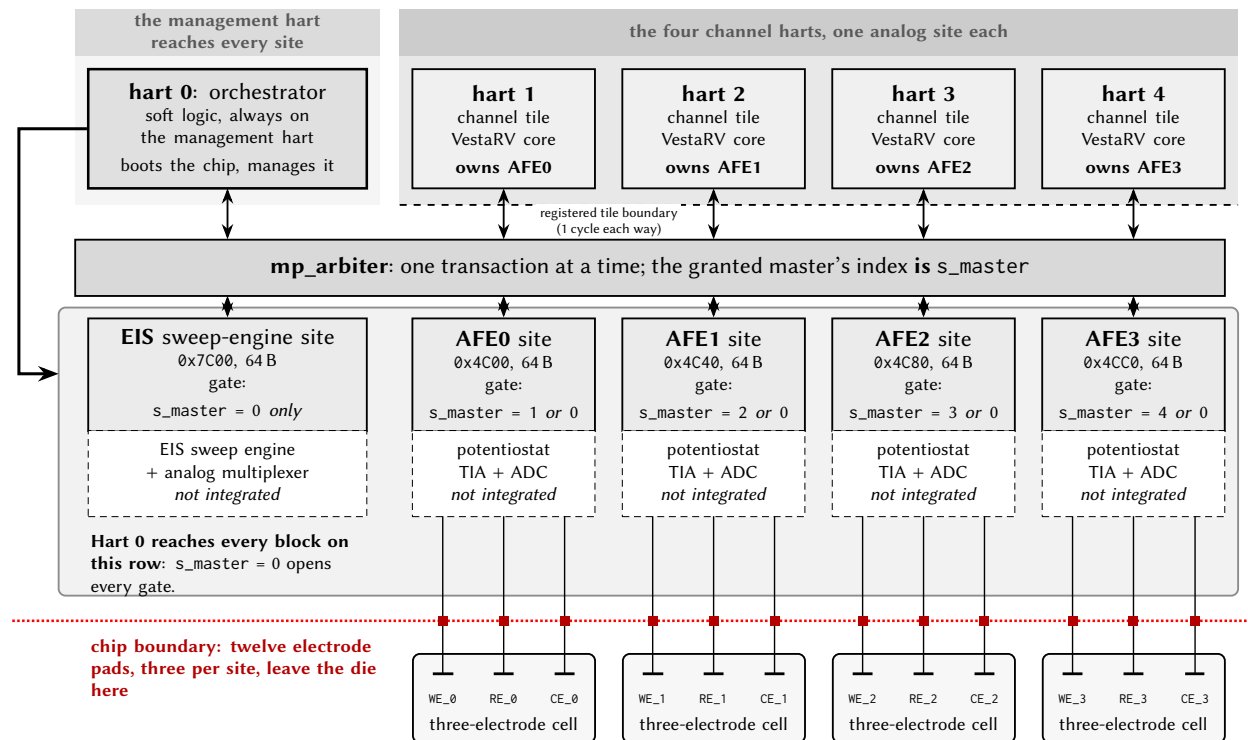


Figure 18: Analog front-end connectivity: who reaches which site, and what leaves the die. Each channel hart owns exactly one measurement site and reaches no other; hart 0, the orchestrator, opens every gate in the row, because the gate compares the arbiter's granted-master index against a single owner and against 0. The analog stages are drawn dashed because they are not yet integrated: what the chip carries today is the digital register site in front of each one (and the pads behind them). Each site brings out a three-electrode cell: WE the working electrode, RE the reference, CE the counter. A fourth pad per site, RE_{2h}, is bonded as well and is not drawn: it is the second sense electrode of the optional four-terminal (Kelvin) configuration.

7.1 Blocks, addresses, and ownership

The five blocks live in otherwise-reserved shared-window space, so they do not disturb the peripheral memory map: the four AFE sites occupy the four 64-byte sub-slots of the reserved page-0 slot 12 at 0x4C00, and the EIS engine occupies the top quarter of the IRQ-router page at 0x7C00. Every block is a 16-word (64-byte) register file reached through the multi-core arbiter like any other shared slave; access is qualified by the *ownership gate* (Section 7.2).

Block	Base	IRQ source	Access allowed when
AFE0	0x4C00	55	s_master = 1 or s_master = 0
AFE1	0x4C40	55	s_master = 2 or s_master = 0
AFE2	0x4C80	55	s_master = 3 or s_master = 0
AFE3	0x4CC0	55	s_master = 4 or s_master = 0
EIS	0x7C00	56	s_master = 0

Here s_master is the granted-master (hart) index that the multi-core arbiter attributes to the in-flight transaction, the same attribution the hardware mutex bank and the IRQ-router CLAIM use. Because each block decodes only its low four address bits, its 16 words fill its 64-byte sub-slot exactly; the EIS block additionally aliases across the 0x7C00–0x7FFF quarter it owns.

7.2 Ownership gating

Each AFE site is owned by one hart (AFE0 to hart 1, AFE1 to hart 2, AFE2 to hart 3, AFE3 to hart 4), and it answers only when s_master is that hart *or* s_master = 0. Hart 0 is the management hart, so it reaches every site (this is what lets it demultiplex the shared interrupt, below); every other hart sees only its own site. The EIS engine is instantiated hart-0-only (s_master = 0), so other harts request a sweep through a software mailbox convention rather than touching it directly. The gate is hardware-enforced inside the slave and keys off s_master alone: there is no way to forge ownership, and no cross-hart data leaks. A *denied* read returns 0 and a *denied* write is silently dropped: no bus error, no stall, no change to the arbiter contract. All registers reset to 0, so a block is a provable no-op (interrupt low, reads 0) until its owner writes it.

7.3 Register map (per block)

All five blocks share one 16-word register layout (they are the same hardware entity, differing only in the ownership gate). Offsets are byte offsets from the block base; each word is 32 bits. Detailed bit fields for the placeholder registers are owned by the analog IP specification and will be filled in when that IP is integrated.

Offset	Name	Access	Description
0x00	CTRL	RW	Control. Placeholder for the analog IP. As a bring-up test hook (until the analog IP drives real events), a write whose data bit 0 is 1 soft-sets IF bit 0, which exercises the block's level-interrupt path end to end.
0x04	DACPAT	RW	DAC excitation-pattern control. Placeholder for the analog IP.
0x08	TIA	RW	Transimpedance-amplifier gain-range select. Placeholder for the analog IP.
0x0C	SWM	RW	Switch-matrix / analog-multiplexer configuration. Placeholder for the analog IP.

0x10	ADCC	RW	ADC control. Placeholder for the analog IP.
0x14	ADCD	RW	ADC data. Placeholder for the analog IP.
0x18	STAT	RW	Status. Placeholder for the analog IP.
0x1C	IF	W1C	Interrupt-flag word. While any bit is set the block drives its level interrupt high; write a 1 to a bit to clear that bit (write-1-to-clear). Reads are side-effect-free. This is the word hart 0 reads to demultiplex which site raised the shared AFE interrupt.
0x20–0x3C	<i>reserved</i>	RW	Scratch / reserved (plain read-write storage).

7.4 Interrupt relay

The AFE/EIS interrupts reuse two vector numbers that the digital-only respin left reserved, so nothing is renumbered: the four AFE sites are OR-combined onto a single shared interrupt at source 55 (formerly the AFE0 vector), and the EIS engine drives source 56 (formerly the SARADC0 vector). Both are delivered through the IRQ router (Section 27) like any other peripheral source, and both are routed, by software convention in the routing rows, to *hart 0 only*.

Routing source 55 to hart 0 is forced, not merely chosen: because the ownership gate lets only hart 0 read all four AFE flag registers, hart 0 is the only hart that can tell which site fired. The relay is: hart 0 takes the external interrupt, CLAIMs source 55 at the router, reads the four IF words to demultiplex the originating site(s), clears the level at each firing site (write-1-to-clear on IF), COMPLETEs source 55, and then notifies the owning hart through the usual CLINT software-interrupt (msip) mailbox. Sensor-rate latency through this relay is immaterial. Source 56 (EIS) is intrinsically hart-0-only and needs no demultiplex.

Giving each AFE site its own interrupt identity (growing the source map, so a site can interrupt its quadrant hart directly without the hart-0 relay) is a deliberately deferred option: it would renumber the CLINT and external-interrupt vectors and disturb the boot-ROM park contract, so it is only worth doing if a future workload needs hart-local AFE interrupt latency.

8 Analog Circuitry

Castalia carries a mixed-signal front end alongside the digital subsystem documented in the preceding chapters. This chapter describes the analog blocks themselves and reports their simulated performance; the memory-mapped registers through which software reaches them are documented with the peripherals in Section 28.

Unless stated otherwise, every figure and table in this chapter is taken from schematic-level simulation of the block’s own testbench (pre-layout, with no extracted parasitics), and is therefore a design-intent characterisation rather than silicon data. Three places run on an extracted netlist and say so where they do: the SAR ADC section, the converter readout of the assembled channel, and the code-level probe of the impedance mode. Each block section names the testbench it came from and the process, temperature and supply points that were simulated, so the coverage behind each number is explicit. Numbers are regenerated directly from the simulator’s results database rather than transcribed by hand.

Blocks documented in this revision

The analog front end is being brought up incrementally. The electrochemical models the benches use are defined once in Section 8.1, and the nonlinear electrode model that cyclic voltammetry needs in Section 8.2. This revision documents the local bias generator, the class-AB amplifier that serves as both the channel’s transimpedance amplifier (A2) and its potentiostat control amplifier (A1), the complete potentiostat pixel system built from a pair of those amplifiers around a three-electrode cell, the assembled channel that replaces that system’s ideal sources with the silicon intended for tapeout, the analog multiplexer fabric through which the converter and the test port see that channel, the 10-bit SAR analog-to-digital converter that closes the channel, the programmable feedback resistor that sets the channel’s transimpedance gain, and the 12-bit R-2R reference DAC with the supply block that feeds it. Section 8.7 documents not a block but a second measurement mode of that channel: impedance spectroscopy performed by the potentiostat itself, with no impedance hardware. Each block section closes with the register settings that observe that block on a real die, and Section 8.11 lists the per-chip constants those measurements produce. Both are design intent: the analog test bus and the electrode isolation switches they use are not yet in the schematics.

Characterisation coverage

Block	Section	Benches	Corners	Monte Carlo
Bias generator	8.3	3	✓ [†]	n/a
Class-AB amplifier	8.4	6	✓ [†]	✓
Potentiostat pixel system	8.5	6	✓ [†]	✓
Assembled channel	8.6	1	✓ [†]	✓
Input multiplexer	8.6.5	2	✓	n/a
Impedance spectroscopy	8.7	1	✓ [†]	✓
SAR ADC	8.8	2	✓	✓
Feedback resistor	8.9	5	✓	✓
Reference DAC	8.10	6	✓ [†]	✓

[†] Corner run with the passive model rows pinned at typical — a lower bound. See below.

Every block now carries a corner run, but a corner run is not the same claim in every row: the six rows marked [†] are cornered over the transistors alone, for the reason set out immediately below. The impedance row is not a block but a mode that adds no devices, and its corner and Monte Carlo runs cover the small-signal measurement only. *n/a* marks a block whose section reports no Monte Carlo because the devices

at issue carry no per-instance mismatch terms: the multiplexer's ninety pass gates per tree, and the bias generator's start-up branch, sit on model cards that give a mismatch run a spread of zero from exactly the devices under test. The converter's Monte Carlo has been run and is reported as a negative result for a related reason (Section 8.8.3); nothing in this chapter quotes a converter gain or offset σ . Corner runs cover process, temperature and supply at the five points listed in each block section. Monte Carlo runs cover process and mismatch together unless a table states mismatch-only; a corner figure and a Monte Carlo σ are different quantities and are reported separately throughout. Design targets, where quoted alongside a measurement, are stated in the block section that reports the measurement.

The pinned-passive caveat. The corner set that serves this chapter moves the sixteen transistor model rows only. The resistor, capacitor and metal-capacitor rows hold their typical sections in every corner, so poly sheet resistance — a $\pm 35\%$ quantity — does not move at all. It sets the programmable feedback resistor of Section 8.9 directly, and it sets the reference resistor of the local bias generator, and through that the tail current, transconductance, bandwidth and supply current of every amplifier in this chapter. Every corner column marked † above is therefore the spread of the transistors alone, and is a lower bound on the true spread.

Measured on the same benches with the passive rows freed: the control amplifier's unity-gain frequency spreads 15.89 MHz to 36.09 MHz against the published 22.36 MHz to 22.91 MHz, a box 36× wider, and its supply current 81.5 μ A to 193.8 μ A against 110.0 μ A to 136.7 μ A, 4.2×; worst-case gain margin falls from 8.10 dB to 7.60 dB. On the transimpedance loop at gain code 63 the feedback resistance spreads -33.4% and $+35.5\%$ against the published $+0.44\%$, a box 157× wider; its -3 dB frequency spreads 8.53 kHz to 16.14 kHz against 10.90 kHz to 11.32 kHz, 18×; and worst-case loop phase margin falls from 71.69° to 61.52° .

The two mixed corners cannot be repaired. The foundry ships the passive families with fast, slow and typical sections only — there is no mixed-corner section for a resistor or a capacitor — so fs and sf had to pin those rows, and the pinned table was then cloned into ff and ss, where a real section does exist. That is why the tables look partly right: the bias generator's fs and sf columns agree with the freed matrix to three digits while its ff and ss columns diverge by 47 % and 25 %, because the two columns that were always going to be pinned are the two that are still correct. Only ff and ss can carry the true spread.

The Monte Carlo has been the only honest process bound in this chapter all along. Its statistical model section varies the resistors, and the 11.45 % process σ it measures on the feedback resistance is $3\sigma \approx 34\%$ — the box the corner set was missing. Every Monte Carlo table and figure in this chapter therefore stands as published. So does the converter of Section 8.8, whose extracted array carries no capacitor model card and whose poly strips are fill with no dc path; the multiplexer measurements of Section 8.6.5, which contain no poly, MIM or MOM device at all; the programmable feedback resistor of Section 8.9, whose corner axis is the resistor family by construction and which is the one section that already shows the $\pm 35\%$ band; the offset and tracking tables, which are matching-limited and move *less* when the passives are freed than the published corner box already allows; and the reference DAC's zero-scale, full-scale and LSB columns, which are ratiometric and identical to six digits pinned or freed.

The corrected corner set. Define it once as a file rather than as a table typed into each setup, so that a corrected table cannot be cloned with its defects: one section per corner, each a complete model-row list. Five sections, named so that the pin is visible from a corner label — `castalia_tt`, `castalia_ff_all` and `castalia_ss_all` with every row moved, plus `castalia_fs_mos` and `castalia_sf_mos`, which cannot move the passives and never will. Verify from the netlist that executed, never from the setup:

```
grep -E 'cor_(res|mim|rtmom)\.scs' <run>/input.scs
```

must show the corner letter on the passive rows. One gap survives the fix: the statistical model section omits the MIM statistics, so MIM capacitor variation is bounded by neither the corner set nor the Monte Carlo today – the 22 pF compensation capacitor of Section 8.4 is the device that makes that matter.

8.1 Electrochemical Models

Every electrode measurement in this chapter uses one of the two lumped models of Figures 19 and 20, valued in Table 12. The electrode impedance sets the feedback factor of both the potentiostat and the transimpedance loop, so those blocks' stability figures depend on it directly.

The three-electrode cell is what the potentiostat drives: into the working electrode, $R_{ct} \parallel C_{dl}$ in series with R_u , about 31 k Ω at DC and falling toward $R_u = 1$ k Ω above roughly 5 Hz. No bench reported here uses the single-interface model.

Both models are linear and frequency-independent, with no Warburg term and no potential dependence, so they give small-signal behaviour about a bias point rather than a full electrochemical response. Both are also nominal: no element carries a spread, so nothing in this chapter accounts for electrode-to-electrode variation, which on a real wound interface is likely to exceed the process spread reported beside it.

Model	Element	Value	Represents
three-electrode cell	R_{ce}	1 k Ω	solution resistance, CE to RE
	R_u	1 k Ω	uncompensated resistance, RE to WE surface
	R_{ct}	30 k Ω	charge transfer at the WE interface
	C_{dl}	1 μ F	double layer at the WE interface
single interface	R_s	10 k Ω	series solution resistance
	R_{ct}	1 M Ω	charge transfer
	C_{dl}	10 nF	double layer

Table 12: Component values of the electrochemical models. The three-electrode values are the subcircuit defaults; the transimpedance benches override them, and the values used are stated with each measurement.

The closed-loop measurements of Section 8.4 override the defaults with $R_u = 100 \Omega$ or 1 k Ω , $R_{ct} = 10$ k Ω or 1 M Ω , and $C_{dl} = 100$ nF or 1 μ F; the value used is stated with each measurement.

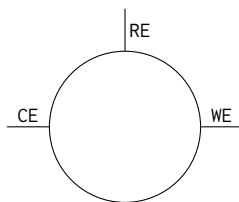


Figure 19: Three-electrode cell model, the load the potentiostat actually drives, shown as the symbol used throughout this chapter. It is a lumped linear network between the counter, reference and working electrodes; its elements and their values are given in Table 12. The model is frequency-independent (no Warburg diffusion term and no potential dependence), so it represents small-signal behaviour about a bias point rather than a full electrochemical response.

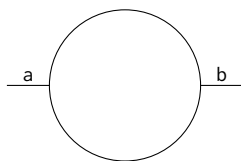


Figure 20: Single electrode–electrolyte interface, presented as a two-terminal element and shown as the same abstracted symbol. Its elements and their values are given in Table 12; they describe a much smaller double layer and a much larger charge-transfer resistance than the three-electrode cell. None of the benches reported in this chapter uses it.

8.2 Voltammetric Electrode Model

This section documents a *simulation model*, not a circuit block and not silicon: a Verilog-A element standing in for the electrode–electrolyte interface when a bench sweeps potential and expects a voltammetric peak.

A linear network cannot produce one: its extrema stay pinned at the switching potentials whatever the element values (Figure 21, dashed). A peak needs Butler–Volmer kinetics, exponential in overpotential about $E^{0'}$, together with diffusional depletion, which turns the exponential rise into a $t^{-1/2}$ decay; kinetics alone give a sigmoid.

The faradaic current is Butler–Volmer in heterogeneous-rate-constant form, emitted anodic-positive to match the transimpedance amplifier:

$$i_f = nFAk^0 \left[C_R(0, t) e^{(1-\alpha)nf(E-E^{0'})} - C_O(0, t) e^{-\alpha nf(E-E^{0'})} \right], \quad f = \frac{F}{RT}, \quad (1)$$

both exponents clamped at $\pm \text{expmax}$ through `limexp`. Surface concentrations follow diffusion into a finite Nernst layer,

$$\frac{\partial C_s}{\partial t} = D \frac{\partial^2 C_s}{\partial x^2}, \quad s \in \{O, R\}, \quad (2)$$

with flux set by i_f at the surface and bulk pinned at the layer edge,

$$D \frac{\partial C_O}{\partial x} \Big|_{x=0} = + \frac{i_f}{nFA}, \quad D \frac{\partial C_R}{\partial x} \Big|_{x=0} = - \frac{i_f}{nFA}, \quad C_s(\delta, t) = C_s^*. \quad (3)$$

The simulator solves the finite-volume form of (2) on an expanding grid of 28 slabs per species,

$$h_j \frac{dC_j}{dt} = \frac{D}{d_{j-1}} (C_{j-1} - C_j) - \frac{D}{d_j} (C_j - C_{j+1}), \quad h_j = h_0 \beta^j, \quad (4)$$

with $h_0 = 0.5 \mu\text{m}$, $\beta = 1.280$ and $\delta = 1.79 \text{ mm}$. Each slab is one electrical node carrying its concentration scaled by $1 \times 10^{-6} \text{ mol cm}^{-3}$, since the raw value solves to noise below the simulator’s voltage tolerance. Parameters are in Table 13, in the CGS units of the source literature.

Benches instantiate it as the cell of Figure 19 with the charge-transfer branch replaced by this element. Only voltammetry needs it; the linear cell remains correct for small-signal impedance work.

Validation. Peak current scales $4.07\times$ between 25 mV s^{-1} and 400 mV s^{-1} against the $4.00\times$ of the $\sqrt{v}$ law, running high against Randles–Ševčík by 0.08% to 1.86% (Table 14). Kinetics walk the Matsuda–Ayabe progression correctly: at 50 mV s^{-1} , $k^0 = 0.1 \text{ cm s}^{-1}$ gives $\Lambda = 18.1$, reversible ($\Delta E_p = 60.0 \text{ mV}$, i_p within 0.11%), and $k^0 = 1 \times 10^{-3} \text{ cm s}^{-1}$ gives $\Lambda = 0.181$, $\Delta E_p = 167 \text{ mV}$, i_p 15.5% low.

Discretisation, not the solver, is the accuracy limit: eight slabs per species is unusable (i_p high by 232%), 16 and above agree within 0.35% , and 28 ships. Dropping the maximum time step from 20 ms to 0.5 ms

Parameter	Symbol	Default	Unit	Quantity
E^0	$E^{0'}$	0.0	V	formal potential of the couple
nel	n	1	—	electrons transferred
alpha	α	0.5	—	transfer coefficient
k^0	k^0	0.1	cm s^{-1}	standard heterogeneous rate constant
Dif	D	5×10^{-6}	$\text{cm}^2 \text{s}^{-1}$	diffusion coefficient, both species
Area	A	0.01	cm^2	electrode area
Cox	C_{O}^*	1×10^{-6}	mol cm^{-3}	bulk concentration, oxidised species
Cred	C_{R}^*	0	mol cm^{-3}	bulk concentration, reduced species
Temp	T	298.15	K	interface temperature
expmax	—	40	—	exponent clamp on the kinetic terms
gpar	—	1×10^{-15}	S	DC shunt across the terminals

Table 13: Parameters of the Verilog-A electrode model, with the defaults the standalone validation runs of Table 14 use. Units are CGS, as in the source literature: a bulk concentration of 1 mmol L^{-1} is $1 \times 10^{-6} \text{ mol cm}^{-3}$ and an area of 1 mm^2 is 0.01 cm^2 . The last two rows are numerical guards, not physical quantities; the exponent clamp is never reached over the $\pm 0.8 \text{ V}$ window of this front end. The grid constants (28 slabs per species, h_0 , β , δ) are compile-time constants of the module, not parameters, and are given in Section 8.2.

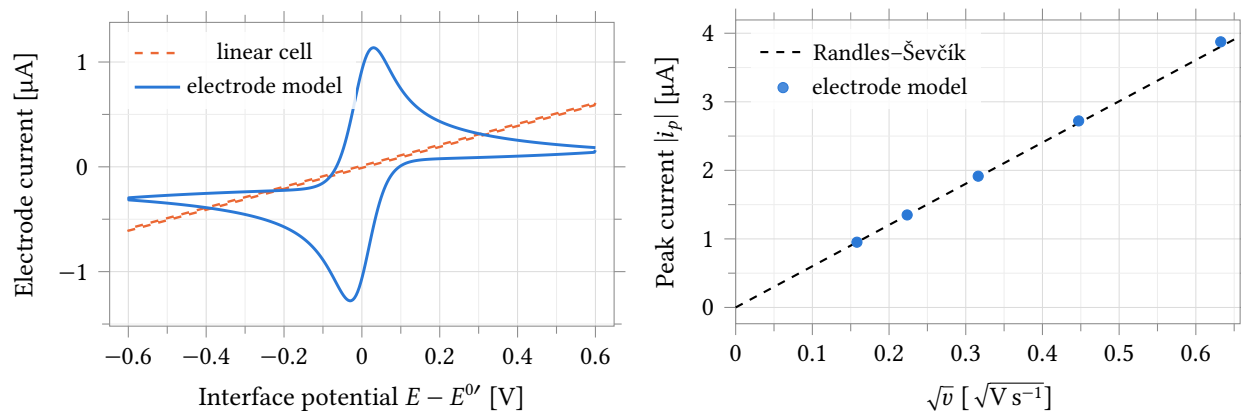


Figure 21: Standalone validation of the electrode model against closed-form voltammetry, ideal potential source, model defaults of Table 13 with $R_s = 500 \Omega$ and $C_{\text{dl}} = 0.2 \mu\text{F}$ in series and in parallel respectively, 25°C . Left: current against interface potential over a $\pm 0.6 \text{ V}$ triangular sweep at 50 mV s^{-1} , second cycle, for the nonlinear model (solid) and for the linear cell of Section 8.1 driven identically (dashed, $R_{\text{ct}} = 1 \text{ M}\Omega$); the linear response is a ramp with its extrema pinned at the switching potentials — its C_{dl} box is only 20 nA wide at this scan rate and invisible here — while the nonlinear one peaks 30 mV either side of $E^{0'}$ and decays afterwards. Right: peak cathodic current of the *first* scan against $\sqrt{v}$ over 25 mV s^{-1} to 400 mV s^{-1} , against the Randles-Ševčík line, of slope $6.015 \mu\text{A}$ per $\sqrt{\text{V s}^{-1}}$; the two panels use different cycles because Randles-Ševčík describes the first scan into an undepleted bulk, while the second cycle is the repeatable steady response and sits 3 % to 5 % below it. Numbers in Table 14.

Scan rate mV s^{-1}	i_p Randles–Ševčík μA	i_p model μA	Error %	Error, $C_{\text{dl}} \rightarrow 0$ %	ΔE_p mV
25	0.951	0.952	+0.08	−0.45	58.78
50	1.345	1.349	+0.32	−0.42	59.85
100	1.902	1.915	+0.67	−0.38	60.41
200	2.690	2.721	+1.16	−0.33	62.33
400	3.804	3.875	+1.86	−0.25	64.74

Table 14: Peak metrics of the electrode model against the Randles–Ševčík law, first cathodic scan, conditions as Figure 21. The last-but-one column repeats the measurement with C_{dl} removed: the remaining −0.45 % to −0.25 % is a scan-rate-independent discretisation bias of the 28-slab grid, so the whole scan-rate dependence of the error is double-layer charging, which the ablation reproduces as $C_{\text{dl}}v/i_p$ to two decimals. Peak separation is inflated by iR drop across $R_s = 500 \Omega$, which grows with scan rate; with R_s and C_{dl} removed it is 57.04 mV at every rate, against Nicholson’s reversible $2.218RT/F = 56.9$ mV.

moves i_p below the fourth significant figure. Roughly 60 sweeps ran without a convergence failure, and the exponentials survive ± 12 V of overpotential with the clamp removed, so the clamp and shunt conductance are insurance.

One design limit follows. At 50 mV s^{-1} on 1 mm^2 , double-layer charging is 0.74 % of the peak at 1 mmol L^{-1} but 37 % of a 27 nA peak at 0.02 mmol L^{-1} : below roughly 0.1 mmol L^{-1} the faradaic peak is buried in charging current rather than in the front end’s noise, and the remedy is a slower scan, not more gain.

First measurement with the rev-2 channel in the loop. The cyclic-voltammetry test is the potentiostat bench of Section 8.5 with the linear cell swapped for the nonlinear one and nothing else changed: ideal $R_f = 35 \text{ k}\Omega$, $C_{\text{dl}} = 0.2 \mu\text{F}$, $R_u = R_{\text{ce}} = 1 \text{ k}\Omega$, electrode at $E^{0'} = 0.3 \text{ V}$, 1 mmol L^{-1} , $k^0 = 0.1 \text{ cm s}^{-1}$, commanded over a 0.75 V to 1.75 V triangle at 2 V s^{-1} , typical process, 37°C for silicon and electrode alike.

The peak measures $8.666 \mu\text{A}$ against $8.74 \mu\text{A}$ predicted ($8.34 \mu\text{A}$ Randles–Ševčík plus $0.40 \mu\text{A}$ of $C_{\text{dl}} dv/dt$), −0.8 %: the nonlinearity survives the loop rather than being reshaped by it. Read as $i_{\text{we}} = (v_{\text{out}} - V_{\text{CM}})/R_f$, the peak moves the output 0.30 V from $V_{\text{CM}} = 1.25 \text{ V}$, 38 % of the $\pm 0.80 \text{ V}$ window, so the sweep is unclipped. The four wound-panel analytes run as corners on the same test (Figure 22).

What these numbers do not cover.

- Nominal transients only: one process point, one temperature, no corner set and no Monte Carlo. The analyte corners vary electrode parameters alone.
- A Verilog-A element carries no mismatch statistics, so every Monte Carlo σ in this chapter holds the electrode nominal and reports circuit variation only.
- Table 14 runs at 25°C ; the in-loop run is the only 37°C result. Temp is a plain model parameter, so a new bench must set it explicitly or the kinetics silently revert to 25°C . D is held at its 25°C value throughout, so the $\sim 30\%$ rise in a real aqueous D over 25°C to 37°C is not modelled.
- Randles–Ševčík assumes semi-infinite diffusion; the model pins bulk at $\delta = 1.79 \text{ mm}$. That holds by $16\times$ at 50 mV s^{-1} ($\sqrt{Dt} = 0.11 \text{ mm}$) but not for a thin unstirred film, for which a $93 \mu\text{m}$ variant attenuates the reverse peak to 0.65 of the forward. Quote δ with any peak from this model.
- k^0 shapes the peak but neither identifies nor quantifies an analyte, and it is the least-supported parameter: no rate constant on carbon is published for these four markers, so the values in use are fitted to a measured ΔE_p . A gap in the literature, not the model.

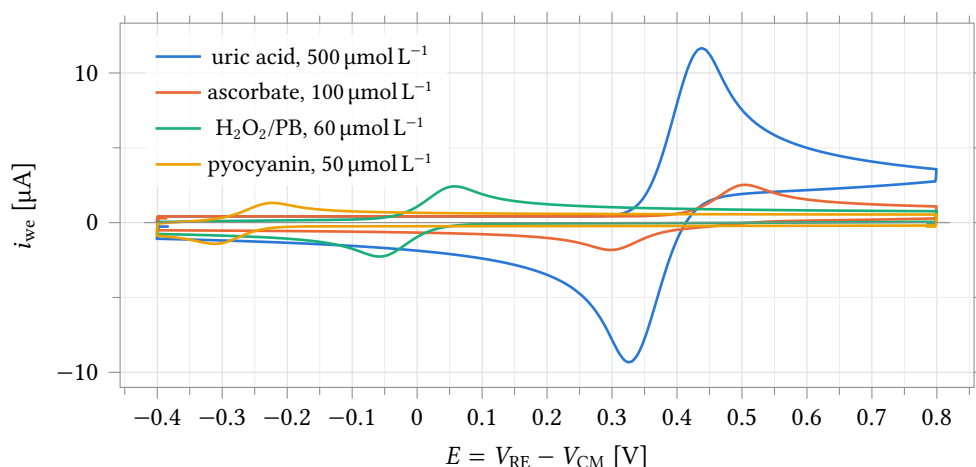


Figure 22: Voltammograms of the four-analyte wound panel through the rev-2 channel, one electrode parameter set per channel, from the cyclic-voltammetry Maestro corner run of Section 8.2 (2 V s^{-1} , $R_f = 35 \text{ k}\Omega$, conditions of the in-loop run above). Peak potential identifies the analyte and peak height its concentration; each trace is one full cycle started at that analyte's non-faradaic vertex. $E^{0'}$ and concentrations are literature-anchored; k^0 is fitted (no measured value exists on carbon for any of the four), and reversal peaks appear for the chemically irreversible couples because following chemistry is not modelled. The two cathodic markers span $-1.4 \mu\text{A}$ to $-2.4 \mu\text{A}$ — unmeasurable by the unidirectional rev-1 front end.

Differential-pulse voltammetry through the channel. The same bench runs a staircase-plus-pulse program at the published assay envelope (Figure 23), every 4 mV tread quantised to the 14-bit DAC LSB of $152.6 \mu\text{V}$ and current sampled 5 ms before each edge. Four reporters give four resolved peaks, apexes within 2 mV and 1 % of the assay targets. Three findings bound how the numbers read:

- *No residual charging baseline.* A blank gives under 1 pA at every step, 45 ms of a 50 ms pulse being far beyond the $200 \text{ nF}/1 \text{ k}\Omega$ time constant, so $\Delta I = I(S_2) - I(S_1)$ cancels charging entirely.
- *The 560 k Ω tap slew-limits on every edge.* A 50 mV step into 200 nF is 10 nC against a $\sim 2.2 \mu\text{A}$ output stage, leaving the applied potential wrong for $\sim 20 \text{ ms}$. Settled samples are unaffected, but the transfer turns superlinear above $\sim 350 \text{ nA}$ (up to 30 % at the top of the tap; proportional within 3 % below it), which is why Figure 24 uses a *measured* transfer.
- *Full scale is consumed by $I(S_2)$, not ΔI .* The largest sampled current is $1.02\text{--}1.04\times$ the differential peak and reaches 84 % of the $\pm 2.06 \mu\text{A}$ tap; do headroom accounting on $I(S_2)$.

Sources. The abbreviations of Table 15 are: B&F, A. J. Bard and L. R. Faulkner, *Electrochemical Methods: Fundamentals and Applications*, 2nd ed., Wiley, 2001; B&S, D. Britz and J. Strutwolf, *Digital Simulation in Electrochemistry*, 4th ed., Springer, 2016; Randles 1947, J. E. B. Randles, *Discuss. Faraday Soc.* 1:11, 1947; Feldberg 1969, S. W. Feldberg, *Electroanal. Chem.* 3:199, 1969; Nicholson 1965, R. S. Nicholson, *Anal. Chem.* 37:1351, 1965; Matsuda & Ayabe 1955, H. Matsuda and Y. Ayabe, *Z. Elektrochem.* 59:494, 1955. The peak-current law is due independently to Randles (*Trans. Faraday Soc.* 44:327, 1948) and Ševčík (*Coll. Czech. Chem. Commun.* 13:349, 1948); the numerical form quoted in Table 15 is the 25°C one of B&F Eq. 6.2.19.

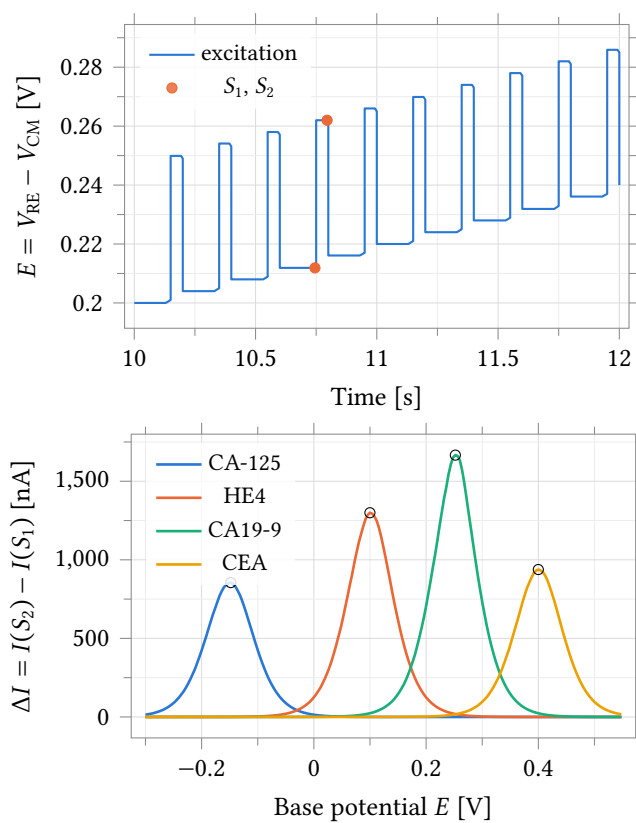


Figure 23: Differential-pulse voltammetry through the rev-2 channel, four-reporter immunoassay panel, one electrode parameter set per reporter ($R_f = 560 \text{ k}\Omega$, $C_{dl} = 200 \text{ nF}$, 37°C). Top: a 2 s excerpt of the excitation — 4 mV treads, 50 mV / 50 ms pulses at a 200 ms period, every level quantised to the 14-bit DAC LSB — with the two current-sampling instants 5 ms before each edge. Bottom: the differential current over the full -0.300 V to 0.548 V scan (213 steps, 42.6 s); peak potential separates the reporters, peak height carries concentration (circles mark the apexes: 855 nA, 1300 nA, 1666 nA and 938 nA). Reporter kinetics as in the cyclic-voltammetry runs; the panel differs from Figure 22 in readout, gain tap and reporter set.

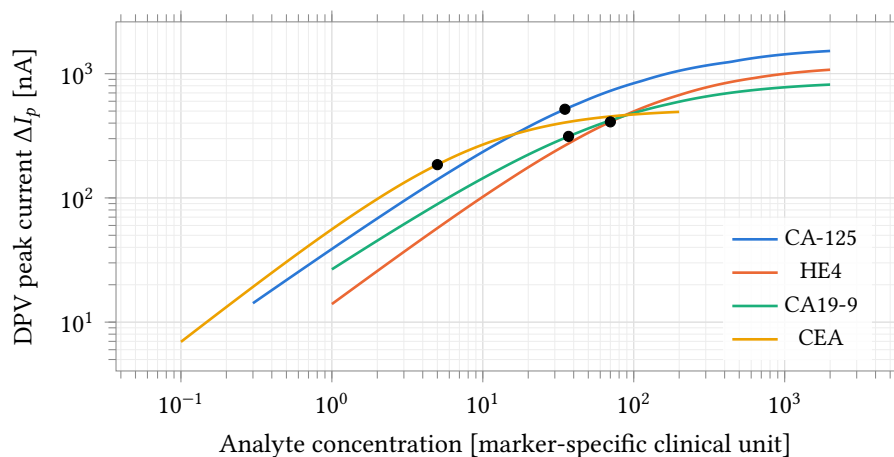


Figure 24: Dose response of the four-reporter panel: published-assay binding curves composed with the current transfer *measured* from the DPV runs of Figure 23 (11 simulated concentrations per reporter, 560 kΩ tap). Filled circles are the clinical decision points through the same chain: CA-125 35 U/mL → 517 nA, HE4 70 pmol L⁻¹ → 409 nA, CA19-9 37 U/mL → 313 nA, CEA 5 ng mL⁻¹ → 185 nA. The concentration axis carries each marker's own clinical unit; only the current axis is common. The binding model is analytic; the electrode and channel behind the current axis are simulated.

Element	Form	Source
<i>Implemented</i>		
Equivalent circuit	$R_s + C_{dl} \parallel Z_f$; the model is Z_f	Randles 1947
Faradaic current	$i = nFAk^0 [C_R(0, t)e^{(1-\alpha)nf\eta} - C_O(0, t)e^{-\alpha nf\eta}]$	B&F Eq. 3.3.11
Mass transport	$\partial C / \partial t = D \partial^2 C / \partial x^2$	B&F Eq. 4.4.2
Surface boundary	$i / (nFA) = D \partial C_O / \partial x _{x=0}$	B&F Eq. 4.4.29
Outer boundary	$C(\delta, t) = C^*$, Nernst diffusion layer	B&F Sec. 1.4.2
Discretisation	box (finite-volume) method	Feldberg 1969; B&S Ch. 3
Space grid	$h_j = h_0 \beta^j$	B&S Ch. 7
<i>Validated against</i>		
Peak current	$i_p = (2.69 \times 10^5) n^{3/2} AD^{1/2} C^* v^{1/2}$	B&F Eq. 6.2.19
Peak separation	$\Delta E_p = 2.218 RT / nF$	Nicholson 1965
Reversibility	$\Lambda = k^0 / \sqrt{\pi D n f v}$	Matsuda & Ayabe 1955

Table 15: Every relation the electrode model implements, in the form it is implemented, with the source it is taken from; the same map is carried in the model's own header. The lower group is not implemented — those are the closed forms the model is validated against in Table 14. Here $\eta = E - E^{0'}$ and $f = F/RT$. Equation numbers are those of the second edition of Bard and Faulkner; the third edition rennumbers.

8.3 Local Wide-Swing Cascode Bias Generator

The local bias generator (schematic cell `anatop_biasgen_local`, drawn from `BiasGenCascWideSwing`) is the cell each analog block instantiates to develop its own cascode bias rails, as distinct from the single chip-level `BiasGenCascWideSwingGlobal` that drives the global `bn! / bp!` nets. It presents four outputs, the NMOS mirror and cascode gates (V_{BNM} , V_{BNC}) and the PMOS mirror and cascode gates (V_{BPM} , V_{BPC}), generated from a resistor-degenerated reference core in the wide-swing cascode arrangement, so the cascode gates sit only one saturation voltage away from the mirror gates and the mirrors keep their output devices in saturation over a wide compliance range. An `En` input gates the cell, and a start-up branch guarantees it leaves the degenerate zero-current state at power-up. A 6-bit trim word, `BiasAdj<5:0>`, moves the reference current so the operating point can be corrected after silicon.

The characterisation below is from schematic-level simulation (no extracted parasitics) of the `BiasGenCascWideSwing_tb` testbench, over the process, temperature and supply points of Table 16. All rail voltages are referred to `vss!`, and the cell is enabled throughout.

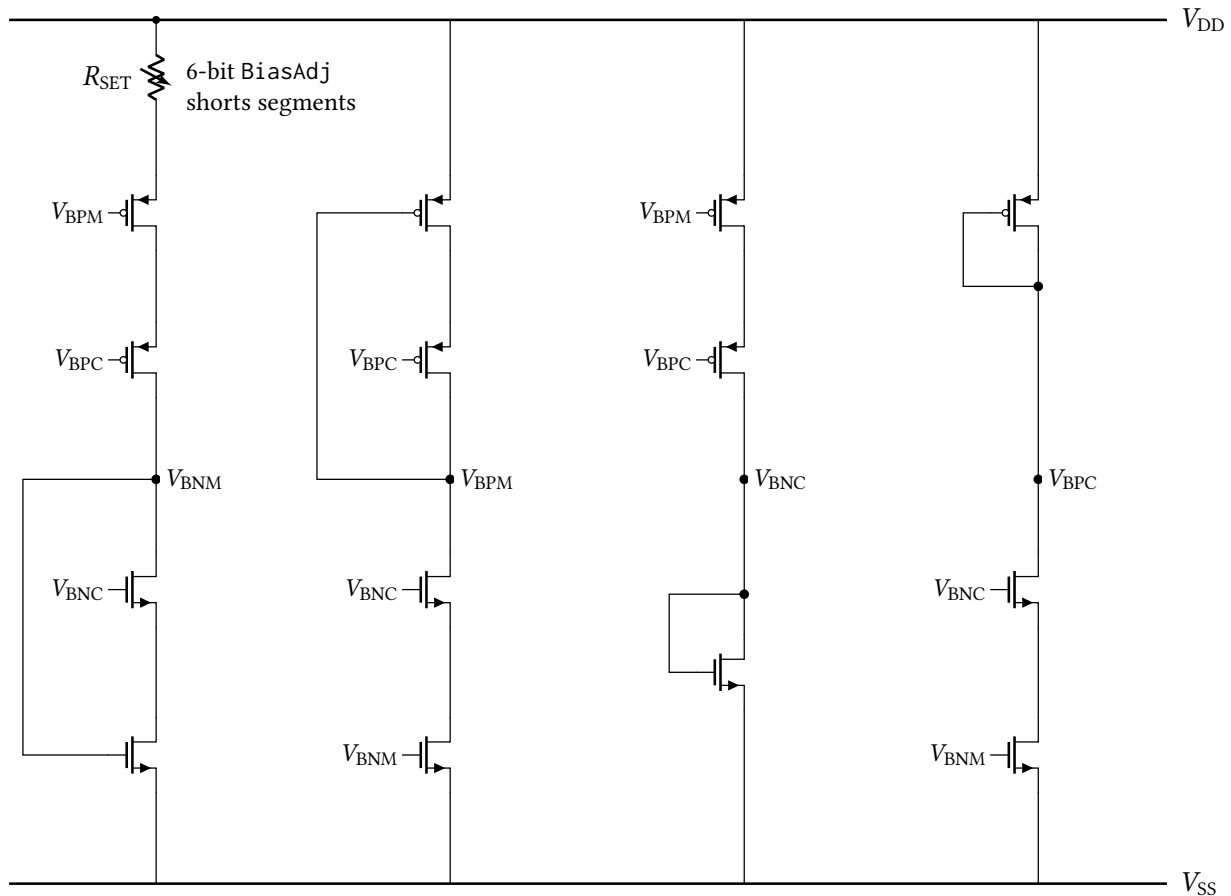


Figure 25: Simplified core of the local wide-swing cascode bias generator, drawn from the `anatop_biasgen_local` schematic. Four self-biased branches develop the four rails: the reference branch (left) passes the current set by the trimmable resistor R_{SET} , and the remaining branches mirror it to produce V_{BPM} , V_{BNC} and V_{BPC} . Each branch's own self-bias (diode) connection is drawn, from the rail it generates back to the gate it drives in the same branch; the cross-branch gate connections are identified by net name rather than drawn as buses. The enable switches, the decoupling capacitors on each rail and the individual segments of the resistor string are omitted for clarity.

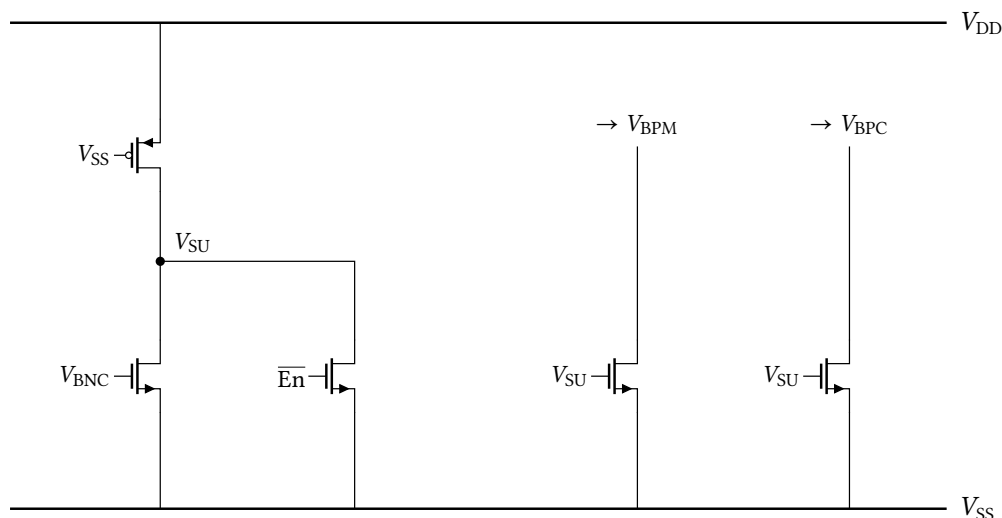


Figure 26: Start-up branch. A gate-grounded, permanently-on weak PMOS pulls V_{SU} up at power-on; while V_{SU} is high it drags V_{BPM} and V_{BPC} down, forcing current into the core and breaking the degenerate zero-current state. Once the core is running, the rising V_{BNC} turns on the left NMOS and collapses V_{SU} , so the branch draws no static current in normal operation, the behaviour visible in the power-up transient of Figure 34. A second device holds V_{SU} low whenever the cell is disabled.

Point	Process	Temperature	Supply
1	tt	27 °C	2.5 V
2	ff	25 °C	2.5 V
3	fs	25 °C	2.5 V
4	sf	25 °C	2.5 V
5	ss	25 °C	2.5 V

Table 16: Process, temperature and supply points simulated for the local bias generator. These points move the sixteen transistor model rows only – the resistor and capacitor rows stay at their typical sections – so every corner spread in this section is a lower bound (pinned-passive caveat, page 69).

Parameter	Unit	tt	ff	fs	sf	ss	Spread
V_{BNM}	mV	614.3	578.9	584.6	644.0	657.5	78.6
V_{BNC}	mV	776.2	765.4	745.4	807.1	805.7	61.7
V_{BPM}	mV	1815.4	1841.7	1804.2	1826.6	1777.6	64.1
V_{BPC}	mV	1589.3	1571.3	1578.4	1600.2	1575.2	28.9
I_{DD}	μA	24.38	35.97	24.23	24.52	18.38	17.59

Table 17: DC operating point of the local bias generator at the nominal trim code ($\text{BiasAdj} = 33$), 25°C , by process corner; the last column is the spread across corners. The degeneration resistor is moved with the process corner here. An earlier corner set held it at typical in every corner and reported a supply-current spread of $0.30\mu\text{A}$: that resistor sets the reference current, so pinning it removed almost all of the block's corner spread, and the fs and sf columns still agree with it because the resistor family has no skew section.

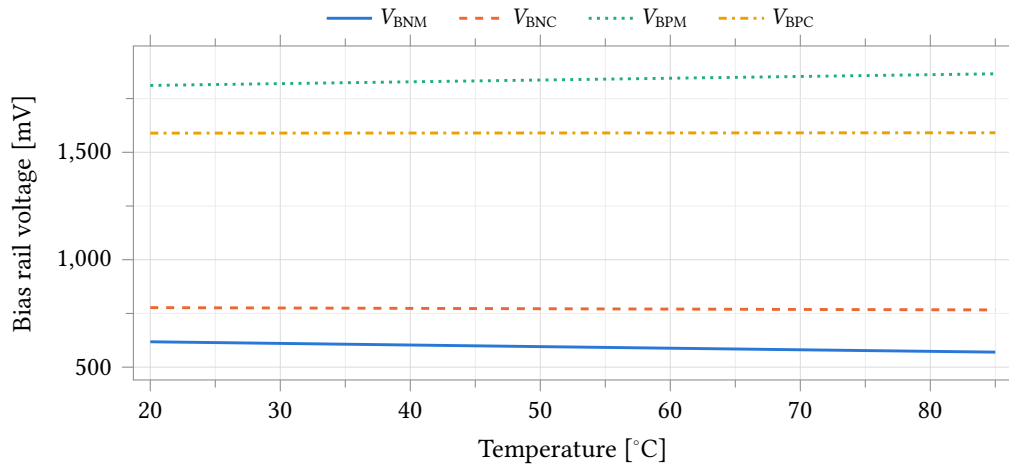


Figure 27: The four local bias rails against temperature at the typical corner. V_{BNM} and V_{BNC} are referred to vss!; V_{BPM} and V_{BPC} sit near the supply.

Parameter	Unit	Min	Mean	Max	Spread
V_{BNM}	mV	569.9	593.8	618.1	48.2
V_{BNC}	mV	766.3	771.5	777.1	10.8
V_{BPM}	mV	1811.2	1838.4	1865.4	54.2
V_{BPC}	mV	1589.2	1590.0	1590.9	1.7
I_{DD}	μA	24.16	25.50	26.81	2.65

Table 18: Typical-corner extremes over the simulated temperature range.

Parameter	Unit	tt	ff	fs	sf	ss
V_{BNM}	ppm/°C	1248	1432	1324	1179	1097
V_{BNC}	ppm/°C	215	308	232	198	134
V_{BPM}	ppm/°C	453	449	455	451	459
V_{BPC}	ppm/°C	17	53	14	19	24
I_{DD}	ppm/°C	1598	1518	1624	1573	1673

Table 19: Temperature coefficient of each bias rail and of the supply current, taken as the total variation over the simulated temperature range normalised to the mean. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

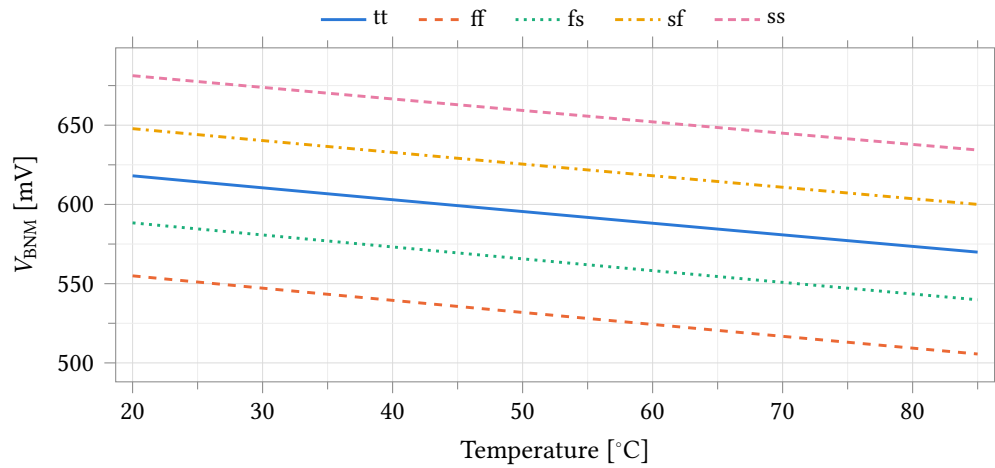


Figure 28: NMOS mirror bias V_{BNM} against temperature over all process corners. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

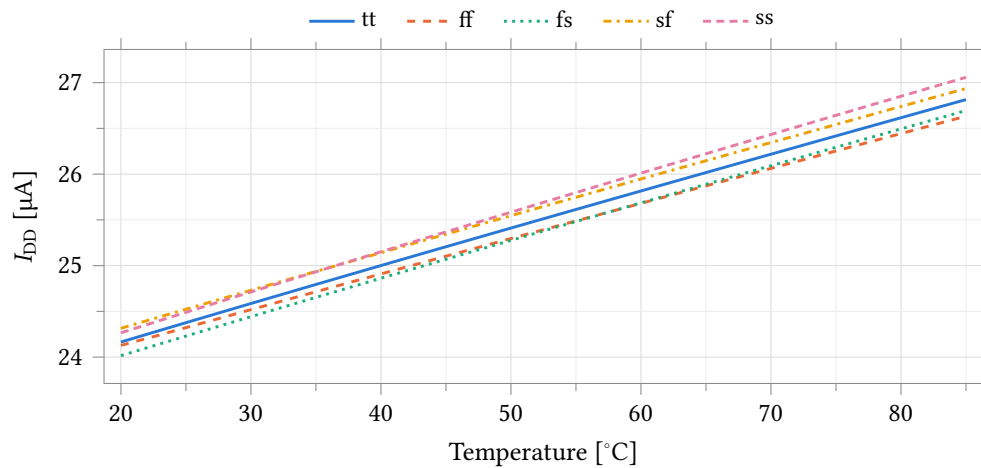


Figure 29: Quiescent supply current of the bias generator against temperature over all process corners. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	tt	ff	fs	sf	ss
V_{BNM}	% V^{-1}	0.30	0.32	0.31	0.28	0.31
V_{BNC}	% V^{-1}	0.53	0.55	0.54	0.50	0.55
$V_{\text{DD}} - V_{\text{BPM}}$	% V^{-1}	0.31	0.33	0.30	0.31	0.32
$V_{\text{DD}} - V_{\text{BPC}}$	% V^{-1}	0.61	0.62	0.60	0.61	0.65
I_{DD}	% V^{-1}	18.74	20.21	18.55	19.23	17.98

Table 20: Line sensitivity: total variation over the simulated supply range, normalised to the mean and to the swept supply span. The PMOS rails are supply-referred (they track V_{DD} by design), so it is the supply-referred drops $V_{\text{DD}} - V_{\text{BPM}}$ and $V_{\text{DD}} - V_{\text{BPC}}$ that are quoted here, not the raw rail voltages, whose sensitivity against v_{ss} ! would simply restate that tracking. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

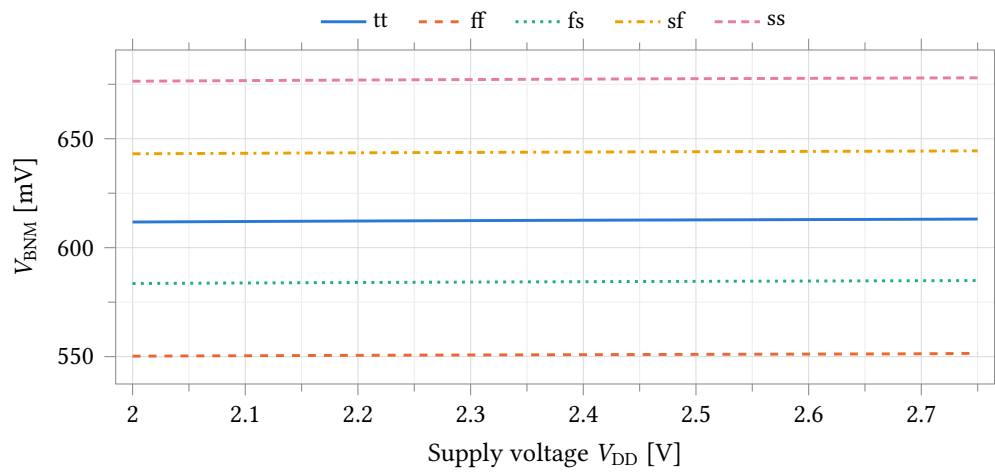


Figure 30: Supply rejection of the NMOS mirror bias: V_{BNM} against V_{DD} over all process corners. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

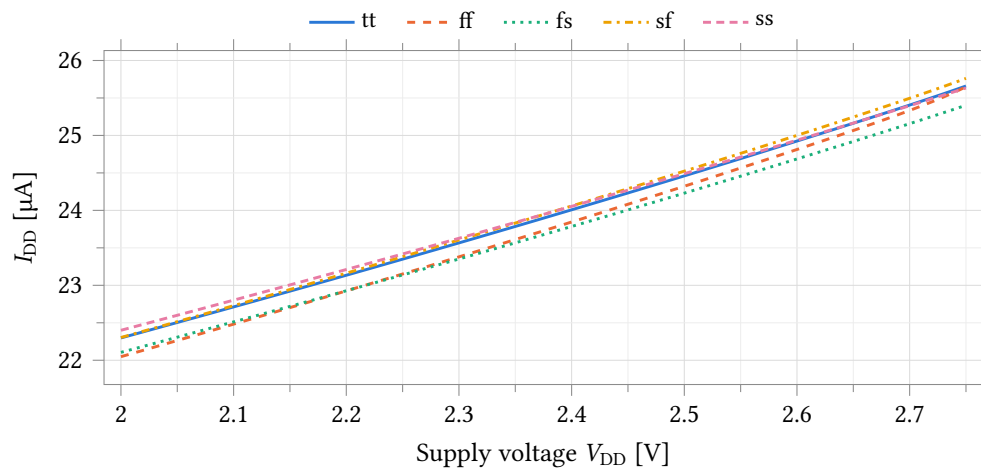


Figure 31: Quiescent supply current against supply voltage over all process corners. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	tt	ff	fs	sf	ss
V_{BNM}	mV	89.6	87.5	88.7	89.7	91.0
V_{BNC}	mV	198.2	190.2	194.8	198.8	203.8
V_{BPM}	mV	98.9	96.4	98.3	98.6	100.7
V_{BPC}	mV	283.2	271.6	281.2	281.4	291.4
I_{DD}	μA	39.87	39.32	39.55	39.81	40.07

Table 21: Trim range: total swing of each quantity between BiasAdj codes 0 and 63. Those are the extremes of the six codes the bench samples. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

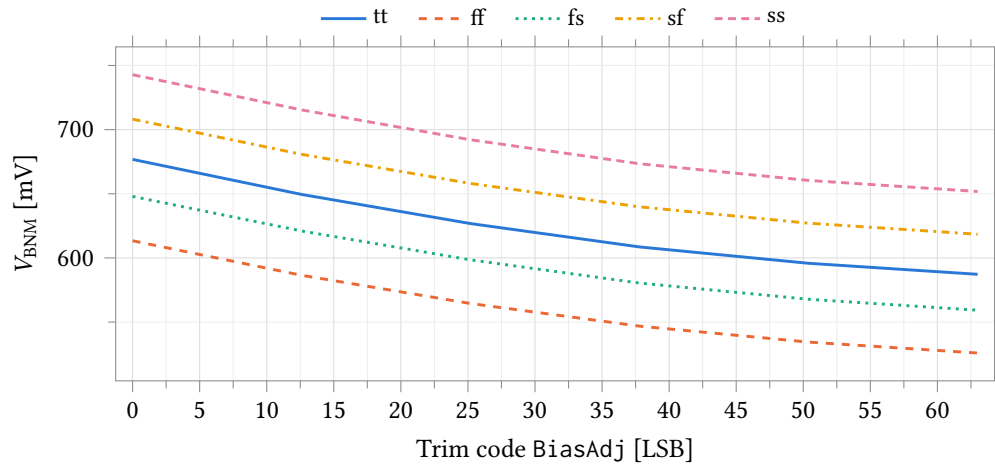


Figure 32: Trim characteristic: V_{BNM} against the BiasAdj word over all process corners. The bench sweeps the trim word as a continuous variable, so only the six codes 0, 12.6, 25.2, 37.8, 50.4 and 63 are sampled: this is the shape of the trim law, not a per-code characterisation. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

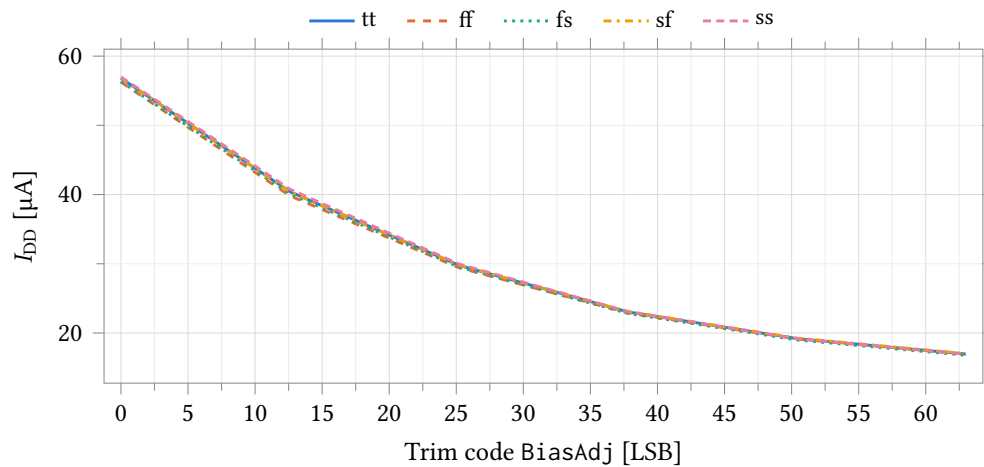


Figure 33: Trim characteristic: quiescent supply current against the BiasAdj word over all process corners, at the same six sampled codes. The current is strongly monotonic in the trim word, falling by a factor of about 3.4 from code 0 to code 63. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	tt	ff	fs	sf	ss
V_{BNM}	μs	0.85	0.77	0.83	0.87	0.91
V_{BNC}	μs	0.95	0.87	0.93	0.97	1.01
V_{BPM}	μs	1.10	1.10	1.10	1.10	1.10
V_{BPC}	μs	1.10	1.10	1.10	1.10	1.10

Table 22: Power-up settling time of each bias rail, defined as the last instant at which the rail is still outside a 1 % band around its final value. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

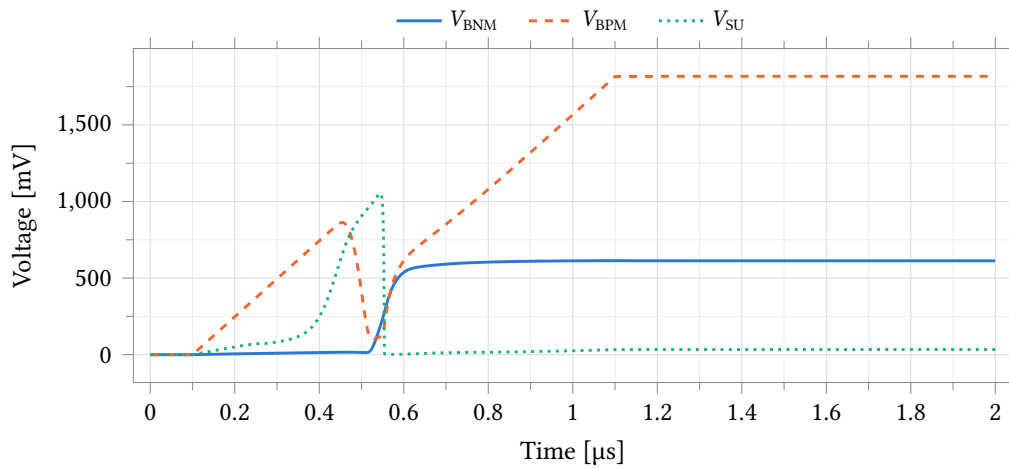


Figure 34: Power-up transient at the typical corner. The start-up node V_{SU} pulls the core out of its zero-current state, after which both mirror rails settle to their steady-state values.

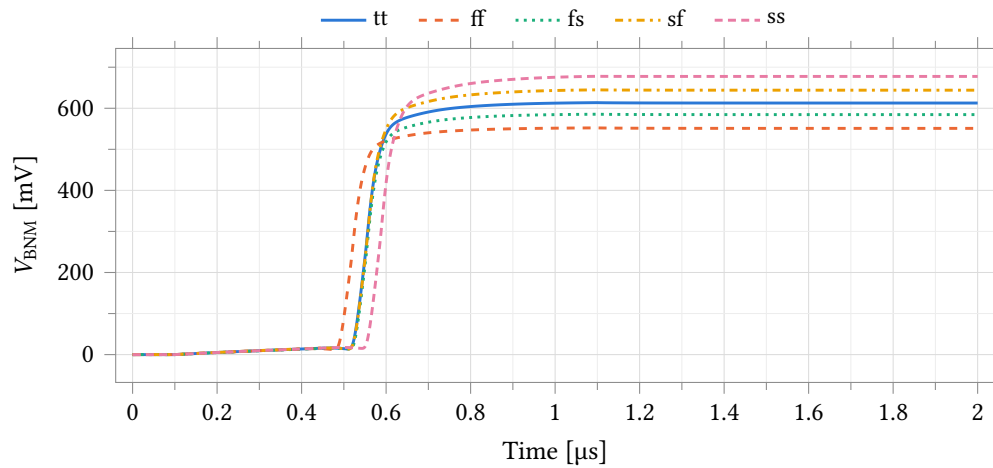


Figure 35: Power-up settling of V_{BNM} over all process corners. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

8.3.1 Start-up

Start-up is verified across 101 corner points — supply ramp rate over five decades, five process corners, four temperatures and three trim codes, plus a brown-out depth sweep and an enable cycle — and all 101 start, none into the degenerate state. The settled operating point is bit-identical across all five decades of ramp rate: the spread of V_{BNM} and of I_{DD} over the six ramp rates is zero to six digits at every process corner. Ramp rate changes when the generator arrives, not where.

Supply ramp	$t_{\text{set}}(V_{\text{BNM}})$	ratio	$t_{\text{set}}(V_{\text{BPC}})$	ratio
100 ns	211 ns	2.11	232 ns	2.32
1 μs	741 ns	0.741	994 ns	0.994
10 μs	4.40 μs	0.440	9.94 μs	0.994
100 μs	43.5 μs	0.435	99.4 μs	0.994
1 ms	436 μs	0.436	994 μs	0.994
10 ms	4.36 ms	0.436	9.94 ms	0.994

Table 23: Settling time of the two extreme bias rails against supply ramp rate, at tt and 25 °C, defined as the last instant outside a 1 % band around the value reached; the ratio is settling time over ramp time. Above 10 μs the block is ramp-limited and the rails settle at a fixed fraction of the ramp; below 1 μs it is circuit-limited, and the floor is 211 ns and 232 ns here, 487 ns and 641 ns at the slow corner.

No brown-out floor exists. The supply is ramped up, held, dipped and returned, at thirteen depths from 2.0 V down to 0 V and at three hold times. Every dip recovers exactly: the ratio of V_{BNM} after to before is 1.000 000 to six digits at every depth and every corner, and the supply current returns to its pre-dip value. How far the rails actually fall is the honest caveat — at 0 V the only discharge path for the rail decoupling is subthreshold leakage, so a short dip is not a power-down and V_{BNM} sags only 72 %. Held 100 ms at 0 V the rails do collapse, to 0.27 mV at the slow hot corner, and recovery is still exact. A recovery edge slower than 1 ms is untested.

The enable input is the stronger test, and it passes. De-asserted, it pulls both NMOS rails to ground and holds the start-up node down, so the cell is forced into the true zero-current state — $V_{\text{BNM}} = 0.000\,00\text{ V}$, against the 96.8 % collapse a 100 ms power-down manages. On re-assert both the rails and the supply current return to 1.000 000 of their previous values at all six cells tested.

One limit on the coverage. The devices that do the starting sit on model cards that carry no per-instance mismatch terms, so a mismatch Monte Carlo of start-up as the cell stands would report zero contribution from exactly the mechanism under test, and none has been run. The results above are corner-based and do not depend on it: the process corner reaches those devices through the model-card parameters they do consume.

8.3.2 Measuring this block on chip

1. Set AFE_DBGOWNSEL = 5; step AFE_DBGGSEL over slots 2–5 for the four global rails. A channel’s own rails are per-channel slots 10 and 11 under AFE_DBGSEL.
2. Read on the buffered pad atb_o with AFE_DBGBUFEN = 1, never on the direct pad: these are gate nets with picoamperes of drive, and the direct path moves what it measures.
3. Compare against the tt column of Table 17: 614.3 mV and 776.2 mV on the NMOS pair, 1815.4 mV and 1589.3 mV on the PMOS pair, corner spread 29 mV to 79 mV. A rail or near-zero on any of the

four means the start-up branch did not fire; no measurement elsewhere on the die is meaningful until it does.

4. Trim from the supply pin, not through the multiplexer: step the 6-bit code and keep the one whose total analog supply current is nearest design — 2.7 % per code over a 39.9 μA span (Table 21), against a 12.5 % Monte Carlo spread the corner runs do not show. The code is per die; $\pm\frac{1}{2}$ code leaves ± 1.4 %.
5. Recovery: chip-level slot 8 drives the reference-current node from atb_d with AFE_DBGDIREN = 1 — the last path to a working amplifier on a die whose generator did not start.

This test fabric is design intent; it is not yet in the schematics.

8.4 Class-AB Amplifier

One cell, anatop_tia_amp_ab (34 transistors; schematic, symbol, layout and extracted views), serves as both amplifiers of a potentiostat channel: the transimpedance amplifier A2 (Section 8.4.2) and the control amplifier A1 (Section 8.4.3). The cell name records its first application; the device is role-independent and the two roles differ only in the network around it. All data in this section is schematic-level, with no extracted parasitics.

The topology is drawn in Figure 36. A complementary rail-to-rail input stage places an NMOS and a PMOS pair in parallel, both active at the 1.25 V common mode used throughout this section; the two pairs fold into a cascode summing stage, which drives a class-AB output stage whose two output devices are both actively driven. Cascode (Ahuja) compensation is taken from the common source node of the PMOS cascodes, C1, and of the NMOS cascodes, C2. Both nodes are brought out as pins and returned to the output by one 500 fF capacitor each, outside the cell, so compensation is set per application without editing the cell.

Bias comes from the local bias generator of Section 8.3 over the global cascode rails, at trim code BiasAdj = 37 throughout this section.

The device-level characterisation below is taken at a 2.5 V supply into a 5 pF load on three benches: the unity-feedback bench of Figure 37, an open-loop bench with the input offset nulled before measurement, and a swept-load drive bench. Coverage is five corner points — the typical statistical point at nominal and 37 °C, and the ff, fs, sf and ss skew corners at 25 °C, uniform across every test of this bench — and a 200-sample Monte Carlo run varying process and mismatch together at 37 °C.

Point	Process	Temperature	Supply
1	tt	37 °C	2.5 V
2	ff	25 °C	2.5 V
3	fs	25 °C	2.5 V
4	sf	25 °C	2.5 V
5	ss	25 °C	2.5 V

Table 24: Process, temperature and supply points simulated for the class-AB amplifier in its control-amplifier role. Point 1 is the typical statistical process point, evaluated with zero statistical offset; points 2–5 are the imported foundry skew corners. Temperature is uniform across the seven benches of this run at every point: 37 °C at point 1 and 25 °C at points 2–5, so the typical point carries the body-temperature condition and the skew corners do not. Every figure and table in this section is schematic-level. These points move the sixteen transistor model rows only — the resistor and capacitor rows stay at their typical sections — so every corner spread in this section is a lower bound (pinned-passive caveat, page 69).

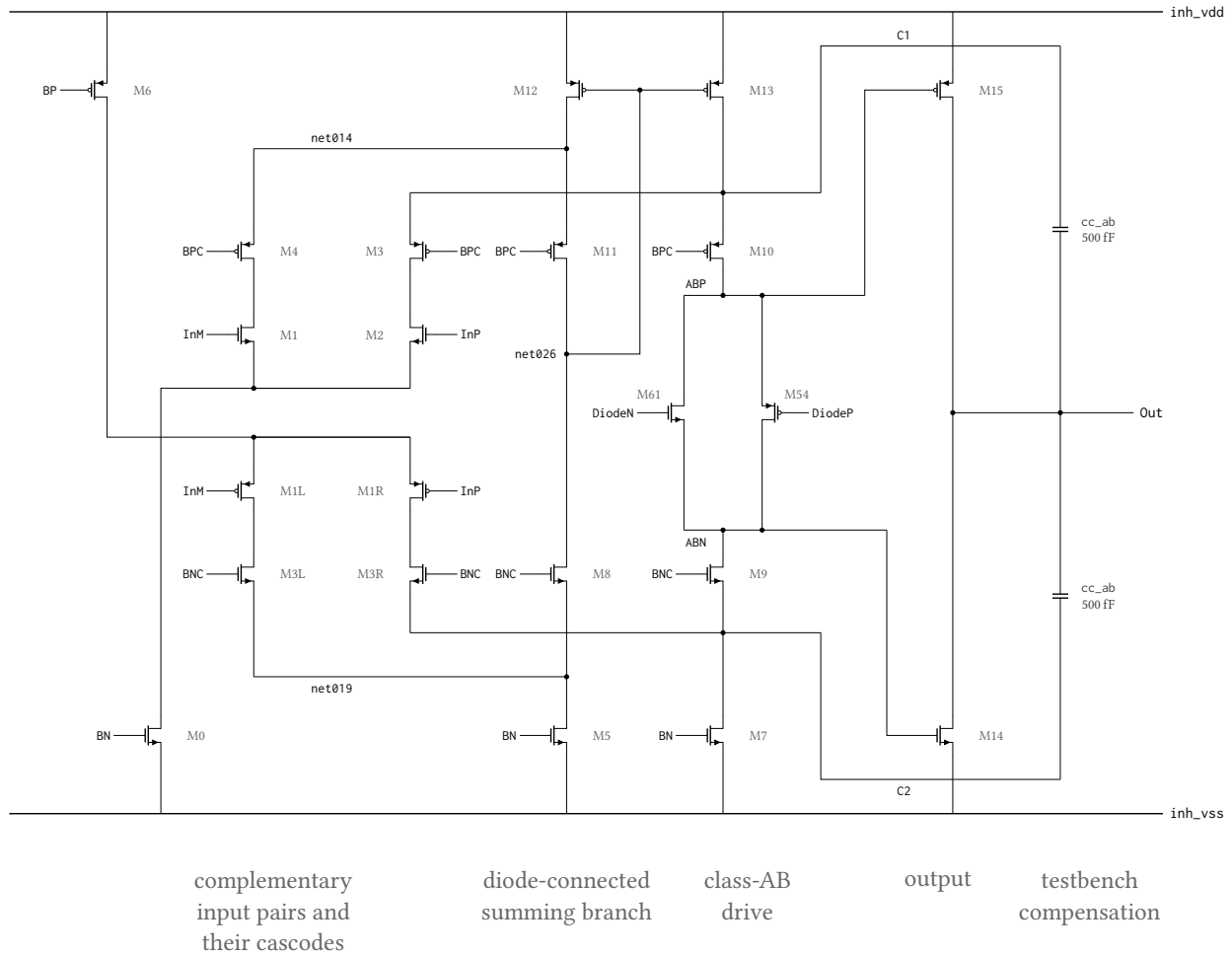


Figure 36: Signal-path topology of the class-AB amplifier `anatop_tia_amp_ab`, drawn from its netlist. The complementary pairs M1/M2 (NMOS, tail M0) and M1L/M1R (PMOS, tail M6) fold into two cascode branches; the InM side terminates in the diode-connected branch M12–M11–M8–M5, whose summing node `net026` sets M13, which restates that current into the InP branch, and the floating pair M54/M61 holds ABP above ABN so both output devices M15 and M14 are actively driven. C1 and C2 are the common source nodes of the PMOS cascodes M10/M3 and of the NMOS cascodes M9/M3R, brought out as pins; the two `cc_ab` capacitors that return them to Out are testbench components outside the cell, 500 fF per side. Simplified: the enable switch and its four power-down devices, and the bias generator that develops DiodeN/DiodeP, are omitted, so 22 of the cell’s 34 transistors are drawn; BP/BPC/BN/BNC are the enabled forms of the `bp!/bpc!/bn!/bnc!` globals and device dimensions are not shown. Identically labelled nets are connected; wires crossing without a filled dot are not.

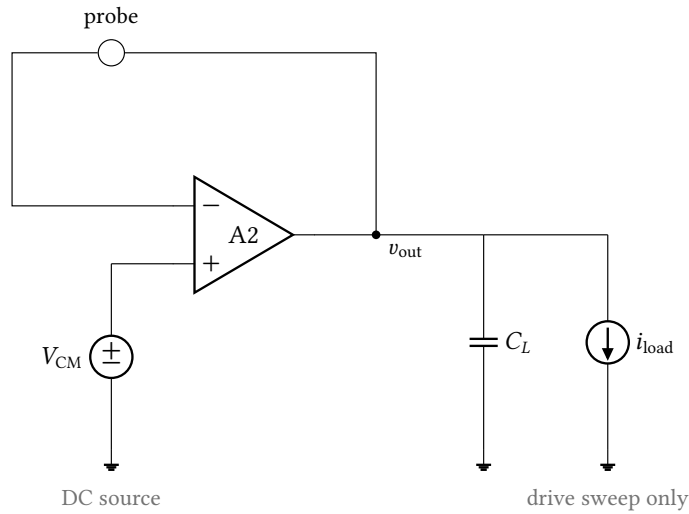


Figure 37: Unity-feedback bench. The amplifier’s output is returned to its inverting input through a current probe and loaded by $C_L = 5$ pF; the loop-gain analysis through that probe gives the amplifier’s own gain and margins, and DC and noise analyses on the same wiring give the offset and output noise. The drive measurement adds the swept load current i_{load} on the same schematic. Supplies, the bias rails and the enable pin are omitted here and in Figure 40.

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Open-loop gain A_{OL}	dB	82.74	80.28	83.30	83.31	86.75	6.47
Unity-gain frequency f_u	MHz	22.474	22.942	22.621	22.587	22.358	0.584
Phase margin	°	73.43	74.26	73.81	73.43	72.76	1.49
Gain margin	dB	8.81	9.75	8.73	8.95	8.12	1.63

Table 25: Amplifier stability in unity-gain feedback by process corner. Unity-feedback bench (Figure 37) at $v_{cm} = 1.25$ V, $C_L = 5$ pF, $C_c = 500$ fF, 2.5 V supply; the typical point is at 37 °C and the four skew corners at 25 °C. Measured through a probe in the feedback wire, so the quantity is the loop gain of the unity-gain configuration and not the amplifier loaded by an application network. Gain margin is read at the first crossing of zero loop phase above the unity-gain frequency. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Open-loop gain A_{OL}	dB	82.77	79.88	83.14	83.37	84.94	5.06
Phase margin	°	73.52	74.70	74.04	72.86	73.34	1.84
Dominant pole $f_{-3\text{dB}}$	kHz	1.592	2.286	1.538	1.494	1.225	1.061
Gain-bandwidth product	MHz	22.781	23.187	22.894	22.939	22.587	0.600

Table 26: Open-loop small-signal response by process corner. The amplifier is driven differentially about mid-supply with no feedback, $C_L = 5\text{ pF}$, $C_c = 500\text{ fF}$, 2.5 V supply, 37°C at the typical point and 25°C at the four skew corners. The input offset is nulled before the sweep and the DC operating point is linear at every point (output 0.60 V to 1.78 V across corners), so the open-loop numbers are valid; they agree with the unity-feedback loop gain of Table 25 to within 1.8 dB and the phase margins to within 0.6° . The -3 dB frequency varies inversely with the DC gain at a unity-gain frequency that is nearly corner-independent. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Input offset voltage	mV	0.622	0.649	0.624	0.569	0.456	0.193

Table 27: Systematic input offset by process corner. The differential input is swept about mid-supply into a 5 pF load, 2.5 V supply, 37°C at the typical point and 25°C at the four skew corners; the entry is the differential input at which the output crosses mid-supply. These are corner figures with zero statistical offset, so they carry only the topology's systematic term: the random offset a manufactured channel carries is the Monte Carlo result of Table 30 and is an order of magnitude larger.

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Load current at 50 mV error, sourcing	mA	2.224	2.564	2.239	2.331	2.051	0.513
Load current at 50 mV error, sinking	mA	-2.159	-2.551	-2.334	-2.099	-1.908	0.643
Closed-loop output resistance	Ω	1.811	1.646	1.862	1.484	1.612	0.378

Table 28: Output drive by process corner. The amplifier is in unity-gain feedback at $v_{cm} = 1.25\text{ V}$ with $C_L = 5\text{ pF}$, $C_c = 500\text{ fF}$, 2.5 V supply and the load current swept -10 mA to 10 mA ; the typical point is at 37°C and the four skew corners at 25°C . The first two entries are the load current at which the output departs from v_{cm} by 50 mV , not a hard current limit; load current is signed positive out of the amplifier, so the sinking entry is negative. Output resistance is the slope of the same curve at zero load and is the quantity that sets load regulation in Table 46. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Supply current	μA	121.03	141.34	114.20	122.81	111.14	30.20
Dissipation	μW	302.57	353.34	285.50	307.03	277.85	75.49

Table 29: Quiescent supply current and dissipation by process corner, at the DC operating point of the open-loop bench with both inputs at mid-supply and no load. 2.5 V supply, 37 °C at the typical point and 25 °C at the four skew corners. The measurement is taken at the supply terminal of the bench and therefore covers the amplifier together with the local bias generator that feeds it; at the typical point the amplifier draws 97 μA of the 121 μA total and the bias generator the remainder. One local bias generator serves one channel, so this is the per-channel budget. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

8.4.1 Monte Carlo

200 samples, process and mismatch together, 37 °C, under the conditions of Section 8.4. Per-channel yield is the fraction of samples inside the specification entered on the test; the four-channel figure is its fourth power, the four channels being nominally identical and assumed independent.

Input offset has a mean of 0.583 mV and $\sigma = 5.201$ mV (Figure 38, Table 30); against the ± 1 mV limit entered on the test the yield is 12.0 % per channel and 0.0 % over four. In unity-gain feedback the phase margin averages 73.40° with a worst sample of 69.21° and the gain margin 8.89 dB with a worst sample of 6.98 dB, so all 200 samples clear the 60° and 6 dB limits; open-loop gain averages 82.64 dB in that configuration (Table 31) and 82.16 dB on the open-loop bench (Table 32). Output noise integrated 0.1 Hz to 100 kHz is 30.44 μV (Table 31). Drive at 50 mV of output error is 2.224 mA sourcing and -2.158 mA sinking, mean $\pm 3\sigma$ of ± 0.252 mA and ± 0.313 mA (Table 34). Supply current, which covers the amplifier together with the local bias generator that feeds it, has a median of 122.81 μA over a 1st-to-99th percentile range of 91.58 μA to 172.09 μA , for a median dissipation of 307.0 μW (Figure 39, Table 33).

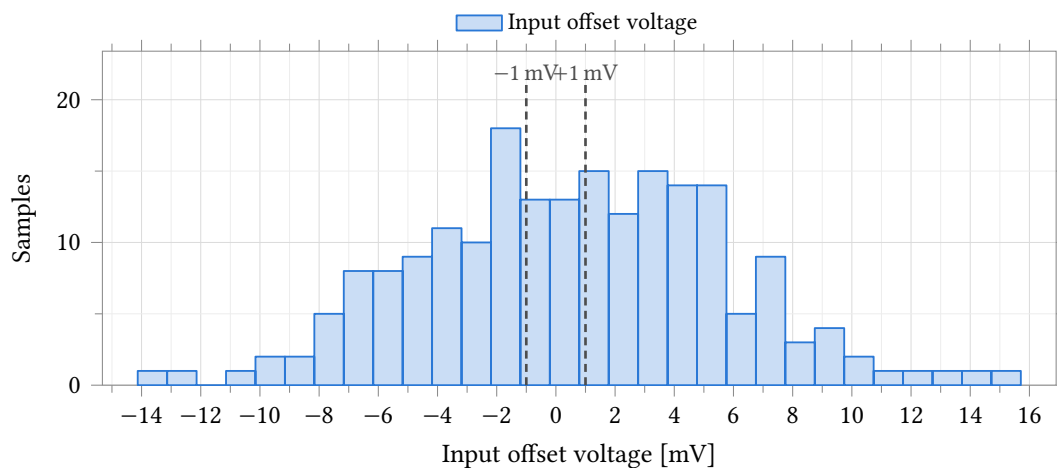


Figure 38: Input offset voltage of the amplifier. **Process and mismatch**, 37 °C, 200 samples, 2.5 V supply, $C_L = 5$ pF, $C_c = 500$ fF, a 5 pF load, differential input swept about mid-supply. The distribution is symmetric about 0.58 mV, so the systematic term of Table 27 is small against the random one, and the ± 1 mV rules are the limits entered on the test; 24 of 200 samples fall inside them. Statistics are in Table 30.

Parameter	Unit	N	Mean	σ	Mean $\pm 3\sigma$	Min	Max	Spec	Yield	4 ch
Input offset voltage	mV	200	0.583	5.201	0.583 ± 15.604	-14.130	15.709	± 1.000	12.0	0.0

Table 30: Monte Carlo input offset voltage. **Process and mismatch**, 37 °C, 200 samples, 2.5 V supply, $C_L = 5$ pF, $C_c = 500$ fF, a 5 pF load. The distribution is symmetric (Figure 38), so mean $\pm 3\sigma$ is quoted here and not for the skewed quantities in the tables that follow. Spec and yield are the ± 1 mV limits entered on the test; the four-channel column assumes independent channels. This run varies process as well as mismatch, so σ is die-to-die and is an upper bound on the within-die channel-to-channel term. The same offset appears at the reference electrode as the zero-current tracking error of Table 49.

Parameter	Unit	N	Mean	σ	p_1	p_{99}	Worst	Spec	Yield	4 ch
Open-loop gain A_{OL}	dB	200	82.64	1.20	79.98	85.52	79.44	≥ 60.00	100.0	100.0
Phase margin	°	200	73.40	1.37	69.60	75.73	69.21	≥ 60.00	100.0	100.0
Gain margin	dB	200	8.89	0.71	7.19	10.35	6.98	≥ 6.00	100.0	100.0
Output noise, 0.1 Hz to 100 kHz	μ V	200	30.44	1.04	28.32	33.11	34.02	≤ 1220.00	100.0	100.0

Table 31: Monte Carlo amplifier stability and noise in unity-gain feedback. **Process and mismatch**, 37 °C, 200 samples, $v_{cm} = 1.25$ V, $C_L = 5$ pF, $C_c = 500$ fF, 2.5 V supply. This is the amplifier alone, with no application network in the loop, and is not comparable with Table 48. Every sample clears all three stability limits entered on the test. Percentiles and the extreme sample are quoted rather than $\pm 3\sigma$ because the margins are not normally distributed. The noise row is the output noise of this configuration integrated 0.1 Hz to 100 kHz, within the 0.1 Hz to 100 MHz span of the analysis.

Parameter	Unit	N	Mean	σ	p_1	p_{99}	Worst	Spec	Yield	4 ch
Open-loop gain A_{OL}	dB	200	82.16	1.06	80.04	84.60	79.50	≥ 40.00	100.0	100.0
Phase margin	°	200	73.62	1.42	69.80	76.21	68.93	≥ 40.00	100.0	100.0
Dominant pole $f_{-3\text{dB}}$	kHz	200	1.723	0.276	1.158	2.453	–	–	–	–
Gain-bandwidth product	MHz	200	22.815	3.566	16.191	31.741	–	–	–	–

Table 32: Monte Carlo open-loop response. **Process and mismatch**, 37 °C, 200 samples, amplifier driven differentially about mid-supply with no feedback, $C_L = 5$ pF, $C_c = 500$ fF, 2.5 V supply. The input offset is nulled per sample from the offset measurement of the same run, leaving the DC operating point linear on every sample; all four quantities are measurable on all 200. Only gain and phase margin carry limits; the two bandwidth rows were not graded and their spec, yield and worst-case columns are therefore empty. Gain agrees with the unity-feedback loop gain of Table 31 to within 0.5 dB of mean. Percentiles are quoted rather than $\pm 3\sigma$: the two bandwidth distributions are right-skewed, both being inversely proportional to a transconductance.

Parameter	Unit	N	Mean	σ	Median	p_1	p_{99}	Min	Max
Supply current	μ A	200	123.42	16.80	122.81	91.58	172.09	86.43	180.31
Dissipation	μ W	200	308.6	42.0	307.0	228.9	430.2	216.1	450.8

Table 33: Monte Carlo quiescent supply current and dissipation. **Process and mismatch**, 37 °C, 200 samples, 2.5 V supply, both inputs at mid-supply, no load. The figure covers one channel's control amplifier together with the local bias generator that feeds it, measured at the supply terminal of the bench; at the typical point the amplifier draws 97 μ A of the 121 μ A total and the bias generator the remainder. Both distributions are the same measurement and are right-skewed (Figure 39), so the median and percentiles are quoted and mean $\pm 3\sigma$ is not. No limit was entered on either output, so the spec and yield columns are omitted rather than left blank.

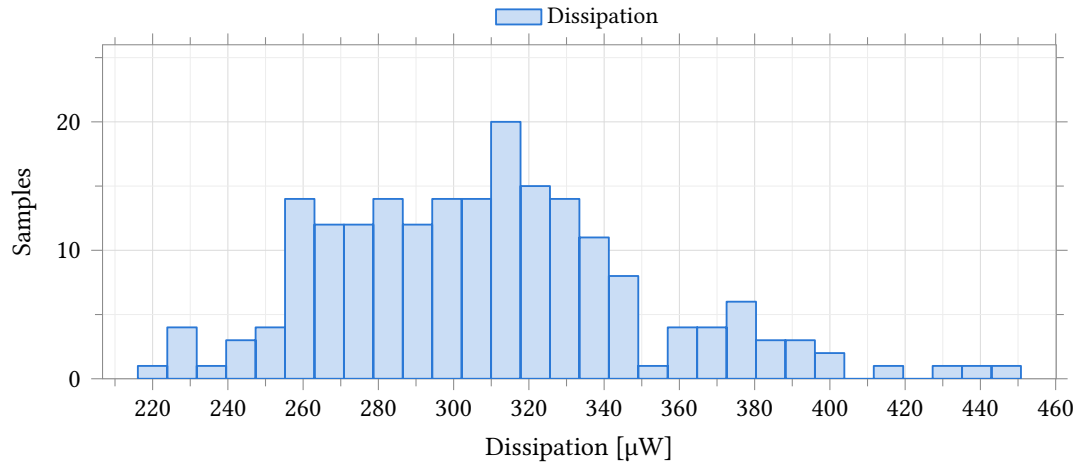


Figure 39: Quiescent dissipation of one channel's control amplifier and its local bias generator together. **Process and mismatch**, 37°C, 200 samples, 2.5 V supply, both inputs at mid-supply, no load. The distribution is unimodal and right-skewed, the tail belonging to the fast-process samples in which every bias branch runs harder; no limit was entered on this output, so none is drawn. The measurement is taken at the supply terminal of the bench, so it is the per-channel budget and not the amplifier alone.

Parameter	Unit	N	Mean	σ	Mean $\pm 3\sigma$	Min	Max	Spec	Yield	4 ch
Load current, sourcing	mA	200	2.224	0.084	2.224 ± 0.252	2.041	2.443	≥ 0.033	100.0	100.0
Load current, sinking	mA	200	-2.158	0.104	-2.158 ± 0.313	-2.511	-1.860	-	-	-
Output resistance	Ω	200	1.850	0.252	1.850 ± 0.757	1.218	2.908	-	-	-

Table 34: Monte Carlo output drive. **Process and mismatch**, 37°C, 200 samples, amplifier in unity-gain feedback at $v_{cm} = 1.25$ V, $C_L = 5$ pF, $C_c = 500$ fF, 2.5 V supply, load current swept -10 mA to 10 mA. The first two entries are the load current at which the output departs from v_{cm} by 50 mV, not a hard current limit; load current is signed positive out of the amplifier, so the sinking entry is negative. All three distributions are symmetric, so mean $\pm 3\sigma$ applies. Every sample clears the sourcing limit entered on the test; the other two rows were not graded.

8.4.2 As the Transimpedance Amplifier (A2)

As A2 the amplifier holds the working electrode at v_{cm} on its inverting input and converts the electrode current across the feedback resistor, $V_{OUT} = v_{cm} - I_{WE}R_f$. Both current polarities pass through R_f , so the ADC sees offset binary about the v_{cm} code. R_f is a 16-tap string spanning 35 k Ω to 560 k Ω .

The bench is Figure 40: $R_f = 560$ k Ω , the only tap characterised here and the only one to which the figures of this part apply, with $C_f = 22$ pF, a 5 pF load and the Randles electrode of Figure 19 at $R_s = 100$ Ω , $R_{ct} = 1$ M Ω , $C_{dl} = 100$ nF. The typical point and the Monte Carlo run are at 37 $^{\circ}$ C, the four skew corners at 25 $^{\circ}$ C.

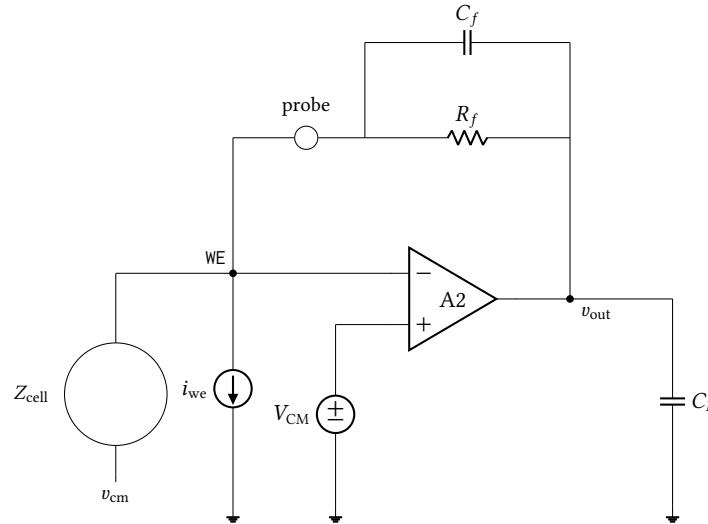


Figure 40: The closed transimpedance loop bench, the only one in this section that closes feedback around the electrode. $R_f \parallel C_f$ returns the output to the working electrode through a current probe, i_{we} is the test source, and Z_{cell} is the Randles cell of Figure 19, returned to v_{cm} . The loop gain, phase margin and crossover are measured through the probe; the DC accuracy, closed-loop transfer and noise figures come from the same wiring.

Point	Process	Temperature	Supply
1	tt	37 $^{\circ}$ C	2.5 V
2	ff	25 $^{\circ}$ C	2.5 V
3	fs	25 $^{\circ}$ C	2.5 V
4	sf	25 $^{\circ}$ C	2.5 V
5	ss	25 $^{\circ}$ C	2.5 V

Table 35: Process, temperature and supply points simulated for the transimpedance application. Point 1 is the typical statistical process point, evaluated with zero statistical offset, at the 37 $^{\circ}$ C on-body temperature; points 2–5 are the imported foundry skew corners at 25 $^{\circ}$ C. Every figure and table in this part is schematic-level and covers only the $R_f = 560$ k Ω feedback tap; nothing here characterises the other fifteen taps of the gain string. These points move the sixteen transistor model rows only — the resistor and capacitor rows stay at their typical sections — so every corner spread in this section is a lower bound (pinned-passive caveat, page 69).

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Output offset at zero current	mV	-0.967	-1.056	-0.987	-0.832	-0.762	0.294
Gain error	ppm	-167.3	-221.9	-150.6	-161.0	-106.3	115.6
Integral non-linearity	LSB	0.021	0.033	0.016	0.023	0.011	0.022

Table 36: DC accuracy by process corner. Closed transimpedance loop bench (Figure 40): $R_f = 560 \text{ k}\Omega$, $C_f = 22 \text{ pF}$, $C_L = 5 \text{ pF}$, $C_c = 500 \text{ fF}$, $\text{BiasAdj} = 37$, $v_{\text{cm}} = 1.25 \text{ V}$, 2.5 V supply, Randles electrode with $R_s = 100 \Omega$, $R_{\text{ct}} = 1 \text{ M}\Omega$, $C_{\text{dl}} = 100 \text{ nF}$. Point tt is the typical statistical point at 37°C ; the four skew corners are at 25°C . Gain error and INL are evaluated over the central ± 0.9 of the full-scale input-current range and are referred to the nominal 2.441 mV ADC LSB; the converter’s measured step is 1.591 mV (Table 73), so multiply the LSB row by 1.535 to read it as converter codes. These are corner figures with zero statistical offset: the offset a manufactured channel carries is the Monte Carlo result of Table 40. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

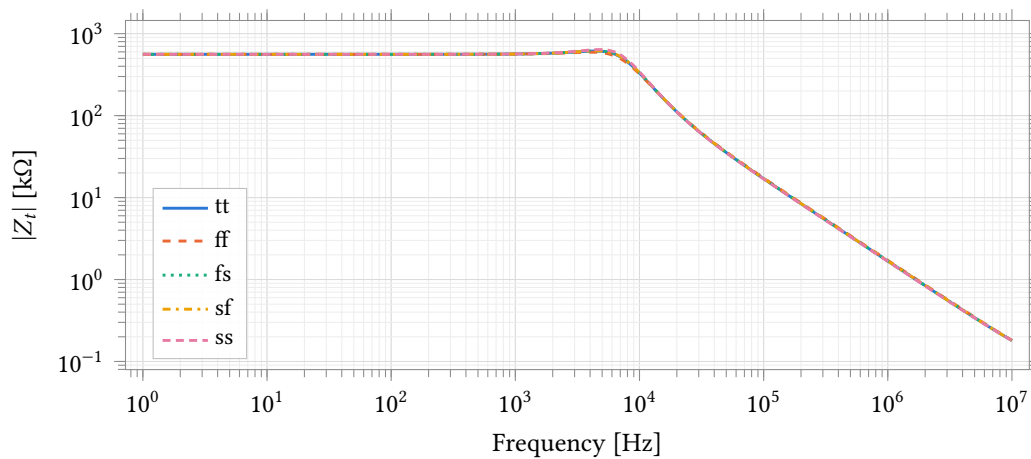


Figure 41: Closed-loop transimpedance magnitude over five process corners, conditions as Figure 42. The AC stimulus is a 1 A current into the working electrode, so the ordinate is transimpedance directly. The feedback resistor is an ideal element here rather than the 16-tap string, which is why the corners separate only near the band edge; the numbers are in Table 37. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Transimpedance at 1 Hz	$\text{k}\Omega$	559.933	559.913	559.937	559.938	559.957	0.044
Deviation from R_f	ppm	-118.8	-155.7	-111.9	-110.7	-76.2	79.4
Signal bandwidth $f_{-3\text{dB}}$	kHz	8.868	8.651	8.927	8.935	9.149	0.498
Gain peaking	dB	0.812	0.523	0.850	0.860	1.156	0.633

Table 37: Small-signal transimpedance by process corner, bench and conditions as Table 36. The feedback resistor is ideal in this measurement, so the deviation of $Z_t(1 \text{ Hz})$ from R_f is set by finite loop gain alone and carries no resistor tolerance. Peaking is referred to $Z_t(1 \text{ Hz})$ and the bandwidth is the -3 dB point of the same curve. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

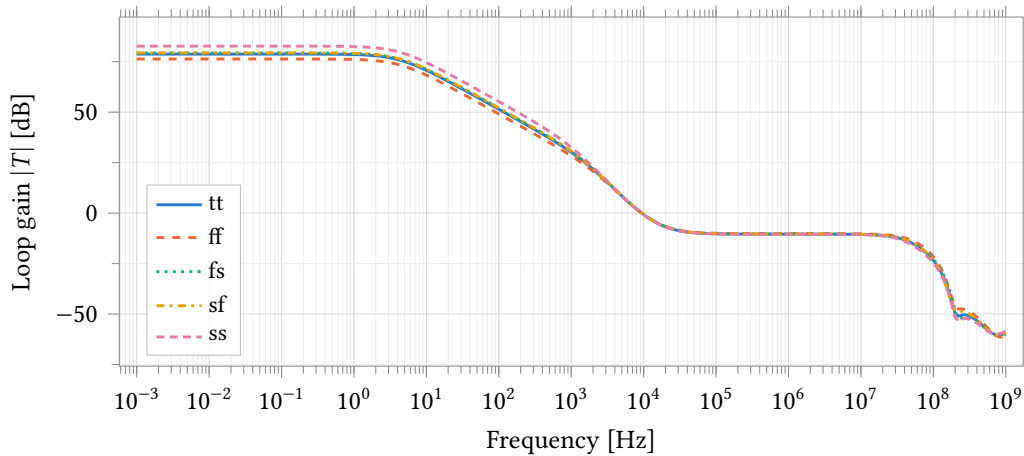


Figure 42: Transimpedance loop gain over five process corners, measured through a probe in series with the feedback network (Figure 40). $R_f = 560 \text{ k}\Omega$, $C_f = 22 \text{ pF}$, $C_L = 5 \text{ pF}$, $C_c = 500 \text{ fF}$, BiasAdj = 37, 2.5 V, Randles electrode with $R_s = 100 \text{ }\Omega$, $R_{ct} = 1 \text{ M}\Omega$, $C_{dl} = 100 \text{ nF}$. tt is at 37°C , the four skew corners at 25°C . This is the loop around the amplifier, the feedback network and the electrode together, not the amplifier's own gain; the two are tabulated separately in Table 38 and Table 25. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Loop gain at DC	dB	78.74	76.32	79.28	79.32	82.69	6.37
Loop crossover	kHz	9.375	9.475	9.428	9.428	9.360	0.115
Phase margin	$^\circ$	76.50	80.30	76.23	76.13	72.78	7.52
Gain margin	dB	21.81	20.67	21.90	21.24	22.49	1.82

Table 38: Transimpedance loop stability by process corner, conditions as Table 36. Measured through the feedback probe of Figure 40, so the loop includes the amplifier, the feedback network and the electrode; the amplifier's own margins are in Table 25. Gain margin is read at the first crossing of zero loop phase above the unity-gain frequency. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Output noise, 0.1 Hz to 1000 Hz	mV	1.690	1.543	1.656	1.647	1.753	0.210
Input current noise, 0.1 Hz to 10 Hz	pA	68.23	62.35	66.73	66.54	70.61	8.27
Input current noise, 0.1 Hz to 100 Hz	pA	340.24	310.83	333.13	331.71	352.48	41.65
Input current noise, 0.1 Hz to 1000 Hz	nA	3.017	2.755	2.957	2.941	3.130	0.375

Table 39: Noise by process corner, conditions as Table 36. The noise analysis runs 0.1 Hz to 10 kHz at 20 points per decade; the three input-referred rows are the output noise integrated to 10 Hz, 100 Hz and 1 kHz and divided by R_f , and no figure is integrated beyond the analysis stop frequency. The input-referred rows therefore scale inversely with the selected gain tap. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Monte Carlo. 200 samples, process and mismatch together, 37 °C, at the conditions above. Output offset at zero electrode current has a mean of -1.08 mV and $\sigma = 8.91$ mV (Figure 43, Table 40); against the ± 5 mV limit entered on the test the yield is 48.0 % per channel and 5.3 % over four. Loop gain at DC averages 78.65 dB, and every sample clears the 45° phase-margin limit, the worst at 64.7° (Figure 44, Table 42). Signal bandwidth averages 8.84 kHz against a 1 kHz floor met on all 200 samples (Table 43). Input current noise integrated 0.1 Hz to 100 Hz is 341.5 pA against the 2 nA channel budget (Figure 45, Table 44); it is referred to the 560 k Ω tap and scales inversely with the tap selected.

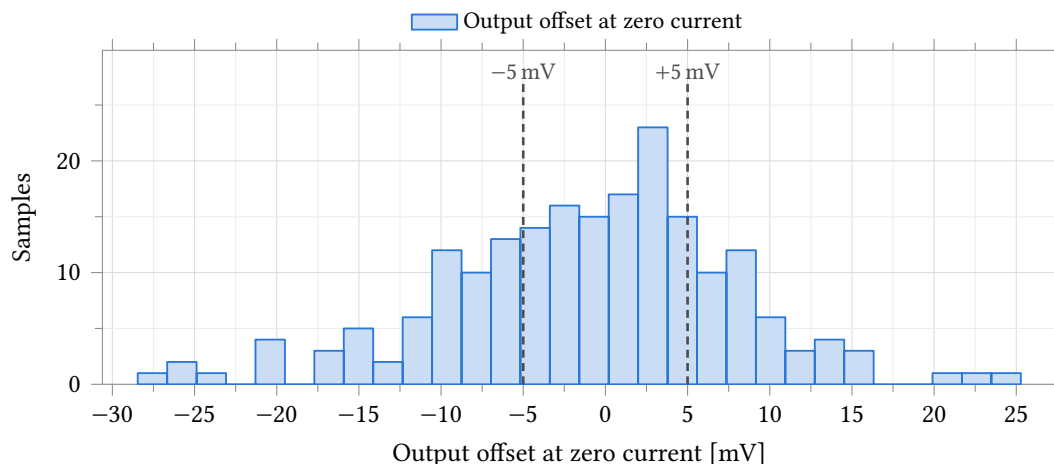


Figure 43: Output offset at zero electrode current, referred to v_{cm} . **Process and mismatch**, 37 °C, 200 samples, $R_f = 560$ k Ω , $C_f = 22$ pF, $C_L = 5$ pF, $C_c = 500$ fF, BiasAdj = 37, 2.5 V. The rules are the ± 5 mV limits entered on the test. $\sigma = 8.91$ mV (sample deviation, as in every table of this section); 96 of 200 samples lie within the limits, 5.3 % across four independent channels. Statistics are in Table 40.

Parameter	Unit	N	Mean	σ	Mean $\pm 3\sigma$	Min	Max	Spec	Yield	4 ch
Output offset at zero current	mV	200	-1.08	8.91	-1.08 ± 26.72	-28.46	25.28	± 5.00	48.0	5.3

Table 40: Monte Carlo output offset at zero electrode current. **Process and mismatch**, 37 °C, 200 samples, $R_f = 560$ k Ω , $C_f = 22$ pF, $C_L = 5$ pF, $C_c = 500$ fF, BiasAdj = 37, 2.5 V, Randles electrode with $R_s = 100$ Ω , $R_{ct} = 1$ M Ω , $C_{dl} = 100$ nF. The distribution is approximately Gaussian, so mean $\pm 3\sigma$ is quoted here and not for the skewed quantities in the tables that follow; the worst sample, -28.5 mV, lies just outside it. Spec and yield are the ± 5 mV limits entered on the test; the four-channel column assumes independent channels. This run varies process as well as mismatch, so σ is die-to-die and is an upper bound on the within-die channel-to-channel term.

8.4.3 As the Potentiostat Control Amplifier (A1)

As A1 the amplifier drives the counter electrode and closes its loop on the reference electrode, holding that electrode at the commanded potential. The commanded potential is swept over the full 0 V to 2.5 V supply range.

The bench is Figure 46: a Randles cell with $R_{CE} = 1$ k Ω , $R_u = 1$ k Ω , $R_{ct} = 10$ k Ω and $C_{dl} = 1$ μ F — the only electrode model simulated — and the working electrode held at 1.25 V. No load current is injected: the cell current follows from the sweep against the cell impedance, and the regulation figures at ± 10 μ A

Parameter	Unit	N	Mean	σ	p_1	p_{99}	Worst	Spec	Yield	4 ch
Gain error	ppm	200	-171.7	23.1	-235.5	-128.0	249.3	± 2000.0	100.0	100.0
Transimpedance at 1 Hz	k Ω	200	559.93	0.01	559.91	559.95	–	–	–	–
Deviation from R_f	ppm	200	-121.3	17.4	-165.2	-89.2	–	–	–	–
Integral non-linearity	LSB	200	0.022	0.006	0.012	0.040	0.051	≤ 0.500	100.0	100.0
Total unadjusted error	LSB	200	2.90	2.34	0.14	10.62	11.72	≤ 4.00	75.0	31.6

Table 41: Monte Carlo transfer accuracy, conditions as Table 40. Gain error, INL and total unadjusted error are evaluated over the central ± 0.9 of the full-scale input-current range and the two LSB rows are referred to the nominal 2.441 mV ADC LSB; the converter’s measured step is 1.591 mV (Table 73), so multiply those two rows by 1.535 to read them as converter codes. Total unadjusted error correlates with $|V_{OS}|$ at $r = +1.000$ across the 200 samples: at this gain tap it is the offset of Table 40 expressed in LSB, not an independent result, and its 4 LSB limit corresponds to an offset of about 9.8 mV, which is why its yield is not the offset yield. Gain error and INL are set by the amplifier and pass on every sample; the feedback resistor is an ideal element here and contributes no tolerance. Percentiles and worst case are quoted rather than $\pm 3\sigma$ because INL and total unadjusted error are right-skewed.

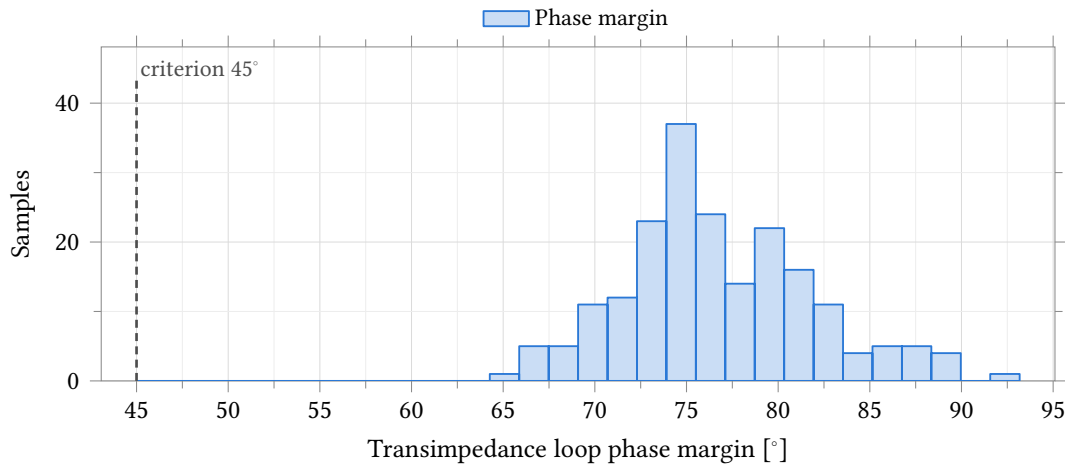


Figure 44: Phase margin of the transimpedance loop, conditions as Figure 43. The loop is measured through the feedback probe of Figure 40 and includes the amplifier, the feedback network and the Randles electrode ($R_s = 100 \Omega$, $R_{ct} = 1 \text{ M}\Omega$, $C_{dl} = 100 \text{ nF}$). All 200 samples clear the 45° criterion, minimum 64.7° . The distribution is right-skewed, so percentiles and the extreme sample are quoted in Table 42 rather than $\pm 3\sigma$.

Parameter	Unit	N	Mean	σ	p_1	p_{99}	Worst	Spec	Yield	4 ch
Loop gain at DC	dB	200	78.65	1.23	75.89	81.26	74.89	≥ 60.00	100.0	100.0
Phase margin	$^\circ$	200	76.7	5.2	66.4	89.6	64.7	≥ 45.0	100.0	100.0
Gain margin	dB	167 / 200	22.06	0.64	20.79	23.70	–	–	–	–
Loop crossover	kHz	200	9.41	0.98	7.56	11.85	–	–	–	–

Table 42: Monte Carlo transimpedance loop stability, conditions as Table 40. Measured through the feedback probe of Figure 40, so the loop includes the amplifier, the feedback network and the electrode. Gain margin reports $N = 167/200$ because the loop phase never reaches the critical value on the other 33 samples and no gain margin exists for them; those samples are held out of the statistics and are not failures. Percentiles and worst case are quoted rather than $\pm 3\sigma$ for the margins, which are not normally distributed.

Parameter	Unit	N	Mean	σ	Mean $\pm 3\sigma$	Min	Max	Spec	Yield	4 ch
Signal bandwidth $f_{-3\text{dB}}$	kHz	200	8.84	0.34	8.84 ± 1.03	7.90	9.73	≥ 1.00	100.0	100.0
Gain peaking	dB	200	0.82	0.23	0.82 ± 0.68	0.30	1.52	≤ 2.00	100.0	100.0

Table 43: Monte Carlo closed-loop bandwidth and gain peaking, conditions as Table 40. Both are read from the transimpedance magnitude with a 1 A AC current into the working electrode; peaking is referred to $Z_t(1\text{ Hz})$. Both distributions are symmetric, so mean $\pm 3\sigma$ applies. Every sample meets the 1 kHz bandwidth floor and the 2 dB peaking ceiling entered on the test.

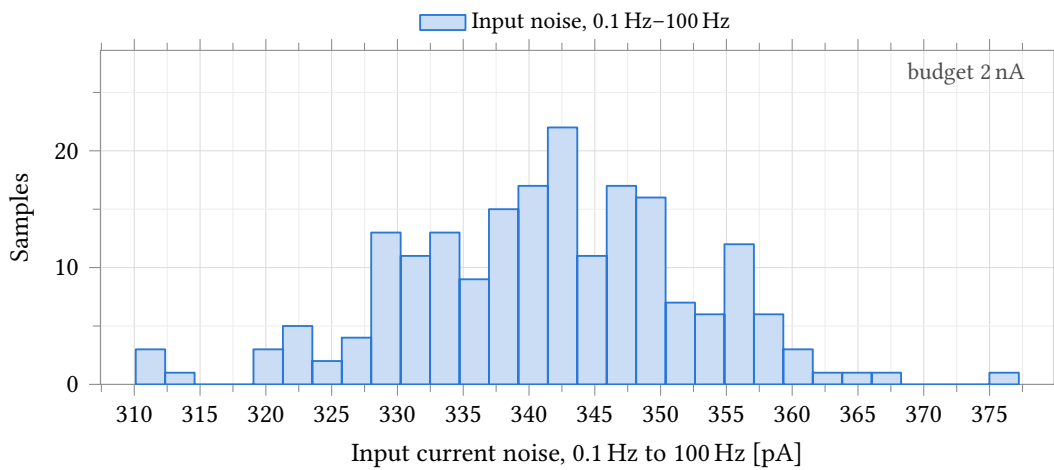


Figure 45: Input-referred current noise integrated 0.1 Hz to 100 Hz, conditions as Figure 43. Output noise divided by R_f , from a noise analysis that runs 0.1 Hz to 10 kHz at 20 points per decade. The 2 nA budget entered on the test is about six times the measured value and is annotated at the axis edge rather than drawn, so that the 310 pA to 377 pA spread stays legible. The figure scales inversely with the selected gain tap.

Parameter	Unit	N	Mean	σ	Mean $\pm 3\sigma$	Min	Max	Spec	Yield	4 ch
Output noise, 0.1 Hz–1 kHz	mV	200	1.696	0.055	1.696 ± 0.165	1.539	1.874	–	–	–
Input noise, 0.1 Hz–10 Hz	pA	200	68.5	2.2	68.5 ± 6.7	62.2	75.8	–	–	–
Input noise, 0.1 Hz–100 Hz	pA	200	341.5	11.0	341.5 ± 33.0	310.1	377.2	≤ 2000.0	100.0	100.0
Input noise, 0.1 Hz–1 kHz	nA	200	3.029	0.098	3.029 ± 0.294	2.749	3.346	–	–	–

Table 44: Monte Carlo noise of the transimpedance stage, conditions as Table 40. The noise analysis runs 0.1 Hz to 10 kHz at 20 points per decade; the three input-referred rows are the output noise integrated to 10 Hz, 100 Hz and 1 kHz and divided by R_f , and no figure is integrated beyond the analysis stop frequency. All four distributions are symmetric, so mean $\pm 3\sigma$ applies. The input-referred rows scale inversely with the selected gain tap; the 2 nA entry on the 100 Hz row is the channel budget.

are read off the sweep where the cell current crosses those values. Temperature is split as for the device tables, 37 °C at the typical point and 25 °C at the four skew corners.

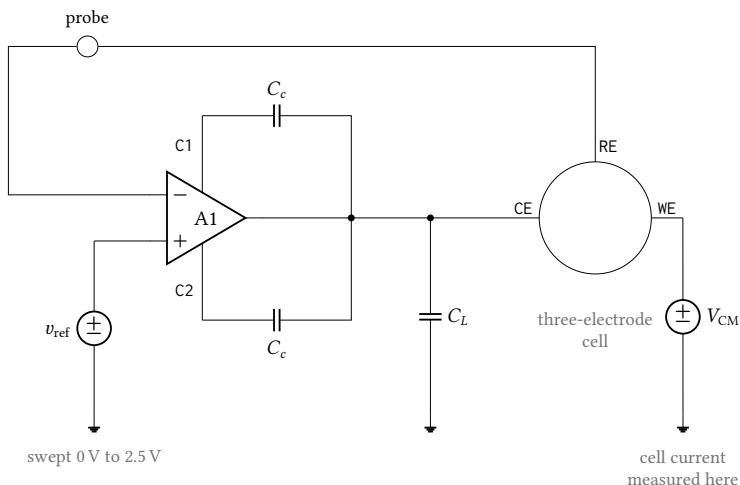


Figure 46: Cell-drive bench. The amplifier drives the counter electrode of a three-electrode cell and closes its loop on the reference electrode, with the commanded potential v_{ref} swept over the full 0 V to 2.5 V supply range. The working electrode is held at V_{CM} by an ideal source: no load current is injected, so the cell current is set by the sweep against the cell impedance and is measured in that source, and the regulation errors quoted at $\pm 10 \mu\text{A}$ are read off the sweep at the points where the cell current reaches those values. The probe in the reference-electrode return measures the loop gain with the cell inside the loop; it is not the amplifier's own gain, and the two are tabulated separately. Cell elements are $R_{\text{ce}} = R_u = 1 \text{ k}\Omega$ and $R_{\text{ct}} = 10 \text{ k}\Omega$ in parallel with $C_{\text{dl}} = 1 \mu\text{F}$ (Figure 19), the load is $C_L = 5 \text{ pF}$, and each external compensation capacitor C_c is 500 fF.

Monte Carlo. 200 samples, process and mismatch together, 37 °C, at the conditions above. Cell-loop gain at DC averages 72.13 dB with a 1st percentile of 68.67 dB, and every sample clears the 60° phase-margin limit, the worst at 80.13° (Figure 48, Table 48). Tracking error at zero cell current — the amplifier's input offset carried to the reference electrode — has a mean of -0.694 mV , a median of -0.267 mV and $\sigma = 5.708 \text{ mV}$; over the fixed 0.4 V to 2.1 V command window the worst magnitude averages 5.494 mV and reaches 23.082 mV, meeting the 10 mV limit on 87.0 % of channels and 57.3 % of four-channel chips (Table 49). With that offset removed, load regulation is $-22.17 \mu\text{V}$ at 10 μA and 23.97 μV at $-10 \mu\text{A}$; all 200 samples lie inside $\pm 1 \text{ mV}$, for 100 % per channel and over four (Table 50). Counter-electrode compliance is 2.4621 V at the ceiling and 25.69 mV at the floor, spreading $\pm 3.9 \text{ mV}$ and $\pm 3.27 \text{ mV}$ at 3σ (Table 51).

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Loop gain at DC	dB	72.24	71.85	72.24	73.57	74.11	2.26
Loop crossover	MHz	10.440	10.835	10.508	10.533	10.223	0.612
Phase margin	°	81.88	82.14	82.05	81.83	81.68	0.45

Table 45: Control-loop stability driving an electrochemical cell, by process corner. Cell-drive bench (Figure 46): Randles cell with $R_{CE} = 1\text{ k}\Omega$, $R_u = 1\text{ k}\Omega$, $R_{ct} = 10\text{ k}\Omega$, $C_{dl} = 1\text{ }\mu\text{F}$, working electrode at 1.25 V, $C_L = 5\text{ pF}$, $C_c = 500\text{ fF}$, 2.5 V supply, 37 °C at the typical point and 25 °C at the four skew corners. The probe sits in the reference-electrode sense wire, so the loop measured here includes the cell; it is a different loop from Table 25 and the two are not comparable. Loop gain is 10 dB lower than the unity-feedback figure because the cell's 12 k Ω loads an open-loop output resistance of about 25 k Ω ; crossover halves with it and the phase margin improves. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

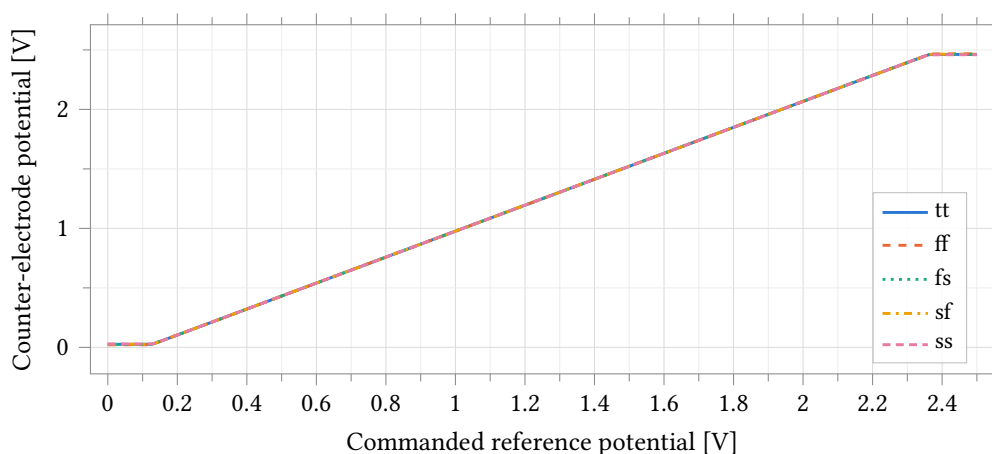


Figure 47: Counter-electrode potential against commanded reference potential over five process corners, cell-drive bench (Figure 46). The amplifier drives a Randles cell with $R_{CE} = 1\text{ k}\Omega$, $R_u = 1\text{ k}\Omega$, $R_{ct} = 10\text{ k}\Omega$, $C_{dl} = 1\text{ }\mu\text{F}$ with the working electrode held at 1.25 V, $C_L = 5\text{ pF}$, $C_c = 500\text{ fF}$, 2.5 V supply; the typical point is at 37 °C and the four skew corners at 25 °C. The central segment rises with slope 12/11, the divider between the 1 k Ω counter-electrode branch and the 11 k Ω remainder of the cell; the flat ends are the amplifier against its rails, and their heights are the compliance limits tabulated in Table 47. Over the whole central segment the cell current is set by the cell resistance and the commanded potential, not by the amplifier.

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Tracking error at zero cell current	μV	-619.5	-676.9	-632.9	-533.2	-488.1	188.7
Worst tracking error over the window	μV	771.5	838.8	794.8	667.8	715.9	171.0
Regulation error at $10\mu\text{A}$	μV	-23.00	-24.25	-23.01	-19.60	-18.22	6.04
Regulation error at $-10\mu\text{A}$	μV	22.63	23.52	22.81	19.08	18.31	5.20

Table 46: Reference-potential tracking by process corner, conditions as Table 45. Tracking error is the reference-electrode potential minus the commanded value. The first row is that error at zero cell current and is the loop’s DC offset; the second is its worst magnitude over the fixed 0.4 V to 2.1 V command window, offset-inclusive. The last two rows are the load-regulation error at $\pm 10\mu\text{A}$ of cell current with the zero-current error subtracted at the same corner, which is the quantity a differential measurement sees; they are two orders of magnitude smaller than the offset that dominates the rows above. The window is fixed for every corner and every sample, so a weak corner cannot report better accuracy over a narrower range.

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Counter-electrode ceiling	V	2.4622	2.4671	2.4627	2.4641	2.4596	0.0075
Counter-electrode floor	mV	25.67	21.88	23.43	26.11	28.28	6.40
Cell current delivered, anodic	μA	101.013	101.422	101.055	101.176	100.796	0.626
Cell current delivered, cathodic	μA	102.028	102.343	102.214	101.991	101.810	0.534

Table 47: Counter-electrode compliance and delivered cell current by process corner, conditions as Table 45. The first two rows are the extremes of the counter-electrode potential over the full 0 V to 2.5 V command sweep (Figure 47); they are measured directly at the electrode and are independent of the loop offset, which is why compliance is published from them. Headroom to each rail is under 40 mV. The current rows are the largest anodic and cathodic currents reached in the same sweep; with $12\text{ k}\Omega$ between the counter and working electrodes they are the compliance limits divided by that resistance, so they are a property of this load and do not transfer to another electrode. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	N	Mean	σ	p_1	p_{99}	Worst	Spec	Yield	4 ch
Loop gain at DC	dB	200	72.13	1.27	68.67	74.79	–	–	–	–
Phase margin	$^\circ$	200	81.88	0.57	80.55	83.02	80.13	≥ 60.00	100.0	100.0
Loop crossover	MHz	200	10.497	1.523	7.649	14.203	–	–	–	–

Table 48: Monte Carlo control-loop stability driving an electrochemical cell. **Process and mismatch**, 37°C , 200 samples, cell-drive bench (Figure 46) with a Randles cell of $R_{\text{CE}} = 1\text{ k}\Omega$, $R_u = 1\text{ k}\Omega$, $R_{\text{ct}} = 10\text{ k}\Omega$, $C_{\text{dl}} = 1\mu\text{F}$, working electrode at 1.25 V, $C_L = 5\text{ pF}$, $C_c = 500\text{ fF}$, 2.5 V supply. The probe sits in the reference-electrode sense wire, so this loop includes the cell and is not comparable with Table 31. All 200 samples clear the phase-margin limit entered on the test with better than 20° to spare, and margin is better with the cell attached than without it: the resistive load costs 10 dB of gain, crossover falls to half the unloaded value. Loop gain and crossover were not graded. Percentiles and the extreme sample are quoted rather than $\pm 3\sigma$; the crossover distribution is right-skewed.

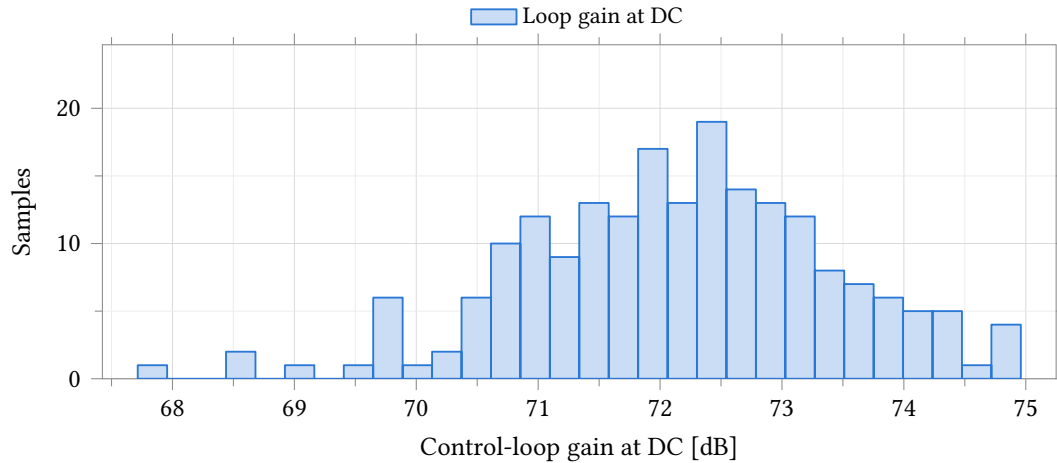


Figure 48: DC gain of the control loop closed around an electrochemical cell. **Process and mismatch**, 37 °C, 200 samples, cell-drive bench (Figure 46) with a Randles cell of $R_{CE} = 1 \text{ k}\Omega$, $R_u = 1 \text{ k}\Omega$, $R_{ct} = 10 \text{ k}\Omega$, $C_{dl} = 1 \text{ }\mu\text{F}$, working electrode at 1.25 V, $C_L = 5 \text{ pF}$, $C_c = 500 \text{ fF}$, 2.5 V supply. The loop is measured through the reference-electrode sense wire and therefore includes the cell, whose 12 k Ω loads an open-loop output resistance of about 25 k Ω and is 10 dB below the unity-feedback gain of Table 31. The spread is 7 dB peak to peak and no sample falls below 67 dB. No limit was entered on this output.

Parameter	Unit	N	Mean	σ	Median	p_1	p_{99}	Worst	Spec	Yield	4 ch
Zero-current error	mV	200	-0.694	5.708	-0.267	-16.532	13.086	–	–	–	–
Worst error in window	mV	200	5.494	4.463	4.423	0.358	21.869	23.082	≤ 10.000	87.0	57.3

Table 49: Monte Carlo reference-potential tracking, conditions as Table 48. Tracking error is the reference-electrode potential minus the commanded value; the first row is that error at zero cell current, the second its worst magnitude over the fixed 0.4 V to 2.1 V command window. Both are offset-inclusive, and the worst-error row tracks the magnitude of the zero-current row at $r = 0.986$ across the 200 samples: what is measured here is the amplifier’s input offset carried to the electrode, not a gain error, and calibrating the offset removes almost all of it. Load regulation with that offset taken out is Table 50. The window is frozen at the same two voltages for every sample, so a weak sample cannot report better accuracy over a narrower range. Both distributions are right-skewed, so the median, percentiles and worst sample are quoted and mean $\pm 3\sigma$ is not.

Parameter	Unit	N	Mean	σ	Mean $\pm 3\sigma$	Min	Max	Spec	Yield	4 ch
Regulation at 10 μA	μV	200	-22.17	27.35	-22.17 $\pm$ 82.04	-99.30	74.00	± 1000.00	100.0	100.0
Regulation at -10 μA	μV	200	23.97	34.41	23.97 $\pm$ 103.22	-76.00	167.80	± 1000.00	100.0	100.0

Table 50: Monte Carlo load regulation of the reference potential, conditions as Table 48. Each entry is the tracking error at $\pm 10 \text{ }\mu\text{A}$ of cell current with the same sample’s zero-current error subtracted, so it is the change the loop allows when the cell starts to draw current and carries no offset term. Both distributions are symmetric, so mean $\pm 3\sigma$ applies, and their means are of opposite sign: the reference potential droops under anodic current and rises under cathodic current, by about 2.3 μV per microamp of cell current. All 200 samples lie inside $\pm 1 \text{ mV}$, the worst by a factor of six; that limit is stated here for the derived quantity because the run graded only the offset-inclusive error. Regulation is more than two hundred times smaller than the raw tracking error of Table 49.

Parameter	Unit	N	Mean	σ	Mean $\pm 3\sigma$	Min	Max
Counter-electrode ceiling	V	200	2.4621	0.0013	2.4621 ± 0.0039	2.4591	2.4654
Counter-electrode floor	mV	200	25.69	1.09	25.69 ± 3.27	22.38	28.94
Cell current delivered, anodic	μA	200	101.011	0.109	101.011 ± 0.327	100.757	101.282
Cell current delivered, cathodic	μA	200	102.026	0.091	102.026 ± 0.272	101.755	102.302

Table 51: Monte Carlo counter-electrode compliance and delivered cell current, conditions as Table 48. The first two rows are the extremes of the counter-electrode potential over the full 0 V to 2.5 V command sweep, measured at the electrode itself and therefore independent of the loop offset; this is the published compliance figure. Headroom is under 35 mV to the upper rail and under 29 mV to the lower one, and each spreads under 7 mV across the 200 samples: compliance is set by the output stage’s saturation and is almost insensitive to matching. The current rows are the largest anodic and cathodic currents reached in the same sweep; with 12 k Ω between the counter and working electrodes they are the compliance limits divided by that resistance and do not transfer to another electrode. All four distributions are symmetric. No limit was entered on any of them, so the spec and yield columns are omitted.

8.4.4 Measuring this block on chip

Input offset is the term that decides yield; a loopback is the only way to see it.

1. Open the reference-electrode isolation switch; close the unity-gain loopback around A1.
2. Select per-channel slot 5 (the amplifier output, inboard of the counter-electrode isolation switch). Read it on atb_d with AFE_DBGDIREN = 1, or on-chip with AFE_ADCSRSEL = 4 and averaging — unbuffered either way: a test buffer built from the same devices carries an offset of the same class.
3. Repeat at three reference codes across the working window. The intercept is the input offset, $\sigma = 5.20$ mV (Table 30); the slope is the finite-loop-gain tracking error.
4. For the transimpedance role: isolate the working electrode and read slot 4 the same way. That reading is output-referred, $\sigma = 8.91$ mV (Table 40); divided by the active gain it is the channel’s zero-current error.

This test fabric is design intent; it is not yet in the schematics.

8.5 Potentiostat Pixel System

The pixel system is the rev-2 grounded-WE two-amp potentiostat: one complete analog channel built from two instances of the class-AB amplifier of Section 8.4 around a three-electrode electrochemical cell (Figure 49). Control amplifier A1 drives the counter electrode (CE) and closes its loop on the reference electrode (RE), holding RE at the commanded potential v_{ref} against whatever counter-electrode drive that requires. Transimpedance amplifier A2 holds the working electrode (WE) at a fixed common-mode potential V_{CM} through the programmable feedback resistor R_f and returns the cell current as a single-ended voltage, $v_{\text{out}} = V_{\text{CM}} + i_{\text{we}} \cdot R_f$, ready for the single-ended 10-bit SAR converter. The electrode is randles_cell (Section 8.1): $R_u = R_{\text{ce}} = 1$ k Ω fixed, R_{ct} and C_{dl} swept over the ranges stated with each measurement below.

Sign convention (measured, firmware-critical). Positive i_{we} flows from the working electrode to ground, and v_{out} rises with it: $dv_{\text{out}}/di_{\text{we}} = +R_f$. This is the orientation this bench actually measures (accuracy slope +1 against injected truth current, Table 53); the TB-07 formula of the system verification plan assumes the opposite source orientation and inverts every sign derived from it. Firmware reading the ADC code must use $i_{\text{we}} = (v_{\text{out}} - V_{\text{CM}})/R_f$, not its negation.

The characterisation below covers, in order: bipolar accuracy against a truth current injected at the working electrode (the rev-2 headline result); the potentiostat control loop; stability across the electrode grid; system noise; CV and step transients; and the Monte Carlo zero-current offset. All data is schematic-level, with no extracted parasitics.

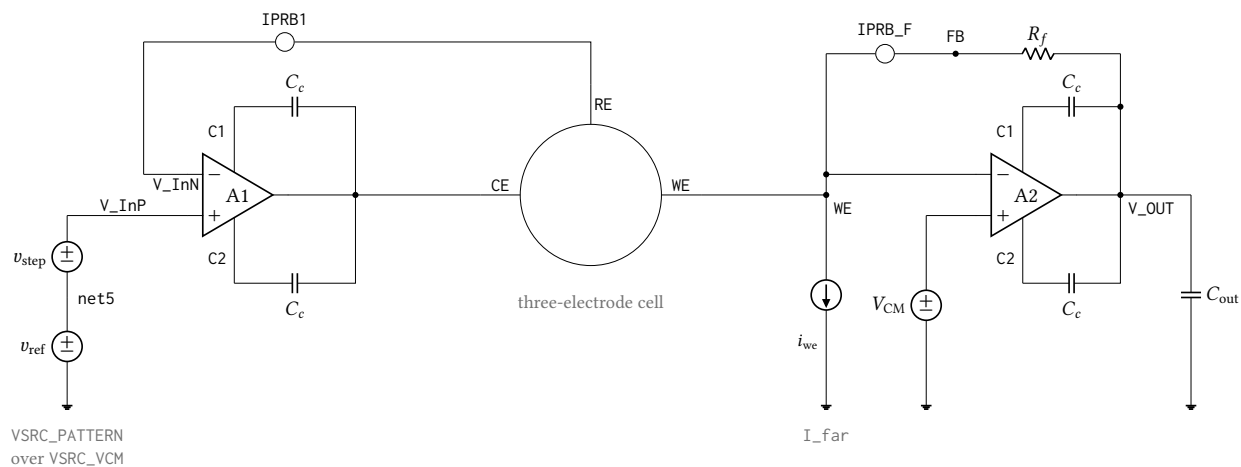


Figure 49: The complete rev-2 potentiostat pixel as simulated, `tb_anatop_pixel` view PotFbTB: control amplifier A1 drives the counter electrode and closes its loop on the reference electrode, while transimpedance amplifier A2 holds the working electrode at V_{CM} and returns the cell current through R_f to V_{OUT} . Both amplifiers are the same `anatop_tia_amp_ab` class-AB cell, each with its own external compensation capacitors $C_c = 500$ fF from pins C1/C2 onto its own output; supplies, E_n and the shared local bias generator are omitted. IPRB1 and IPRB_F are 0 V current probes inserted only so that `stb` can break the control loop and the transimpedance loop respectively – they add no impedance and no offset. I_{far} injects the accuracy-truth current with positive i_{we} flowing from the working electrode to ground, so $v_{out} = V_{CM} + i_{we} \cdot R_f$ and the measured current is the probe current in IPRB_F; the opposite source orientation inverts the sign of every accuracy number. Cell elements are $R_{ce} = R_u = 1$ k Ω fixed with R_{ct} and C_{dl} swept over the electrode grid, R_f is the programmable feedback resistor (35 k Ω to 560 k Ω), and $C_{out} = 5$ pF.

8.5.1 Bipolar Accuracy

Figure 50 sweeps the truth current i_{we} through zero at the operational $R_f = 35$ k Ω tap. The error crosses zero smoothly, with no kink between the sourcing and sinking halves of the sweep: the same TIA feedback path carries current in both directions, so there is no crossover distortion of the kind a complementary NMOS/PMOS mirror front end would show at zero current. The crossing slope is $-1/T$ exactly (finite loop gain, not a nonlinearity), and the crossing offset is -1.1 nA at the operational 1 M Ω electrode.

Table 53 gives the operational accuracy at $R_f = 35$ k Ω into the 10 k Ω electrode: 0.41–0.61 LSB worst-case error across the five process corners. Table 54 gives the operational accuracy at the maximum gain tap, $R_f = 560$ k Ω into the 1 M Ω electrode operational at that gain: 0.253–0.376 LSB. These are the two gain taps with a matched operational electrode simulated; the 140 and 560 k Ω rows measured against the 10 k Ω electrode (Table 55) are a finite-loop-gain demonstration only and are not comparable accuracy figures – see that table’s caption.

At 35 k Ω the error is range-dependent: 0.41–0.61 LSB over the full ± 33 μ A span, improving to 0.21–0.30 LSB over the ± 2.3 μ A subrange. The growth toward full scale is the $1/T$ gain error of the transimpedance loop; the 560 k Ω figure is already a full-scale measurement at its gain.

Point	Process	Temperature	Supply
81	tt	37 °C	2.5 V
82	ff	25 °C	2.5 V
83	fs	25 °C	2.5 V
84	sf	25 °C	2.5 V
85	ss	25 °C	2.5 V

Table 52: Process, temperature and supply points simulated for the potentiostat pixel system (design point 17 of the archived Interactive.20 run: $R_f = 35\text{ k}\Omega$, $R_{ct} = 10\text{ k}\Omega$, $C_{dl} = 100\text{ nF}$). Point 1 is the typical statistical process point at 37 °C; points 2–5 are the imported foundry skew corners at 25 °C. Every figure and table in this section is schematic-level. These points move the sixteen transistor model rows only – the resistor and capacitor rows stay at their typical sections – so every corner spread in this section is a lower bound (pinned-passive caveat, page 69).

Every LSB in this subsection is referred to the nominal 2.441 mV converter step; the measured step is 1.591 mV (Table 73), so multiply by 1.535 to read these errors as converter codes.

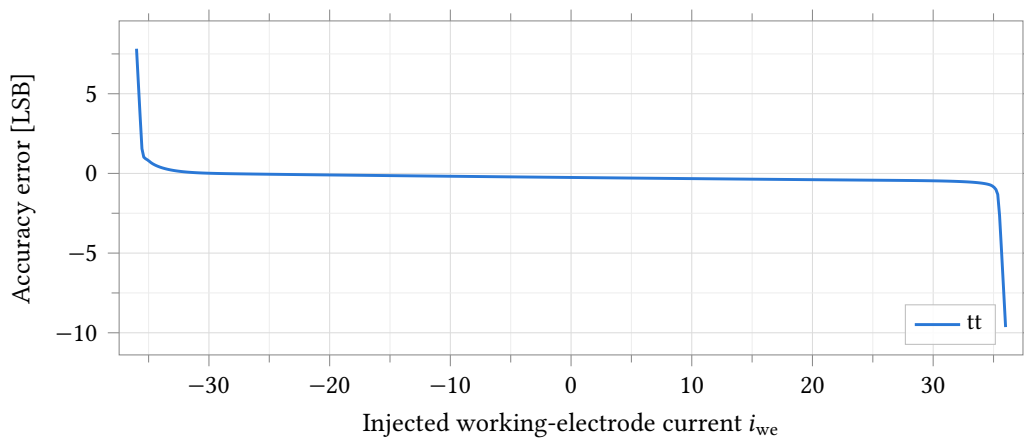


Figure 50: Bipolar accuracy error against injected working-electrode current, $R_f = 35\text{ k}\Omega$, $R_{ct} = 10\text{ k}\Omega$ electrode, typical process point, 37 °C, full $\pm 36\text{ }\mu\text{A}$ sweep (full scale $\pm 33\text{ }\mu\text{A}$). The zero crossing is smooth with no kink or discontinuity between the sourcing and sinking halves of the sweep: the error's slope through zero is $-1/T$ exactly (Table 53) and the same single anatop_tia_amp_ab TIA feedback path serves both current directions, unlike the rev-1 PMOS-only mirror it replaces. The error axis is referred to the nominal 2.441 mV step; the converter's measured step is 1.591 mV (Table 73), so multiply by 1.535 to read it as converter codes.

Finite-loop-gain error away from the operational electrode. Table 55 sweeps a small band around zero at all four gain taps with the electrode held at the stiff 10 k Ω cell used for the control-loop bench. The 140, 315 and 560 k Ω rows are not the accuracy of those taps – 10 k Ω is not the electrode those taps are designed against – but they isolate the mechanism: the zero-crossing slope error tracks $-1/T$ within 4 % at every tap, so the growth in worst-case error with R_f is finite closed-loop gain, not a distortion that appears only at high gain.

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Worst full-scale error	LSB	0.539	0.606	0.521	0.483	0.409	0.197
Zero-crossing error	nA	-17.704	-19.342	-18.087	-15.235	-13.949	5.393
Zero-crossing slope error		-0.00052	-0.00060	-0.00051	-0.00046	-0.00038	0.00022

Table 53: Bipolar accuracy at the $R_f = 35\text{ k}\Omega$ gain tap, the operational point at the $R_{ct} = 10\text{ k}\Omega$ electrode, by process corner. Truth current i_{we} swept $-36\text{ }\mu\text{A}$ to $36\text{ }\mu\text{A}$ (full scale is $33\text{ }\mu\text{A}$), error referred to the nominal 2.441 mV ADC LSB and clipped to $\pm$ full scale before taking the worst case. Over the full range the worst-case error is $0.41\text{--}0.61$ LSB across corners; over the $\pm 2.3\text{ }\mu\text{A}$ subrange it improves to $0.21\text{--}0.30$ LSB, the error growing toward full scale as the finite transimpedance loop gain ($\varepsilon \propto 1/T$). The converter's measured step is 1.591 mV , not the nominal 2.441 mV (Table 73); multiply every LSB figure here by 1.535 to read it as converter codes. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Worst full-scale error	LSB	0.333	0.376	0.331	0.293	0.253	0.123
Zero-crossing error	nA	-1.107	-1.209	-1.130	-0.952	-0.872	0.337
Zero-crossing slope error		-0.00012	-0.00016	-0.00011	-0.00011	-0.00008	0.00008

Table 54: Bipolar accuracy at the $R_f = 560\text{ k}\Omega$ gain tap, the operational point at the $R_{ct} = 1\text{ M}\Omega$ electrode, by process corner – **the headline rev-2 accuracy figure**. Truth current swept $\pm 2.3\text{ }\mu\text{A}$ (full scale is $2.06\text{ }\mu\text{A}$), error referred to the nominal 2.441 mV ADC LSB and clipped to $\pm$ full scale before taking the worst case. The converter's measured step is 1.591 mV , not the nominal 2.441 mV (Table 73); multiply every LSB figure here by 1.535 to read it as converter codes. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

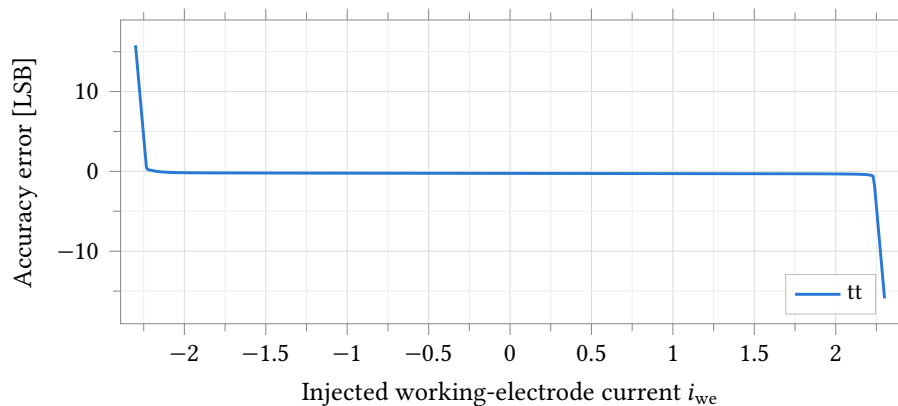


Figure 51: Accuracy error at the operational max-gain point, $R_f = 560\text{ k}\Omega$ into $R_{ct} = 1\text{ M}\Omega$, typical process point, 37°C . The truth current sweep is $\pm 2.3\text{ }\mu\text{A}$, just beyond the $2.06\text{ }\mu\text{A}$ full-scale current at this gain tap; the crossing at zero is smooth, with no kink, over the full swept range shown. The error axis is referred to the nominal 2.441 mV step; the converter's measured step is 1.591 mV (Table 73), so multiply by 1.535 to read it as converter codes.

R_f	Worst error over the sweep [LSB]	Zero-crossing error [nA]	Zero-crossing slope error ($= -1/T$)
35000	0.271	-17.704	-0.00052
140000	0.407	-4.426	-0.00118
315000	0.935	-1.967	-0.00233
560000	3.101	-1.107	-0.00394

Table 55: Finite-loop-gain error against feedback resistance at the typical statistical process point, 37 °C, truth current swept $-2.3\text{ }\mu\text{A}$ to $2.3\text{ }\mu\text{A}$ with $R_{ct} = 10\text{ k}\Omega$ held fixed across the whole sweep. **The 315 and 560 k Ω rows are NOT the accuracy of those gain taps** – $R_{ct} = 10\text{ k}\Omega$ is a stiff, non-operational electrode at those gains (the operational electrode is $1\text{ M}\Omega$, Table 54). They demonstrate that the residual error is finite closed-loop gain alone: the zero-crossing slope error is $-1/T$ to within 4 % at every tap, and the worst-case error scales with R_f as T falls. The LSB column is referred to the nominal 2.441 mV step; the converter’s measured step is 1.591 mV (Table 73), so multiply it by 1.535 to read it as converter codes.

8.5.2 Control Loop

The commanded reference potential v_{ref} is swept 0 V to 2.5 V ; Table 56 gives the resulting loop stability and Table 57 the tracking error. Loop gain exceeds 71.8 dB and crossover exceeds 21.2 MHz at every corner, both well past the $60\text{ dB}/60^\circ$ limits entered on the bench; the worst tracking error stays at or below 0.814 mV over the fixed 0.4 V to 2.1 V evaluation window at every corner. Table 58 and Table 59 give the compliance: the reference electrode tracks over 0.054 V to 2.44 V (a 10 mV error criterion, offset-inclusive) and the loop delivers $\pm 52.5\text{ }\mu\text{A}$ into the cell at every corner, read through the TIA feedback probe since no separate truth-current source is active on this bench.

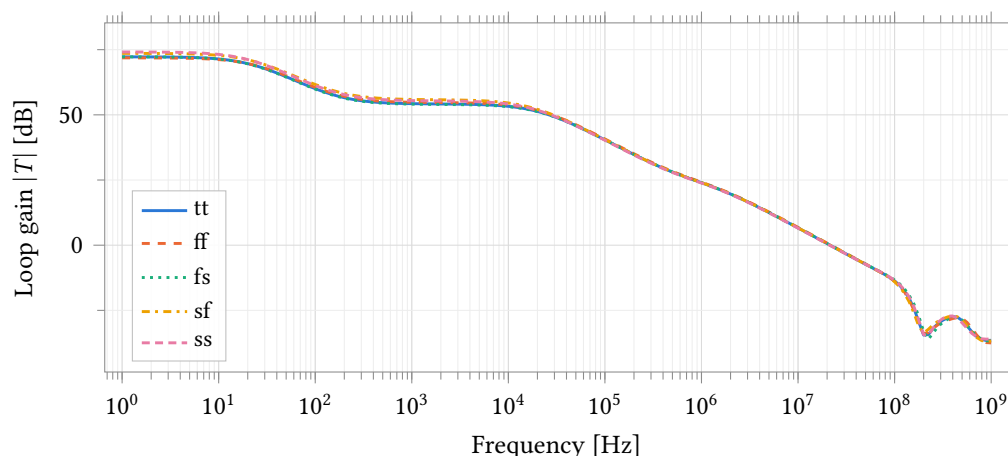


Figure 52: Potentiostat control-loop gain over five process corners, measured through the 0 V probe IPRB1 in the RE feedback path (Figure 49). $R_f = 35\text{ k}\Omega$, cell $R_{ce} = R_u = 1\text{ k}\Omega$, $R_{ct} = 10\text{ k}\Omega$, $C_{dl} = 100\text{ nF}$, 2.5 V supply. This is the full loop – A1, A2, both feedback networks and the electrode together – and is not the open-loop amplifier gain of Section 8.4. tt is at 37°C , the four skew corners at 25°C . The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Loop gain at DC	dB	72.24	71.85	72.25	73.57	74.11	2.26
Loop crossover	MHz	21.475	21.959	21.595	21.578	21.276	0.683
Phase margin	°	79.51	79.18	79.98	79.09	79.81	0.89

Table 56: Potentiostat control-loop stability by process corner, conditions as Figure 52. Every corner exceeds the 60° phase-margin and 60 dB DC-gain limits entered on the bench by a wide margin. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Zero-current tracking error	mV	−0.620	−0.677	−0.633	−0.533	−0.488	0.189
Worst tracking error over the window	mV	0.748	0.814	0.773	0.646	0.692	0.168
Worst tracking-error slope	mV V ^{−1}	1.735	0.541	1.534	2.022	2.442	1.901

Table 57: Reference-potential tracking by process corner. Cell-drive bench (Figure 49), v_{ref} swept 0 V to 2.5 V, $R_f = 35 \text{ k}\Omega$, cell $R_{\text{ce}} = R_u = 1 \text{ k}\Omega$, $R_{\text{ct}} = 10 \text{ k}\Omega$, $C_{\text{dl}} = 100 \text{ nF}$, 2.5 V supply. Tracking error is the reference-electrode potential minus the commanded value; the worst-error row is evaluated over the fixed 0.4 V to 2.1 V design-variable window ($v_{\text{ref_lo}}/v_{\text{ref_hi}}$), the same window for every corner, and is offset-inclusive. All five corners clear the 1 mV re_err_max and 50 mV V^{−1} re_err_slope limits entered on the bench.

Parameter	Unit	tt	ff	fs	sf	ss	Spread
RE tracking range (10 mV crit.)	V	2.382	2.386	2.383	2.382	2.379	0.007
TIA compliance range (10 mV crit.)	V	0.805	0.806	0.805	0.805	0.804	0.002

Table 58: RE tracking range and TIA compliance range by process corner, conditions as Table 57. Both are the width of the region over which the respective error stays inside a 10 mV threshold; the threshold is on the total error including the loop’s static offset, so a corner with more offset reports a narrower range for the same underlying loop gain – read these next to Table 57 rather than as an independent compliance figure. The TIA range is $\pm 35 \mu\text{A} \times 35 \text{ k}\Omega$ at the 10 k Ω electrode used here.

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Counter-electrode ceiling	V	2.4804	2.4829	2.4806	2.4814	2.4790	0.0039
Counter-electrode floor	mV	13.15	11.20	11.99	13.38	14.49	3.29
Cell current delivered, anodic	μA	52.501	52.595	52.530	52.517	52.444	0.151
Cell current delivered, cathodic	μA	52.496	52.591	52.526	52.513	52.439	0.153
Supply current	μA	218.71	250.19	204.17	230.66	196.65	53.54

Table 59: Counter-electrode compliance, delivered cell current and supply current by process corner, conditions as Table 57. Cell current is measured through the TIA feedback probe IPRB_F, which carries the entire working-electrode current by KCL since no truth current is injected on this bench; both directions clear the 33 μA limits entered on the test by more than 50 %. Supply current covers both amplifiers, both local bias generators and the shared global bias generator. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

8.5.3 Stability Across the Electrode Grid

Table 60 gives the transimpedance-loop phase margin at the typical process point over all twelve $R_f \times R_{ct} \times C_{dl}$ cells simulated. Phase margin stays inside 89.3° to 93.3° over the entire twelve-cell grid and all five process corners – every combination simulated, not only the typical point shown here. No cell needs an explicit compensation capacitor C_F across the electrode range simulated: the loop is unconditionally stable at both gain taps, both electrode resistances and all three double-layer capacitances tested.

R_f	C_{dl} [μ F]	R_{ct} [k Ω]	TIA loop gain at 1 Hz [dB]	TIA loop phase margin [$^\circ$]
35000	0.10	10	66.57	91.36
35000	1.00	10	66.56	91.50
35000	1.00	1000	81.71	91.50
35000	10.00	10	65.47	91.51
35000	10.00	1000	70.80	91.51
35000	0.10	1000	82.23	91.36
560000	0.10	10	48.07	90.21
560000	0.10	1000	78.52	90.21
560000	1.00	10	48.05	92.30
560000	1.00	1000	70.70	92.30
560000	10.00	10	46.68	92.51
560000	10.00	1000	51.44	92.51

Table 60: Transimpedance-loop stability across the electrode grid at the typical statistical process point (37°C , zero statistical offset). Randles electrode, $R_u = R_{ce} = 1\text{ k}\Omega$ fixed, C_{dl} and R_{ct} swept over the twelve $R_f \times C_{dl} \times R_{ct}$ combinations simulated (rows grouped by R_f). Phase margin stays inside 89.3° to 93.3° over all twelve cells and all five process corners of this grid (verified from the same extracted data, not tabulated in full here for width) – the transimpedance loop needs no explicit compensation capacitor C_F across the electrode range simulated.

8.5.4 System Noise

Table 61 gives output and input-referred noise over the same electrode grid. Output noise spans 0.129 mV to 11.97 mV across the full twelve-cell grid; at $R_f = 35\text{ k}\Omega$ it stays under 0.75 mV everywhere, but at $R_f = 560\text{ k}\Omega$ it grows to 2.0 mV to 12 mV – 1.3 to 7.5 ADC LSB on the converter’s measured 1.591 mV step (Table 73). The growth tracks C_{dl} , not R_{ct} : the closed-loop noise gain is $e_n \cdot (1 + R_f/Z_{\text{electrode}})$, and $|Z_{\text{electrode}}|$ falls with C_{dl} over the integration band, so a larger double-layer capacitance raises the noise gain even though it does not change R_{ct} . Input-referred current noise at $R_f = 560\text{ k}\Omega$ reaches 3.6 nA to 21 nA across the grid against a 2 nA design budget – exceeded by a factor of two to ten at the maximum gain tap. This mechanism is invisible to the standalone TIA bench of Section 8.4.2, which uses an ideal feedback resistor with no electrode network in the loop; publishing it here, with the mechanism, is why the pixel-system noise grid exists.

8.5.5 Transients

Figure 53 is a cyclic-voltammetry sweep at the 1 M Ω electrode: the working-electrode current stays within the loop’s linear region for the entire sweep, unclipped (Table 62). Figure 55 is the current response to a 400 mV step in the commanded reference potential at the same electrode. **The edges rail transiently by**

R_f	C_{dl} [μ F]	R_{ct} [$k\Omega$]	Output noise, 0.1 Hz to 1000 Hz [mV]	Input-referred current noise [nA]
35000	0.10	10	0.186	5.309
35000	1.00	10	0.477	13.640
35000	1.00	1000	0.474	13.534
35000	10.00	10	0.743	21.242
35000	10.00	1000	0.746	21.308
35000	0.10	1000	0.141	4.031
560000	0.10	10	2.739	4.891
560000	0.10	1000	2.187	3.905
560000	1.00	10	7.299	13.033
560000	1.00	1000	7.287	13.013
560000	10.00	10	11.343	20.255
560000	10.00	1000	11.402	20.360

Table 61: System output and input-referred noise across the electrode grid, conditions and process point as Table 60. Output noise integrated 0.1 Hz to 1000 Hz; input-referred current noise is the same integral divided by R_f and is therefore not comparable across the two R_f blocks by itself. At $R_f = 560 k\Omega$ noise grows with C_{dl} , not R_{ct} : the closed-loop noise gain is $e_n \cdot (1 + R_f/Z_{\text{electrode}})$ and $|Z_{\text{electrode}}|$ falls with C_{dl} over the 0.1 kHz to 1 kHz integration band, which the standalone TIA bench of Section 8.4.2 (ideal R_f , no electrode network) cannot show.

design: the bench exercises the step edge itself, and the current briefly saturates against the loop’s slew-limited transient (Table 63) before settling to the value the 1 M Ω cell impedance sets. Read the transient extremes as edge behaviour, not a sustained current rating.

Parameter	Unit	Min	Max	Spread
Commanded RE potential	V	0.749	1.749	1.000
Working-electrode current	μ A	−20.473	20.440	40.913

Table 62: Extremes of the CV sweep and its current response, typical process point, conditions as Figure 54. The current stays inside the loop’s linear region for the whole sweep – unclipped, unlike the 10 k Ω electrode used for the accuracy and control-loop measurements elsewhere in this section.

8.5.6 Monte Carlo: Zero-Current Offset

200 samples, **mismatch only**, nominal process at 25 °C, the 10 k Ω electrode, one 200-sample sub-run at each of the four gain taps. The reference-electrode offset $\sigma(v_{re,zero}) = 5.72$ mV (Table 66) is the clean, gain-independent statistic: it is the control loop’s own tracking offset carried to the electrode, identical to five significant figures at all four gain taps. The output-referred current offset $\sigma(i_{zero})$ is nearly gain-independent in its own right, 0.729 μ A to 0.843 μ A across the four taps (Table 65) – consistent with $\sigma(i_{zero}) \approx \sigma(v_{re,zero})/R_{cell}$. Referred to the nominal 2.441 mV ADC LSB the same offset current scales with R_f : $\sigma \approx 12/43/96/167$ LSB at 35/140/315/560 k Ω (Figure 56, Figure 57). At 560 k Ω some samples reach the output rails (Table 64) – this bounds the range a vcm-mux offset calibration must cover. Applying the same 5.72 mV loop offset to the operational 1 M Ω electrode of Table 54 predicts ≈ 1.3 LSB at 560 k Ω – a channel-to-channel spread on top of, not instead of, the corner-level finite-gain error of that table. Both LSB figures

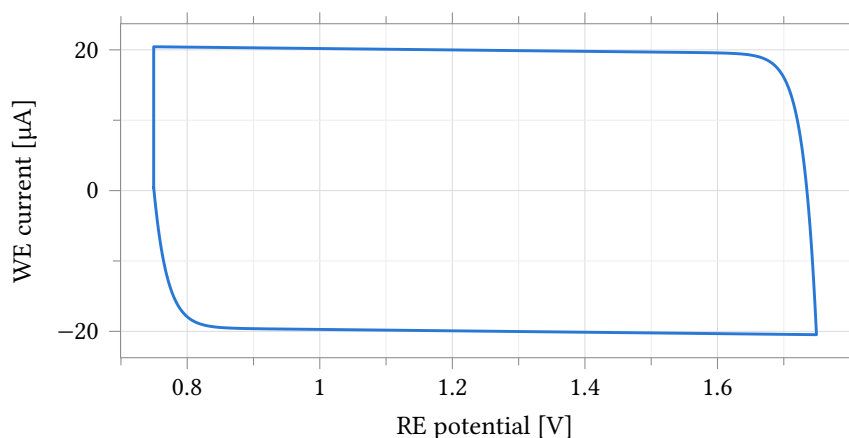


Figure 53: Working-electrode current against commanded reference-electrode potential over one triangular voltammetry sweep, typical process point, 37 °C, $R_f = 35 \text{ k}\Omega$, $R_{ct} = 1 \text{ M}\Omega$ electrode. The trace is an ellipse-like loop rather than a single-valued curve because the double-layer capacitance C_{dl} carries a current proportional to dv_{re}/dt in addition to the resistive charge-transfer current; the two straight sweep segments (RE potential linear in time, Figure 54) are why the loop closes into two nearly-parallel branches rather than a smooth curve. The current stays unclipped over the whole sweep (Table 62).

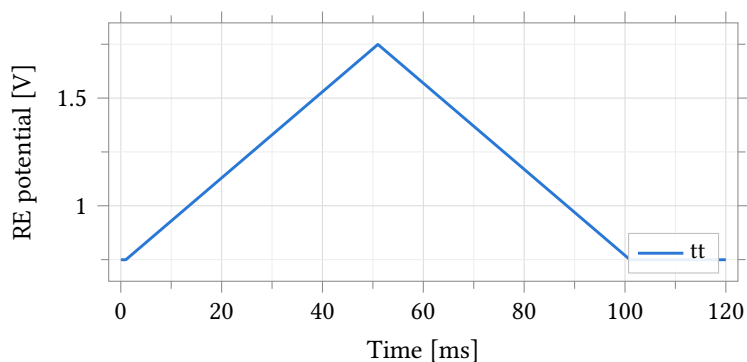


Figure 54: Commanded reference-electrode potential against time for the triangular voltammetry sweep, typical process point, 37 °C, $R_f = 35 \text{ k}\Omega$, $R_{ct} = 1 \text{ M}\Omega$ electrode. Paired with the current response of Figure 53.

Parameter	Unit	Min	Max	Spread	Settling
Working-electrode current	μA	-35.456	35.315	70.771	49.001
Counter-electrode potential	V	1.204	1.695	0.491	49.002

Table 63: Extremes and settling of the step response, conditions as Figure 55. `settle` is the last exit from a $\pm 1 \%$ band around the final value; the extremes are transient peaks reached during the rail-limited edge and are not a sustained operating rating.

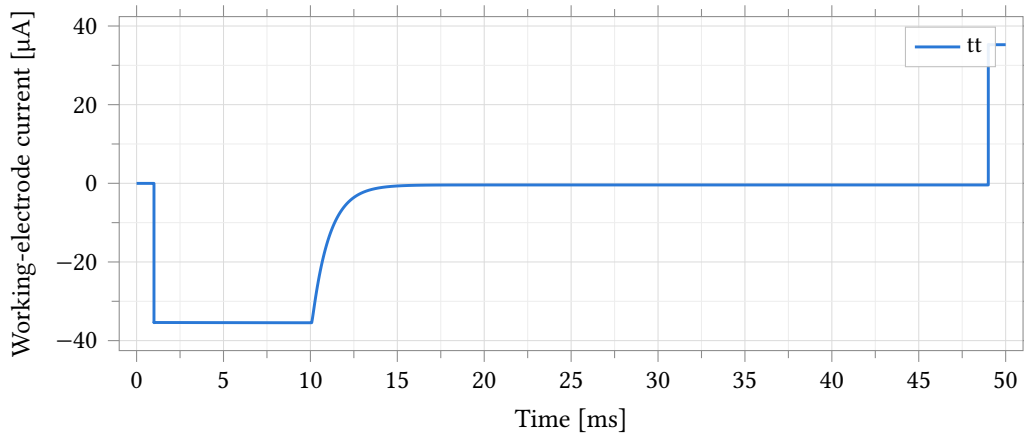


Figure 55: Step response of the working-electrode current to a 400 mV step in the commanded reference potential, typical process point, 37 °C, $R_f = 35 \text{ k}\Omega$, $R_{ct} = 1 \text{ M}\Omega$ electrode, 1 μs edge. The transient overshoots and briefly rails against the loop’s slew-limited edges immediately after the step – by design, the bench exercises the edge rather than a settled operating condition – before settling to the value set by the cell impedance.

are on the nominal step; the converter’s measured step is 1.591 mV (Table 73), so multiply by 1.535 to read them as converter codes.

Parameter	Unit	N	Mean	σ	Min	Max	Spec	Yield
Zero-current offset, $R_f = 560 \text{ k}\Omega$	μA	200	-0.002	0.729	-1.856	2.231	-	-
Zero-current offset in LSB, $R_f = 560 \text{ k}\Omega$	LSB	200	-0.49	167.33	-425.83	511.74	± 1.00	0.5
Output voltage at zero current, $R_f = 560 \text{ k}\Omega$	V	200	1.249	0.408	0.211	2.499	-	-

Table 64: Monte Carlo zero-current offset at $R_f = 560 \text{ k}\Omega$, $R_{ct} = 10 \text{ k}\Omega$ electrode. **Mismatch only**, 25 °C nominal process, 200 samples. $\sigma(i_{\text{zero}}) = 0.729 \mu\text{A}$ here against 0.729 μA to 0.843 μA across all four gain taps (Table 65) – nearly gain-independent, as expected for an amplifier-referred current offset. `vout_zero` ranges 0.21 V to 2.50 V, i.e. some samples reach the output rails at this gain – this bounds the range a `vcm-mux` offset calibration must cover. `i_zero_lsb` and the ± 1 LSB limit entered on the test are referred to the nominal 2.441 mV step, 1.535 times the converter’s measured 1.591 mV step (Table 73); only the yield against that literal limit is quoted, not a four-channel figure, because the limit itself is not gain-scaled (see Figure 57).

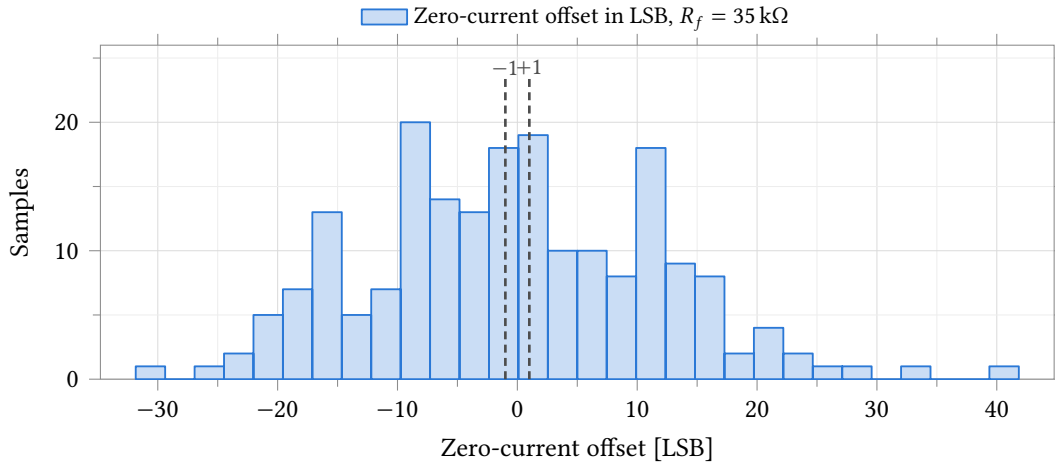


Figure 56: Zero-current output offset at $R_f = 35\text{ k}\Omega$, in nominal 2.441 mV ADC LSB. **Mismatch only**, 25 °C nominal process, 200 samples, $R_{ct} = 10\text{ k}\Omega$ electrode. The limit shown is the ± 1 LSB entered on the test in Assembler – comparable in kind across the four gain taps but mechanically harder to meet as R_f rises, since the underlying offset is closer to gain-independent in current (Table 64). The converter’s measured step is 1.591 mV, not the nominal 2.441 mV (Table 73); multiply the axis and the limit by 1.535 to read them as converter codes. Statistics are in Table 64.

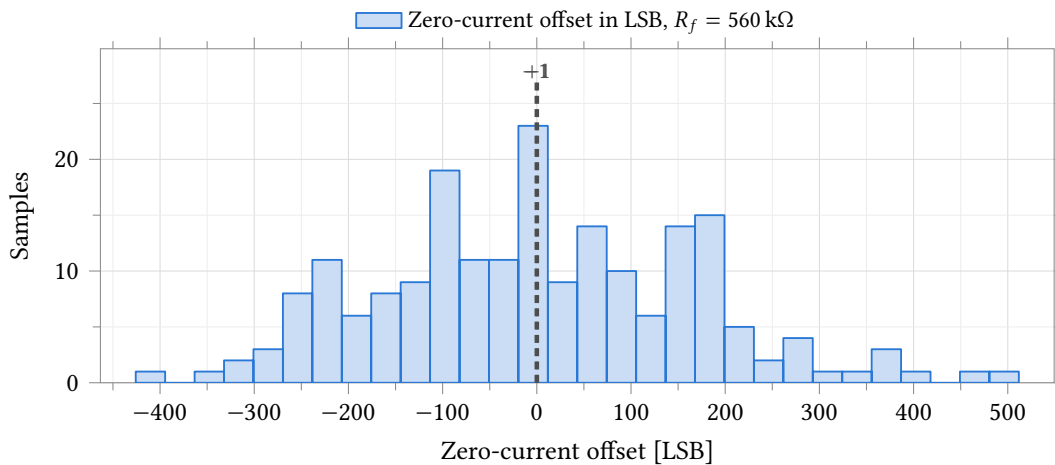


Figure 57: Zero-current output offset at $R_f = 560\text{ k}\Omega$, conditions as Figure 56 but at $R_{ct} = 10\text{ k}\Omega$ (the same stiff electrode as the finite-loop-gain demonstration of Table 55, NOT the 1 M Ω operational electrode of Table 54). Some samples reach the sweep’s output rails (Table 64); the LSB spread here is 14 $\times$ wider than at 35 k Ω for the same underlying current offset, since the same offset current is multiplied through a 16 $\times$ larger R_f .

Parameter	Unit	N	Mean	σ	Min	Max
Zero-current offset, $R_f = 35\text{ k}\Omega$	μA	200	-0.019	0.843	-2.221	2.918

Table 65: Monte Carlo zero-current offset at $R_f = 35\text{ k}\Omega$, conditions as Table 64. $\sigma = 0.843\text{ }\mu\text{A}$; the four-gain range 0.729 μA to 0.843 μA (35k highest, 560k lowest) is the full spread of this same statistic recomputed independently at each of the four staged gain blocks (140k: 0.758 μA , 315k: 0.742 μA) – all four traceable to `skill/px_mc_slice.py` and the corresponding `px_mc_g*_mc.csv` files.

Parameter	Unit	N	Mean	σ	Min	Max
RE zero-current offset, $R_f = 35\text{ k}\Omega$	mV	200	-0.686	5.718	-18.623	16.457

Table 66: Monte Carlo reference-electrode zero-current offset, the gain-independent statistic. **Mismatch only**, 25 °C nominal process, 200 samples. $\sigma = 5.7176\text{ mV}$, identical to five significant figures at all four gain taps (verified: 5.71757, 5.71757, 5.71758, 5.71759 mV at 35k/140k/315k/560k) – this is the control loop’s own reference-tracking offset (Table 57), which does not depend on which TIA gain tap is selected. At the operational $1\text{ M}\Omega$ electrode this offset predicts $\sigma(i_{\text{zero}}) \approx \sigma(v_{\text{os}})/R_{\text{cell}}$; using the measured 5.72 mV against the operational electrode’s own $R_{\text{ct}} + R_u \approx 1.001\text{ M}\Omega$ gives $\approx 5.7\text{ nA} \approx 1.3\text{ LSB}$ at 560 k Ω – consistent with, and about 3.5 $\times$ larger than, the corner-level finite-gain error of Table 54.

8.5.7 Measuring this block on chip

Two nodes separate the four faults that all read as a stuck converter code.

1. Select slot 5: the counter-electrode drive must sit mid-rail and follow the reference code.
2. Select slot 6, the reference-electrode buffer output: it must equal the commanded reference within 16 mV (3σ of A1’s offset).
3. Triage on that pair, not on the converter (it reports a rail for both faults): counter electrode railed with the summing node at common mode = control-loop or electrode fault; counter electrode unrailed with the summing node off common mode = transimpedance fault.
4. Read the summing node itself at slot 7 only through its series guard and with the mid node clamped: it is a virtual ground at 0.824 nA per LSB at the top gain code, and an ungranted site must stay doubly isolated from it.
5. Zero-current constant: open the working-electrode isolation switch (cell current is zero by construction), read the transimpedance output at every gain code the channel will use, subtract the converter’s zero code. Uncalibrated spread: $\sigma = 12.1\text{--}167.3\text{ LSB}$ over the four taps (Section 8.5.6), on the nominal 2.441 mV step; multiply by 1.535 for the converter’s measured 1.591 mV step (Table 73).

This test fabric is design intent; it is not yet in the schematics.

8.6 Assembled Channel

This is the cell intended for tapeout: one complete analog channel with every ideal source of Section 8.5 replaced by the silicon that will drive it on die. The topology and the sign convention are unchanged – A1 on the counter electrode, A2 holding the working electrode at the common mode through R_f – so this section reports only what the assembly changes.

- The potential program and the common mode come from two 12-bit R-2R DACs (Section 8.10), one per channel, at 610.35 μV per code. Both ladders run from the 2.5 V analog rail directly; the cell carries no reference-DAC supply regulator.
- R_f is the 320-code programmable ladder of Section 8.9, not an ideal resistor, and carries a 22.0 pF compensation capacitor across it, without which the assembled loop does not close (Section 8.6.6).
- The 10-bit converter of Section 8.8 closes the channel behind a 16:1 input multiplexer whose slots select the transimpedance output, the common-mode rail, the commanded potential and the analog test point. A unity-gain buffer stands between that multiplexer and the converter input, and a second buffers the test-port multiplexer output, so the class-AB amplifier of Section 8.4 appears four times per channel: A1, A2 and the two followers. Each buffer carries its own 492 fF metal–insulator–metal compensation pair, as does A1.
- Bias comes from the local generator of Section 8.3 at trim code 37.

The counter, reference and working electrodes leave the cell directly: there are no isolation switches, no reference-electrode buffer and no on-chip gain reference resistor, so the fabric several of the calibration steps of Section 8.11 assume is not yet present.

All data below is schematic-level, at the nominal process point unless a corner is named, 37 °C and a 2.5 V analog supply. It comes from three dc and transient tests — a sweep of the potential code, a transient with the electrode inert, and a sweep of all 320 gain codes — a netlist-level staircase voltammetry run (Section 8.6.3), and loop-gain, noise, transient-noise, supply-current and 200-sample Monte Carlo runs on the netlist those tests produce (Sections 8.6.6–8.6.9). Section 8.6.4 is the exception and is stated there: the converter’s capacitor array exists only in layout, so every converter number comes from a netlist in which the extracted converter replaces the schematic one.

8.6.1 Potential Axis

The 2 mV tracking specification is written over the 0.4 V to 2.1 V commanded window that Section 8.5 uses; Section 8.6.9 gives the reason and the full-range figure is reported alongside it throughout. The reference electrode follows the commanded DAC output to 0.672 mV over that window, and to 1.419 mV over the full 0 V to 2.42 V code range — +0.402 mV at code 0, –1.419 mV at code 3968. The ideal-source measurement is 0.814 mV over the same window. Two terms make it up, both static: the ladder’s own error on a 610.35 μ V step, which is –181 μ V at the top code, and A1’s –620 μ V input offset, which is constant across the range. A two-point fit per channel over the potential code axis absorbs the pair and leaves a residual well inside the 2 mV budget of Section 8.4 — inside the window; at the ends of the code range A1’s input pair leaves its common-mode range and the error stops being a constant any fit can absorb (Section 8.6.9). The offset-and-slope constant of Table 97 is what stores it. The common-mode DAC sits 262 μ V below 1.25 V at mid-code and every current referred to that node inherits the error.

The step is 610.35 μ V, 2.5 V/4096, not the 528.96 μ V of Section 8.10, because in this cell both ladders run from the 2.5 V analog rail directly: the channel carries no reference-DAC supply regulator. The 25.8 mV of code-dependent reference movement measured in Section 8.10.1 is therefore absent from every potential number here. It is 40 $\times$ the tracking error above and would dominate the potential axis if a regulator were ever put in the path — the reason to read 0.672 mV as a missing stimulus rather than a result.

8.6.2 Zero-Current Axis

With the electrode inert the working electrode sits 619.64 μ V below the common mode, and it is the same 619.64 μ V at every one of the 320 gain codes: 0.3 nV of spread. No current flows in R_f , so the output carries A2’s input offset alone and nothing that scales with the tap. **One zero measurement therefore calibrates every gain code** — and it holds per sample under mismatch, not only at this typical point (Section 8.6.9). It is the result the per-gain-code current zero of Table 97 rests on, and it is measured through the common-mode slot of the converter multiplexer, which the input buffer of Section 8.6.4 makes readable.

What survives that calibration is input bias current. The output drifts a further 3.91 μ V between $N = 0$ and $N = 319$ while R_f grows by 1.914 M Ω : 2.04 pA into A2’s inverting input, 0.5 % of the 0.41 μ A full scale at maximum gain and negligible at every code below it.

Referred to current, the uncalibrated zero is 34.4 nA at $N = 0$, 1.57 nA at $N = 63$ and 0.323 nA at $N = 319$. The last two are already inside the 2 nA input-referred noise budget, so at high gain the uncalibrated offset is not what limits a reading; at the bottom code it is 34.4 nA on a 40.7 μ A full scale and below one converter code either way (Section 8.6.4).

Quantity	Value	Conditions
Reference tracking error	0.672 mV	dc sweep over the specified 0.4 V to 2.1 V window (§8.6.9); 0.814 mV with an ideal source (§8.5)
Reference tracking error, full range	1.419 mV	0 V to 2.42 V, all 4096 codes; reported, not specified
Reference DAC step	610.35 μ V	2.5 V/4096 from an ideal rail; -181μ V endpoint error at code 3968
Common-mode output	1.249 738 V	mid-code, -262μ V from 1.25 V
Zero-current output, $v_{\text{out}} - V_{\text{CM}}$	-619.64μ V	inert electrode, $N = 0$
Spread of that zero over gain codes	0.3 nV	all 320 codes
A2 input bias current	2.04 pA	3.91μ V of drift over $\Delta R_f = 1.914 \text{ M}\Omega$
Uncalibrated current zero	34.4 / 1.57 / 0.323 nA	$N = 0 / 63 / 319$
Converter-input zero	1.248 499 V	inert electrode, readout slot; -619.6μ V from the transimpedance output, which is the input buffer's own offset
Feedback resistance, in situ	37.16 k Ω	$N = 3, 37^\circ\text{C}$; median, p10–p90 36.95 k Ω to 37.44 k Ω
Readout against the in-line probe	$\leq 39.4 \text{ nA}$	331 treads above 100 nA, on 7.7 μ A peaks, i.e. 0.5 %
Voltammetric peak potential shift	0.3 / 2.3 mV	forward / reverse, against the ideal-source run of §8.2
Sampled peak height	$-11.2 / -13.9 \%$	forward / reverse, same reference; staircase sampling, not an error
Converter agreement with its own bench	$\leq 3.3 \text{ LSB}$	extracted converter, 0.62 V to 1.87 V
Current per converter LSB	88.5 / 0.825 nA	$N = 0 / 319$; 1.593 mV LSB over the measured 1.63 V span, on the nominal ladder. The measured in-loop R_f puts the bottom code at 81.2 nA
Calibration-slot disturbance	1.75 mV	peak on the transimpedance output over a 21 μ s record with the common-mode slot selected; settles in 0.54 μ s, residual below 1 nV
Transimpedance phase margin	75.7°	worst of 24 electrode $\times$ gain points, 1.07 dB peaking; both buffer loops $\geq 78.4^\circ$
Closed-loop bandwidth	85.8 / 3.4 kHz	$N = 0 / 319$, $C_{\text{dl}} = 100 \text{ nF}$, with the 22.0 pF compensation capacitor
Input-referred noise, one tread	1.5 nA to 2.9 nA	2 ms average; 0.036–1.78 LSB as the gain code rises
Channel power	1.323 mW	extracted converter; 487.1 μ W of it is the two unity-gain buffers
Zero-current output, σ	5.488 mV	200 Monte Carlo samples, process and mismatch (§8.6.9); $\pm 2 \text{ mV}$ budget
Reference tracking error, σ	5.287 mV	same run, mid-code; 5.72 mV on the flat bench (§8.5.6)
Converter-input zero, σ	8.040 mV	same run; the transimpedance offset and the buffer's own, uncorrelated. It cancels in the zero subtraction (§8.6.9)

Table 67: Headline measurements on the assembled channel, nominal process point, 37°C , 2.5 V supply. Rows 1–9 are dc; rows 10–13 come from the staircase voltammetry run of Figure 58; rows 14–16 and the power figure need the extracted converter, everything else in the channel staying at schematic level; rows 17–20 are the small-signal, noise and power runs of Sections 8.6.6–8.6.8; the final three are Monte Carlo, and are narrower than the built channel's for the reason Section 8.6.10 gives. The two tracking figures are optimistic by construction: both ladders run from the 2.5 V rail, so the 25.8 mV of code-dependent reference movement Section 8.10.1 measures on the supply block is not exercised here.

8.6.3 Cyclic Voltammetry Through the Complete Chain

Figure 58 is the electrode of Section 8.2 measured through the whole assembled channel: the potential program is a real 12-bit code sequence, the applied potential is servo'd by A1, and the current is read back

through the programmable resistor at gain code 3. The programme is a triangle staircase, ± 0.5 V about the common mode in 4 mV treads of 6 or 7 codes, 2 ms per tread — 2 V s^{-1} — 501 treads over one cycle, sampled once per tread 50 μs before the next code change.

Potential. The peaks land within 0.3 mV (forward) and 2.3 mV (reverse) of the same electrode driven by an ideal source. The potential axis therefore survives 610 μV quantisation, the ladder's own error, the common-mode DAC's $-262 \mu\text{V}$ and A1's $-620 \mu\text{V}$ offset and still puts the redox features where an ideal source puts them — which is what identifies an analyte.

Current. Sampled peak heights are 11.2 % and 13.9 % below the ideal-source run. A tread sampled at its end has shed the double-layer charging current that a continuous ramp carries, and the diffusion layer relaxes over each 2 ms hold, so the two numbers measure different things about the same electrode rather than disagreeing: kept raw, the same waveform peaks 26 % and 42 % *above* the ideal-source run on the charging tails.

Feedback resistance, measured in the loop. A current probe in the feedback branch makes R_f a measurement rather than an assumption: $(v_{\text{out}} - v_{\text{we}})/i_f = 37.16 \text{ k}\Omega$ median over 124 921 points, p10–p90 36.95 k Ω to 37.44 k Ω , against 36.0 k Ω from the nominal law and 37.62 k Ω measured two-terminal at 25 °C (Section 8.9) — 1.2 % below the latter at 37 °C.

Readout. Over the 331 treads carrying more than 100 nA the voltage readout $(v_{\text{out}} - V_{\text{CM}} - V_{\text{os}})/R_f$ — the arithmetic firmware performs — reproduces the probe current to within 39.4 nA on 7.7 μA peaks, 0.5 %, inside the ± 0.65 % spread of R_f itself. No systematic term is left over. V_{os} has to be supplied from a genuine zero-current measurement (Section 8.6.2): estimated instead from the pre-scan quiet period, where this electrode already passes 27 nA, it comes out +0.38 mV and silently subtracts a standing current from the whole voltammogram.

One cycle, one gain code, one scan rate, at nominal, with the converter clock held static.

8.6.4 Converter Readout

The converter's charge-redistribution array exists only in layout, so at schematic level it makes no bit decisions and the code the assembled channel reports is not a conversion. The numbers below come from a netlist in which the extracted converter replaces the schematic one and nothing else changes.

Against known voltages the channel's transfer matches the standalone characterisation of Section 8.8 to within 3.3 LSB across 0.62 V to 1.87 V, and three conversions in one run repeat to 2 LSB; on the four levels re-measured through the buffered multiplexer described below the agreement is 0.4 LSB. One LSB is 1.593 mV over the converter's measured 1.63 V span, so what it resolves in current is set by R_f alone: 88.5 nA at $N = 0$, 4.02 nA at $N = 63$ and 0.825 nA at $N = 319$. The zero calibration consequently cannot be taken at the bottom gain code, where the whole 34.4 nA offset is sub-LSB: measured there the code corresponds to -73 nA of apparent current on a -34 nA signal, which is quantisation and not a reading. It must be taken at a high-gain tap, where one LSB is smaller than the offset being measured — which the gain-independent zero of Section 8.6.2 makes free.

The multiplexer no longer disturbs the channel. A unity-gain buffer stands between the multiplexer and the converter, so the sample switch draws its charge from the buffer and the multiplexer drives only a gate. Selecting the common-mode slot moves that rail by 0.834 mV and the transimpedance output by 1.75 mV — 1.1 LSB — both recovered inside one conversion, the output settling in 0.54 μs to a residual below

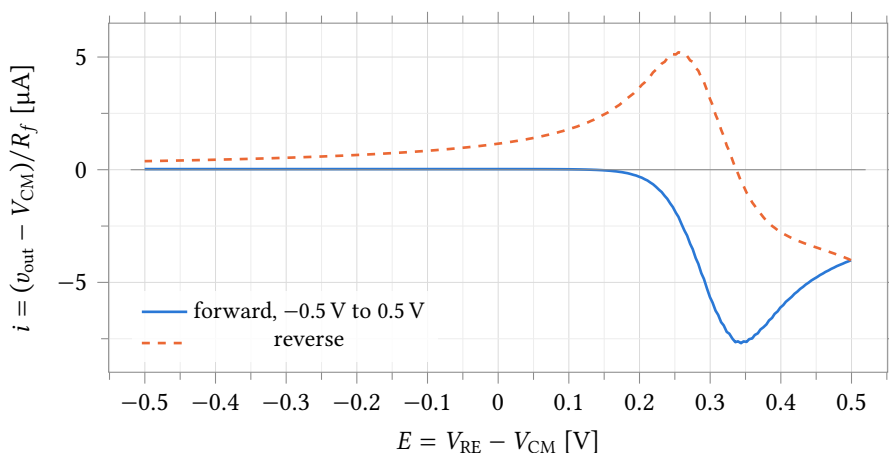


Figure 58: Staircase cyclic voltammogram taken through the complete assembled channel — real 12-bit potential codes, real programmable feedback resistor at $N = 3$ (37.16 k Ω measured in the loop) — of the electrode of Section 8.2 at 2 V s⁻¹, 37 °C, nominal process point. One point per 4 mV tread, sampled 50 μ s before the next code change, so the double-layer charging current has decayed and what remains is faradaic; the sign convention is the measured one of Section 8.5, $dv_{\text{out}}/di_{\text{we}} = +R_f$, so oxidation reads negative. Peak potentials sit within 0.3 mV and 2.3 mV of the same electrode driven by an ideal source; sampled peak heights are 11.2 % and 13.9 % lower, which is the staircase sampling and not a channel error (Table 67).

1 nV. Without the buffer the same measurement dragged the common mode 518 mV and threw the output 1233 mV, with 149 mV still un-recovered at the end of a 21 μ s record: it never settled, at every conversion. **The gain-independent zero measurement of Section 8.6.2 is therefore executable on this cell.** The commanded-potential slot was broken the same way and worse on the residual (−99.9 mV at the end of the record) and now peaks at 0.207 mV; the readout slot is 51 $\times$ quieter as a side effect, its 5.6 mV sampling kick now landing on the buffer’s output where the channel does not see it. Both buffer loops are stable — 78.4° of phase margin with 17.8 dB of gain margin at the converter, 80.9° and 20.3 dB at the test port, and neither moves with the gain code.

The buffer’s offset cancels in the subtraction. The buffer is the same amplifier as A2 at the same operating point, so it contributes that amplifier’s own input offset, −619.6 μ V, directly at the converter input. One buffer downstream of the multiplexer rather than one per source is what makes this free: every slot is read through it, so the offset subtracts out of a zero-referenced reading exactly and only its variation with input level survives. Over 100 Monte Carlo samples that residual is 0.62 mV mean and 2.07 mV worst across the full ± 0.8 V readout swing, and 0.031 mV mean, 0.11 mV worst over ± 150 mV — against a 5.58 mV σ on the offset itself.

Bit 9 is inverted (firmware-critical). The raw output bus carries its most significant bit inverted — the two’s-complement form of the offset-binary code Section 8.8 describes. Every published converter number, including the transfer of Table 73, is the corrected code raw + 512 − 1024 b_9 , so firmware must invert bit 9 before using the bus. Read raw, the transfer is a plausible straight line that wraps at the common-mode code — a silent 512-code error over half the range.

8.6.5 Input Multiplexer

Both the converter and the test port see the channel through a 16:1 transmission-gate tree, and twenty select bits reach those trees through 1.0 V-to-2.5 V level shifters. The fabric is in the signal path, so its leakage, its on-resistance and its switching behaviour are channel specifications rather than debug conveniences.

Off-state leakage meets its budget conditionally. A slot's leakage is set by the voltage across the one off switch it shares with its level-1 sibling, and the dependence has a knee rather than a slope: at 40 °C it is 0.05 pA anywhere within 1.00 V of the sibling, 0.37 pA at 1.125 V and 24.7 pA at 1.25 V (Table 68). At 85 °C the same three points are 0.05, 8.5 and 298 pA. The mechanism is gate-induced drain leakage in the off pass device, whose gate sits at a supply rail, so the drain-gate field reaches its onset exactly as the slot-to-sibling difference passes 1.1 V; junction leakage is bounded separately at 0.047 pA and is not the limit. Against a 20 pA budget the fabric therefore passes with 280× margin while neighbouring slots sit near one another, and misses by 1.2× at 40 °C and 15× at 85 °C when a slot's sibling sits at a rail.

The tree earns its place, measurably. Only the level-1 sibling appears in the answer: holding the other fourteen inputs at ground and at 1.25 V gives 24.71 and 24.86 pA for the same slot, and every slot crosses the same four gates, so no slot is privileged. A flat sixteen-way star has no such structure — every off switch faces the shared node, so a sense slot's leakage would be set by whichever slot is currently selected, a quantity the measurement itself changes. The costs are four gates in series and the switching behaviour below.

On-resistance peaks high, not mid-rail. It is 3.68 kΩ at 1.80 V and 37 °C, worst 5.03 kΩ at the slow hot corner, against a 6 kΩ limit. The hump sits above mid-rail because the two pass devices are the same width, which makes the PMOS the weaker half. It does not threaten the sampling aperture by four orders of magnitude: the multiplexer drives a gate, whose input capacitance measures 23.4 fF, and a 1.25 V slot change settles to half a converter code in 2.36 ns. Read at the test pad instead, into 1.2 kΩ and 50 pF, the same change takes 1.42 μs — irrelevant for a meter, decisive for a procedure that assumes a settled node. Coupling from unselected slots simulates at 114 nV peak-to-peak for a full-scale aggressor step, but that is a model floor: a cut-off device is given no source-to-drain path, so the physical number is set by routing and needs an extracted view.

Make-before-break is real. Each two-way cell drives its pass gates from a select bit and its locally inverted copy, so during any transition both conduct. A toggle of the lowest select bit does this in all eight level-1 cells at once, shorting all eight sibling pairs; the higher bits short the group outputs. Measured on a parked slot with 0.94 V across its pair, 119 fC is transferred per toggle at 1 ns select edges, ≈ 0.13 fC per millivolt of pair difference, rising to 830 fC at 20 ns edges. A change that toggles only the top select bit moves the same probe by 0 fC, which is what makes the effect per-bit and predictable rather than diffuse. Select-bit skew is the sharper hazard: 5 ns of skew between select bits swings the output the full 1.25 V through a third, unintended slot and stretches settling from 2.4 to 6.3 ns; 50 ns of skew stretches it to 51 ns. A reading taken too soon after a slot change is not a noisy version of the right slot — it is another slot's voltage. Removing this would need a decoder and a non-overlap generator, which is the device count the tree exists to avoid, so it is handled by procedure:

1. **Keep every unselected slot near the common mode**, and in no case more than 1 V from its level-1 sibling. Tying spares to local ground satisfies neither. The level-1 pairs are fixed by the tree — (0, 1), (2, 3) and so on — and the channel as drawn violates the rule: the test-port tree has all sixteen inputs undriven and the converter tree eleven of sixteen, and one of those undriven slots is

the level-1 sibling of the commanded-potential output, into which it dumps its accumulated charge on every toggle of the lowest select bit.

2. **Wait after any slot change before sampling**, for the settling time plus the worst-case skew between the four select bits of that tree, and drive those four bits from one shifter instance so the skew is bounded.

Level shifters. Non-inverting, output high at 2.5 V and low below 0.6 μV , threshold 0.52 to 0.55 of the 1.0 V rail, propagation 1.2 ns to 2.5 ns. The figure to carry into the power budget is the crowbar: a select input parked at its own threshold — a floating select bus, a half-driven pin, a slow ramp — draws 2.5 μA per channel, and twenty select bits is 51 μA , on the analog rail rather than the digital one. With the 1.0 V domain unpowered and the analog domain live, no select line exceeds 283 mV against a ≈ 550 mV pass threshold and no switch closes that would not have closed anyway; a parked slot moves by at most 0.52 mV when the 1.0 V rail comes up.

Slot-to-slot on-resistance matching is not measurable here. Every path crosses the same four gates, and nominal on-resistance is identical across slots to seven digits; a 50-sample mismatch run returns $\sigma = 0$ exactly, because all ninety pass devices per tree sit on model cards that carry no per-device mismatch. The concern is bounded rather than measured: at 1 μA the on-resistance drops 1.7 mV, so at the femtoampere-to-nanoampere currents this fabric carries a 20 % mismatch is under 1 μV . Slot-dependent on-resistance is not a measurement-error mechanism for this path.

Parameter	25 °C	37 °C	40 °C	85 °C	ss 85 °C	ff -40 °C
Leakage, sibling at V_{CM} [pA]	0.062	0.061	0.061	0.048	0.047	0.069
Leakage, sibling at ground [pA]	9.17	20.6	24.7	300	73.7	0.167
Leakage, worst neighbour [pA]	19.2	42.1	50.8	582.6	85.7	1.37
R_{on} at V_{CM} [Ω]	1680	1739	1753	1971	2588	1041
R_{on} peak [Ω]	3572	3682	3709	4117	5029	2415

Table 68: Off-state leakage and on-resistance of one 16:1 input multiplexer tree, schematic level. Leakage is the current out of one slot with the tree selecting elsewhere, and is quoted against the position of that slot's level-1 sibling; the worst-neighbour row sweeps the sibling over the whole rail. On-resistance is measured with 1 μA forced out of the tree output, and its peak sits at 1.80 V rather than mid-rail. The leakage budget is 20 pA and the on-resistance limit 6 k Ω .

8.6.6 Compensation and Stability

The transimpedance stage carries a 22.0 pF capacitor across R_f , a four-unit metal–insulator–metal array occupying 10 853 μm^2 per channel. All three capacitors in that stage carry mismatch statistics. Every compensation capacitor in the channel is now a real metal–insulator–metal device: A1's pair and each buffer's are 492 fF, against the 500 fF the ideal elements they replaced carried.

It exists because the assembled loop did not close without it. With the programmable ladder and the real reference in the loop — neither of which Section 8.5.3 had — 6 of 24 electrode-by-gain grid points failed, falling to 35.95° of phase margin with 4.54 dB of peaking at the nominal corner and to 30.00° with 5.96 dB at the slow one. With the capacitor all 24 points pass: the worst cell is $C_{\text{dl}} = 100$ nF, $R_{\text{ct}} = 1$ M Ω , $N = 63$ at 75.71° and 1.068 dB, and it held 72.37° and 1.379 dB at the slow corner — the one figure here measured on the previous revision of the cell, the five-corner run not having been repeated. Gain margin is at least 20.72 dB

everywhere. The grid spans $C_{dl} = 100 \text{ nF} / 1 \mu\text{F} / 10 \mu\text{F}$, $R_{ct} = 10 \text{ k}\Omega / 1 \text{ M}\Omega$ and $N = 0 / 63 / 191 / 319$, and the two worst cells were re-run at five corners.

The capacitor also lands in the control loop, because it ties the transimpedance output to the working electrode and therefore sits inside A1's path through the electrode: A1's gain margin rises from 21 dB to 36.5 dB to 39.1 dB and its crossover falls from 18 MHz to 6.0 MHz; its phase margin, 68.1° to 71.4° before, is now 79.7° to 81.7° . At minimum gain the control loop's phase never reaches zero, so its gain margin is unbounded.

Making A1's compensation pair a real device rather than an ideal element costs 0.4 dB to 0.7 dB of transimpedance gain margin and buys 0.36° of transimpedance phase margin and 1.0 dB of control-loop gain margin. Peaking, transimpedance and bandwidth do not move at any printed digit. The two multiplexer buffers sit outside both loops and appear in neither; their own loops close at 78.4° and 80.9° of phase margin with 17.8 dB and 20.3 dB of gain margin, unchanged across gain codes.

The cost is closed-loop bandwidth, which falls by a third to a half at every code: 116.6 kHz to 85.8 kHz at $N = 0$, 13.6 kHz to 10.0 kHz at $N = 63$ and 5.4 kHz to 3.4 kHz at $N = 319$. Against the 2 ms staircase tread of Section 8.6.3 even the slowest of those settles two decades early, so no measurement in this section is bandwidth-limited.

The five-corner rows exercise the transistors alone: neither R_f nor the compensation capacitor corners (pinned-passive caveat, page 69). The 72.37° slow-corner margin quoted above is therefore not the worst case — freeing the passive rows costs the transimpedance loop 10° of phase margin on the amplifier bench, and both quantities the compensation is sized against, R_f and the capacitor, move with them.

8.6.7 Noise and the Sampled Readout

A flat current-noise budget cannot be written for this channel. One converter LSB is 81.2 nA at $N = 0$ and 0.828 nA at $N = 319$, so the 2 nA rms budget of Sections 8.4 and 8.5 is 0.025 LSB at one end of the ladder and 2.4 LSB at the other. The budget is therefore stated against the code in use: the rms of the readout current, after averaging every converter sample taken inside one staircase tread, must stay below one LSB at that code. A tread is 2.00 ms and holds 1738 samples at the converter's 869 kHz sample rate.

Measured that way, the channel passes at $N = 0$ (0.036 LSB) and $N = 63$ (0.423 LSB) and misses at $N = 191$ (1.13 LSB) and $N = 319$ (1.78 LSB). Figure 59 shows why: the averaged floor is 1.5 nA to 1.9 nA at every code above the bottom one and 2.9 nA at $N = 0$, while one LSB falls 98× across the ladder. What survives a 2 ms average is the low-frequency input-referred noise of A2, which is a property of the amplifier and not of R_f , so over this band the top gain codes are noise-limited and not quantisation-limited. Section 8.9 reaches the opposite verdict for the same codes, integrating only the amperometric decade where that amplifier noise falls an order below the current quantum: which term binds is set by the measurement bandwidth, not by the gain code, so neither figure carries meaning without its band. The compensation capacitor of Section 8.6.6 does not change any of this: it divides output noise and transimpedance by the same factor, and the input-referred figures are unchanged by it to four digits.

Measured at the converter input rather than at the transimpedance output, the buffer of Section 8.6.4 adds 4.0 % at $N = 0$ and nothing measurable above it. At minimum gain the $19.6 \text{ k}\Omega$ transimpedance makes the buffer's own voltage noise a visible fraction once referred to the input; from $N = 63$ up the transimpedance swamps it. The 4 % lands on a tread-averaged 0.036 LSB, 35× inside the quantiser.

The electrode sets the floor at larger area. At $C_{dl} = 1 \mu\text{F}$ the tread-averaged noise is 11.8 nA to 13.7 nA and at $10 \mu\text{F}$ 48 nA to 76 nA; a $10 \mu\text{F}$ electrode cannot be read to one LSB in one tread at any gain code, and the tread must lengthen or the electrode shrink.

Averaging is not $\sqrt{N}$. Eighteen transient-noise records — three gain codes, six seeds, 20 ms each, strobed at the converter's sample period — put $\sqrt{N}$ wrong by up to 6.7× and in both directions: at high gain

the noise sits three decades below Nyquist and consecutive samples correlate at $\rho(1) = 0.988$, so averaging ten of them buys 2.5 % rather than $3.16\times$, while at $N = 0$ adjacent samples anticorrelate and the average beats $\sqrt{N}$. The exact Dirichlet-kernel integral of the measured spectrum reproduces the measurement to 0.4 % to 1.8 % and is the predictor used throughout this section.

No anti-alias filter is required. Replacing the sampler with an ideal brickwall at the 434.5 kHz Nyquist frequency multiplies the per-sample rms by 1.054 at $N = 0$ and by 1.001–1.003 at every other code. At most 10.5 % of the per-sample variance folds in from above Nyquist, and only at minimum gain, where it moves the tread-averaged figure from 0.036 LSB to 0.027 LSB. An analog anti-alias filter would buy nothing measurable; a post-transimpedance pole placed in the signal band is a different question, and the tread average already answers it.

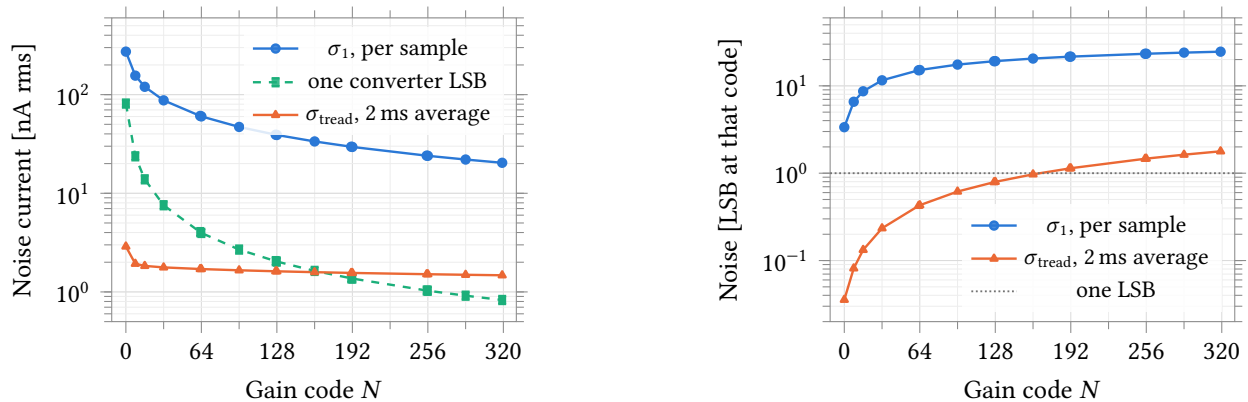


Figure 59: Input-referred current noise of the assembled channel against gain code, at the nominal electrode ($C_{\text{dl}} = 100 \text{ nF}$, $R_{\text{ct}} = 1 \text{ M}\Omega$), nominal process point, 37°C , integrated from 0.1 Hz to 10 MHz and referred by the measured in-loop transimpedance. σ_1 is one converter sample; σ_{tread} is the average of the 1738 samples taken inside one 2 ms staircase tread, by the exact Dirichlet integral of the same spectrum — *not* $\sigma_1/\sqrt{N}$, which the transient-noise measurement of Section 8.6.7 finds wrong by up to $6.7\times$. Left, in current: the averaged floor is 1.5 nA to 2.9 nA and barely moves with gain, while one LSB falls $98\times$, so the two cross at $N \approx 165$ and every code above it is noise- rather than quantisation-limited. Right, the same referred to the LSB at each code; a single sample is never better than 3.4 LSB, which is why the readout average is part of the specification and not an option. The feedback capacitor does not appear here: it divides output noise and transimpedance by the same factor and leaves the input-referred quantity unchanged to four digits.

8.6.8 Power and Corners

One channel draws 1.323 mW at the nominal process point, $528.70 \mu\text{A}$ from the 2.5 V analog rail and $1.04 \mu\text{A}$ from the 1.0 V converter core, against a 1 mW target. Over the four process corners the analog rail spans $432 \mu\text{A}$ to $914 \mu\text{A}$, so the fast corner misses the $400 \mu\text{A}$ rail budget by $2.3\times$. That range is a lower bound, and by a wide factor: the rail is dominated by bias-derived branches, the reference resistor that sets them is poly, and the corner set holds the passive model rows at typical (pinned-passive caveat, page 69). Freeing them widens the amplifier bench's own supply box by $4.2\times$.

Four amplifiers carry it: A1, A2 and the two multiplexer buffers, at $243.8 \mu\text{W}$ each and 86 % of the 1.135 mW that rolls up per block. The rest is the local bias generator at $59.3 \mu\text{W}$, the two ladders at $88.8 \mu\text{W}$ and the converter at $12.3 \mu\text{W}$. The rollup does not close against the 1.323 mW the rails deliver, because it does not follow the inherited supply ports; the rail figure is the authoritative one.

The channel is $319.6\ \mu\text{W}$ cheaper than the revision before it, and the two effects cancel almost exactly: deleting the two reference-DAC supply regulators returned $806.8\ \mu\text{W}$ that drove nothing, and the two buffers cost $487.1\ \mu\text{W}$. The regulators were dead load — both ladders draw $44.4113\ \mu\text{W}$ before and after, to every printed digit — so the disturbance fix of Section 8.6.4 was bought out of waste rather than out of budget.

All twenty dc-reachable specs — reference tracking, the potential range, the common mode, the zero-current offsets and the gain-code sweep — pass at all five corners. Only one moved: the converter-input voltage now sits $620\ \mu\text{V}$ lower at every corner, the input buffer's own offset, leaving $3.50\ \text{mV}$ of margin to its window where there was $4.12\ \text{mV}$. That is weaker evidence than it looks: the corner rows keep the passive model rows at typical (page 69), and for a block whose accuracy is set by matching they exercise almost nothing. Monte Carlo is the real test, and Section 8.6.9 spreads several of these same quantities by more than their budget.

8.6.9 Monte Carlo

200 samples of process and mismatch together, at the nominal process point and 37°C , on the tapeout cell itself: the statistical model section resolves 1264 random variables and all 200 samples converge. Table 69 is the result. The corner runs of Section 8.6.8 cannot show it: they hold the resistor and capacitor models typical, and every quantity here is set by matching.

Mismatch is the whole of it. A process-only run leaves 0.68% of σ on the zero-current output — 0.005% of the variance — and under 1.5% on every other quantity, and the two runs add in quadrature to the combined σ within 0.4% . Six further mismatch-only runs, each suppressing one matched structure, locate it: the transimpedance amplifier A2 alone sets the current axis — suppressing it collapses σ of the zero-current output to $0.034\ \text{mV}$ — and the control amplifier A1 alone sets the potential axis, $0.042\ \text{mV}$. The two R-2R ladders move only the common mode and the DAC gain; the 320-code feedback ladder and the local bias generator move no dc quantity by more than the sampling error of a 100-sample run. The feedback ladder cannot enter: the readout is $V_{\text{CM}} - IR_f$, so R_f is a gain a constant divides out while the two amplifiers are a raw additive input offset. That offset is what the flat bench of Section 8.5 measured mismatch-only at $5.72\ \text{mV}$; the assembled channel with process *and* mismatch reproduces it at $5.29\ \text{mV}$ to $5.49\ \text{mV}$, so assembly and process add nothing. The die-common term being that small, the four channels on a die are very nearly independent. The attribution and ablation runs behind this paragraph were taken on the previous revision of the cell; the re-baseline moves both amplifier-set spreads by under 5% and contradicts none of it.

The converter input is the one node with a wider spread than the transimpedance output: $\sigma = 8.040\ \text{mV}$ against $5.488\ \text{mV}$, because the input buffer contributes its own offset, $\sigma = 5.325\ \text{mV}$, uncorrelated with A2's at $r = 0.055$. It does not reach a calibrated reading. Both the signal and the zero are read through that one buffer, so its offset subtracts out and only its variation with input level survives: $0.62\ \text{mV}$ mean and $2.07\ \text{mV}$ worst of 100 samples over the full readout swing, $0.031\ \text{mV}$ and $0.11\ \text{mV}$ over $\pm 150\ \text{mV}$ (Section 8.6.4).

The zero-current output is gain-independent *per sample*. Its σ agrees to five significant figures at $N = 0, 63$ and 319 and its mean to $4\ \mu\text{V}$, and within one sample the three taps agree to at worst $5.4\ \mu\text{V}$. One stored constant therefore calibrates all 320 gain codes on a real channel, not only at the typical point where Section 8.6.2 measured it — the premise the whole current-axis calibration rests on.

Calibration. Every spread above is a per-channel additive constant, so the two constants of Section 8.11 remove it, each measured through the channel's own converter and so carrying that path's $0.65\ \text{mV}$ rms floor. The reference-electrode floor of Table 69 is that same offset seen from the other side: the electrode cannot be driven below ground, so the floor is A1's offset rectified at zero commanded potential.

The tracking specification is the working window. The 2 mV reference-tracking limit is specified over the 0.4 V to 2.1 V commanded range, not over the full 0 V to 2.42 V code sweep. Inside that window the tracking error is a pure constant per channel: after the mid-code value is removed the residual is 0.36 mV at 0.63 V and falls further toward mid-code (Figure 60). Outside it A1's input pair is running out of common-mode range, the error stops being a constant, and the same residual is 3.4 mV at 0.31 V — which no stored constant reaches. On the same samples and the same calibration the worst tracking error averages 0.885 mV over the window and 5.78 mV over the full sweep. The difference is a choice of range, not a circuit difference, and the electrochemistry does not use the outermost 0.4 V at the bottom or 0.32 V at the top.

Stability is the one axis that needs nothing. A separate 200-sample run at the worst electrode-by-gain point of Section 8.6.6 gives 74.11° of transimpedance phase margin, $\sigma = 3.07^\circ$ and a worst sample of 64.35° — 9.5 standard deviations above the 45° limit. Gain margins are 21.57 ± 0.88 dB and 38.17 ± 0.64 dB. The compensation capacitor is not a marginal fix that mismatch undoes. That run is the previous revision's and has not been repeated; at the same grid point the nominal phase margin did not move and the nominal gain margin fell 0.5 dB.

Quantity	Mean	σ	Specification
Zero-current output, $v_{\text{out}} - V_{\text{CM}}$	−1.004 mV	5.488 mV	± 2 mV
Reference tracking error, mid-code	−0.637 mV	5.287 mV	± 2 mV
Reference tracking error, worst over 0.4 V to 2.1 V	4.766 mV	3.494 mV	< 2 mV
Reference tracking error, worst over 0 V to 2.42 V	9.534 mV	5.939 mV	< 2 mV
Zero-current offset referred to current, $N = 0$	−55.8 nA	304.9 nA	± 100 nA
Same at maximum gain, $N = 319$	2.308 nA	1.729 nA	< 2 nA
Reference-electrode floor	4.100 mV	5.809 mV	< 5 mV
Reference-electrode ceiling	2.4205 V	9.97 mV	> 2.4 V
Common-mode output	1.249 75 V	0.769 mV	1.245 V to 1.255 V
Converter-input voltage	1.248 15 V	8.040 mV	1.245 V to 1.255 V
Reference DAC step	610.22 μ V	0.184 μ V	—
Transimpedance phase margin	74.11°	3.07°	$> 45^\circ$
Control-loop phase margin	79.94°	1.06°	$> 45^\circ$

Table 69: Monte Carlo distributions on the assembled channel: 200 samples, process and mismatch together, nominal process point, 37°C , 2.5 V supply, 1264 random variables, all 200 converged. The specification column is the budget each quantity is designed against, uncalibrated; the two constants that remove the additive part of these spreads are in Section 8.11. The two tracking rows are the same measurement over two ranges — the 0.4 V to 2.1 V row is the specification (Section 8.6.1), the full-sweep row is reported because the bench writes it that way. The converter-input row is the widest here and the least consequential: it carries the input buffer's own offset as well as A2's, and both cancel in a zero-referenced reading (Section 8.6.4). Phase margins come from a separate 200-sample run at the worst electrode-by-gain point of Section 8.6.6, taken on the previous revision of the cell and not repeated; the nominal phase margin at that point is unchanged.

8.6.10 Not Yet Verified

- **Devices outside the statistics.** Of the 742 mismatch-capable device instances in the executed netlist, 587 carry mismatch, four have it explicitly switched off and 151 sit on raw model cards. No matched structure is excluded: the four are bias-rail decoupling capacitors, out by intent because they decouple rather than match, and the 151 are the converter's sample switch and level shifters,

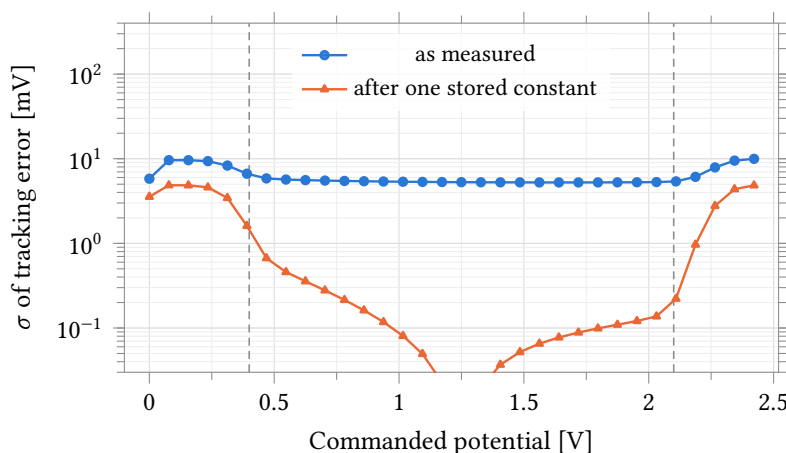


Figure 60: Spread of $V_{RE} - V_{PATTERN}$ against the commanded potential on the assembled channel, 200 samples, process and mismatch, nominal process point, 37 °C: as measured, and after the same sample's mid-code value is subtracted — the residual one stored per-channel constant leaves behind. That residual is zero at mid-code by construction and falls off the bottom of the axis there. Dashed lines mark the specified 0.4 V to 2.1 V window; interior to it the error is a pure constant to 0.36 mV, while outside it A1's input pair is leaving its common-mode range and no constant reaches the remainder.

the multiplexer pass gates, the bias generator's start-up branch and the digital cells of the successive-approximation logic. Two structures were admitted after Table 69 was measured, and neither moves it: capacitor mismatch is invisible to a dc campaign by construction, and the divider that sets the converter's threshold is isolated from the readout by the buffer. That divider is worth its own note — with its matching disabled it reported a threshold of exactly 1.250 000 V in all 200 samples, because process cancels in a ratio of identical resistors and a matched pair switched off reads as an ideal reference rather than a disabled one. Enabled it carries 0.968 mV, 0.607 of a converter code — dormant while the array is unpopulated, and a converter offset as soon as a code is scored.

- **The converter at schematic level.** Its capacitor array exists only in the extracted view (Section 8.6.4), so the code the schematic tests report is stuck and is deliberately left unspecified.
- **Current read as a code.** No run has taken a real electrode current through the on-chip converter. The voltammogram of Section 8.6.3 reads current from an in-line probe and from the measured transimpedance; the converter figures of Section 8.6.4 come from known voltages applied at the multiplexer. The two have not been closed into one measurement.
- **Layout parasitics on the compensation node.** The 10 853 μm^2 of capacitor added by Section 8.6.6 sits on the transimpedance summing node, the one node in the channel where stray capacitance costs phase margin. No stability or noise figure here carries its wiring parasitics: the stage has no extracted view.
- **Wiring.** All sixteen test-port multiplexer inputs are undriven, so the debug path selects between floating nodes on demand — and its output now carries a buffer, which spends 243 μW of Section 8.6.8 driving them. Either the inputs get something to look at or the buffer goes; as built it is the second-largest consumer in the channel. Eleven of the converter multiplexer's sixteen slots are undriven the same way, and one of them is the level-1 sibling of the commanded-potential output, so it injects its accumulated charge into that output on every toggle of the lowest select bit (Section 8.6.5). The fix is one wire per input, to the common mode rather than to local ground.

8.6.11 Measuring this block on chip

The channel's own converter multiplexer carries this sequence; only the last step needs the tester.

1. Invert bit 9 of the raw converter bus before anything else reads it. Every step below is that number.
2. Select the transimpedance output (slot 0) with the working electrode isolated and take the current zero at a *high-gain* code. The zero is gain-independent to 0.3 nV over all 320 taps, so one measurement calibrates every code — but at the bottom code one LSB is 88.5 nA against 34 nA of offset and the measurement is quantisation noise.
3. Take the common-mode reference for that zero through slot 1. Sampling it moves the common mode by 0.83 mV and the transimpedance output by 1.1 LSB, both recovered within 0.54 μ s — well inside one conversion, so consecutive readings do not interact.
4. Read the commanded potential at slot 2 to separate a DAC fault from an amplifier fault. A1's own contribution is not observable: slot 3 is reserved for a buffered reference-electrode tap that does not exist, and the raw electrode must not be sampled.
5. Absolute potential scale and transimpedance gain still need the production tester at the reference-DAC pad and the working-electrode pad (Sections 8.10 and 8.9). The in-loop feedback resistance of Section 8.6.3 has no on-die counterpart — it is read from a simulation probe.

8.7 Impedance Spectroscopy

The channel measures impedance with no impedance hardware. A small excitation rides on the reference-DAC setpoint of Section 8.10, A1 applies it at the reference electrode, A2 and the converter read the resulting current through the same programmable feedback resistor a dc measurement uses (Section 8.9), and firmware demodulates the potential and current records in quadrature against its own reference. The excitation is a toggle between two DAC codes, so the mode adds a clock divider and an accumulate to the channel of Section 8.6 and nothing else.

Everything below is schematic-level on the assembled channel's own netlist at a 2.5 V analog supply. At the nominal process point and 37 °C: 30 small-signal runs, five electrode loads against six gain codes over 0.1 Hz–100 kHz; 24 runs against pure resistors chosen to hold the ratio $R_f/|Z|$ fixed across the sweep; and 17 transient runs in which the twelve potential-code lines are driven individually, so the excitation is produced by the real ladder rather than by an ideal source. Across process and temperature: the first two of those repeated over the chapter's four process corners at 25 °C and over 0 °C to 85 °C on the nominal models, 252 runs. Across a wafer: eight Monte Carlo runs of the small-signal measurement at the nominal corner, six of 200 samples with process and mismatch together and two of 100 that separate them. The demodulator is the firmware one — an I/Q multiply-accumulate over an integer number of excitation periods, no window, no detrending. The electrodes are the linear cell model of Section 8.1 with $R_u = 1$ k Ω throughout; *fresh*, *typical* and *fouled* name $R_{ct} \parallel C_{dl}$ of 1 M $\Omega \parallel 100$ nF, 100 k $\Omega \parallel 1$ μ F and 10 k $\Omega \parallel 10$ μ F.

The pinned-passive caveat of page 69 governs every corner number in this section. Neither the ladder that is R_f nor the capacitor that sets the feedback pole corners here. Cornering the poly as well moves R_f by –33.4 % and +35.5 % and the feedback pole at the top gain code to 2.40 kHz to 6.61 kHz, against the ± 1.2 % the corner set alone reports. The Monte Carlo does vary the poly, and measures the same quantity as $\sigma = 11.5$ % (Section 8.7.1).

8.7.1 Accuracy

A known 11.000 k Ω resistor reads 11 003 Ω at 0.000° – +0.029 % – flat from 0.1 Hz to 1 kHz at the bottom gain code, and +0.030 % at 5 kHz. On an electrode the error is set by the noise gain $1 + R_f/|Z|$ at the frequency being measured, not by the electrode's chemistry (Figure 61): the fresh case at gain code 63 is the most accurate of the three below 100 Hz at 0.005 % and the worst above 1 kHz – 2.8 % at 5 kHz, and

1.84 % to 3.68 % across the corner set — because its impedance falls from 999 k Ω to R_u across the sweep while R_f does not move. The fouled case, at the bottom gain code throughout, stays within 0.44 % and 1.02° from 0.1 Hz to 20 kHz.

What limits accuracy, in falling order. First the uncalibrated feedback capacitor: on the fresh electrode at gain code 63 and 5 kHz a stored dc feedback resistance costs 3.8 % of magnitude and 15° of phase on top of the loop's own 2.8 %, because it ignores the pole the 22 pF compensation capacitor puts in the feedback impedance. Measuring a known impedance at every frequency and taking the ratio removes that term entirely — 2.33 % at the same point, below the loop error itself, the two errors partly cancelling. Second, a magnitude mismatch between that calibration impedance and the load: a 999 k Ω load calibrated against 11 k Ω at the same gain code carries a fixed -0.27 %, exactly the difference of the two loop errors (0.278 against 0.003 %). At a fixed process point that is a correction; across a wafer it is a spread, and the requirement below. Third, the loop error itself, which scales exactly with the noise gain: 0.120, 0.330, 0.660 and 1.322 % at $R_f/|Z| = 7.3, 20, 40$ and 80. Last, the reference-electrode tap, worth 0.02 % to 0.2 % of magnitude — the difference between reading the applied potential at the electrode and assuming the commanded one arrives. On this axis it is not the limiting term; what earns it its multiplexer input is the excitation, where the same assumption costs up to 3.5 % (Section 8.7.3).

Across corners the error spans a factor of two. It is monotone in the amplifiers' transconductance, so 85 °C is the worst point of the set for every load and every gain code and the slow corner the best: the calibration resistor reads 0.023 % to 0.041 % about its nominal 0.029 %, and the typical electrode at gain code 20 and 1 kHz spans 0.70 % to 1.40 % over the process corners and 0.88 % to 1.46 % over 0 °C to 85 °C. Temperature is the larger axis, and the two do not add, because the fast corner and 85 °C lie at the same end. Phase moves far less: the fresh case at gain code 63 and 5 kHz spans 4.57° to 4.79° where its magnitude error doubles, because phase is set by pole positions and this corner set does not move them — for the same reason it does not move R_f , which is the caveat of Section 8.7.

Across a wafer the spread is the feedback resistance, and it is process rather than mismatch. At the bottom gain code, 200 samples give $\sigma = 11.45$ % with process variation alone against 0.41 % with mismatch alone; at gain code 63 the same quantity is 11.93 %, and the fresh electrode's 2.8 % at 5 kHz becomes 3.06 ± 0.92 %. The poly sheet resistance moves all four channels of a die together, so no amount of averaging removes it and a single measurement does; what a single measurement cannot remove, the channel-to-channel residual, is that 0.41 %. That orders the ways firmware can turn two converter readings into an impedance. **A stored nominal resistance law is unusable as a calibration constant:** it puts the whole spread on the answer — $\sigma = 10.2$ % of magnitude at gain code 0 and 12.1 % at code 63, individual dies reading -32 % to $+39$ % below the feedback pole — on top of the 9 % nominal offset the law already carries at the bottom code. **One dc measurement of the feedback resistance per chip removes essentially all of it:** $\sigma = 0.0006$ %, four orders of magnitude better and indistinguishable from measuring the current at every frequency, which is the bound and would need a current reference the die does not carry. That is a fourth independent reason for the calibration impedance. Phase escapes the argument entirely — the feedback resistance is a real scale factor, so the stored-law and dc-calibrated architectures report identical phase in every case measured, and a chip with no gain calibration at all still reports phase to the accuracy above.

The calibration impedance must be switchable across decades. Against a standard matched to the load, the ratiometric measurement reaches the bound above in mean and in σ at every frequency, and beats it at dc where the two loop errors are identical and cancel outright: $\sigma = 0.0002$ % against the bound's

own 0.0006 % on the typical load and 0.0053 % on the fouled one. Against a standard 9.2 times away in magnitude it instead *injects* spread, $\sigma = 0.18\%$, 300 times worse than the dc calibration it replaces. Above the feedback pole, where a dc calibration cannot help at all, it is the only thing that does: on the fresh electrode at gain code 63 and 5 kHz a stored dc resistance gives $7.03 \pm 1.84\%$ and the ratio $2.56 \pm 0.80\%$ even against a standard 91 times away — 2.7 times less bias and 2.3 times less spread. The fixed -0.27% above is therefore a distribution, $-0.27 \pm 0.10\%$, and matching the calibration impedance to the decade being measured is a requirement on the hardware, not a refinement of the firmware.

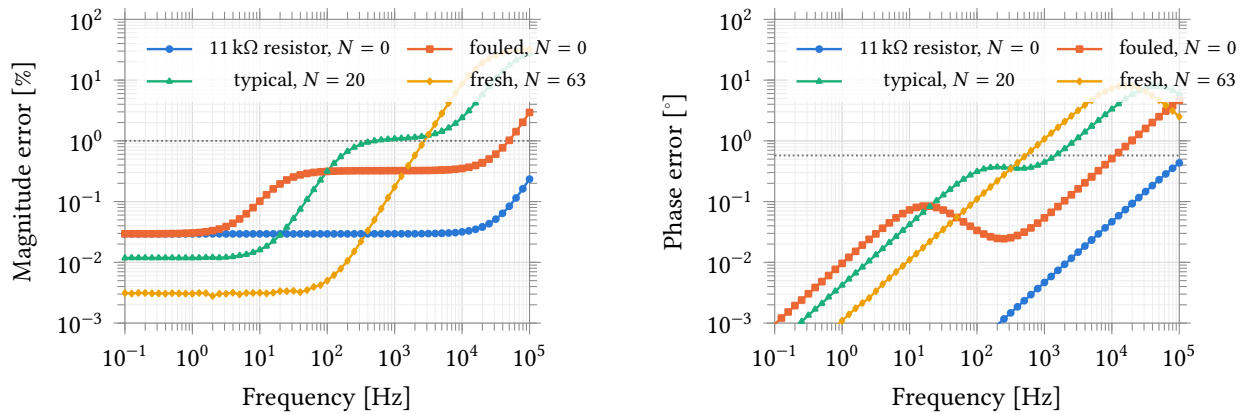


Figure 61: Measured impedance error against frequency — magnitude (left) and phase (right) — for the calibration resistor and the three electrode cases, each at the gain code that ranges it, with the applied potential read at the reference electrode and the current measured. Error is taken against the analytic impedance of the load, so these are the chain’s errors and not the model’s; dotted lines are the 1 % and 0.573° gates. Each curve turns up where the electrode’s impedance has collapsed toward $R_u = 1\text{ k}\Omega$ and the noise gain $1 + R_f/|Z|$ has therefore risen, which is why the highest-gain case is the most accurate below 100 Hz and the first to degrade above 1 kHz. The gate crossings here are not the bandwidth specification: the ratio $R_f/|Z|$ moves with frequency on a real electrode, and the specification comes from the fixed-ratio measurement of Section 8.7.2.

8.7.2 Coverage and Bandwidth

At the measured excitation the channel covers $236\ \Omega$ to $14.79\text{ M}\Omega$, 4.80 decades (Table 70). Each gain code spans $639\times$, 2.81 decades, between the impedance at which the excitation’s peak current fills the $\pm 0.80\text{ V}$ compliance and the impedance at which the fundamental of the output falls to one converter code. Adjacent codes overlap by more than a decade, so **there are no impedance seams**: the 320-tap ladder tiles that span as a continuum, and the only seams in the sweep are the ladder’s own carry codes (Section 8.9), which move a window edge by far less than the overlap. The frequency and impedance limits do not collide either — a capacitive interface’s impedance falls as frequency rises, so a sweep walks down the table as it walks up in frequency, and the high-gain codes whose feedback pole sits between 3.7 and 11 kHz are only selected at the low-frequency end. That pole is measured here and confirms the 22 pF compensation capacitor to better than 1 %: 52.0 kHz against 52.2 kHz predicted at gain code 20, 3.75 kHz against 3.76 kHz at code 319.

The capacitor is confirmed; the pole is not a constant. It moves by $\pm 1.2\%$ over the corner set, which holds the poly at typical, and by 2.40 kHz to 6.61 kHz at code 319 once the poly is cornered too — because the pole is $1/2\pi R_f C_F$ and R_f is an 11 % process quantity (Section 8.7.1). The window itself translates with R_f for the same reason: it holds to $232\ \Omega$ to $243\ \Omega$ and $14.58\text{ M}\Omega$ to $14.98\text{ M}\Omega$ over the corner set, 4.79 to 4.81 decades everywhere, but a poly-cornered die runs from $158\ \Omega$ to $20.0\text{ M}\Omega$. Every

tap is the same poly, so the span is unchanged and one dc calibration recovers the absolute edges.

The band edge is a phase specification, not a magnitude one. Presenting each gain code with a pure resistor of exactly $R_f/40$ freezes the noise gain across the sweep and measures the edge directly. The dc error then scales exactly with that ratio — 0.120, 0.330, 0.660, 1.322 % at 7.3, 20, 40, 80 — so the form of the design equation holds, and it holds across corners too, at 0.52 % to 0.92 % about the nominal 0.660 % at ratio 40, with only the amplifier’s open-loop gain moving. Its bandwidth prediction does not hold: the ± 1 % magnitude edge measures 9.1 kHz to 22.7 kHz across the gain codes against 5.56 kHz predicted, 1.6 to 4 $\times$ pessimistic, while the $\pm 0.573^\circ$ phase edge measures 5.53 kHz to 6.93 kHz — the predicted value. The same model under-predicts the error itself by 42 % where the noise gain is near unity (0.029 measured against 0.021 %), which at that size changes nothing.

The band holds across corners, and the phase gate stays the binding one. Repeating the fixed-ratio measurement over the seven corner points moves the phase gate by -4.3 % at worst (Table 71): 5.29 kHz at the slow corner against 5.53 kHz nominal, and 5.29 kHz to 7.53 kHz over every corner and gain code. The magnitude edge spans 8.4 kHz to 27.1 kHz against 9.1 kHz to 22.7 kHz nominal, its minimum at 85 °C and gain code 102, still 1.6 $\times$ above the phase gate at that worst point. **The usable band is therefore 0.1 Hz to 5.3 kHz set by phase**, with magnitude in specification to at least 8.4 kHz — a conclusion the corners make firmer, since the two gates are worst at opposite ends of the set.

N	R_f measured	vs law	$ Z _{\min}$	$ Z _{\max}$	Feedback pole
0	19.63 k Ω	+9.06 %	236 Ω	150.9 k Ω	> 100 kHz ^a
20	138.7 k Ω	+0.52 %	1.67 k Ω	1.066 M Ω	52.0 kHz
63	394.8 k Ω	-0.29 %	4.75 k Ω	3.035 M Ω	18.2 kHz
102	628.4 k Ω	-0.25 %	7.56 k Ω	4.830 M Ω	11.5 kHz
191	1.160 M Ω	-0.38 %	13.9 k Ω	8.912 M Ω	6.24 kHz
319	1.924 M Ω	-0.40 %	23.1 k Ω	14.79 M Ω	3.75 kHz

^a Outside the 100 kHz sweep; the element values put it at 368 kHz.

Table 70: Impedance window and measured feedback pole against gain code, at the delivered 19.24 mV excitation. R_f is the transimpedance measured in the closed loop at 0.1 Hz, compared with the ladder’s nominal law. The window runs from the impedance at which the excitation’s peak current fills the ± 0.80 V output compliance to the impedance at which the fundamental of the output falls to one converter code; the peak sets the lower edge because the peak is what rails, and the fundamental sets the upper edge because the fundamental is what the demodulator reads. Adjacent windows overlap by more than a decade. The last column is the -3 dB frequency of the measured transimpedance, i.e. the real 22 pF feedback pole. Every column but the first scales with R_f , and R_f is neither a constant nor a corner-free quantity: the +9.06 % at $N = 0$ is +7.1 % to +12.1 % across the process corners and carries $\sigma = 11.0$ % across dies (Section 8.7.1), so this table is one die and the window it gives is recovered per chip by one dc calibration.

8.7.3 Excitation and Averaging

The excitation is delivered as commanded only when its two codes lie inside one octave. A 32-code toggle asks for 19.531 mV peak to peak; measured at the DAC output it arrives 0.03 % to 0.06 % high when neither code crosses a power-of-two boundary and 0.15 % to 1.51 % low when the pair straddles one (Table 72). **The whole amplitude error is the carry**: a pair that flips one or two bits delivers what it was told to within

N	$\pm 1\%$ magnitude			$\pm 0.573^\circ$ phase		
	Nominal	Worst	at	Nominal	Worst	at
0	22.7 kHz	10.9 kHz	85 °C	5.53 kHz	5.29 kHz	ss
20	14.1 kHz	12.8 kHz	85 °C	5.76 kHz	5.58 kHz	ss
63	9.85 kHz	9.09 kHz	85 °C	6.93 kHz	6.47 kHz	ss
102	9.10 kHz	8.37 kHz	85 °C	> 100 kHz	> 100 kHz	—
191	13.9 kHz	11.9 kHz	85 °C	> 100 kHz	> 100 kHz	—
319	> 100 kHz	> 100 kHz	—	> 100 kHz	> 100 kHz	—

Table 71: Band edges per gain code at the nominal point and at the worst of seven corner points — the four process corners at 25 °C, plus 0 °C and 85 °C on the nominal models — with each gain code presented a pure resistor of exactly $R_f/40$ so that the noise gain is frozen across the sweep. The magnitude gate is where the error crosses $\pm 1\%$ and the phase gate where it crosses $\pm 0.573^\circ$; a dash means the error never reaches that gate inside the 100 kHz sweep. The two gates move in opposite directions, so the worst magnitude edge and the worst phase edge are never the same die. The frozen ratio makes these edges insensitive to a global shift in R_f by construction, which is why they survive the poly caveat of Section 8.7 and the impedance window of Table 70 does not.

0.06 %, a pair that flips eight bits is out by 1.5 %, and at the 4 mV excitation a carry at the most significant bit costs -3.5% . It is frequency-flat — the same pair delivers the same amplitude at 100 Hz, 1 kHz and 10 kHz in every printed digit — so it is a static property of the code pair, removed either by choosing the pair or by calibrating it once.

Two levels beat a synthesised sine. For the same code excursion the two-level square delivers $1.28\times$ the fundamental of a stepped sine, against $4/\pi = 1.273$ predicted, and the measured impedance is identical for both to five significant figures — $11\,003.45\,\Omega$ against $11\,003.52\,\Omega$ at 8, 16 and 32 steps — because the demodulator is ratiometric and the excitation’s shape cancels. The sine is also the less accurate excitation, -2.4% of amplitude against the square’s -1.5% , since each of its steps visits a different code carrying its own linearity error. The square costs less firmware and delivers 2.2 dB more signal.

Averaging buys $\sqrt{N}$, and then stops at a bias. Process gain is exactly $\sqrt{N}$ over four decades of record length: at the bottom gain code an $11\,\text{k}\Omega$ load reads $\sigma/|Z| = 1.20\%$ from a single 1 ms cycle and 0.0093% from 16 384 cycles, every factor of four in record length being 6.02 dB to the last digit. Underneath it sits a deterministic quantisation bias that averaging does not reduce at all: $+1.94\%$ at that bottom code, where the output fundamental is 13.7 converter codes, and -0.56% at the code that ranges the same load, where it is 275. The current channel’s quantisation is dithered by 3.4 codes of front-end noise and averages away; the potential channel carries no front-end noise, so its quantisation is periodic with the excitation and lands in the phasor as a bias. It cancels exactly in a ratio against a known impedance — a third independent reason for the calibration impedance, after the feedback pole of Section 8.7.1 and the code-pair error above.

8.7.4 Rules for a Record

A calibration impedance is required, for the four independent reasons above, and it must be switchable across decades. These six rules are what a record needs beyond it, and each carries the measured cost of ignoring it.

<i>Pair straddles a carry</i>				<i>Both codes inside one octave</i>			
Code	Pair	Delivered	Error	Code	Pair	Delivered	Error
256	240/272	19.503 mV	−0.147 %	272	256/288	19.537 mV	+0.031 %
512	496/528	19.461 mV	−0.360 %	528	512/544	19.537 mV	+0.031 %
1024	1008/1040	19.378 mV	−0.783 %	1040	1024/1056	19.537 mV	+0.028 %
2048	2032/2064	19.236 mV	−1.511 %	2064	2048/2080	19.538 mV	+0.035 %
3072	3056/3088	19.407 mV	−0.637 %	3088	3072/3104	19.541 mV	+0.047 %
3584	3568/3600	19.481 mV	−0.257 %	3600	3584/3616	19.543 mV	+0.058 %

Table 72: Delivered excitation against commanded, measured at the reference DAC output with the potential bus driven bit by bit. Both halves of the table ask for the same 19.531 mV peak to peak — a 32-code toggle at 610.35 μ V per code — and differ only in whether the two codes straddle a power-of-two boundary. The error is a static property of the code pair: it does not change with excitation frequency, and it is the reason a code pair is chosen rather than an amplitude.

1. **Choose an excitation code pair that does not cross a major carry.** Both codes inside one octave delivers the commanded amplitude to 0.06 %; a pair spanning eight bits is out by 1.5 %, and at the 4 mV excitation a carry at the most significant bit costs 3.5 % (Table 72).
2. **Use the two-level excitation, not a synthesised sine.** The measured impedance is the same to five significant figures and the square delivers 1.28 $\times$ the fundamental for the same interface excursion.
3. **Start the record only after the electrode’s own dc transient has decayed.** A load with a 0.99 ms step time constant, recorded from 1 ms after the excitation starts, reads +16 % at 1 kHz and +74 % at 10 kHz. Nothing in the record shows it: the two-level source steps the mean by half the excitation when the programme starts, and the per-cycle phasors agree to 0.02 % while the answer is wrong by tens of percent.
4. **Sample on a grid commensurate with the excitation,** by deriving the converter strobe and the excitation from the same divider. A grid 0.35 samples per period out of step costs 0.6 % on a reactive load — and reads correctly on a resistor, so a resistive self-check does not catch it.
5. **Average with a boxcar rather than by discarding samples.** On a reactive load the n -th current harmonic is $Z(f)/(nZ(nf))$ of the fundamental rather than $1/n$, so the high orders fold into the fundamental bin: plain decimation to 135 samples per period is 1.3 $\times$ worse than a boxcar and not monotonic in rate, and a record kept at 256 samples per period reads 5 % high where the small-signal analysis says 0.03 %.
6. **Measure the feedback resistance on every chip; never compute it.** The law $18\text{ k}\Omega + 6\text{ k}\Omega N$ is already 9.1 % low at $N = 0$ at the nominal point — the closed-switch stack of Section 8.9, and within 0.5 % at every code above $N = 20$ (Table 70) — and it carries $\sigma = 11\%$ of process spread on top of that, so storing it costs $\sigma = 10.2\%$ of measured magnitude. One dc measurement per chip takes that to 0.0006 % (Section 8.7.1). Gain constants are per die, not per design.

8.7.5 Not Yet Verified

- **The runnable bench is not the whole channel.** Its feedback element is an ideal resistor set to the measured ladder value, which the campaign shows is worth 0.019 % in the loop, and it carries no feedback capacitor. It is therefore faithful below the feedback pole and optimistic above a few kilohertz at the high-gain codes: at gain code 63 it reports 2.73 % where the complete channel mea-

tures 2.84 % at 1 kHz, and 3.17 % against 5.18 % at 5 kHz. It passes all 35 specs over the seven corner points, but its ideal feedback element moves 0.01 % across them where the real ladder moves 4.6 %, so it bounds the amplifiers and not the channel.

- **Current read as a code, demodulated.** The extracted converter does convert a live impedance record — three conversions decoded while the transimpedance output moves 317 mV peak to peak — so the readout is alive, but no demodulated result exists at code level. Two obstacles, one structural: the converter multiplexer has no reference-electrode input (Section 8.6.4), so the potential channel can only be the commanded potential; and a coherent record of four excitation cycles at 64 conversions per cycle costs about 7.5 hours of simulation per multiplexer slot.
- **The reference-modulation term is unmodelled, not absent.** The supply regulator whose output moved 25.8 mV with the code word (Section 8.10.1) has been deleted from the channel, and in this deck the ladder takes its reference from an ideal 2.5 V rail; what the excitation table measures is the ladder's own linearity between two codes and nothing else. On silicon the term returns in proportion to the impedance between the regulator and the ladder, which is a fifth reason the calibration impedance is not optional.
- **What the corner and Monte Carlo runs still do not cover.** Three things. The corner table is the pinned one of page 69, so R_f does not corner and every corner figure here is a lower bound: with the poly cornered too, R_f moves -33.4% and $+35.5\%$, the feedback pole at the top gain code becomes 2.40 kHz to 6.61 kHz, and the error above that pole moves by $6\times$ — the fresh electrode at gain code 63 and 10 kHz reads 3.19 %, 8.64 % and 19.97 % at the fast, nominal and slow poly. Below the pole nothing moves, because a global shift in R_f moves the noise gain and the transimpedance together. The statistical model section likewise holds the feedback capacitor at typical, so no run in this section spreads the 22 pF. And only the small-signal measurement was repeated: the delivered excitation, the demodulator and the code-level readout remain nominal-point results.
- **The electrode carries no temperature coefficient.** The cell model is a linear $R \parallel C$, so the 0°C to 85°C runs move the amplifiers and not the interface. On silicon the charge-transfer resistance is strongly Arrhenius, which moves the load; since the error is a function of the noise gain and not of the chemistry that is a gain-ranging problem rather than an accuracy one — but that is an argument here, not a measurement.

8.8 SAR Analog-to-Digital Converter

`anatop_pixel_adc` is a 10-bit single-ended charge-redistribution SAR converter (Figure 62). One instance per channel digitises the transimpedance output of Section 8.5: $v_{\text{out}} = V_{\text{CM}} + i_{\text{we}} \cdot R_f$ enters the converter and leaves as a 10-bit code, offset-binary about the code that corresponds to V_{CM} . The capacitor array is both the sampling network and the feedback DAC; its layout is Figure 63.

Conversion is externally clocked from a 20 MHz trigger clock, and the macro advances one bit trial per falling edge of that clock. The bench steps the input from 0 V to 2.5 V in 0.25 V steps and reads the code at the settled point of the first completed conversion, where the converter's ready output is high at every step.

The delivered stimulus converts four times too slowly. The clock pattern the converter peripheral emits carries three pulse groups per 1150 ns — clear, sample, convert — but holds the convert phase statically high, so it presents a single falling edge where the ten bit trials need ten. The ten trials therefore consume the falling edges of the following three groups, and ready asserts every 4.600 μs : 217.4 kSa/s, not the 869.6 kSa/s the 1150 ns group period implies. Replacing the held level with ten 20 MHz pulses in the same 500 ns window and changing nothing else closes each conversion in 1150 ns and returns *identical codes* at every input level. The defect is in the stimulus, not the converter: the fix is one commented-out

line of the peripheral's register-transfer source, and both protocols are reported separately throughout Section 8.8.2.

Coverage. All data in this section comes from a *post-extraction* netlist — Quantus, C-only coupled parasitic extraction of the ADC macro — and is, with the converter readout of Section 8.6.4 that splices this same macro into the channel, the only extracted data in this chapter; every other section is schematic-level. The distinction is not incidental: the capacitor array exists only in layout, so a schematic-level simulation of this block would not be a simulation of this converter. The transfer, timing and power figures are measured at five process corners (Table 74) and the nominal point is 37 °C, 2.5 V analog and 1.0 V digital supplies. A 100-sample process-and-mismatch Monte Carlo also exists, and its result is that this netlist cannot produce a matching distribution at all (Section 8.8.3); no gain or offset σ is quoted anywhere below. The 0.25 V sweep step is 157 codes. A second sweep of the unclipped window in 0.0625 V steps, 39 codes apart, carries the linearity figures, and the corner runs use a 20-point staircase at 0.1 V, 63 codes apart. No grid resolves individual codes, so nothing here is a DNL or INL measurement.

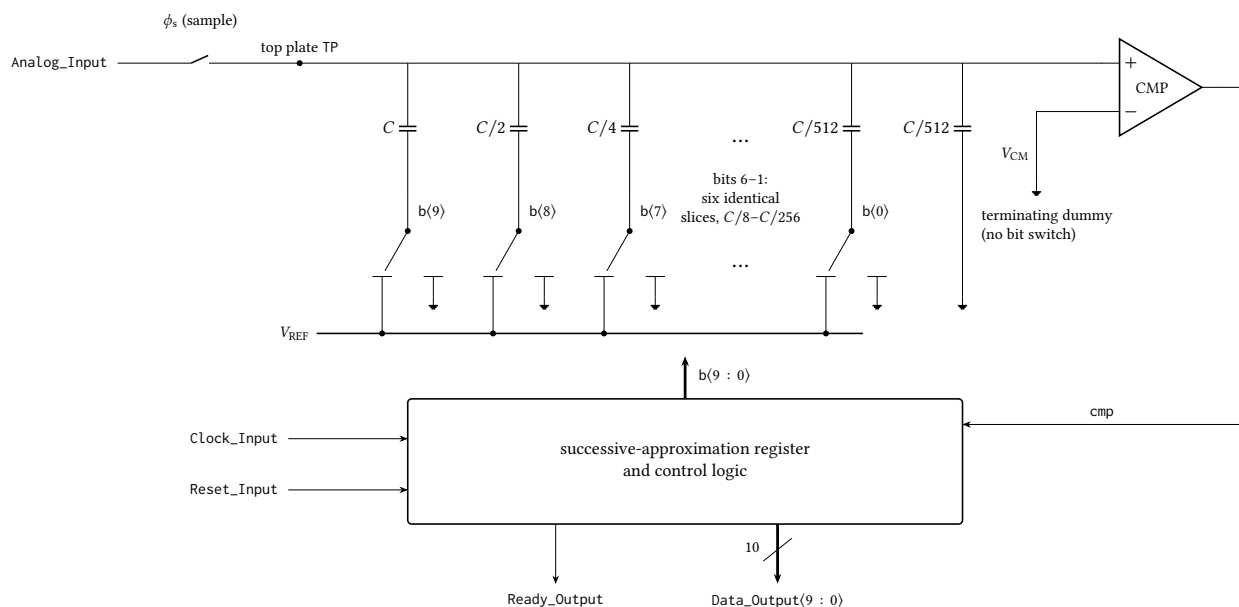


Figure 62: Generic single-ended charge-redistribution SAR ADC, drawn to explain the architecture of the 10-bit converter rather than to reproduce its implemented transistor-level schematic. The input is sampled through ϕ_s onto the top plate of a binary-weighted array; the logic then switches each bottom plate from V_{REF} to ground one bit at a time, most significant first, and keeps or discards the trial from the comparator decision, so the conversion takes ten comparator cycles and the top plate converges to V_{CM} . Bits 6–1 are elided: six further slices identical to those drawn sit between $C/4$ and $C/512$, and the terminating dummy $C/512$ carries no bit switch, which makes the total array $2C$ and the LSB weight $C/512$. Only the external port names are those of the implemented block; the sampling network, comparator topology, clock generation and reset behaviour are representative, and no device dimensions are stated.

8.8.1 Transfer Function

The converter is monotonic across the whole sweep, reading code 0 at 0 V and code 1023 at 2.5 V (Figure 64). Codes climb 1.535 times faster than ideal: a least-squares fit to the 24 unclipped points of a 0.0625 V sweep over 0.5 V to 2.0 V gives 628.64 codes per volt against the ideal 409.60, so the effective full-scale input span is

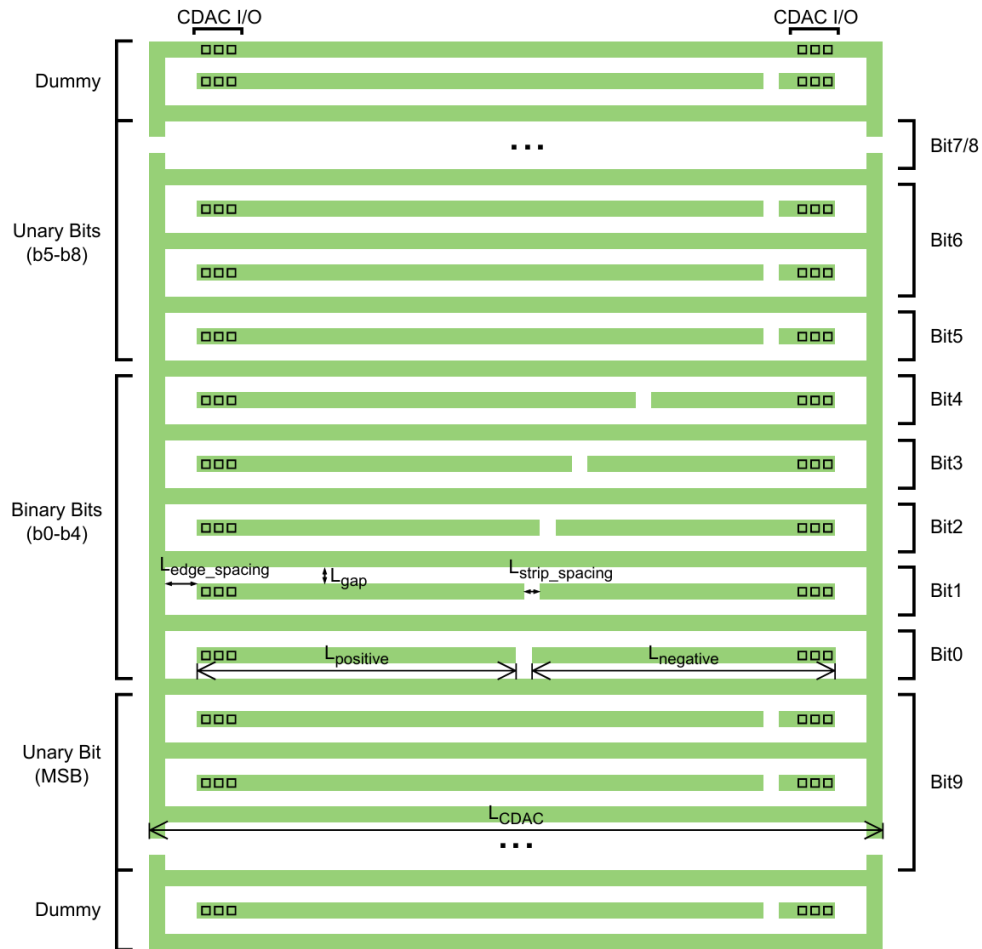


Figure 63: Floor plan of the 10-bit segmented capacitor array of anatop_pixel_adc. The five lower bits b0–b4 are binary-weighted within single strips, each strip split into a positive and a negative section whose boundary sets the weight; the upper bits b5–b8 and the MSB b9 are unary, built from repeated identical strips, and every row terminates in CDAC I/O at both ends. Unary segmentation of the large weights makes their ratio a count of identical unit strips rather than the dimension of one large device, so it is set by matching; the dummy rows at the top and bottom of the array give the outermost active strips the same local surroundings as interior ones. The array exists only in layout, which is why this section’s data is taken from an extracted netlist.

1.627 V rather than the 2.5 V supply span, and the effective LSB is 1.591 mV rather than 2.441 mV (Table 73). Extrapolating that fit puts code 0 at 0.441 V and code 1023 at 2.068 V, a window centred on 1.255 V — the 1.25 V channel common mode. The converter clips outside that window, losing 34.9 % of the intended input range, split between the two ends rather than concentrated at one: a full-scale transimpedance swing about V_{CM} is truncated symmetrically.

Mechanism. The 1.535 factor is a product of two, and the smaller of them is the parasitic. The array is a *differencing* capacitor DAC: each bit drives two plates in antiphase, so the charge it injects is proportional to the difference of the two, and it is that difference which is binary — to four figures over the 512:1 weight range — while the physical plates are not; the positive and negative segments of each bit row are visible in Figure 63. Building the sub-LSB weights as small differences of much larger plates is what makes them matchable, and the price is that all of that plate loads the top node: the summed physical plate is 1.235 times the summed weight. The extracted top-plate parasitic and the comparator input add a further 1.243, and $1.235 \times 1.243 = 1.535$. The parasitic alone is 0.226 of the physical array, not the 0.535 of the array a single-mechanism reading infers. The slope then closes in one expression,

$$m = \frac{C_{top,total}}{V_{ref} C_{unit}},$$

giving 627.8 codes per volt against the 628.59 the corner staircase measures (Table 74), 0.13 %.

Seven perturbation runs separate the terms. Two known capacitances added to the top plate predict the perturbed slope from one fitted C_{top} to better than 0.03 %, with nothing else in the converter changed. Lowering the analog supply scales the slope by exactly the reference ratio and moves the window centre with it, while forcing the comparator's reference alone moves the centre and leaves the slope unchanged to 0.003 % — so the span is set by the reference and the centre by the comparator's own supply-halving resistor divider, and the 1.25 V centre is that divider rather than an offset. Halving and then removing the antiphase plates moves the slope the predicted way and destroys the binary weighting at the same time, taking the worst residual about the fitted line from 0.50 LSB to 7.02 and 10.36 LSB. Nothing here is a layout defect that a parasitic-free layout would remove: the larger factor is the topology.

Departure from the fitted line is at most 0.52 LSB over the fitted range (Figure 65). That is a residual of a 24-point fit at 39 codes per step, not a linearity figure.

Referred to its own range instead of the nominal one the picture separates (Table 73). The usable window is 0.441 V to 2.068 V, 1.627 V wide against the 2.5 V span and centred on the common mode (Figure 66); 34.9 % of the nominal range is unreachable, split 0.441 V low and 0.432 V high. Across that window the effective LSB is 1.591 mV rather than 2.441 mV, and each code sits within 0.523 LSB of the measured line, 0.294 LSB rms (Figure 67). The two defects are therefore separable: the range is short by a third, and within what remains the transfer is straight to better than half a code at the sampled inputs.

8.8.2 Supply Current and Conversion Energy

Power depends on which clock protocol the converter is driven with, so both are reported (Table 75). On the stimulus as delivered a conversion takes 4.600 μ s and costs 12.95 μ W, 59.6 pJ per conversion; at the design rate the same converter draws 18.09 μ W and 20.8 pJ. Four times the throughput for 40 % more power is the expected trade — the static term is carried four times longer in the slow case — and the energy per conversion is the figure that changes, by 2.9 $\times$.

Power corners by 2.7 $\times$ where the transfer does not (Table 74): 15.8 μ W to 43.1 μ W across the five process points, the slow corner alone a factor of two above the other four. That signature is leakage and bias rather than switching, consistent with the 1.0 V core logic slowing down.

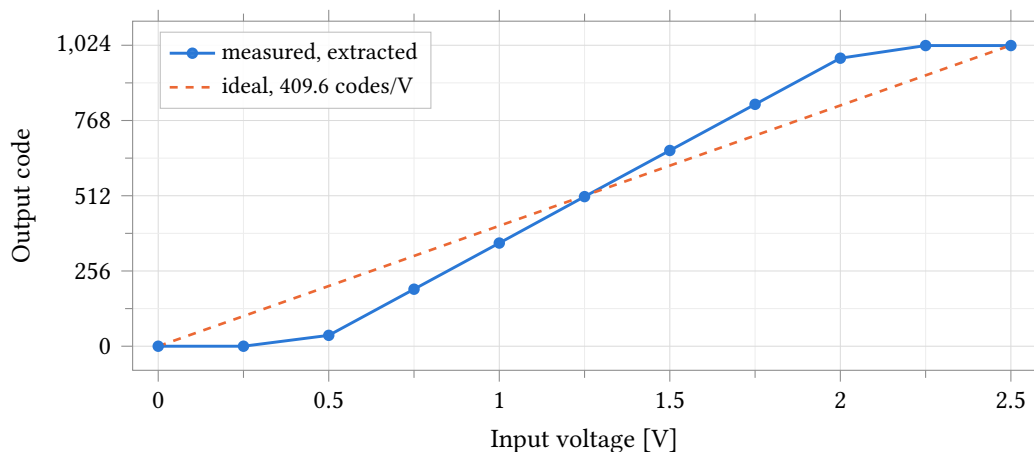


Figure 64: Output code against input voltage for the extracted anatop_pixel_adc, nominal corner, 37 °C, 2.5 V analog and 1.0 V digital supplies, with the ideal 409.6 codes/V line overlaid. The transfer is monotonic and reaches both code limits, but its slope is 628.64 codes/V — 1.535 times ideal — so the flat segments below 0.44 V and above 2.07 V are the code limits, not the ends of the sweep. Markers are the eleven simulated points; the 0.25 V step is 157 codes, so the curve between them is interpolation and carries no linearity information.

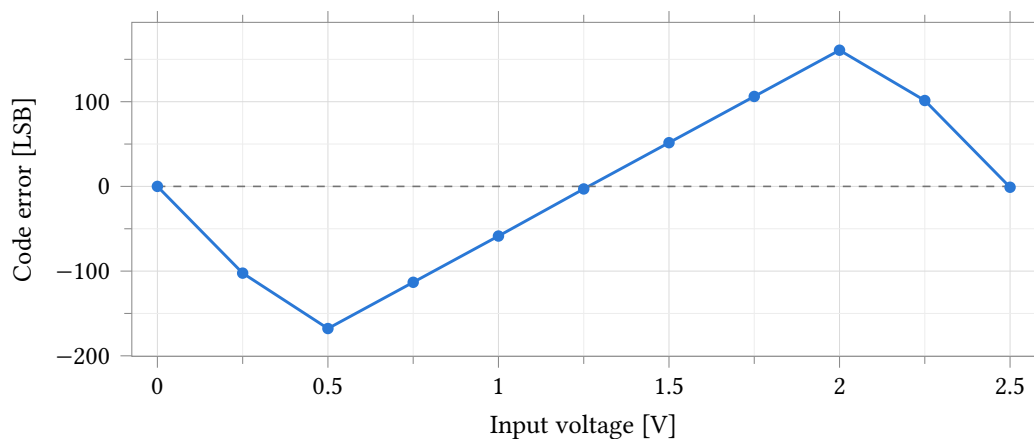


Figure 65: Code error against input voltage — measured code minus the ideal 409.6 codes/V line — from the same run as Figure 64. The error is the 1.535 slope ratio, not nonlinearity: it reaches -167.8 LSB at 0.5 V and 160.8 LSB at 2.0 V, crossing zero near the 1.25 V common mode where the measured and ideal lines intersect. It also returns near zero at 0 V and 2.5 V, where the clipped output coincides with the ideal endpoints; those two points are not accuracy. Referred to the fitted line instead, the residual over 0.5 V to 2.0 V is at most 0.52 LSB.

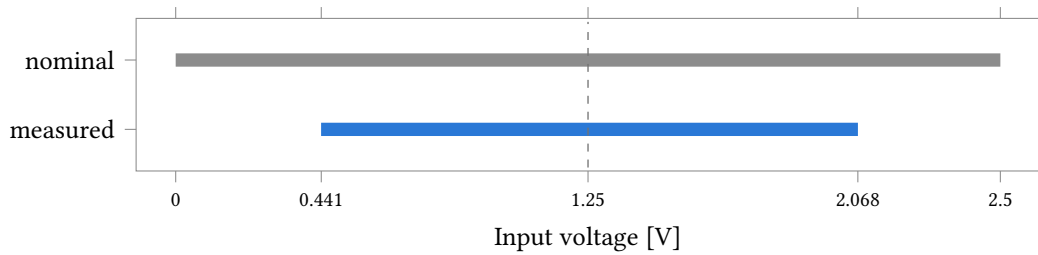


Figure 66: Usable input window of the converter against the nominal 0 V to 2.5 V supply span. The measured window runs 0.441 V to 2.068 V, 1.627 V wide, so 34.9 % of the nominal span is unreachable — 0.441 V at the bottom and 0.432 V at the top, near enough symmetric about the 1.25 V common mode (dashed). Outside that window the output sits at a code rail, so a transimpedance swing wider than ± 0.81 V about V_{CM} is truncated rather than digitised.

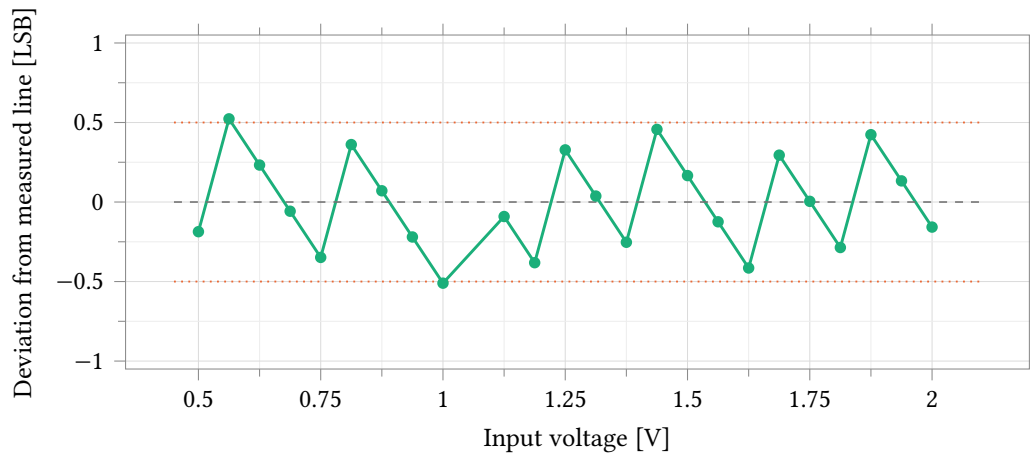


Figure 67: Deviation of each measured code from the converter's own best-fit line, 628.64 codes/V, over a 0.0625 V sweep of the unclipped window. Referred to its own scale rather than the nominal one the converter is straight to 0.523 LSB worst case and 0.294 LSB rms, so the 168 LSB of Figure 65 is gain error alone and carries no linearity information. The worst point sits just outside the ± 0.5 LSB dotted guides. Twenty-four points 39 codes apart bound the deviation at the sampled inputs; a DNL or INL measurement needs every code and this grid does not provide one. A twenty-fifth point at 1.0625 V is absent because its simulation returned an error.

Parameter	Unit	Value
Fitted slope, 0.5 V to 2.0 V	codes/V	628.64
Ideal slope, 1024/2.5 V	codes/V	409.60
Slope ratio, measured / ideal		1.535
Effective LSB	mV	1.591
Nominal LSB, 2.5 V/1024	mV	2.441
Effective full-scale input span	V	1.627
Input at code 0, extrapolated	V	0.441
Input at code 1023, extrapolated	V	2.068
Centre of usable window	V	1.255
Input range lost to clipping	%	34.9
Range utilisation	%	65.1
Worst residual about the fit	LSB	0.52
RMS residual about the fit	LSB	0.29

Table 73: Transfer parameters of the extracted `anatop_pixel_adc` from test `adc_dc`, nominal corner, 37 °C, 2.5 V analog and 1.0 V digital supplies. The slope is a least-squares fit to the 24 unclipped points of a 0.0625 V sweep over 0.5 V to 2.0 V; the code-0, code-1023 and centre voltages are extrapolations of that fit and are not measured points. The worst and rms residuals are the departure of those 24 points from the fit, not an INL: the step is 39 codes. Every entry is reproduced to 0.10 % at all five process corners (Table 74); no mismatch spread attaches to any entry, and none can be measured from this netlist (Section 8.8.3).

Parameter	Unit	tt 37 °C	ff	fs	sf	ss
Fitted slope	codes/V	628.59	628.59	628.40	628.73	628.13
Effective LSB	mV	1.5909	1.5909	1.5914	1.5905	1.5920
Input at code 0	V	0.4408	0.4408	0.4407	0.4408	0.4399
Usable span	V	1.6274	1.6274	1.6280	1.6271	1.6286
Worst residual	LSB	0.495	0.495	0.490	0.477	0.484
RMS residual	LSB	0.294	0.294	0.277	0.288	0.302
Monotonic points		20/20	20/20	20/20	20/20	20/20
Conversion period	μs	1.150	1.150	1.150	1.150	1.150
Analog supply current	μA	6.360	4.787	5.453	8.437	15.429
Total power	μW	18.09	16.77	15.77	23.74	43.06
Energy per conversion	pJ	20.8	19.3	18.1	27.3	49.5

Table 74: Transfer, timing and power of the extracted converter at five process corners: typical at 37 °C and the four skew corners at 25 °C, on a 20-point staircase at 0.1 V (63 codes apart) fitted over 0.55 V to 1.95 V. The transfer is invariant to 0.10 % because it is a ratio of extracted fixed capacitances, which the corner rows do not move; the residuals are departures from that fit at 63 codes per step, not an INL. Supply current is averaged over one whole conversion at the design-rate protocol of Section 8.8.2, and depends on the sweep step because the switching term dominates.

Currents are averaged over one whole conversion, and they depend on the sweep step: the nominal point reads $6.36\text{ }\mu\text{A}$ on a 63-code step and $12.56\text{ }\mu\text{A}$ on a 220-code step, because the switching term dominates. A converter current quoted without its code step is not a comparable number.

Parameter	Unit	As delivered	Design rate	Corner range
Trigger clock	MHz	20	20	20
Falling edges per conversion		10	10	10
Conversion period	μs	4.600	1.150	1.150
Conversion rate	kSa/s	217.4	869.6	869.6
Analog supply current, 2.5 V	μA	4.737	6.360	4.79–15.43
Digital supply current, 2.5 V	nA	2.0	—	—
Digital supply current, 1.0 V	μA	1.09	—	—
Total power	μW	12.95	18.09	15.77–43.06
Energy per conversion	pJ	59.6	20.8	18.1–49.5
Fitted slope	codes/V	628.85	628.59	628.13–628.73
Input at code 0	V	0.4410	0.4408	0.4399–0.4408

Table 75: Conversion timing, supply current and conversion energy of the extracted converter on the two clock protocols of Section 8.8, nominal corner, 37°C . Currents are averaged over one whole conversion rather than sampled, and on a 63-code sweep step: the switching term dominates, so the same point reads $12.56\text{ }\mu\text{A}$ on a 220-code step, and any quoted current must name its step. The corner range is the design-rate protocol across the five corners of Table 74. Over a whole conversion the digital 2.5 V rail carries leakage only.

8.8.3 Monte Carlo: What This Netlist Cannot Measure

A 100-sample process-and-mismatch Monte Carlo of the extracted converter runs clean and measures nothing about matching. Two of the three input levels return the same code in all 100 samples; the third dithers between the two codes adjacent to the boundary it already sits on. A mismatch-only control returns $\sigma = 0$ on every level and on the supply current, and the full run reproduces the process-only run to every printed digit.

Three structures account for it, and none is fixable in post-processing. Extraction flattens away the inline model wrappers in which the kit’s per-instance mismatch parameters are declared, so the converter’s transistors sit on raw model cards that the statistical block has no path to — the simulator reports five of its six mismatch parameters as non-referenced. The capacitor array is extracted linear capacitors with no statistical model of any kind, so the matching that actually sets a charge-redistribution converter’s linearity is not represented at all. The one remaining matched structure, the resistor divider that sets the comparator’s reference, ships with its mismatch flag cleared, which multiplies the model’s mismatch terms to zero.

The σ on this converter’s gain and offset are therefore unmeasured, not small. Nothing below should be read as a matching result. What the run does establish is the analog bias spread, which is process and is real: 11.4 % on the analog supply current and 10.5 % on total power, $10.5\text{ }\mu\text{W}$ to $16.4\text{ }\mu\text{W}$, with conversion timing identical in every sample.

Forcing the reference divider’s mismatch on bounds the one contribution the netlist can express: 0.821 mV , 0.52 LSB of 1σ on the converter zero, ± 1 code, and 0.012 % on gain — the divider moves the comparison threshold, not the array’s charge balance. That is small against the three-code offset the nom-

inal converter already carries and smaller still than the channel's own loop offset. Measuring the array itself needs a different bench: a schematic capacitor DAC built from the kit's statistical metal-fringe capacitor device rather than from extracted linear elements. A precedent bench for exactly that exists in the legacy library and has not been run against this array.

8.8.4 Measuring this block on chip

The converter is characterised end to end with no analog pad measurement — and its gain has never been checked in system.

1. Gain and intercept: set `AFE_ADCSRCSEL = 4`, drive an external ramp onto `atb_d` (`AFE_DBGDIREN = 1`, `AFE_DBGBUFEN = 0`), fit slope and intercept per die. The $1.535\times$ slope, 0.441 V intercept and 1.591 mV effective LSB of Table 73 hold to 0.10 % across process corners in simulation; their die-to-die spread is neither measured nor simulable (Section 8.8.3), which is what makes this step per-die rather than per-lot.
2. Zero code: set `AFE_ADCSRCSEL = 1` and average the common-mode rail. That code is the origin of the current axis and carries the converter's offset and the common-mode value together — the combination a current reading needs.
3. Dither before averaging: with the electrode isolated the amplifier output noise is tens of microvolts, below one code, so step the common-mode DAC over three of its $529\text{ }\mu\text{V}$ codes (together they span one converter code); only then does averaging reach the quantisation floor.
4. Observe the sampling node itself at per-channel slot 9 between conversions only.

This test fabric is design intent; it is not yet in the schematics.

8.9 Programmable feedback resistor

`anatot_tia_rprog` is the transimpedance gain element of the rev-2 channel: a digitally programmable resistor that sets R_F between the TIA output and the working-electrode virtual ground. Its law is

$$R = 18\text{ k}\Omega + 6\text{ k}\Omega \cdot N, \quad N = \text{bin}(\text{ResEn} < 5 : 0 >) + 64 \cdot \text{popcount}(\text{ThEn} < 3 : 0 >),$$

$N = 0 \dots 319$, so every $6\text{ k}\Omega$ value from $18\text{ k}\Omega$ to $1.932\text{ M}\Omega$ is reachable and the map is monotonic by construction. The ladder is a series chain of identical poly units — a three-unit fixed base, six binary-weighted segments of 1 to 32 units, and four thermometer segments of 64 units — each switched segment shunted by a CMOS bypass transmission gate (the thermometer gates at twice the binary gates' width; Figure 68). A control bit at 1 opens its gate and inserts the segment; terminals are A, B and the ten control bits, with the gate supplies arriving by inherited connection.

The endpoints follow from the converter, not from the ladder. The SAR's effective input window is 1.63 V , $\pm 0.815\text{ V}$ about V_{CM} (§8.8), essentially coincident with the TIA's measured $\pm 0.80\text{ V}$ compliance. At $N = 0$ ($19.6\text{ k}\Omega$ typical) that window carries $\pm 40.7\text{ }\mu\text{A}$, covering the $\pm 33\text{ }\mu\text{A}$ application target; at $N = 319$ ($1.931\text{ M}\Omega$) the range is $\pm 0.41\text{ }\mu\text{A}$ with a current quantum of 0.82 nA , an order above the 49 pA the loop actually delivers in the amperometric band (Section 8.9.1), so at maximum gain the resolution floor is quantisation rather than noise. Larger R_F would buy nothing: the noise floor does not shrink with R_F , and headroom does — at $V_{\text{CM}} = 1.25\text{ V}$ on 2.5 V the top code can carry at most $\approx 0.65\text{ }\mu\text{A}$ before the terminal voltage rails.

All data is schematic-level; the cell has no layout view yet. The bench forces 500 nA through A–B with B held at V_{CM} and reads R from the terminal voltage; one swept variable decodes into all ten control bits, and the decode was verified against an independent per-code deck to $0.18\text{ }\Omega$ over all 320 codes.

Corners move every code by the same sheet-resistance factor, so matching — the property that decides whether a 64-unit thermometer segment can invert a step against 63 binary units — is invisible to them.

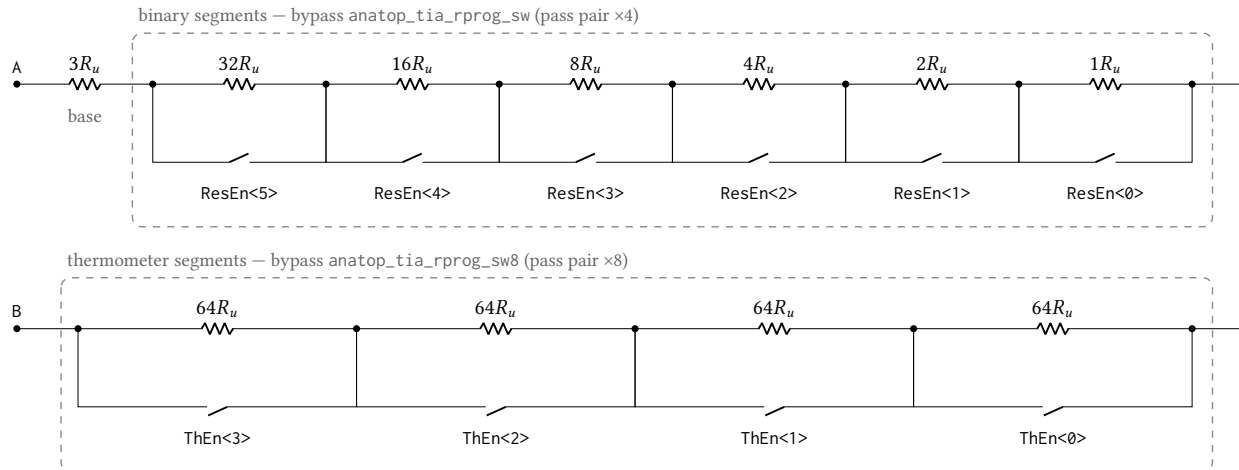


Figure 68: Segment and bypass-switch structure of anatop_tia_rprog, transcribed from the executed spec-run netlist. The unit R_u is one series poly resistor; from terminal A the chain runs through the never-switched three-unit base, the six binary-weighted segments of 32 down to 1 unit, and the four identical 64-unit thermometer segments to terminal B. Each switched segment is shunted by a CMOS transmission gate with a local inverter: the control pin EnBar is the ResEn/ThEn bit itself and drives the PMOS pass gate directly and the NMOS pass gate through the inverter, so a bit at 1 turns both pass devices off and inserts the segment. The pass pair has multiplicity 4 in the binary switch (`_sw`) and 8 in the thermometer switch (`_sw8`), which halves the thermometer gates' on-resistance where up to six binary gates close at a seam. Three grounded dummy units accompany the 322 active units for matching.

Point	Process	Temperature	Supply
1	typ	27 °C	2.5 V
2	ff-res	27 °C	2.5 V
3	ss-res	27 °C	2.5 V
4	ff -40C	-40 °C	2.5 V
5	ss 125C	125 °C	2.5 V

Table 76: Process and temperature points simulated for the programmable feedback resistor. Point 1 is the typical statistical process point (castalia_typ_25) at 27 °C. Points 2–3 skew the poly resistor family alone (ff_res/ss_res with typical MOS, 27 °C) and points 4–5 skew MOS and resistor together at the temperature extremes (ff_25+ff_res at -40 °C, ss_25+ss_res at 125 °C), so the ladder term and the switch-resistance term of each code can be told apart. The `_res` family has no sf/fs sections, so no cross corner exists to run. This is a corner run and carries no device mismatch.

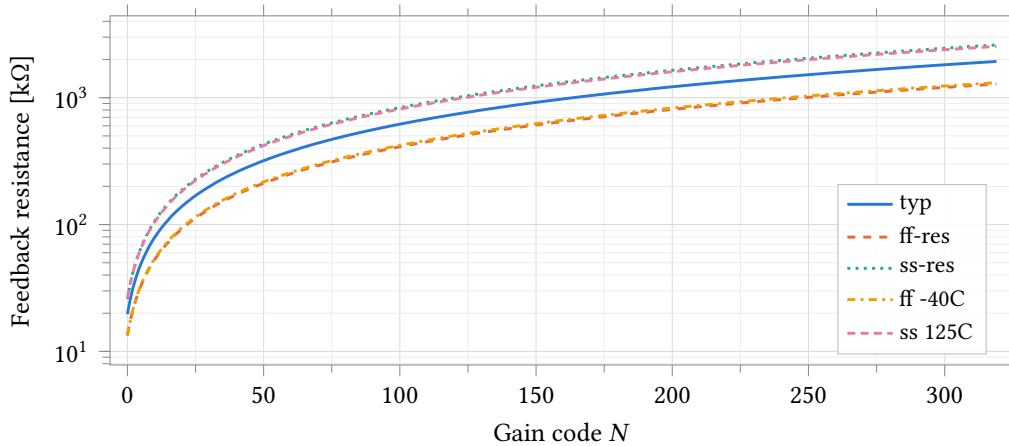


Figure 69: Measured resistance against gain code at all five corners, 500 nA forced through the A–B terminals with B held at $V_{CM} = 1.25$ V. The law is $R = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N$ with N the binary value of `ResEn<5:0>` plus 64 per set `ThEn<3:0>` thermometer bit; every corner is strictly monotonic over all 320 codes, including the four thermometer seams. The corner families move the absolute value by -33 to $+35$ %, which is the poly sheet-resistance tolerance and must be absorbed by gain calibration, not by code choice.

Parameter	Unit	typ	ff-res	ss-res	ff -40C	ss 125C	Spread
$R(N=0)$	k Ω	19.648	13.579	25.947	13.263	26.407	13.143
$R(N=63)$	k Ω	396.1817	262.8186	534.6769	269.5412	520.7830	271.8583
$R(N=64)$	k Ω	403.6009	268.0920	544.3539	274.4177	531.2663	276.2619
$R(N=319)$	M Ω	1.9308	1.2802	2.6064	1.3137	2.5369	1.3262
Seam step 63→64	k Ω	7.419	5.273	9.677	4.876	10.483	5.607
Step, min	Ω	5576	3574	7668	3812	7250	4094
Step, max	Ω	8509	6445	10618	5687	11659	5972
Step, mean	Ω	5991.0	3970.5	8089.1	4076.7	7870.0	4118.6

Table 77: Code-space summary by corner, 500 nA test current. `step_min/step_max` are taken from the point-to-point derivative of the 320-code sweep and `step_mean` from the endpoints; `step_min` is positive at every corner, which is the monotonicity statement. The seam step (code 63 to 64, one 64-unit thermometer segment in, six binary segments out) exceeds a normal step everywhere. At ss 125C the top code reaches 2.553 V at the forced 500 nA — 53 mV over the rail; a 250 nA control run bounds the cost of that at 0.0017 % on the top code and ≤ 0.036 % anywhere, so the values are reported as measured.

Parameter	Unit	typ
Chord resistance	k Ω	19.639
Worst deviation	mV	1.998
Worst deviation	ADC LSB	1.255
Secant- R modulation	%	0.504

Table 78: Drive linearity at the bottom code ($N = 0$), typical corner, sweep ± 90 % of the ± 0.8 V-window full scale. The chord is the straight line through the sweep endpoints; the worst deviation of 1.998 mV (1.26 ADC LSB) sits at mid-range, where the closed bypass gates run at lower on-resistance than at the endpoints that pin the chord. The companion runs at $N = 63$ and $N = 319$ measure $5.03 \mu\text{V}$ (0.0032 LSB) and $4.87 \mu\text{V}$ (0.0031 LSB) respectively (run directories `simruns/tb_anatop_tia_rprog_lin_n63,_n319`).

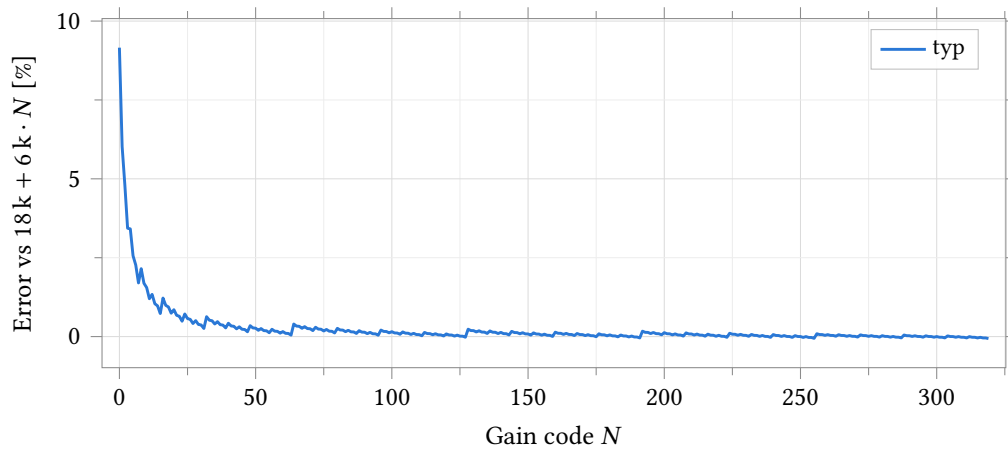


Figure 70: Typical-corner deviation from the ideal code law. The +9.2 % at $N = 0$ is the series resistance of the ten closed bypass switches (1.65 k Ω , about 165 Ω each), which decays as $1/R$ once real ladder segments dominate; by $N = 319$ the error is -0.063 %. The ripple is the carry structure: at each binary carry and at each thermometer seam the set of closed switches changes, so the R_{on} term steps while the ladder term is exact.

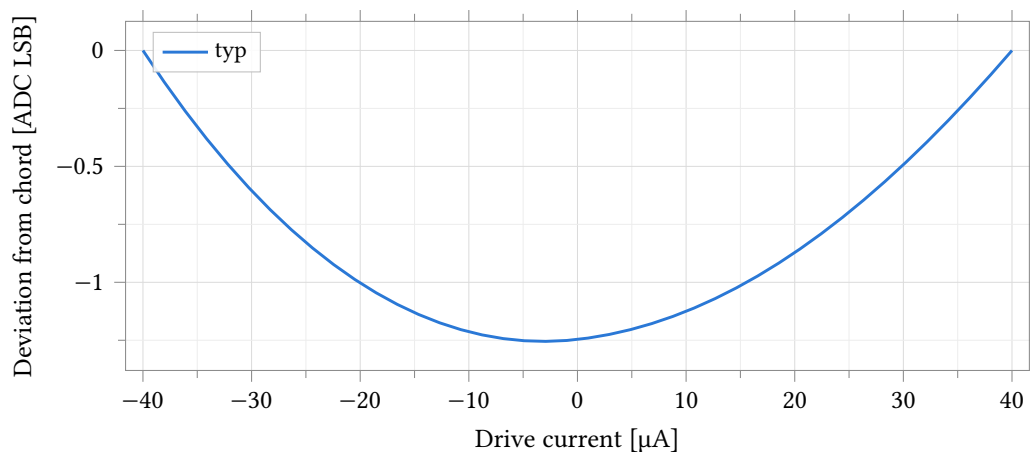


Figure 71: Drive nonlinearity of the bottom code ($N = 0$, $R \approx 19.6$ k Ω): voltage deviation from the endpoint chord over ± 90 % of full scale (± 40 μ A), typical corner, expressed in 1.592 mV ADC LSB. All ten bypass gates are closed at this code and carry the full signal current; their on-resistance rises with the terminal's excursion from V_{CM} , so the chord — pinned at the sweep endpoints where that excursion is largest — overestimates the mid-range voltage by up to 1.25 LSB, and the deviation is bowl-shaped with a slight skew from the gates' voltage asymmetry. The same measurement at $N = 63$ and $N = 319$, where the ladder dominates and current is 20–100 $\times$ smaller, stays below 0.004 LSB and is tabulated, not plotted.

The Monte Carlo run therefore grades the seam directly, per sample, alongside the absolute spread at four codes.

Every graded code carries a $\pm 5\%$ absolute window about its typical value, and the seam a monotonicity limit at zero; all five yield 100.0 % over the 200 samples (Table 80). The margins are not close: the nearest limit sits 14.2σ from the mean at $N=0$, 54σ at $N=63$, 55σ at $N=64$ and 134σ at $N=319$, and the worst seam sample is 14.7σ above zero. Mismatch is therefore nowhere near yield-limiting for this block; the items that need engineering attention are the deterministic ones graded in Table 81 – bottom-code linearity and slow-corner top-code compliance.

All four seams, all 320 codes. A second 200-sample run sweeps the whole code space in every sample, so each sample is graded at all four thermometer seams and none can contribute to one seam and not another (Table 79). **No sample is non-monotonic anywhere:** zero codes with a negative step, in 200 samples across 320 codes. The nearest approach to zero is not a seam but the smallest ordinary step, 5470Ω in the worst sample, 137σ from zero on its own distribution; the four seam steps clear zero by 14.7 to 17.7 standard deviations. The worst of them is the first, and predictably so: it is the only seam at which a thermometer segment switches in with no thermometer segment already in circuit, so its step is the smallest and its relative spread the largest.

The ratio the seams rest on measures well. Sixty-four thermometer units against sixty-three binary units read 1.0197 against an ideal 1.015 873 – 0.38 % high, which is switch resistance and does not cancel between the two differences – with a spread of 0.132 %, i.e. 0.13 % on one thermometer segment. It is the spread and not the offset that decides monotonicity, and the offset calibrates out.

Seam	Mean k Ω	σ Ω	σ/μ %	Worst sample k Ω	Margin σ
$N = 63 \rightarrow 64$	7.424	504.0	6.79	5.878	14.7
$N = 127 \rightarrow 128$	8.003	458.6	5.73	6.590	17.5
$N = 191 \rightarrow 192$	8.494	481.3	5.67	7.214	17.7
$N = 255 \rightarrow 256$	8.250	468.5	5.68	6.782	17.6

Table 79: Monte Carlo of the four thermometer seam steps, 200 mismatch samples, seed 12345, low-discrepancy sampling, nominal process, 25 °C, 500 nA test current. Each sample sweeps all 320 codes, so every sample is graded at every seam. The last column is the margin from a negative step, in standard deviations of that seam’s own distribution; the minimum step anywhere in the code space is 137σ from zero.

The bench carries 14 pass/fail specifications, graded by the simulator against every corner and every Monte Carlo sample (Table 81). The limits are screening bounds chosen from the measured data: the endpoint windows allow the $\pm 35\%$ poly sheet-resistance span plus margin, so a violation flags a broken device or model, not process spread. Twelve of the fourteen pass at every corner and every sample.

The two exceptions are honest and bounded. `v_top` exceeds its 2.45 V limit at the two slow-resistor corners (2.553 V at ss-res, 2.518 V at ss 125C): at +35 % resistance the fixed 500 nA test current rails the 2.5 V supply at the top codes. That is the measurement drive running out of compliance at the slow corner, not a device defect – the same compliance ceiling already documented as the $\sim 0.65\mu\text{A}$ top-code operating limit – and the 250 nA control run bounds its effect on the reported resistances at $\leq 0.036\%$ (Table 77). `lsb_err` fails its 0.5-LSB budget at four of five corners (1.24 typical, 3.60 worst at ss 125C, passing only at ff -40C with 0.41): this is the bottom-code switch-Ron bowl of Figure 71, largest where the gates are slowest and hottest. It is accepted as documented – the chord-resistance error `r_lin_err_pct` stays below

Parameter	Unit	N	Mean	σ	Yield
$R(N=0)$	k Ω	200	19.650	0.071	100.0
$R(N=63)$	k Ω	200	396.4515	0.3608	100.0
$R(N=64)$	k Ω	200	403.8759	0.3644	100.0
$R(N=319)$	M Ω	200	1.93212	0.00071	100.0
Seam step 63→64	k Ω	200	7.424	0.504	100.0

Table 80: Monte Carlo of the programmable resistor at four codes plus the per-sample seam step. **Mismatch only**, model section `castalia_pm_25`, nominal process, 200 samples, seed 12345, low-discrepancy sampling, 25 °C, 500 nA test current. Relative spread falls with code as units accumulate: $\sigma/\mu = 0.359\%$ at $N=0$ (three ladder units plus ten switch Rons), 0.091 % at $N=63$, 0.037 % at $N=319$; the gain ratio $R(319)/R(0)$ spreads 0.354 %. Yield grades each code against a $\pm 5\%$ absolute window about its typical value and the seam against zero (limits in Table 81): all five yield 100.0 % over the 200 samples, with the nearest limit 14.2σ from the mean ($N=0$).

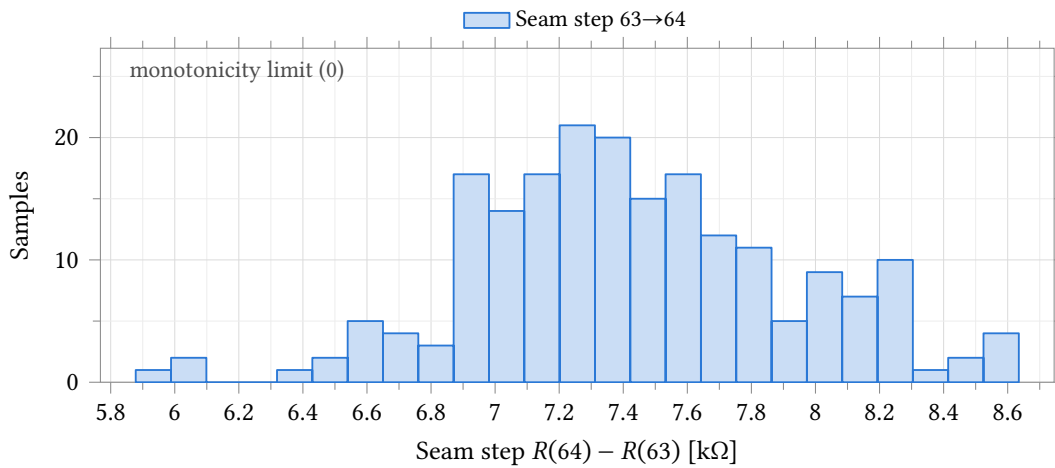


Figure 72: Per-sample step across the first thermometer seam (code 63 to 64: one 64-unit segment switches in while all six binary segments switch out), the code transition most exposed to mismatch. **Mismatch only**, model section `castalia_pm_25`, nominal process, 200 samples, seed 12345, low-discrepancy sampling, 25 °C. Mean 7.42 k Ω , $\sigma = 504 \Omega$, worst sample 5.88 k Ω — 14.7σ above the monotonicity limit at zero, so no realistic mismatch closes the seam.

1 % everywhere (0.12–0.53 %), and the fix, if bottom-code linearity ever matters, is wider bypass gates. One bench note: the linearity sweep’s drive source includes the i test term ($i = i_{\text{test}} + f \cdot 0.8 / R_{\text{ideal}}$, mirroring the netlist source exactly), so the chord is taken about the same operating point the code sweep uses; against that corrected drive the typical worst deviation reads 1.24 LSB where Table 78 reported 1.26.

Specification	Unit	Limit	Measured	Verdict
<i>Code sweep, 320 codes, five corners</i>				
R at $N=0$	k Ω	12–28	13.26–26.41	pass
R at $N=319$	M Ω	1.25–2.65	1.280–2.606	pass
Step, min	k Ω	> 1	3.57–7.67	pass
Step, mean	k Ω	3.8–8.2	3.97–8.09	pass
Non-monotonic steps	count	0	0	pass
Error vs $18\text{ k} + 6\text{ k} \cdot N$, max	%	< 55	9.2–46.7	pass
$V(A)$ at top code	V	< 2.45	1.89–2.55	marginal ^a
<i>Drive linearity at $N=0$, five corners</i>				
Chord deviation, max	ADC LSB	< 0.5	0.41–3.60	fail ^b
Secant- R error	%	< 1	0.12–0.53	pass
<i>Monte Carlo, 200 mismatch samples, typical</i>		Nominal	Std. dev.	
$R(N=0)$	k Ω	19.650	0.071	pass, 200/200
$R(N=63)$	k Ω	396.452	0.361	pass, 200/200
$R(N=64)$	k Ω	403.876	0.364	pass, 200/200
$R(N=319)$	M Ω	1.93212	0.00071	pass, 200/200
Seam step 63→64	k Ω	7.424	0.504	pass, 200/200

^a Exceeded at ss-res (2.553 V) and ss 125C (2.518 V): test-drive compliance, not a device defect (see text).

^b Fails at four of five corners; passes at ff -40C (0.41). Bottom-code switch-Ron bowl, accepted (see text).

Table 81: Specification grading: the 14 limits carried by the bench and the verdict. Corner rows list the limit and the measured extremes across all five corners of Table 76; the endpoint and step windows are screening bounds ($\pm 35\%$ corner span plus margin). Monte Carlo rows list the nominal (mean) and standard deviation over the 200 mismatch samples (typical process, 25 °C), each graded against a $\pm 5\%$ absolute-accuracy window about its typical value and the seam against zero.

8.9.1 In-loop noise

Inside the transimpedance loop the ladder is not a noise contributor. Its own Johnson noise is 0.0008 % to 0.0065 % of the input-referred noise power in the dc-amperometric band, four to five orders below the amplifier (Table 82), and replacing the whole switched ladder by a single ideal resistor of the measured value changes the answer by 0.02 % at the worst code. The roughly three hundred open bypass switches contribute nothing measurable. Loop stability is unchanged by the noise run — phase margin 74.9° to 92.9°, gain margin 21.65 dB to 21.79 dB — so these are the operating points the stability grid already reports.

No noise figure for this block is meaningful without its band. Over 0.1 Hz to 10 Hz the input-referred current noise is 48.9 pA to 1060 pA depending on gain code. Over 0.1 Hz–1 kHz it is 3.00 nA at every code, and over 0.1 Hz–10 kHz 26.7 nA at every code. That is not a ladder defect: above the frequency at which the feedback impedance exceeds the electrode impedance the noise gain is set by the electrode capacitance, so the input-referred noise becomes independent of R_f — which is what the identical 132 fA/ $\sqrt{\text{Hz}}$ spot noise at 1 kHz across six decades of R_f shows. The same design is 49 pA or 27 nA according to where the integral stops, so the band edges are fixed design variables rather than tracked to each code’s own bandwidth — letting them follow would make the slow, high-gain codes report less noise

merely for being slower.

Against the 2 nA input-referred budget in the amperometric band, codes $N \geq 63$ sit 25× to 40× under it and the bottom code 1.9× under. At the top code the measured 48.9 pA is 17× below the 0.824 nA current quantum, so the resolution floor at maximum gain is quantisation, not noise.

N	R_f	Input-referred noise, rms		Ladder share of noise power
		0.1 Hz to 10 Hz	0.1 Hz–1 kHz	
0	19.63 k Ω	1060.5 pA	3.304 nA	0.0008 %
63	394.8 k Ω	81.4 pA	3.005 nA	0.0065 %
127	777.1 k Ω	60.0 pA	3.004 nA	0.0061 %
191	1.159 M Ω	53.6 pA	3.004 nA	0.0051 %
255	1.542 M Ω	50.6 pA	3.003 nA	0.0043 %
319	1.924 M Ω	48.9 pA	3.003 nA	0.0037 %

Table 82: Input-referred current noise of the assembled transimpedance loop against gain code, 37 °C, nominal process, at a 100 nF double-layer capacitance and the 22 pF compensation capacitor. Noise is referred through the measured transimpedance rather than by an input-referral option, which does not reach the netlist from this setup. The last column is the feedback resistor’s own Johnson noise as a fraction of the noise power in the 0.1 Hz to 10 Hz band. The two bands differ because the noise gain above the feedback/electrode impedance crossover is set by the electrode, not by R_f .

Three limits on what this section claims. First, the resistance, linearity and mismatch data are two-terminal measurements of the ladder alone; only noise and loop stability have been measured with the ladder inside the transimpedance loop (Section 8.9.1), at one electrode and one process corner, and settling with the real switches is still open. Second, the Monte Carlo is mismatch-only at 25 °C; no process-plus-mismatch statistics exist. Third, the bottom code is the block’s weak point, and all three of its symptoms share one root cause: the 1.65 k Ω closed-switch stack (+9.2 % at $N = 0$), the 1.26-LSB drive nonlinearity, and the 4× larger relative mismatch spread are all the ten bypass gates in series with a three-unit ladder. Absolute gain at any code calibrates out against the corner and mismatch spread; if the bottom code ever needs uncalibrated accuracy, the fix is wider bypass gates, not more ladder. Integration into anatop_tia (whose legacy 16-bit tia_gain bus becomes these ten control bits) is pending, as is layout.

8.9.2 Measuring this block on chip

1. Force $\pm$ full-scale current into the working-electrode pin with no cell present; read the transimpedance output at per-channel slot 4 on atb_d, one gain code at a time. Scale the drive per gain code or the high-gain codes rail; skip the bottom code — its drive nonlinearity is 1.26 LSB typical and no two-point slope sees it.
2. Read at the pad, not through the converter: the converter window truncates samples that rail, and the two-point slope wanted here is the end-to-end constant in amperes per code, absorbing the resistor’s ± 35 % sheet tolerance (Table 77), the 1.65 k Ω closed-switch stack and the converter’s own gain in one number.
3. Switch resistance: slots 7 and 8 straddle the working-electrode isolation switch; their difference under the same forced current is that switch’s $i \cdot R_{on}$, which calibration stores rather than minimises.
4. Gain refresh without equipment: switch the on-chip reference resistor between the counter-electrode node and the summing node, take the same two-point slope, and scale the stored constant by the change in that ratio. Sheet resistance cancels; matching alone remains, 0.04 % to 0.36 % (Table 80).

This test fabric is design intent; it is not yet in the schematics.

8.10 Reference DAC

Two cells produce a channel's reference potential: `anatot_pixel_dacR2R12`, a 12-bit voltage-mode R-2R DAC, and `anatot_pixel_dac_psu`, the supply block that feeds it. All data in this section is schematic-level, with no extracted parasitics; layout and extracted views exist for both cells and have not been simulated.

The ladder is drawn in Figure 73. Its terminals are `A<11:0>`, `En` and `Out`; the supply arrives by inherited connection and netlists as `inh_vdd/inh_vss`, so there is no supply pin on the symbol. Each code bit is gated through an `AND2X8MA10TH` whose output high level is the ladder reference, so whatever drives `inh_vdd` sets full scale directly. Every element of the ladder is the same unit resistor — R is two units in series, $2R$ four — which makes the binary weighting a ratio of identical devices and the converter's linearity a matching property rather than a design margin. A nodal solve of the transcribed network gives $V_{\text{out}}/V_{\text{ref}} = D/4096$ exactly at every code, so none of the nonlinearity reported below is the ratio design.

Three testbench views drive the two cells. `DacTB` puts the DAC on an ideal 2.5 V rail and measures the ladder alone. `DacPSUTB` exercises the supply block by itself against a current load. `DacWPSUTB` runs both together, the DAC's inherited supply resolved to `VddDac` by a `netSet` property, and is the block as delivered.

Coverage is five corner points — the typical statistical point and the `ff`, `fs`, `sf` and `ss` skew corners — and a 200-sample Monte Carlo run varying process and mismatch together. Temperature is 25 °C on both DAC benches throughout; the four supply-block benches run at 27 °C at the typical point and 25 °C at the skew corners. The two DAC benches carry tightened solver tolerances (`reltol` = 1×10^{-6} , `vabstol` = 1×10^{-9} , `iabstol` = 1×10^{-15}): the LSB through the supply block is 529 μV and the ADE defaults permit several LSB of error per DC solve, of which DNL is a difference.

Two defects dominate this section and they are independent. The ladder is matching-limited and fails its own linearity limits on an ideal rail (Section 8.10.2). Separately, the supply block modulates the reference by 25.8 mV peak to peak across the code range against a quarter-LSB budget of 133 μV (Section 8.10.1). The second is in the corner data, at 25.7 mV to 29.6 mV at every one of the five points; the first is not, and cannot be, because a process corner moves every ladder resistor together.

8.10.1 Reference movement and the supply-block requirement

Divide the reference out and the ladder's own transfer is well behaved; leave it in and the same sweep is -37 LSB of INL and -10.8 LSB of DNL (Table 86). The difference is entirely reference movement, and the mechanism is not the one a reader expects.

In an R-2R the binary weighting comes from ladder attenuation, not from branch current: every $2R$ branch switched to the reference draws a comparable current. The current drawn from `VddDac` therefore tracks the bit *pattern*, not the code value. An ideal-ladder model gives a reference current of zero at code 0, $0.25 V_{\text{ref}}/R$ at code 2048 and $0.333 V_{\text{ref}}/R$ at code 4095, with a maximum of $1.444 V_{\text{ref}}/R$ at code 1365 = `0b010101010101` — the alternating-bit pattern, and exactly where the measured `VddDac` minimum sits, in every corner but `ff` and in all 200 Monte Carlo samples. A one-parameter fit $V_{\text{DDDAC}} = V_0 - R_{\text{out}} I_{\text{ref}}$ accounts for 94 % of the variance of the measured curve, at a correlation of -0.97 and a residual of 931 μV rms against a 25.70 mV spread. Figure 77 is that curve. It closes on the measured nonlinearity: droop times code weight accounts for -36.9 LSB of the -37.1 LSB of raw INL, so 99.5 % of the delivered block's static error is reference modulation and not a ladder error.

Two consequences follow that are easy to get wrong. First, the -10.86 LSB DNL is real non-monotonicity of the block as delivered and not a numerical artifact: tightening `reltol` from 1×10^{-3} to 1×10^{-6} , `vabstol` from 1×10^{-6} to 1×10^{-9} and `iabstol` from 1×10^{-12} to 1×10^{-15} moves `VddDac` by at most 193 μV anywhere

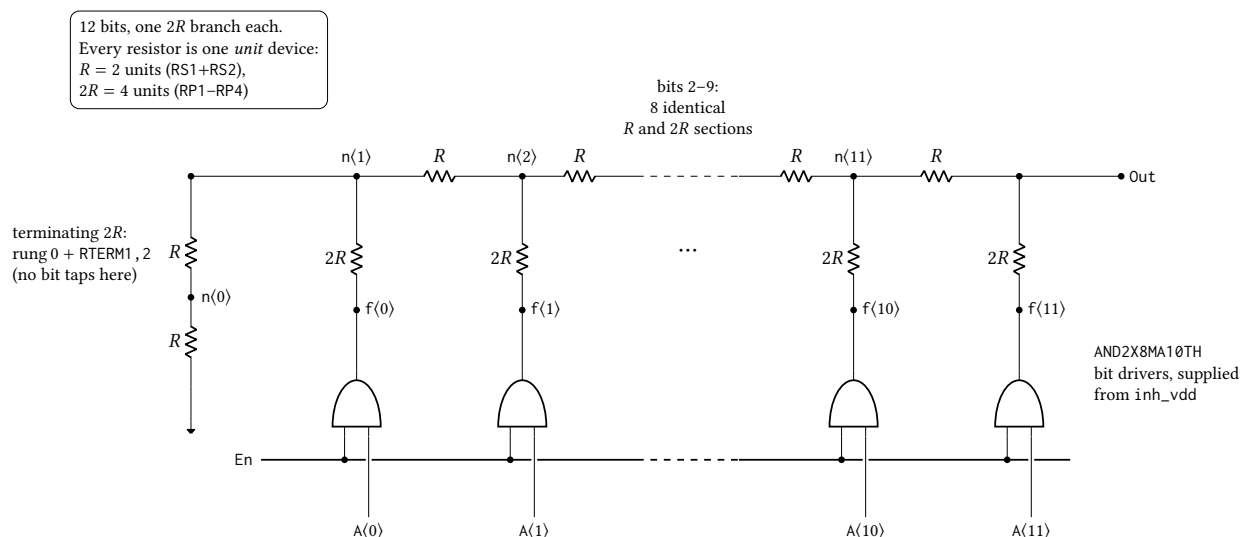


Figure 73: The 12-bit voltage-mode R-2R ladder of `anatot_pixel_dacR2R12`, transcribed from the `DacNonLinTB` netlist. Bit k drives tap $n\langle k+1 \rangle$ through its own $2R$ branch, so the twelve taps are $n\langle 1 \rangle$ – $n\langle 11 \rangle$ plus `Out`, and the MSB branch lands directly on `Out` at the top of the ladder; $n\langle 0 \rangle$ carries no bit, its rung and `RTERM1,2` together forming the terminating $2R$. Bits 2–9 are elided: eight further sections identical to those drawn sit between $n\langle 2 \rangle$ and $n\langle 11 \rangle$. Every element is the same unit resistor, so the network itself is exact and gives $\text{Out} = (\text{code}/4096) V_{\text{ref}}$; the reference is the *output high level* of the `AND2X8MA10TH` drivers, which sit in series with each $2R$ branch and bound the ladder in place of the rails — measured zero scale is $90 \mu\text{V}$ above V_{SS} and full scale $103 \mu\text{V}$ below the ideal $(4095/4096) V_{\text{ref}}$ on a 2.5 V rail. Device dimensions are proprietary and are not stated.

Point	Process	Temperature	Supply
1	tt	25 °C	2.5 V
2	ff	25 °C	2.5 V
3	fs	25 °C	2.5 V
4	sf	25 °C	2.5 V
5	ss	25 °C	2.5 V

Table 83: Process, temperature and supply points simulated for the 12-bit R-2R DAC and its supply block. Point 1 is the typical statistical process point evaluated at nominal; points 2–5 are the imported foundry skew corners. Temperature is not uniform across this run: the two DAC benches are at 25 °C at all five points, the four supply-block benches (load regulation, line regulation, output impedance, startup) at 27 °C at point 1 and 25 °C at points 2–5; the column below reports the DAC benches. Solver tolerances split the same way — the DAC benches run $\text{reltol} = 1 \times 10^{-6}$, $\text{vabstol} = 1 \times 10^{-9}$, $\text{iabstol} = 1 \times 10^{-15}$, the supply-block benches the ADE defaults 1×10^{-3} , 1×10^{-6} , 1×10^{-12} . This is a corner run and carries no device mismatch; R-2R linearity is a matching property, so every nonlinearity figure in this section is a systematic term and none of it is a yield statement. These points move the sixteen transistor model rows only — the resistor and capacitor rows stay at their typical sections — so every corner spread in this section is a lower bound (pinned-passive caveat, page 69).

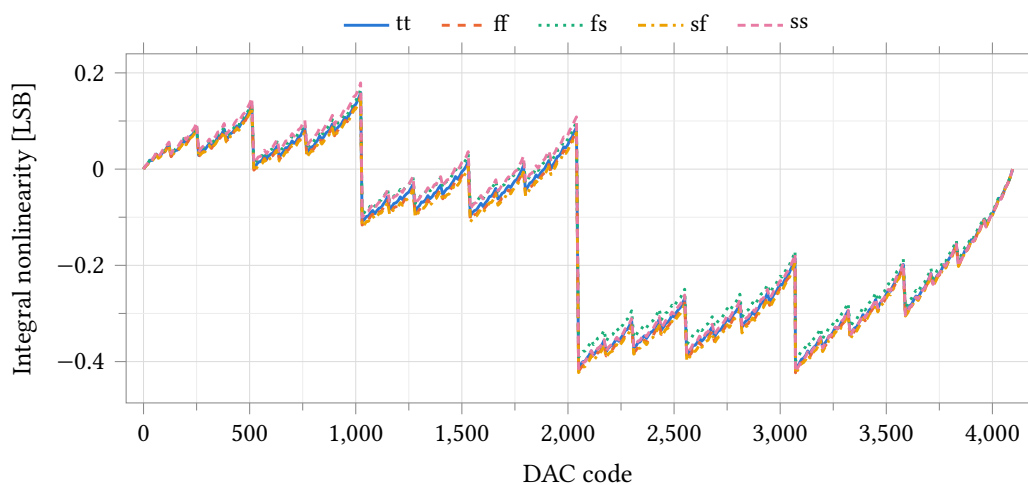


Figure 74: Integral nonlinearity against code over five process corners, ladder on an ideal 2.5 V rail, 25 °C, $\text{reltol} = 1 \times 10^{-6}$. INL is taken from the endpoint line through each corner's own zero- and full-scale outputs, so it is zero at codes 0 and 4095 by construction. The positive extreme is code 1023, +0.15 to +0.18 LSB; the negative extreme is shared by the major carries at codes 2048 and 3072, which sit within 0.006 LSB of each other at -0.39 to -0.43 LSB at every corner. The code of each extreme is the same at all five corners: the shape is set by the ladder, not by the process. Corner run, no device mismatch — R-2R linearity is a matching property, so this envelope is the systematic term alone.

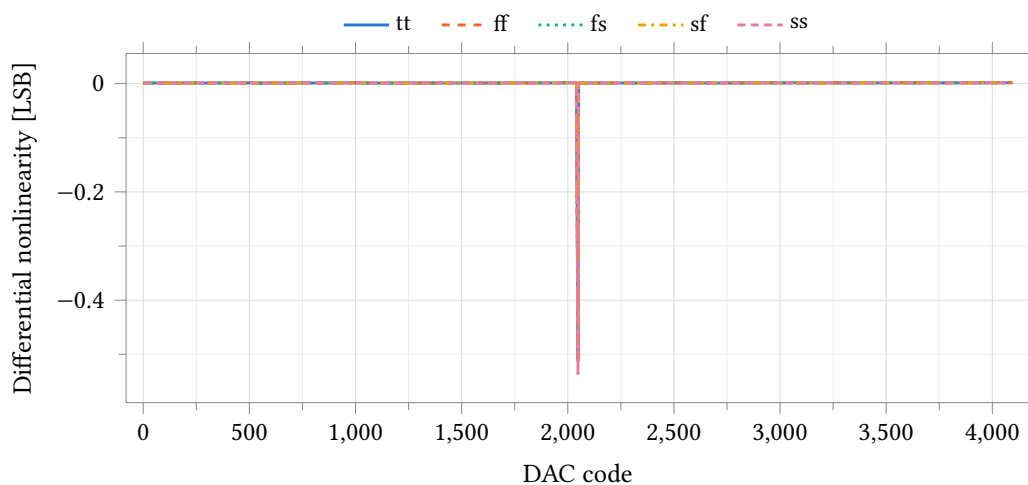


Figure 75: Differential nonlinearity against code over five process corners, conditions as Figure 74. DNL is the first difference of the transfer divided by the endpoint LSB, minus one. The worst step is the mid-scale major carry at code 2048, -0.50 to -0.54 LSB; the sub-carries measure -0.27 at code 1024, -0.23 at 3072, -0.14 at 512 and -0.13 LSB at 2560. The 4095-point curve is stride-decimated for the plot, so the comb drawn is a sample of the code grid and not every tooth of it; the extreme is preserved exactly. The converter is monotonic at every corner, with 0.46 LSB of margin to the -1 LSB limit at worst.

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Zero-scale output	μV	89.98	745.16	75.05	156.22	41.54	703.62
Full-scale output	V	2.499287	2.498719	2.499228	2.499299	2.499329	0.000610
LSB size	μV	610.305	610.006	610.294	610.291	610.327	0.321
INL, most positive	LSB	0.159	0.153	0.166	0.150	0.179	0.029
INL, most negative	LSB	-0.417	-0.423	-0.399	-0.425	-0.421	0.027
INL, worst magnitude	LSB	0.417	0.423	0.399	0.425	0.421	0.027
DNL, most positive	LSB	0.002	0.002	0.002	0.002	0.002	0.000
DNL, most negative	LSB	-0.511	-0.512	-0.500	-0.511	-0.535	0.036
DNL, worst magnitude	LSB	0.511	0.512	0.500	0.511	0.535	0.036

Table 84: Static accuracy of the ladder on an ideal 2.5 V rail by process corner, DacTB, 25 °C, all 4096 codes, $\text{reltol} = 1 \times 10^{-6}$. The ideal endpoints for this bench are 2.499 390 V full scale and 610.35 μV per LSB ($2.5 \times 4095/4096$); the measured LSB is within 0.35 μV of that at every corner. INL and DNL are referred to the endpoint line through each corner’s own zero- and full-scale outputs, so gain and offset are removed and only shape is reported. The bench saves the output node only and records no current, so no supply current is measured on it; the supply block’s quiescent current is Table 93. Corner run, no device mismatch: these are systematic terms. The scale columns stand: zero scale, full scale and the LSB are ratiometric and identical to six digits whether the passive model rows are pinned at typical or freed. The INL and DNL columns are a lower bound, because the pass-gate on-resistance is a transistor quantity against a poly ladder — freed, worst DNL is -0.755 and worst INL -0.617 LSB (pinned-passive caveat, page 69).

and leaves the structure of the curve intact. Second, endpoint-anchored INL makes the block look worse still, because codes 0 and 4095 both sit near the top of the droop curve and the whole bow therefore lands in INL at maximum leverage.

The requirement on the supply block follows from the DAC rather than from an assumed target. A quarter-LSB reference budget is $0.25 \times 528.96 \mu\text{V} = 133 \mu\text{V}$, against the 25.8 mV the ladder’s own code-dependent current actually produces: the supply’s output impedance has to fall by a factor of 193, to about 5 Ω . Measured over the 200 Monte Carlo samples it is 894 Ω small-signal at zero load (Table 96) and 1133 Ω as a chord across the full 0 μA to 100 μA load sweep (Table 92); the typical corner gives 902 Ω and 1137 Ω . That is what the topology of Figure 76 gives: the block is a shunt clamp, not a regulated loop, so its output impedance is the $1/g_m$ of the shunt device at a quiescent current of 160 μA and is flat from DC to 100 kHz. Closing the factor of 193 is a topology change, not a trim. VddDac is the ladder reference today, not merely a gated supply for the digital side: Out/VddDac is 0.499 890 at code 2048 and 0.999 733 at code 4095, against an ideal $4095/4096 = 0.999\,756$.

The supply-side entries that follow are one defect measured five ways, not five problems: vddac_pp, the raw dnl_min, load_reg, dvddac_load and zout_max are all the same output impedance. load_reg and zout_lf correlate at $r = 0.97$ across the 200 samples, and vddac_pp against the raw dnl_min at $r = -0.94$.

8.10.2 Matching

R-2R linearity is a matching property, so it is the Monte Carlo run and not the corner set that measures it. On an ideal 2.5 V rail, over 200 samples varying process and mismatch together, INL is 1.49 LSB mean against a 1 LSB limit and the worst sample reaches 3.45 LSB; DNL reaches -6.61 LSB (Table 87). 21.0 % of samples meet $\text{INL} < 1 \text{ LSB}$ and 40.0 % are monotonic ($\text{DNL} > -1 \text{ LSB}$); 17.0 % meet both, which over four independent channels is a 0.08 % chip yield.

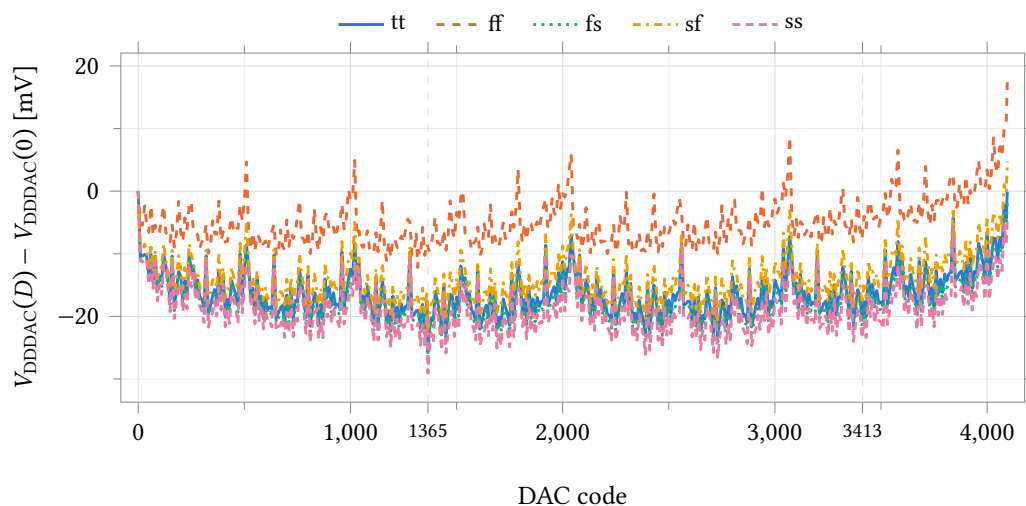


Figure 77: Ladder reference V_{DDAC} referred to its value at code 0, against DAC code, over five process corners, DacWPSUTB at 25 °C with the external load current set to zero. The ladder is the only load on the supply block and its current is code-dependent, so the reference the ladder divides moves 25.7 mV to 29.6 mV peak-to-peak with no external load at all; the absolute levels, which differ by 357 mV across corners, are Table 85. The two marked codes are alternating-bit patterns: 1365 = 0b010101010101 is the global minimum at four of the five corners (1317 at ff), and 3413 = 1365 + 2048 droops 23.4 mV at 2.5× the code weight and therefore sets the worst raw INL of Figure 78. Reference droop times code accounts for −36.9 of the −37.1 LSB measured there, so the raw nonlinearity of the delivered block is reference modulation and not a ladder error.

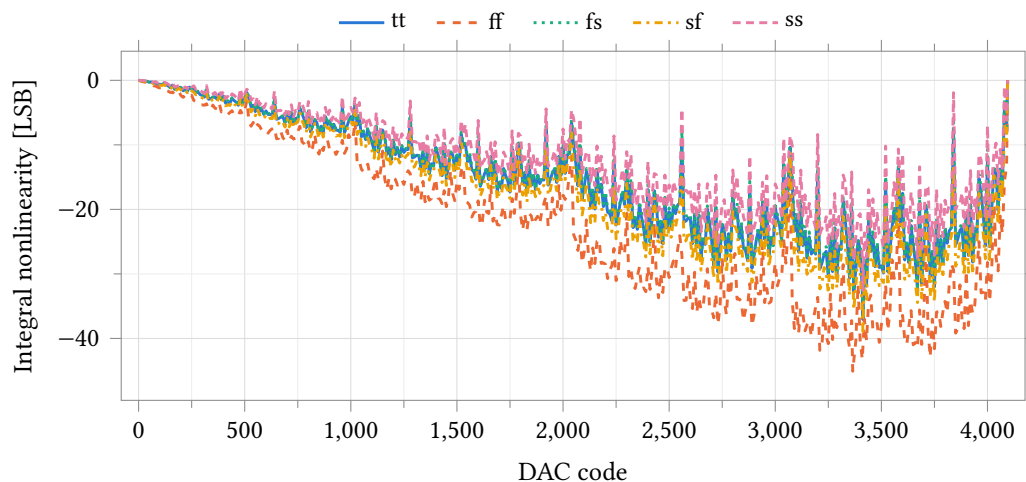


Figure 78: Integral nonlinearity against code over five process corners with the ladder referenced to V_{ddDac} from the supply block, zero external load, 25 °C. Endpoint-line INL as in Figure 74; the vertical scale is tens of LSB, not tenths. The bow reaches −37.1 LSB at tt (code 3413) and −45.1 LSB at ff (code 3365) and tracks the reference droop of Figure 77 scaled by code. Dividing each code's output by the reference measured at that code removes it and leaves 0.43 to 0.45 LSB (Table 86), so the ladder is unchanged from the ideal-rail case.

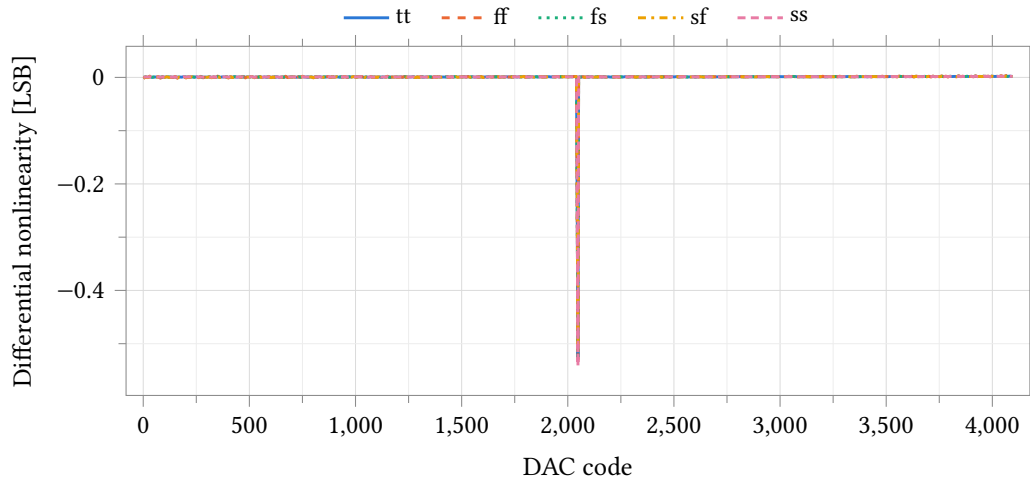


Figure 79: Ratiometric differential nonlinearity against code over five process corners: each code’s output is divided by the reference measured at that code before the LSB and the first difference are taken, so the reference droop divides out and the ladder alone is measured. Conditions as Figure 78; the 4095-point curve is stride-decimated for the plot as in Figure 75. The major-carry hierarchy is the ideal-rail one — -0.53 at code 2048, then -0.28 at 1024, -0.24 at 3072, -0.14 at 512 and -0.13 LSB at 2560 — and the worst step matches Figure 75 to within 0.03 LSB. Raw DNL on the same bench is -9.8 to -18.4 LSB (Table 86); the difference is the reference step between adjacent codes and nothing else.

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Full-scale output	V	2.16484	1.99313	2.10312	2.22263	2.34569	0.35256
LSB size	μV	528.646	486.684	513.576	542.747	572.810	86.125
INL, worst (raw)	LSB	37.064	45.087	37.145	38.890	32.546	12.541
DNL, most negative (raw)	LSB	-10.864	-18.356	-11.542	-9.765	-10.836	8.592
INL, worst (ratio)	LSB	0.442	0.450	0.438	0.443	0.432	0.018
DNL, most negative (ratio)	LSB	-0.525	-0.530	-0.525	-0.518	-0.543	0.024
DNL, worst (ratio)	LSB	0.525	0.530	0.525	0.518	0.543	0.024
Zero-scale offset (ratio)	LSB	0.077	0.325	0.058	0.139	0.051	0.273
Gain error (ratio)	ppm	-41.3	-170.9	-46.0	-56.0	-30.6	140.2

Table 86: Static accuracy with the ladder referenced to V_{ddDac} from the supply block, DacWPSUTB , zero external load, 25°C , all 4096 codes, $\text{reltol} = 1 \times 10^{-6}$. The first four rows are the block as delivered, reference included; the last five divide the output by the measured V_{DDDAC} at each code and so measure the ladder alone. The pair attributes the error: raw INL of 32.5 to 45.1 LSB collapses to 0.43 to 0.45 LSB ratiometric and raw DNL of -9.8 to -18.4 LSB to -0.52 to -0.54 LSB, so 99% of the raw nonlinearity is the reference modulation of Figure 77. Gain error is referred to the ideal ratiometric full scale $4095/4096 = 0.999756$; it is not an absolute-voltage gain error, which the line regulation of Table 93 dominates. Corner run, no device mismatch.

The failure is matching and nothing else. Full scale, LSB and gain error are set by the rail and by ladder attenuation, not by device ratios, and they barely move across the same 200 samples: σ is $1.8\mu\text{V}$ on a 2.499 V full scale and 0.5 ppm on a -76.9 ppm gain error (Table 88). All of the spread in Table 87 is therefore resistor mismatch in the ladder.

Referencing the ladder to VddDac instead of an ideal rail does not change this. The ratiometric figures of Table 89 — 1.51 LSB mean INL, -1.75 LSB mean DNL, the same 40.0% monotonic — reproduce the ideal-rail figures to within 2% , because dividing the reference out removes the supply block and leaves the same ladder. The raw columns beside them, 36.8 LSB and -10.8 LSB , carry the reference movement of Section 8.10.1 on top. The two defects are additive and independent, and either alone fails the block.

The corner set sees none of it. Ratiometric INL spans 0.432 LSB to 0.450 LSB across ff, fs, sf and ss — a 4% spread — while the Monte Carlo samples of the same quantity scatter from 0.35 LSB to 3.49 LSB . A process corner moves every ladder resistor in the same direction, which a ratio cancels; the corner table is context and not a linearity result.

INL and DNL are both skewed (Fisher skew 0.74 and -1.11), so the tables quote the median, the first and ninety-ninth percentiles and the worst sample rather than $\text{mean} \pm 3\sigma$.

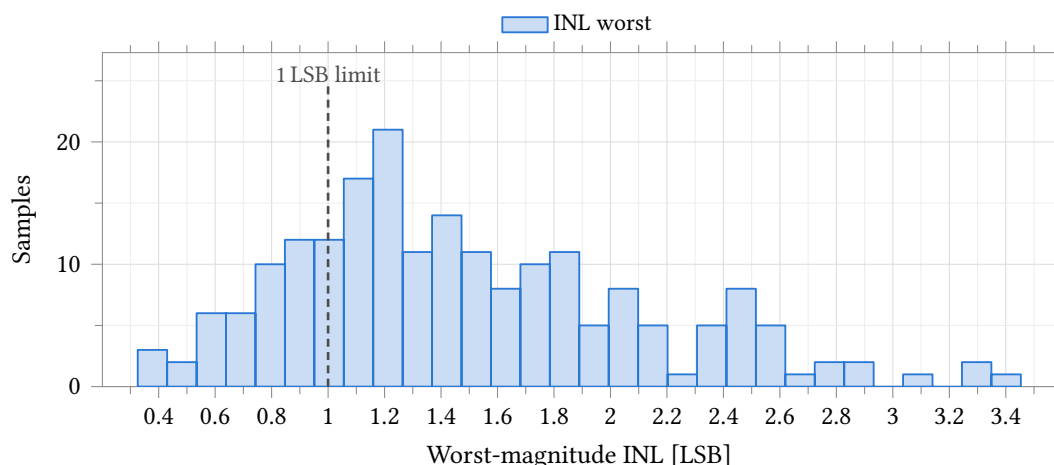


Figure 80: Integral nonlinearity of the R - $2R$ ladder driven from an ideal 2.5 V reference. **Process and mismatch**, model section `castalia_pm_25`, 200 samples, seed 12345, low-discrepancy sampling, 25°C , all 4096 codes. The dashed rule is the 1 LSB limit entered on the test; 42 of 200 samples fall inside it and the worst reaches 3.453 LSB . The distribution is right-skewed (Fisher skew $+0.74$), so Table 87 quotes the median and percentiles rather than $\text{mean} \pm 3\sigma$. LSB size, full scale and gain error are fixed to a part in 10^6 on this bench (Table 88), so the whole of this spread is resistor mismatch in the ladder and none of it is the reference.

8.10.3 Scope of these measurements

All data in this section is schematic-level. Layout and `av_extracted` views exist for both cells and have not been simulated, so no number here carries interconnect parasitics — which matter for an R - $2R$ ladder, whose accuracy is a ratio of resistances that layout can degrade.

Settling time and glitch impulse were not measured. The code bus is driven by a `busset12` model that holds a static code, so every measurement in this section is a sequence of DC solves. Code-step transients, and with them settling time, glitch impulse and the major-carry glitch that an R - $2R$ is prone to, require a different stimulus and are not characterised.

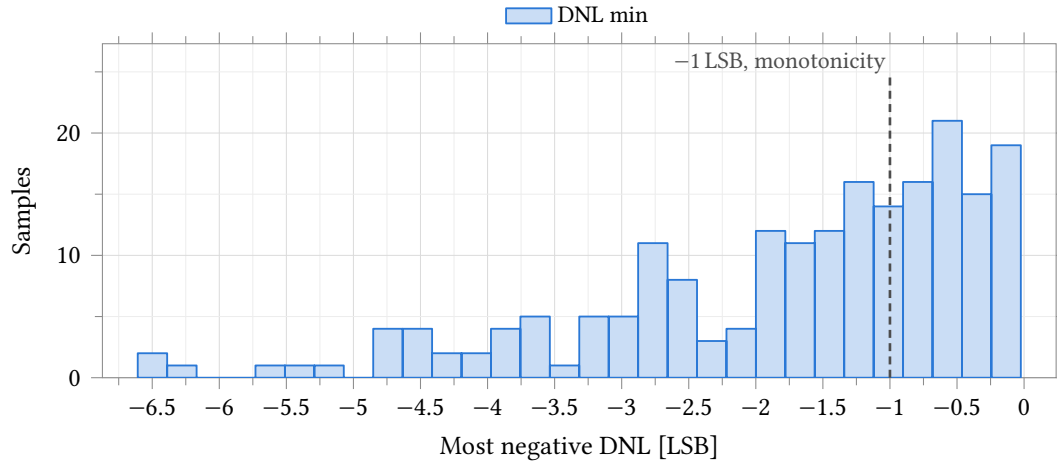


Figure 81: Worst negative differential nonlinearity, same bench, samples and conditions as Figure 80. The dashed rule at -1 LSB is the monotonicity limit entered on the test: 80 of 200 samples clear it, so 40 % of channels are monotonic and $0.40^4 = 2.6$ % of four-channel chips are monotonic on all four. The distribution is left-skewed (skew -1.11) with a tail to -6.608 LSB, so the median -1.318 LSB and the percentiles of Table 87 describe it and mean $\pm 3\sigma$ does not.

Parameter	Unit	N	Mean	σ	Median	p_1	p_{99}	Worst	Spec	Yield	4 ch
INL worst	LSB	200	1.490	0.625	1.363	0.397	3.299	3.453	≤ 1.000	21.0	0.2
INL max	LSB	200	1.165	0.616	1.028	0.171	2.942	–	–	–	–
INL min	LSB	200	-1.490	0.625	-1.363	-3.299	-0.397	–	–	–	–
DNL min	LSB	200	-1.737	1.437	-1.318	-6.240	-0.062	-6.608	≥ -1.000	40.0	2.6
DNL max	LSB	200	1.078	1.109	0.663	0.007	4.655	–	–	–	–
DNL worst	LSB	200	2.267	1.326	1.915	0.415	6.240	6.608	≤ 0.750	7.0	0.0

Table 87: Monte Carlo static linearity of the R - $2R$ ladder on an ideal 2.5 V reference. **Process and mismatch**, model section `castalia_pm_25`, 200 samples, seed 12345, low-discrepancy sampling, 25°C , all 4096 codes, $\text{reltol } 1 \times 10^{-6}$; the yield columns are per cent of samples and of four independent channels. The ladder fails on its own: 42 of 200 samples hold INL below 1 LSB, 80 of 200 are monotonic, and both together on 34 of 200 — 17.0 % of channels and 0.08 % of four-channel chips. Only the three worst-case rows carry limits entered on the test; `inl_max`, `inl_min` and `dnl_max` were not graded and their spec, yield and worst-case cells are therefore empty. Worst-magnitude INL is the negative extreme on all 200 samples, so the first and third rows are one measurement with opposite sign, while `inl_max` is the independent positive extreme. All three graded rows are skewed (Figures 80 and 81), so the median and percentiles are quoted and mean $\pm 3\sigma$ is not.

Parameter	Unit	N	Mean	σ	Min	Max
LSB size	μV	200	610.3046	0.0005	610.3040	610.3050
Full-scale output	V	200	2.499290	0.000002	2.499280	2.499290
Zero-scale offset	LSB	200	0.1474	0.0000	0.1474	0.1474
Gain error	ppm	200	-76.91	0.50	-78.15	-75.48

Table 88: Monte Carlo scale factors, same bench and samples as Table 87. **Process and mismatch**, `castalia_pm_25`, 200 samples, 25 °C. On an ideal reference these four are set by the rail and by the ladder’s attenuation ratio, not by matching: LSB size spreads 0.5 nV on 610.3 μV , full scale 1.8 μV on 2.5 V, and gain error 0.50 ppm on -76.91 ppm. All of the linearity spread of Table 87 is therefore ladder mismatch and none of it is a gain or offset term. No limit was entered on any of these outputs, so the spec, yield and worst-case columns are omitted rather than left blank; mean $\pm 3\sigma$ is not quoted either, because at a part in 10^6 the distribution shapes are solver residue and not device statistics.

Parameter	Unit	N	Mean	σ	Median	p_1	p_{99}	Worst	Spec	Yield	4 ch
INL worst	LSB	200	1.514	0.627	1.388	0.418	3.326	3.491	≤ 1.000	20.5	0.2
DNL min	LSB	200	-1.747	1.441	-1.324	-6.252	-0.063	-6.629	≥ -1.000	40.0	2.6
DNL worst	LSB	200	2.271	1.328	1.913	0.412	6.252	6.629	≤ 0.750	7.0	0.0
INL worst, raw	LSB	200	36.776	2.308	36.911	30.562	41.460	-	-	-	-
DNL min, raw	LSB	200	-10.816	0.833	-10.837	-12.522	-8.906	-12.700	≥ -1.000	0.0	0.0

Table 89: Monte Carlo linearity with the ladder referenced to `VddDac` from the on-chip supply block: ratiometric rows first, raw rows below. **Process and mismatch**, `castalia_pm_25`, 200 samples, 25 °C, all 4096 codes, no external load on `VddDac`; yields are per cent of samples and of four channels. The ratiometric rows normalise each code to the reference at that code and so measure the ladder alone; they reproduce Table 87 sample for sample, at $r = 1.000$ on both INL and DNL, and their yields are 20.5 % and 40.0 % against 21.0 % and 40.0 % there. The raw rows include the reference and are 24 and 6 times worse; the run’s own `inl_from_ref`, which is the INL predicted from the reference error alone, averages 36.770 LSB against the raw 36.776 LSB, so the reference accounts for essentially all of the difference and no second matching mechanism is needed to explain it. Raw `inl_lsb` carries no entered limit and its spec, yield and worst-case cells are empty; raw `dnl_min` was graded and no sample meets it. The three ratiometric rows are skewed (+0.74, -1.10, +0.89), so the median and percentiles are quoted throughout.

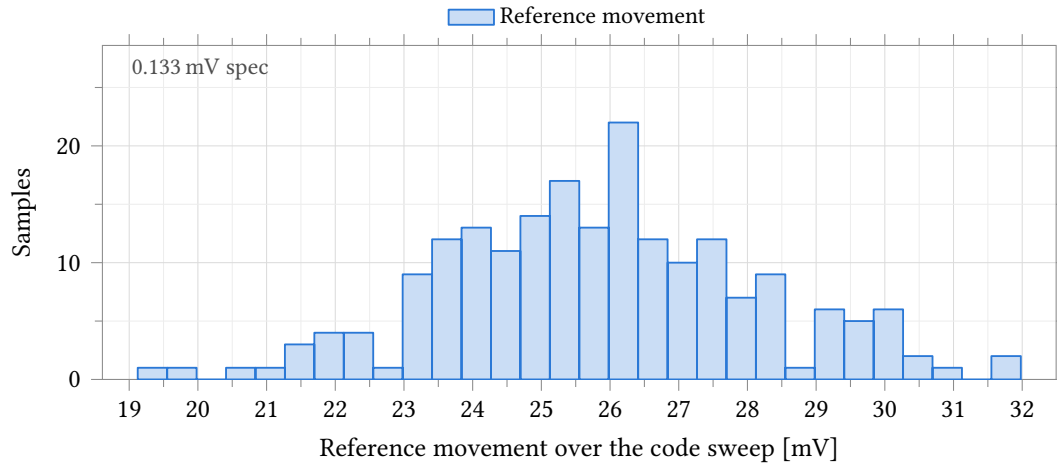


Figure 82: Peak-to-peak movement of the DAC reference $V_{dd}DAC$ across the 4096-code sweep, with the ladder referenced to the on-chip supply block instead of an ideal rail. **Process and mismatch**, `castalia_pm_25`, 200 samples, 25 °C, no external load on $V_{dd}DAC$. The quarter-LSB budget of 133 μV entered on the test sits about 200 times below the data, so it is annotated at the left axis edge rather than drawn and the distribution is not compressed into a single bar; the closest sample misses it by a factor of 144. Meeting it needs the supply's output impedance down by a factor of 193, to about 5 Ω , against 894 Ω small-signal (Table 96) and 1133 Ω as a 0 μA to 100 μA chord (Table 92).

Parameter	Unit	N	Mean	σ	Mean $\pm 3\sigma$	Min	Max	Spec	Yield	4 ch
Reference movement	mV	200	25.835	2.302	25.835 $\pm$ 6.905	19.124	31.980	≤ 0.133	0.0	0.0
Reference min	V	200	2.1412	0.0554	2.1412 $\pm$ 0.1663	1.9821	2.3066	–	–	–
Reference max	V	200	2.1671	0.0571	2.1671 $\pm$ 0.1712	2.0030	2.3348	–	–	–
LSB size	μV	200	528.96	13.88	528.96 $\pm$ 41.64	488.91	569.53	–	–	–
Full scale	V	200	2.1661	0.0568	2.1661 $\pm$ 0.1705	2.0021	2.3323	–	–	–

Table 90: Monte Carlo behaviour of the $V_{dd}DAC$ reference itself, same bench and samples as Table 89; only `vddac_pp` carries an entered limit, so the other four rows have empty spec, yield and worst-case cells. **Process and mismatch**, `castalia_pm_25`, 200 samples, 25 °C; yields are per cent of samples and of four channels. Full scale is 2.166 V and the LSB 529.0 μV rather than 2.5 V and 610.3 μV , because the supply block delivers about 2.15 V; the reference then moves 25.8 mV peak to peak across the code sweep against the 133 μV quarter-LSB budget, which no sample meets. The excursion is not monotonic in code: in an $R-2R$ ladder the binary weighting comes from attenuation, so every $2R$ branch switched to the reference draws a comparable current and the reference current follows the bit pattern rather than the code value — a nodal solve of this topology gives $I_{ref} = 0$ at code 0, $0.250 V_{ref}/R$ at 2048 and 0.333 at 4095, with a maximum of 1.444 at code 1365 (0b010101010101), which is exactly where `vddac_min` lands on all 200 samples, and a one-parameter fit $V = V_0 - R_{out}I_{ref}$ accounts for 94 % of the excursion. All five distributions are near-symmetric (skew ≤ 0.13), so mean $\pm 3\sigma$ applies.

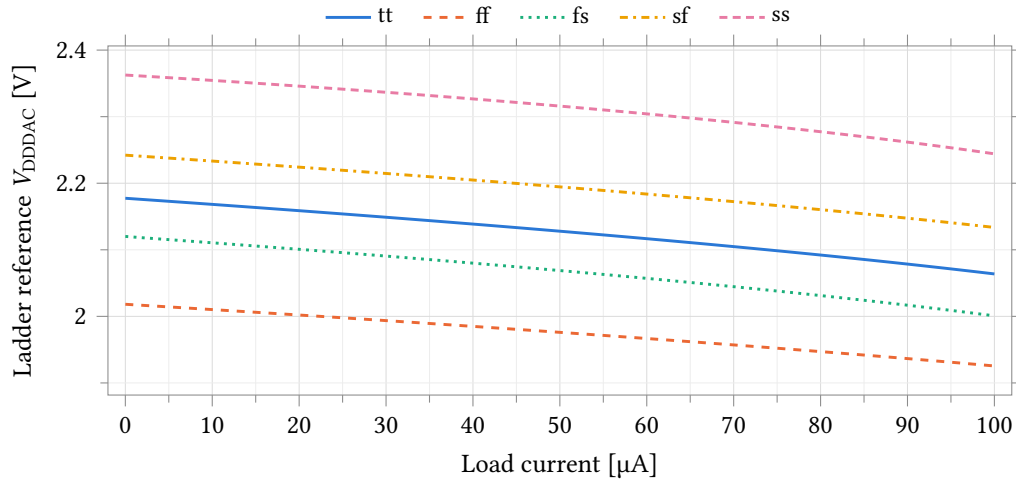


Figure 83: V_{DDAC} against external load current over five process corners, supply block alone, 27 °C at point 1 and 25 °C at points 2–5, ADE default tolerances. There is no feedback loop around the output, so the slope is the block’s own output resistance and the reference falls 92.7 mV to 119.3 mV across the 0 μ A to 100 μ A sweep. The curve steepens with load, which is why the chord in Table 91 exceeds the small-signal $|Z_{out}|$ of Table 95. This sweep is an external load only: the ladder is not connected on this bench, and its own code-dependent draw is Figure 77.

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Reference at zero load	V	2.1774	2.0182	2.1201	2.2422	2.3625	0.3444
Reference at 100 μ A	V	2.0638	1.9255	2.0008	2.1336	2.2442	0.3187
Load-induced drop	mV	113.667	92.664	119.287	108.585	118.361	26.623
Load regulation	Ω	1136.7	926.6	1192.9	1085.9	1183.6	266.2

Table 91: Supply-block load regulation by process corner: V_{DDAC} against an external load current swept 0 μ A to 100 μ A, DacPSUTB, 27 °C at point 1 and 25 °C at points 2–5, ADE default tolerances. Load regulation is the chord across the whole sweep, not the slope at the origin, and runs 18 % to 51 % above the small-signal $|Z_{out}|$ of Table 95. The ladder is not connected on this bench; its own code-dependent current is Figure 77. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	N	Mean	σ	Mean $\pm 3\sigma$	Min	Max	Spec	Yield	4 ch
Reference, no load	V	200	2.1785	0.0595	2.1785 $\pm$ 0.1784	2.0053	2.3495	–	–	–
Reference, full load	V	200	2.0653	0.0593	2.0653 $\pm$ 0.1778	1.9003	2.2462	–	–	–
Load droop	mV	200	113.281	9.335	113.281 $\pm$ 28.006	90.214	138.458	≤ 0.133	0.0	0.0
Output resistance	Ω	200	1132.8	93.4	1132.8 $\pm$ 280.1	902.1	1384.6	≤ 3.0	0.0	0.0

Table 92: Monte Carlo load regulation of the DAC supply block. **Process and mismatch**, castalia_pm_25, 200 samples, 27 °C, VddDac loaded 0 μ A to 100 μ A, ADE default tolerances – looser than the two DAC benches, which run 25 °C and $reltol\ 1 \times 10^{-6}$; yields are per cent of samples and of four channels. VddDac droops 113.3 mV over that sweep, a chord output resistance of 1132.8 Ω against the 3 Ω limit entered on the test; no sample meets either graded limit, and the two reference-voltage rows were not graded. The last two rows are one measurement scaled by the load current, and both are near-symmetric (skew +0.29), so mean $\pm 3\sigma$ applies. Together with the reference movement of Table 90 and the small-signal impedance of Table 96 this is one defect measured three ways, not three problems.

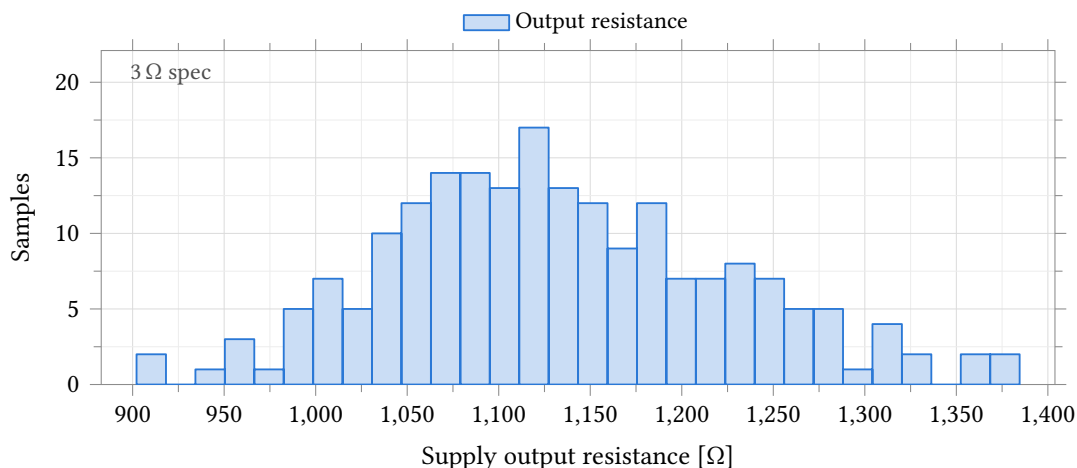


Figure 84: Output resistance of the DAC supply block, taken as the chord $\Delta V / \Delta I$ over a $0 \mu\text{A}$ to $100 \mu\text{A}$ load sweep on V_{DDDAC} . **Process and mismatch**, castalia_pm_25, 200 samples, 27°C . The 3Ω limit entered on the test is two orders of magnitude below the data and is annotated at the left axis edge rather than drawn; no sample meets it. This is not a second defect: the reference movement of Figure 82 is this impedance multiplied by the ladder's own code-dependent reference current.

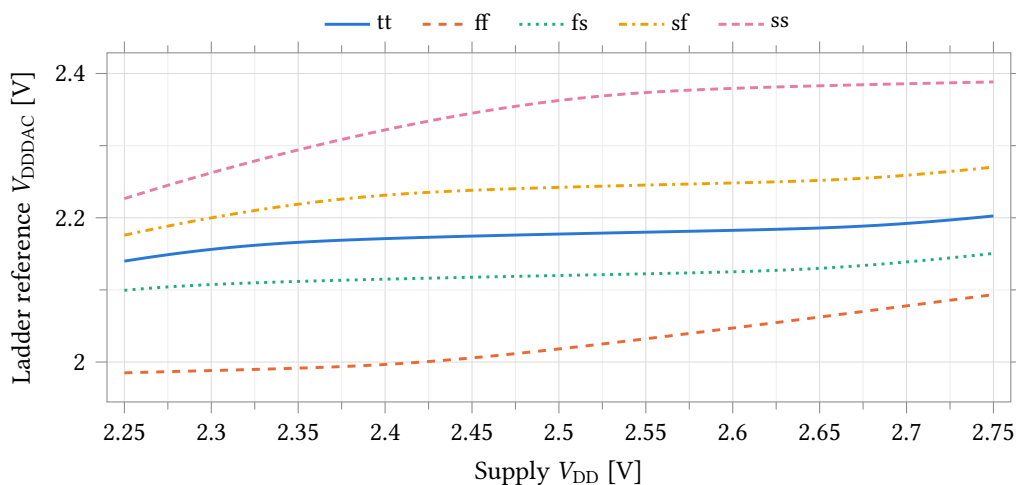


Figure 85: V_{DDDAC} against the supply, swept 2.25 V to 2.75 V , over five process corners at zero external load, 27°C at point 1 and 25°C at points 2–5. The slope is the line regulation tabulated in Table 93; it is not constant across the sweep, so the tabulated value is the chord between the two endpoints. The corner spread of the reference itself, 344 mV at nominal supply, is larger than anything the supply sweep produces.

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Reference at 2.5 V	V	2.1774	2.0182	2.1200	2.2422	2.3625	0.3444
Reference change, 2.25 V to 2.75 V	mV	62.67	108.46	51.11	94.46	161.70	110.59
Line regulation	V V ⁻¹	0.1253	0.2169	0.1022	0.1889	0.3234	0.2212
Quiescent supply current	μA	158.05	194.10	152.07	161.96	129.18	64.91

Table 93: Supply-block line regulation and quiescent current by process corner: V_{DDDAC} with the supply swept 2.25 V to 2.75 V at zero external load, DacPSUTB, 27 °C at point 1 and 25 °C at points 2–5. With no feedback loop, 0.102 V V⁻¹ to 0.323 V V⁻¹ of any supply movement reaches the reference, and every absolute output voltage of the DAC moves with it; the ratiometric rows of Table 86 are what survives. The current is read at the bench supply terminal and covers the supply block only — the ladder is not connected on this bench. The quiescent-current column is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	N	Mean	σ	Median	p ₁	p ₉₉	Worst	Spec	Yield	4 ch
Line shift	mV	200	78.750	23.770	75.740	38.686	144.474	157.026	≤ 5.000	0.0	0.0
Line regulation	V V ⁻¹	200	0.1575	0.0475	0.1515	0.0774	0.2889	–	–	–	–
Quiescent current	μA	200	159.56	12.97	159.41	134.34	193.35	–	–	–	–
Dissipation	μW	200	398.89	32.42	398.52	335.85	483.38	–	–	–	–

Table 94: Monte Carlo line regulation and quiescent power of the DAC supply block. **Process and mismatch**, castalia_pm_25, 200 samples, 27 °C, supply swept 2.25 V to 2.75 V, VddDac unloaded, ADE default tolerances; yields are per cent of samples and of four channels. VddDac follows the supply by 78.8 mV over that 0.5 V span, a rejection of only 0.158 V V⁻¹, against the 5 mV limit entered on the test that no sample meets. Only dvddac_line was graded; the other three rows have empty spec, yield and worst-case cells, the last two being the same measurement scaled by the 2.5 V supply. All four are right-skewed (+0.58, +0.58, +0.45, +0.45), so the median and percentiles are quoted and mean ± 3σ is not.

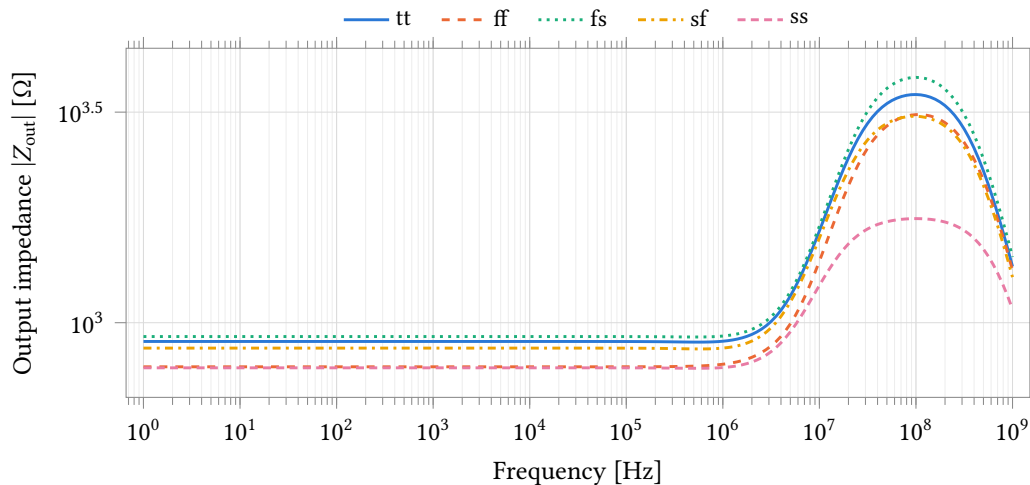


Figure 86: Magnitude of the supply block’s output impedance against frequency over five process corners, 27 °C at point 1 and 25 °C at points 2–5. The AC load source carries unit magnitude, so $|V_{DDDAC}|$ is the impedance in ohms. It is flat at 781 Ω to 927 Ω from 1 Hz to 100 kHz and peaks at 1.8 kΩ to 3.8 kΩ near 100 MHz. This is the small-signal impedance at zero load; the large-signal chord over the 0 μA to 100 μA sweep is 18 % to 51 % higher (Table 91).

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Output impedance at 1 Hz	Ω	902.5	785.9	927.1	870.3	781.5	145.6
Output impedance at 1 kHz	Ω	902.5	785.9	927.1	870.3	781.5	145.6
Output impedance at 100 kHz	Ω	902.2	786.0	926.9	870.0	781.3	145.6
Output impedance, peak	Ω	3482.1	3120.4	3825.2	3092.0	1766.9	2058.3
Frequency of the peak	MHz	100.0	112.2	100.0	100.0	100.0	12.2

Table 95: Supply-block small-signal output impedance by process corner, DacPSUTB AC sweep from 1 Hz to 1 GHz at 20 points per decade, 27 °C at point 1 and 25 °C at points 2–5. The AC load source carries unit magnitude, so $|V_{DDDAC}|$ reads directly in ohms. The impedance is flat to 100 kHz, within 0.3 Ω of the 1 Hz value at every corner, and peaks three decades higher in frequency at two to four times the flat value. Grid points are 12 % apart, so the peak frequency is located only to that resolution and the ff entry is one grid step from the other four. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 69).

Parameter	Unit	N	Mean	σ	Median	p_1	p_{99}	Worst	Spec	Yield	4 ch
$Z_{out, LF}$	Ω	200	894.3	46.6	893.8	779.4	990.8	–	–	–	–
$Z_{out, peak}$	Ω	200	3402.1	299.5	3442.8	2369.2	3853.9	3872.2	≤ 10.0	0.0	0.0
Peak frequency	MHz	200	98.94	6.23	100.00	89.13	112.20	–	–	–	–

Table 96: Monte Carlo small-signal output impedance of the DAC supply block, AC swept 1 Hz to 1 GHz. **Process and mismatch**, castalia_pm_25, 200 samples, 27 °C, VddDac unloaded, ADE default tolerances; yields are per cent of samples and of four channels. The impedance is flat at 894.3 Ω to within 0.4 Ω from 1 Hz to 100 kHz, and this is the value that sets the reference movement of Table 90, the ladder’s code-dependent current being a DC excursion. $zout_max$ is the peak over the whole band and lands at 98.9 MHz, loop peaking well above anything the ladder excites; it is the only graded row here and no sample meets its 10 Ω limit. $zout_max$ is strongly left-skewed (skew -1.82), so the median and percentiles are quoted; $zout_lf$ and the peak frequency carry no entered limit and their spec, yield and worst-case cells are empty.

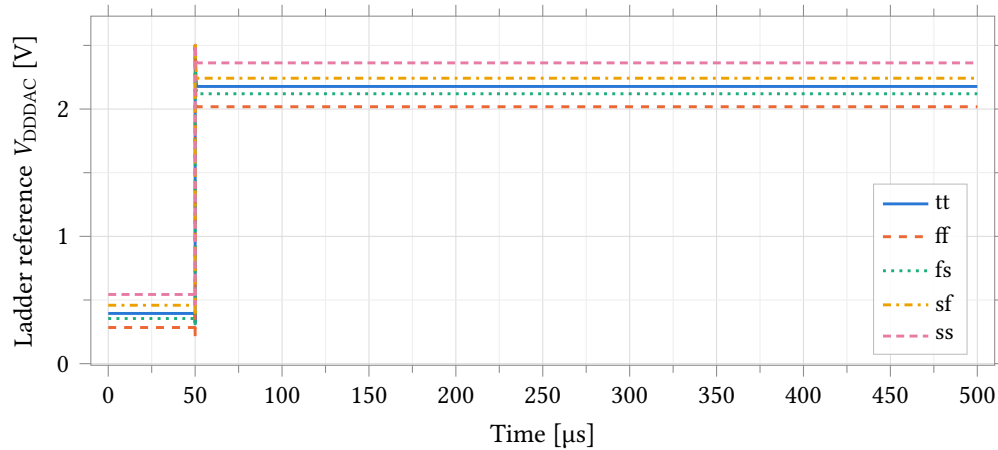


Figure 87: V_{DDDAC} against time over five process corners, PSUStartupTB: En rises at 50 μs into a 500 μs window at zero external load, 27 $^{\circ}\text{C}$ at point 1 and 25 $^{\circ}\text{C}$ at points 2–5. With En low the node sits at 0.28 V to 0.54 V; at the 1 ns enable edge the pass device is driven hard on and the node is pulled to the 2.5 V supply rail for 64 ns to 100 ns before falling back. It reaches the zero-load DC value of Table 91 to within 140 μV at every corner and holds it to the end of the window. No settling-time or overshoot figure of merit is published from this bench: the design variable t_{win} sets both the En pulse width and the settlingTime evaluation window, so the two cannot be specified independently.

No settling or overshoot figure of merit is quoted for the supply block. The corner run’s startup transient is valid and is plotted in Figure 87: $V_{\text{dd}Dac}$ reaches its zero-load DC value to within 140 μV at every corner and holds it. What cannot be published from it is a number, because the bench’s t_{win} variable sets both the En pulse width and the settlingTime evaluation window, so the two cannot be specified independently. The Monte Carlo run of the same test is separately unusable: its enable pulse is delayed 1 s against a 500 μs window, so En never asserts and every startup output it reports grades the disabled state. Note from the figure that the reference is pulled to the full 2.5 V supply rail for 64 ns to 100 ns at the enable edge; any code held through an enable event sees that excursion.

Disabled-state supply current was not measured in this run. The enable bench is not part of the corner or Monte Carlo runs reported here. The last measurement of it, 59.45 μA against a 1 μA target, comes from a superseded setup and is not quoted as a result; it is a separate finding from the reference problem of Section 8.10.1 and needs re-measuring.

Which supply the ladder was referenced to is stated on every table and figure: the DacTB results are against an ideal 2.5 V rail and characterise the ladder, and the DacWPSUTB results are against $V_{\text{dd}Dac}$ and characterise the block as delivered. The corner run carries no mismatch information and the Monte Carlo run varies process and mismatch together; the two are different quantities and are never totalled.

8.10.4 Measuring this block on chip

1. Select per-channel slot 2 with $\text{AFE_DBGISOEN} = 1$ and read the reference output on atb_d . The isolation switch is required, not optional: leakage on an R-2R output is a code-dependent error.
2. Absolute anchor: take three codes with a voltmeter to fix the DAC’s offset and step in volts. Nothing on the die can supply that scale; it anchors the channel’s whole potential axis.
3. Sweep all 4096 codes into the converter with $\text{AFE_ADCSRCSEL} = 2$ (75 ms per channel at sixteen averages and the design conversion rate, four times that on the clock pattern as delivered — Section 8.8.2) and fit the one-parameter droop of Section 8.10.1: it removes 94 % of the 25.8 mV reference move-

ment. Do not store the per-code residual table — through this converter each entry carries as much quantisation as the error it corrects (0.07 mV rms for 4 KiB per channel) — but run the sweep regardless: it is where a broken ladder shows.

4. Chip-level slot 7 reads the DAC supply node directly — the only place the 25.8 mV code-dependent excursion is confirmed or refuted on silicon. Chip-level slot 6 reads the shared common-mode DAC, forceable through AFE_VCMISOEN.

This test fabric is design intent; it is not yet in the schematics.

8.11 Per-Chip Calibration

Every correction is a stored digital constant except one, the bias generator's 6-bit trim code. Table 97 is the complete set of knobs: what each corrects, how it is measured, and when it is applied. The measurement itself is described in each block's section, under *Measuring this block on chip*.

Only two constants are absolute — one voltage at the reference DAC output, one current forced at the working electrode — and both need the production tester. The other eight are zeros and ratios the channel measures against itself through its own converter, which is what lets the power-up refresh run with no equipment.

Order matters twice. At test, apply the bias trim before anything else is measured: it moves every transconductance behind every constant that follows. In a reading, subtract the zero code, scale by the gain constant for the active gain code, then subtract the drop across the working-electrode switch and, where an impedance sweep has supplied it, the uncompensated electrolyte resistance. Potential-axis corrections apply where the waveform is tabulated, not per sample.

Residual. Each zero is the difference of two conversions, so its floor is 0.65 mV rms, two quantisations of the converter's 1.591 mV step; the common-mode DAC has to dither the input to reach it. On the potential axis the residual is 1.38 mV rms — that floor, the 0.93 mV droop-fit residual and 0.79 mV of uncorrected ladder INL in quadrature — against 5.71 mV of uncalibrated σ . That holds over the specified 0.4 V to 2.1 V window only: outside it the tracking error is no longer a constant and no stored constant removes it (Section 8.6.9). On the current axis it is 0.71 mV rms — 0.45 LSB on the converter's measured 1.591 mV step (Table 73) — plus 0.1 % to 0.4 % of reading, against 12.1 LSB to 167.3 LSB uncalibrated, which are on the nominal 2.441 mV step. A constant removes the current zero only while that zero stays inside the converter window, which requires $R_f/R_{\text{cell}} < 47.5$; the 560 k Ω tap into a 10 k Ω electrode (Section 8.5.6) is outside it and those samples are recoverable by nothing.

Out of reach. The reference DAC's -6.6 LSB of DNL is lost command resolution, up to 3.5 mV at the worst code, and no constant creates a code the ladder does not produce. A reference-electrode junction potential is indistinguishable from A1's offset, so only a zero taken *wet* against that electrode absorbs it. The temperature dependence of every constant here is unquantified: no block in this chapter has a temperature-resolved Monte Carlo, so how far a power-up re-measurement tracks a 37 °C operating point is not known.

Constant	Scope	Corrects	Measured by	Applied
Bias trim code, 6 bit	chip	Bias-current spread, $\sigma/\mu = 12.5\%$ in MC	Step the code; keep the one whose total analog supply current is nearest design (Tab. 21)	Test; reloaded at power-up before anything else is measured
Reference DAC offset and step	channel	Absolute potential scale; full scale 2.166 V, $\sigma = 56.8$ mV (Tab. 90)	Tester voltmeter at three codes, direct test pad	Test only
Reference droop parameter	channel	25.8 mV pp movement of the reference with code	4096-code sweep into the converter; one-parameter fit (§8.10.1)	Test only; applied where the waveform is built
Converter zero code	channel	Converter offset and the common-mode value together	Convert the common-mode rail and average	Test, power-up, and between potential steps
A1 offset and tracking slope	channel	Applied-potential error, $\sigma = 5.20$ mV (Tab. 30)	A1 in unity-gain loopback, three reference codes (§8.4)	Test and power-up; applied where the waveform is built
Current zero, per gain code	channel	Zero-current offset, $\sigma = 12.1\text{--}167.3$ LSB (§8.5.6), that row on the nominal 2.441 mV step	Working electrode isolated; read at each gain code	Test and power-up
Transimpedance gain, per gain code	channel	$\pm 35\%$ resistor tolerance, switch stack and converter gain, in one number (Tab. 77)	Tester forces $\pm$ full-scale current at the working-electrode pad	Test; rescaled at power-up by the ratio below
Internal gain ratio, per gain code	channel	Drift of the gain constant	Same slope against an on-chip reference resistor (§8.9)	Power-up
Working-electrode switch R_{on}	channel	$i \cdot R_{on}$: 3.3 mV at 33 μ A through 100 Ω	Force current; read both sides of the switch	Test; applied per sample

Table 97: Per-chip calibration constants — the complete set of digital knobs. *Test* constants need external equipment and are written once; *power-up* constants are re-measured on-die at every start and overwrite the stored copy. At eight gain codes per channel the whole block is under 128 bytes per channel, so it is staged from flash with a channel hart’s image rather than read back during a measurement. LSB is the converter’s measured 1.591 mV step (Table 73) unless a row states otherwise. No calibration has been executed, in simulation or on silicon, and the isolation switches and test multiplexer these measurements need are design intent, not schematics.

9 Startup and Reset Behavior

To place the MCU in reset mode, drive the `resetn` pin LOW (asserted). To take the MCU out of reset mode, let the `resetn` pin go high (deasserted). The `resetn` pin has an internal 47 k Ω pullup resistor which constantly attempts to pull the `resetn` pin HIGH. If the user does not wish to use the `resetn` pin, it is safe to leave it unconnected.

When the MCU is first powered up, a power-on reset (POR) circuit resets the MCU. As the core voltage, beginning at 0 V, begins to climb, the POR circuit does not assert the internal reset. Once the core voltage reaches a particular value (which is temperature and process dependent, but typically around 0.7 V), the POR circuit asserts the internal reset signal for about 20 ns and then deasserts it. If either the `resetn` pin or the POR reset signal is asserted, the MCU will be in reset mode. In reset, the memory mapped registers and CPU state variables are all reset to their powerup values.

When the MCU comes out of reset, it immediately begins to execute the bootloader stored in the ROM, setting the program counter to address 0x0000. If the `BOOT` pin is LOW, the bootloader launches a Forth interpreter which communicates over UART0 at 115,200 baud (see Section 13 for details concerning the Forth interpreter). If the `BOOT` pin is HIGH, the bootloader reads and executes the program stored on an external SPI flash on SPI0 (see Section 15 for details on how to program the MCU). When the bootloader reads the program stored in the SPI flash, it dumps the contents of the SPI flash directly into RAM, completely filling the RAM from beginning to end using an address-to-address copy. After the bootloader has copied the program stored in the SPI flash into RAM, it jumps (moves the program counter) to address 0x8200 and begins executing the code.

Only hart 0 executes the boot ROM and the SPI-flash boot sequence described above. Harts 1–4 come out of reset executing from their private RAM and park until hart 0 releases them through the boot mailboxes in shared RAM; see Section 4.3.

Note that the boot ROM reprograms the clock system during boot, so the SMCLK source at application entry is a boot-ROM implementation detail; in particular, it may be left on the slow LFXT (32.768 kHz) source. Applications that use SMCLK-domain peripherals (the UARTs, SPIs, I²Cs, or a timer clock-source switch) should therefore always select their desired SMCLK source in the SYSTEM peripheral first, rather than relying on the boot-time clock state; see Section 20.1 and the clock handoff note in the TIMERx chapter.

10 Interrupts

Table 98: Interrupt Vectors

Priority	Interrupt Source
0	CPU Internal Timer
1	EBREAK, ECALL, or Illegal Instruction
2	Bus Error (unaligned memory access)
0	SYSTEM
1	GPI00
9	SPI0
11	SPI1
13	UART0
16	TIMER0
22	TIMER1
28	GPI01
36	GPI02
44	GPI03
52	UART1
57	I2C0
70	I2C1
83	CLINT
94	NFC0
98	GPI04
106	GPI05
120	NPU

The VestaRV CPU implements an interrupt system with 121 interrupt vectors, numbered 0–120, whose sources are enumerated in the table above. The numbering is frozen: a vector belongs to the block that owns it whether or not this configuration instantiates that block, so vectors whose block is absent are reserved and counted with the rest rather than shifting the ones above them. Two of the 121 are the CLINT pair (83, software/inter-processor interrupt; 84, timer interrupt) added for multi-core operation; the other 119 sit on the IRQROUTER side of the fabric — the peripheral and system sources, the reserved slots frozen among them, and meip’s own slot. Each hart takes three interrupt lines: its own CLINT software and timer interrupts (vectors 83 and 84, hardwired enabled, delivered on dedicated per-hart wires) and the external interrupt meip (vector 85, a slot of its own that no source pends on), which delivers *all* peripheral and system sources through the IRQROUTER’s claim/complete mechanism: the vector-85 handler reads the router’s CLAIM register to discover and claim the pending source, dispatches on the returned vector number, and writes it back to complete (see the IRQROUTER chapter). Which sources reach which hart is programmed per hart in the router’s routing rows, identical for every hart, hart 0 included. All peripheral sources are masked by default upon MCU reset and must be explicitly routed in software before use; source priority is fixed (the lowest pending vector number is claimed first). Additionally, peripherals generating interrupts must be configured to enable interrupt generation at the peripheral level.

Figure 88 is that whole fabric in one view: the source population of the table above collapsed by type, the IRQROUTER as the spine every one of those sources terminates in, and the five harts with the three lines each of them takes. It is the drawing to read before the vector table, because it is where the one asymmetry lives: meip arrives from the router, and msip and mtip do not go near it.

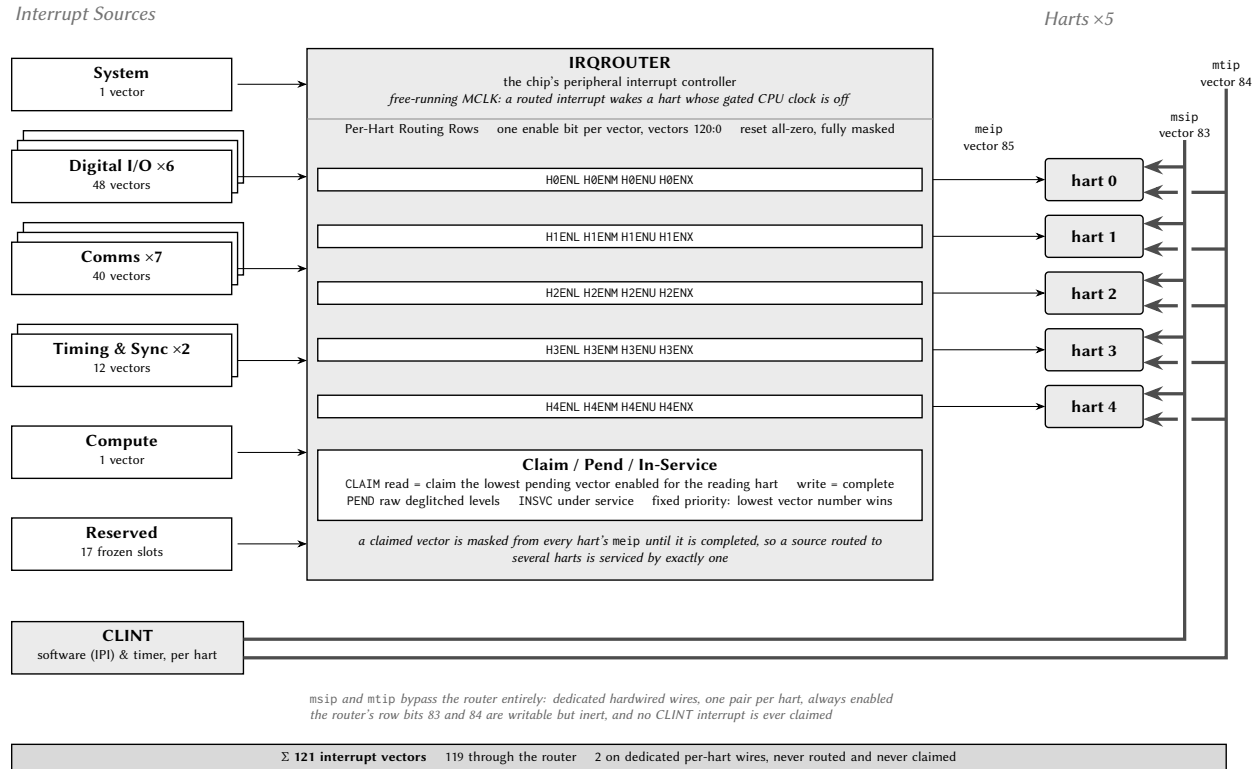


Figure 88: Castalia interrupt fabric. Every one of this configuration's 121 interrupt vectors is on this page: the left column collapses Table 98's source population into types, each with the number of blocks that feed it and the number of vectors it owns, and the reserved slots are counted with the rest because the vector numbering is frozen — a dropped or not-yet-built block keeps its slot rather than shifting the ones above it. All of those terminate in the IRQROUTER, which holds one routing row per hart — one enable bit per vector, all zero out of reset — and one claim/complete stage; each row is drawn at the height of the hart it belongs to, so a row's meip wire runs straight across into that hart. **The two heavy rails are the point of the figure:** the CLINT's msip and mtip are per-hart *hardwired* levels that run under the router, up the far margin, and into each hart from the opposite side. They are never enabled, never pending, never claimed and never completed; the router's row bits at those two vector numbers are writable but inert, and its CLAIM never returns them. A hart's interrupt service routine clears an msip by writing its MSIP register and an mtip by advancing its MTIMECMP, not by completing a claim (Section 24). Both blocks run on the free-running MCLK, which is what lets an interrupt wake a hart whose gated CPU clock is off. The register layout behind all of it is Section 27's; no address appears here. Everything drawn is generated from this configuration.

10.1 Interrupt Vector Table

The interrupt vector table (IVT) is located at the base of RAM, beginning at address `0x8000`. Unlike traditional pointer-based interrupt vector tables, the VestaRV IVT contains direct jump instructions to the interrupt service routines (ISRs). Each entry in the IVT is a 4-byte RISC-V jump instruction (e.g., `j handler`) that vectors execution immediately to the corresponding ISR when an interrupt occurs. This architecture eliminates the indirection overhead of loading function pointers and provides deterministic, low-latency interrupt response.

While the VestaRV hardware does not mandate a specific IVT structure, it is strongly recommended to populate the IVT with jump instructions during system initialization. The automatic code generation tools produce linker scripts and initialization code that configure the IVT at the beginning of RAM. The memory map header files provide macros to facilitate proper ISR placement and IVT entry configuration.

10.2 Interrupt Entry, Exit, and Recursion

When an interrupt is taken, the hardware saves a minimal context: it pushes the return program counter (one 32-bit word) onto the system stack at `sp - 4`, decrements `sp` by 4, and vectors execution through the IVT entry for the interrupt. **No other state is saved by hardware.** Software is responsible for saving and restoring every general-purpose register the interrupt service routine (ISR) uses. The generated runtime provides an interrupt entry wrapper (in the startup code) that saves the general-purpose registers on the stack (about 120 bytes per interrupt level), calls the C-level handler, restores the registers, and returns with the `retirq` instruction, so C-coded ISRs need no special handling. Hand-written assembly ISRs must perform the equivalent save/restore themselves. Executing `retirq` pops the hardware-saved program counter, increments `sp` by 4, and resumes execution at the interrupted location.

Because the very first thing interrupt entry does is a store at `sp - 4`, **the stack pointer must point at valid, writable private memory before interrupts are enabled.** This applies to every hart: a freshly released tile hart must load its stack pointer before unmasking interrupts or going to sleep (see Section 4).

Interrupt handling is non-recursive: a running ISR is never preempted, and further pending interrupts (including a pending CLINT interrupt during a vector-85 service) are taken after the current ISR returns. Stack allocation must cover one interrupt level: the 4-byte hardware-saved return address plus the handler's register frame (about 120 bytes for the runtime wrapper) and the ISR's local variables. (The single-core chip's priority-tier and recursion machinery was retired together with the SYSTEM interrupt registers; source arbitration now happens in the IRQROUTER.)

10.3 Sleep Mode and Interrupt Wake-Up

If an interrupt occurs while the CPU is in sleep mode, the processor will automatically wake from sleep and service the interrupt. Upon execution of the `retirq` instruction, the CPU will return to sleep mode unless the ISR or woken code explicitly reconfigures the clock and power settings. During sleep mode, the CPU clock is gated to conserve power, but all register contents and the program counter are retained. If the clock source for MCLK is disabled during sleep, any enabled interrupt will automatically power up and enable the source clock for the duration of interrupt servicing.

Clock and power management, including sleep mode configuration, are controlled through the SYSTEM peripheral. Note that if SMCLK is disabled, peripherals may be unable to assert interrupt signals, potentially preventing wake from sleep.

11 CPU Instructions and Assembly

The Castalia MCU integrates the VestaRV CPU core, a custom 32-bit RISC-V processor implementing the RV32IMAC instruction set with bit manipulation extensions (Zba, Zbb, Zbc, Zbs). This configuration provides integer arithmetic (I), hardware multiplication and division (M), atomic operations (A), compressed 16-bit instructions (C), and enhanced bit manipulation capabilities. The RISC-V instruction set architecture (ISA) defines the binary encoding of machine instructions, their operational semantics, and the register architecture used for computation and control flow.

11.1 CPU Registers

There are 32 CPU registers defined in the RISC-V ISA. Though the user can technically use any register for any purpose (except for `x0/zero`, which is read-only and is always equal to `0x00000000`), it is generally advisable to use each register for its intended purpose and follow the convention that the RISC-V ISA describes. The convention was created to create a standard use for each register so that assembly functions can be written that will be compatible with any RISC-V assembly program. The RISC-V compiler always follows this convention. Therefore, if the user also follows the convention, then the user's assembly code will be drop-in compatible with other compiled code (such as C code). If the user does not follow the convention, then it may cause many runtime errors.

Each register has a specific purpose and a designated “saver” that is allowed to write to it. The saver is either the caller (the code that calls a function) or the callee (the function that was called). This allows for communication between caller and callee such that arguments can be passed from caller to callee and returned data can be sent back from callee to caller. For example, the registers `a0-a7` are intended for function arguments and are written to by the caller. Registers `s0-s11` are preserved across function calls and are written to by the callee. Furthermore, registers `t0-t6` are temporary register that can be used for any purpose and are written to by the caller.

In assembly, registers can be referred to interchangeably by their register number or name (e.g. you can either type `x1` or `ra` to refer to the same register). Generally, one should only use the register names since they give hints as to the purpose of the register.

A CPU instruction can have up to two source register `rs1` and `rs2`, and up to one destination register `rd`. `rs1`, `rs2`, and `rd` can be any of the 32 CPU registers (e.g. `sp`, `t6`, `zero`, etc.). The source registers are used as “arguments” to the instruction and are not modified by the instruction, and the destination register is modified by the instruction as its output.

A list of all 32 CPU registers is provided in Table 99.

Table 99: CPU Registers

Register	Name	Description	Saver
<code>x0</code>	<code>zero</code>	Hard-wired zero (read-only)	N/A
<code>x1</code>	<code>ra</code>	Return Address	Caller
<code>x2</code>	<code>sp</code>	Stack Pointer	Callee
<code>x3</code>	<code>gp</code>	Global Pointer	–
<code>x4</code>	<code>tp</code>	Thread Pointer	–
<code>x5</code>	<code>t0</code>	Temporary/Alternate Link Register	Caller
<code>x6-x7</code>	<code>t1-t2</code>	Temporary Registers	Caller
<code>x8</code>	<code>s0/fp</code>	Saved Register/Frame Pointer	Callee
<code>x9</code>	<code>s1</code>	Saved Register	Callee
<code>x10-x11</code>	<code>a0-a1</code>	Function Arguments/Return Values	Caller

Register	Name	Description	Saver
x12-x17	a2-a7	Function Arguments	Caller
x18-x27	s2-s11	Saved Registers	Callee
x28-x31	t3-t6	Temporary Registers	Caller

11.2 Load/Store Memory Instructions

These instructions interface between the memory bus (containing RAM, ROM, and memory-mapped registers) and the CPU registers. They are used to load data from memory into a CPU register or store data from a CPU register into memory. The immediate parameter (i) refers to a hard-coded 12-bit constant that is stored inside the CPU instruction itself and is added to the value of one of the registers. All load/store instructions take 2 MCLK cycles to execute.

- Load Signed Byte: `lb rd, I(rs1)` : loads the signed byte (8 bits) stored at memory location pointed to by (rs1 + I) into the destination CPU register rd
- Load Unsigned Byte: `lbu rd, I(rs1)` : loads the unsigned byte (8 bits) stored at memory location pointed to by (rs1 + I) into the destination CPU register rd
- Load Signed Half Word: `lh rd, I(rs1)` : loads the signed half word (16 bits) stored at memory location pointed to by (rs1 + I) into the destination CPU register rd
- Load Unsigned Half Word: `lhu rd, I(rs1)` : loads the unsigned half word (16 bits) stored at memory location pointed to by (rs1 + I) into the destination CPU register rd
- Load Word: `lw rd, I(rs1)` : loads the signed or unsigned word (32 bits) stored at memory location pointed to by (rs1 + I) into the destination CPU register rd
- Store byte: `sb rs1, I(rs2)` : stores the signed or unsigned byte (8 bits) from source CPU register 1 rs1 into the memory location pointed to by (rs2 + I)
- Store half word: `sh rs1, I(rs2)` : stores the signed or unsigned half word (16 bits) from source CPU register 1 rs1 into the memory location pointed to by (rs2 + I)
- Store word: `sw rs1, I(rs2)` : stores the signed or unsigned word (32 bits) from source CPU register 1 rs1 into the memory location pointed to by (rs2 + I)

11.3 Shift Instructions

These instructions perform bit shift operations (e.g. <<). The immediate parameter (I) refers to a hard-coded 12-bit constant that is stored inside the CPU instruction itself and is used as the number of bits to shift. Each shift instruction takes 1 MCLK cycle to execute.

- Shift Left: `sll rd, rs1, rs2` : Logical shifts the value in rs1 left by rs2 bits and stores the result in rd.
- Shift Left Immediate: `slli rd, rs1, I` : Logical shifts the value in rs1 left by I bits and stores the result in rd.
- Shift Right: `srl rd, rs1, rs2` : Logical shifts the value in rs1 right by rs2 bits and stores the result in rd.

- Shift Right Immediate: `slli rd, rs1, I` : Logical shifts the value in `rs1` right by `I` bits and stores the result in `rd`.
- Shift Right Arithmetic: `sll rd, rs1, rs2` : Arithmetic shifts the value in `rs1` right by `rs2` bits and stores the result in `rd`.
- Shift Right Arithmetic Immediate: `slli rd, rs1, I` : Arithmetic shifts the value in `rs1` right by `I` bits and stores the result in `rd`.

11.4 Arithmetic Instructions

These instructions perform arithmetic operations. The immediate parameter (`I`) refers to a hard-coded 12-bit signed integer that is stored inside the CPU instruction itself and is added to the operation (except for in the `auipc` instruction). These instructions take 1 MCLK cycle to execute.

- Addition: `add rd, rs1, rs2` : Adds the signed/unsigned integer values in `rs1` and `rs2` together and stores the sum in `rd`
- Addition Immediate: `addi rd, rs1, I` : Adds the signed/unsigned integer value in `rs1` with the immediate 12-bit signed integer `I` together and stores the sum in `rd`
- Subtraction: `sub rd, rs1, rs2` : Subtracts the signed/unsigned integer values in `rs1` from `rs2` and stores the result in `rd` (i.e. $rd \leftarrow rs1 - rs2$)
- Load Upper Immediate: `lui rd, I` : Loads the 20-bit immediate value `I` into the upper 20 bits of the destination CPU register `rd` (the lower 12 bits are all cleared to 0s)
- Add Upper Immediate to PC: `auipc rd, I` : Adds the 12-bit signed integer immediate `I` to the program counter to jump the program's execution to a location `I` away from the current program counter. Note that `I` can be negative.

11.5 Bitwise Instructions

These instructions perform bitwise operations. The immediate parameter (`I`) refers to a hard-coded 12-bit signed integer that is stored inside the CPU instruction itself and is used as one of the operands of the bitwise operations. These instructions take 1 MCLK cycle to execute.

- Bitwise exclusive OR: `xor rd, rs1, rs2` : Performs the bitwise XOR of `rs1` and `rs2` and stores the result in `rd`.
- Bitwise exclusive OR Immediate: `xori rd, rs1, I` : Performs the bitwise XOR of `rs1` and the immediate `I` and stores the result in `rd`.
- Bitwise OR: `or rd, rs1, rs2` : Performs the bitwise OR of `rs1` and `rs2` and stores the result in `rd`.
- Bitwise OR Immediate: `ori rd, rs1, I` : Performs the bitwise OR of `rs1` and the immediate `I` and stores the result in `rd`.
- Bitwise AND: `and rd, rs1, rs2` : Performs the bitwise AND of `rs1` and `rs2` and stores the result in `rd`.
- Bitwise AND Immediate: `andi rd, rs1, I` : Performs the bitwise AND of `rs1` and the immediate `I` and stores the result in `rd`.

11.6 Comparison Instructions

These instructions perform comparison operations. The results of these instructions are either true (0x00000001) or false (0x00000000). The immediate parameter (I) refers to a hard-coded 12-bit signed or unsigned integer that is stored inside the CPU instruction itself and is used as one of the operands of the bitwise operations. These instructions take 1 MCLK cycle to execute.

- Set when Less Than: `slt rd, rs1, rs2`: If $rs1 < rs2$ (using signed integers), the destination register `rd` is set to 0x00000001, otherwise it is set to 0x00000000.
- Set when Less Than Immediate: `slti rd, rs1, I`: If $rs1 < I$ (using signed integers), the destination register `rd` is set to 0x00000001, otherwise it is set to 0x00000000.
- Set when Less Than Unsigned: `sltu rd, rs1, rs2`: If $rs1 < rs2$ (using unsigned integers), the destination register `rd` is set to 0x00000001, otherwise it is set to 0x00000000.
- Set when Less Than Immediate Unsigned: `sltiu rd, rs1, I`: If $rs1 < I$ (using unsigned integers), the destination register `rd` is set to 0x00000001, otherwise it is set to 0x00000000.

11.7 Branch Instructions

These instructions are conditional instructions that branch (jump) the program execution to a new location if the condition is met, or proceed on to the next instruction like normal if the condition is not met. The immediate parameter (I) refers to a hard-coded 12-bit signed integer that is stored inside the CPU instruction itself and is added to the program counter if the condition is met. Usually, in assembly, the program writer can simply provide the *label* to the desired branch location/function in the program (e.g. `labelToFunction`;) rather than its actual memory location, which is usually not known before being built. These instructions take 1 MCLK cycle to execute regardless of whether the branch is taken or not.

- Branch if Equal: `beq rs1, rs2, I`: Branches to $PC + I$ if $rs1 = rs2$, proceeds to the next instruction like normal otherwise.
- Branch if Not Equal: `bne rs1, rs2, I`: Branches to $PC + I$ if $rs1 \neq rs2$, proceeds to the next instruction like normal otherwise.
- Branch if Less Than: `blt rs1, rs2, I`: Branches to $PC + I$ if $rs1 < rs2$ (signed integers), proceeds to the next instruction like normal otherwise.
- Branch if Greater Than or Equal: `bge rs1, rs2, I`: Branches to $PC + I$ if $rs1 \geq rs2$ (signed integers), proceeds to the next instruction like normal otherwise.
- Branch if Less Than Unsigned: `bltu rs1, rs2, I`: Branches to $PC + I$ if $rs1 < rs2$ (unsigned integers), proceeds to the next instruction like normal otherwise.
- Branch if Greater Than or Equal Unsigned: `bgeu rs1, rs2, I`: Branches to $PC + I$ if $rs1 \geq rs2$ (unsigned integers), proceeds to the next instruction like normal otherwise.

11.8 Jump and Link Instructions

The jump and link instructions write the address of the next consecutive instruction in the program to the destination register, but then jump the program counter to another location in memory. These instructions

are used to call new functions while preserving the “return address”, or the memory location of the instruction to go back to after the function returns. The immediate parameter (I) refers to a hard-coded 20-bit signed integer that is stored inside the CPU instruction itself and is used to modify the program counter. Usually, in assembly, the program writer can simply provide the *label* to the desired location/function in the program (e.g. `labelToFunction:`) rather than its actual memory location, which is usually not known before being built. All jump and link instructions take 1 MCLK cycle to execute.

- **Jump and Link:** `jal rd, I` Stores the return address to rd (which is usually the return address register ra) and then adds I to the program counter, effectively jumping to PC + I.
- **Jump and Link Register:** `jalr rd, rs1, I` Stores the return address to rd (which is usually the return address register ra) and then sets the program counter to rs1 + I, effectively jumping to rd1 + I.

11.9 Atomic Memory Operations (A Extension)

The VestaRV core implements the RISC-V Atomic (A) extension, providing atomic read-modify-write operations for multiprocessor synchronization. On the Castalia MCU these instructions are the standard way to build locks and shared counters in the arbitrated shared RAM: an AMO instruction holds the shared-window arbiter grant across its whole read-modify-write pair, and a failed `sc.w` has its write suppressed by the reservation unit, so both are atomic with respect to all five harts (see Section 4.4). Do not use atomic instructions on hardware mutex addresses (see the MUTEX peripheral chapter). All atomic memory operations operate on 32-bit words and take 5 MCLK cycles to execute.

- **Load Reserved:** `lr.w rd, (rs1)` : Loads a 32-bit word from memory address in rs1 into rd and reserves the memory location for atomic access.
- **Store Conditional:** `sc.w rd, rs2, (rs1)` : Conditionally stores the 32-bit word in rs2 to the memory address in rs1. Returns 0 in rd on success, nonzero on failure.
- **Atomic Swap:** `amoswap.w rd, rs2, (rs1)` : Atomically swaps the 32-bit word at memory address in rs1 with the value in rs2, storing the original memory value in rd.
- **Atomic Add:** `amoadd.w rd, rs2, (rs1)` : Atomically adds rs2 to the 32-bit word at memory address in rs1, storing the original memory value in rd.
- **Atomic XOR:** `amoxor.w rd, rs2, (rs1)` : Atomically performs bitwise XOR of rs2 with the 32-bit word at memory address in rs1, storing the original memory value in rd.
- **Atomic AND:** `amoand.w rd, rs2, (rs1)` : Atomically performs bitwise AND of rs2 with the 32-bit word at memory address in rs1, storing the original memory value in rd.
- **Atomic OR:** `amoor.w rd, rs2, (rs1)` : Atomically performs bitwise OR of rs2 with the 32-bit word at memory address in rs1, storing the original memory value in rd.
- **Atomic Minimum:** `amomin.w rd, rs2, (rs1)` : Atomically compares rs2 with the 32-bit signed word at memory address in rs1, storing the minimum value to memory and the original memory value in rd.
- **Atomic Maximum:** `amomax.w rd, rs2, (rs1)` : Atomically compares rs2 with the 32-bit signed word at memory address in rs1, storing the maximum value to memory and the original memory value in rd.

- Atomic Minimum Unsigned: `amominu.w rd, rs2, (rs1)` : Atomically compares `rs2` with the 32-bit unsigned word at memory address in `rs1`, storing the minimum value to memory and the original memory value in `rd`.
- Atomic Maximum Unsigned: `amomaxu.w rd, rs2, (rs1)` : Atomically compares `rs2` with the 32-bit unsigned word at memory address in `rs1`, storing the maximum value to memory and the original memory value in `rd`.

11.10 Bit Manipulation Instructions (Zb* Extensions)

The VestaRV core implements several bit manipulation extensions (Zba, Zbb, Zbc, Zbs) that provide efficient operations for bit-level data manipulation, cryptographic primitives, and address generation. All bit manipulation instructions execute in 1 MCLK cycle.

11.10.1 Address Generation (Zba)

- Shift Left Add: `sh1add rd, rs1, rs2` : Shifts `rs1` left by 1 bit, adds `rs2`, and stores result in `rd`. Equivalent to `rd = (rs1 « 1) + rs2`.
- Shift Left Add 2: `sh2add rd, rs1, rs2` : Shifts `rs1` left by 2 bits, adds `rs2`, and stores result in `rd`. Equivalent to `rd = (rs1 « 2) + rs2`.
- Shift Left Add 3: `sh3add rd, rs1, rs2` : Shifts `rs1` left by 3 bits, adds `rs2`, and stores result in `rd`. Equivalent to `rd = (rs1 « 3) + rs2`.

11.10.2 Basic Bit Manipulation (Zbb)

- AND with Inverted: `andn rd, rs1, rs2` : Performs bitwise AND of `rs1` with the bitwise complement of `rs2`, storing result in `rd`.
- OR with Inverted: `orn rd, rs1, rs2` : Performs bitwise OR of `rs1` with the bitwise complement of `rs2`, storing result in `rd`.
- XOR with Inverted: `xnor rd, rs1, rs2` : Performs bitwise XOR of `rs1` with the bitwise complement of `rs2`, storing result in `rd`.
- Count Leading Zeros: `clz rd, rs1` : Counts the number of consecutive zero bits starting from the most significant bit in `rs1`, storing the count in `rd`.
- Count Trailing Zeros: `ctz rd, rs1` : Counts the number of consecutive zero bits starting from the least significant bit in `rs1`, storing the count in `rd`.
- Count Population: `cpop rd, rs1` : Counts the number of set bits (1s) in `rs1`, storing the count in `rd`.
- Maximum: `max rd, rs1, rs2` : Stores the maximum of `rs1` and `rs2` (signed comparison) in `rd`.
- Maximum Unsigned: `maxu rd, rs1, rs2` : Stores the maximum of `rs1` and `rs2` (unsigned comparison) in `rd`.
- Minimum: `min rd, rs1, rs2` : Stores the minimum of `rs1` and `rs2` (signed comparison) in `rd`.

- Minimum Unsigned: `minu rd, rs1, rs2` : Stores the minimum of `rs1` and `rs2` (unsigned comparison) in `rd`.
- Sign Extend Byte: `sext.b rd, rs1` : Sign-extends the least significant byte in `rs1` to 32 bits, storing result in `rd`.
- Sign Extend Halfword: `sext.h rd, rs1` : Sign-extends the least significant halfword (16 bits) in `rs1` to 32 bits, storing result in `rd`.
- Zero Extend Halfword: `zext.h rd, rs1` : Zero-extends the least significant halfword in `rs1` to 32 bits, storing result in `rd`.
- Rotate Left: `rol rd, rs1, rs2` : Rotates `rs1` left by the number of bits specified in the lower 5 bits of `rs2`, storing result in `rd`.
- Rotate Right: `ror rd, rs1, rs2` : Rotates `rs1` right by the number of bits specified in the lower 5 bits of `rs2`, storing result in `rd`.
- Rotate Right Immediate: `rori rd, rs1, shamt` : Rotates `rs1` right by `shamt` bits, storing result in `rd`.
- OR Combine: `orc.b rd, rs1` : Sets each byte in `rd` to 0xFF if the corresponding byte in `rs1` is nonzero, otherwise sets to 0x00.
- Byte Reverse: `rev8 rd, rs1` : Reverses the byte order in `rs1`, storing result in `rd` (converts between big-endian and little-endian).

11.10.3 Carryless Multiplication (Zbc)

- Carryless Multiply Low: `clmul rd, rs1, rs2` : Performs carryless multiplication of `rs1` and `rs2`, storing the lower 32 bits of the result in `rd`. Useful for CRC calculations and cryptographic operations.
- Carryless Multiply High: `clmulh rd, rs1, rs2` : Performs carryless multiplication of `rs1` and `rs2`, storing the upper 32 bits of the result in `rd`.
- Carryless Multiply Reversed: `clmulr rd, rs1, rs2` : Performs carryless multiplication with a bit-reversed result, storing bits [62:31] of the 64-bit carryless product in `rd`.

11.10.4 Single-Bit Manipulation (Zbs)

- Bit Clear: `bclr rd, rs1, rs2` : Clears the bit in `rs1` at the position specified by the lower 5 bits of `rs2`, storing result in `rd`.
- Bit Clear Immediate: `bclri rd, rs1, shamt` : Clears the bit in `rs1` at position `shamt`, storing result in `rd`.
- Bit Extract: `bext rd, rs1, rs2` : Extracts the bit in `rs1` at the position specified by the lower 5 bits of `rs2`, storing the single bit (zero-extended) in `rd`.
- Bit Extract Immediate: `bexti rd, rs1, shamt` : Extracts the bit in `rs1` at position `shamt`, storing the single bit (zero-extended) in `rd`.

- Bit Invert: `binv rd, rs1, rs2`: Inverts the bit in `rs1` at the position specified by the lower 5 bits of `rs2`, storing result in `rd`.
- Bit Invert Immediate: `binvi rd, rs1, shamt`: Inverts the bit in `rs1` at position `shamt`, storing result in `rd`.
- Bit Set: `bset rd, rs1, rs2`: Sets the bit in `rs1` at the position specified by the lower 5 bits of `rs2`, storing result in `rd`.
- Bit Set Immediate: `bseti rd, rs1, shamt`: Sets the bit in `rs1` at position `shamt`, storing result in `rd`.

11.11 Pseudo-Instructions

The RISC-V assembler also includes several pseudo-instructions for convenience. These pseudo-instructions are actually special cases of regular instructions, and can all be expressed as regular instructions.

- No operation: `nop` → `addi zero, zero, 0`
- Move: `mv rd, rs1` → `addi rd, rs1, 0`
- Bitwise Complement/Inversion: `not rd, rs1` → `xor rd, rs1, -1`
- Set if Equal to Zero: `seqz rd, rs1` → `sltiu rd, rs1, 1`
- Set if Not Equal to Zero: `snez rd, rs1` → `sltu rd, zero, rs1`
- Set if Less Than Zero: `sltz rd, rs1` → `slt rd, rs1, zero`
- Set if Greater Than Zero: `sgtz rd, rs1` → `slt rd, zero, rs1`
- Jump (without linking): `j I` → `jal zero, I`
- Return: `ret` → `jalr zero, ra, 0`
- Call function: `call I` → `auipc t1 I; jalr ra t1 I`
- Load immediate: `li rd, I` (*this could compile to several instructions*)
- Branch if Greater Than: `bgt rs1, rs2, I` → `blt rs2, rs1, I`
- Branch if Less Than or Equal To: `ble rs1, rs2, I` → `bge rs2, rs1, I`
- Branch if Greater Than Unsigned: `bgtu rs1, rs2, I` → `bltu rs2, rs1, I`
- Branch if Less Than or Equal To Unsigned: `bleu rs1, rs2, I` → `bgeu rs2, rs1, I`
- Branch if Equal to Zero: `beqz rs1, I` → `beq rs1, zero, I`
- Branch if Not Equal to Zero: `bnez rs1, I` → `bne rs1, zero, I`
- Branch if Less Than or Equal To Zero: `blez rs1, I` → `bge zero, rs1, I`

- Branch if Greater Than or Equal To Zero: `bgez, rs1, I` → `bge rs1, zero, I`
- Branch if Less Than Zero: `bltz rs1, I` → `blt rs1, zero, I`
- Branch if Greater Than Zero: `bgtz rs1, I` → `blt zero, rs1, I`

12 Core Performance and Instruction Timing

The VestaRV core is a multi-cycle machine on a single unified instruction and data bus, but it issues the fetch of the next instruction during the execute cycle of the current one. Most instructions therefore retire at one per MCLK cycle, and the core’s measured cycles-per-instruction (CPI) is much closer to 1 than the term “multi-cycle” on its own would suggest: over the benchmark suite of Section 12.3 it is 1.260. This section collects the per-instruction timings quoted throughout Section 11, adds the classes that section does not cover, and reports the CPI actually measured on that suite.

Every figure below is for one hart executing out of its own TCM with no shared-bus contention. An access that crosses the arbiter into the shared window, or that is made while another hart holds the grant, adds stall cycles on top of these numbers (Section 4). The core’s clock is gated while it waits, so a stalled cycle is still an MCLK cycle in which nothing retires and it is charged to CPI like any other.

12.1 Instruction Timing

Table 100 gives the cost of each instruction class. A row for an optional extension applies only when that extension is configured for this build; a disabled extension’s instructions raise an illegal-instruction trap instead of executing. Atomic memory operations are the one class not re-measured here — the bare-core bench has no second master — and their 5-cycle cost is as stated in Section 11.

Table 100: Measured instruction timing, in MCLK cycles

Instruction class	MCLK cycles
Arithmetic and logical (add, andi, lui, auipc)	1
Shift (sll, srl, sra and immediate forms)	1
Comparison (slt, sltu and immediate forms)	1
Bit manipulation (Zba, Zbb, Zbc, Zbs)	1
Multiply (M): mul, mulh, mulhu, mulhsu	1
Divide and remainder (M): div, divu, rem, remu	36
Load and store (lw, lh, lb, sw, sh, sb)	2
Conditional branch, whether taken or not taken	1
Jump and link (jal, jalr)	1
CSR read (csrr)	1
Atomic memory operations (A), per Section 11	5
<i>Additional</i> penalty: 32-bit instruction straddling a word boundary, reached sequentially	0
<i>Additional</i> penalty: the same, reached by a taken branch or jump	+1

Three entries are worth drawing out. Hardware multiply is single-cycle, so mul costs no more than add and there is no reason to hand-expand a multiplication into shifts and adds. Divide and remainder share one iterative divider and are, by a wide margin, the most expensive instructions the core executes; where the divisor is a constant, the compiler’s multiply-and-shift expansion is far faster than div and is worth checking for in the disassembly of any hot loop. And a conditional branch costs the same whether it is taken or not, because the core resolves the condition and issues the redirected fetch in the same cycle: there is no branch predictor, and no misprediction penalty to design around.

The two straddling rows are one instruction class split by how the core arrives at it, and Section 12.2 is about that split.

12.2 Compressed Instructions and Fetch Alignment

The core fetches one 32-bit word per bus cycle. With the C extension enabled instructions are only guaranteed 2-byte aligned, so a 32-bit instruction may straddle a word boundary, and the core must have both halves before it can decode. Compressed (2-byte) instructions and word-aligned 32-bit instructions are never affected.

The core covers most of that case in hardware, with a fetch-ahead. When it is about to step *sequentially* onto a straddling 32-bit instruction, it can already see that instruction's lower half in the word it is holding, and the fetch its ordinary sequencing would issue is for the word it is holding already — a bus cycle that would carry nothing new. The fetch-ahead spends that cycle on the *next* word instead and latches the lower half at the same edge, so the straddling instruction dispatches in one cycle and costs nothing at all. The whole mechanism is one flip-flop, it issues no bus cycle the core would not have issued one cycle later, and it changes cycle counts only, never an architectural result. It is the `core.fetchAhead` knob of Table 1 (the `ENABLE_IF_AHEAD` generic on the core), it is *on* in this build, and turning it off is a performance choice and nothing else.

What the fetch-ahead cannot cover is a *taken* control transfer whose target is a half-word-aligned 32-bit instruction. It arms only on a sequential advance, because that is the only case in which the core can prove — from a half-word it can already see — what it is about to fetch; after a taken branch or jump the core arrives at the target holding nothing, and genuinely needs two fetches before it can decode. That case still costs one extra cycle, and it is the whole of the residual reported below. Both rows of Table 100 are measured directly: a straight-line block in which every 32-bit instruction straddles runs at 1.000 cycles per instruction, while a block of a taken branch, a straddling 32-bit target and a compressed instruction — three instructions that cost one cycle each on their own — takes 4 cycles.

Table 101 isolates what the C extension still costs by rebuilding the whole benchmark suite for RV32IMA instead of RV32IMAC. The compiler emits the identical instruction sequence in both builds — the retired-instruction counts match exactly, benchmark for benchmark — and only the encoding differs, so the entire difference in cycle count is the straddling-fetch penalty that survives the fetch-ahead. Across the suite it is 0.031 CPI, or 2.2% of all cycles.

Table 101: Straddling fetches that survive the fetch-ahead: the same code built with and without the C extension

Benchmark	CPI (RV32IMAC)	CPI (RV32IMA)	Penalty, CPI	Cycles lost
dhrystone	1.504	1.481	0.022	1.5%
median	1.398	1.397	0.000	0.0%
memcpy	1.726	1.726	0.000	0.0%
multiply	1.163	1.014	0.148	12.8%
qsort	1.398	1.330	0.068	4.8%
rsort	1.412	1.412	0.000	0.0%
towers	1.682	1.681	0.002	0.1%
vvadd	1.374	1.374	0.000	0.0%
Aggregate	1.443	1.412	0.031	2.2%

The aggregate row weights each benchmark by its retired-instruction count, and covers the eight benchmarks of Table 102 that were rebuilt both ways — `spmv` was not, which is why this aggregate differs from the one in that table.

Read the spread against the paragraph above it. Four of the eight benchmarks pay nothing measurable: their straddling instructions are all reached sequentially, and the hardware absorbs every one. What is left is concentrated in short, branch-dense loops, and `multiply` is the extreme case: its inner loop is seven

instructions long and, in the measured build, its head is a half-word-aligned 32-bit and, so the loop's own backward branch lands on a straddling instruction on every one of its iterations, and one loop head accounts for the entire 12.8%.

That is also the whole of the alignment advice, and it is narrower than it used to be. Aligning a hot loop's *body* no longer buys anything — the fetch-ahead already covers straight-line code, and a body of compressed instructions never straddled in the first place. What still pays is aligning branch *targets*: loop heads, and the entry of a hot function reached by a call. In C, `-falign-loops=4` does exactly that across a translation unit, and it complements hand alignment rather than replacing it: rebuilding `multiply` with it moves that one loop head onto a word boundary and recovers the entire residual, bringing the benchmark back to the RV32IMA figure in the table above, while an assembly hot spot or a computed jump target is still something you align yourself with `.p2align 2` on the label. None of this is an argument against the C extension, which is configured on by default and earns its place by shrinking code substantially — decisive when a hart's private TCM is 8 KiB.

12.3 Measured CPI

Table 102 reports CPI over nine benchmarks derived from the `riscv-tests` benchmark suite, compiled `-O2` for RV32IMAC with the Zb* extensions enabled. The measured region is each benchmark's own timed kernel, so it excludes program startup, the cache-warming pass, and result verification.

Table 102: Measured CPI by benchmark (timed kernel, single hart, no bus contention)

Benchmark	Instructions retired	CPI
<code>spmv</code>	802,447	1.143
<code>multiply</code>	20,892	1.163
<code>vvadd</code>	2,408	1.374
<code>median</code>	4,015	1.398
<code>qsort</code>	123,205	1.398
<code>rsort</code>	146,548	1.412
<code>dhrystone</code>	200,523	1.504
<code>towers</code>	3,747	1.682
<code>memcpy</code>	11,017	1.726
Aggregate	1,314,802	1.260

The aggregate is the total cycle count over the total retired-instruction count, so each benchmark contributes in proportion to the work it does. Read against Table 100, the spread has a simple explanation: the further a workload sits above 1.0, the more of its instruction mix is loads and stores, which cost two cycles each on the unified bus. Fetch alignment barely enters into it any more — `multiply` is register-bound and runs at 1.163, of which 0.148 is the one unaligned loop head of Section 12.2 and the rest is its own work — and `towers` and `memcpy` sit at the other end because they are dominated by memory traffic. Branches move none of it, since they are free.

12.4 Measuring Your Own Code

The core implements the `cycle/cycleh` and `instret/instreth` counters, and their machine-mode aliases `mcycle` and `minstret`, in every configuration. `cycle` counts MCLK cycles, free-running and independent of whether the core is stalled; `minstret` counts instructions actually retired. Bracketing a region with `rdcycle` and `rdinstret` and taking the differences therefore gives that region's exact cycle count and CPI on hardware, with no external instrumentation:

```

1 unsigned c0, i0, c1, i1;
2 __asm__ volatile ("rdcycle %0; rdinstret %1" : "=r"(c0), "=r"(i0));
3 hot_loop();
4 __asm__ volatile ("rdcycle %0; rdinstret %1" : "=r"(c1), "=r"(i1));
5 /* cycles = c1 - c0, instructions = i1 - i0 */

```

Note that `time` and `timeh` are *not* implemented in any configuration — no real-time source is wired to the core — and reading either raises an illegal-instruction trap. Use `cycle` for elapsed time, converting through the known MCLK frequency, or a timer peripheral for wall-clock measurement.

12.5 Measurement Methodology

The numbers in this section are measured on the RTL, not estimated. Each benchmark is run to completion in a VHDL simulation of the bare core with a one-cycle-latency memory and no bus back-pressure, which is the timing a single hart sees executing from its own TCM. Cycles are counted on the free-running core clock, so any cycle in which the core's clock is gated is still counted. Instructions are counted from the core's internal instruction-retire strobe — the same signal that increments `minstret` — so the instruction count is architectural rather than an estimate derived from the simulation. The core under measurement is configured as this build ships it, fetch-ahead included.

The per-class costs in Table 100 come from paired micro-kernels: a loop containing 64 copies of the instruction under test is run against an otherwise identical empty loop, and the difference in cycles divided by the difference in retired instructions gives the marginal cost of that instruction alone, with the loop's own overhead subtracted out. Every micro-kernel body is word-aligned, so no class is charged a fetch cost that is not its own, with two deliberate exceptions: the two kernels that measure the straddling rows are built to straddle. One is a straight-line block of four instructions, two of them straddling 32-bit ones and every one of them reached sequentially, which measures 1.000 cycles per instruction; the other is a taken branch onto a straddling 32-bit target followed by a compressed instruction, three instructions whose measured classes cost one cycle each, which measures 4 cycles for the block.

13 Forth Interpreter

When the MCU comes out of reset and the BOOT pin is LOW, the MCU boots to an interactive Forth interpreter based on the Forth language. Forth is a stack-based language where variables are pushed to a stack in LIFO (last in, first out) order. When a variable is set, it is pushed to the top of the stack (TOS). When a variable is used, it is popped (or consumed) from the TOS. Functions that take arguments pop them from the TOS. Functions that have return values push them to the TOS (after first popping its arguments). Variables are not assigned names, and the only way of distinguishing them from one another is by their order on the stack. All variables are treated as 32-bit signed integers. Variables may be pushed to the stack through the command line by simply typing the variable value into the interpreter as a plain text integer value or as a plain text hexadecimal value. Hexadecimal values must be preceded by the string `0x` without spaces in between (e.g. `0xA4`), but the value may be upper or lower case and does not need leading zeros. Variables are not pushed or popped and functions are not executed until a newline (or line feed) `\n` character is received, at which point the entire received line will be executed in order. New functions may be declared, and may use the stack and any already defined functions to perform their procedures. If the Forth interpreter does not recognize an input as a variable or a function, it prints the `?` char over the UART.

The Forth interpreter contained in the ROM of the MCU communicates over UART0 at 115,200 baud. The baud may be changed by reconfiguring SMCLK and the UART0 peripheral. When the Forth interpreter first starts up, it prints the string `myshkin rv4th-rom!\n\n>`, and is then ready to receive characters over the UART. The Forth interpreter does not perform error handling on stack underflows. In the case of a stack underflow, it simply uses the value 0 as the TOS. By default, the echo is enabled, which repeats all characters received by the Forth interpreter back to the host. This is useful for typing into a terminal emulator manually, but may be switched off with the command `0 echo`.

Several useful built-in Forth functions are listed below. The syntax is read as follows: The Forth function keyword is surrounded by quotations (though you do not actually type the quotes into the Forth interpreter), and is followed by the list of arguments and return values in parentheses. Left of the double dash is the list of arguments in the order that they would be typed into the Forth interpreter (reverse stack order), and right of the double dash is the list of return values in reverse stack order. Reverse stack order means that the rightmost item is at the TOS, and items to the left of it are below it on the stack. Arguments are always popped from the stack before the function executes, and return values are always pushed to the stack after the function finishes.

13.1 Memory Access Functions

- `@ ( addr -- )`

Read directly from the memory address and push the value to the TOS

Arguments: (in order from TOS down, reverse text order):

1. `addr:` (TOS) The address to read from (must be aligned to a 4-byte (32-bit) boundary)

Return values: none

- `! ( x addr -- )`

Write directly to memory address

Arguments: (in order from TOS down, reverse text order):

1. `addr:` (TOS) The address to write to (must be aligned to a 4-byte (32-bit) boundary)

2. x: The value to write to the memory address

Return values: none

- **+**! (x addr --)

Add to value, direct memory access (*addr += x)

Arguments: (in order from TOS down, reverse text order):

1. addr: (TOS) The address to write to (must be aligned to a 4-byte (32-bit) boundary)
2. x: The value to add to the value at the memory address

Return values: none

- **sbi** (bitnum addr --)

Set bit, direct memory access (*addr |= (1 « bitnum))

Arguments: (in order from TOS down, reverse text order):

1. addr: (TOS) The address to write to (must be aligned to a 4-byte (32-bit) boundary)
2. bitnum: The bit number

Return values: none

- **cbi** (bitnum addr --)

Clear bit, direct memory access (*addr &= ~(1 « bitnum))

Arguments: (in order from TOS down, reverse text order):

1. addr: (TOS) The address to write to (must be aligned to a 4-byte (32-bit) boundary)
2. bitnum: The bit number

Return values: none

- **mask** (bitmask x addr --)

Change only masked bits, direct memory access (*addr = (*addr & ~bitmask) | (x & bitmask))

Arguments: (in order from TOS down, reverse text order):

1. addr: (TOS) The address to write to (must be aligned to a 4-byte (32-bit) boundary)
2. x: The value to write (only masked bits will be written)
3. bitmask: The bit mask. Only bits with 1s can be changed

Return values: none

13.2 Print and Input Functions

- **.** (x --)

Prints the TOS as an integer

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) The value to print

Return values: none

- `h. ( x -- )`

Prints the TOS as a hex string

Arguments: (in order from TOS down, reverse text order):

1. `x`: (TOS) The value to print

Return values: none

- `echo ( bool -- )`

Change the echo state, which controls whether the Forth interpreter repeats all of its received characters

Arguments: (in order from TOS down, reverse text order):

1. `bool`: (TOS) If 0, does not echo. Otherwise, it does

Return values: none

- `emit ( x -- )`

Prints the char at the TOS (integers are mapped to their ASCII counterparts)

Arguments: (in order from TOS down, reverse text order):

1. `x`: (TOS) The ASCII value of the char to print

Return values: none

- `key ( -- char )`

Waits for a char over UART and pushes the ASCII value of the char to the TOS

Arguments: none

Return values: (in order from TOS down, reverse text order):

1. `char`: (TOS) The char that was received over UART

- `list ( -- )`

Prints all currently available functions

Arguments: none

Return values: none

13.3 Arithmetic Functions

- `+ ( y x -- ret )`

Adds two numbers ($y + x$)

Arguments: (in order from TOS down, reverse text order):

1. `x`: (TOS) Signed integer
2. `y`: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **-** (y x -- ret)

Subtracts two numbers (y - x)

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- ***** (y x -- retprodhi retprodlo)

Multiplies two numbers (y * x)

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. retprodlo: (TOS) Lower 32 bits of the return product
2. retprodhi: Upper 32 bits of the return product

- **/%** (y x -- retdiv retrem)

Divides two numbers with remainder (y / x and y % x)

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. retrem: (TOS) The remainder (or modulo) result
2. retdiv: The integer division result

- **neg** (x -- ret)

Returns the negative (twos complement) of x

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **abs** (x -- ret)

Absolute value ($|x|$)

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

13.4 Bitwise Logic Functions

- **~** (x -- ret)

Bitwise complement/inversion ($\sim x$)

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **|** (y x -- ret)

Bitwise OR ($y | x$)

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **&** (y x -- ret)

Bitwise AND ($y \& x$)

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **^** (y x -- ret)

Bitwise XOR ($y \wedge x$)

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- `*2 ( x -- ret )`

Shift left 1 bit ($x \ll 1$)

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- `/2 ( x -- ret )`

Shift right 1 bit ($x \gg 1$)

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- `sll ( x bits -- ret )`

Shift left logical ($x \ll \text{bits}$)

Arguments: (in order from TOS down, reverse text order):

1. bits: (TOS) Signed integer
2. x: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- `srl ( x bits -- ret )`

Shift right logical ($x \gg \text{bits}$), no sign extension

Arguments: (in order from TOS down, reverse text order):

1. bits: (TOS) Signed integer
2. x: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

13.5 Comparison Functions

- **>** (y x -- ret)

Greater than. Returns 1 if $y > x$, returns 0 otherwise.

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) 1 if true, 0 if false

- **<** (y x -- ret)

Less than. Returns 1 if $y < x$, returns 0 otherwise.

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) 1 if true, 0 if false

- **==** (y x -- ret)

Equals. Returns 1 if $y == x$, returns 0 otherwise.

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) 1 if true, 0 if false

13.6 Boolean Functions

- **not** (x -- ret)

Logical not (!x). Returns 1 if x is equal to 0, returns 0 otherwise

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **and** (y x -- ret)

Logical and (y && x). Returns 1 if y and x are both true (nonzero), returns 0 otherwise

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **or** (y x -- ret)

Logical or ($y \parallel x$). Returns 1 if y or x are true (nonzero), returns 0 otherwise

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

13.7 Stack Manipulation Functions

- **dup** (x -- x x)

Duplicates the TOS (pops the TOS and then pushes the value to the TOS twice)

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Value to duplicate

Return values: (in order from TOS down, reverse text order):

1. x: (TOS) The duplicated value
2. x: The original TOS

- **drop** (x --)

Pops the TOS and sets it as the internal Forth i value. Usually used to delete the TOS.

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) The value that is dropped

Return values: none

- **swap** (y x -- x y)

Swaps the TOS with the value below the TOS

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Old TOS
2. y: Old value below the TOS

Return values: (in order from TOS down, reverse text order):

1. y: (TOS) Old value below the TOS
2. x: Old TOS

- **over** (y x -- y x y)

Copies the value below the TOS and pushes it to the TOS

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed Integer
2. y: Signed Integer

Return values: (in order from TOS down, reverse text order):

1. y: (TOS) Signed Integer
2. x: Signed Integer
3. y: Signed Integer

- **tuck** (y x -- x y x)

Copies the TOS and inserts it two after the TOS

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed Integer
2. y: Signed Integer

Return values: (in order from TOS down, reverse text order):

1. x: (TOS) Signed Integer
2. y: Signed Integer
3. x: Signed Integer

- **roll** (n --)

Removes value at index n of the stack and pushes it to the TOS (zero indexed)

Arguments: (in order from TOS down, reverse text order):

1. n: (TOS) Stack index (zero indexed)

Return values: none

- **pick** (n --)

Copies value at index n of the stack and pushes it to the TOS (zero indexed)

Arguments: (in order from TOS down, reverse text order):

1. n: (TOS) Stack index (zero indexed)

Return values: none

- **depth** (-- ret)

Gets size of stack and pushes it to the top of the stack (making the new stack size the value of TOS + 1)

Arguments: none

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

13.8 SPI Flash R/W and Programming Functions

- **fr** (bin length addr --)

Reads data from the SPI flash.

Arguments: (in order from TOS down, reverse text order):

1. addr: (TOS) The address in the SPI flash to begin reading from
2. length: The number of bytes to read
3. bin: If true (nonzero), transmits binary data, otherwise transmits ASCII hex string

Return values: none

- **fe** (conf addr -- eraseSuccessful)

Erases a 256-byte page from the SPI flash memory. The start address must be 256-byte aligned.

Arguments: (in order from TOS down, reverse text order):

1. addr: (TOS) The start address of the page to erase in the SPI flash. Must be 256-byte aligned.
2. conf: If equal to 123, erases the page and returns 1, otherwise does not erase and returns 0

Return values: (in order from TOS down, reverse text order):

1. eraseSuccessful: (TOS) Nonzero if successful, zero if failed

- **fw** (bin addr --)

Writes data to the SPI flash. Must write exactly 256 bytes at a time. The start address must be 256-byte aligned.

Arguments: (in order from TOS down, reverse text order):

1. addr: (TOS) The address in the SPI flash to begin writing to. Must be 256-byte aligned.
2. bin: If true (nonzero), expects to receive binary data, otherwise expects to receive an ASCII hex string

Return values: none

13.9 Miscellaneous Functions

- **rst** (--)

Performs a soft reset (turns on all memory cells and jumps to address 0 in the ROM, does not reset clocks or hardware)

Arguments: none

Return values: none

- **clk** (meastime clkselect -- freq)

Measures selected clock frequency

Arguments: (in order from TOS down, reverse text order):

1. clkselect: (TOS) Clock select. 0 = SMCLK; 1 = MCLK; 2 = LFXT; 3 = HFXT

2. meastime: Measurement time. 0 = 1 s; 1 = 0.5 s; 2 = 0.25 s; 3 = 0.125 s

Return values: (in order from TOS down, reverse text order):

1. freq: (TOS) Selected clock frequency in Hz

- **min** (y x -- ret)

Returns minimum value of y and x

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **max** (y x -- ret)

Returns maximum value of y and x

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **swpb** (x -- ret)

Swap bits 15 downto 8 with bits 7 downto 0 of TOS, also causes bits 31 downto 16 to all be 0

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **swphw** (x -- ret)

Swap upper 16 bits with lower 16 bits of TOS

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

13.10 Interpreter Limits

The interpreter keeps all of its mutable state in the private TCM of the hart that runs it, hart 0. That TCM is 8 KiB, of which the interrupt vector table takes the first 484 bytes and the C stack is reserved the top 1 KiB, so the interpreter's arrays are sized to what is left. The capacities are therefore fixed by the ROM and are not configurable at run time:

- Data (math) stack: 128 cells of 32 bits.
- Return (address) stack: 64 cells of 32 bits. Each active `do` loop uses three of them, so loop nesting is bounded well below that.
- Dictionary: 64 user-defined words, 512 bytes total of word names including the separating space after each, and 1024 opcodes of compiled body shared by all of them.
- Input line buffer: 128 characters.
- Word buffer: 64 characters, so a single word name is at most 63 characters.

Only three of these are checked at run time: math stack underflow, input line overflow and word buffer overflow. Exceeding the dictionary capacities or underflowing the return stack is not detected and corrupts interpreter state. The interpreter is intended as an interactive monitor and a flash programming path, not as a development environment for large programs; a program that outgrows these limits belongs in the SPI flash boot mode described in Section 15.

Note also that the interpreter's receive path is a polled loop with no flow control. A host that delivers a line back to back at 115,200 baud can lose characters, because the interpreter echoes each character as it takes it. Typing at a terminal, or a host that leaves a character time of idle line between frames, does not.

13.11 Creating New Forth Functions

To create a new Forth function on-the-fly, simply use the syntax `: funcname definition ;` where `funcname` is the name of the new function and `definition` is the body of the function. The new function may be called at any time after the semicolon (even before a new line, though it will not actually be executed until a new line) by typing the function name, just like all other Forth functions. New Forth functions can only add or remove variables from the stack and call other defined Forth functions. The arguments to custom Forth functions are the initial values on the stack that the custom function pops off the TOS. Likewise, the return values are any new values the custom function pushes to the TOS.

Listed below is an example program that increments the TOS by one when it is called:

```
: inc 1 + ;
```

13.12 Conditionals

The `if then` functions implement a conditional statement which may only be used inside a user defined function. The conditional is used with the syntax:

```
if procedureIfTrue then
```

inside a function (do not forget the space and semicolon which must come after `forthinlinethen`). The `if` function pops the TOS and then executes whatever is defined within `procedureIfTrue` if and only if the TOS is true (nonzero). The `then` function marks the end of the conditional procedure. For example, a function to print the text `cool` if the TOS is equal to 5 might look like this:

```
: cool? 5 == if 108 111 111 99 emit emit emit emit then ;
```

The `else` keyword may be inserted after `procedureIfTrue` if you want to execute something if the conditional is false. The syntax then becomes:

```
if procedureIfTrue else procedureIfFalse then
```

13.13 Loops

Forth has the equivalent of C-like while loops (or, more accurately, `do while` loops) with its `begin until` loops, which may only be executed inside a user defined function. The syntax is:

```
begin loopProcedure until
```

which first executes the loop procedure. When it gets to the `until` statement, it then checks if the TOS is true (nonzero) or false (zero). If the TOS is false, it loops back to the beginning of the loop. If true, it exits the loop and continues on.

Forth also has the equivalent of C-like for loops with its `do loop` functions. Again, this may only be used inside a user defined function. It uses the syntax:

```
stop start do loopProcedure loop
```

where `stop` is the index the loop stops on (it does NOT execute the loop procedure when the index is equal to `stop`), `start` is the first loop index, and `loopProcedure` is any functions or variables that you want to execute inside the loop. There is a special function `i` that may only be called from within a loop which pushes the loop index to the TOS. The functions `j` and `k` behave similarly, except they return the indices of the second and third nested loops respectively when using nested loops. The `i` function always returns the index of the deepest (or most frequent) loop.

Below is an example Forth program that prints the numbers from 0 to 9 using a `do loop` procedure:

```
: looptest 10 0 do i . loop ;
```

14 Software Development and Build Process

14.1 The RISC-V Toolchain and its Extensions

To compile software for the VestaRV CPU core, you will need the RISC-V GNU toolchain, an open source compiler suite supporting multiple programming languages. The base instruction set, RV32I (Integer), provides fundamental integer arithmetic operations, excluding multiplication, division, and modulo. All RV32I instructions are 32 bits in length without exception.

The Castalia MCU's VestaRV core implements several standard and extended instruction sets beyond the base RV32I:

- **M Extension (Integer Multiplication and Division):** Adds hardware multiply (`mul`), divide (`div`), and modulo/remainder (`rem`) instructions with both signed and unsigned variants. Without this extension, the compiler must emulate these operations in software, significantly impacting performance.
- **A Extension (Atomic Instructions):** Provides atomic memory operations including load-reserved/store-conditional (`lr.w/sc.w`) and atomic read-modify-write operations (`amoadd.w`, `amoswap.w`, etc.). While primarily designed for multiprocessor synchronization, these instructions ensure atomic operations in interrupt-driven single-core systems.
- **C Extension (Compressed Instructions):** Introduces 16-bit compressed encodings for frequently used instructions, reducing program size by approximately 20-30% according to RISC-V documentation. Uncompressed instructions on non-4-byte-aligned addresses may incur additional execution cycles. This extension significantly improves code density without sacrificing functionality.

- **Zba Extension (Address Generation):** Provides shift-and-add instructions (sh1add, sh2add, sh3add) optimized for efficient array indexing and pointer arithmetic, particularly beneficial for accessing elements in arrays with 2, 4, or 8-byte strides.
- **Zbb Extension (Basic Bit Manipulation):** Includes instructions for bit counting (clz, ctz, cpop), logical operations with inverted operands (andn, orn, xnor), min/max operations, sign/zero extension, rotations, and byte manipulation. These instructions accelerate bit-level algorithms, cryptographic primitives, and data packing/unpacking.
- **Zbc Extension (Carryless Multiplication):** Implements carryless multiplication instructions (clmul, clmulh, clmulr) essential for efficient CRC calculations, error detection codes, and certain cryptographic operations such as Galois field arithmetic.
- **Zbs Extension (Single-Bit Manipulation):** Offers single-bit set, clear, invert, and extract operations (bset, bclr, binv, bext) with both register and immediate variants, facilitating efficient manipulation of bit fields and flags.

When compiling for the Castalia MCU, specify the complete architecture string `-march=rv32imac_zba_zbb_zbc_zbs` to enable all supported extensions. Individual extensions may be selectively disabled by omitting them from the `-march` flag if desired for compatibility testing or code size optimization.

14.2 Getting the RISC-V Toolchain on Linux

There are two primary methods for obtaining the RISC-V toolchain on Linux: installing a prebuilt binary distribution or building from source. For most users, the prebuilt xPack distribution is recommended due to its simplicity and time savings.

14.2.1 Option 1: Prebuilt xPack Toolchain (Recommended)

The xPack project provides precompiled RISC-V GCC toolchains with support for all standard extensions including RV32IMAC and bit manipulation (Zb*). This is the fastest and simplest installation method.

1. Download the latest xPack RISC-V Embedded GCC release from:

<https://github.com/xpack-dev-tools/riscv-none-elf-gcc-xpack/releases>

Select the appropriate package for your system (typically `xpack-riscv-none-elf-gcc*-linux-x64.tar.gz` for 64-bit Linux systems).

2. Create a directory for the toolchain and extract the archive:

```
mkdir -p ~/riscv-toolchain
```

```
tar -xf xpack-riscv-none-elf-gcc-*.tar.gz -C ~/riscv-toolchain/
```

3. Set the toolchain path as an environment variable. Add the following to your `~/.bashrc` or `~/.profile` file:

```
export RISCVC_TOOLCHAIN_DIR=~/riscv-toolchain/xpack-riscv-none-elf-gcc-13.2.0-2
```

```
export PATH=$RISCVC_TOOLCHAIN_DIR/bin:$PATH
```

Replace 13.2.0-2 with your downloaded version number.

4. Reload your shell configuration or open a new terminal:

```
source ~/.bashrc
```

5. Verify the installation:

```
riscv-none-elf-gcc --version
```

You should see the GCC version information confirming successful installation.

14.2.2 Option 2: Building from Source

Building the toolchain from source provides maximum flexibility and ensures the latest features, but requires significantly more time and disk space. You can find the source code for the RISC-V toolchain at <https://github.com/riscv/riscv-gnu-toolchain>, which also contains instructions for building the toolchain. The following procedure builds a toolchain with full support for RV32IMAC and bit manipulation extensions.

First, install the required build dependencies on Ubuntu or Linux Mint:

```
sudo apt install autoconf automake autotools-dev curl libmpc-dev  
libmpfr-dev libgmp-dev gawk build-essential bison flex texinfo gperf  
libtool patchutils bc zlib1g-dev libexpat-dev
```

The instructions are as follows:

1. Switch to your home directory with

```
cd ~
```

2. Make a new directory with

```
mkdir riscv-download
```

and go into it with

```
cd riscv-download
```

3. Download the RV32IMC toolchain with

```
git clone --recursive https://github.com/riscv/riscv-gnu-toolchain riscv-gnu-toolchain  
-rv32imc
```

4. Enter into the new directory with

```
cd riscv-gnu-toolchain-rv32imc
```

5. Create a build directory with

```
mkdir build
```

Enter into the new build directory with `cd build`

6. Configure the build with

```
../configure --with-arch=rv32imc --prefix=/opt/riscv/rv32imc
```

7. Begin the build process with

```
sudo make
```

This will take quite a while. If you wish to speed up the process, you can allocate more CPU cores to the build process by using `make -jn` instead of `make` where `n` can be replaced by the number of

CPU cores you want to use to build the toolchain.). Once the build process has finished successfully, the toolchain will be located at `/opt/riscv/rv32imc`

8. Add a permanent environment variable to your system that defines the location of the `riscv` directory (perhaps in your `.bashrc` script) with something like

```
export RISCVC=/opt/riscv
```

You may also wish to add the `/opt/riscv/rv32imc/bin` directory to your `PATH`, though this is optional.

9. You may now safely delete the `riscv-download` directory with

```
sudo rm -rf ~/riscv-download
```

14.3 Getting the RISC-V Toolchain on Windows

The RISC-V toolchain has no official support for Windows, but several options exist to get it working. For Windows 10 users, it is recommended to install the Windows Subsystem for Linux (WSL) in Ubuntu mode, which can be found in the Microsoft Store app. Then, in WSL, follow the installation instructions in Section 14.2. You must then use WSL to compile all software for the RISC-V MCU. If you have an older version of Windows, it is suggested that you use an Ubuntu or Linux Mint virtual machine to build the RISC-V toolchain and compile your MCU software.

14.4 Compilation

The RISC-V toolchain includes a GCC compiler. Thus, the compilation process is similar to other toolchains that use GCC. To compile a source file into an object file, use:

```
riscv32-unknown-elf-gcc -c $CFLAGS srcFile -o objectFile.o
```

or for the xPack toolchain:

```
riscv-none-elf-gcc -c $CFLAGS srcFile -o objectFile.o
```

where `$CFLAGS` should be replaced by any compilation flags, `srcFile` is the source code file (C, C++, assembly, or other GCC-supported languages), and `objectFile.o` is the output object file. The compiler determines which language is used in a source file by the file extension (`.c` is for C, `.cpp` is for C++, `.S` is for assembly code which must be preprocessed, and `.s` is for assembly code which is not preprocessed).

There are several compilation flags that are suggested:

- `-march=rv32imac_zba_zbb_zbc_zbs` Architecture flag. This specifies which ISA extensions are enabled during compilation. For the Castalia MCU with full VestaRV capabilities, use the complete architecture string as shown to enable the base integer instructions (I), hardware multiplication/division (M), atomic operations (A), compressed instructions (C), and all bit manipulation extensions (Zba, Zbb, Zbc, Zbs). Individual extensions may be selectively disabled by omitting them from the string (e.g., `-march=rv32imc` for basic support without atomics or bit manipulation, or `-march=rv32imac` to include atomics but exclude bit manipulation extensions).
- `-mabi=ilp32` Application Binary Interface (ABI) flag. Specifies the calling convention and data model. Use `ilp32` for integer-only ABI without floating point support, which is appropriate for the VestaRV core.
- `-Os` Optimize for program size flag. Attempts to minimize the compiled program size, which is particularly important for memory-constrained embedded systems.

- `-O<number>` Optimization level flag. Controls compiler optimization aggressiveness. Common values are `-O0` (no optimization, useful for debugging), `-O1` (basic optimizations), `-O2` (recommended for production), `-O3` (aggressive optimizations, may increase code size), and `-Og` (optimize for debugging experience).
- `-I <include_directory>` Include directory flag. Adds the specified directory to the list of directories searched for header files. Any files in the include directory may be included with the `#include <filename.h>` preprocessor directive.
- `-g` GDB debugger enable flag. Includes debugging symbols in the compiled output. Adding the `-g` flag to both the compiler and linker commands will allow disassembly listings (`.lst` files) to show the correspondence between source code and generated assembly instructions.
- `-Wall -Wextra` Warning flags. Enable comprehensive compiler warnings to catch potential issues during development. The `-Wall` flag enables common warnings, while `-Wextra` adds additional checks.

Example compilation command for the Castalia MCU:

```
riscv-none-elf-gcc -c \
-march=rv32imac_zba_zbb_zbc_zbs \
-mabi=ilp32 -O2 -Wall -Wextra \
-I./include -g \
main.c -o main.o
```

You must compile every source file that you wish to use in your project. If your project is in C or C++, it is suggested that you use the provided `start.S` assembly file to call your main function. This assembly file needs to have its first instruction at address `0x8200`, so it must be the first file to be given to the linker. You must have one and only one `int main()` function throughout all of your source files in a single project. For C++ projects, you need to start the main function with `extern "C" int main()` to prevent the C++ compiler from name-mangling the main function.

14.5 Linking

Once all source files have been compiled, they must be linked to produce the output program binary file, the `.elf` file. The linker assigns memory addresses to every variable and function. The MCU contains no writable non-volatile memory, so the program and the variables all share the RAM (with a notable exception for the bootloader and Forth interpreter stored on the ROM). The linker must be aware of the RAM and ROM start address and size, the interrupt vector table (IVT) start address and size (along with the address of each ISR pointer within the IVT), the first program instruction address that the bootloader jumps to after loading the program into RAM, and the address of the master interrupt handler.

Defining the aforementioned memory segments and addresses is done using a linker script. It is highly suggested to use the provided linker script `MCU.ld` and its include files and tweak it to suit your need.

To execute the linker, use:

```
riscv32-unknown-elf-gcc $LFLAGS $OBJECTS -o program.elf
```

where `$LFLAGS` is a list of linker flags, `$OBJECTS` is a list of all the compiled object files needed for the project, and `program.elf` is the output ELF binary containing the program.

There are several linker flags that are suggested:

- `-march=rv32imac_zba_zbb_zbc_zbs` Architecture flag. This specifies which extensions are enabled in the linking process. The VestaRV core supports the RV32IMAC instruction set plus the Zba,

Zbb, Zbc, and Zbs bit manipulation extensions. This flag must match the architecture flags used during compilation.

- `-flto` Link-time optimization flag. This instructs the linker to perform optimizations to the final binary.
- `-nostdlib` Disables the use of the standard library for linking, decreasing software bloat. However, using the flag precludes using standard library functions such as software emulation of integer multiplication and division, floating point numbers, initialization routines, memory allocation, etc. It is suggested that you only use this flag if you have a specific need to reduce the software size.
- `-T linker_script.ld` Path to the linker script
- `-Map mapfile.map` Path to the output map file, which contains the symbol table for your project with their addresses
- `-L linkerfiles_dir` Path to the linker scripts directory that contains any files that are included in the main linker script
- `-g` GDB debugger enable flag. Adding the `-g` flag to both the compiler and linker commands will allow the `.lst` file to show the assembly commands your code compiles into.

14.6 Intel Hex File Generation

Once the final ELF file has been created from the linker, an Intel hex file can be created, which converts the binary program data into an easy-to-read ASCII file. Furthermore, the Python package `IntelHex` can read the file, allowing you to use Python to upload a new program to the SPI flash. A Python program that utilizes `IntelHex` to upload new programs, called `forthprog.py`, is provided.

14.7 A Note on Using C++

C++ generates some extra data sections that C and assembly do not use. In particular, it generates the `.eh_frame` section, which is used for exception handling. You may not need exception handling, in which case you can pass in the compiler flags:

`-fno-exceptions -fno-unwind-tables -fno-asynchronous-unwind-tables`

which will significantly reduce the size of `.eh_frame`. If you wish to eliminate it entirely, you can create a phony section in the linker script at an unused memory address and point the `.eh_frame` section to it.

15 Uploading Program to SPI Flash

15.1 SPI Flash Boot Process

In order to boot in SPI flash mode, there must first be a valid program stored in the SPI flash memory in the proper location. In SPI flash mode, the bootloader reads the data in the SPI flash memory starting at address 0x8200 and copies every 32-bit word into RAM, also starting at address 0x8000. This particular address is the beginning of the RAM on the MCU. The bootloader stops copying from the SPI flash once completely fills the RAM.

Note the distinction between copying 32-bit words as opposed to 8-bit bytes. Though bytes and words are both MSB-first, they are also transmitted starting with the lowest address to the greatest address from the SPI flash. If the bootloader copied every 8-bit byte, the transmission sequence it would expect would be:

byte0, byte1, byte2, byte3, byte4, byte5, byte6, byte7 ...

However, because the bootloader expects MSB-first 32-bit words, the sequence must be:

byte3, byte2, byte1, byte0, byte7, byte6, byte5, byte4 ...

Therefore, the program data on the SPI flash must follow the 32-bit word sequence.

15.2 Upload Process

The Forth boot mode has a mechanism for writing data to the SPI flash, which may be used to upload a new program for use in the SPI flash boot mode. When in Forth boot mode, the flash write `fw` function writes a page of data (which is exactly 256 bytes long and begins on an address that is a multiple of 256) to the SPI flash. The Forth function is used with the syntax:

```
bin addr fw
```

If `bin` is true (nonzero), the command expects 256 bytes of binary data to be transmitted over the UART. If false, it expects a 512-character long ASCII hex string representing the data to be transmitted over the UART. The `addr` argument must be the address of the beginning of the 256-byte page, and must be aligned to a 256-byte boundary.

To execute the command, a newline (or line feed) character must be transmitted to the MCU over the UART. Once the command begins executing, it initiates some handshaking to ensure a proper transaction. The MCU prints a `$` character to indicate it is ready for the page data to be transmitted. After that, you must transmit the page data in the format specified by the `bin` argument. Remember to transmit the data in the proper order (byte3, byte2, byte1, byte0, byte7, byte6, byte5, byte4 ...). Once the page data has been transmitted, the MCU will compute a CRC value upon the data using the CRC16_CDMA2000 algorithm (polynomial = 0xC857, initial CRC value = 0xFFFF, no final XOR, no data reflection, no output reflection) and then transmit the CRC value as a 4-character long hex string. If this value matches your own CRC computation (or if you do not care and simply want it to write the data), transmit a single `Y` character without anything else to tell the MCU to write the data to the SPI flash. Any other character will abort the process, leaving the SPI flash memory unchanged. You are technically supposed to wait an average of 10 ms (25 ms at the greatest) before writing more data to the SPI flash. However, the substantial handshaking and transmission times likely fulfill this requirement, and it is likely that no delay is needed.

If the entire page is blank, you may instead use the flash erase `fe` function to erase the page, making every byte in the page equal to 0xFF. This is much quicker than writing data to the SPI flash, and can speed up programming times substantially for smaller programs. The syntax to use it is:

```
conf addr fe
```

If the `conf` parameter is equal to 123, the function will erase the 256-byte page at address `addr` (which must be 256-byte aligned) and then return 1. If not equal to 123, it will not erase the page and will return 0. You are technically supposed to wait an average of 6 ms (25 ms at the greatest) before another erase

or write operation. Because the erase function is much quicker than the write function, it may become necessary to delay between erases at higher bauds.

You may read back the data stored on the SPI flash with the `fr` function. See Section 13.8 for details.

In order to write an entire program to the SPI flash, you must write/erase a total of 32 pages (each with a length of 256 bytes) beginning with page address 0x8000. After the data is written, you may execute the newly uploaded program by putting the MCU in reset, setting the BOOT pin to SPI flash mode, and taking the chip out of reset. Or, you can simply set the BOOT pin and move the program counter to address 0x0000, which may be done with the `rst` Forth function.

It is highly suggested to use the provided Python program `forthprog.py` to upload new programs to the SPI flash, which takes an Intel hex file containing the new program and sends it to the MCU and the SPI flash using the process described above.

16 GPIOx Peripheral

The MCU has several GPIOx peripherals, each of which controls a set of general purpose input/output (GPIO) pins. The VestaRV implementation supports GPIO ports with 8, 16, or 32 pins; every port on this device has 8 pins. Each GPIO peripheral (typically called a port) constitutes a parallel port, allowing all pins in the port to be updated simultaneously. Each GPIO pin may also be configured and updated individually, allowing for design flexibility. Each GPIO pin is referred to as Px.y, where x refers to the port number and y refers to the corresponding bit in the GPIO registers that controls the pin. For example, P2.5 refers to port 2, bit 5. Like every peripheral on this multi-core device, the GPIO ports live in the shared peripheral window and are accessible to all harts through the bus arbiter.

Every GPIO pin on the MCU can be in one of two modes: primary/GPIO mode, or alternate function (peripheral) mode. In alternate function mode, each pin further selects *which* of up to 8 alternate functions (AF0 through AF7) governs it, using the pin's field in the PxAFS register. Section 2 details the alternate functions available at each GPIO pin.

16.1 Primary/GPIO Mode Functionality

In primary/GPIO mode, the GPIO pin state is controlled by the GPIO registers PxDIR, PxOUT, and PxREN. The PxDIR register determines if the pin is in input or output mode, the PxOUT register determines if the pin outputs a logic high or logic low level when the pin is in output mode, and the PxREN register enables or disables the internal pullup/pulldown resistor. Table 103 shows the various states available to every GPIO pin.

PxDIR[y]	PxOUT[y]	PxREN[y]	Px.y State
0	-	0	High impedance input, resistor disabled
0	-	1	Pullup/pulldown resistor enabled
1	0	-	Output, strongly driven low
1	1	-	Output, strongly driven high

Table 103: GPIO Configurations

The PxIN register always reads the digital state of the pin, regardless of whether it is in input or output mode, and regardless of whether it is in primary/GPIO mode or alternate function mode. The pin Px.y is at logic low level if PxIN[y] is 0, and is at logic high level if it is 1. Note that PxIN is latched on the falling edge of the memory enable signal to ensure stable readings during memory access.

16.2 Register Access Convenience Features

The GPIO peripheral provides convenient set, clear, and toggle registers for efficient bit manipulation without read-modify-write operations:

- PxOUTS: Writing 1 to a bit sets the corresponding output pin high. Writing 0 has no effect. Reading returns the current PxOUT value.
- PxOUTC: Writing 1 to a bit clears the corresponding output pin low. Writing 0 has no effect. Reading returns the inverted PxOUT value.
- PxOUTT: Writing 1 to a bit toggles the corresponding output pin. Writing 0 has no effect. Reading returns the current PxOUT value.

These registers eliminate race conditions in interrupt-driven or multi-threaded code where multiple code paths may modify GPIO outputs.

16.3 Alternate Function (Peripheral) Mode

Most GPIO pins have one or more alternate functions, each controlled by one of the peripherals in the MCU. Two registers together determine what drives a pin:

- PxSEL selects the pin's *mode*: if PxSEL[y] is 0, Px.y is in primary/GPIO mode; if it is 1, Px.y is in alternate function mode.
- PxAFS selects *which* alternate function governs the pin while it is in alternate function mode. Each pin owns a 4-bit field in PxAFS (pin y occupies bits 4y+3 down to 4y; only the low 3 bits are implemented). A field value of *n* selects alternate function AF*n*.

PxAFS resets to 0, so out of reset every pin in alternate function mode drives its AF0 function. AF0 is the pin's classic (legacy) peripheral function, and software that only ever touches PxSEL behaves exactly as it did before multiple alternate functions existed.

When in alternate function mode, the selected peripheral function seizes control over the pin's DIR, OUT, and REN controls, overriding PxDIR[y], PxOUT[y], and PxREN[y]. For example, when the SCK1 pin is in alternate function mode with AF0 selected, the SPI1 peripheral controls the state of the pin, and the corresponding PxDIR[y], PxOUT[y], and PxREN[y] registers have no effect until PxSEL[y] returns to 0. If the pin's PxAFS field selects an AF plane for which the pin has no function defined, the pin becomes a high-impedance input.

To make board layout easy, the alternate-function planes are populated generously with a shared pool of relocatable *output* functions fanned across all four ports: the UART0/UART1 transmitters, the TIMER0/TIMER1 compare (PWM) outputs, and the SPI1 clock and master-out. Each of these output functions is therefore reachable on roughly two dozen pins, so a peripheral output can be routed to whichever pin is most convenient for the PCB. Because there is exactly one instance of each peripheral resource, a given function is *selected* on at most one pin at a time (like the Raspberry Pi's fixed alternate-function map): populating many pins provides placement choice, not duplicate peripherals. Functions belonging to a peripheral that a configuration omits simply leave the pool: their plane slots become unassigned (high-impedance input). Section 2 tabulates every pin's full AF plane assignment.

A handful of plane slots carry *bidirectional bus completions* rather than pool outputs, so that whole buses (not just their output halves) can land on alternate pins: both I2C buses relocate as pairs (SDA0/SCL0 on P1.6/7 or P2.6/7, and SDA1/SCL1 on P2.2/3 or P1.4/5), RX0 on P3.5 AF2 pairs with TX0 on P3.4 AF2 to form a complete relocated UART0, and MIS01 on P3.6 AF7 joins SCK1/MOSI1 on P3.4/P3.5 AF7 to complete an SPI1 master bus on the DTP pins.

For example, to output the T0CMP0 compare waveform (TIMER0 PWM) on pin P1.0 (where it is alternate function 1) instead of on its home pin P2.0:

1. Write 1 to the pin's PxAFS field: P1AFS |= 0x1 (field of pin 0)
2. Set the pin to alternate function mode: P1SEL[0] = 1

Some pins with alternate functions may only be used as inputs by the peripheral. For example, timer capture pins like T0CAP0 must be manually configured as inputs in PxDIR before use, even when PxSEL[y] is set to enable the alternate function.

16.3.1 Relocatable Peripheral Inputs

Peripheral *input* functions (for example UART receivers, timer captures, and the I2C data/clock lines) that appear at more than one pin observe the pad selected by PxAFS *regardless of* PxSEL, mirroring the always-visible behavior of PxIN: a peripheral input is routed from a pin whenever that pin's PxAFS field selects the function's AF plane, and from its home (AF0) pin otherwise. If the same input function is selected at more than one pin simultaneously, a fixed priority applies: candidate locations are checked in a fixed order (the newest alternate location first, then the older alternate, then home; the pin function tables in Section 2 list each function's locations), and the highest-priority selected pin sources the input. Selecting the same input on two pins at once is a software configuration error; the priority merely makes the outcome deterministic. Because input routing ignores PxSEL, a pin can be left in GPIO output mode while its PxAFS field routes the pad into a peripheral input, a useful software loopback for self-test.

Note that output functions may be mapped to (and driven on) multiple pins simultaneously by selecting the function's plane on each pin; each pin's mux is independent.

16.4 Pin Interrupts

The VestaRV GPIO implementation provides individual interrupt outputs for each GPIO pin. Each pin can generate an interrupt independently, allowing for flexible interrupt handling through the system's interrupt vector table. Pin interrupts remain available in alternate function mode in addition to any interrupts the governing peripheral may generate.

To enable interrupts on a pin:

1. Unmask the corresponding GPIO pin interrupt in the system interrupt controller
2. Set PxIE[y] to 1 for the desired pin Px.y
3. Configure the interrupt edge using PxIES[y]: 0 for low-to-high (rising edge), 1 for high-to-low (falling edge)

When the pin Px.y experiences the edge selected by PxIES[y] while PxIE[y] is enabled, the interrupt flag PxIF[y] is immediately set to 1 and the corresponding pin-specific ISR is executed. The interrupt flag must be cleared by writing 1 to PxIF[y] before the ISR returns, otherwise the interrupt will re-trigger.

Important: Modifying PxIES[y] while PxIE[y] is enabled may immediately generate a spurious interrupt. Always disable the pin interrupt (PxIE[y] = 0) before changing PxIES[y].

The PxIF register is also latched on the falling edge of the memory enable signal to ensure stable readings during interrupt service routines.

16.5 Register Description

16.5.1 PIN Register

Register memory slot: 0, offset: 0 bytes, size: 1 byte

GPIO read pin register. Each bit corresponds to the input logic state of the GPIO pin of the same number. The register is latched on the falling edge of the memory enable signal. Reading a 0 in a bit indicates a logic low pin state; reading a 1 indicates a logic high state. This register always reflects the pin state regardless of pin direction or peripheral select settings.

7	6	5	4	3	2	1	0
IN[7]	IN[6]	IN[5]	IN[4]	IN[3]	IN[2]	IN[1]	IN[0]
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)

16.5.2 POUT Register

Register memory slot: 1, offset: 4 bytes, size: 1 byte

GPIO output drive register. Each bit corresponds to the output logic state of the GPIO pin of the same number. Only has an effect if the pin is configured as an output in PxDIR and is set to GPIO (primary) mode in PxSEL. Write a 0 to the desired bit to make the corresponding pin output a logic low value; write a 1 to output a logic high value.

7	6	5	4	3	2	1	0
OUT[7]	OUT[6]	OUT[5]	OUT[4]	OUT[3]	OUT[2]	OUT[1]	OUT[0]
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

16.5.3 POUTS Register

Register memory slot: 2, offset: 8 bytes, size: 1 byte

GPIO output drive set register. Each bit corresponds to the output logic state of the GPIO pin of the same number. Only has an effect if the pin is configured as an output in PxDIR and is in GPIO (primary) mode in PxSEL. Write a 1 to the desired bit to set the corresponding pin (make the pin output a logic high value). Writing a 0 has no effect. Reading this register is equivalent to reading the output drive register PxOUT.

7	6	5	4	3	2	1	0
OUTS[7]	OUTS[6]	OUTS[5]	OUTS[4]	OUTS[3]	OUTS[2]	OUTS[1]	OUTS[0]
rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)

16.5.4 POUTC Register

Register memory slot: 3, offset: 12 bytes, size: 1 byte

GPIO output drive clear register. Each bit corresponds to the output logic state of the GPIO pin of the same number. Only has an effect if the pin is configured as an output in PxDIR and is in GPIO (primary) mode in PxSEL. Write a 1 to the desired bit to clear the corresponding pin (make the pin output a logic low value). Writing a 0 has no effect. Reading this register yields the inversion of the output drive register PxOUT.

7	6	5	4	3	2	1	0
OUTC[7]	OUTC[6]	OUTC[5]	OUTC[4]	OUTC[3]	OUTC[2]	OUTC[1]	OUTC[0]
rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)

16.5.5 POUTT Register

Register memory slot: 4, offset: 16 bytes, size: 1 byte

GPIO output drive toggle register. Each bit corresponds to the output logic state of the GPIO pin of the same number. Only has an effect if the pin is configured as an output in PxDIR and is in GPIO (primary) mode in PxSEL. Write a 1 to the desired bit to toggle the corresponding pin state. Writing a 0 has no effect. Reading this register is equivalent to reading the output drive register PxOUT.

7	6	5	4	3	2	1	0
OUTT[7]	OUTT[6]	OUTT[5]	OUTT[4]	OUTT[3]	OUTT[2]	OUTT[1]	OUTT[0]
rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)

16.5.6 PDIR Register

Register memory slot: 5, offset: 20 bytes, size: 1 byte

GPIO pin direction register. Each bit corresponds to the GPIO pin of the same number. Only has an effect if the pin is configured in GPIO (primary) mode in PxSEL. Write a 0 to the desired bit to set the corresponding pin to input mode; write a 1 to set to output mode.

7	6	5	4	3	2	1	0
DIR[7]	DIR[6]	DIR[5]	DIR[4]	DIR[3]	DIR[2]	DIR[1]	DIR[0]
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

16.5.7 PIF Register

Register memory slot: 6, offset: 24 bytes, size: 1 byte

GPIO interrupt flag register. Each bit corresponds to the GPIO pin of the same number. The register is latched on the falling edge of the memory enable signal. Reading a 0 in a bit indicates there is no pending interrupt for the corresponding pin; reading a 1 indicates there is a new interrupt pending for the corresponding pin. Write a 1 to each bit for which you wish to clear the interrupt flag. Writing 0 has no effect.

7	6	5	4	3	2	1	0
IF[7]	IF[6]	IF[5]	IF[4]	IF[3]	IF[2]	IF[1]	IF[0]
rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)

16.5.8 PIES Register

Register memory slot: 7, offset: 28 bytes, size: 1 byte

GPIO interrupt edge select register. Each bit corresponds to the GPIO pin of the same number. Write a 0 to the desired bit to set the corresponding pin interrupt to trigger on low-to-high (rising) edge; write a 1 to set to high-to-low (falling) edge triggering.

7	6	5	4	3	2	1	0
IES[7]	IES[6]	IES[5]	IES[4]	IES[3]	IES[2]	IES[1]	IES[0]
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

16.5.9 PIE Register

Register memory slot: 8, offset: 32 bytes, size: 1 byte

GPIO interrupt enable register. Each bit corresponds to the GPIO pin of the same number. Write a 0 to the desired bit to disable the pin interrupt; write a 1 to enable the pin interrupt. Each pin has an individual interrupt output that connects to the system interrupt vector table. Interrupts function in both GPIO (primary) and secondary function (peripheral) modes.

7	6	5	4	3	2	1	0
IE[7]	IE[6]	IE[5]	IE[4]	IE[3]	IE[2]	IE[1]	IE[0]
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

16.5.10 PSEL Register

Register memory slot: 9, offset: 36 bytes, size: 1 byte

GPIO peripheral select register. Each bit corresponds to the GPIO pin of the same number. Write a 0 to the desired bit to set the corresponding pin to GPIO (primary) mode; write a 1 to set the pin to alternate function (peripheral) mode. When a pin is in alternate function (peripheral) mode, the governing peripheral takes control of the pin output, direction, and resistor enable states, and the PxOUT, PxDIR, and PxREN registers have no effect on the pin. WHICH alternate function governs the pin is selected by the pin's field in the PxAFS register: at reset all PxAFS fields are 0, selecting alternate function 0 (AF0). Pin interrupts remain available when in alternate function (peripheral) mode in addition to any interrupts the governing peripheral may generate. If a pin has no alternate function defined at the selected PxAFS plane, setting PxSEL to 1 will configure the pin as a high-impedance input.

7	6	5	4	3	2	1	0
SEL[7]	SEL[6]	SEL[5]	SEL[4]	SEL[3]	SEL[2]	SEL[1]	SEL[0]
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

16.5.11 PREN Register

Register memory slot: 10, offset: 40 bytes, size: 1 byte

GPIO resistor enable register. Each bit corresponds to the GPIO pin of the same number. Only has an effect if the pin is configured in GPIO (primary) mode in PxSEL. Write a 0 to the desired bit to disable the pin pullup/pulldown resistor; write a 1 to enable the pin pullup/pulldown resistor.

7	6	5	4	3	2	1	0
REN[7]	REN[6]	REN[5]	REN[4]	REN[3]	REN[2]	REN[1]	REN[0]
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

16.5.12 PAFS Register

Register memory slot: 11, offset: 44 bytes, size: 1 byte

GPIO alternate function select register. One 4-bit field per pin (pin y occupies bits $4y+3$ down to $4y$; only the low 3 bits of each field are implemented, the top bit is reserved and reads 0). When a pin is in alternate function (peripheral) mode (PxSEL bit = 1), the value of its PxAFS field selects WHICH alternate function (AF0-AF7) controls the pin. Each pin's available alternate functions are listed in the pin configuration table in Section 2. The register resets to 0, so every pin comes out of reset selecting its AF0 (legacy) function. Selecting an AF plane with no function defined for the pin configures the pin as a high-impedance input. While a pin is in GPIO (primary) mode its PxAFS field has no effect on the pad, but it still routes relocatable peripheral INPUTS: a peripheral input function relocated to this pin (e.g. a UART receiver or timer capture) observes the pin whenever the PxAFS field selects it, regardless of PxSEL.

7	6	5	4	3	2	1	0
Unused	AFS1			Unused	AFS0		
	rw-(0)	rw-(0)	rw-(0)		rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
Unused	Bit 31	
AFS7	Bits 30-28	Alternate function select for pin 7 (0 = AF0 ... 7 = AF7)
Unused	Bit 27	
AFS6	Bits 26-24	Alternate function select for pin 6 (0 = AF0 ... 7 = AF7)
Unused	Bit 23	
AFS5	Bits 22-20	Alternate function select for pin 5 (0 = AF0 ... 7 = AF7)
Unused	Bit 19	
AFS4	Bits 18-16	Alternate function select for pin 4 (0 = AF0 ... 7 = AF7)
Unused	Bit 15	
AFS3	Bits 14-12	Alternate function select for pin 3 (0 = AF0 ... 7 = AF7)
Unused	Bit 11	
AFS2	Bits 10-8	Alternate function select for pin 2 (0 = AF0 ... 7 = AF7)
Unused	Bit 7	
AFS1	Bits 6-4	Alternate function select for pin 1 (0 = AF0 ... 7 = AF7)
Unused	Bit 3	
AFS0	Bits 2-0	Alternate function select for pin 0 (0 = AF0 ... 7 = AF7)

16.5.13 PTASK Register

Register memory slot: 12, offset: 48 bytes, size: 1 byte

GPIO event-fabric task pin-select register. Each bit corresponds to the GPIO pin of the same number and selects whether that pin participates in the EVFAB0 event-fabric output tasks: when the fabric fires the port's OUT-SET task, every pin whose PxTASK bit is 1 has its PxOUT bit set; when it fires the OUT-CLR task, every selected pin has its PxOUT bit cleared. The two task pulses are applied AFTER the CPU register write in the same cycle (a task wins its own pins against a coincident PxOUT write) and CLR is applied after SET (a same-cycle set+clear on an overlapping pin resolves to CLEAR, the safe direction). A toggle task is deliberately not offered. Resets to 0, so no pin is fabric-driven out of reset. In a configuration WITHOUT the event fabric (peripherals.eventFabric false) the register is still readable and writable but has no effect, because both task inputs are tied inactive. Only the port's implemented pins have bits; the rest read 0.

7	6	5	4	3	2	1	0
TASK[7]	TASK[6]	TASK[5]	TASK[4]	TASK[3]	TASK[2]	TASK[1]	TASK[0]
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

17 SPIx Peripheral

The Serial Peripheral Interface (SPI) peripheral is a high speed, full-duplex, synchronous, multipoint serial interface. It transfers one bit at a time. The SPI bus requires three wires (SCK, MOSI, and MISO), along with one CS wire for every slave. The main features of the SPI peripheral are listed below:

- Master or slave mode (SPI0 is master-only; SPI1, in configurations that include it, supports both master and slave modes)
- LSB-first or MSB-first transfers
- 8-, 16-, or 32-bit transfer lengths
- Master mode supports SPI modes 0-3
- Slave mode supports SPI modes 1 and 3 (CPHA = 1 only)
- Optional byte swap for multi-byte transfers with systems of different endiannesses
- Programmable transfer rate with configurable baud rate divider
- Continuous transfer operation to reduce dead time between frames
- Interrupts for transfer complete and TX register empty
- Flash extended memory feature on SPI0 for memory-mapped read access to external SPI flash (the boot flash), available to hart 0

17.1 SPI Pins

The SPI bus has four pins: SCKx, MOSIx, MISOx, and CSx (present only on SPI1). Figure 89 shows how they are wired on Castalia: SPI0 to the external boot flash, and SPI1 to whatever else the board carries.

SCKx is the serial clock pin, which governs when each data bit is both latched and updated. It must be attached to the SCK pin on every other device that shares the SPI bus. When the corresponding PxSEL[y] bit is a 1, the SPIx peripheral seizes the DIR, OUT, and REN controls of SCKx.

MOSIx is the master out slave in pin, which transfers each data bit from the master to the slave. It is an output when the SPI port is enabled and in master mode, otherwise it is a high impedance input. It must be attached to the MOSI pin on every other device that shares the SPI bus. When the corresponding PxSEL[y] bit is a 1, the SPIx peripheral seizes the DIR, OUT, and REN controls of MOSIx.

MISOx is the master in slave out pin, which transfers each data bit from the slave to the master. It is an output when the SPI port is enabled, in slave mode, and the CSx pin is asserted (low), otherwise it is a high impedance input. It must be attached to the MISO pin on every other device that shares the SPI bus. When the corresponding PxSEL[y] bit is a 1, the SPIx peripheral seizes the DIR, OUT, and REN controls of MISOx.

CS1 is the chip select pin for the SPI1 peripheral (SPI0 is master-only and has no CS input). It is a high impedance input that waits to be asserted (pulled low) by an external master when SPI1 is in slave mode, which begins the SPI transfer. One CS wire needs to be connected between a GPIO pin of the master and the CS pin of the slave for every slave on the bus. Each slave CS wire must be attached to a dedicated GPIO pin on the master in a one-to-one fashion. The SPI1 peripheral never seizes control of the CS1 pin (regardless of the state of the corresponding PxSEL[y] bit), and as such it must be configured as a high impedance input manually.

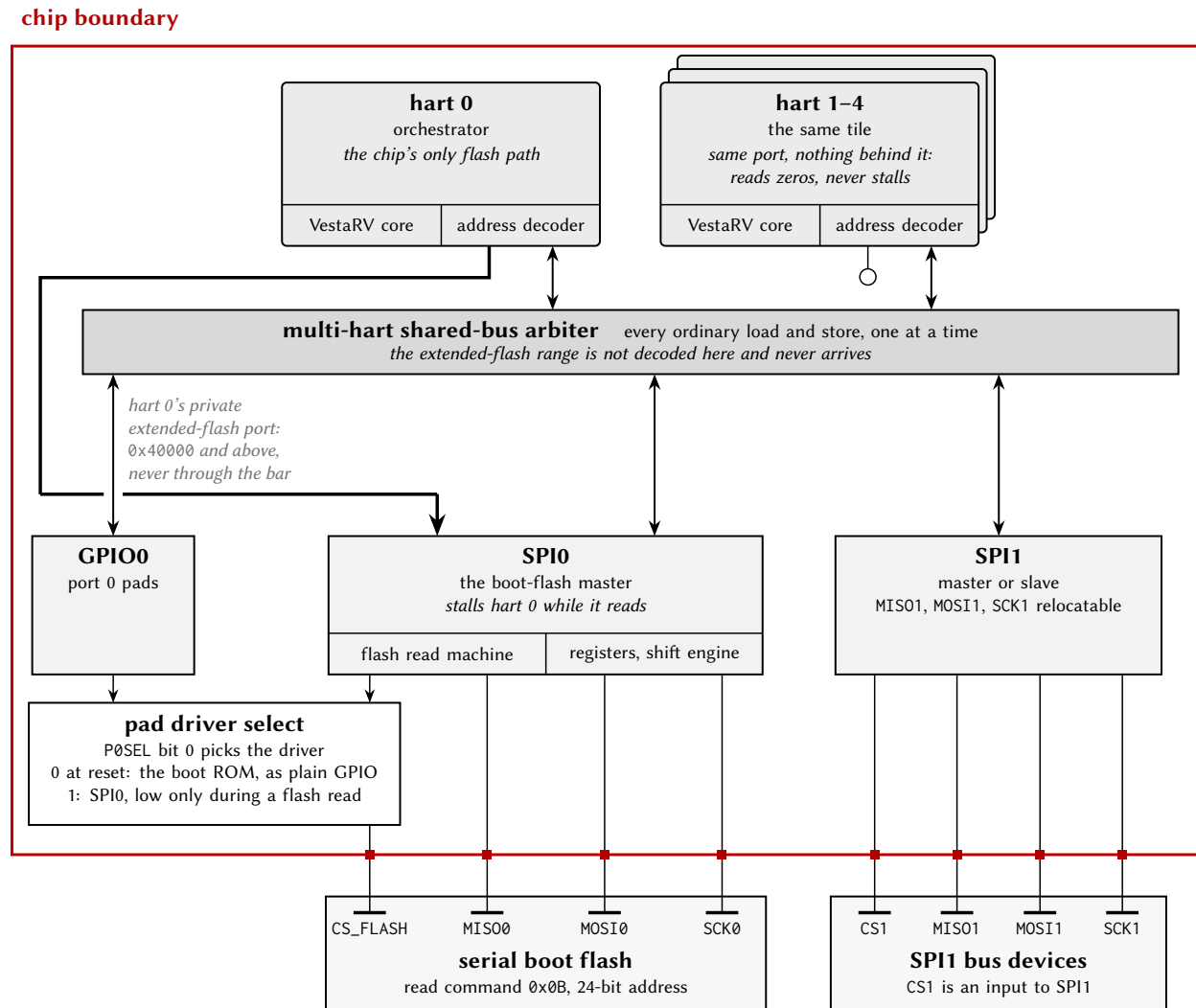


Figure 89: SPI connection diagram: SPI0 and the boot flash. Everything drawn is generated from this configuration. SPI0's registers are an ordinary arbiter slave, so every hart reaches them the same way, through the bar. The *flash* is different: hart 0's own address decoder claims every address at or above 0x40000 — the strict complement of the shared window — and takes it out of the tile on a private port that never reaches the bar and is never arbitrated, which is what makes execute-in-place possible (Section 17.10). SPI0 turns each such access into a flash read, sending the command byte and then $(\text{CPU address} + \text{SPI0FOS}) \bmod 2^{24}$, and stalls hart 0 until the word comes back. Every other hart has the same decoder and the same port with nothing behind it, so its accesses in that range read zeros and do *not* stall. CS_FLASH has two drivers and the pad's PxSEL bit chooses between them: it resets to 0, a plain GPIO output driven high, which is how the boot ROM clocks the flash before any of this is set up. SPI1, in configurations that include it, is a separate block with its own pins and no flash path of any kind.

SPI0 additionally owns the dedicated CS_FLASH pin, the chip select of the external boot flash. When its PxSEL[y] bit is a 1, SPI0 drives CS_FLASH automatically as part of the flash extended memory protocol; when the bit is a 0 it is an ordinary GPIO output, which is how the boot ROM toggles it during the boot sequence.

Note that several of SPI1's signals are also members of the shared alternate-function pool, so they can be relocated onto other GPIO pins through the PxAFS plane selects. Figure 89 names the relocatable set for this configuration — it is read off the same alternate-function matrix the GPIO chapter tabulates (Table 5), not written down twice — and SPI0's pins are not in that pool at all: the boot flash is wired to fixed pads.

17.2 Shared Access on Castalia

The SPI peripherals live in the shared peripheral window behind the multi-core arbiter (see Section 4), so any hart can use them, each at its original memory-mapped slot: SPI0 at 0x4200, and SPI1 (in configurations that include it) at 0x4300. SPI0 is the boot-flash master: its bus pins are wired to the external SPI flash, it is used by the boot ROM to load programs, and it provides the flash extended memory feature.

The arbiter makes every individual register access atomic: one hart's read or write of SPIxTX, SPIxSR, and so on completes as a whole before any other hart's access is served. Transfer-level ownership, however, is software's responsibility: if two harts push frames at the same time, their frames interleave on the wire and their received words race for SPIxRX. Software that shares an SPI port between harts should serialize whole transfers (or whole packets) with a lock: claim a hardware mutex (MUTEX bank) or an LR/SC session lock in shared RAM, perform the transfer, then release. See Section 4.4.

Beware that some SPI register reads have side effects that now fire on *any* hart's access: reading SPIxRX clears the SPITCIF flag. A hart must therefore never read SPIxRX speculatively while another hart owns the transfer; it would consume the owner's transfer-complete event. When ownership is ambiguous, clear flags explicitly by writing 1s to SPIxSR instead of relying on the read strobe.

The SPI interrupt lines (transfer complete, TX register empty) are wired to hart 0's interrupt enables in the SYSTEM peripheral and can additionally be routed to any other hart through the IRQROUTER peripheral. A hart's interrupt service routine must clear the interrupt flag (through the shared window) before returning.

Note that the SPI core runs in the SMCLK clock domain, and the boot ROM may leave SMCLK sourced from LFXT (32.768 kHz) when it jumps to the application; in that state SCK runs at a small fraction of that frequency and transfers appear to hang. Always configure the SMCLK source in the SYSTEM peripheral (Section 20.1) before using an SPI port, rather than relying on the boot-time clock state.

The flash extended memory region is the one exception to shared access: it is decoded on hart 0's private bus, not through the arbiter. Only hart 0 can read the flash through it; another hart's access to that address range reads zeros (see Section 17.10).

17.3 Generalized SPI Transfer Process

When no transfer is occurring, the SPI bus is idle. The transfer begins when the master asserts (pulls low) the CS wire attached to the desired slave. Immediately, the master seizes control of the MOSI wire by making its MOSI pin an output, and the slave does the same for the MISO wire. Note that the master always has control of the SCK and CS wires, which are configured as outputs. Then, the master issues SCK clock pulses in increments of 8, 16, or 32 (one frame of data). On every transition of SCK, both the master and slave alternate between updating and latching the values of MOSI and MISO, the master updating MOSI or latching MISO, and the slave updating MISO or latching MOSI. When the wire is updated, the next bit of data is pushed out of the TX shift register and written to the wire. When the wire is latched, that bit is read from the wire and pushed into the RX shift register. As the SPI port is full-duplex, the master and slave

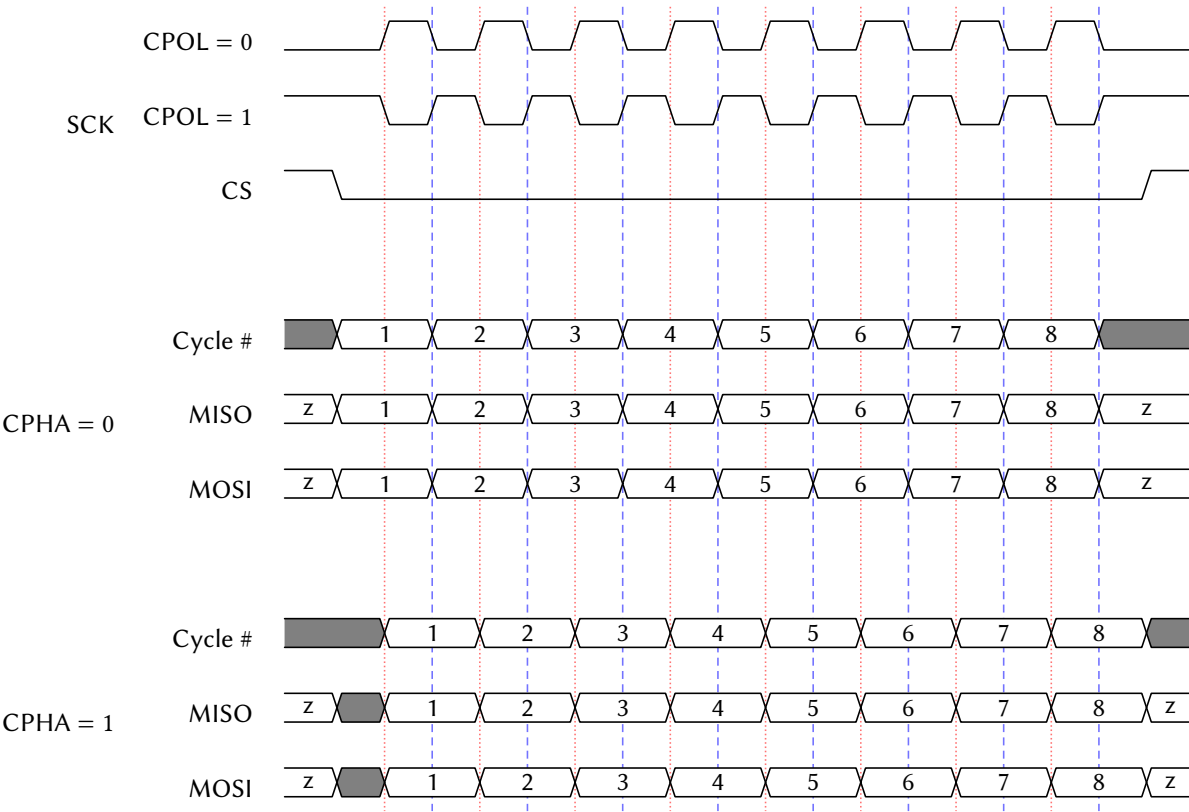


Figure 90: SPI Timing Diagram

both send data to the other simultaneously. After each frame of data has been transferred, both the master and slave RX shift registers contain their respective received words (the 8-, 16-, or 32-bit data transmitted in the frame). At that point, the master may either continue the process by transferring another frame, or cease the transfer by deasserting (pulling high) the CS wire. The frames that the master issues while the slave's CS wire is low is called a packet, and the master may issue as many frames in a packet as it wants before deasserting CS.

The order of the update/latch process, as well as the polarity of the SCK wire, are determined by two 1-bit parameters: clock polarity (CPOL) and clock phase (CPHA). CPOL determines the idle state and transition direction of SCK, while CPHA determines when the data in MOSI and MISO is updated and latched. These parameters together are referred to as the SPI mode, or SPIMODE, shown in Table 105. Note that this SPI peripheral supports all four SPI modes while in master mode, but only supports SPI modes 1 and 3 while in slave mode (CPHA = 1). Figure 90 shows a SPI timing diagram for all four SPI modes, with a single-frame packet with 8-bit frames.

SPIMODE	CPOL	CPHA	SCK Idle	Data Update Edge	Data Latch Edge
0	0	0	Low	Trailing/Falling	Leading/Rising
1	0	1	Low	Leading/Rising	Trailing/Falling
2	1	0	High	Trailing/Rising	Leading/Falling
3	1	1	High	Leading/Falling	Trailing/Rising

Table 105: SPIMODE Key

Note that during the very first frame of a transfer, the slave is required to send data to the master on the MISO wire, regardless of if it has any valid data to send. If it has no data to send, a slave will issue a “dummy” word, which the master must ignore.

17.4 Bit and Byte Ordering

The SPI peripheral has the ability to transfer the most significant bit (MSB) first or the least significant bit (LSB) first using SPIMSB. When SPIMSB is 0, the LSB is transferred first, and when SPIMSB is 1, the MSB is transferred first.

For 16- and 32-bit transfers, the byte ordering may optionally be swapped using SPITXSB and SPIRXSB, which swap the bytes being transmitted and received, respectively. For example, during a 16-bit transfer, if SPITXSB is enabled, then byte 1 of SPITX will be transmitted followed by byte 0. If SPIRXSB is enabled, then the first byte received will be placed in byte 1 of SPIRX, and the next 8 bits received will be placed in byte 0. In a 32-bit transfer, the byte order would thus be 3, 2, 1, 0. Figure 91 illustrates this ordering.

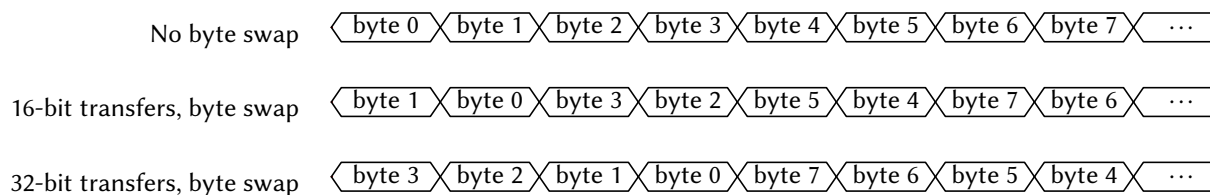


Figure 91: SPI Byte Ordering

Note that the bits in the frame are ordered MSB-first or LSB-first *before* the byte swap operation takes place. As such, a 16-bit, MSB-first transfer with byte swapping activated would transmit the bits 7 down to 0, and then 15 down to 8. Figure 92 shows the bit ordering for a 16-bit transfer.

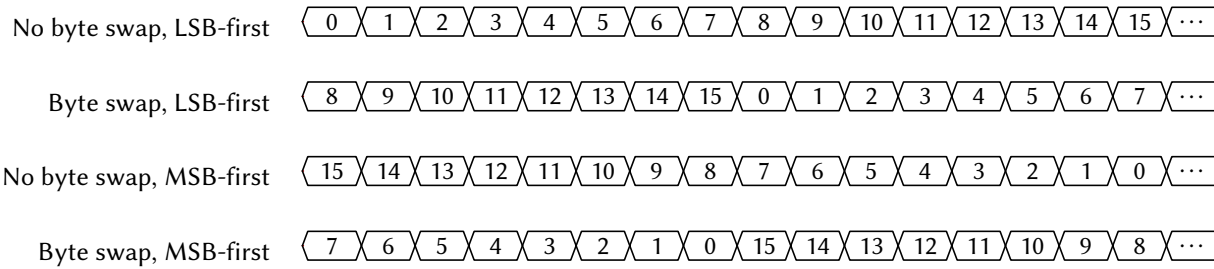


Figure 92: SPI Bit Ordering (16-bit transfer)

17.5 SPI Baud Clock

When in master mode, the frequency of the SCKx clock pin is determined by the frequency of SMCLK and the SPI baud control SPIBR, given by the expression:

$$f_{\text{SCK}} = \frac{f_{\text{SMCLK}}}{2 \times (1 + \text{SPIBR})}$$

Note that the maximum frequency of SCKx is half that of SMCLK.

17.6 Transmit Buffer Register Queue

Writing to the SPI transmit buffer register SPIxTX queues up the word written to it as the next frame to transmit. If in master mode and a transfer is not currently ongoing (i.e., SCKx is not clocking), the SPI peripheral immediately begins transmitting the word written to SPIxTX, and SPITEIF is set immediately. If a transfer is currently ongoing, the SPI peripheral waits for the current frame to finish transferring. As soon as it finishes, the SPI peripheral begins transmitting the new frame written to SPIxTX and sets SPITEIF. If in slave mode, the SPI peripheral latches and begins transferring the last word written to SPIxTX and sets SPITEIF whenever the master starts the first clock cycle of SCKx.

The TX queue allows the SPI peripheral to constantly transfer data without any dead time in between frames, ensuring maximum transfer speed. However, the user must take care not to overflow the queue, which happens if the SPIxTX is written two or more times during a single frame transfer. Queue overflows can be prevented by waiting for the SPITEIF flag to be 1 or SPIBUSY to be 0 before writing to SPIxTX.

17.7 Flags and Interrupts

The SPI peripheral has two interrupt flags (SPITCIF and SPITEIF) and one status indicator (SPIBUSY). If the corresponding interrupts are enabled in the SPIxCR register and the SPIx ISR is not masked, the interrupt flags will generate interrupts when set to 1. Both interrupt flags must be cleared before the ISR returns to prevent spurious re-execution.

The SPITCIF interrupt flag is set when a SPI transfer completes. It can be cleared by writing 1 to it or by reading the SPIxRX register (see the shared-access note above: the read-clear fires on any hart's read).

The SPITEIF interrupt flag is set when the SPIxTX transmit buffer becomes empty and ready to accept new data. In master mode without an ongoing transfer, it sets immediately when writing to SPIxTX. During an active transfer, it sets when the current frame completes and the queued data begins transmission. In slave mode, it sets when the master begins the first SCK clock cycle. This flag must be cleared by writing 1 to it.

The SPIBUSY indicator reflects the peripheral's transfer state. In master mode, it is 1 during active frame transmission (SCK actively clocking) and 0 when idle. Note that SPIBUSY can be 0 between frames

if no new data is queued before the previous frame completes. In slave mode, SPIBUSY is 1 when the CSx pin is asserted (low) and 0 when deasserted (high).

Note that flash extended memory reads (Section 17.10) are handled entirely in hardware and do not set SPITCIF or SPITEIF.

17.8 Master Mode Transfer Procedure

To use the SPI peripheral in master mode, the user must first initialize the SPIx peripheral by configuring the SPIxCR register. Of note, SPIEN must be 1 and SPISM must be 0 to enable the SPI peripheral in master mode. Then, the GPIO pins multiplexed with SCKx, MISOx, and MOSIx must be set to secondary/peripheral mode with their respective bits in the PxSEL registers. Finally, the GPIO pins attached to each slave's CS pin need to be set to output a logic high voltage (note that these CS pins are *not* the same as SPI1's own CS1 pin, which is used when this device is in slave mode).

To initiate a SPI transfer, the master should first clear the SPI status register SPIxSR by writing 0 to it, and then must assert (pull low) the slave's CS pin. Then, the master may begin transferring as many frames as it wants in the packet. To send a frame, the master must first check if the SPIxTX register is empty (i.e. ready to accept a new word to queue for transfer) by ensuring that either SPITEIF is 1 or SPIBUSY is 0. Then, the master may write the next word to transmit to SPIxTX. Next, if SPITCIF is 1, the master should read the value of the SPIxRX register to read the value the slave transmitted on the prior frame. Then, the master must clear the status register and may either start another frame or end the packet by deasserting (pulling high) the CS wire.

A simplified procedure for producing master mode SPI transfers is shown in Listing 1 that does not use the TX queue for continuous transfers.

```

1  /** Includes **/
2  #include <MemoryMap.h>
3
4  /** Function Declarations **/
5  uint32_t spi_transfer(uint32_t tx_data)
6
7  /** Main Function **/
8  int main()
9  {
10     // Initialize the SPI peripheral
11     SPIxCR = SPIEN;
12
13     // Configure the SCKx, MOSIx, and MISOx pins
14     SCKx_PxSEL |= SCKx_BIT;
15     MOSIx_PxSEL |= MOSIx_BIT;
16     MISOx_PxSEL |= MISOx_BIT;
17
18     // Configure the GPIO pin attached to the slave's CS pin as an output high
19     PxSEL &= ~(BITy);
20     PxOUT |= BITy;
21     PxDIR |= BITy;
22
23     // Assert the CS pin
24     PxOUT &= ~(BITy);
25
26     // Do a single SPI transfer
27     uint32_t tx_data = 57;
28     uint32_t rx_data;
29
30     rx_data = spi_transfer(tx_data);
31
32     // Deassert the CS pin
33     PxOUT |= BITy;
34
35     return 0;

```



```

36 }
37
38 /** Function Definitions */
39 uint32_t spi_transfer(uint32_t tx_data)
40 {
41     // Begin a SPI frame by writing the transmit data to SPIxTX
42     SPIxTX = tx_data;
43
44     // Wait for the frame to finish transmitting
45     while (SPIxSR & SPIBUSY) {}
46
47     // Return the received data from the slave
48     return SPIxRX;
49 }

```

Listing 1: Example Program for SPI Master Transfers

17.9 Slave Mode Transfer Procedure

Slave mode is available only on SPI1, in configurations that include it (SPI0 is master-only). To use SPI1 in slave mode, the user must first initialize the peripheral by configuring the SPI1CR register. Of note, SPIEN must be 1 and SPISM must be 1 to enable the SPI peripheral in slave mode. Then, the GPIO pins multiplexed with SCK1, MIS01, and MOSI1 must be set to secondary/peripheral mode with their respective bits in the PxSEL registers. Finally, the CS1 pin must be set as a high impedance input in GPIO mode to allow the master to assert this SPI1 peripheral.

Immediately after initialization, a word should be written to the SPI1TX register, which will be transmitted to the master when the master asserts the CS1 pin and begins sending SCK pulses. Whatever data is contained in SPI1TX will always be transmitted to the master when the master begins each frame regardless of if the slave has updated the value of SPI1TX, so care must be taken to always put valid data in the SPI1TX register before the master begins the next frame. Once the master begins sending a frame, the SPITEIF flag will become 1, indicating that the next word to transmit to the master should be written to SPI1TX. At the end of the frame, the SPITCIF flag will become 1, and the data in SPI1RX will contain the data received from the master. SPI1RX should then be read, and the status register SPI1SR must be cleared. Note that the SPI1RX register must be read before the end of the next frame, else the data will be overwritten and unrecoverable.

There is no dedicated interrupt indicating when the CS1 pin is asserted or deasserted, but this functionality can be achieved using the ordinary GPIO pin interrupts.

17.10 External SPI Flash Extended Memory Access

The SPI0 peripheral includes a flash extended memory feature that enables memory-mapped read access to the external SPI flash memory. When enabled via the SPIFEN control bit, this feature allows hart 0 to read from the SPI flash using standard load instructions (and to execute code in place from it), as if the flash were directly addressable ROM on the memory bus. This capability is designed to support programs larger than the internal RAM capacity, up to the full size of the SPI flash.

The flash region is decoded on hart 0's private bus, *not* through the multi-core arbiter: memory accesses by hart 0 at or above the extended flash base address (0x40000) are automatically translated to SPI flash read operations. The other harts have no path to the flash; their accesses in this range read zeros (without stalling). The SPI0 peripheral manages the flash read protocol transparently, including chip select control (via CS_FLASH), command sequencing, and data retrieval; the hart is stalled for the duration of each flash access.

Important: Using flash extended memory significantly reduces execution speed compared to RAM-based execution, as each 32-bit read requires multiple SCK clock cycles to retrieve data from the external flash (and SCK itself runs from SMCLK). This feature is best suited for read-only data, lookup tables, or code sections that are not performance-critical.

17.10.1 Flash Address Translation

The SPI0 peripheral translates CPU memory addresses to 24-bit SPI flash addresses using the SPI0FOS (SPI Flash Offset) register. Only the low 24 bits of the CPU address participate; the actual flash address accessed is computed as:

$$\text{Flash Address} = (\text{CPU Address} + \text{SPI0FOS}) \bmod 2^{24}$$

This offset mechanism allows flexible mapping of flash contents to different virtual addresses in the CPU's address space. The 24-bit address wraps at 0xFFFFF, discarding any overflow bits. By modifying SPI0FOS, software can remap flash regions without physically moving data in the flash memory. For example, to make flash address 0 appear at the extended flash base address itself, set SPI0FOS to the two's complement of the base address modulo 2^{24} .

17.10.2 Flash Initialization Procedure

Before enabling the flash extended memory feature, the SPI flash must be properly initialized. The initialization sequence depends on the specific flash chip, but typically includes:

1. Configure SPI0 in master mode with appropriate SPI mode (typically SPIMODE3), 8-bit transfers initially, and SCK frequency within flash specifications
2. Set SCK0, MOSI0, and MISO0 pins to secondary/peripheral mode via PxSEL
3. Configure the flash chip select pin (CS_FLASH) as GPIO output, initially high (deasserted)
4. Wake the flash from deep power-down mode (command varies by flash chip, often 0xAB); note that the boot ROM deliberately puts the flash *into* deep power-down at the end of every SPI flash boot, so this step is required after a flash boot
5. Configure flash-specific features such as page size or addressing mode using vendor-specific commands
6. Transfer control of CS_FLASH to the SPI peripheral by setting it to secondary/peripheral mode
7. Enable flash extended memory by setting SPIFEN = 1 in SPI0CR

Once initialized, the flash extended memory feature operates transparently. Read operations by hart 0 in the designated address range automatically trigger SPI flash transactions without software intervention.

17.11 SPI0 Boot Behavior

The SPI0 peripheral is used during the boot process to load programs from the external SPI flash memory. All harts reset into the shared boot ROM (see Section 9); hart 0 then samples the BOOT pin to decide the boot mode. If BOOT is low, the interactive Forth interpreter is launched from ROM over UART0. If BOOT is high, the boot ROM uses SPI0 to read the program image from the attached SPI flash, copies its segments into RAM (the load window is 0x8000–0xFFFFC), puts the flash into deep power-down, switches SMCLK to

LFXT, and jumps to the program entry point at 0x8200. The other harts remain parked in the ROM until launched through the boot-ROM loader (see Section 4).

During the boot sequence the ROM drives CS_FLASH (as a GPIO output) and clocks the flash, and the flash drives the MISO0 line. If other slave devices share the SPI0 bus, they must not drive MISO0 during the boot sequence. This conflict can occur if a slave's chip select (CS) pin is controlled by a GPIO pin that defaults to high-impedance input at reset, potentially allowing the slave's CS to float low (asserted state).

To prevent bus contention during boot:

- Use GPIO pins with pull-up resistors or start-high reset behavior to control slave CS pins
- Ensure all non-flash slaves on the SPI0 bus have their CS pins driven high during MCU reset
- Consider dedicating SPI0 exclusively to the boot flash, using SPI1 (in configurations that include it) for other SPI devices
- Add external pull-up resistors to slave CS lines if GPIO reset behavior is insufficient

After the boot sequence completes and the program begins executing, SPI0 becomes available for normal SPI operations, including the flash extended memory feature if enabled by software.

17.12 Register Description

17.12.1 SPICR Register

Register memory slot: 0, offset: 0 bytes, size: 4 bytes

SPI control register

31	30	29	28	27	26	25	24
Unused							
23	22	21	20	19	18	17	16
Unused				FEN	SM	TXSB	RXSB
				rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
BR							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
EN	MSB	TCIE	TEIE	DL		CPOL	CPHA
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
Unused	Bits 31-20	
FEN	Bit 19	SPI Flash extended memory enable. When enabled, allows native read-only memory access to the SPI flash memory via the system memory bus. The SPI peripheral automatically handles flash read commands and provides transparent memory-mapped access. Available only on SPI0; reads as 0 on SPI1. 0 Flash extended memory disabled 1 Flash extended memory enabled
SM	Bit 18	SPI slave mode select. Configures the SPI peripheral for master or slave operation. SPI0 is master-only; this bit has no effect on SPI0. SPI1 supports both master and slave modes. 0 Master mode 1 Slave mode
TXSB	Bit 17	SPI TX swap bytes. Swaps the byte order in 16- and 32-bit transmissions. In 32-bit transmissions, bytes 3 and 0 are swapped and bytes 2 and 1 are swapped. In 16-bit transmissions, bytes 1 and 0 are swapped. Does not affect 8-bit transmissions. 0 Bytes not swapped 1 Bytes swapped
RXSB	Bit 16	SPI RX swap bytes. Swaps the byte order in 16- and 32-bit receptions. In 32-bit receptions, bytes 3 and 0 are swapped and bytes 2 and 1 are swapped. In 16-bit receptions, bytes 1 and 0 are swapped. Does not affect 8-bit receptions. 0 Bytes not swapped

		1 Bytes swapped
BR	Bits 15-8	SPI clock (SCK) baud rate control for master mode. Baud rate = $SMCLK / (2 * (1 + SPIBR))$. For example, with SMCLK at 24 MHz: SPIBR=0 gives 12 MHz, SPIBR=1 gives 8 MHz, SPIBR=2 gives 6 MHz, SPIBR=5 gives 4 MHz, SPIBR=11 gives 2 MHz, SPIBR=23 gives 1 MHz.
EN	Bit 7	SPI enable. When disabled, all SPI operations cease and the peripheral is held in reset state. 0 Disabled 1 Enabled
MSB	Bit 6	Bit endianness select. Determines whether data is transmitted and received MSB-first or LSB-first. 0 LSB-first 1 MSB-first
TCIE	Bit 5	SPI transmit complete interrupt enable. Interrupt triggers when a full SPI transfer (transmit and receive) completes. 0 Interrupt disabled 1 Interrupt enabled
TEIE	Bit 4	SPI transmit register empty interrupt enable. Interrupt triggers when SPIxTX register is empty and ready to accept new data for the next transfer. 0 Interrupt disabled 1 Interrupt enabled
DL	Bits 3-2	SPI transmission data length select. Determines the number of bits transferred per SPI transaction. 00 SPIDL_8: 8-bit transfers 01 SPIDL_16: 16-bit transfers 10 SPIDL_32: 32-bit transfers 11 SPIDL_RES: Reserved (do not use)
CPOL	Bit 1	SPI clock (SCK) polarity. Determines the idle state of the SCK line. 0 SCK idles low (SPIMODE0 or SPIMODE1) 1 SCK idles high (SPIMODE2 or SPIMODE3)
CPHA	Bit 0	SPI clock (SCK) phase. Determines when data is sampled relative to the SCK edge. In slave mode, only SPICPHA=1 is supported. 0 Data sampled on leading edge, shifted on trailing edge (SPIMODE0 or SPIMODE2) 1 Data shifted on leading edge, sampled on trailing edge (SPIMODE1 or SPIMODE3)

17.12.2 SPISR Register

Register memory slot: 1, offset: 4 bytes, size: 1 byte

SPI status register. Provides real-time status of SPI transfer operations and interrupt flags.

7	6	5	4	3	2	1	0
Unused					BUSY	TCIF	TEIF
					r-(0)	rw1-(0)	rw1-(0)

Bitfield	Range	Description
<i>Unused</i>	Bits 7-3	
BUSY	Bit 2	Indicates whether a SPI transfer is currently in progress. In master mode, set when a transfer starts and cleared when complete. In slave mode, set when chip select is asserted (driven low) and cleared when deasserted. 0 SPI is idle 1 SPI transfer in progress
TCIF	Bit 1	SPI transfer complete interrupt flag. Set when a SPI transfer completes. Must be cleared by writing 1 to this bit or by reading SPIxRX register. 0 No transfer completed 1 Transfer completed
TEIF	Bit 0	SPI transmit register empty interrupt flag. Set when SPIxTX register is empty and ready to accept new data. The data will not be transmitted until any current transfer completes. Must be cleared by writing 1 to this bit. 0 SPIxTX not empty 1 SPIxTX empty and ready

17.12.3 SPITX Register

Register memory slot: 2, offset: 8 bytes, size: 4 bytes

SPI transmit buffer register. In master mode, writing to this register initiates a new SPI transfer. In slave mode, writing to this register queues data to be transmitted during the next master-initiated transfer. Actual number of bits transmitted is determined by SPIDL setting.

31	30	29	28	27	26	25	24
TX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
TX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
TX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
TX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
TX	Bits 31-0	

17.12.4 SPIRX Register

Register memory slot: 3, offset: 12 bytes, size: 4 bytes

SPI receive buffer register. Contains the data received during the most recent SPI transfer. Reading this register also clears the SPITCIF flag. Valid data width depends on SPIDL setting; unused upper bits read as 0.

31	30	29	28	27	26	25	24
RX							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
23	22	21	20	19	18	17	16
RX							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
15	14	13	12	11	10	9	8
RX							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
7	6	5	4	3	2	1	0
RX							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)

Bitfield	Range	Description
RX	Bits 31-0	

17.12.5 SPIFOS Register

Register memory slot: 4, offset: 16 bytes, size: 4 bytes

SPI Flash memory address offset. This 24-bit value is added to memory access addresses when flash extended memory mode is enabled (SPIFEN=1). Allows remapping of flash memory to different virtual addresses. The addition wraps around at 0x00FFFFFF. Available only on SPI0; reads as 0 on SPI1.

31	30	29	28	27	26	25	24
Unused							
23	22	21	20	19	18	17	16
FOS							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
FOS							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
FOS							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
<i>Unused</i>	Bits 31-24	
FOS	Bits 23-0	

18 UARTx Peripheral

The Universal Asynchronous Receiver/Transmitter (UART) is an asynchronous, full-duplex serial interface with hardware parity support. The main features of the UART are listed below:

- Full-duplex, independent receive and transmit operation
- Asynchronous (no serial clock wire)
- Two wires, one for TX and one for RX
- Communicates with one external device
- 8-bit data frames
- Adjustable baud generator with 16× oversampling
- Hardware parity generation and checking (even or odd)
- Error detection: framing error, parity error, and receive overflow flags
- Three separate interrupt sources: receive complete, transmit complete, and TX register empty
- Latched register reads for noise immunity

18.1 UART Pins

The UARTx peripheral has two pins: RXx and TXx.

The TXx pin transmits each data bit from this UARTx peripheral to the external device. It must be attached to the RX pin of the external device. When the corresponding PxSEL[y] bit is a 1, the UARTx peripheral seizes control of the DIR, OUT, and REN controls of TXx, making it an output.

The RXx pin receives each data bit from the external device. It must be attached to the TX pin of the external device. When the corresponding PxSEL[y] bit is a 1, the UARTx peripheral seizes control of the DIR, OUT, and REN controls of RXx, making it a high impedance input.

The UART transmitter and receiver are independent of one another. Thus, transmissions and receptions need not occur at the same time or with the same external device. Indeed, both pins do not necessarily need to be used if one-way communications are all that is needed.

A connection diagram for the UART is displayed in Figure 93.

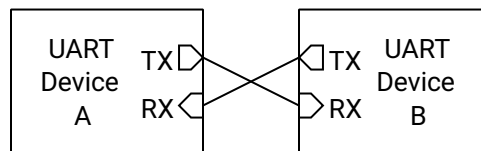


Figure 93: UART Connection Diagram

18.2 Shared Access on Castalia

The UARTs live in the shared peripheral window behind the multi-core arbiter (see Section 4), so any hart can use them, each at its original memory-mapped slot: UART0 at 0x4400, and UART1 (in configurations that include it) at 0x4500. UART0 is the *console UART*: it carries the boot-time Forth console.

The arbiter makes every individual register access atomic: one hart’s read or write of UARTxTX, UARTxSR, and so on completes as a whole before any other hart’s access is served. Message-level ownership, however, is software’s responsibility: if two harts transmit at the same time, their *bytes* interleave on the wire. Software that shares a UART between harts should serialize whole messages with a lock: claim a hardware mutex (MUTEX bank) or an LR/SC session lock in shared RAM, transmit the message, then release. See Section 4.4.

The UART interrupt lines (receive complete, TX register empty, transmit complete) are wired to hart 0’s interrupt enables in the SYSTEM peripheral and can additionally be routed to any other hart through the IRQROUTER peripheral. A hart’s interrupt service routine must clear the interrupt flag (through the shared window) before returning.

Note that the UART core runs in the SMCLK clock domain, and the boot ROM may leave SMCLK sourced from LFXT (32.768 kHz) when it jumps to the application; in that state UART frames take roughly 10 ms each. Always configure the SMCLK source in the SYSTEM peripheral (Section 20.1) before using a UART, rather than relying on the boot-time clock state.

18.3 Generalized UART Transmission Process

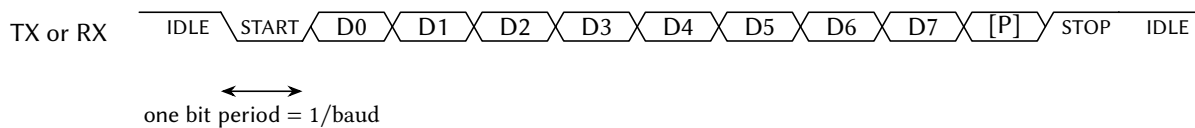


Figure 94: UART Data Frame Timing Diagram

Each UART wire is unidirectional, with the TX pin of the transmitting device driving the logic level of the wire and the RX pin of the receiving device reading the level of the pin. The only distinction between the two wires is the direction that data travels, and as such, this section will only depict one wire.

When no UART transmission is occurring, the UART wire is idle, and the TX pin drives the UART wire high. To start a transmission, the TX pin transmits a start bit by driving the wire low for $T_B = 1/\text{baud}$ seconds. It then begins transmitting each data bit, LSB-first, every T_B seconds until all 8 bits have been transmitted. Then, it optionally transmits a parity bit, which helps detect transmission errors. Finally, it transmits a stop bit by driving the wire high for T_B seconds. After that point, it may begin another transmission by sending another start bit, or it can let the wire be idle.

Figure 94 shows a timing diagram for the UART transmission process. Note that the parity bit [P] is optional.

18.4 UART Baud Generation

Both devices communicating over the UART must agree on the baud rate, or the number of bits transmitted per second. In the absence of a clock wire, the UART uses the baud rate to sample the RX wire every $T_B = 1/\text{baud}$ seconds, recording the logic levels in sequence. The transmitter must send bits at that same rate. Any significant deviation of the baud rate can result in bit errors.

The UART uses 16× oversampling for improved noise immunity and accurate bit detection. The baud rate for the UARTx peripheral is configured through the 12-bit UARTxBR register:

$$\text{UART baud} = \frac{f_{\text{SMCLK}}}{16 \times (\text{BR} + 1)}$$

For example, with SMCLK running at 24 MHz:

- BR = 12 gives 115,385 baud (within 0.2 % of standard 115,200)
- BR = 25 gives 57,692 baud (within 0.2 % of standard 57,600)
- BR = 51 gives 28,846 baud (within 0.2 % of standard 28,800)

18.5 Parity Bit

A UART frame has an optional parity bit after the last data bit and before the stop bit. The purpose of the parity bit is to verify that the data received is indeed the data that was transmitted. If the received parity bit does not match the expected parity bit, then the receiver knows that the received data has been corrupted.

If the parity bit is enabled with UPEN, the transmitter calculates the parity of the byte to transmit by taking the exclusive OR of all 8 bits in the transmitted byte, the result of which is then put through a final exclusive OR gate with the parity select bit PSEL:

$$p = b_7 \oplus b_6 \oplus \dots \oplus b_1 \oplus b_0 \oplus \text{PSEL}$$

When PSEL is 0, the parity is said to be “even”, and when it is 1, “odd”. This operation is functionally counting how many bits are 1s in the transmitted byte. When even parity is selected, the parity bit is 0 when the number of 1s is even and 1 when the number of 1s is odd. That result is inverted when odd parity is selected.

On the receiver end, the parity bit is capable of detecting a single bit error in the transmission. That is, if one of the eight data bits (or the parity bit itself) is incorrectly received, then the parity error flag PEF will detect a bit error. However, if two bits are incorrectly received, then the parity error flag will not detect the bit error. Note that if the number of bit errors is an odd number, the parity error flag will detect the error, but will not when the number of bit errors is even.

If a parity bit is enabled on the transmitter side, it must also be enabled on the receiver side with UPEN.

18.6 Transmit Buffer Register Queue

Writing to the UART transmit buffer register UARTxTX queues up the byte written to it as the next frame to transmit. If a transmission is not currently ongoing, the UARTx peripheral immediately begins transmitting the byte written to UARTxTX, and TEIF is set immediately. If a transfer is currently ongoing, the UARTx peripheral waits for the current frame to finish transferring. As soon as it finishes, the UARTx peripheral begins transmitting the new frame written to UARTxTX and sets TEIF.

The TX queue allows the UARTx peripheral to constantly transfer data without any dead time in between frames, ensuring maximum transfer speed. However, the user must take care not to overflow the queue, which happens if the UARTxTX is written two or more times during a single frame transfer. Queue overflows can be prevented by waiting for the TEIF flag to be 1 or TXBF to be 0 before writing to UARTxTX.

18.7 Flags and Interrupts

The UARTx peripheral has five status flags and three interrupt flags in the UARTxSR register. If the corresponding interrupts are enabled in the UARTxCR register and the UARTx ISR is not masked, then the three

interrupt flags will generate an interrupt whenever they are equal to 1. Any and all interrupt flags must be cleared before the UARTx ISR returns, else the ISR will mistakenly re-execute immediately upon the prior return.

The UARTxSR and UARTxRX registers are latched on the falling edge of the memory enable signal for improved noise immunity during reads.

The UART receiver busy flag RXBF shows when the UARTx peripheral is currently receiving a byte of data when it is 1. Likewise, TXBF indicates that a transmission is ongoing or pending when 1.

The UART framing error flag FEF indicates that a framing error occurred on the last reception when it is a 1. A framing error typically occurs if the UART transmitter settings do not match the receiver settings, such as baud rate, number of bits to transmit, or the presence/absence of a parity bit. Specifically, a framing error is detected if the receiver does not detect a high level at the expected stop bit time. If FEF is a 1 after a reception, that reception must be treated as unrecoverable, corrupted data. FEF is cleared when the UARTxRX register is read.

The UART parity error flag PEF indicates that a parity error occurred on the last reception when it is a 1. A parity error can only happen if the parity bit is enabled by setting UPEN to 1. A parity error occurs when the parity bit appended to the received data byte does not match the expected value calculated from the received data. If PEF is a 1 after a reception, that reception must be treated as corrupted data. PEF is cleared when the UARTxRX register is read.

The UART receive data overflow flag OVF indicates that the user did not read the UARTxRX register in time before another byte of data was received by the UARTx peripheral when it is a 1. In this case, the previous received data has been lost and must be treated as incomplete. OVF is cleared when the UARTxRX register is read.

The UART receive complete interrupt flag RCIF is set to 1 immediately when the UARTx peripheral finishes receiving a frame of data from the transmitter. The new data byte becomes available in the UARTxRX register as soon as RCIF is set. This is also the moment that FEF, PEF, and OVF flags will be updated, if the proper conditions are met. RCIF is cleared when the UARTxRX register is read or by writing a 1 to this bit in UARTxSR.

The UART transmit register empty interrupt flag TEIF is set the moment the UART TX buffer register UARTxTX becomes empty (i.e., is ready to queue up the next byte to transmit). If a frame is not currently being transmitted, the flag is set immediately when a new byte is written to UARTxTX. If a frame is currently being transmitted, it is set as soon as the current transfer is completed, whereupon the value in the UARTxTX register begins to be transmitted. Write a 1 to this bit in UARTxSR to clear TEIF.

The UART transmit complete interrupt flag TCIF is set the moment the UART transmission completes (all bits sent) and the transmitter becomes idle. Write a 1 to this bit in UARTxSR to clear TCIF.

18.8 Transmit Procedure

To transmit a byte of data via the UART, first initialize the UARTx peripheral by configuring the UARTxCR register and setting the baud rate with the UARTxBR register. Of note, the UART enable bit UEN must be a 1 to activate the UART. When UEN is 0, the UART peripheral is held in reset state. The GPIO pin multiplexed with TXx must be set to secondary/peripheral mode in the corresponding PxSEL register.

To transmit a byte of data over the UART, first clear the status register by reading it and discarding its value. Then, wait until the UART is not transmitting (when TXBF is 0) or the UART TX buffer is empty (when TEIF is 1). Next, write the byte to transmit to the UARTxTX register. The byte will be finished transmitting once TXBF is 0.

A simplified procedure for transmitting a byte over the UART is shown in Listing 2.

```
1  /** Includes **/
2  #include <MemoryMap.h>
```

```

3
4  /** Function Definitions */
5  void uartx_putc(char tx_data)
6  {
7      // Wait for the transmitter to finish any previous transmissions
8      while (UARTxSR & TXBF) { }
9
10     // Send the new data byte
11     UARTxTX = tx_data;
12 }

```

Listing 2: Example Function for UART Transmissions

18.9 Receive Procedure

To receive a byte of data via the UART, first initialize the UARTx peripheral by configuring the UARTxCR register and setting the baud rate with the UARTxBR register. Of note, the UART enable bit UEN must be a 1 to activate the UART. The GPIO pin multiplexed with RXx must be set to secondary/peripheral mode in the corresponding PxSEL register.

Once done setting up the UART, wait for the UART receive complete interrupt flag RCIF to be 1. Then, read the received byte contained in the UARTxRX register. Reading UARTxRX automatically clears the RCIF, FEF, PEF, and OVF flags.

A simplified procedure for receiving a byte over the UART is shown in Listing 3.

```

1  /** Includes */
2  #include <MemoryMap.h>
3
4  /** Function Definitions */
5  char uartx_getc()
6  {
7      // Wait for the UART to receive a new character
8      while ((UARTxSR & RCIF) == 0) { }
9
10     // Return the new character that is in the receive buffer register
11     return UARTxRX;
12 }

```

Listing 3: Example Function for UART Reception

18.10 UART0 Boot Behavior

When booting in Forth interpreter mode, the UART0 peripheral provides an interactive command processor with an external device, such as computer with a USB to UART device attached. When the MCU is powered up or reset and the BOOT pin is low, the Forth interpreter will launch over UART0, which can be used to program or debug the MCU.

Section 9 elaborates on the power up and reset behavior of the MCU.

18.11 Register Description

18.11.1 UARTCR Register

Register memory slot: 0, offset: 0 bytes, size: 1 byte

UART control register. Controls UART enable, parity configuration, and interrupt enable bits. When UCR.EN is disabled, the UART peripheral is held in reset state.

7	6	5	4	3	2	1	0
Unused		EN	PEN	PSEL	CIE	TEIE	TCIE
		rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
Unused	Bits 7-6	
EN	Bit 5	UART enable. When disabled, the UART peripheral is in reset state and the transmitter is idle. 0 UART disabled 1 UART enabled
PEN	Bit 4	Parity enable. Enables parity generation on transmit and parity checking on receive. 0 Parity disabled 1 Parity enabled
PSEL	Bit 3	Parity select. Selects even or odd parity when parity is enabled. 0 Even parity 1 Odd parity
CIE	Bit 2	Receive complete interrupt enable. Enables interrupt generation when a byte is successfully received. 0 RX complete interrupt disabled 1 RX complete interrupt enabled
TEIE	Bit 1	Transmit empty interrupt enable. Enables interrupt generation when the transmit buffer becomes empty and ready for new data. 0 TX empty interrupt disabled 1 TX empty interrupt enabled
TCIE	Bit 0	Transmit complete interrupt enable. Enables interrupt generation when transmission is complete and the transmitter is idle. 0 TX complete interrupt disabled 1 TX complete interrupt enabled

18.11.2 UARTSR Register

Register memory slot: 1, offset: 4 bytes, size: 1 byte

UART status register. Contains receiver and transmitter status flags and interrupt flags. Error flags (FEF, PEF, OVF, RCIF) are cleared by reading UARTxRX. Interrupt flags (RCIF, TEIF, TCIF) can also be cleared by writing a 1 to the respective bit.

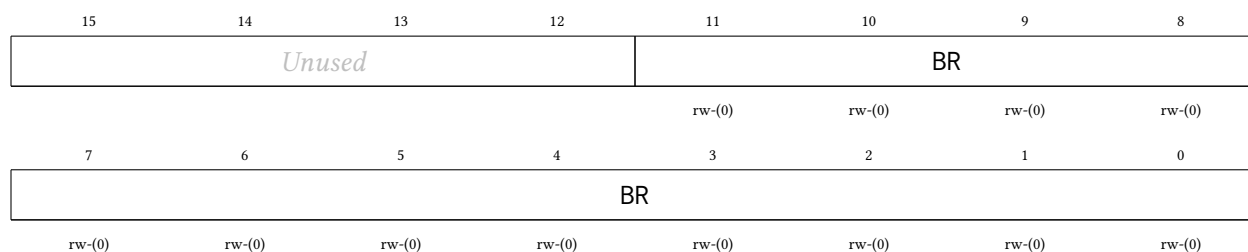
7	6	5	4	3	2	1	0
RXBF	TXBF	FEF	PEF	OVF	RCIF	TEIF	TCIF
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	rw1-(0)	rw1-(0)	rw1-(0)

Bitfield	Range	Description
RXBF	Bit 7	Receiver busy flag. Set while the UART receiver is actively receiving a byte. 0 Receiver idle 1 Reception in progress
TXBF	Bit 6	Transmitter busy flag. Set while the UART transmitter is actively transmitting a byte or has a pending transmission. 0 Transmitter idle 1 Transmission in progress
FEF	Bit 5	Framing error flag. Set when the stop bit is not detected (RX line not high at expected stop bit time). Cleared by reading UARTxRX. 0 No framing error 1 Framing error detected on last reception
PEF	Bit 4	Parity error flag. Set when received parity does not match expected parity (when parity is enabled). Cleared by reading UARTxRX. 0 No parity error 1 Parity error detected on last reception
OVF	Bit 3	Receive overflow flag. Set when a new byte is received before the previous byte in UARTxRX was read by the processor. Cleared by reading UARTxRX. 0 No receive data overrun 1 Receive data overflow detected
RCIF	Bit 2	Receive complete interrupt flag. Set when a byte is successfully received and placed in UARTxRX. Cleared by reading UARTxRX or writing a 1 to this bit. 0 No pending RX complete interrupt 1 RX complete interrupt pending
TEIF	Bit 1	Transmit empty interrupt flag. Set when the transmit buffer is empty and ready to accept new data. Write a 1 to this bit to clear it. 0 No pending TX empty interrupt 1 TX empty interrupt pending
TCIF	Bit 0	Transmit complete interrupt flag. Set when transmission is complete (all bits sent) and the transmitter is idle. Write a 1 to this bit to clear it. 0 No pending TX complete interrupt 1 TX complete interrupt pending

18.11.3 UARTBR Register

Register memory slot: 2, offset: 8 bytes, size: 2 bytes

UART baud rate register. Configures the baud rate divisor for the UART. The UART uses 16× oversampling.

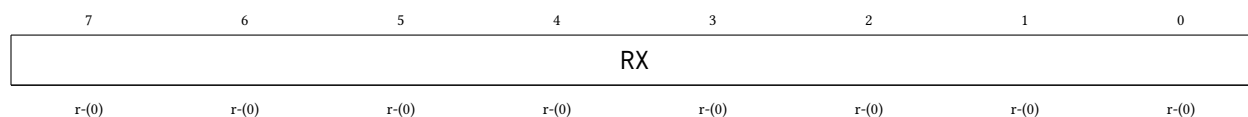


Bitfield	Range	Description
<i>Unused</i>	Bits 15-12	
BR	Bits 11-0	Baud rate divisor. UART baud rate = $SMCLK \div (16 \times (BR + 1))$. For example, with $SMCLK = 48\text{ MHz}$: $BR = 25$ gives 115,200 baud; $BR = 51$ gives 57,600 baud; $BR = 103$ gives 28,800 baud.

18.11.4 UARTRX Register

Register memory slot: 3, offset: 12 bytes, size: 1 byte

UART receive buffer register. Contains the last received byte. Reading this register clears the FEF, PEF, OVF, and RCIF flags in UARTxSR. The value is latched on the falling edge of the memory enable signal.

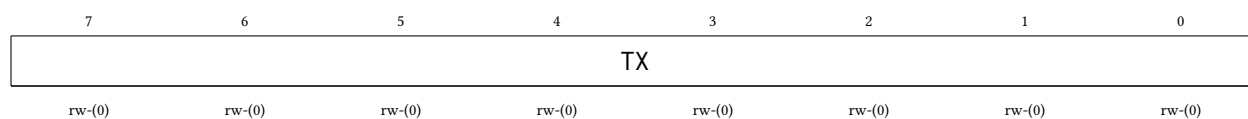


Bitfield	Range	Description
RX	Bits 7-0	Received data byte

18.11.5 UARTTX Register

Register memory slot: 4, offset: 16 bytes, size: 1 byte

UART transmit buffer register. Writing to this register loads the byte to transmit and initiates transmission. Do not write to this register while TXBF is set in UARTxSR.



Bitfield	Range	Description
TX	Bits 7-0	Transmit data byte

19 TIMERx Peripheral

The TIMERx peripheral is a 32-bit general purpose timer/counter unit with glitch-free clock source switching. Its main features are listed below:

- 32-bit timer values, from 0 to 4,294,967,295
- Memory mapped register to read and write current timer value
- Three output compare units, two of which can generate pulse width modulation (PWM) waveforms
- Two input capture units for precise timing measurements of external signals
- Four clock sources: SMCLK, MCLK, LFXT, HFXT
- Configurable clock divider ($\div 1$ to $\div 32,768$)
- Glitch-free clock source multiplexer
- Six separate interrupts: 2 capture, 3 compare, 1 overflow
- Latched register reads for noise immunity

The timers live in the shared peripheral page behind the multi-core arbiter (TIMER0 at 0x4600, and TIMER1, in configurations that include it, at 0x4700), so any hart can own and operate a timer (see Section 4). Register accesses are serialized by the arbiter; a timer's interrupts are wired to hart 0's enables in the SYSTEM peripheral and can be routed to any hart through the IRQROUTER peripheral.

19.1 Timer Pins

The TIMERx peripheral has four pins: TxCMP0, TxCMP1, TxCAP0, and TxCAP1.

The TxCMP0 pin outputs the PWM waveform generated by the output compare unit 0. When the corresponding PxSEL[y] bit is a 1, the TIMERx peripheral seizes the DIR, OUT, and REN controls of the pin, forcing it to be in output mode. Similarly, TxCMP1 outputs the PWM waveform of output compare unit 1.

The TxCAP0 pin is an input that waits for a pin state change and then notifies the input capture unit 0, allowing precise timing of pin changes. The corresponding PxSEL[y] bit has no effect on this pin, since its secondary/peripheral function only ever acts as an input. Similarly, TxCAP1 is attached to input capture unit 1.

19.2 Clock Sources and Divider

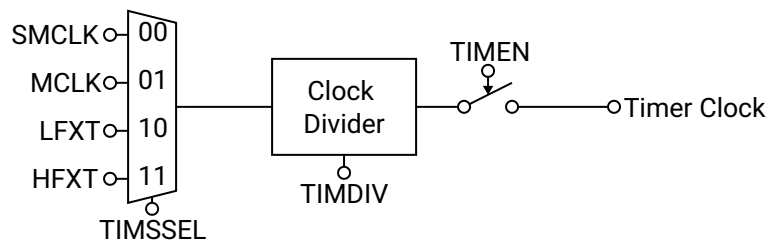


Figure 95: TIMERx Clock Diagram

The clock source for the TIMERx peripheral can be selected using SSEL from four options: SMCLK, MCLK, LFXT (low frequency external clock), and HFXT (high frequency external clock). A glitch-free clock multiplexer built into the clock source selector prevents spurious clock transitions on the timer clock, allowing the user to switch clock sources freely even while the timer is enabled. However, switching clocks will cause the timer clock to be inactive for several clock cycles of the clock that is being switched from or to, whichever has a smaller frequency.

Clock handoff after reset: the glitch-free multiplexer powers up parked on the SMCLK slice, and releasing a slice takes three clock edges *of the clock being released* before the newly selected source can engage. The practical consequence: the first source selection after reset waits for three SMCLK edges, however slow SMCLK currently is. Since the boot ROM may leave SMCLK sourced from LFXT (32.768 kHz) when it jumps to the application, selecting a timer clock source right after boot can stall the timer for roughly 92 μ s, even when the selected source is MCLK or HFXT. The safe handoff procedure is:

1. Configure SMCLK to a fast source in the SYSTEM peripheral (see Section 20.1) before touching the timer.
2. After enabling the timer, poll TIMxVAL until it has visibly incremented, rather than assuming the timer counts immediately. Two back-to-back reads taken right after enabling can both legitimately return the same value while the handoff completes.

After the clock multiplexer, the timer clock frequency may optionally be reduced with a clock divider controlled by DIV. The timer value register TIMxVAL increments by 1 on every rising edge of the divided timer clock when TIMERx is enabled with TEN.

A clock diagram for the TIMERx peripheral is shown in Figure 95.

19.3 Starting, Halting, Setting, and Reading the Timer

To start the timer, set the TEN bit to 1, whereupon the timer will begin incrementing by 1 with each rising edge of the divided timer clock. To halt the timer at its present value, clear the TEN bit to 0. The timer will retain its value until it is enabled again, its value is set manually, or the MCU is reset. If the timer is reenabled, it will begin counting from the timer value just before it was reenabled.

The timer value may be manually set at any time by writing the new 32-bit timer value to the TIMxVAL register. The timer value is updated immediately, regardless of if the timer is enabled or disabled.

The timer value may be read at any time by reading its value from the TIMxVAL register, regardless of if the timer is enabled or disabled. The TIMxVAL register is latched on the falling edge of the memory enable signal for stable reads during timer operation.

19.4 Overflow and Rollover Behavior

When the timer is enabled and its 32-bit value reaches its maximum of 4,294,967,295, the value will roll over back to zero on the next rising edge of the timer clock and the timer overflow interrupt flag OVIF will be set. The timer will then continue incrementing as normal.

Optionally, the user may configure the timer to roll over before the 32-bit limit is reached by setting the new rollover value in the TIMxCMP2 register and setting the CMP2RST bit to 1. In this mode, when the timer value becomes equivalent to the value of TIMxCMP2, the timer value will roll over back to zero on the next rising edge of the timer clock and then continue incrementing as normal, as depicted in Figure 96. This mechanism is used for setting the period of the PWM signals for the output compare units. Note: the timer rolls over back to zero when its value is *equal* to that of TIMxCMP2, but not if its value is greater than TIMxCMP2. This becomes important if the user manually sets the timer value above TIMxCMP2, or if

the user sets TIMxCMP2 below the current timer value, whereupon the timer will continue counting all the way to the 32-bit maximum value.

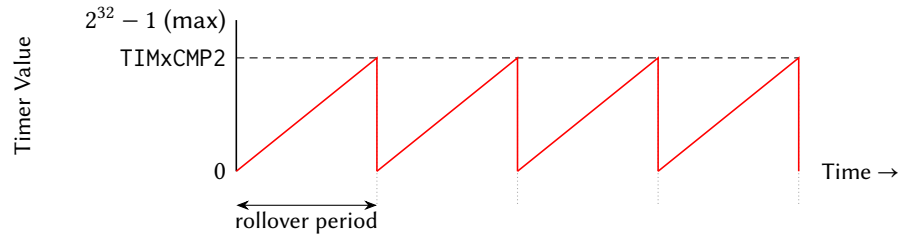


Figure 96: Timer Rollover Operation

19.5 Output Compare Units and Pulse Width Modulation

The TIMERx peripheral contains two output compare units that can be used to create two pulse width modulation (PWM) output signals on GPIO pins. Each output compare unit has an output compare register (TIMxCMP0 and TIMxCMP1), which is continually compared to the timer value TIMxVAL. The corresponding output compare pins (TxCMP0 and TxCMP1) toggle when either the timer value TIMxVAL equals the output compare register TIMxCMPy, or the timer rolls over. The CMP0IH and CMP1IH bits control the initial state of the TxCMP0 and TxCMP1 pins when the timer value is less than the output compare register. Figure 97 provides an example of PWM operation using an output compare unit.

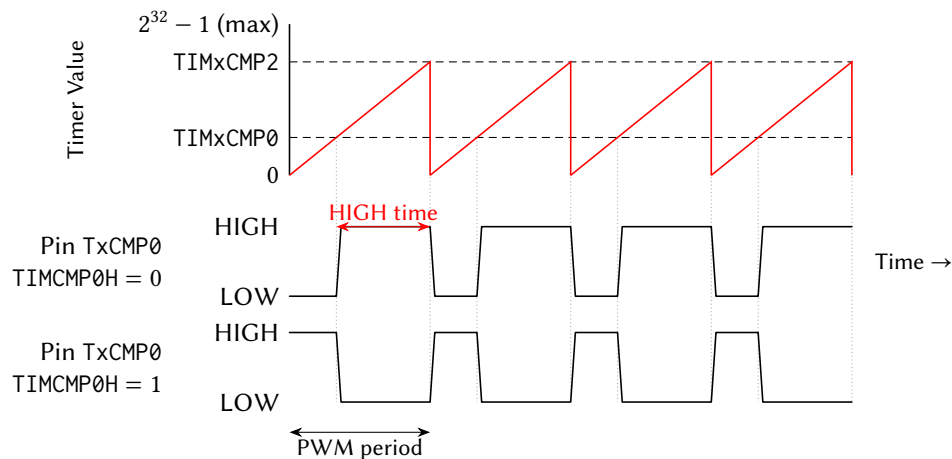


Figure 97: Timer Output Compare and PWM Operation

19.6 Input Capture Units

The TIMERx peripheral contains two input capture units which allow the precise timestamping of pin change events. When enabled with CAPyEN, the input capture unit waits for the TxCAPy pin to toggle in the appropriate edge direction. The CAPyFE bit selects either rising edge or falling edge sensitivity. When the pin toggles in the appropriate edge direction, the input capture unit stores the current timer value in the TIMxCAPy register and sets the CAPyIF interrupt flag. The CAPyIF flag must be cleared before the next input capture event by writing a 1 to it in the TIMxSR register.

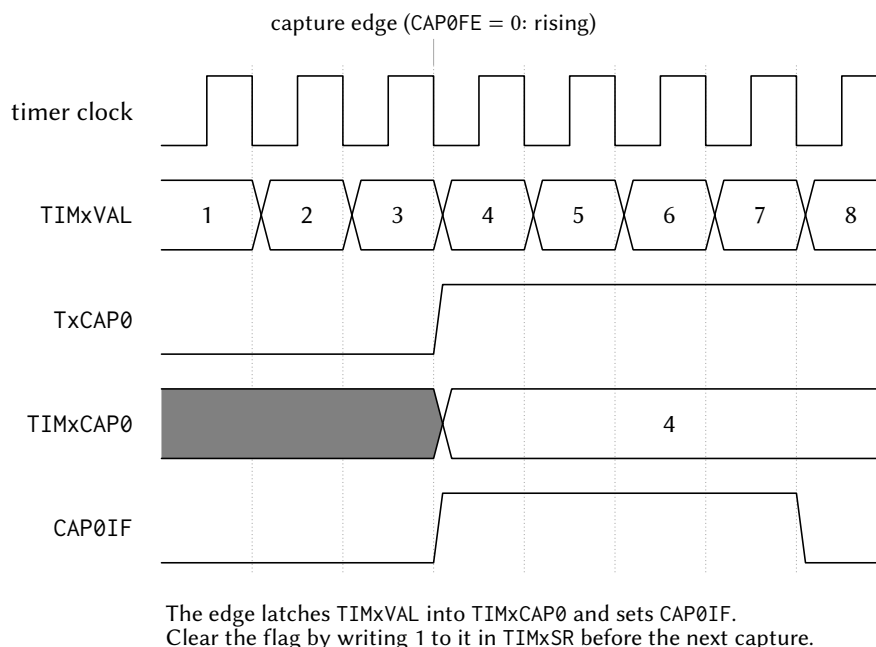


Figure 98: Input capture. The selected edge on TxCAP0 latches the live TIMxVAL into TIMxCAP0 and sets CAP0IF; the flag must be cleared by writing 1 to it in TIMxSR before the next capture event.

Note: changing the value of CAPyFE while CAPyEN is enabled can inadvertently cause false input capture triggers. The user should not change CAPyFE while CAPyEN is enabled.

19.7 Flags and Interrupts

The TIMEx peripheral has two indicator bits and six interrupt flags in the timer status register TIMxSR. The TIMxSR register is latched on the falling edge of the memory enable signal for stable reads.

The two indicator bits CMP0OUT and CMP1OUT show the current value of the output compare pins TxCMP0 and TxCMP1 respectively, regardless of if the corresponding GPIO pins are set to primary/GPIO mode or secondary/peripheral mode.

The timer value overflow interrupt flag OVIF is set when the 32-bit timer value overflows from $2^{32} - 1$ back to 0. Note that when CMP2RST is enabled, the timer resets at the TIMxCMP2 value and does not overflow to $2^{32} - 1$, so OVIF will not be set in this mode. The overflow interrupt request is generated when both OVIF and OVIE are set. OVIF is cleared by writing a 1 to it in the status register TIMxSR.

The output compare interrupt flags CMP0IF, CMP1IF, and CMP2IF are set when the timer value equals the respective output compare value (TIMxCMP0, TIMxCMP1, and TIMxCMP2). Each compare interrupt request is generated when both the interrupt flag (CMPyIF) and the corresponding interrupt enable bit (CMP0IE, CMP1IE, or CMP2IE) are set. Each CMPyIF flag is cleared by writing a 1 to it in the status register TIMxSR. Note that CMP2IF is only set when CMP2RST is enabled and the timer resets at the TIMxCMP2 value.

The input capture interrupt flags CAP0IF and CAP1IF are set when a new input capture event occurs on the TxCAP0 and TxCAP1 pins, respectively. The input capture register TIMxCAPy can be read as soon as CAPyIF is set. Each capture interrupt request is generated when both the interrupt flag (CAPyIF) and the corresponding interrupt enable bit (CAP0IE or CAP1IE) are set. Each CAPyIF flag is cleared by writing a 1 to it in the status register TIMxSR.

19.8 Register Description

19.8.1 TIMCR Register

Register memory slot: 0, offset: 0 bytes, size: 4 bytes

Timer/Counter control register. Configures clock source, clock divider, capture/compare settings, and interrupt enables.

31	30	29	28	27	26	25	24
Unused							
23	22	21	20	19	18	17	16
Unused				DIV			
				rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
CMP1IH	CMP0IH	CAP1FE	CAP0FE	CAP1EN	CAP0EN	SSEL	
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
CMP2RST	EN	CAP1IE	CAP0IE	OVIE	CMP2IE	CMP1IE	CMP0IE
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
Unused	Bits 31-20	
DIV	Bits 19-16	Timer/Counter clock divider. Divides the clock source selected by SSEL to generate the timer clock for TIMxVAL increments. 0000 DIV_1: /1 (no division) 0001 DIV_2: /2 0010 DIV_4: /4 0011 DIV_8: /8 0100 DIV_16: /16 0101 DIV_32: /32 0110 DIV_64: /64 0111 DIV_128: /128 1000 DIV_256: /256 1001 DIV_512: /512 1010 DIV_1024: /1,024 1011 DIV_2048: /2,048 1100 DIV_4096: /4,096 1101 DIV_8192: /8,192 1110 DIV_16384: /16,384 1111 DIV_32768: /32,768
CMP1IH	Bit 15	Compare 1 initial PWM output level. Sets the TxCMP1 pin level when TIMxVAL < TIMxCMP1. 0 PWM output starts LOW

		1	PWM output starts HIGH
CMP0IH	Bit 14		Compare 0 initial PWM output level. Sets the TxCMP0 pin level when TIMxVAL < TIMxCMP0.
		0	PWM output starts LOW
		1	PWM output starts HIGH
CAP1FE	Bit 13		Capture 1 edge select. Selects which edge on TxCAP1 pin triggers a capture event.
		0	Capture on rising edge
		1	Capture on falling edge
CAP0FE	Bit 12		Capture 0 edge select. Selects which edge on TxCAP0 pin triggers a capture event.
		0	Capture on rising edge
		1	Capture on falling edge
CAP1EN	Bit 11		Capture 1 enable. Enables input capture on TxCAP1 pin.
		0	Capture 1 disabled
		1	Capture 1 enabled
CAP0EN	Bit 10		Capture 0 enable. Enables input capture on TxCAP0 pin.
		0	Capture 0 disabled
		1	Capture 0 enabled
SSEL	Bits 9-8		Timer/Counter clock source select. Glitch-free multiplexer prevents spurious transitions when switching sources.
		00	SSEL_SMCLK: SMCLK
		01	SSEL_MCLK: MCLK
		10	SSEL_LFXT: LFXT (low frequency crystal)
		11	SSEL_HFXT: HFXT (high frequency crystal)
CMP2RST	Bit 7		Timer/Counter reset on Compare 2 enable. Resets TIMxVAL to 0 when TIMxVAL equals TIMxCMP2.
		0	Free-running mode (resets at $2^{32}-1$)
		1	Resets on TIMxVAL = TIMxCMP2
EN	Bit 6		Timer/Counter enable. When disabled, timer clock is gated off and TIMxVAL holds its value.
		0	Timer disabled
		1	Timer enabled
CAP1IE	Bit 5		Capture 1 interrupt enable. Enables interrupt when CAP1IF flag is set.
		0	Interrupt disabled
		1	Interrupt enabled
CAP0IE	Bit 4		Capture 0 interrupt enable. Enables interrupt when CAP0IF flag is set.
		0	Interrupt disabled
		1	Interrupt enabled
OVIE	Bit 3		Timer/Counter overflow interrupt enable. Enables interrupt when OVIF flag is set.
		0	Interrupt disabled
		1	Interrupt enabled
CMP2IE	Bit 2		Compare 2 interrupt enable. Enables interrupt when CMP2IF flag is set.
		0	Interrupt disabled
		1	Interrupt enabled
CMP1IE	Bit 1		Compare 1 interrupt enable. Enables interrupt when CMP1IF flag is set.

		0	Interrupt disabled
		1	Interrupt enabled
CMP0IE	Bit 0	Compare 0 interrupt enable. Enables interrupt when CMP0IF flag is set.	
		0	Interrupt disabled
		1	Interrupt enabled

19.8.2 TIMSR Register

Register memory slot: 1, offset: 4 bytes, size: 1 byte

Timer/Counter status register. Contains current compare output levels and interrupt flags. The register is latched on the falling edge of the memory enable signal for stable reads.

7	6	5	4	3	2	1	0
CMP1OUT	CMP0OUT	CAP1IF	CAP0IF	OVIF	CMP2IF	CMP1IF	CMP0IF
r-(0)	r-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)

Bitfield	Range	Description
CMP1OUT	Bit 7	Current value of the Compare 1 output pin TxCMP1. Reflects the actual PWM output level. 0 TxCMP1 output LOW 1 TxCMP1 output HIGH
CMP0OUT	Bit 6	Current value of the Compare 0 output pin TxCMP0. Reflects the actual PWM output level. 0 TxCMP0 output LOW 1 TxCMP0 output HIGH
CAP1IF	Bit 5	Capture 1 interrupt flag. Set when TxCAP1 pin triggers a capture event. Write 1 to clear. 0 No pending capture 1 interrupt 1 Capture 1 interrupt pending
CAP0IF	Bit 4	Capture 0 interrupt flag. Set when TxCAP0 pin triggers a capture event. Write 1 to clear. 0 No pending capture 0 interrupt 1 Capture 0 interrupt pending
OVIF	Bit 3	Timer/Counter overflow interrupt flag. Set when timer overflows from $2^{32}-1$ to 0. Write 1 to clear. 0 No pending overflow interrupt 1 Overflow interrupt pending
CMP2IF	Bit 2	Compare 2 interrupt flag. Set when TIMxVAL equals TIMxCMP2. Write 1 to clear. 0 No pending compare 2 interrupt 1 Compare 2 interrupt pending
CMP1IF	Bit 1	Compare 1 interrupt flag. Set when TIMxVAL equals TIMxCMP1. Write 1 to clear. 0 No pending compare 1 interrupt 1 Compare 1 interrupt pending

CMP0IF	Bit 0	Compare 0 interrupt flag. Set when TIMxVAL equals TIMxCMP0. Write 1 to clear.
	0	No pending compare 0 interrupt
	1	Compare 0 interrupt pending

19.8.3 TIMVAL Register

Register memory slot: 2, offset: 8 bytes, size: 4 bytes

Timer/Counter value register. Reads the current timer count. Writes immediately update the timer value regardless of enable state. The value is latched on the falling edge of the memory enable signal for stable reads.

31	30	29	28	27	26	25	24
VAL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
VAL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
VAL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
VAL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
VAL	Bits 31-0	Current timer count value

19.8.4 TIMCMP0 Register

Register memory slot: 3, offset: 12 bytes, size: 4 bytes

Timer/Counter Compare 0 threshold register. When TIMxVAL equals this value, CMP0IF is set and the TxCMP0 PWM output toggles.

31	30	29	28	27	26	25	24
CMP0							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
CMP0							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
CMP0							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
CMP0							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
CMP0	Bits 31-0	Compare 0 threshold value

19.8.5 TIMCMP1 Register

Register memory slot: 4, offset: 16 bytes, size: 4 bytes

Timer/Counter Compare 1 threshold register. When TIMxVAL equals this value, CMP1IF is set and the TxCMP1 PWM output toggles.

31	30	29	28	27	26	25	24
CMP1							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
CMP1							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
CMP1							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
CMP1							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
CMP1	Bits 31-0	Compare 1 threshold value

19.8.6 TIMCMP2 Register

Register memory slot: 5, offset: 20 bytes, size: 4 bytes

Timer/Counter Compare 2 threshold register. When TIMxVAL equals this value, CMP2IF is set. If CMP2RST is enabled, timer resets to 0 (PWM period control).

31	30	29	28	27	26	25	24
CMP2							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
CMP2							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
CMP2							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
CMP2							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
CMP2	Bits 31-0	Compare 2 threshold value (PWM period)

19.8.7 TIMCAP0 Register

Register memory slot: 6, offset: 24 bytes, size: 4 bytes

Timer/Counter Capture 0 value register. Automatically latches TIMxVAL when TxCAP0 pin edge triggers a capture event. The captured value is latched on the falling edge of the memory enable signal for stable reads.

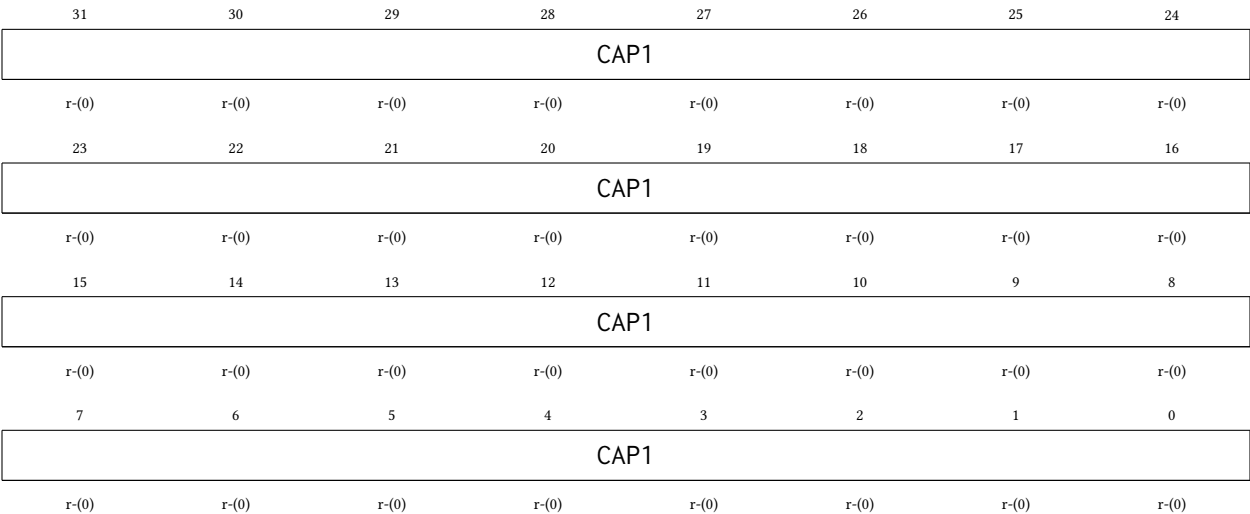
31	30	29	28	27	26	25	24
CAP0							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
23	22	21	20	19	18	17	16
CAP0							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
15	14	13	12	11	10	9	8
CAP0							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
7	6	5	4	3	2	1	0
CAP0							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)

Bitfield	Range	Description
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CAP0	Bits 31-0	Captured timer value from TxCAP0 event
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19.8.8 TIMCAP1 Register

Register memory slot: 7, offset: 28 bytes, size: 4 bytes
Timer/Counter Capture 1 value register. Automatically latches TIMxVAL when TxCAP1 pin edge triggers a capture event. The captured value is latched on the falling edge of the memory enable signal for stable reads.



Bitfield	Range	Description
CAP1	Bits 31-0	Captured timer value from TxCAP1 event

20 SYSTEM Peripheral

The SYSTEM peripheral manages a potpourri of system-level blocks, including the MCU clocks, the watchdog timer, and a CRC data verification module.

Its effects are global (MCLK clocks all harts and SMCLK clocks the peripherals), so by convention only hart 0 (the management hart) reconfigures the clocks, after quiescing the other harts. Interrupt routing, masking and delivery are **not** managed here: every hart, hart 0 included, enables peripheral interrupts through its IRQROUTER row and takes them via the claim/complete mechanism, and the CLINT software/timer interrupts (vectors 83 and 84) are hardwired enabled in every hart. (The single-core chip's IRQENx/IRQPRIx/IRQCR registers are retired; their register slots are reserved and read as zero.)

20.1 System Clocks and Clock Management

The MCU has four clock sources: HFXT, LFXT, DCO0, and DCO1. HFXT and LFXT are external clock sources, which must be provided to the MCU on the HFXT and LFXT pin functions (Section 2), respectively. DCO0 and DCO1 are internal, programmable frequency clock sources provided by digitally controllable oscillator circuits.

The Castalia MCU is designed and characterised for operation with a maximum clock frequency of 24 MHz. HFXT should therefore not exceed 24 MHz. LFXT is suggested to be exactly 32.768 kHz to provide an accurate method of measuring time. The frequencies of DCO0 and DCO1 are configured with the DCO0BIAS and DCO1BIAS registers; higher bias values produce higher frequencies.

Note on overclocking: It is possible to operate the Castalia MCU beyond 24 MHz by supplying a higher-frequency signal on the HFXT pin function or by configuring the internal DCOs to produce frequencies above 24 MHz. As with any digital circuit operated beyond its rated frequency, doing so may produce setup-time violations, incorrect computational results, and unpredictable behaviour. Operation above 24 MHz is **not guaranteed** and is undertaken entirely at the user's risk.

Two clocks, the main clock (MCLK) and the submain clock (SMCLK), form the roots of the MCU clock tree, shown in Figure 99. SMCLKSEL picks SMCLK's base from the four sources above. MCLKSEL picks MCLK's from three of them and, on code 01, from SMCLK itself: that input is taken after the SMCLK divider but *before* the SMCLKOFF gate, so a chip running MCLK out of SMCLK keeps its main clock when SMCLK is switched off for the peripherals. Either clock may then be divided again with CLKDIVCR. MCLK drives all five harts, the multi-core arbiter and the always-on blocks, while SMCLK drives the serial engines of most peripherals; the register interface of every peripheral is clocked from MCLK, so a peripheral stays readable with SMCLK stopped. Both clocks use glitch-free multiplexers to prevent spurious transitions when switching sources.

MCLK cannot be turned off, but SMCLK and individual source clocks can be independently disabled via SYSCLKCR. DCO0 and DCO1 are off by default and must be explicitly powered on by setting DCO0ON or DCO1ON. HFXT and LFXT are enabled by default and may be disabled with HFXTOFF and LFXTOFF. If a source clock is currently selected by MCLK or SMCLK, that source will be automatically kept enabled regardless of its disable bit in SYSCLKCR.

20.2 Power Saving Features

The MCU has several power-saving features.

The CPU is the most power-hungry part of the MCU, and its dynamic power consumption is proportional to the CPU clock frequency. Reducing the frequency of MCLK is the primary avenue for reducing total dynamic power consumption. This can be done by utilising the MCLK clock divider in CLKDIVCR, by switching MCLK to a lower-frequency source via MCLKSEL, or by reducing the frequency of the currently-

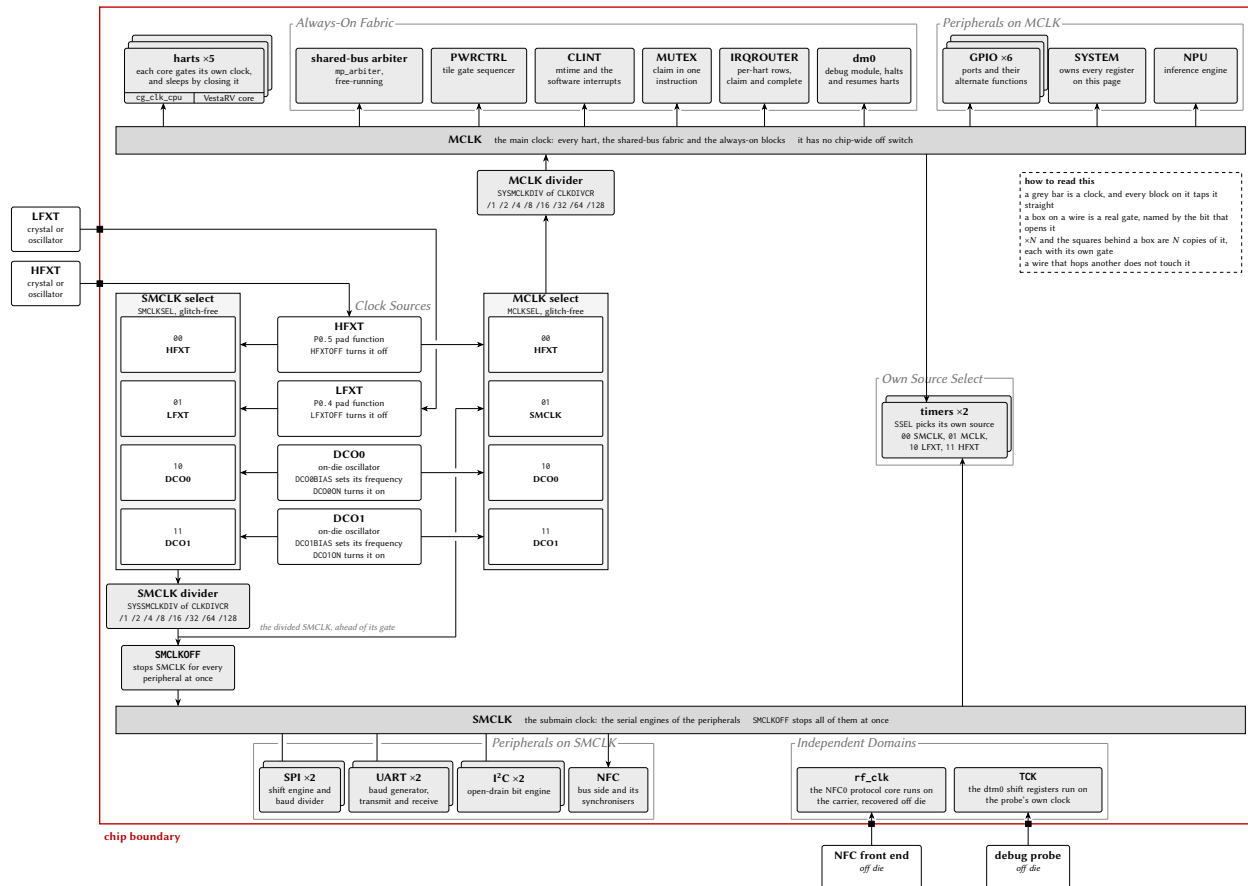


Figure 99: The MCU clock system, generated from this configuration. The layout is a mirror: the sources stand in one column, in the order SMCLKSEL enumerates them, the SMCLK chain runs left out of that column and down into its own bar, and the MCLK chain runs right out of it and up into its own, so every source leg is a straight horizontal at its own row. Each select box prints *every* code it has, with the name of what that code picks, and those rows are built from the register model itself rather than drawn by hand. One code is not a source: MCLKSEL 01 takes SMCLK, tapped after the SMCLK divider and *before* the SMCLKOFF gate, which is why stopping SMCLK never stops MCLK; it is the one wire in the drawing that has to leave its row, and it hops the legs it crosses rather than touching them. A source that either select is using stays enabled whatever its own bit says. Every block on a bar taps it straight up or straight down; the sleep gate is drawn where the hardware puts it, inside each core, so the stack of five harts is a stack of five gates. A block with a select of its own names every code it has and taps the two bars it can reach, so the crystal sources it may also pick are named in its box rather than drawn as two more wires across the page. The blocks under *Independent Domains* are on neither bar: they are clocked from outside this tree entirely, and SYSCLKCR does not reach them. The register interface of every peripheral is clocked from MCLK even where its engine runs on SMCLK, which is why a peripheral stays readable with SMCLK stopped.

selected source. The internal DCOs offer a wide range of programmable frequencies through their bias settings and can be used as a low-power clock source when HFXT is disabled.

Another method to save power is to turn off unused clocks. Remember, however, that HFXT is the reset selection of both MCLKSEL and SMCLKSEL, so a valid clock signal is required on the HFXT pin function while the MCU boots. Failure to provide one may cause the chip to boot improperly, or not at all: the board must therefore hold the external oscillator driving HFXT enabled across reset, rather than gating it from a pin the MCU itself only drives after boot. After boot, the user may switch MCLK to another source and then safely disable HFXT.

The static (leakage) power can be reduced by powering off unused on-chip memory blocks with BLOCKPWR. The ROM and RAM blocks include power-gating circuitry that physically disconnects their supply rail. The ROM can be powered down by setting SYSROMOFF, and the block RAMs by setting SYSRAM0OFF and SYSRAM10FF, respectively. In addition, BLOCKPWR gates the first four shared bulk-RAM banks (banks 0–3) individually, one bit per bank, so unused banks may be powered down while the bank holding live stack or payload data stays on; any bank beyond the fourth is permanently powered. Each powered-down block or bank reduces leakage. Any access to a powered-down block or bank will fail and produce undefined results. All blocks and banks power up on at reset, and the entire contents of a RAM block or bank are lost when it is powered down (there is no retention), so data are undefined when it is powered back on until explicitly overwritten. Care must be taken to track the power state of each block and bank, and to keep at least the bank in active use powered, to avoid undefined behaviour.

20.3 CPU Sleep Mode

Another way to conserve power is to run the CPU only when needed and switch it off when idle. Because the CPU accounts for the majority of dynamic power, duty-cycling its clock can yield significant savings. Peripherals can continue to operate while the CPU sleeps and may wake it when events occur. Long-term average CPU power is:

$$P_{\text{CPU}} = P_{\text{CPU}, 100\%} \times \frac{t_{\text{on}}}{t_{\text{on}} + t_{\text{off}}}$$

Sleep mode is per hart: each hart can execute the custom sleep instruction independently, which gates off that hart's CPU clock and reduces its dynamic power to zero. A sleeping hart can only be woken by an interrupt (typically a CLINT software interrupt or a peripheral interrupt routed to it through the IRQROUTER) or a system reset. When an interrupt fires, the CPU clock is re-enabled for the duration of that interrupt service routine (ISR). Once all pending interrupts have been serviced the CPU returns to sleep, unless one of the ISRs executed the custom wake instruction, in which case the CPU remains awake and returns to the point in the main program where it went to sleep.

A code example to put the CPU to sleep is included in Listing 4.

```

1  /** Includes **/
2  #include <MemoryMap.h>
3  #include <irq.h>
4
5  /** Interrupt Service Routine Declaration Macros **/
6  RVISR(IRQ_TIMER1_VECTOR, ISR_TIMER1)
7
8  /** Main Function **/
9  int main()
10 {
11     // Set up TIMER1 here
12
13     // Enable interrupts by unmasking them
14     enable_all_interrupts();
15
16     // Go to sleep
17     cpu_sleep();

```

```

18
19     // Insert code to execute after ISR wakes the CPU
20
21     // Wait forever
22     while (1) {}
23 }
24
25 /** Interrupt Service Routines */
26 void ISR_TIMER1()
27 {
28     // Do ISR processing here
29
30     // Wake the CPU
31     cpu_wake();
32 }

```

Listing 4: Example program to use CPU sleep mode

20.4 CRC Module

The CRC module enables the computation of a CRC16 checksum over an array of bytes. The checksum can be used to verify the integrity of data in a communication channel or in memory.

The module implements the CRC16-CDMA2000 standard with the following parameters:

- CRC width: 16 bits
- Polynomial: 0xC857
- Initial value: 0xFFFF
- No output XOR
- No input reflection
- No output reflection

To compute a CRC over a byte array, first write 0xFFFF to CRCSTATE to initialise the accumulator. Then write each byte of the array sequentially to CRCDATA; each write updates the running CRC. After the last byte has been written, read CRCSTATE to obtain the final CRC16 value.

20.5 Watchdog Timer

The watchdog timer improves system reliability by automatically taking corrective action if software fails to service the watchdog within a configurable timeout period. The watchdog is controlled by WDTCR, which is write-protected and must be unlocked via WDPASS before any configuration change.

The watchdog uses a 24-bit up-counter that increments on every MCLK cycle when enabled via SYSWDTEN. The timeout period is selected by SYSWDTCDIV, which chooses which bit of the counter triggers the watchdog event. A watchdog event occurs on the 0-to-1 transition of the selected bit, giving a timeout of $2^{\text{WDTCDIV}+16}$ MCLK cycles. At 24 MHz the timeout ranges from approximately 2.73 ms (WDTCDIV = 0000, bit 16) to approximately 89.5 s (WDTCDIV = 1111, bit 31).

On a watchdog event, the response depends on the configuration:

- If SYSWDTIE is set **and** the watchdog vector (vector 0) is routed to some hart in the IRQROUTER, the watchdog interrupt is delivered through the claim/complete mechanism and the servicing hart executes its handler. If SYSWDTHWRST is also set, the system resets after that hart completes the claim.

- If SYSWDTIE is clear, or vector 0 is not routed to any hart, the system resets immediately on timeout provided SYSWDTHWRST is set.

To prevent timeout, software must periodically write the clear password (0xA0C8A620) to WDTPASS, which resets the counter to zero. To modify WDTCR, first write the unlock password (0x5F3759DF) to WDTPASS; this opens a 64-MCLK-cycle window during which WDTCR is writable.

The WDTSR status register holds two sticky flags. SYSWDTRF is set when the last system reset was caused by the watchdog and persists across resets; it is cleared by writing 1 to the bit. SYSWDTIF is set when a watchdog timeout event occurs and is likewise cleared by writing 1. The current counter value is readable at any time from WDTVAL; the register reads as zero when the watchdog is disabled.

20.6 Register Description

20.6.1 SYSCLKCR Register

Register memory slot: 0, offset: 0 bytes, size: 2 bytes

System clock control register

15	14	13	12	11	10	9	8
Unused							DCO1ON
rw-(0)							
7	6	5	4	3	2	1	0
DCO0ON	HFXTOFF	LFXTOFF	SMCLKOFF	SMCLKSEL		MCLKSEL	
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
Unused	Bits 15-9	
DCO1ON	Bit 8	Digitally controlled oscillator 1 (DCO1) power enable. Set to power on DCO1. DCO1 is automatically kept on if it is currently selected as the source for MCLK or SMCLK, regardless of this bit. 0 DCO1 powered off 1 DCO1 powered on
DCO0ON	Bit 7	Digitally controlled oscillator 0 (DCO0) power enable. Set to power on DCO0. DCO0 is automatically kept on if it is currently selected as the source for MCLK or SMCLK, regardless of this bit. 0 DCO0 powered off 1 DCO0 powered on
HFXTOFF	Bit 6	High frequency external crystal clock disable. Cannot be disabled if it is currently selected as the source for MCLK or SMCLK. 0 HFXT enabled 1 HFXT disabled
LFXTOFF	Bit 5	Low frequency external crystal clock disable. Cannot be disabled if it is currently selected as the source for SMCLK. 0 LFXT enabled 1 LFXT disabled
SMCLKOFF	Bit 4	Submain clock disable. Globally and unconditionally disables SMCLK, gating all peripherals clocked from SMCLK. 0 SMCLK enabled 1 SMCLK disabled
SMCLKSEL	Bits 3-2	Submain clock source select 00 SMCLKSEL_HFXT: High Frequency Crystal Clock 01 SMCLKSEL_LFXT: Low Frequency Crystal Clock 10 SMCLKSEL_DCO0: Digitally Controlled Oscillator 0 11 SMCLKSEL_DCO1: Digitally Controlled Oscillator 1
MCLKSEL	Bits 1-0	Main clock source select (also CPU clock source select) 00 MCLKSEL_HFXT: High Frequency Crystal Clock 01 MCLKSEL_SMCLK: Submain Clock

- 10 MCLKSEL_DC00: Digitally Controlled Oscillator 0
 11 MCLKSEL_DC01: Digitally Controlled Oscillator 1

20.6.2 CLKDIVCR Register

Register memory slot: 1, offset: 4 bytes, size: 1 byte

MCLK and SMCLK clock divider control register. Configures clock division for main and submain clocks using glitch-free multiplexers.

7	6	5	4	3	2	1	0
Unused		SMCLKDIV			MCLKDIV		
rw-(0)		rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
Unused	Bits 7-6	
SMCLKDIV	Bits 5-3	SMCLK clock division selection. Division is applied after clock source selection through glitch-free divider multiplexer. 000 SYSSMCLKDIV_1: /1 (no division) 001 SYSSMCLKDIV_2: /2 010 SYSSMCLKDIV_4: /4 011 SYSSMCLKDIV_8: /8 100 SYSSMCLKDIV_16: /16 101 SYSSMCLKDIV_32: /32 110 SYSSMCLKDIV_64: /64 111 SYSSMCLKDIV_128: /128
MCLKDIV	Bits 2-0	MCLK clock division selection. Division is applied after clock source selection through glitch-free divider multiplexer. 000 SYSMCLKDIV_1: /1 (no division) 001 SYSMCLKDIV_2: /2 010 SYSMCLKDIV_4: /4 011 SYSMCLKDIV_8: /8 100 SYSMCLKDIV_16: /16 101 SYSMCLKDIV_32: /32 110 SYSMCLKDIV_64: /64 111 SYSMCLKDIV_128: /128

20.6.3 BLOCKPWR Register

Register memory slot: 2, offset: 8 bytes, size: 1 byte

Block power control register. Controls power gating for on-chip memory blocks. All bits reset to 0 (every block powered). DP-S3: bits 6:3 gate the four low shared bulk-RAM banks individually; a gated bank LOSES ITS CONTENTS (no retention) and stops responding, so software must keep the bank holding its stack, mailboxes, or live payload powered, and must treat a re-powered bank as uninitialized (the boot-ROM zero-fill contract is write-before-read anyway). In configurations with more than four banks, banks 4 and up are hardwired always-on.

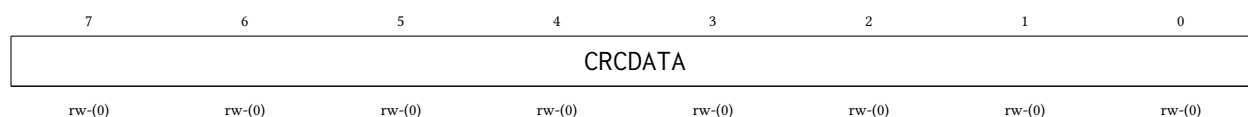
7	6	5	4	3	2	1	0
<i>Unused</i>	SHB30FF	SHB20FF	SHB10FF	SHB00FF	RAM10FF	RAM00FF	ROMOFF
	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
<i>Unused</i>	Bit 7	
SHB30FF	Bit 6	Shared bulk-RAM bank 3 (0x1C000-0x1FFFF) power control. When set, the bank is powered off: contents are LOST and accesses no longer respond. Reduces static power (a gated sram1p16k halves its leakage). 0 Bank 3 powered on 1 Bank 3 powered off (contents lost)
SHB20FF	Bit 5	Shared bulk-RAM bank 2 (0x18000-0x1BFFF) power control. When set, the bank is powered off: contents are LOST and accesses no longer respond. Reduces static power (a gated sram1p16k halves its leakage). 0 Bank 2 powered on 1 Bank 2 powered off (contents lost)
SHB10FF	Bit 4	Shared bulk-RAM bank 1 (0x14000-0x17FFF) power control. When set, the bank is powered off: contents are LOST and accesses no longer respond. Reduces static power (a gated sram1p16k halves its leakage). 0 Bank 1 powered on 1 Bank 1 powered off (contents lost)
SHB00FF	Bit 3	Shared bulk-RAM bank 0 (0x10000-0x13FFF) power control. When set, the bank is powered off: contents are LOST and accesses no longer respond. Reduces static power (a gated sram1p16k halves its leakage). 0 Bank 0 powered on 1 Bank 0 powered off (contents lost)
RAM10FF	Bit 2	RAM block 1 power control. When set, RAM block 1 is powered off. All data becomes undefined and the block no longer responds to memory access. Reduces static power consumption. 0 RAM block 1 powered on 1 RAM block 1 powered off
RAM00FF	Bit 1	RAM block 0 power control. When set, RAM block 0 is powered off. All data becomes undefined and the block no longer responds to memory access. Reduces static power consumption. 0 RAM block 0 powered on 1 RAM block 0 powered off
ROMOFF	Bit 0	ROM power control. When set, boot ROM is powered off. ROM no longer responds to read access. Reduces static power consumption. 0 ROM powered on 1 ROM powered off

20.6.4 CRCDATA Register

Register memory slot: 3, offset: 12 bytes, size: 1 byte

CRC input data register. Write the next byte of data to this register to update the CRC calculation. Uses CRC16-CDMA2000 polynomial 0xC857.

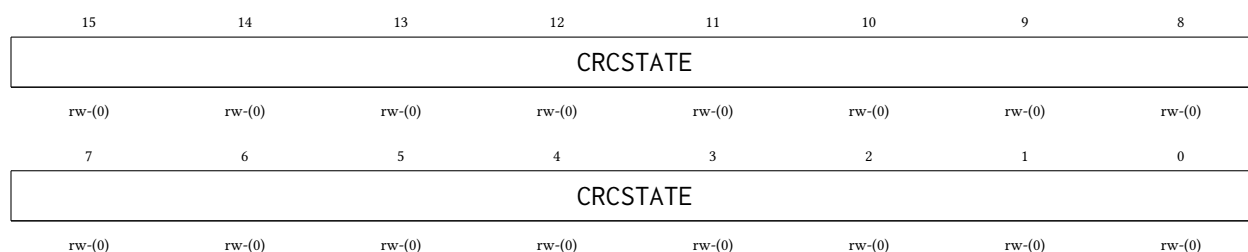


Bitfield	Range	Description
CRCDATA	Bits 7-0	CRC data input byte. Writing to this register feeds the byte into the CRC calculation and updates CRCSTATE.

20.6.5 CRCSTATE Register

Register memory slot: 4, offset: 16 bytes, size: 2 bytes

CRC state register. Contains the current CRC16 calculation result. Write to this register to initialize or restart the CRC calculation. Read to obtain the computed CRC16 checksum.



Bitfield	Range	Description
CRCSTATE	Bits 15-0	CRC state value. Initialize to 0xFFFF before starting CRC calculation. Read after processing all data bytes to get final CRC16 checksum.

20.6.6 WDTPASS Register

Register memory slot: 12, offset: 48 bytes, size: 4 bytes

Watchdog timer password register. Write-only register for two security functions: (1) Write 0x5F3759DF to unlock WDTCR for 64 MCLK cycles, enabling writes to watchdog configuration. (2) Write 0xA0C8A620 to clear watchdog counter to 0, preventing timeout. Reading always returns 0.

31	30	29	28	27	26	25	24
WDTPASS							
w-(0)	w-(0)	w-(0)	w-(0)	w-(0)	w-(0)	w-(0)	w-(0)
23	22	21	20	19	18	17	16
WDTPASS							
w-(0)	w-(0)	w-(0)	w-(0)	w-(0)	w-(0)	w-(0)	w-(0)
15	14	13	12	11	10	9	8
WDTPASS							
w-(0)	w-(0)	w-(0)	w-(0)	w-(0)	w-(0)	w-(0)	w-(0)
7	6	5	4	3	2	1	0
WDTPASS							
w-(0)	w-(0)	w-(0)	w-(0)	w-(0)	w-(0)	w-(0)	w-(0)

Bitfield	Range	Description
WDTPASS	Bits 31-0	Watchdog password. Write 0x5F3759DF (unlock password) to enable WDTCTCR writes for 64 MCLK cycles. Write 0xA0C8A620 (clear password) to reset watchdog counter to 0.

20.6.7 WDTCTCR Register

Register memory slot: 13, offset: 52 bytes, size: 1 byte

Watchdog timer control register. This register is protected and requires password unlock via WDTPASS before writing. Configures watchdog operation mode and timeout period.

7	6	5	4	3	2	1	0
WDTEN	Unused	WDTCDIV				WDTIE	WDTHWRST
rw-(0)		rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
WDTEN	Bit 7	Watchdog timer enable. When set, watchdog counter increments on MCLK. Watchdog generates interrupt and/or reset when counter bit selected by WDTCDIV transitions from 0 to 1. Register is write-protected; unlock with WDTPASS first. 0 Watchdog disabled 1 Watchdog enabled
Unused	Bit 6	
WDTCDIV	Bits 5-2	Watchdog timer clock divider select. Selects which bit of the 24-bit counter triggers watchdog event. Event occurs when selected bit transitions from 0 to 1. Timeout period = $2^{(WDTCDIV+16)}$ MCLK cycles. 0000 SYSWDTCDIV_65536: Bit 16: 65,536 MCLK cycles 0001 SYSWDTCDIV_131072: Bit 17: 131,072 MCLK cycles 0010 SYSWDTCDIV_262144: Bit 18: 262,144 MCLK cycles

		0011	SYSWDTCDIV_524288: Bit 19: 524,288 MCLK cycles
		0100	SYSWDTCDIV_1048576: Bit 20: 1,048,576 MCLK cycles
		0101	SYSWDTCDIV_2097152: Bit 21: 2,097,152 MCLK cycles
		0110	SYSWDTCDIV_4194304: Bit 22: 4,194,304 MCLK cycles
		0111	SYSWDTCDIV_8388608: Bit 23: 8,388,608 MCLK cycles
		1000	SYSWDTCDIV_16777216: Bit 24: 16,777,216 MCLK cycles
		1001	SYSWDTCDIV_33554432: Bit 25: 33,554,432 MCLK cycles
		1010	SYSWDTCDIV_67108864: Bit 26: 67,108,864 MCLK cycles
		1011	SYSWDTCDIV_134217728: Bit 27: 134,217,728 MCLK cycles
		1100	SYSWDTCDIV_268435456: Bit 28: 268,435,456 MCLK cycles
		1101	SYSWDTCDIV_536870912: Bit 29: 536,870,912 MCLK cycles
		1110	SYSWDTCDIV_1073741824: Bit 30: 1,073,741,824 MCLK cycles
		1111	SYSWDTCDIV_2147483648: Bit 31: 2,147,483,648 MCLK cycles
WDTIE	Bit 1	Watchdog timer interrupt enable. When set, watchdog timeout raises the watchdog interrupt level (vector 0), delivered to whichever hart the IRQROUTER routes it to. After the servicing hart completes the claim (IRQROUTER CLAIM/COMPLETE), the system resets if SYSWDTHWRST is set. If the vector is not routed to any hart, the reset (when enabled) occurs immediately on timeout.	
		0	Watchdog interrupt disabled
		1	Watchdog interrupt enabled
WDTHWRST	Bit 0	Watchdog hardware reset enable. When set, watchdog timeout causes system reset. If WDTIE is set, reset occurs after interrupt service routine completes. If WDTIE is cleared or IRQ is not enabled, reset occurs immediately on timeout.	
		0	Watchdog reset disabled
		1	Watchdog reset enabled

20.6.8 WDTSR Register

Register memory slot: 14, offset: 56 bytes, size: 1 byte

Watchdog timer status register. Contains flags indicating watchdog reset and interrupt events. Flags are cleared by writing 1 to the respective bit.

7	6	5	4	3	2	1	0
Unused						WDTIF	WDTRF
						rw1-(0)	rw1-(0)

Bitfield	Range	Description
Unused	Bits 7-2	
WDTIF	Bit 1	Watchdog timer interrupt flag. Set when watchdog timeout occurs and WDTIE is enabled. Cleared by writing 1 to this bit. If not cleared before next timeout, indicates watchdog event occurred.
	0	No watchdog interrupt pending
	1	Watchdog interrupt occurred

WDTRF	Bit 0	Watchdog timer reset flag. Set when system reset was caused by watchdog timer. Persists across resets until cleared by software. Cleared by writing 1 to this bit.
	0	Reset not caused by watchdog
	1	Reset caused by watchdog

20.6.9 WDTVAl Register

Register memory slot: 15, offset: 60 bytes, size: 4 bytes

Watchdog timer value register. Read-only register containing current watchdog counter value. Counter increments on MCLK when watchdog is enabled. Returns 0 when watchdog is disabled.

31	30	29	28	27	26	25	24
Unused							
23	22	21	20	19	18	17	16
WDTVAl							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
15	14	13	12	11	10	9	8
WDTVAl							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
7	6	5	4	3	2	1	0
WDTVAl							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)

Bitfield	Range	Description
Unused	Bits 31-24	
WDTVAl	Bits 23-0	Watchdog counter value. 24-bit up-counter that increments on MCLK cycles. Watchdog event occurs when bit selected by WDTCDIV transitions from 0 to 1.

20.6.10 DCO0BIAS Register

Register memory slot: 16, offset: 64 bytes, size: 2 bytes

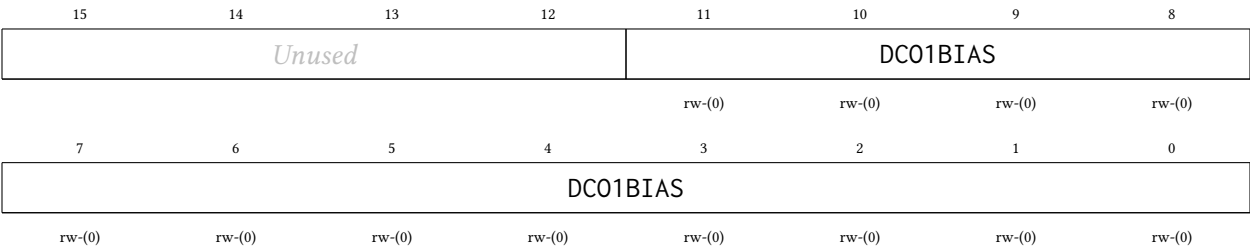
Digitally controlled oscillator 0 bias register. Controls DCO0 output frequency through bias voltage adjustment. Higher bias values generally produce higher frequencies.

15	14	13	12	11	10	9	8
Unused				DCO0BIAS			
				rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
DCO0BIAS							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
<i>Unused</i>	Bits 15-12	
DCO0BIAS	Bits 11-0	DCO0 bias adjustment value. 12-bit bias control for DCO0 frequency tuning. Default value loaded from constants on reset.

20.6.11 DC01BIAS Register

Register memory slot: 17, offset: 68 bytes, size: 2 bytes
Digitally controlled oscillator 1 bias register. Controls DCO1 output frequency through bias voltage adjustment. Higher bias values generally produce higher frequencies.



Bitfield	Range	Description
<i>Unused</i>	Bits 15-12	
DCO1BIAS	Bits 11-0	DCO1 bias adjustment value. 12-bit bias control for DCO1 frequency tuning. Default value loaded from constants on reset.

21 NPU Peripheral

The NPU is a fixed-point neural network processing unit built around a single multiply-accumulate datapath and a configurable sequencer. One engine provides **four datapath modes**, selected per computation by the NPUMODE field (NPUCR bits 22:20): a fully-connected (dense) MLP layer (mode 0, the legacy datapath and the reset default), a one-dimensional convolution (mode 1), a binary XNOR-popcount layer (mode 2), and a general matrix-matrix multiply (mode 3). The peripheral operates on data staged in the shared NPU staging RAM (the 16 KiB SRAM occupying 0xC000–0xFFFF), which is multiplexed between the harts (reaching it through the multi-core arbiter) and the NPU's own compute port. Software is responsible for staging operands into the staging RAM, configuring the NPU, and reading the results after the computation is complete.

The NPU's *register bus* is in the shared peripheral page (see Section 4), so any hart may configure the NPU, start a computation, and poll for completion. The *data path* is likewise shared: all operands live in the shared NPU staging RAM at 0xC000–0xFFFF, and any hart may stage operands there or retrieve results through the arbiter. While a computation is running, the in-NPU port multiplexer switches the staging RAM over to the NPU's compute port for the duration of the THINK, and **no hart is put to sleep**. Because the staging RAM belongs to the NPU during a run, software of any hart **must not read or write 0xC000–0xFFFF while a computation may be active**: poll NPUCR bit 16 (NPUTHINK) and wait for it to read 0 before touching the staging RAM.

Figure 100 draws the whole of that arrangement in one picture: the two wires a hart reaches the NPU on, the port multiplexer that decides which side drives the staging RAM's single port, the three vector pointers as taps into that one word array, and the engine behind the port.

Data Formats

All non-binary NPU operands are fixed-point numbers, each stored in its own 32-bit SRAM word:

- **Inputs** (MLP/convolution inputs, GEMM A elements): signed Q0.24 format (25 bits: 1 sign bit + 24 fractional bits), representing values in $[-1, 1)$. Stored in bits 24:0 of a 32-bit SRAM word; bits 31:25 are ignored on read. Bit 24 is the sign bit.
- **Weights** (MLP/convolution weights, GEMM B elements): signed Q7.24 format (32 bits: 1 sign bit + 7 integer bits + 24 fractional bits), representing values in $[-128, 128)$.
- **Outputs**: signed Q7.24 format by default; the activation functions produce narrower ranges (see *Activation Functions* below). XNOR mode packs its 1-bit operands 32 per word (see mode 2 below) and emits ± 1.0 Q7.24 output words.

The accumulator is Q7.24 with **per-step round-to-nearest and saturation at ± 128** : every multiply-accumulate step rounds and clips individually, in the fixed sequencer order. Sums whose intermediate values exceed ± 128 clip; this is benign for normalized data but is a real ceiling for long dot products with large coefficients.

Datapath Modes

The count registers NPUNI and NPUNN (NPUCR bits 15:8 and 7:0, each holding count – 1) and the per-mode configuration words NPUCFG1/NPUCFG2 are reinterpreted by each mode; the start-address registers NPUIVSAR/NPUWSAR/NPUOVSAR always hold *word indices within the staging RAM* (byte offset from 0xC000 divided by 4).

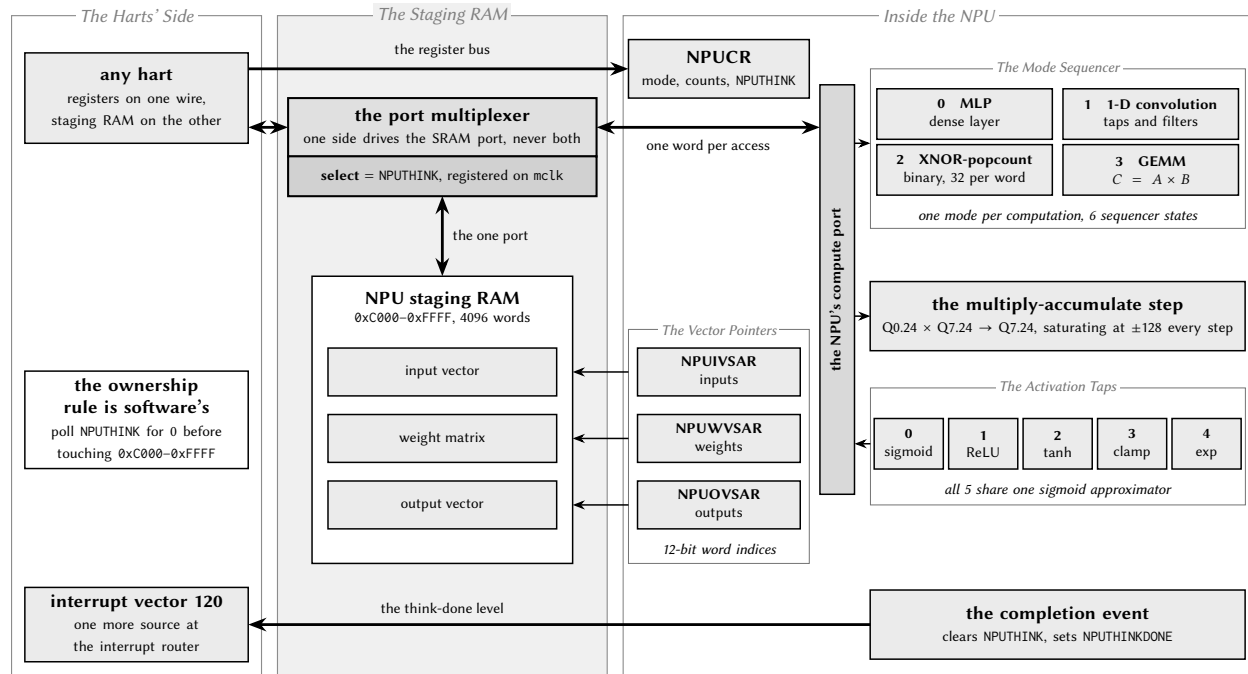


Figure 100: The NPU datapath, and the staging RAM it borrows. The picture draws the shape; this caption carries the contracts. The staging RAM is one 16 KiB single-port SRAM macro, so the whole middle band is about which side of it is driving that one port. The multiplexer's select is NPUTHINK re-clocked on the free-running chip clock, which is why the port changes hands a full cycle after a hart's start write lands and a full cycle after the engine's completion pulse: a hart's in-flight access can never be cut in half by the switch. What the hardware does *not* do is stop a hart reaching into the window while the engine has it — no hart is put to sleep, the access completes at the arbiter in the ordinary three cycles, and what comes back is whatever the SRAM is holding for the engine — which is why the rule in the left band is software's and the poll on NPUTHINK is the whole of it. NPUIVSAR, NPUWVSAR and NPUOVSAR are word indices into that one array, not addresses of three separate memories; firmware places and sizes all three, and the sequencer bounds-checks none of it, so an oversized working set wraps and corrupts its neighbours. Behind the port there is one engine: one mode per computation, latched into run shadows at the NPUTHINK edge, with mode codes 4–7 running as mode 0; one multiply-accumulate step that rounds to nearest and saturates on every single step rather than once at the end; and one piecewise-quadratic sigmoid approximator that all five activation taps are shift-and-add wrappers around, with activation codes 5–7 reading as sigmoid and $\text{NPUAEN} = 0$ writing the raw accumulator whatever NPUACTF says. The completion pulse does three things at once — it clears NPUTHINK, which is what hands the port back one clock edge later; it sets the sticky, write-1-to-clear NPUTHINKDONE in NPUSR; and the interrupt is the *level* of that flag gated by NPUTDIE, on the free-running clock.

- **Mode 0: MLP (legacy, reset default).** One dense layer: NPUNI + 1 inputs at NPUIVSAR, NPUNN + 1 output neurons. The weight matrix at NPUWVSAR holds one contiguous row per neuron; if bias is enabled (NPUBEN = 1) each row begins with one bias weight (multiplied by an implicit input of 1.0) followed by the NPUNI + 1 synaptic weights. NPUCFG1/NPUCFG2 are unused.
- **Mode 1: 1-D convolution.** NPUNI = taps $K - 1$, NPUNN = filters $C_{out} - 1$. NPUCFG1 packs stride, dilation, input length, and input channel count; NPUCFG2 holds the host-computed output length L_{out} (the sequencer *trusts* it; the ‘valid’/‘same’ padding convention is a firmware decision). Inputs are channel-major at NPUIVSAR; weights are filter-major (bias first per filter when NPUBEN = 1); outputs are written filter-major. See the register descriptions for the exact field packing.
- **Mode 2: XNOR-popcount (binary).** 1-bit activations and weights packed 32 per word, LSB-first (bit i of the layer lives at bit $i \bmod 32$ of word $i \div 32$). NPUNI = packed words per neuron – 1; the *exact* bit count $K \leq 4096$ lives in NPUCFG2 bits 12:0, and hardware masks the unused tail bits of each last packed word out of the popcount. A neuron fires when $2 \cdot \text{popcount}(\text{XNOR}(a, w)) - K \geq$ the signed threshold in NPUCFG1, emitting +1.0 (0x01000000) else –1.0 (0xFF000000). NPUBEN and NPUAEN must be 0 in this mode.
- **Mode 3: GEMM.** $C = A \times B$ with A ($M \times K$, row-major at NPUIVSAR, Q0.24 elements in $[-1, 1)$), B ($K \times N$, **column-major** at NPUWVSAR, Q7.24), and C ($M \times N$, row-major at NPUOVSAR). $K - 1$ in NPUNI, $N - 1$ in NPUNN, $M - 1$ in NPUCFG1 bits 7:0. The working set $M \cdot K + K \cdot N + M \cdot N$ must fit the 4096-word staging RAM; larger problems tile by re-pointing the start-address registers between computations (there is **no hardware accumulation across tiles**, so K must fit in one run). Because C is row-major, a result matrix chains directly as the next layer’s A operand with no repacking (activated outputs are valid Q0.24 inputs; see below). NPUBEN must be 0 in this mode.

The sequencer trusts its configuration: there are no hardware bounds checks. A working set that exceeds the 4096-word staging RAM, or count/length fields inconsistent with the staged data, wraps the 12-bit word addresses and silently corrupts neighbouring staging-RAM regions; the engine never hangs, but the results (and possibly unrelated staged data) are garbage. Firmware owns operand sizing and tiling.

Activation Functions

When NPUAEN = 1, each output is passed through the activation selected by NPUACTF (NPUCR bits 25:23) before being written; NPUAEN = 0 writes the raw Q7.24 accumulator regardless of NPUACTF. All activations share the one hardware sigmoid approximator (a piecewise-quadratic logistic function):

- **0, sigmoid** (reset default): $\sigma(x)$, output Q0.24 in $[0, 1)$; saturates for $|x| \geq 4$.
- **1, ReLU**: $\max(0, x)$, output Q7.24. Note that a ReLU output at or above 1.0 is *not* a valid Q0.24 next-layer input; chain through normalized weights or use the clamp.
- **2, tanh** approximation, computed as $2\sigma(2x) - 1$ through the same sigmoid hardware: output in $[-1, 1)$, sign-extended, whose low 25 bits are a valid Q0.24 next-layer input. Saturates for $|x| \geq 2$ (half the sigmoid’s linear region, a consequence of the input doubling).
- **3, clamp** (hardtanh): the identity saturated to the Q0.24 rails $[-1, 1)$, sign-extended, a bounded linear activation whose output always chains as a Q0.24 input.
- **4, exponential approximation** $\widetilde{\exp}(x) = 2\sigma(x)$, output Q1.24 in $[0, 2)$, for the softmax leg: hardware emits the per-element scores only, and **firmware performs the sum and normalization**.

Subtracting the maximum logit before the run is recommended; the maximum element then maps to exactly 1.0.

- Codes 5–7 are reserved and behave as 0 (sigmoid).

The activation selection is latched when THINK is set, so a mid-computation write to NPUACTF (or NPUMODE) takes effect at the *next* computation. XNOR mode (mode 2) ignores the activation path entirely; its ± 1.0 encoding is fixed.

Operation

1. Stage the operands into the staging RAM per the selected mode's layout and write their word indices to NPUIVSAR and NPUWVSAR.
2. Write the desired output word index to NPUOVSAR.
3. Write the per-mode configuration (NPUCFG1/NPUCFG2) if the mode uses it.
4. Configure NPUCR: NPUNI, NPUNN, NPUMODE, NPUBEN, NPUAEN, and NPUACTF as required by the mode.
5. Set NPUTHINK (NPUCR bit 16) to 1 to start the computation. The staging RAM is now owned by the NPU; no hart may access $0xC000-0xFFFF$ until the computation completes.
6. Poll NPUCR bit 16 (NPUTHINK) until it reads 0, indicating that the computation is complete and the staging RAM has been released back to the harts, or take the think-done interrupt (below).
7. Read the outputs from the staging RAM at the word index stored in NPUOVSAR.

Run times scale with the mode's loop product (e.g. a GEMM costs on the order of $5 \cdot M \cdot N \cdot K$ NPU clock cycles; a convolution $5 \cdot C_{out} \cdot L_{out} \cdot C_{in} \cdot K$). Full-scope convolutions can reach hundreds of milliseconds at 24 MHz; size polling budgets (or use the think-done interrupt) accordingly. XNOR mode is the cheapest per output by roughly an order of magnitude (32 synapses per fetched word pair).

Think-done interrupt (vector 120)

As an alternative to polling, the NPU raises the *think-done* interrupt (interrupt vector 120) when a computation completes. The NPUTHINKDONE flag (NPUSR bit 0) is set once per computation, by the same internal event that self-clears NPUTHINK, and is *sticky*: it remains set until software clears it by writing a 1 to NPUSR bit 0 (write-1-to-clear; writing 0 has no effect, and reading NPUSR never clears it). The interrupt is a level, asserted while both NPUTHINKDONE and the enable bit NPUTDIE (NPUCR bit 19) are set; it is routed, claimed, and completed through the interrupt router like every other peripheral source. NPUTDIE resets to 0, so software that polls is unaffected by the interrupt's existence. Every mode uses the same completion event and the same vector.

A typical interrupt-driven flow arms NPUTDIE, starts the computation, and sleeps; the handler clears NPUTHINKDONE (write-1-to-clear) before completing the interrupt at the router. **The staging-RAM ownership contract is unchanged by the interrupt:** the staging RAM belongs to the NPU while a computation may be active, and no hart may access $0xC000-0xFFFF$ until NPUCR bit 16 (NPUTHINK) reads 0. Taking the think-done interrupt implies the computation has finished, but any hart *other* than the one servicing the interrupt must still consult NPUTHINK before touching the staging RAM.

21.1 Register Description

21.1.1 NPUCR Register

Register memory slot: 0, offset: 0 bytes, size: 4 bytes

NPU control register

31	30	29	28	27	26	25	24
Unused				XPk	WPK	ACTF	
				rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
ACTF	MODE			TDIE	BEN	AEN	THINK
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw1-(0)
15	14	13	12	11	10	9	8
NI							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
NN							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
Unused	Bits 31-28	
XPk	Bit 27	Packed input format enable. When cleared (reset default) each input element occupies its own 32-bit staging word, exactly as before this bit existed. When set, the input vector is stored as TWO 16-bit Q0.15 elements per word – element 2i in bits 15:0, element 2i+1 in bits 31:16 – so the vector at NPUIVSAR occupies half as many words, and element i is read from word NPUIVSAR + i/2. Range is unchanged at [-1, +1) but resolution drops from Q0.24 to Q0.15: firmware packs by arithmetic-shifting each Q0.24 input right by 9. The output vector is NOT packed (outputs are full-width accumulator or activation words), so a layer whose outputs feed the next layer’s inputs must either leave this bit clear or repack between THINKs. Ignored in XNOR-popcount mode, which already packs 32 one-bit activations per word. The setting is latched when THINK is set; a mid-THINK write takes effect at the next THINK. 0 One input element per 32-bit word (reset default) 1 Two 16-bit Q0.15 input elements per 32-bit word

WPK	Bit 26	Packed weight format enable. When cleared (reset default) each synaptic weight occupies its own 32-bit staging word, exactly as before this bit existed. When set, the weight matrix is stored as TWO 16-bit Q3.12 weights per word – element $2i$ in bits 15:0, element $2i+1$ in bits 31:16 – so a weight block occupies half as many words and a maximal 256x256 layer needs 8 re-stagings of the 4096-word staging RAM instead of 16. The element walk is unchanged in every mode, including the convolution filter-block reload and the GEMM per-row reload, so the same weight ORDER is used either way; only the storage layout changes. Q3.12 narrows the representable weight range from [-128, +128] to [-8, +8] and coarsens the resolution from 2^{-24} to 2^{-12}: firmware packs by arithmetic-shifting each Q7.24 weight right by 12 and MUST SATURATE, because a weight outside [-8, +8] has no packed representation and the hardware only ever sees the already-packed half. The bias weight (see NPUBEN) rides the same stream and is subject to the same range. Ignored in XNOR-popcount mode, which already packs 32 one-bit weights per word. The setting is latched when THINK is set; a mid-THINK write takes effect at the next THINK. 0 One weight per 32-bit word (reset default) 1 Two 16-bit Q3.12 weights per 32-bit word
ACTF	Bits 25-23	Post-accumulator activation function select, effective when NPUAEN = 1 (P4 architecture family; NPUAEN = 0 is always raw passthrough regardless of this field). 0 = logistic sigmoid approximation (the legacy activation; reset default), output Q0.24 in [0,1). 1 = ReLU ($\max(0, x)$), output Q7.24 – note a ReLU output at or above 1.0 is not a valid Q0.24 next-layer input. 2 = tanh approximation computed as $2 \cdot \text{sigmoid}(2x) - 1$ through the same sigmoid hardware, output in [-1,1) sign-extended (the low 25 bits are a valid Q0.24 next-layer input); saturates for $x \geq 2$. 3 = clamp (hardtanh): the identity saturated to the Q0.24 rails [-1, 1), sign-extended. 4 = exponential approximation $2 \cdot \text{sigmoid}(x)$, output Q1.24 in [0,2), for the softmax leg – hardware emits per-element scores only, firmware performs the sum and normalize (subtracting the maximum logit first is recommended, mapping the maximum to exactly 1.0). Codes 5-7 are reserved and behave as 0 (sigmoid). The selection is latched when THINK is set; a mid-THINK write takes effect at the next THINK.

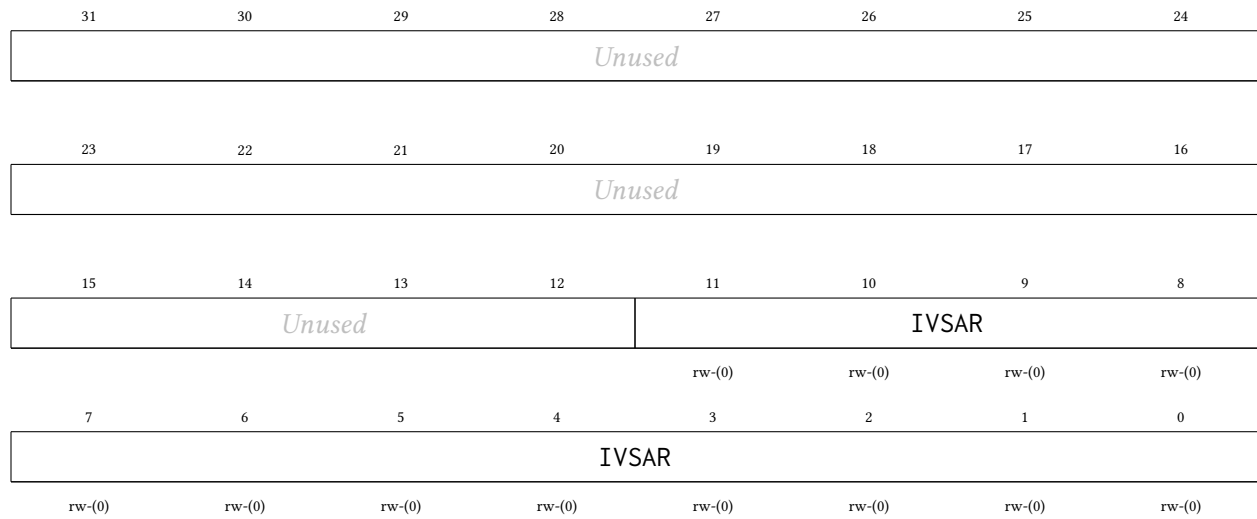
MODE	Bits 22-20	Datapath mode select for the NPU architecture family (P4). 0 = multilayer-perceptron mode (the legacy datapath; reset default). 1 = one-dimensional convolution mode. 2 = XNOR-popcount binary mode (1-bit activations/weights packed 32 per word, LSB-first; NPUBEN and NPUAEN must be 0 in this mode). 3 = general matrix-multiply (GEMM) mode: $C = A \times B$ with A ($M \times K$, row-major at NPUIVSAR, Q0.24 elements in $[-1,1]$), B ($K \times N$, column-major at NPUWVSAR, Q7.24), C ($M \times N$, row-major at NPUOVSAR, Q7.24 or Q0.24 through the sigmoid when NPUAEN=1); K-1 in NPUNI, N-1 in NPUNN, M-1 in NPUCFG1 bits 7:0; NPUBEN must be 0 in this mode; the working set $M \times K + K \times N + M \times N$ must fit the 4096-word staging RAM (larger problems tile by re-pointing the start-address registers between THINKs; there is no hardware accumulation across tiles). Codes 4-7 are reserved. Reserved and unimplemented codes behave as mode 0.
TDIE	Bit 19	Think-done interrupt enable. When set, NPUSR.THINKDONE drives the NPU think-done interrupt (vector 120). When cleared (reset default) the NPU is polling-only, exactly as before the interrupt existed. 0 Interrupt disabled (polling only) 1 Interrupt enabled
BEN	Bit 18	Bias enable. When set, the first weight of each output neuron's row in the weight matrix is used as a bias term: it is multiplied by an implicit input of 1.0 and accumulated before the synaptic weights. 0 Disabled 1 Enabled
AEN	Bit 17	Activation function enable. When set, the logistic sigmoid approximation activation function is applied to the accumulator output. When cleared, the raw accumulator output is used (linear/identity). 0 Disabled (linear output) 1 Enabled (logistic sigmoid approximation)
THINK	Bit 16	NPU computation start and status bit. Write 1 to start the NPU. Self-clears when the computation is complete. Poll this bit to determine when results are ready. 0 Idle (computation complete or not started) 1 Running (write 1 to start)
NI	Bits 15-8	Number of inputs in the input vector minus 1. The actual number of inputs is NPUNI + 1.
NN	Bits 7-0	Number of output neurons minus 1. The actual number of outputs is NPUNN + 1.

21.1.2 NPUIVSAR Register

Register memory slot: 1, offset: 4 bytes, size: 4 bytes

Input vector start word index within the shared NPU staging RAM: the byte offset from the start of the staging RAM (0xC000) divided by 4. For example, an input vector at byte address 0xC100 has word index 0x40. Each input is a signed Q0.24 value stored in bits 24:0 of its 32-bit SRAM word; bits 31:25 are ignored. Bit 24 is the sign bit. The input at index 0 is at word index NPUIVSAR. The input at index 1 is at word index NPUIVSAR + 1. The rest of the inputs follow in consecutive words. This one-input-per-word layout is what NPUCR.NPUXPK = 0 (the reset default) selects; with NPUXPK = 1 the vector is packed two 16-bit

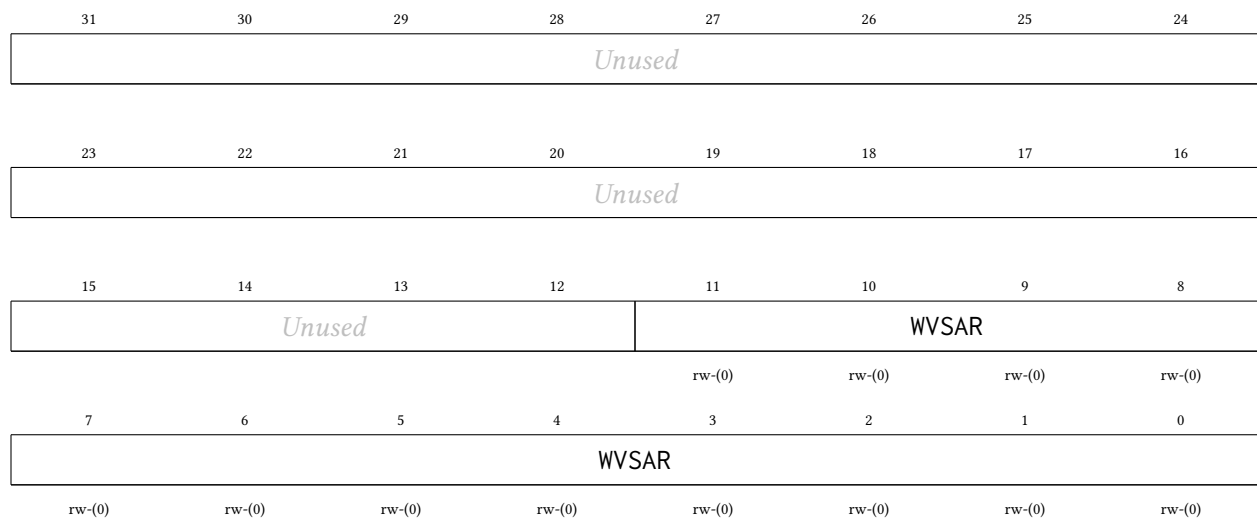
Q0.15 inputs per word and input index i is read from word index $\text{NPUIVSAR} + i/2$, low half first.



21.1.3 NPUWVSAR Register

Register memory slot: 2, offset: 8 bytes, size: 4 bytes

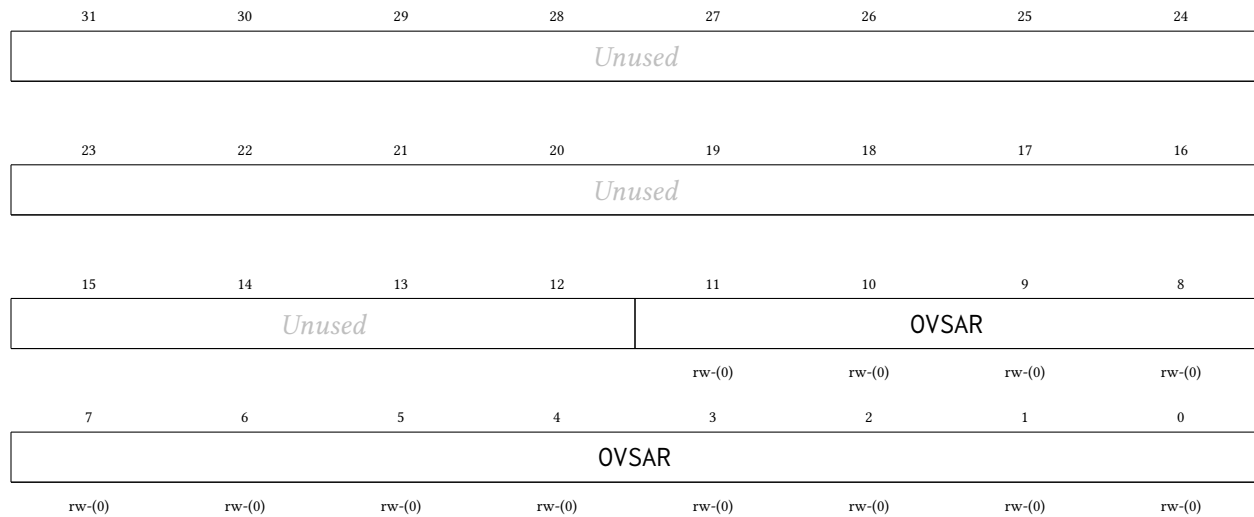
Synaptic weight matrix start word index within the shared NPU staging RAM: the byte offset from the start of the staging RAM (0xC000) divided by 4. Each weight is a signed Q7.24 value occupying all 32 bits of its SRAM word; bit 31 is the sign bit. Weights are stored row-major, one per 32-bit word, in the following order: for each output neuron (0 through NPUNN), if bias is enabled (NPUBEN = 1), the first word in the row is the bias weight (multiplied by an implicit input of 1.0), followed by NPUNI + 1 synaptic weights for inputs 0 through NPUNI. If bias is disabled, each row contains NPUNI + 1 synaptic weights only. This one-weight-per-word layout is what NPUCR.NPUWPK = 0 (the reset default) selects; with NPUWPK = 1 the same weight sequence is packed two 16-bit Q3.12 weights per word, low half first, halving the word count of the block.



21.1.4 NPUOVSAR Register

Register memory slot: 3, offset: 12 bytes, size: 4 bytes

Output vector start word index within the shared NPU staging RAM: the byte offset from the start of the staging RAM (0xC000) divided by 4. Each output is a signed Q7.24 value occupying all 32 bits of its SRAM word; bit 31 is the sign bit. The output at index 0 is written to word index NPUOVSAR. The output at index 1 is at word index NPUOVSAR + 1, and so on.



21.1.5 NPUSR Register

Register memory slot: 4, offset: 16 bytes, size: 4 bytes

NPU status register. THINKDONE is write-1-to-clear (write a 1 to bit 0 to clear it; writing 0 leaves it unchanged) and is never cleared by a read. The NPU think-done interrupt (vector 120) is THINKDONE and NPUCR.NPUTDIE. Clearing THINKDONE does not affect NPUCR.NPUTHINK, which self-clears at completion as before; the staging-RAM ownership contract is unchanged (no hart may access 0xC000-0xFFFF until NPUCR bit 16 reads 0 again).



Bitfield	Range	Description
<i>Unused</i>	Bits 31-1	
THINKDONE	Bit 0	Think-done flag. Set once per computation, when the NPU finishes (the same event that self-clears NPUCR.NPUTHINK); sticky. Drives the think-done interrupt (vector 120) when NPUCR.NPUTDIE is set. Write 1 to clear. On a same-cycle collision between a completing computation and a write-1-to-clear, the set wins (a completion is never lost). 0 No completed computation pending 1 A computation has completed

21.1.6 NPUCFG1 Register

Register memory slot: 5, offset: 20 bytes, size: 4 bytes

NPU mode configuration word 1. The interpretation depends on NPUCR.NPUMODE. Multilayer-perceptron mode (0): unused, no effect. One-dimensional convolution mode (1): bits 3:0 = stride S (1-15), bits 7:4 = dilation D (1-15), bits 23:8 = input length L per channel in samples, bits 31:24 = number of input channels minus 1 (the actual channel count C_{in} is this field + 1). XNOR-popcount mode (2): the full 32-bit signed firing threshold THRESH. A neuron fires (output +1.0) when $2 * \text{popcount}(\text{XNOR}(\text{activations}, \text{weights})) - K$ is greater than or equal to THRESH, else outputs -1.0. GEMM mode (3): bits 7:0 = M - 1, the number of A rows minus 1 (the actual row count M is this field + 1); bits 31:8 are ignored in this mode.

31	30	29	28	27	26	25	24
CFG1							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
CFG1							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
CFG1							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
CFG1							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

21.1.7 NPUCFG2 Register

Register memory slot: 6, offset: 24 bytes, size: 4 bytes

NPU mode configuration word 2. The interpretation depends on NPUCR.NPUMODE. Multilayer-perceptron mode (0): unused, no effect. One-dimensional convolution mode (1): bits 15:0 = output length Lout per filter in samples. Lout is computed by the host (the sequencer trusts this value; the valid/same padding convention is a firmware decision) and every referenced input sample index $j*S + k*D$ must be less than L. XNOR-popcount mode (2): bits 12:0 = K, the exact input bit count (1-4096); NPUNI must hold $\text{ceil}(K/32) - 1$, the packed words per neuron, and when $K \bmod 32$ is nonzero the unused high bits of each last packed word are masked out of the popcount by hardware. GEMM mode (3): unused, no effect. Bits 31:16 are reserved and read 0.

31	30	29	28	27	26	25	24
Unused							
23	22	21	20	19	18	17	16
Unused							
15	14	13	12	11	10	9	8
CFG2							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
CFG2							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

22 PWRCTRL Peripheral

The power controller manages the switchable MTCMOS power domains of the channel tiles: harts 1–4, which is every hart but hart 0. Each tile is implemented as its own header-switched power domain: PMOS header switches sit between the always-on supply grid and the tile’s local rail, isolation cells clamp every signal leaving the tile while it is unpowered, and a per-tile hardware sequencer guarantees the controls can only ever be applied in the legal order. Hart 0 (the orchestrator, which owns the SPI boot path, the console UART, and the CLINT) is soft logic in the always-on centre band with no header switches of its own, so it has no gate bit, is permanently powered and cannot be gated. The main features are listed below:

- One independently switchable power domain per channel tile (harts 1–4; PWRCR gate mask 0x1E)
- Single-bit gate/ungate control per tile; a hardware sequencer orders isolation, reset, and the header-switch enable correctly in both directions
- Cold gating: a gated tile consumes no dynamic or switched-leakage power and loses **all** state, including its TCM
- Wake equals boot: an ungated tile cold-boots through the shared boot ROM, parks in WFI, and is reloaded with the standard loader-row + msip launch sequence
- Read-only per-tile sequencer state for software polling
- All domains power up ON at reset: chip boot is unaffected and the controller is inert until software gates a tile

22.1 Usage

To gate tile hart h (1–4), first ensure the tile is parked or otherwise quiesced; typically it is sitting in its boot-ROM WFI park, or software has confirmed it finished its work. Then set the gate bit and, if required, wait for the sequencer to report the domain off:

```
li      t0, PWRCTRL_BASE_ADDR
lw      t1, PWRCR(t0)
ori     t1, t1, (1 << h)      # request gate of tile h
sw      t1, PWRCR(t0)
gate_poll:
lw      t2, PWRSR(t0)
srli    t2, t2, (4 * h)
andi    t2, t2, 0xF
li      t3, 3                  # 3 = OFF
bne     t2, t3, gate_poll
```

The gate bits are one field: PWRCR decodes bits 1–4 (mask 0x1E), and bit 0 (hart 0) is reserved, reads 0 and ignores writes. A blanket sw of 0x1E therefore gates every channel tile at once and leaves hart 0 running; use full-word stores, because the write is qualified by byte lane 0 only.

To wake the tile, clear the same bit and wait for the state nibble to return to 0 (ON). The tile then re-executes the shared boot ROM from address 0, parks in WFI, and is relaunched exactly like at chip power-on: write the loader row (source, length, entry) and send the tile an msip.

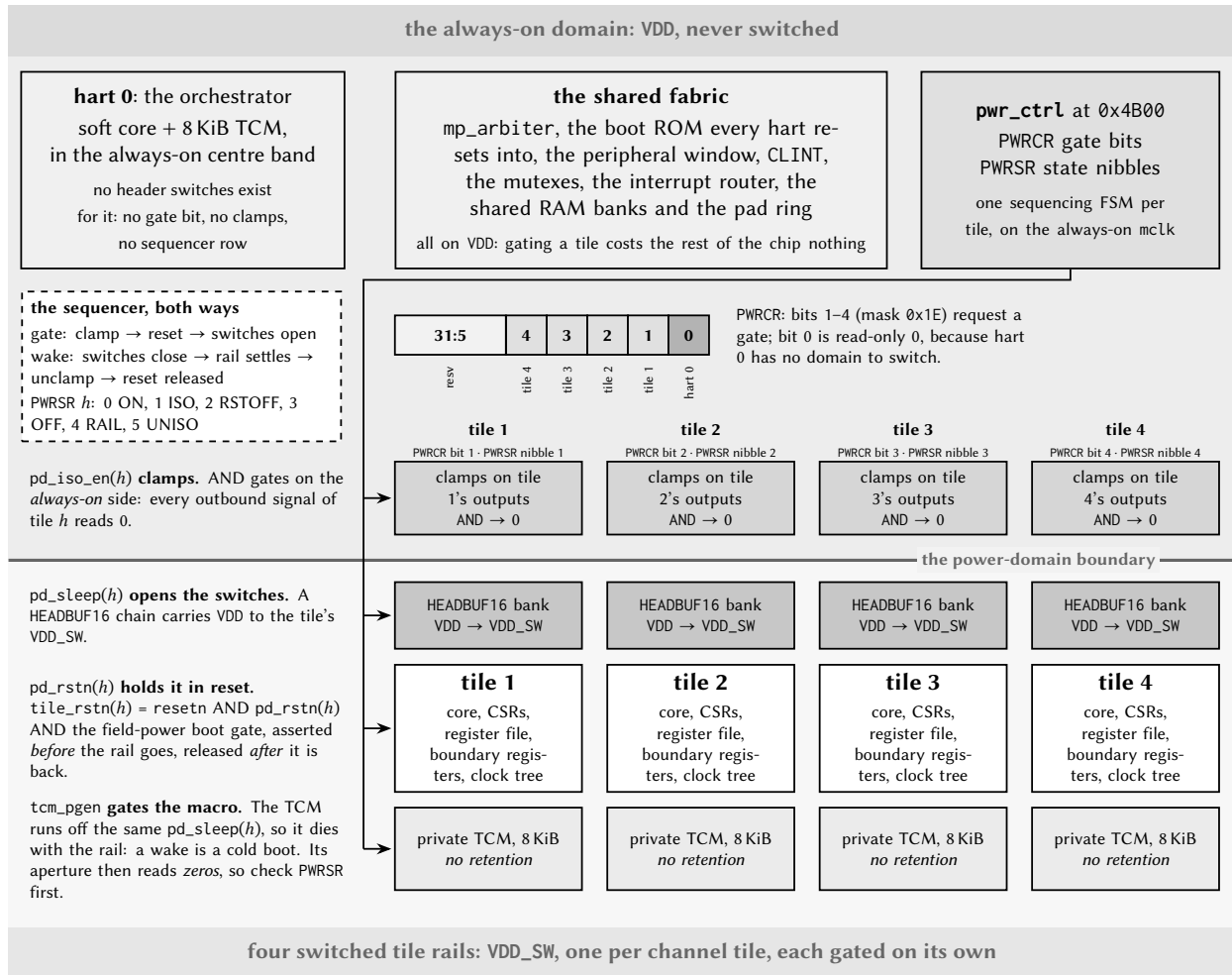


Figure 101: The chip's power domains. Everything the rest of the chip needs (the arbiter, the boot ROM, the peripherals, the shared RAM, the pad ring and hart 0) sits on the unswitched VDD rail; each channel tile has a rail of its own. Setting PWRCR bit h walks the four layers of tile h downwards: the clamps assert on the always-on side of the boundary, the tile is held in reset, and only then do the header switches open. Clearing the bit walks them back up, and because nothing in the tile retains state, what comes back up is a cold boot.

Because gating is *cold*, everything on the tile dies with the rail: the register file, CSRs, and the private TCM contents. Any data the tile must keep across a power cycle has to be staged to shared RAM before gating. The isolation clamps drive every outbound tile signal to its inactive level, which is identical to the tile's reset state, so the rest of the chip observes a gated tile simply as a silent one.

The sequencer never aborts a transition: a gate-bit change made mid-sequence takes effect when the current sequence completes. The rail-settle delay on wake is sized for the header-switch daisy chain of one tile domain; software does not need to add its own delays beyond polling PWRSR.

22.2 Field-Powered Boot Gate and Wake Sources

In configurations wired for field power (the harvested-energy NFC operating mode) the power controller additionally supervises *when* the chip is allowed to start executing, so that the harts are held quiescent until the harvested supply can actually carry a transaction. Two package pins participate: a power-good input (PGOOD), driven by an external supply supervisor, and a boot-mode strap that selects harvested boot. On the LQFP-100 package these are the two spare pins P6.7 (PGOOD, pin 69) and P6.6 (strap, pin 68). Both pads reset to inputs with their pull-downs enabled, so an unconnected pin reads as “power not good” and “normal boot”: a board that does not use the feature behaves exactly like earlier revisions, and the whole mechanism is a provable no-op with the inputs tied inactive.

Two registers expose the feature. PWRWAKE (read/write, offset 0x14, reset 0) arms and configures the boot gate; PWRSTS (read-only, offset 0x18) reports the synchronized live pin states, the once-sampled strap value, and the current gate state. Both sit at fixed offsets above the PWRSR words, so this part of the map does not depend on the hart count. All three pad-side inputs are asynchronous and are double-flop synchronized inside the controller on the always-on clock domain.

The boot gate is a hold-in-reset. An internal release signal is ANDed into *every* hart's outer reset (hart 0 included, which gains a reset fold it does not otherwise have), so a held hart fetches nothing and issues no shared-bus request; the rest of the chip cannot distinguish it from a hart still in power-on-reset. When the release conditions are met the harts leave reset and run the ordinary boot ROM, with no software-visible partial state.

The strap drives policy; software ORs in overrides. The strap is sampled exactly once, a few master-clock cycles after reset release, and sets the *hardware defaults* of the gate; the PWRWAKE bits then OR in software overrides. This resolves the chicken-and-egg of a gate that holds the very cores that would program it: a harvested board self-arms with zero firmware (arm the gate, make PGOOD a release condition, and re-hold on brownout), while a normally-powered board leaves the gate disarmed. Software may independently arm the gate, add PGOOD and/or the NFC *field-detect* input as release conditions, force a release, or choose between a one-shot latched release and re-holding whenever a release condition later drops.

Field detect is a wake path, not an interrupt. The NFC field-detect signal, asserted when a reader's RF field is present, can be selected as a gate release condition, in which case it brings the chip out of the boot hold and consumes no interrupt vector. (The same signal remains separately available as an ordinary interrupt to firmware that is already running; the two uses are independent.)

Brownout is a cold boot. With re-hold enabled, a PGOOD deassertion re-asserts the boot gate and its later return releases the harts into a fresh boot ROM run. Because gating is cold and the chip keeps no retention, a full cold boot is the honest model of losing and regaining the harvested rail, and is exactly what re-hold produces.

The same silicon supports two field-power schemes. In the *batteryless* scheme the gate is the mechanism: the chip holds in reset until PGOOD, cold-boots the harvested-boot ROM path (see the multi-core boot chapter), serves the reader, and re-holds when the field collapses. In the more robust *supercap duty-cycled* scheme an external buffer stays charged and the gate is left disarmed after the initial boot; the service loop sleeps between transactions and is resumed by the ordinary field-detect interrupt, never re-entering reset.

22.2.1 Exact mechanism and timing

All of the boot-gate state lives on the always-on master-clock domain and evaluates once per MCLK cycle. Writing the effective policy as logic (PWRWAKE bit names on the right):

```
strap_harvest = STRAP_VALID and STRAP
arm    = PWGATEEN  or strap_harvest
p_rls  = PWRLSPGOOD or strap_harvest
f_rls  = PWRLSFIELD
rehold = PWREHOLD  or strap_harvest
release = PWSWRLS or (p_rls and PGOOD_live) or (f_rls and FIELD_live)
hold    = arm and not (rehold ? release : release_latched)
pgood_rstn = not hold                                (registered; reset value = released)
```

The external inputs PGOOD_live and FIELD_live are the double-flop-synchronized pad and field-detect levels (two MCLK cycles of latency); release_latched is a sticky OR of every past release, giving the one-shot behaviour when PWREHOLD = 0. The gate output itself is registered, so releases and re-holds are glitch-free and take effect one cycle after their condition resolves.

The strap is sampled exactly once: a free-running counter starts at reset release and, after the settle delay (eight MCLK cycles in this implementation, comfortably beyond the two-cycle synchronizer and the board pull's settling during the much longer power-on-reset), latches the synchronized strap level and sets PWSTRAPVLD. On a strapped (harvested) board the whole self-arm therefore completes roughly eleven cycles after reset release. The gate is *released* during that sampling window (deliberately, so that a normal board's boot is never delayed), which means a harvested board's harts may begin their first boot-ROM fetch and then be re-held. That is harmless by construction: the hold is the same outer reset that cleared the harts at power-on, the boot-fetch tracking state resets with it, and the eventual release simply runs a clean cold boot. The same argument makes every hold/release transition safe at any moment: a held hart's bus interface sits at its reset values, which the shared-bus arbiter already treats as "no request" (the cold-gate isolation contract of the tile domains).

Note the asymmetry between the two PWRWAKE policies. With PWREHOLD = 1 the hold *tracks* the release condition: PGOOD dropping re-asserts the hold on the next cycle, and the harts re-enter reset immediately; there is no grace period, because without retention there is nothing a grace period could save. With PWREHOLD = 0 the first release is final until the next chip reset; use this when supervising software, not the gate, should own brownout policy. An armed gate with *no* release source enabled holds indefinitely until PWSWRLS; this is intentional and is used by the production test to prove the gate is load-bearing.

22.2.2 Programming sequences

Harvested board (zero firmware). Strap P6.6 high, wire the supervisor's power-good output to P6.7. Nothing else: the hardware defaults arm the gate, wait on PGOOD, and re-hold on brownout. The first instruction any hart executes happens after the supply is declared good.

Software-armed gate (test or supervised systems). From any running hart: write PWRWAKE = PWGATEEN | PWRLSPGOOD (optionally | PWREHOLD); the gate now follows PGOOD. To add the RF field as a release source, set PWRLSFIELD. To force a release regardless of pins, set PWSWRLS. Poll PWRSTS.PWBOOTHOLD to observe the gate; note that software can only observe the gate *released* or arm it for the *next* hold; while the hold is active there is, by design, nobody executing.

Supercap duty-cycled service loop (gate disarmed). Boot normally, then: gate the unused tile harts through PWRRCR; route the NFC field-detect interrupt (vector 94) to the serving hart in its interrupt-router enable row; provision and arm the NFC block in LISTEN with auto-read; optionally divide MCLK down;

sleep with WFI. A reader arriving raises field-detect, the router delivers the external interrupt, and the hart resumes with all state intact; the boot gate plays no part. PWRSTS.PWFIELDLIV remains available for polling designs.

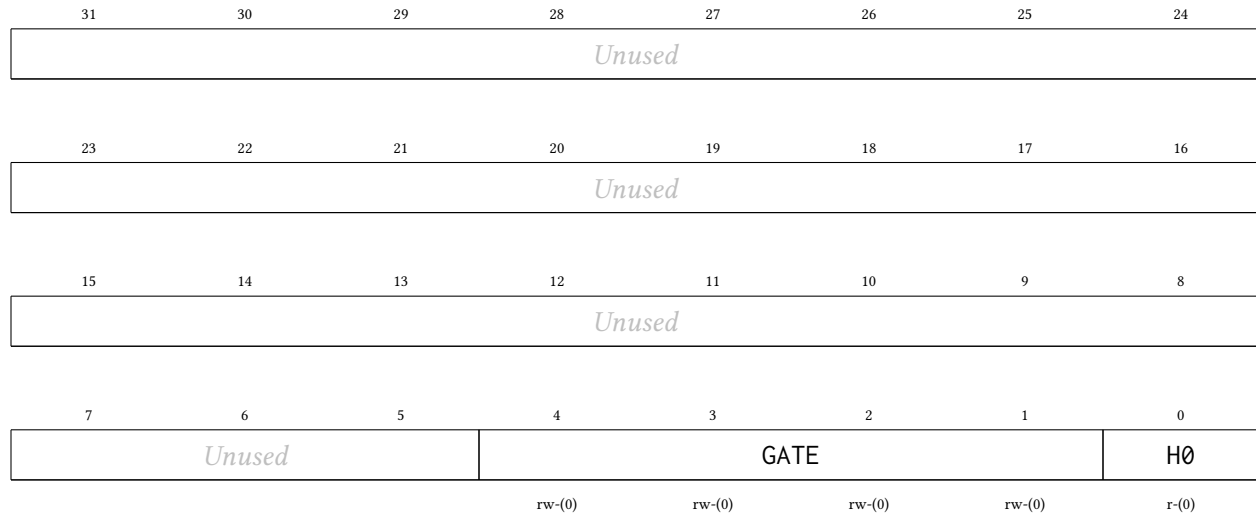
Choosing bank power (service-loop floor). In either scheme the deepest useful power state keeps one shared-RAM bank (stack and payload) and the serving hart powered: gate the tiles via PWRCR, gate the unused shared banks and memories via BLOCKPWR (see the SYSTEM chapter: gated macros halve their leakage; contents are lost), and divide the clock. Dynamic power scales with the divider; the floor that remains is the leakage of the always-on fabric and whatever memories stay powered.

22.3 Register Description

22.3.1 PWRCR Register

Register memory slot: 0, offset: 0 bytes, size: 4 bytes

Power gate control. Setting GATE bit h powers tile hart h down (isolation, reset, rail off); clearing it powers the tile back up and cold-boots it. A request made while the sequencer is mid-sequence is honored when the sequence completes (no aborts). Bit 0 (hart 0) is reserved: always-on, reads 0, writes ignored.

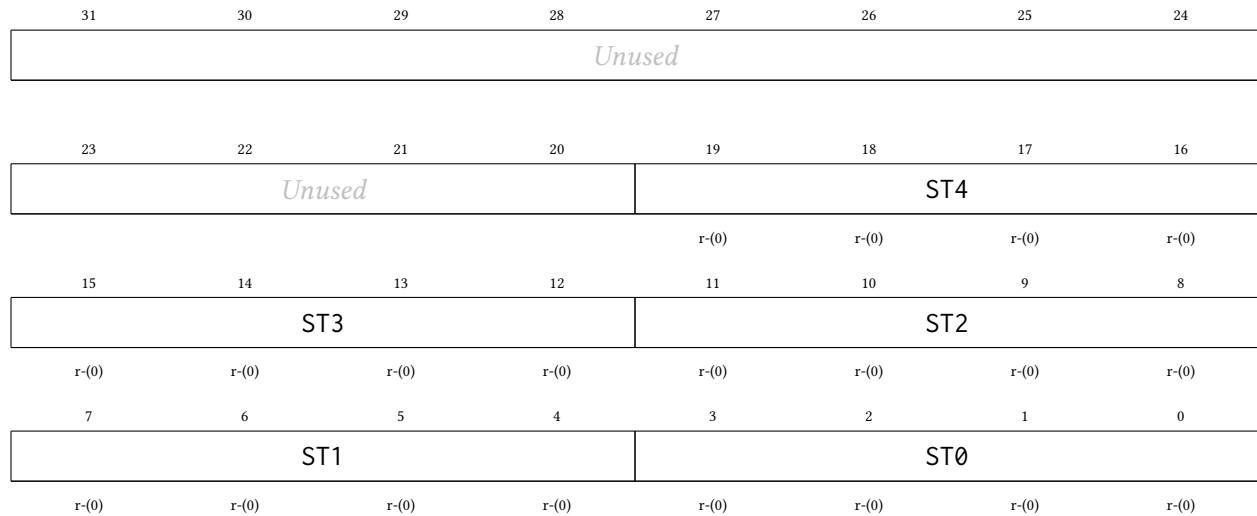


Bitfield	Range	Description
Unused	Bits 31-5	
GATE	Bits 4-1	Gate request per tile hart: bit h = 1 powers tile hart h down, 0 powers it up (cold boot). Poll PWRSR for sequencer completion. 0000 PWRGATE_NONE: All tile harts powered
H0	Bit 0	Hart 0 is always-on: reads 0, writes ignored.

22.3.2 PWRSR Register

Register memory slot: 1, offset: 4 bytes, size: 4 bytes

Power sequencer state, one read-only nibble per hart (bits 4h+3:4h). 0 = ON, 1 = ISO (clamps asserting), 2 = RSTOFF (reset held, rail dying), 3 = OFF (gated), 4 = RAIL (waking, rail settling), 5 = UNISO (clamps releasing). Hart 0's nibble always reads 0. A tile is safely gated when its nibble reads 3, and fully awake (booting or parked in the ROM) when it returns to 0.

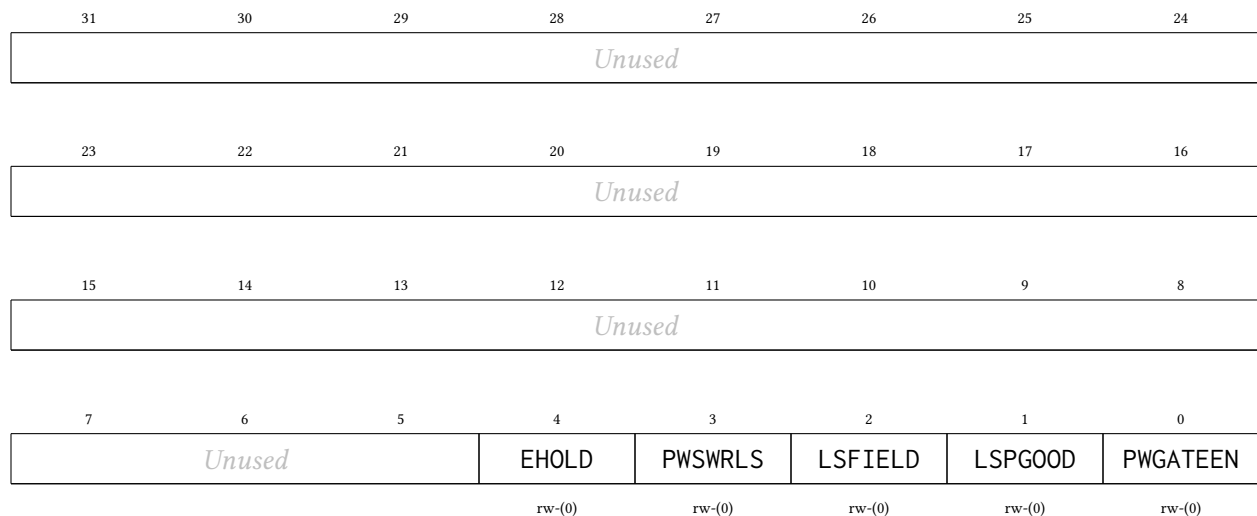


Bitfield	Range	Description
<i>Unused</i>	Bits 31-20	
ST4	Bits 19-16	Tile hart 4 sequencer state.
ST3	Bits 15-12	Tile hart 3 sequencer state.
ST2	Bits 11-8	Tile hart 2 sequencer state.
ST1	Bits 7-4	Tile hart 1 sequencer state.
ST0	Bits 3-0	Hart 0 state: always 0 (ON, always-on domain).

22.3.3 PWRWAKE Register

Register memory slot: 5, offset: 20 bytes, size: 4 bytes

Boot-gate / wake-source control (DP-S3 field-powered mode). The harvested-boot STRAP drives the hardware defaults (a strapped board self-arms the gate, waits on PGOOD, and re-holds on brownout with no software involvement); these bits OR-IN software overrides on top. Resets to 0 = no software override = the gate is released on every normal boot. Write with full-word stores (byte-lane-0 qualified, like PWRCLR).



Bitfield	Range	Description
<i>Unused</i>	Bits 31-5	
EHOLD	Bit 4	Re-hold policy override: 1 = re-assert the boot hold when the release condition drops (brownout re-hold; PGOOD return then cold-boots the harts through the shared ROM). 0 = one-shot: once released, stays released (see PWRSTS.PWRRLSLATCH). The strap ORs this in on a harvested board.
PWSWRLS	Bit 3	Software-forced release: 1 releases an armed gate unconditionally (test/override, or "software says proceed"). With no release source enabled, an armed gate holds until this bit, the negative-control proof that the gate is load-bearing.
LSFIELD	Bit 2	1 = the synchronized NFC field level (PWRSTS.PWFIELDLIV) is a release condition, the field_detect WAKE source: a reader arriving releases the boot gate. No IRQ vector is involved. Reads as tied-0 field when NFC is absent or fieldPower is off.
LSPGOOD	Bit 1	1 = the synchronized PGOOD pad level (PWRSTS.PWPGOODLIV) is a release condition. The strap ORs this in on a harvested board (it always waits on the supply supervisor).
PWGATEEN	Bit 0	Software boot-gate arm: 1 arms the HOLD-IN-RESET gate (every hart's outer reset held until a release condition fires). The harvested-boot strap ORs this in, so a strapped board arms with no software.

22.3.4 PWRSTS Register

Register memory slot: 6, offset: 24 bytes, size: 4 bytes

Boot-gate / wake-source status (read-only). The synchronized live pad levels, the one-shot strap sample (THE bootrom harvested-boot branch bit), and the gate state.

31	30	29	28	27	26	25	24
<i>Unused</i>							
23	22	21	20	19	18	17	16
<i>Unused</i>							
15	14	13	12	11	10	9	8
<i>Unused</i>							
7	6	5	4	3	2	1	0
<i>Unused</i>		RLSLATCH	PWBOOTHOLD	PWSTRAPVLD	PWSTRAP	PWFIELDLIV	PWPGOODLIV
		r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)

Bitfield	Range	Description
<i>Unused</i>	Bits 31-6	

RLSLATCH	Bit 5	One-shot release has latched (sticky until reset; only meaningful with PWRE-HOLD = 0).
PWBOOTHOLD	Bit 4	Current gate state: 1 = the boot gate is holding every hart in reset (pgood_rstn asserted).
PWSTRAPVLD	Bit 3	Strap sample complete (the one-shot sample lands a few mclk after reset release; poll before consuming PWSTRAP).
PWSTRAP	Bit 2	Latched harvested-boot strap sample: 1 = harvested boot (the bootrom skips the SPI-flash copy and runs the ROM-resident service loop), 0 = normal SPI boot. Sampled ONCE after reset; mid-run strap changes are ignored.
PWFIELDLIV	Bit 1	Synchronized NFC field_detect level (0 when NFC is absent or fieldPower is off).
PWPGOODLIV	Bit 0	Synchronized PGOOD pad level (P6.7). Reads 0 = power-not-good when the pin is unconnected (reset pull-down).

23 I2Cx Peripheral

The I2Cx peripherals are general-purpose I²C interfaces, each able to act as a bus master or as an addressed slave. Master and slave are driven through two separate sets of register bits sharing one pin pair, so a port can be configured for either role (or, with both enabled, participate in a multi-master bus). The main features are listed below:

- Master transmitter and master receiver modes, with START, repeated-START and STOP generation (I2CMST, I2CMSP) and a programmable bus clock divider (I2CMDIV)
- Multi-master arbitration: losing the bus releases it and raises I2CMARB
- Slave transmitter and slave receiver modes with a 7-bit address (I2CxAR), a per-bit address mask (I2CxAMR) for responding to a range of addresses, and an optional general-call response (I2CGCE)
- Optional slave clock stretching (I2CSCS): the port holds SCLx low through the ACK bit until firmware releases it, trading bus latency for lossless flow control
- Separate transmit and receive data registers for each role (I2CxMTX/I2CxMRX, I2CxSTX/I2CxSRX)
- Individually maskable interrupt sources for transfer complete, arbitration loss, NACK received, address match, START and STOP detection, and receive overrun

23.1 The Wire Protocol

Both roles speak the same two-wire protocol. SDAx and SCLx are open-drain and require external pull-up resistors: a device drives a bit low by pulling the line down and releases it to let the pull-up present a one. Because every device can only pull down, two devices driving at once cannot damage each other, and that is what makes both multi-master arbitration and slave clock stretching possible on the same pair of wires.

A transfer is framed by two conditions that are deliberately illegal during data: **START** is a falling edge on SDAx while SCLx is high, and **STOP** is a rising edge on SDAx while SCLx is high. Between them, SDAx may only change while SCLx is low, so every data bit is stable across the whole SCLx high period and any receiver may sample it there. After the address byte and after each data byte, the transmitter releases SDAx for one clock and the receiver answers by pulling it low for an ACK or leaving it high for a NACK. Figure 102 shows one complete byte write.

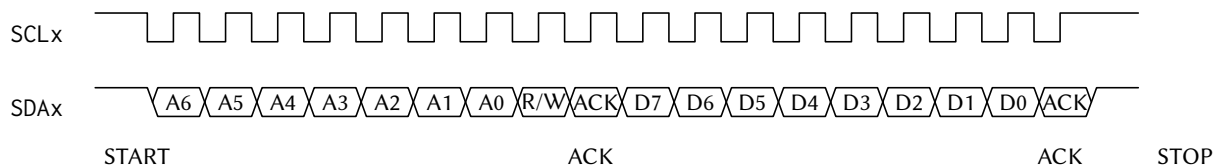


Figure 102: One I2C byte write on the wire: START, the 7-bit address and direction bit, the addressed device's ACK, one data byte, its ACK, then STOP. START and STOP are the two transitions of SDAx that occur while SCLx is high; every data bit is stable across the SCLx high period.

The address byte carries the 7-bit slave address in its most significant bits and the direction in its least significant bit: 0 selects a write (master transmitter), 1 a read (master receiver). A master that wants to change direction or address without giving up the bus issues a repeated START instead of a STOP.

23.2 Clock Domain

The I2Cx cores are clocked from SMCLK, and I2CMDIV divides that down to the bus rate. The boot ROM may leave SMCLK sourced from LFXT (32.768 kHz), which would put the bus in the low-kilohertz range; configure the SMCLK source through SYSCLOCKR (see the SYSTEM chapter) before relying on any I²C timing, rather than inheriting whatever the boot path left behind.

The step-by-step register sequences for each of the four roles are given with the register descriptions below.

23.3 Register Description

23.3.1 I2CCR Register

Register memory slot: 0, offset: 0 bytes, size: 4 bytes

I2C control register

31	30	29	28	27	26	25	24
Unused							
23	22	21	20	19	18	17	16
Unused		MEN	SEN	SN	SCS	GCE	MDIV
rw-(0)		rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
MDIV		SAIE	STXEIE	SOVFIE	SNRIE	SXCIE	
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MSTSIE	MSPSIE	MARBIE	MTXEIE	MNRIE	MXCIE	STRIE	SPRIE
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
Unused	Bits 31-22	
MEN	Bit 21	I2C master enable. When enabled, this device awaits a command to send a start condition with I2CMST, and then begins acting as a master until commanded to send a stop condition with I2CMSP. If master mode is also enabled on this device, then this device will act as a slave until commanded to send a start condition with I2CMST, whereupon it will begin acting as a master. Once the master transfer is complete, it will resume acting as a slave. 0 Disabled 1 Enabled
SEN	Bit 20	I2C slave enable. When enabled, this device behaves as an I2C slave and begins listening for its address. If master mode is also enabled on this device, then this device will act as a slave until commanded to send a start condition with I2CMST, whereupon it will begin acting as a master. Once the master transfer is complete, it will resume acting as a slave. 0 Disabled 1 Enabled
SN	Bit 19	I2C slave NACK next byte received. When enabled, this slave will reply with a NACK whenever it receives its address or whenever it receives a byte from a master. When disabled, it will send an ACK in those situations. If clock stretching is enabled, this bit can be changed in accordance with the desired ACK/NACK reply before allowing a rising edge of SCL. 0 ACK

		1	NACK
SCS	Bit 18	I2C slave clock stretching enable. When enabled, this slave will hold the SCL line low during the ACK phase of the transmission to allow this slave more time. Note that the master will be left waiting for the ACK/NACK as long as this bit is set to 1.	
		0	Disabled
		1	Enabled
GCE	Bit 17	I2C slave general call enable. When enabled, this slave will be addressed if a global call is issued on the bus.	
		0	Disabled
		1	Enabled
MDIV	Bits 16-13	I2C master mode clock divider. The master mode finite state machine clock source is SMCLK, which is divided by a factor of $4 * 2^{I2CMDIV}$.	
		0000	I2CMDIV_1: /1 (no division)
		0001	I2CMDIV_2: /2
		0010	I2CMDIV_4: /4
		0011	I2CMDIV_8: /8
		0100	I2CMDIV_16: /16
		0101	I2CMDIV_32: /32
		0110	I2CMDIV_64: /64
		0111	I2CMDIV_128: /128
		1000	I2CMDIV_256: /256
		1001	I2CMDIV_512: /512
		1010	I2CMDIV_1024: /1,024
		1011	I2CMDIV_2048: /2,048
		1100	I2CMDIV_4096: /4,096
		1101	I2CMDIV_8192: /8,192
		1110	I2CMDIV_16384: /16,384
		1111	I2CMDIV_32768: /32,768
SAIE	Bit 12	I2C slave mode addressed interrupt enable	
		0	Interrupt disabled
		1	Interrupt enabled
STXEIE	Bit 11	I2C slave transmit register empty interrupt enable	
		0	Interrupt disabled
		1	Interrupt enabled
SOVFIE	Bit 10	I2C slave receive register overflow interrupt enable	
		0	Interrupt disabled
		1	Interrupt enabled
SNRIE	Bit 9	I2C slave mode NACK received from master interrupt enable	
		0	Interrupt disabled
		1	Interrupt enabled
SXCIE	Bit 8	I2C slave mode transfer complete interrupt enable	
		0	Interrupt disabled
		1	Interrupt enabled
MSTSIE	Bit 7	I2C master mode start condition sent interrupt enable	
		0	Interrupt disabled
		1	Interrupt enabled

MSPSIE	Bit 6	I2C master mode stop condition sent interrupt enable 0 Interrupt disabled 1 Interrupt enabled
MARBIE	Bit 5	I2C master mode arbitration error interrupt enable 0 Interrupt disabled 1 Interrupt enabled
MTXEIE	Bit 4	I2C master transmit register empty interrupt enable 0 Interrupt disabled 1 Interrupt enabled
MNRIE	Bit 3	I2C master mode NACK received interrupt enable 0 Interrupt disabled 1 Interrupt enabled
MXCIE	Bit 2	I2C master transfer complete interrupt enable 0 Interrupt disabled 1 Interrupt enabled
STRIE	Bit 1	I2C start received interrupt enable 0 Interrupt disabled 1 Interrupt enabled
SPRIE	Bit 0	I2C stop received interrupt enable 0 Interrupt disabled 1 Interrupt enabled

23.3.2 I2CFCR Register

Register memory slot: 1, offset: 4 bytes, size: 1 byte

I2C flow control register. Writing a 1 to a bit in this register initiates or queues the associated command. Writing a 0 to a bit does nothing. Reading this register always returns the value 0.

7	6	5	4	3	2	1	0
Unused				SC	MST	MSP	MRB
				w1-(0)	w1-(0)	w1-(0)	w1-(0)

Bitfield	Range	Description
Unused	Bits 7-4	
SC	Bit 3	I2C slave continue command. When clock stretching is enabled in slave mode, set this bit to tell the slave to continue with the ACK/NACK phase of the current byte by releasing SCL. This may only be set if clock stretching is enabled, slave mode is enabled, and the slave transfer complete flag has just been set. 0 No effect 1 Send a start condition

MST	Bit 2	I2C master send start condition command. Set this bit to 1 to send a start condition or a repeated start condition. Once this bit is set, this master is required to send at least one address frame before initiating this command again. If another master has control of the bus when this bit is set, this master will wait until the bus is idle before seizing control of it and sending the start condition. If this master is busy with a transaction when this bit is set, then it will send a restart condition immediately after it finishes the transaction. 0 No effect 1 Send a start condition
MSP	Bit 1	I2C master send stop condition command. Set this bit to 1 while this master is in control of the bus to send a stop condition. This master is required to have already sent a start condition and at least one address frame before initiating this command. If this master is busy with a transaction when this bit is set, then it will send the stop condition immediately after it finishes the transaction. 0 No effect 1 Send a stop condition
MRB	Bit 0	I2C master read byte command. Set this bit to 1 while in master receiver mode to read one byte from the slave. This master is required to have already sent the slave address and received an ACK from the slave before initiating this command. 0 No effect 1 Read next byte

23.3.3 I2CSR Register

Register memory slot: 2, offset: 8 bytes, size: 2 bytes

I2C status register

15	14	13	12	11	10	9	8
BS	MCB	STM	SA	STXE	SOVF	SNR	SXC
r-(0)	r-(0)	r-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)
7	6	5	4	3	2	1	0
MSTS	MSPS	MARB	MTXE	MNR	MXC	STR	SPR
rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)

Bitfield	Range	Description
BS	Bit 15	I2C bus state indicator. This bit cannot be cleared by writing to the status register. 0 The I2C bus is idle 1 The I2C bus is active
MCB	Bit 14	I2C master controls bus indicator. This bit cannot be cleared by writing to the status register. 0 This master does not control the bus 1 This master controls the bus

STM	Bit 13	I2C slave transmitter mode indicator. Indicates for which mode this slave has been addressed. Only valid if I2CSA is 1. This bit cannot be cleared by writing to the status register. 0 Slave receiver mode 1 Slave transmitter mode
SA	Bit 12	I2C slave mode addressed interrupt flag. Indicates that this slave has been addressed by another master. Write a 1 to this bit to clear it. 0 No pending interrupt 1 Pending interrupt
STXE	Bit 11	I2C slave transmit register empty interrupt flag. This bit is set when this slave latches the data stored in the masslaveter transmit register to indicate that the slave transmit register is ready to accept another byte and queue it for transmission. Write a 1 to this bit to clear it. 0 No pending interrupt 1 Pending interrupt
SOVF	Bit 10	I2C slave receive register overflow interrupt flag. Indicates that this slave has failed to read one or more bytes from the I2CxSRX register before they were overwritten by another transmission. Write a 1 to this bit to clear it. 0 No pending interrupt 1 Pending interrupt
SNR	Bit 9	I2C slave mode NACK received from master interrupt flag. This bit is set in slave transmitter mode if the master responds with a NACK after this slave sends it a byte of data. This bit is not set if the master responds with an ACK. Write a 1 to this bit to clear it. 0 No pending interrupt 1 Pending interrupt
SXC	Bit 8	I2C slave mode transfer complete interrupt flag. This bit is set after this slave receives a byte of data from a master, but before this slave sends an ACK/NACK. Write a 1 to this bit to clear it. 0 No pending interrupt 1 Pending interrupt
MSTS	Bit 7	I2C master mode start condition sent interrupt flag. This flag is set after this master sends a start condition or repeated start condition. Write a 1 to this bit to clear it. 0 No pending interrupt 1 Pending interrupt
MSPS	Bit 6	I2C master mode stop condition sent interrupt flag. This flag is set after this master sends a stop condition. Write a 1 to this bit to clear it. 0 No pending interrupt 1 Pending interrupt
MARB	Bit 5	I2C master mode arbitration loss interrupt flag. This bit is set when this master detects that the value it tried to write to SDA is being overridden by another master. After it detects the arbitration loss, this master releases control of the bus. Write a 1 to this bit to clear it. 0 No pending interrupt 1 Pending interrupt

MTXE	Bit 4	I2C master transmit register empty interrupt flag. This bit is set when this master latches the data stored in the master transmit register to indicate that the master transmit register is ready to accept another byte and queue it for transmission. Write a 1 to this bit to clear it. 0 No pending interrupt 1 Pending interrupt
MNR	Bit 3	I2C master mode NACK received interrupt flag. This flag is set in master transmitter mode after ACK/NACK bit is sent if the slave sends a NACK. It is not set if the slave sends an ACK. Write a 1 to this bit to clear it. 0 No pending interrupt 1 Pending interrupt
MXC	Bit 2	I2C master transfer complete interrupt flag. In master transmitter mode, this flag is set after this master has sent the data byte and the slave has sent an ACK/NACK. In master receiver mode, this flag is set after the slave has sent the data byte and before this master sends an ACK/NACK. Write a 1 to this bit to clear it. 0 No pending interrupt 1 Pending interrupt
STR	Bit 1	I2C start condition received interrupt flag. This flag is set whenever a start condition or repeated start condition is detected on the bus, regardless of if this master or another sent it. Write a 1 to this bit to clear it. 0 No pending interrupt 1 Pending interrupt
SPR	Bit 0	I2C stop condition received interrupt flag. This flag is set whenever a stop condition condition is detected on the bus, regardless of which device sent it. Write a 1 to this bit to clear it. 0 No pending interrupt 1 Pending interrupt

23.3.4 I2CMTX Register

Register memory slot: 3, offset: 12 bytes, size: 1 byte

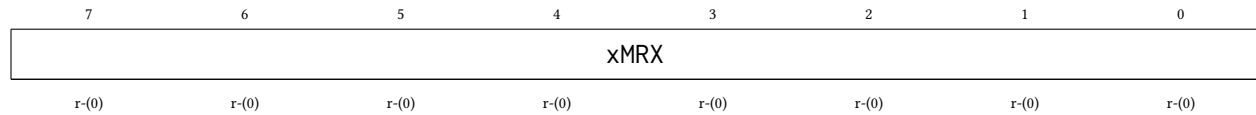
I2C master transmit register. Write the desired slave address and read/write bit to this register after sending a start condition to begin a transmission with a slave. Note that the desired slave address must occupy the upper seven bits and the read/write bit must occupy the least significant bit. If the read bit is 0, the master enters master transmitter mode. If the read bit is 1, the master enters master receiver mode. Write a byte of data to this register after sending the address frame or a data frame to send that byte of data to the slave. If this master is busy with a transmission when this register is written, it will send the byte after it finishes the transmission.



23.3.5 I2CMRX Register

Register memory slot: 4, offset: 16 bytes, size: 1 byte

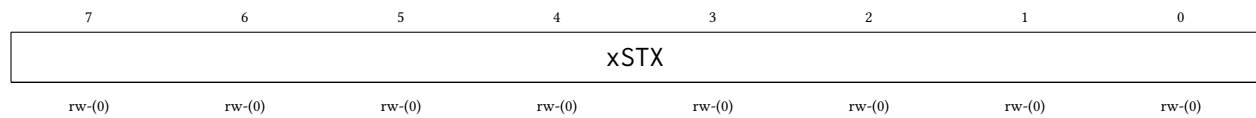
I2C master receive register. When in master receiver mode, read this register after the master transfer complete interrupt flag (I2CMXC) is set to get the received data byte.



23.3.6 I2CSTX Register

Register memory slot: 5, offset: 20 bytes, size: 1 byte

I2C slave transmit register. When in slave transmitter mode, write to this register after the slave addressed flag (I2CSA) or the slave transaction complete flag (I2CSXC) has been set to queue the next byte to send to the master.



23.3.7 I2CSRX Register

Register memory slot: 6, offset: 24 bytes, size: 1 byte

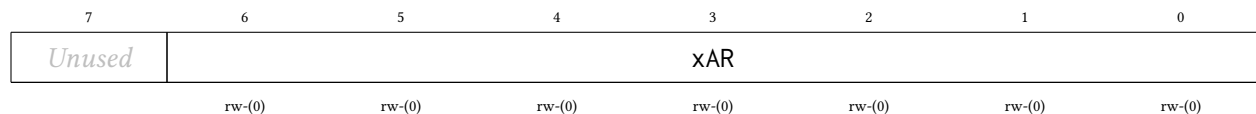
I2C slave receive register. When in slave receiver mode, read this register after the slave transaction complete flag (I2CSXC) has been set to get the data byte sent from the master. Note that if this slave fails to clear the status register before another byte is received, the slave receive overflow flag (I2CSOVF) will be set.



23.3.8 I2CAR Register

Register memory slot: 7, offset: 28 bytes, size: 1 byte

I2C this slave address register. When in slave mode, any master that sends an address frame containing this address will activate this slave.



23.3.9 I2CAMR Register

Register memory slot: 8, offset: 32 bytes, size: 1 byte

I2C this slave address mask register. Any bit set to 1 in this register indicates that the corresponding bit in the slave address register is a wildcard. Only the slave address register bits that correspond to 0s in this register will be compared to the received slave address.

7	6	5	4	3	2	1	0
Unused	xAMR						
	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

24 CLINT Peripheral

The Core-Local Interruptor (CLINT) provides the two interrupt sources that make multi-core software possible: per-hart *software interrupts* (the inter-processor interrupt, or IPI, mechanism) and per-hart *timer interrupts* driven by a shared free-running 64-bit counter. It lives in the shared window behind the multi-core arbiter, so **any hart can raise, clear, or program any hart's interrupts**. The main features are listed below:

- One software interrupt (IPI) register per hart (MSIP0–MSIP4), writable by any hart
- Shared free-running 64-bit MTIME counter, incrementing once per MCLK cycle
- One 64-bit compare register per hart (MTIMECMP0–MTIMECMP4); hart h 's timer interrupt is asserted while $MTIME \geq MTIMECMP_h$
- Level-sensitive interrupt delivery into interrupt vectors 83 (software) and 84 (timer)
- Wakes harts sleeping via the custom sleep instruction, enabling zero-power hart parking
- Compare registers reset to all-ones, so no timer interrupt fires until one is programmed

The CLINT runs on the free-running MCLK, the same clock as each hart's interrupt handler, so a software interrupt can wake a hart whose gated CPU clock is currently off.

24.1 Software Interrupts (IPIs)

Writing 1 to MSIP $_h$ raises the software-interrupt level into hart h (interrupt vector 83); writing 0 clears it. Because the interrupt is **level-sensitive**, hart h 's interrupt service routine must write 0 to its own MSIP $_h$ register before returning, or the interrupt immediately re-triggers.

The canonical use is waking a parked hart: the parked hart enables interrupt vector 83 and executes the custom sleep instruction (see Section 20.3); another hart writes 1 to the sleeper's MSIP register. The sleeper's interrupt service routine runs, clears its MSIP, and must execute the custom wake instruction before returning; otherwise the hart goes back to sleep when the service routine returns.

24.2 Timer Interrupts

MTIME is a single 64-bit counter shared by all five harts. Each hart has its own 64-bit compare value; the timer interrupt level for hart h (interrupt vector 84) is asserted while $MTIME \geq MTIMECMP_h$ and cleared by moving MTIMECMP $_h$ into the future (or by wrapping it back to all-ones to disarm it).

Because the registers are 32 bits wide, 64-bit values are accessed as low/high pairs. To read a coherent 64-bit time: read MTIMEH, then MTIMEL, then MTIMEH again, and retry if the two MTIMEH reads differ. To program a compare value without a spurious interrupt: write all-ones to MTIMECMPxH, then the new low half to MTIMECMPxL, then the real high half to MTIMECMPxH.

24.3 Addressing Note

The CLINT lives in shared-window page 1 at base address 0x5000. The register layout scales with the hart count: the software-interrupt registers are word-mapped from the base (MSIP $_h$ at 0x5000 + 4 h), the shared MTIME low/high pair begins at 0x5020, and hart h 's MTIMECMP $_h$ low/high pair is at 0x5030 + 8 h . The block decodes only the low bits of the address, so its registers alias every 128 bytes throughout 0x5000–0x5FFF. Always use the canonical addresses.

24.4 Register Description

24.4.1 MSIP0 Register

Register memory slot: 0, offset: 0 bytes, size: 4 bytes

Hart 0 software interrupt (IPI) register. Any hart may write it through the shared window: writing 1 raises the software interrupt level into hart 0 (interrupt vector 83); writing 0 clears it. The interrupt is level-sensitive, so hart 0's service routine must write 0 here before returning.



Bitfield	Range	Description
Unused	Bits 31-1	
MSIPH0	Bit 0	Software interrupt (IPI) level for hart 0.
		0 No software interrupt pending
		1 Software interrupt raised

24.4.2 MSIP1 Register

Register memory slot: 1, offset: 4 bytes, size: 4 bytes

Hart 1 software interrupt (IPI) register. Any hart may write it through the shared window: writing 1 raises the software interrupt level into hart 1 (interrupt vector 83); writing 0 clears it. The interrupt is level-sensitive, so hart 1's service routine must write 0 here before returning.



Bitfield	Range	Description
Unused	Bits 31-1	
MSIPH1	Bit 0	Software interrupt (IPI) level for hart 1. 0 No software interrupt pending 1 Software interrupt raised

24.4.3 MSIP2 Register

Register memory slot: 2, offset: 8 bytes, size: 4 bytes

Hart 2 software interrupt (IPI) register. Any hart may write it through the shared window: writing 1 raises the software interrupt level into hart 2 (interrupt vector 83); writing 0 clears it. The interrupt is level-sensitive, so hart 2’s service routine must write 0 here before returning.



Bitfield	Range	Description
<i>Unused</i>	Bits 31-1	
MSIPH2	Bit 0	Software interrupt (IPI) level for hart 2.
	0	No software interrupt pending
	1	Software interrupt raised

24.4.4 MSIP3 Register

Register memory slot: 3, offset: 12 bytes, size: 4 bytes

Hart 3 software interrupt (IPI) register. Any hart may write it through the shared window: writing 1 raises the software interrupt level into hart 3 (interrupt vector 83); writing 0 clears it. The interrupt is level-sensitive, so hart 3's service routine must write 0 here before returning.



Bitfield	Range	Description
<i>Unused</i>	Bits 31-1	
MSIPH3	Bit 0	Software interrupt (IPI) level for hart 3.
	0	No software interrupt pending
	1	Software interrupt raised

24.4.5 MSIP4 Register

Register memory slot: 4, offset: 16 bytes, size: 4 bytes

Hart 4 software interrupt (IPI) register. Any hart may write it through the shared window: writing 1 raises the software interrupt level into hart 4 (interrupt vector 83); writing 0 clears it. The interrupt is level-sensitive, so hart 4's service routine must write 0 here before returning.

31	30	29	28	27	26	25	24
<i>Unused</i>							
23	22	21	20	19	18	17	16
<i>Unused</i>							
15	14	13	12	11	10	9	8
<i>Unused</i>							
7	6	5	4	3	2	1	0
<i>Unused</i>							MSIPH4
rw-(0)							

Bitfield	Range	Description
<i>Unused</i>	Bits 31-1	
MSIPH4	Bit 0	Software interrupt (IPI) level for hart 4.
		0 No software interrupt pending
		1 Software interrupt raised

24.4.6 MTIMEL Register

Register memory slot: 8, offset: 32 bytes, size: 4 bytes

Machine time counter, lower 32 bits. mtime is a free-running 64-bit counter shared by all five harts that increments once per MCLK cycle. It is writable for initialization; a write merges only the addressed 32-bit half.

31	30	29	28	27	26	25	24
MTIMEL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
MTIMEL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
MTIMEL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MTIMEL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
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MTIMEL	Bits 31-0	mtime bits 31:0.
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24.4.7 MTIMEH Register

Register memory slot: 9, offset: 36 bytes, size: 4 bytes

Machine time counter, upper 32 bits. Read MTIMEH, then MTIMEL, then MTIMEH again (retry if the two MTIMEH reads differ) to obtain a coherent 64-bit time value.

31	30	29	28	27	26	25	24
MTIMEH							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
MTIMEH							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
MTIMEH							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MTIMEH							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
MTIMEH	Bits 31-0	mtime bits 63:32.

24.4.8 MTIMECMP0L Register

Register memory slot: 12, offset: 48 bytes, size: 4 bytes

Hart 0 timer compare register, lower 32 bits. The timer interrupt level for hart 0 (interrupt vector 84) is raised while mtime >= mtimecmp0. Resets to all-ones (no interrupt). To set a new compare value without a spurious interrupt, first write all-ones to MTIMECMP0H, then the new lower half, then the real upper half.

31	30	29	28	27	26	25	24
MTIMECMP0L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
MTIMECMP0L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
MTIMECMP0L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MTIMECMP0L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
MTIMECMP0L	Bits 31-0	mtimecmp0 bits 31:0.

24.4.9 MTIMECMP0H Register

Register memory slot: 13, offset: 52 bytes, size: 4 bytes

Hart 0 timer compare register, upper 32 bits.

31	30	29	28	27	26	25	24
MTIMECMP0H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
MTIMECMP0H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
MTIMECMP0H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MTIMECMP0H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
MTIMECMP0H	Bits 31-0	mtimecmp0 bits 63:32.

24.4.10 MTIMECMP1L Register

Register memory slot: 14, offset: 56 bytes, size: 4 bytes

Hart 1 timer compare register, lower 32 bits. The timer interrupt level for hart 1 (interrupt vector 84) is raised while `mtime >= mtimecmp1`. Resets to all-ones (no interrupt). To set a new compare value without a spurious interrupt, first write all-ones to `MTIMECMP1H`, then the new lower half, then the real upper half.

31	30	29	28	27	26	25	24
MTIMECMP1L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
MTIMECMP1L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
MTIMECMP1L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MTIMECMP1L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
MTIMECMP1L	Bits 31-0	mtimecmp1 bits 31:0.

24.4.11 MTIMECMP1H Register

Register memory slot: 15, offset: 60 bytes, size: 4 bytes

Hart 1 timer compare register, upper 32 bits.

31	30	29	28	27	26	25	24
MTIMECMP1H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
MTIMECMP1H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
MTIMECMP1H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MTIMECMP1H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
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MTIMECMP1H	Bits 31-0	mtimecmp1 bits 63:32.
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24.4.12 MTIMECMP2L Register

Register memory slot: 16, offset: 64 bytes, size: 4 bytes

Hart 2 timer compare register, lower 32 bits. The timer interrupt level for hart 2 (interrupt vector 84) is raised while mtime >= mtimecmp2. Resets to all-ones (no interrupt). To set a new compare value without a spurious interrupt, first write all-ones to MTIMECMP2H, then the new lower half, then the real upper half.

31	30	29	28	27	26	25	24
MTIMECMP2L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
MTIMECMP2L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
MTIMECMP2L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MTIMECMP2L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
MTIMECMP2L	Bits 31-0	mtimecmp2 bits 31:0.

24.4.13 MTIMECMP2H Register

Register memory slot: 17, offset: 68 bytes, size: 4 bytes

Hart 2 timer compare register, upper 32 bits.

31	30	29	28	27	26	25	24
MTIMECMP2H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
MTIMECMP2H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
MTIMECMP2H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MTIMECMP2H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
MTIMECMP2H	Bits 31-0	mtimecmp2 bits 63:32.

24.4.14 MTIMECMP3L Register

Register memory slot: 18, offset: 72 bytes, size: 4 bytes

Hart 3 timer compare register, lower 32 bits. The timer interrupt level for hart 3 (interrupt vector 84) is raised while mtime >= mtimecmp3. Resets to all-ones (no interrupt). To set a new compare value without a spurious interrupt, first write all-ones to MTIMECMP3H, then the new lower half, then the real upper half.

31	30	29	28	27	26	25	24
MTIMECMP3L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
MTIMECMP3L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
MTIMECMP3L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MTIMECMP3L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
MTIMECMP3L	Bits 31-0	mtimecmp3 bits 31:0.

24.4.15 MTIMECMP3H Register

Register memory slot: 19, offset: 76 bytes, size: 4 bytes

Hart 3 timer compare register, upper 32 bits.

31	30	29	28	27	26	25	24
MTIMECMP3H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
MTIMECMP3H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
MTIMECMP3H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MTIMECMP3H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
MTIMECMP3H	Bits 31-0	mtimecmp3 bits 63:32.

24.4.16 MTIMECMP4L Register

Register memory slot: 20, offset: 80 bytes, size: 4 bytes

Hart 4 timer compare register, lower 32 bits. The timer interrupt level for hart 4 (interrupt vector 84) is raised while mtime >= mtimecmp4. Resets to all-ones (no interrupt). To set a new compare value without a spurious interrupt, first write all-ones to MTIMECMP4H, then the new lower half, then the real upper half.

31	30	29	28	27	26	25	24
MTIMECMP4L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
MTIMECMP4L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
MTIMECMP4L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MTIMECMP4L							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
MTIMECMP4L	Bits 31-0	mtimecmp4 bits 31:0.

24.4.17 MTIMECMP4H Register

Register memory slot: 21, offset: 84 bytes, size: 4 bytes

Hart 4 timer compare register, upper 32 bits.

31	30	29	28	27	26	25	24
MTIMECMP4H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
MTIMECMP4H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
MTIMECMP4H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MTIMECMP4H							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
MTIMECMP4H	Bits 31-0	mtimecmp4 bits 63:32.

25 MUTEX Peripheral

The hardware mutex bank provides sixteen word-mapped advisory locks with **single-instruction** cross-hart mutual exclusion: no LR/SC retry loop and no reservation state. Atomicity comes for free from the multi-core arbiter, which serializes whole shared-window transactions. The main features are listed below:

- 16 independent mutexes, one 32-bit word each
- Claim with one load: reading a free mutex returns 0 *and claims it* in the same transaction
- Reading a held mutex returns the owner's marker (hartid + 1) without disturbing it
- Release with one store of 0
- Force-release supported: any hart may write 0, so a supervisor can recover a dead hart's mutex
- Ownership cannot be forged: nonzero writes are ignored
- All mutexes reset to free

25.1 Usage

To claim mutex i , read it once and test the result:

```
lw      t0, (MUTEXi)      # t0 == 0 -> you now hold the mutex
bnez    t0, retry         # t0 == h+1 -> held by hart h, back off and retry
```

To release a mutex you hold:

```
sw      x0, (MUTEXi)      # owner := free
```

Re-reading a mutex you already hold returns your own marker (mhartid + 1) and does not disturb the lock, so an accidental double claim-read is harmless.

When a claim fails, spin with a hart-scaled backoff and a bounded retry count (see the Multi-Core Architecture chapter): five harts re-reading a contested mutex in lockstep waste arbiter bandwidth and can starve useful work.

25.2 Rules

- These are **advisory** locks: the hardware does not block a hart that ignores the protocol and accesses the protected resource directly. Correct software claims the mutex before touching the resource it guards.
- **Never** access a mutex address with LR/SC or AMO instructions; use only plain lw/sw. A suppressed store-conditional arrives at the bank indistinguishable from a claim-read and will corrupt the ownership protocol.
- Release (write 0) is deliberately *not* qualified by owner so that a supervisory hart can force-release the mutex of a hung or dead hart. Correct software releases only mutexes it holds.

The bank occupies shared-window page 2 at base address 0x6000; the sixteen mutexes are word-mapped, with mutex i at $0x6000 + 4i$. The bank decodes only the low bits of the address, so its words alias throughout the page. Always use the canonical addresses.

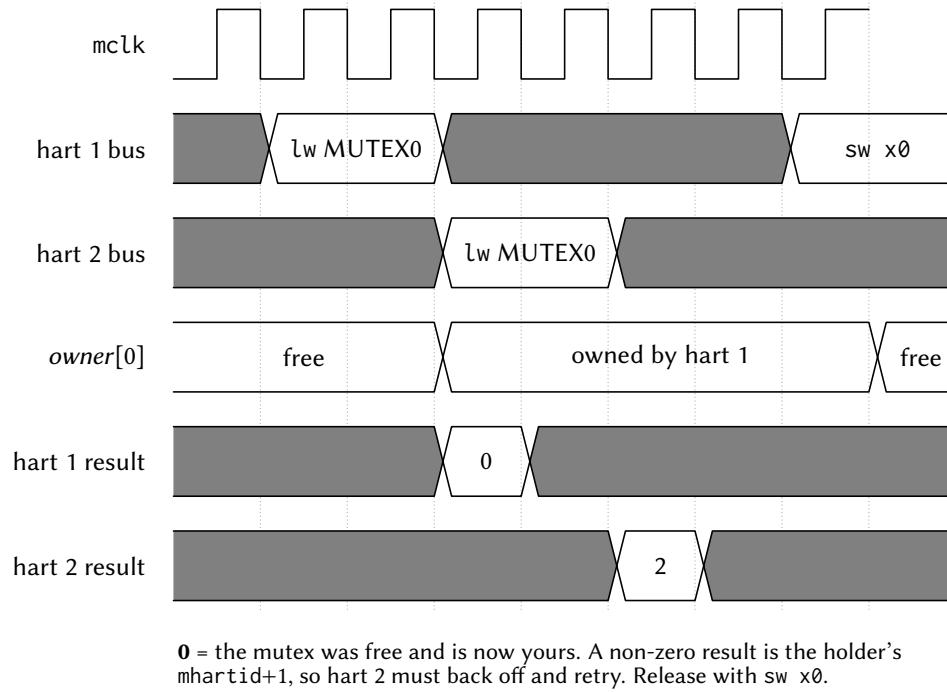


Figure 103: Two harts racing for the same mutex. The arbiter serializes the two claim reads, so the return-old-and-claim happens atomically with no retry loop: the hart the arbiter serves first reads 0 and thereby owns the mutex, and the other reads the winner's marker (mhartid + 1) and must back off.

25.3 Register Description

25.3.1 MUTEX0 Register

Register memory slot: 0, offset: 0 bytes, size: 4 bytes

Hardware mutex 0. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.

31	30	29	28	27	26	25	24
OWN0							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
OWN0							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
OWN0							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
OWN0							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
OWN0	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h. 00000000000000000000000000000000 MTXOWN0_FREE: Mutex free (a read returning this value claims the mutex) 00000000000000000000000000000001 MTXOWN0_H0: Held by hart 0 00000000000000000000000000000010 MTXOWN0_H1: Held by hart 1 00000000000000000000000000000011 MTXOWN0_H2: Held by hart 2 00000000000000000000000000000100 MTXOWN0_H3: Held by hart 3 00000000000000000000000000000101 MTXOWN0_H4: Held by hart 4

25.3.2 MUTEX1 Register

Register memory slot: 1, offset: 4 bytes, size: 4 bytes

Hardware mutex 1. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.

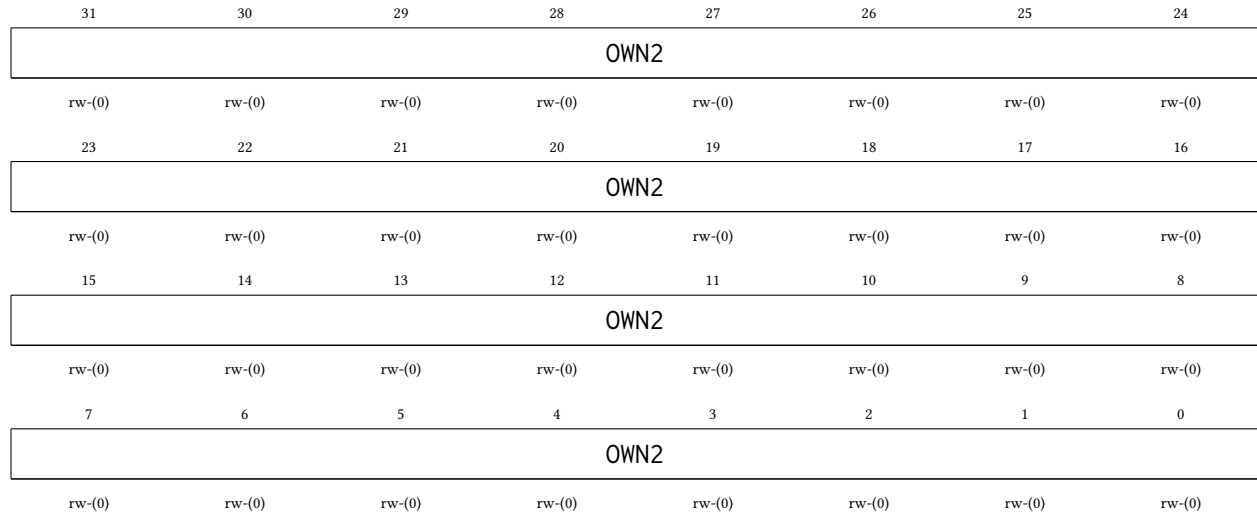
31	30	29	28	27	26	25	24
OWN1							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
OWN1							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
OWN1							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
OWN1							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
OWN1	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h. 00000000000000000000000000000000 MTXOWN1_FREE: Mutex free (a read returning this value claims the mutex) 00000000000000000000000000000001 MTXOWN1_H0: Held by hart 0 00000000000000000000000000000010 MTXOWN1_H1: Held by hart 1 00000000000000000000000000000011 MTXOWN1_H2: Held by hart 2 00000000000000000000000000000100 MTXOWN1_H3: Held by hart 3 00000000000000000000000000000101 MTXOWN1_H4: Held by hart 4

25.3.3 MUTEX2 Register

Register memory slot: 2, offset: 8 bytes, size: 4 bytes

Hardware mutex 2. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.



Bitfield	Range	Description
OWN2	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h.
		00000000000000000000000000000000 MTXOWN2_FREE: Mutex free (a read returning this value claims the mutex)
		00000000000000000000000000000001 MTXOWN2_H0: Held by hart 0
		00000000000000000000000000000010 MTXOWN2_H1: Held by hart 1
		00000000000000000000000000000011 MTXOWN2_H2: Held by hart 2
		00000000000000000000000000000100 MTXOWN2_H3: Held by hart 3
		00000000000000000000000000000101 MTXOWN2_H4: Held by hart 4

25.3.4 MUTEX3 Register

Register memory slot: 3, offset: 12 bytes, size: 4 bytes

Hardware mutex 3. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.

31	30	29	28	27	26	25	24
OWN3							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
OWN3							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
OWN3							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
OWN3							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
OWN3	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h. 00000000000000000000000000000000 MTXOWN3_FREE: Mutex free (a read returning this value claims the mutex) 00000000000000000000000000000001 MTXOWN3_H0: Held by hart 0 00000000000000000000000000000010 MTXOWN3_H1: Held by hart 1 00000000000000000000000000000011 MTXOWN3_H2: Held by hart 2 00000000000000000000000000000100 MTXOWN3_H3: Held by hart 3 00000000000000000000000000000101 MTXOWN3_H4: Held by hart 4

25.3.5 MUTEX4 Register

Register memory slot: 4, offset: 16 bytes, size: 4 bytes

Hardware mutex 4. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.

31	30	29	28	27	26	25	24
OWN4							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
OWN4							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
OWN4							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
OWN4							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
OWN4	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h. 00000000000000000000000000000000 MTXOWN4_FREE: Mutex free (a read returning this value claims the mutex) 00000000000000000000000000000001 MTXOWN4_H0: Held by hart 0 00000000000000000000000000000010 MTXOWN4_H1: Held by hart 1 00000000000000000000000000000011 MTXOWN4_H2: Held by hart 2 00000000000000000000000000000100 MTXOWN4_H3: Held by hart 3 00000000000000000000000000000101 MTXOWN4_H4: Held by hart 4

25.3.6 MUTEX5 Register

Register memory slot: 5, offset: 20 bytes, size: 4 bytes

Hardware mutex 5. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.

31	30	29	28	27	26	25	24
OWN5							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
OWN5							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
OWN5							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
OWN5							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
OWN5	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h. 00000000000000000000000000000000 MTXOWN5_FREE: Mutex free (a read returning this value claims the mutex) 00000000000000000000000000000001 MTXOWN5_H0: Held by hart 0 00000000000000000000000000000010 MTXOWN5_H1: Held by hart 1 00000000000000000000000000000011 MTXOWN5_H2: Held by hart 2 00000000000000000000000000000100 MTXOWN5_H3: Held by hart 3 00000000000000000000000000000101 MTXOWN5_H4: Held by hart 4

25.3.7 MUTEX6 Register

Register memory slot: 6, offset: 24 bytes, size: 4 bytes

Hardware mutex 6. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.

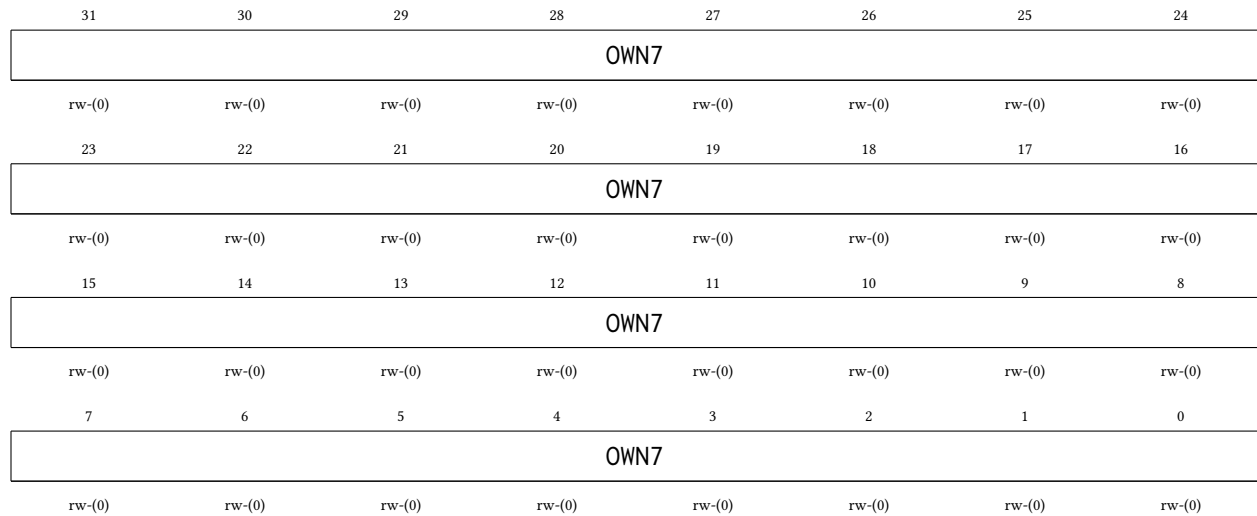
31	30	29	28	27	26	25	24
OWN6							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
OWN6							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
OWN6							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
OWN6							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
OWN6	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h. 00000000000000000000000000000000 MTXOWN6_FREE: Mutex free (a read returning this value claims the mutex) 00000000000000000000000000000001 MTXOWN6_H0: Held by hart 0 00000000000000000000000000000010 MTXOWN6_H1: Held by hart 1 00000000000000000000000000000011 MTXOWN6_H2: Held by hart 2 00000000000000000000000000000100 MTXOWN6_H3: Held by hart 3 00000000000000000000000000000101 MTXOWN6_H4: Held by hart 4

25.3.8 MUTEX7 Register

Register memory slot: 7, offset: 28 bytes, size: 4 bytes

Hardware mutex 7. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.



Bitfield	Range	Description
OWN7	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h.
	00000000000000000000000000000000	MTXOWN7_FREE: Mutex free (a read returning this value claims the mutex)
	00000000000000000000000000000001	MTXOWN7_H0: Held by hart 0
	00000000000000000000000000000010	MTXOWN7_H1: Held by hart 1
	00000000000000000000000000000011	MTXOWN7_H2: Held by hart 2
	00000000000000000000000000000100	MTXOWN7_H3: Held by hart 3
	00000000000000000000000000000101	MTXOWN7_H4: Held by hart 4

25.3.9 MUTEX8 Register

Register memory slot: 8, offset: 32 bytes, size: 4 bytes

Hardware mutex 8. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.

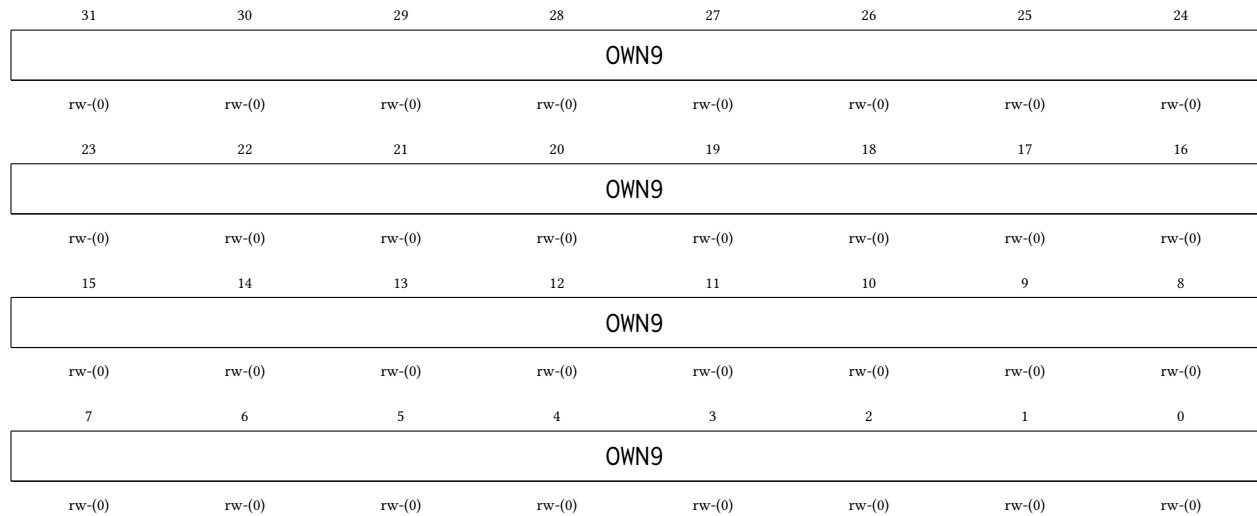
31	30	29	28	27	26	25	24
OWN8							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
OWN8							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
OWN8							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
OWN8							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
OWN8	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h. 00000000000000000000000000000000 MTXOWN8_FREE: Mutex free (a read returning this value claims the mutex) 00000000000000000000000000000001 MTXOWN8_H0: Held by hart 0 00000000000000000000000000000010 MTXOWN8_H1: Held by hart 1 00000000000000000000000000000011 MTXOWN8_H2: Held by hart 2 00000000000000000000000000000100 MTXOWN8_H3: Held by hart 3 00000000000000000000000000000101 MTXOWN8_H4: Held by hart 4

25.3.10 MUTEX9 Register

Register memory slot: 9, offset: 36 bytes, size: 4 bytes

Hardware mutex 9. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.



Bitfield	Range	Description
OWN9	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h. 00000000000000000000000000000000 MTXOWN9_FREE: Mutex free (a read returning this value claims the mutex) 00000000000000000000000000000001 MTXOWN9_H0: Held by hart 0 00000000000000000000000000000010 MTXOWN9_H1: Held by hart 1 00000000000000000000000000000011 MTXOWN9_H2: Held by hart 2 00000000000000000000000000000100 MTXOWN9_H3: Held by hart 3 00000000000000000000000000000101 MTXOWN9_H4: Held by hart 4

25.3.11 MUTEX10 Register

Register memory slot: 10, offset: 40 bytes, size: 4 bytes

Hardware mutex 10. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.

31	30	29	28	27	26	25	24
OWN10							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
OWN10							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
OWN10							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
OWN10							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
OWN10	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h. 00000000000000000000000000000000 MTXOWN10_FREE: Mutex free (a read returning this value claims the mutex) 00000000000000000000000000000001 MTXOWN10_H0: Held by hart 0 00000000000000000000000000000010 MTXOWN10_H1: Held by hart 1 00000000000000000000000000000011 MTXOWN10_H2: Held by hart 2 00000000000000000000000000000100 MTXOWN10_H3: Held by hart 3 00000000000000000000000000000101 MTXOWN10_H4: Held by hart 4

25.3.12 MUTEX11 Register

Register memory slot: 11, offset: 44 bytes, size: 4 bytes

Hardware mutex 11. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.

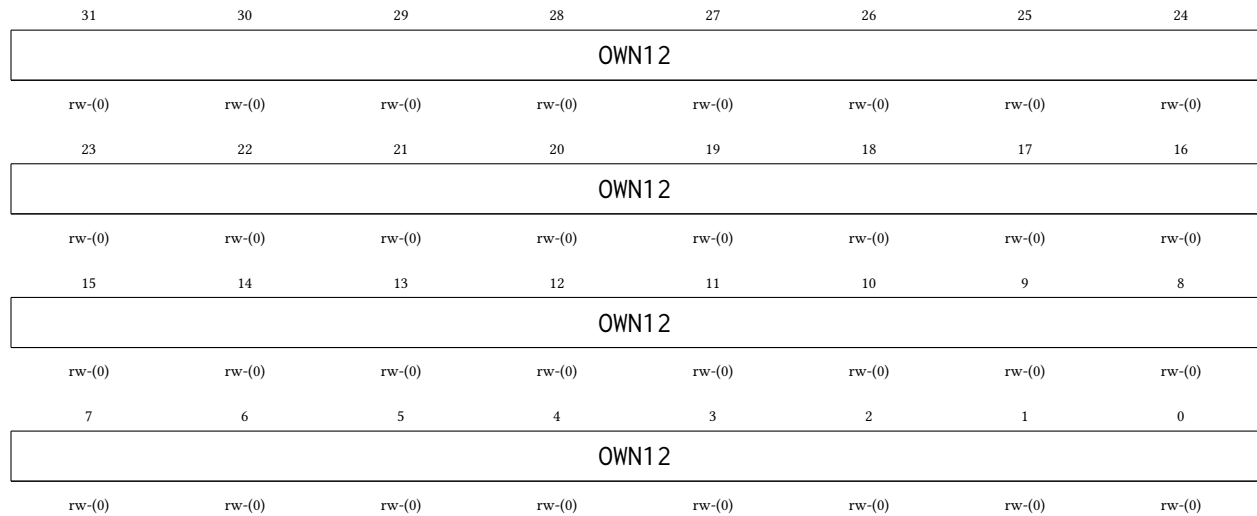
31	30	29	28	27	26	25	24
OWN11							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
OWN11							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
OWN11							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
OWN11							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
OWN11	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h. 00000000000000000000000000000000 MTXOWN11_FREE: Mutex free (a read returning this value claims the mutex) 00000000000000000000000000000001 MTXOWN11_H0: Held by hart 0 00000000000000000000000000000010 MTXOWN11_H1: Held by hart 1 00000000000000000000000000000011 MTXOWN11_H2: Held by hart 2 00000000000000000000000000000100 MTXOWN11_H3: Held by hart 3 00000000000000000000000000000101 MTXOWN11_H4: Held by hart 4

25.3.13 MUTEX12 Register

Register memory slot: 12, offset: 48 bytes, size: 4 bytes

Hardware mutex 12. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.



Bitfield	Range	Description
OWN12	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h.
		00000000000000000000000000000000 MTXOWN12_FREE: Mutex free (a read returning this value claims the mutex)
		00000000000000000000000000000001 MTXOWN12_H0: Held by hart 0
		00000000000000000000000000000010 MTXOWN12_H1: Held by hart 1
		00000000000000000000000000000011 MTXOWN12_H2: Held by hart 2
		00000000000000000000000000000100 MTXOWN12_H3: Held by hart 3
		00000000000000000000000000000101 MTXOWN12_H4: Held by hart 4

25.3.14 MUTEX13 Register

Register memory slot: 13, *offset:* 52 bytes, *size:* 4 bytes

Hardware mutex 13. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.

31	30	29	28	27	26	25	24
OWN13							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
OWN13							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
OWN13							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
OWN13							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
OWN13	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h. 00000000000000000000000000000000 MTXOWN13_FREE: Mutex free (a read returning this value claims the mutex) 00000000000000000000000000000001 MTXOWN13_H0: Held by hart 0 00000000000000000000000000000010 MTXOWN13_H1: Held by hart 1 00000000000000000000000000000011 MTXOWN13_H2: Held by hart 2 00000000000000000000000000000100 MTXOWN13_H3: Held by hart 3 00000000000000000000000000000101 MTXOWN13_H4: Held by hart 4

25.3.15 MUTEX14 Register

Register memory slot: 14, offset: 56 bytes, size: 4 bytes

Hardware mutex 14. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.

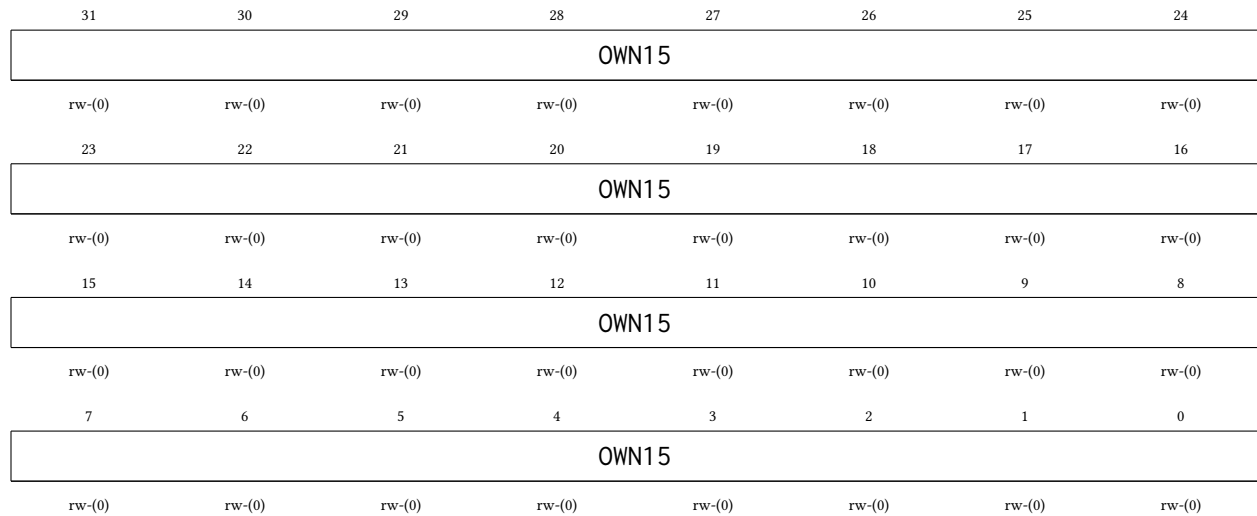
31	30	29	28	27	26	25	24
OWN14							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
OWN14							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
OWN14							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
OWN14							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
OWN14	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h. 00000000000000000000000000000000 MTXOWN14_FREE: Mutex free (a read returning this value claims the mutex) 00000000000000000000000000000001 MTXOWN14_H0: Held by hart 0 00000000000000000000000000000010 MTXOWN14_H1: Held by hart 1 00000000000000000000000000000011 MTXOWN14_H2: Held by hart 2 00000000000000000000000000000100 MTXOWN14_H3: Held by hart 3 00000000000000000000000000000101 MTXOWN14_H4: Held by hart 4

25.3.16 MUTEX15 Register

Register memory slot: 15, offset: 60 bytes, size: 4 bytes

Hardware mutex 15. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.



Bitfield	Range	Description
OWN15	Bits 31-0	Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h.
		00000000000000000000000000000000 MTXOWN15_FREE: Mutex free (a read returning this value claims the mutex)
		00000000000000000000000000000001 MTXOWN15_H0: Held by hart 0
		00000000000000000000000000000010 MTXOWN15_H1: Held by hart 1
		00000000000000000000000000000011 MTXOWN15_H2: Held by hart 2
		00000000000000000000000000000100 MTXOWN15_H3: Held by hart 3
		00000000000000000000000000000101 MTXOWN15_H4: Held by hart 4

26 NFCx Peripheral

The NFC controller (NFCx) is an ISO/IEC 14443 Type A (14443A) tag / card-emulation digital protocol engine. It emulates a passive contactless tag to an external reader: it recovers the reader-to-tag frames, runs the tag transaction state machine, and load-modulates the tag's response. The 13.56 MHz RF analog front-end is *off-die*: this block is strictly digital and presents only a small digital interface to an external AFE (a demodulated receive envelope, a field-detect level, an AFE-derived carrier clock, and the load-modulation drive / listen-enable outputs). The register read path is registered with no read side effects. The main features of the NFC controller are listed below:

- ISO 14443A tag emulation: REQA / WUPA to ATQA, bit-frame anticollision by 4-byte UID to SAK, then a Type-2 READ that auto-answers from a firmware-filled payload window (NFCAUTOREAD)
- Miller decode of the reader-to-tag frames and Manchester / subcarrier (fc/16) load-modulation of the tag response
- Byte framing with odd parity and the ISO 14443A CRC_A (reflected 0x8408, init 0x6363)
- A provisioned 4-byte UID (NFCxUID), programmable ATQA and SAK (NFCxCFG), and hardware BCC derivation
- Two 64-byte indexed windows behind one index / data pair (NFCxIDX, NFCxDATA): the payload window the reader collects (firmware-filled) and the raw received-frame buffer (firmware-read)
- Register-programmable protocol timing (NFCxTIM): the bit-period ETU, the subcarrier divisor, and the Frame Delay Time; a bench compresses all three for fast simulation
- Registered read path: register reads have *no* side effects
- Four interrupt sources: RF field detected, reader frame received, tag response transmit done, and RX CRC / parity error

26.1 Digital AFE Boundary

NFCx does not touch the 13.56 MHz carrier. It connects to an external analog front-end through a digital interface: a demodulated receive envelope (1 = field, 0 = a reader pause), an asynchronous field-present level, and the AFE carrier-derived clock `rf_clk` that clocks the entire protocol core, plus a load-modulation drive output, a transmit-active window, and an AFE listen-power enable. Because these signals cross into the memory-bus domain, the block spans three clock domains: the gated bus clock (`ClkMem`), a free-running `SMCLK` reference that hosts the clock-domain-crossing synchronizers and the write-1-to-clear retirement, and `rf_clk`. Quasi-static configuration (UID, ATQA, SAK, timing) is latched transaction-locally so a multi-bit value is never sampled mid-update; the received-frame bytes cross by a flag-gated data handshake (the RX buffer is written before `NFCRXFRAMEF` is raised). As with the SPI, UART, and I2C peripherals, software must set `SYSCLKCR` to 0 (`SMCLK` from `HFXT`) before using the controller.

26.2 Provisioning and the Payload Window

Before enabling the tag, program its identity: write the 4-byte UID to `NFCxUID` and the answer-to-request / select-acknowledge bytes to `NFCxCFG` (`NFCATQA`, `NFCSAK`). Fill the record the reader will collect through the indexed payload window: select it with `NFCIDXSEL = 0` in `NFCxIDX`, set `NFCIDX` to the starting byte,

optionally set NFCIDXAINC for streaming, and write successive bytes to NFCxDATA. Set NFCEN to run the protocol core and NFCLISTEN to arm the tag to respond, along with any interrupt enables in NFCxCR.

With NFCAUTOREAD set (its reset value), hardware answers a Type-2 READ directly from the payload window with no firmware intervention. When a reader frame arrives for firmware (an unrecognized command, or NFCAUTOREAD cleared), NFCRXFRAMEF is set and the frame is surfaced through NFCxRXST (command byte, length, and the CRC / parity result) and the RX frame buffer (NFCIDXSEL = 1 in NFCxIDX, read through NFCxDATA). The status flags NFCFIELD, NFCRXFRAMEF, NFCTXDONEF, NFCCRCERRF, and NFCPARERRF are write-1-to-clear: write a 1 to a bit to clear it. Reads of the status, receive-inspection, and window-data registers have no side effects.

The protocol timing register NFCxTIM carries the bit-period (NFCETU), the subcarrier half-period (NFCSUBCDIV), and the Frame Delay Time (NFCFDT), all counted in `rf_clk` ticks. The reset values follow the real 13.56 MHz grid; a simulation bench writes them small to compress the protocol time.

26.3 Shared Access on Castalia

In configurations that include it, NFC0 lives in the shared peripheral window behind the multi-core arbiter (see Section 4) in the mutex-bank page, sub-slot 2 at 0x6200, so any hart can use it. Its four interrupt sources occupy vectors 94 through 97 (field detected, reader frame received, transmit done, and CRC / parity error, in that order), delivered through the interrupt router like every other shared-peripheral source. These sources sit above the external-interrupt (meip) delivery vector, which stays fixed at vector 85, and above the I3C sources at vectors 86 through 93.

26.4 Register Description

26.4.1 NFCCR Register

Register memory slot: 0, offset: 0 bytes, size: 4 bytes

NFC control register. Enables the protocol core, arms the tag to respond, selects the auto-answer path, and holds the interrupt enables and the carrier-division note. Resets to 0 except NFCAUTOREAD, which resets to 1. Program the identity and timing registers before setting NFCEN.

31	30	29	28	27	26	25	24
Unused							
23	22	21	20	19	18	17	16
Unused				RFDIV			
				rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
Unused			AUTOREAD	CRCIE	TXIE	RXFIE	FIELDIE
			rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
Unused					HALTCLR	LISTEN	EN
					rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
Unused	Bits 31-20	
RFDIV	Bits 19-16	Carrier-division note. Documents the assumed division from the 13.56 MHz carrier to rf_clk (informational; the off-die AFE supplies rf_clk).
Unused	Bits 15-13	
AUTOREAD	Bit 12	Auto-answer READ path (resets to 1). When set, hardware answers a Type-2 READ command directly from the payload window without firmware intervention. When clear, standard frames are surfaced to firmware through NFCxRXST and the RX buffer for a firmware-composed response. 0 Firmware-handled reads 1 Hardware auto-answer
CRCIE	Bit 11	CRC / parity-error interrupt enable. When set, an RX CRC or parity error (NFCCRCERRF or NFPCPARERRF) drives interrupt vector 97. 0 Interrupt disabled 1 Interrupt enabled
TXIE	Bit 10	Transmit-done interrupt enable. When set, the NFCTXDONEF flag drives interrupt vector 96. 0 Interrupt disabled 1 Interrupt enabled
RXFIE	Bit 9	Frame-received interrupt enable. When set, the NFCRXFRAMEF flag drives interrupt vector 95. 0 Interrupt disabled

		1 Interrupt enabled
FIELDIE	Bit 8	Field-detect interrupt enable. When set, the NFCFIELDIF flag drives interrupt vector 94. 0 Interrupt disabled 1 Interrupt enabled
<i>Unused</i>	Bits 7-3	
HALTCLR	Bit 2	Halt-clear pulse. Writing 1 forces the transaction FSM from the HALT state back to IDLE (a self-clearing, write-1-style event pulse); it reads 0. 0 No action 1 Force HALT to IDLE
LISTEN	Bit 1	Listen arm. When set, the tag responds to reader commands once a field is present; when clear, the tag stays deaf even in a field. 0 Deaf (no response) 1 Armed to respond
EN	Bit 0	NFC enable. When 0 the rf_clk protocol core is held in reset; it does NOT wipe the UID, CFG, payload window, or status flags (disable-preserves-data rule). Set to 1 to run the tag engine. 0 Disabled (protocol core in reset) 1 Enabled

26.4.2 NFCSR Register

Register memory slot: 1, offset: 4 bytes, size: 4 bytes

NFC status register. NFCBUSY, NFCFIELDLIVE, NFCHALTED, and NFCSTATE are read-only live status; the five event flags (bits 1-5) are write-1-to-clear (write a 1 to a bit to clear it; writing 0 leaves it unchanged). The volatile bits are captured by the registered pre-latch read.

31	30	29	28	27	26	25	24
<i>Unused</i>							
23	22	21	20	19	18	17	16
<i>Unused</i>							
15	14	13	12	11	10	9	8
<i>Unused</i>				STATE			
				r-(0)	r-(0)	r-(0)	r-(0)
7	6	5	4	3	2	1	0
HALTED	FIELDLIVE	PARERRF	CRCERRF	TXDONEF	RXFRAMEF	FIELDIF	BUSY
r-(0)	r-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	rw1-(0)	r-(0)

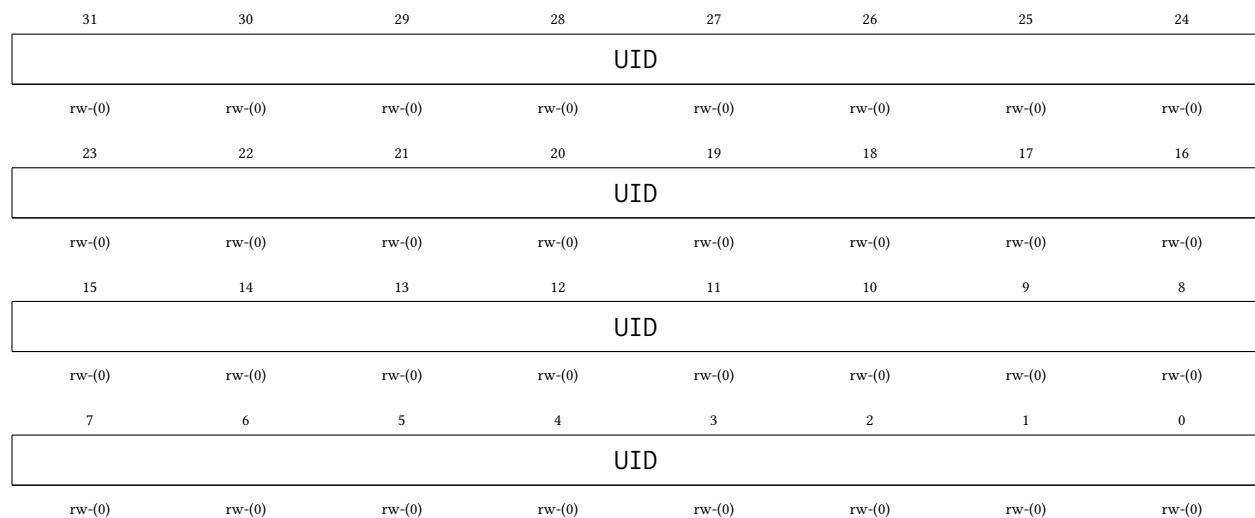
Bitfield	Range	Description
<i>Unused</i>	Bits 31-12	

STATE	Bits 11-8	Transaction FSM state: 0 = POWER_OFF, 1 = IDLE, 2 = READY, 3 = ACTIVE, 4 = HALT.
HALTED	Bit 7	Halted. Reads 1 while the transaction FSM is in the HALT state (the tag has been HLTA / halted by the reader). 0 Not halted 1 FSM halted
FIELDLIVE	Bit 6	Field-live level. The synchronized RF field-present level (1 while a reader field is on the tag). 0 No field 1 Field present
PARERRF	Bit 5	RX parity-error flag. Set when a received frame fails an odd-parity check; drives vector 97 (folded with NFCCRCERRF) when NFCCRCIE is set. Write 1 to clear. 0 No event 1 RX parity error
CRCERRF	Bit 4	RX CRC-error flag. Set when a received frame fails the CRC_A check; drives vector 97 when NFCCRCIE is set. Write 1 to clear. 0 No event 1 RX CRC error
TXDONEF	Bit 3	Transmit-done flag. Set when the tag response end-of-frame has been sent; drives vector 96 when NFCTXIE is set. Write 1 to clear. 0 No event 1 Tag response sent
RXFRAMEF	Bit 2	Reader-frame-received flag. Set when a reader frame has landed for firmware (its CRC / parity result is summarized in NFCxRXST); drives vector 95 when NFCRXFIE is set. Write 1 to clear. 0 No event 1 Reader frame received
FIELDF	Bit 1	Field-detect flag. Set on a synchronized rising edge of the RF field-present input; drives vector 94 when NFCFIELDIE is set. Write 1 to clear. 0 No event 1 RF field detected
BUSY	Bit 0	Busy. Reads 1 while a reader frame is being received or a tag response is in flight. 0 Idle 1 RX or TX in progress

26.4.3 NFCUID Register

Register memory slot: 2, offset: 8 bytes, size: 4 bytes

Provisioned single-size 4-byte tag UID, used by bit-frame anticollision. Byte order on air is UID byte 0 first: UID[7:0] is the first byte, UID[31:24] the last. Hardware derives the BCC check byte. Latched transaction-locally at the start of each transaction.

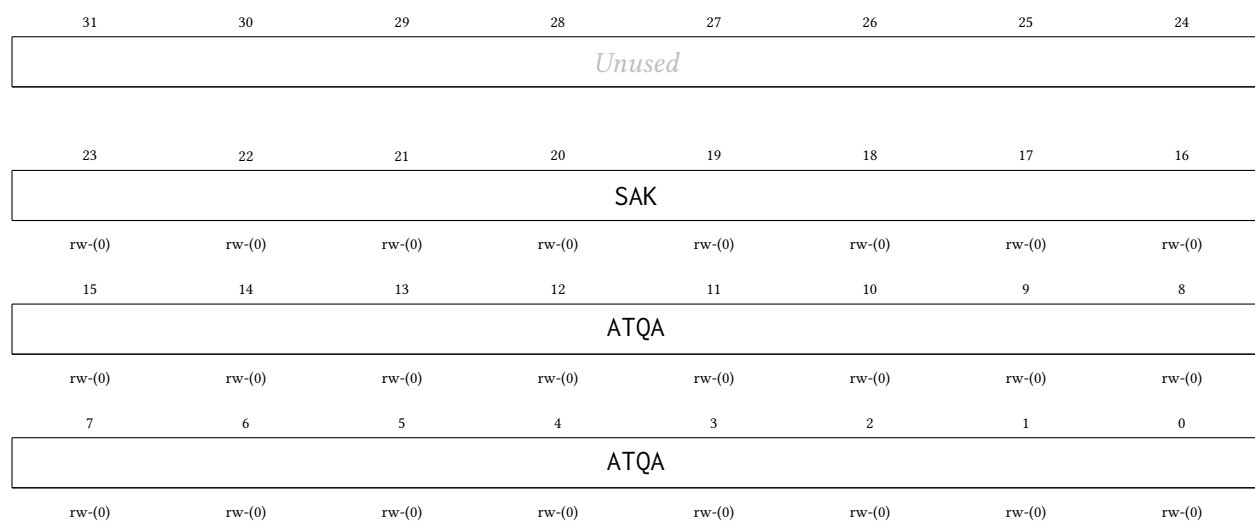


Bitfield	Range	Description
UID	Bits 31-0	4-byte UID (UID0 in bits 7:0, first on air).

26.4.4 NFCCFG Register

Register memory slot: 3, offset: 12 bytes, size: 4 bytes

Tag identity response configuration: the ATQA answer-to-request word and the SAK select-acknowledge byte. Latched transaction-locally.

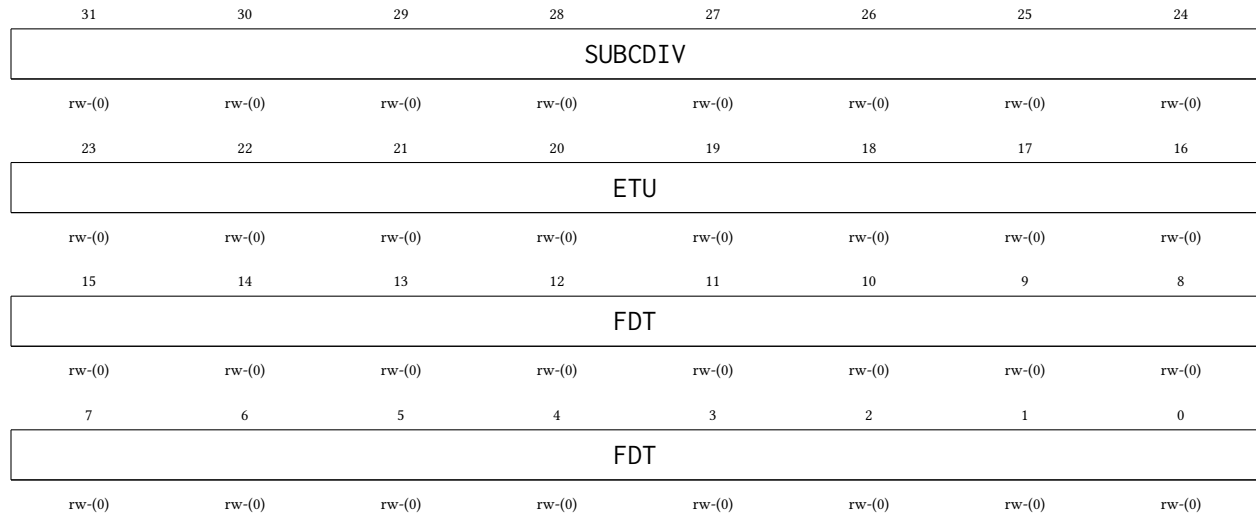


Bitfield	Range	Description
<i>Unused</i>	Bits 31-24	
SAK	Bits 23-16	Select Acknowledge byte (resets to 0x00) returned after the final anticollision level.
ATQA	Bits 15-0	Answer To Request, Type A (resets to 0x0044). Bits 7:0 are the first byte on air.

26.4.5 NFCTIM Register

Register memory slot: 4, offset: 16 bytes, size: 4 bytes

Protocol timing divisors, all in `rf_clk` ticks, latched transaction-locally so a mid-count reload never glitches. The reset values are the real 13.56 MHz grid; a bench compresses all three for fast simulation.

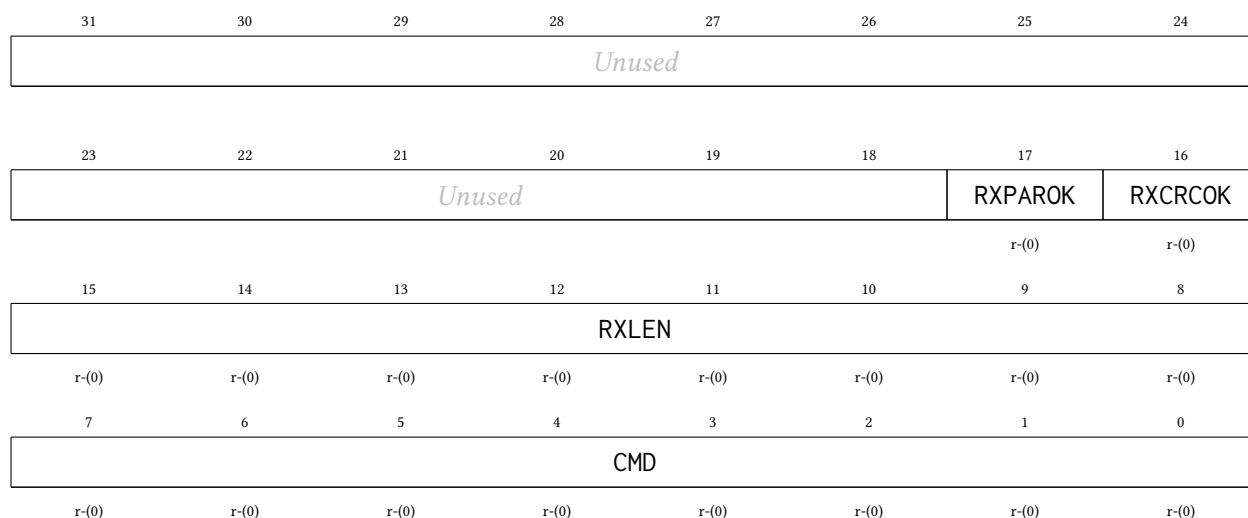


Bitfield	Range	Description
SUBCDIV	Bits 31-24	Subcarrier half-period, in <code>rf_clk</code> ticks (resets to 8, giving the <code>fc/16</code> load-modulation subcarrier).
ETU	Bits 23-16	Elementary Time Unit (bit period), in <code>rf_clk</code> ticks (resets to 128).
FDT	Bits 15-0	Frame Delay Time, in <code>rf_clk</code> ticks (resets to approximately 1236, the ISO 14443A tag response grid). The composed response is released when the FDT down-counter reaches zero.

26.4.6 NFCRXST Register

Register memory slot: 5, offset: 20 bytes, size: 4 bytes

Received-frame inspection, for firmware-handled frames (`NFCAUTOREAD` = 0 or an unrecognized command). Read side-effect-free; the frame byte payload is read separately through the indexed RX buffer (`NFCxIDX` / `NFCxDATA` with `NFCIDXSEL` = 1). Pre-latched.

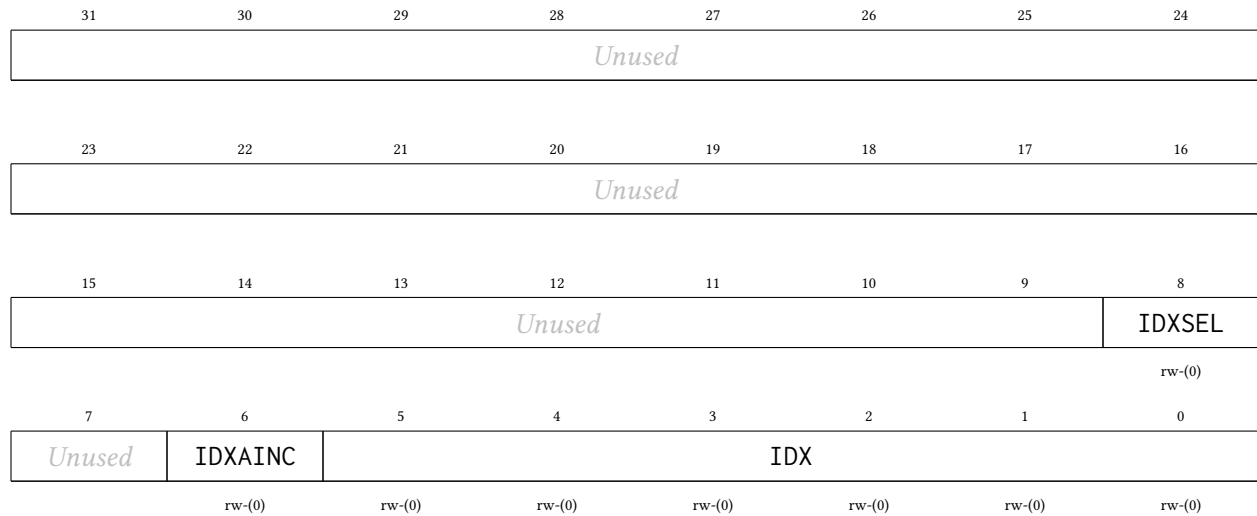


Bitfield	Range	Description
<i>Unused</i>	Bits 31-18	
RXPAROK	Bit 17	Parity OK for the last received frame. 0 Parity bad 1 Parity good
RXCRCOK	Bit 16	CRC OK for the last received frame. 0 CRC bad 1 CRC good
RXLEN	Bits 15-8	Received length in bytes, including the CRC.
CMD	Bits 7-0	Command byte: the first byte of the last received standard frame.

26.4.7 NFCIDX Register

Register memory slot: 6, offset: 24 bytes, size: 4 bytes

Byte-index pointer into one of the two 64-byte windows accessed through NFCxDATA. The index persists across accesses; NFCIDXSEL selects which window. Mirrors the indexed-window idiom used by I3C's Device Address Table.

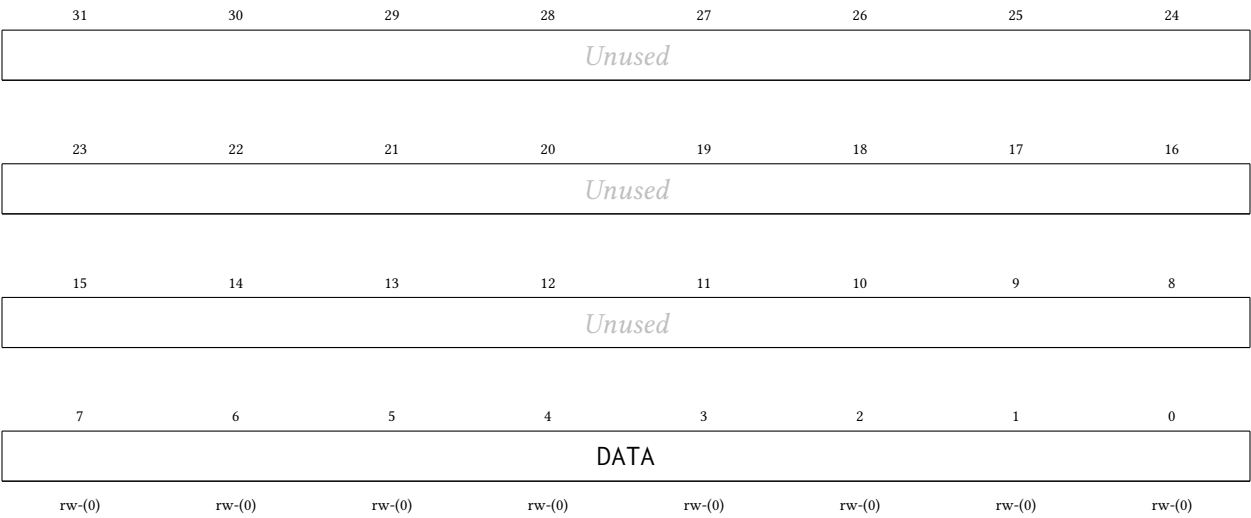


Bitfield	Range	Description
<i>Unused</i>	Bits 31-9	
IDXSEL	Bit 8	Window select for NFCxDATA. 0 1 Payload TX window (64 B, firmware-filled, HW-read) RX frame buffer (64 B, HW-filled, firmware-read)
<i>Unused</i>	Bit 7	
IDXAINC	Bit 6	Auto-increment. When set, NFCIDX increments after each NFCxDATA access (streaming). 0 1 Index held Auto-increment after each access
IDX	Bits 5-0	Byte index (0 to 63) into the selected window.

26.4.8 NFCDATA Register

Register memory slot: 7, offset: 28 bytes, size: 4 bytes

The byte at NFCIDX in the window selected by NFCIDXSEL. The payload TX window is read/write (firmware fills the record the reader collects; hardware reads it for the auto-answer READ); the RX frame buffer is read-only (hardware fills it from the reader frame; writes are ignored). Auto-increments per NFCIDXAINC. Pre-latched on read (no side effects).

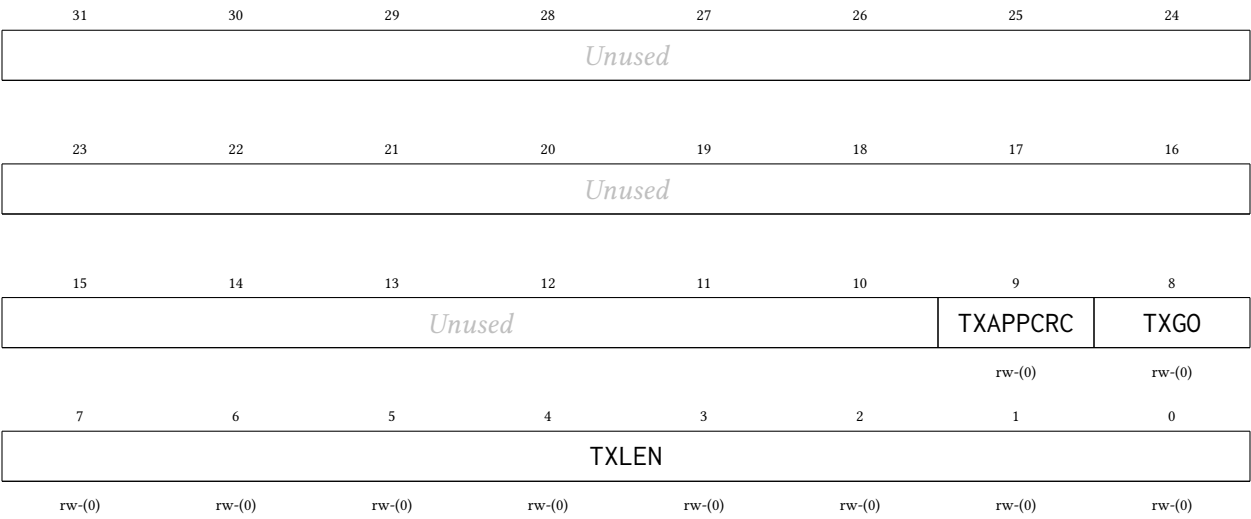


Bitfield	Range	Description
Unused	Bits 31-8	
DATA	Bits 7-0	Window data byte at the current index.

26.4.9 NFCTXCTL Register

Register memory slot: 8, offset: 32 bytes, size: 4 bytes

Firmware-composed response control (used when NFCAUTOREAD = 0 or for a vendor command). INERT in this MVP (auto-answer is the default path); present in the frozen register map so the firmware-WRITE stage bolts on without a map change.



Bitfield	Range	Description
Unused	Bits 31-10	

TXAPPCRC	Bit 9	Append CRC_A to the firmware-composed response. 0 No CRC appended 1 Append CRC_A
TXGO	Bit 8	Launch a firmware-composed response (a lane-0 write is a held-level launch, suppressed unless the FSM is in ACTIVE and awaiting a firmware reply). 0 No launch 1 Send response
TXLEN	Bits 7-0	Number of payload bytes to send from the payload TX window.

26.4.10 NFCDBG Register

Register memory slot: 9, offset: 36 bytes, size: 4 bytes

Debug telemetry / bench cross-checks: frame counters. Read-only; resets to 0.

31	30	29	28	27	26	25	24
TXFRAMECNT							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
23	22	21	20	19	18	17	16
TXFRAMECNT							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
15	14	13	12	11	10	9	8
RXFRAMECNT							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
7	6	5	4	3	2	1	0
RXFRAMECNT							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)

Bitfield	Range	Description
TXFRAMECNT	Bits 31-16	Transmitted-frame count.
RXFRAMECNT	Bits 15-0	Received-frame count.

27 IRQROUTER Peripheral

The interrupt router is the chip’s peripheral interrupt controller. Every deglitched peripheral interrupt level terminates here, and delivery to the harts follows the claim/complete model of a RISC-V platform-level interrupt controller (PLIC): the router raises a single external-interrupt wire (meip, interrupt vector 85) to each hart, and the servicing hart reads the CLAIM register to discover, and atomically claim, the pending source. Routing is per-hart and per-vector: software can steer any peripheral’s interrupt to whichever hart currently owns that peripheral. For example, a hart that has taken over TIMER1 through the shared peripheral page can also receive TIMER1’s interrupts. The main features are listed below:

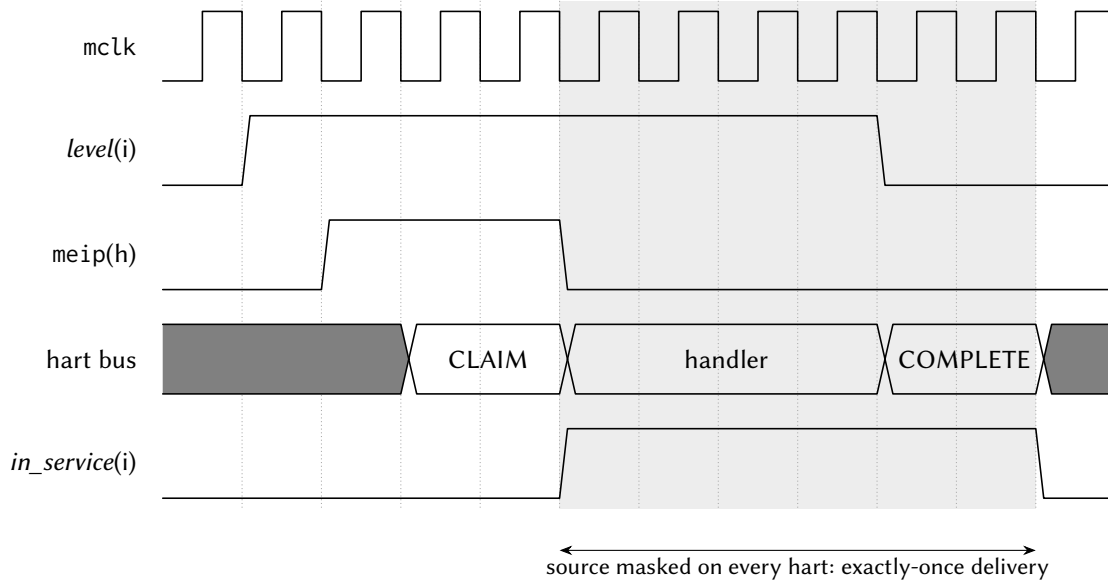
- One row of four 32-bit routing/enable registers per hart, covering interrupt vectors 0–120 (one enable bit per vector)
- Claim/complete delivery: one meip wire per hart (vector 85), CLAIM read = atomic claim of the lowest pending enabled vector, CLAIM write = complete
- Exactly-once delivery: a claimed vector is masked from every hart’s meip until completed, so a vector routed to several harts is serviced by exactly one of them
- Programmable by *any* hart through the shared window; claims are attributed to the reading hart by the shared-bus arbiter
- Fixed priority: the lowest pending vector number wins
- Read-only PENDx (raw levels) and INSVCx (under-service) status words for debugging and recovery
- Resets fully masked (all zeros): the router is inert until software programs it
- The CLINT vectors (83 and 84) are never delivered through meip (each hart receives its own msip/mtip on dedicated hardwired wires), so their row bits are writable but have no effect
- Runs on the free-running MCLK, so a routed interrupt can wake a hart whose gated CPU clock is off

Figure 88 draws this block in its place in the chip: every interrupt source on the die on one side, one routing row per hart and the claim/complete stage in the middle, and the harts on the other — including the two CLINT levels, which are drawn going *round* the router rather than through it, because that is what the last two bullets above mean.

27.1 Delivery Model

Hart h ’s meip wire is asserted while some vector v is pending (deglitched level high), enabled in hart h ’s row (bit v), and not under service. The hart takes vector 85 through its interrupt vector table like any other interrupt; its handler then:

1. Reads CLAIM. The router returns the lowest such v for the reading hart and marks it under service, hiding it from every hart’s meip. A return value of 0xFFFFFFFF means nothing was pending for this hart (for example, another hart claimed the source first); the handler treats the interrupt as spurious and simply returns.
2. Services the source and *clears the interrupt level at the peripheral* (the usual level-interrupt contract).



The handler clears the level at the peripheral *before* the dispatcher completes; if the level is still high at COMPLETE the source simply re-pends.
The handler cell stands for many mclk cycles of software.

Figure 104: Claim/complete delivery of one peripheral interrupt. The router raises meip while the source is pending and enabled; the CLAIM read marks it under service, which hides it from *every* hart until the completing write. That masking window is what makes delivery exactly-once even when several harts enable the same source.

3. Writes v back to CLAIM (complete). The vector becomes deliverable again; if its level is still asserted (a new event), it pends again immediately.

Completion is deliberately not qualified by owner: a supervising hart can complete a vector on behalf of a hart that died mid-service (the INSVCx words show which vectors are stuck). Every hart, hart 0 included, uses this same path: the single-core chip's vectored controller in the SYSTEM peripheral is retired, and row 0 is hart 0's live routing row.

Note that the vector packing of the HxENU registers is contiguous (bit b of HxENU is vector $64 + b$, for $b = 0 \dots 31$, i.e. up to vector 95), unlike the write-packing quirk of the retired single-core IRQENU register.

A typical hand-off of a peripheral to hart 2 looks like:

1. Hart 2 (or any hart) sets the peripheral's vector bits in row 2, that is in whichever of hart 2's four enable words (H2ENL first, one bit per vector) hold the vectors concerned.
2. The previous owner clears the same bits in its own row.
3. Hart 2 starts operating the peripheral through the shared peripheral page; its vector-85 handler dispatches on the claimed vector number.

The routing rows live at the block base (16 bytes per hart); the CLAIM register and the status words sit at fixed offsets +0x800, +0x810 and +0x820, independent of the hart count.

27.2 Register Description

27.2.1 H0ENL Register

Register memory slot: 0, offset: 0 bytes, size: 4 bytes

Hart 0 interrupt routing register, vectors 31:0. Each bit enables delivery of the corresponding interrupt vector to hart 0 via its meip wire (vector 85) and the CLAIM/COMPLETE mechanism.

31	30	29	28	27	26	25	24
H0ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H0ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H0ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H0ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H0ENL	Bits 31-0	Interrupt enable bits for vectors 31:0, routed to hart 0.

27.2.2 H0ENM Register

Register memory slot: 1, offset: 4 bytes, size: 4 bytes

Hart 0 interrupt routing register, vectors 63:32.

31	30	29	28	27	26	25	24
H0ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H0ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H0ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H0ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H0ENM	Bits 31-0	Interrupt enable bits for vectors 63:32, routed to hart 0.

27.2.3 H0ENU Register

Register memory slot: 2, offset: 8 bytes, size: 4 bytes

Hart 0 interrupt routing register, vectors 95:64. Unlike the retired SYSTEM IRQENU register of the single-core chip, the packing here is contiguous in both directions. Bits 19 and 20 correspond to the CLINT vectors 83 and 84, which are delivered on dedicated hardwired wires and never through meip: these two bits are writable but have no effect.

31	30	29	28	27	26	25	24
H0ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H0ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H0ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H0ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H0ENU	Bits 31-0	Interrupt enable bits for vectors 95:64, routed to hart 0.

27.2.4 H0ENX Register

Register memory slot: 3, offset: 12 bytes, size: 4 bytes

Hart 0 interrupt routing register, vectors 120:96 (bits 24:0; upper bits read as 0).

31	30	29	28	27	26	25	24
<i>Unused</i>							H0ENX
rw-(0)							
23	22	21	20	19	18	17	16
H0ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H0ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H0ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
<i>Unused</i>	Bits 31-25	
H0ENX	Bits 24-0	Interrupt enable bits for vectors 120:96, routed to hart 0.

27.2.5 H1ENL Register

Register memory slot: 4, offset: 16 bytes, size: 4 bytes

Hart 1 interrupt routing register, vectors 31:0. Each bit enables delivery of the corresponding interrupt vector to hart 1 via its meip wire (vector 85) and the CLAIM/COMPLETE mechanism.

31	30	29	28	27	26	25	24
H1ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H1ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H1ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H1ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H1ENL	Bits 31-0	Interrupt enable bits for vectors 31:0, routed to hart 1.

27.2.6 H1ENM Register

Register memory slot: 5, offset: 20 bytes, size: 4 bytes

Hart 1 interrupt routing register, vectors 63:32.

31	30	29	28	27	26	25	24
H1ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H1ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H1ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H1ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H1ENM	Bits 31-0	Interrupt enable bits for vectors 63:32, routed to hart 1.

27.2.7 H1ENU Register

Register memory slot: 6, offset: 24 bytes, size: 4 bytes

Hart 1 interrupt routing register, vectors 95:64. Unlike the retired SYSTEM IRQENU register of the single-core chip, the packing here is contiguous in both directions. Bits 19 and 20 correspond to the CLINT vectors 83 and 84, which are delivered on dedicated hardwired wires and never through meip: these two bits are writable but have no effect.

31	30	29	28	27	26	25	24
H1ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H1ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H1ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H1ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H1ENU	Bits 31-0	Interrupt enable bits for vectors 95:64, routed to hart 1.

27.2.8 H1ENX Register

Register memory slot: 7, offset: 28 bytes, size: 4 bytes

Hart 1 interrupt routing register, vectors 120:96 (bits 24:0; upper bits read as 0).

31	30	29	28	27	26	25	24
Unused							H1ENX
rw-(0)							
23	22	21	20	19	18	17	16
H1ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H1ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H1ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
Unused	Bits 31-25	
H1ENX	Bits 24-0	Interrupt enable bits for vectors 120:96, routed to hart 1.

27.2.9 H2ENL Register

Register memory slot: 8, offset: 32 bytes, size: 4 bytes

Hart 2 interrupt routing register, vectors 31:0. Each bit enables delivery of the corresponding interrupt vector to hart 2 via its meip wire (vector 85) and the CLAIM/COMPLETE mechanism.

31	30	29	28	27	26	25	24
H2ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H2ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H2ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H2ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H2ENL	Bits 31-0	Interrupt enable bits for vectors 31:0, routed to hart 2.

27.2.10 H2ENM Register

Register memory slot: 9, offset: 36 bytes, size: 4 bytes

Hart 2 interrupt routing register, vectors 63:32.

31	30	29	28	27	26	25	24
H2ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H2ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H2ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H2ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H2ENM	Bits 31-0	Interrupt enable bits for vectors 63:32, routed to hart 2.

27.2.11 H2ENU Register

Register memory slot: 10, offset: 40 bytes, size: 4 bytes

Hart 2 interrupt routing register, vectors 95:64. Unlike the retired SYSTEM IRQENU register of the single-core chip, the packing here is contiguous in both directions. Bits 19 and 20 correspond to the CLINT vectors 83 and 84, which are delivered on dedicated hardwired wires and never through meip: these two bits are writable but have no effect.

31	30	29	28	27	26	25	24
H2ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H2ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H2ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H2ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H2ENU	Bits 31-0	Interrupt enable bits for vectors 95:64, routed to hart 2.

27.2.12 H2ENX Register

Register memory slot: 11, offset: 44 bytes, size: 4 bytes

Hart 2 interrupt routing register, vectors 120:96 (bits 24:0; upper bits read as 0).

31	30	29	28	27	26	25	24
Unused							H2ENX
							rw-(0)
23	22	21	20	19	18	17	16
H2ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H2ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H2ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
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<i>Unused</i>	Bits 31-25
H2ENX	Bits 24-0 Interrupt enable bits for vectors 120:96, routed to hart 2.

27.2.13 H3ENL Register

Register memory slot: 12, offset: 48 bytes, size: 4 bytes

Hart 3 interrupt routing register, vectors 31:0. Each bit enables delivery of the corresponding interrupt vector to hart 3 via its meip wire (vector 85) and the CLAIM/COMPLETE mechanism.

31	30	29	28	27	26	25	24
H3ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H3ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H3ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H3ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H3ENL	Bits 31-0	Interrupt enable bits for vectors 31:0, routed to hart 3.

27.2.14 H3ENM Register

Register memory slot: 13, offset: 52 bytes, size: 4 bytes

Hart 3 interrupt routing register, vectors 63:32.

31	30	29	28	27	26	25	24
H3ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H3ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H3ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H3ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H3ENM	Bits 31-0	Interrupt enable bits for vectors 63:32, routed to hart 3.

27.2.15 H3ENU Register

Register memory slot: 14, offset: 56 bytes, size: 4 bytes

Hart 3 interrupt routing register, vectors 95:64. Unlike the retired SYSTEM IRQENU register of the single-core chip, the packing here is contiguous in both directions. Bits 19 and 20 correspond to the CLINT vectors 83 and 84, which are delivered on dedicated hardwired wires and never through meip: these two bits are writable but have no effect.

31	30	29	28	27	26	25	24
H3ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H3ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H3ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H3ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H3ENU	Bits 31-0	Interrupt enable bits for vectors 95:64, routed to hart 3.

27.2.16 H3ENX Register

Register memory slot: 15, offset: 60 bytes, size: 4 bytes

Hart 3 interrupt routing register, vectors 120:96 (bits 24:0; upper bits read as 0).

31	30	29	28	27	26	25	24
<i>Unused</i>							H3ENX
rw-(0)							
23	22	21	20	19	18	17	16
H3ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H3ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H3ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
<i>Unused</i>	Bits 31-25	
H3ENX	Bits 24-0	Interrupt enable bits for vectors 120:96, routed to hart 3.

27.2.17 H4ENL Register

Register memory slot: 16, offset: 64 bytes, size: 4 bytes

Hart 4 interrupt routing register, vectors 31:0. Each bit enables delivery of the corresponding interrupt vector to hart 4 via its meip wire (vector 85) and the CLAIM/COMPLETE mechanism.

31	30	29	28	27	26	25	24
H4ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H4ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H4ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H4ENL							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H4ENL	Bits 31-0	Interrupt enable bits for vectors 31:0, routed to hart 4.

27.2.18 H4ENM Register

Register memory slot: 17, offset: 68 bytes, size: 4 bytes

Hart 4 interrupt routing register, vectors 63:32.

31	30	29	28	27	26	25	24
H4ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H4ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H4ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H4ENM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H4ENM	Bits 31-0	Interrupt enable bits for vectors 63:32, routed to hart 4.

27.2.19 H4ENU Register

Register memory slot: 18, offset: 72 bytes, size: 4 bytes

Hart 4 interrupt routing register, vectors 95:64. Unlike the retired SYSTEM IRQENU register of the single-core chip, the packing here is contiguous in both directions. Bits 19 and 20 correspond to the CLINT vectors 83 and 84, which are delivered on dedicated hardwired wires and never through meip: these two bits are writable but have no effect.

31	30	29	28	27	26	25	24
H4ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
H4ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H4ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H4ENU							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
H4ENU	Bits 31-0	Interrupt enable bits for vectors 95:64, routed to hart 4.

27.2.20 H4ENX Register

Register memory slot: 19, offset: 76 bytes, size: 4 bytes

Hart 4 interrupt routing register, vectors 120:96 (bits 24:0; upper bits read as 0).

31	30	29	28	27	26	25	24
Unused							H4ENX
rw-(0)							
23	22	21	20	19	18	17	16
H4ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
H4ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
H4ENX							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
Unused	Bits 31-25	
H4ENX	Bits 24-0	Interrupt enable bits for vectors 120:96, routed to hart 4.

27.2.21 CLAIM Register

Register memory slot: 512, offset: 2048 bytes, size: 4 bytes

Interrupt claim/complete register (M19). READ = claim: atomically returns the lowest pending interrupt vector that is enabled in the READING hart's routing row and not already under service, and marks it under service (masking it from every hart's meip). Returns 0xFFFFFFFF if nothing is pending for the reader; a handler seeing that value treats the interrupt as spurious and simply returns. WRITE = complete: write the claimed vector number to end its service; the vector becomes deliverable again (and pends immediately if its level is still asserted, so clear the level at the peripheral BEFORE completing). Completion is deliberately not qualified by owner, so a supervisor hart can complete on behalf of a hung hart (recovery); written values that are not valid vector numbers, including a stored 0xFFFFFFFF, are ignored.

31	30	29	28	27	26	25	24
CLAIM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
23	22	21	20	19	18	17	16
CLAIM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
15	14	13	12	11	10	9	8
CLAIM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
CLAIM							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Bitfield	Range	Description
CLAIM	Bits 31:0	Read: claimed vector number (0xFFFFFFFF = none pending for this hart). Write: vector number to complete.

27.2.22 PENDL Register

Register memory slot: 516, offset: 2064 bytes, size: 4 bytes

Raw pending interrupt levels, vectors 31:0 (read-only; deglitched peripheral levels before enable/claim masking). Debug and polling aid.

31	30	29	28	27	26	25	24
PENDL							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
23	22	21	20	19	18	17	16
PENDL							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
15	14	13	12	11	10	9	8
PENDL							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
7	6	5	4	3	2	1	0
PENDL							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)

Bitfield	Range	Description
PENDL	Bits 31:0	Deglitched interrupt levels for vectors 31:0.

27.2.23 PENDM Register

Register memory slot: 517, offset: 2068 bytes, size: 4 bytes

Raw pending interrupt levels, vectors 63:32 (read-only).

31	30	29	28	27	26	25	24
PENDM							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
23	22	21	20	19	18	17	16
PENDM							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
15	14	13	12	11	10	9	8
PENDM							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
7	6	5	4	3	2	1	0
PENDM							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)

Bitfield	Range	Description
PENDM	Bits 31-0	Deglitched interrupt levels for vectors 63:32.

27.2.24 PENDU Register

Register memory slot: 518, offset: 2072 bytes, size: 4 bytes

Raw pending interrupt levels, vectors 95:64 (read-only).

31	30	29	28	27	26	25	24
PENDU							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
23	22	21	20	19	18	17	16
PENDU							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
15	14	13	12	11	10	9	8
PENDU							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
7	6	5	4	3	2	1	0
PENDU							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)

Bitfield	Range	Description
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PENDU	Bits 31-0	Deglitched interrupt levels for vectors 95:64.
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27.2.25 PENDX Register

Register memory slot: 519, offset: 2076 bytes, size: 4 bytes

Raw pending interrupt levels, vectors 120:96 (bits 24:0, read-only; upper bits read as 0). Completes the raw-level readback for the fourth enable word (added with the peripheral-library program; before it, sources above 95 were routable and claimable but absent from the debug readback).

31	30	29	28	27	26	25	24
Unused							PENDX
r-(0)							
23	22	21	20	19	18	17	16
PENDX							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
15	14	13	12	11	10	9	8
PENDX							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
7	6	5	4	3	2	1	0
PENDX							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)

Bitfield	Range	Description
Unused	Bits 31-25	
PENDX	Bits 24-0	Deglitched interrupt levels for vectors 120:96.

27.2.26 INSVCL Register

Register memory slot: 520, offset: 2080 bytes, size: 4 bytes

Under-service (claimed, not yet completed) flags, vectors 31:0 (read-only). Debug and recovery visibility: a stuck bit here means a hart claimed the vector and never completed it; any hart can recover by writing the vector number to CLAIM.

31	30	29	28	27	26	25	24
INSVCL							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
23	22	21	20	19	18	17	16
INSVCL							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
15	14	13	12	11	10	9	8
INSVCL							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
7	6	5	4	3	2	1	0
INSVCL							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)

Bitfield	Range	Description
INSVCL	Bits 31-0	Under-service flags for vectors 31:0.

27.2.27 INSVCM Register

Register memory slot: 521, offset: 2084 bytes, size: 4 bytes

Under-service flags, vectors 63:32 (read-only).

31	30	29	28	27	26	25	24
INSVCM							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
23	22	21	20	19	18	17	16
INSVCM							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
15	14	13	12	11	10	9	8
INSVCM							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
7	6	5	4	3	2	1	0
INSVCM							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)

Bitfield	Range	Description
INSVCM	Bits 31-0	Under-service flags for vectors 63:32.

27.2.28 INSVCU Register

Register memory slot: 522, offset: 2088 bytes, size: 4 bytes

Under-service flags, vectors 95:64 (read-only).

31	30	29	28	27	26	25	24
INSVCU							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
23	22	21	20	19	18	17	16
INSVCU							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
15	14	13	12	11	10	9	8
INSVCU							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
7	6	5	4	3	2	1	0
INSVCU							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)

Bitfield	Range	Description
INSVCU	Bits 31-0	Under-service flags for vectors 95:64.

27.2.29 INSVCX Register

Register memory slot: 523, offset: 2092 bytes, size: 4 bytes

Under-service flags, vectors 120:96 (bits 24:0, read-only; upper bits read as 0). Completes the under-service readback for the fourth enable word.

31	30	29	28	27	26	25	24
Unused							INSVCX
r-(0)							
23	22	21	20	19	18	17	16
INSVCX							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
15	14	13	12	11	10	9	8
INSVCX							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)
7	6	5	4	3	2	1	0
INSVCX							
r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)	r-(0)

Bitfield	Range	Description
Unused	Bits 31-25	
INSVCX	Bits 24-0	Under-service flags for vectors 120:96.

28 Register and Peripheral Memory Map

28.1 GPI00 Peripheral

Base address: 0x4000 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
P0IN	0x4000	0	0	1
P0OUT	0x4004	1	4	1
P0OUTS	0x4008	2	8	1
P0OUTC	0x400C	3	12	1
P0OUTT	0x4010	4	16	4
P0DIR	0x4014	5	20	1
P0IF	0x4018	6	24	4
P0IES	0x401C	7	28	1
P0IE	0x4020	8	32	1
P0SEL	0x4024	9	36	1
P0REN	0x4028	10	40	1
P0AFS	0x402C	11	44	4
P0TASK	0x4030	12	48	4

28.2 GPI01 Peripheral

Base address: 0x4100 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
P1IN	0x4100	0	0	1
P1OUT	0x4104	1	4	1
P1OUTS	0x4108	2	8	1
P1OUTC	0x410C	3	12	1
P1OUTT	0x4110	4	16	4
P1DIR	0x4114	5	20	1
P1IF	0x4118	6	24	4
P1IES	0x411C	7	28	1
P1IE	0x4120	8	32	1
P1SEL	0x4124	9	36	1
P1REN	0x4128	10	40	1
P1AFS	0x412C	11	44	4
P1TASK	0x4130	12	48	4

28.3 SPI0 Peripheral

Base address: 0x4200 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
SPI0CR	0x4200	0	0	4
SPI0SR	0x4204	1	4	1
SPI0TX	0x4208	2	8	4
SPI0RX	0x420C	3	12	4

SPI0FOS	0x4210	4	16	4
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28.4 SPI1 Peripheral

Base address: 0x4300 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
SPI1CR	0x4300	0	0	4
SPI1SR	0x4304	1	4	1
SPI1TX	0x4308	2	8	4
SPI1RX	0x430C	3	12	4
SPI1FOS	0x4310	4	16	4

28.5 UART0 Peripheral

Base address: 0x4400 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
UART0CR	0x4400	0	0	1
UART0SR	0x4404	1	4	1
UART0BR	0x4408	2	8	2
UART0RX	0x440C	3	12	1
UART0TX	0x4410	4	16	1

28.6 UART1 Peripheral

Base address: 0x4500 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
UART1CR	0x4500	0	0	1
UART1SR	0x4504	1	4	1
UART1BR	0x4508	2	8	2
UART1RX	0x450C	3	12	1
UART1TX	0x4510	4	16	1

28.7 TIMER0 Peripheral

Base address: 0x4600 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
TIM0CR	0x4600	0	0	4
TIM0SR	0x4604	1	4	1
TIM0VAL	0x4608	2	8	4
TIM0CMP0	0x460C	3	12	4
TIM0CMP1	0x4610	4	16	4
TIM0CMP2	0x4614	5	20	4
TIM0CAP0	0x4618	6	24	4
TIM0CAP1	0x461C	7	28	4

28.8 TIMER1 Peripheral

Base address: 0x4700 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
TIM1CR	0x4700	0	0	4
TIM1SR	0x4704	1	4	1
TIM1VAL	0x4708	2	8	4
TIM1CMP0	0x470C	3	12	4
TIM1CMP1	0x4710	4	16	4
TIM1CMP2	0x4714	5	20	4
TIM1CAP0	0x4718	6	24	4
TIM1CAP1	0x471C	7	28	4

28.9 GPIO2 Peripheral

Base address: 0x4800 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
P2IN	0x4800	0	0	1
P2OUT	0x4804	1	4	1
P2OUTS	0x4808	2	8	1
P2OUTC	0x480C	3	12	1
P2OUTT	0x4810	4	16	4
P2DIR	0x4814	5	20	1
P2IF	0x4818	6	24	4
P2IES	0x481C	7	28	1
P2IE	0x4820	8	32	1
P2SEL	0x4824	9	36	1
P2REN	0x4828	10	40	1
P2AFS	0x482C	11	44	4
P2TASK	0x4830	12	48	4

28.10 SYSTEM Peripheral

Base address: 0x4900 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
SYSCLKCR	0x4900	0	0	2
CLKDIVCR	0x4904	1	4	1
BLOCKPWR	0x4908	2	8	1
CRCDATA	0x490C	3	12	1
CRCSTATE	0x4910	4	16	2
WDTPASS	0x4930	12	48	4
WDTCR	0x4934	13	52	1
WDTSR	0x4938	14	56	1
WDTVAL	0x493C	15	60	4
DC00BIAS	0x4940	16	64	2
DC01BIAS	0x4944	17	68	2

28.11 NPU Peripheral

Base address: 0x4A00 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
NPUCR	0x4A00	0	0	4
NPUIVSAR	0x4A04	1	4	4
NPUVVSAR	0x4A08	2	8	4
NPUOVSAR	0x4A0C	3	12	4
NPUSR	0x4A10	4	16	4
NPUCFG1	0x4A14	5	20	4
NPUCFG2	0x4A18	6	24	4

28.12 PWRCTRL Peripheral

Base address: 0x4B00 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
PWRCR	0x4B00	0	0	4
PWRSR	0x4B04	1	4	4
PWRWAKE	0x4B14	5	20	4
PWRSTS	0x4B18	6	24	4

28.13 GPI03 Peripheral

Base address: 0x4D00 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
P3IN	0x4D00	0	0	1
P3OUT	0x4D04	1	4	1
P3OUTS	0x4D08	2	8	1
P3OUTC	0x4D0C	3	12	1
P3OUTT	0x4D10	4	16	4
P3DIR	0x4D14	5	20	1
P3IF	0x4D18	6	24	4
P3IES	0x4D1C	7	28	1
P3IE	0x4D20	8	32	1
P3SEL	0x4D24	9	36	1
P3REN	0x4D28	10	40	1
P3AFS	0x4D2C	11	44	4
P3TASK	0x4D30	12	48	4

28.14 I2C0 Peripheral

Base address: 0x4E00 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
I2C0CR	0x4E00	0	0	4
I2C0FCR	0x4E04	1	4	1

I2C0SR	0x4E08	2	8	2
I2C0MTX	0x4E0C	3	12	1
I2C0MRX	0x4E10	4	16	1
I2C0STX	0x4E14	5	20	1
I2C0SRX	0x4E18	6	24	1
I2C0AR	0x4E1C	7	28	1
I2C0AMR	0x4E20	8	32	1

28.15 I2C1 Peripheral

Base address: 0x4F00 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
I2C1CR	0x4F00	0	0	4
I2C1FCR	0x4F04	1	4	1
I2C1SR	0x4F08	2	8	2
I2C1MTX	0x4F0C	3	12	1
I2C1MRX	0x4F10	4	16	1
I2C1STX	0x4F14	5	20	1
I2C1SRX	0x4F18	6	24	1
I2C1AR	0x4F1C	7	28	1
I2C1AMR	0x4F20	8	32	1

28.16 CLINT Peripheral

Base address: 0x5000 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
MSIP0	0x5000	0	0	4
MSIP1	0x5004	1	4	4
MSIP2	0x5008	2	8	4
MSIP3	0x500C	3	12	4
MSIP4	0x5010	4	16	4
MTIMEL	0x5020	8	32	4
MTIMEH	0x5024	9	36	4
MTIMECMP0L	0x5030	12	48	4
MTIMECMP0H	0x5034	13	52	4
MTIMECMP1L	0x5038	14	56	4
MTIMECMP1H	0x503C	15	60	4
MTIMECMP2L	0x5040	16	64	4
MTIMECMP2H	0x5044	17	68	4
MTIMECMP3L	0x5048	18	72	4
MTIMECMP3H	0x504C	19	76	4
MTIMECMP4L	0x5050	20	80	4
MTIMECMP4H	0x5054	21	84	4

28.17 MUTEX Peripheral

Base address: 0x6000 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
MUTEX0	0x6000	0	0	4
MUTEX1	0x6004	1	4	4
MUTEX2	0x6008	2	8	4
MUTEX3	0x600C	3	12	4
MUTEX4	0x6010	4	16	4
MUTEX5	0x6014	5	20	4
MUTEX6	0x6018	6	24	4
MUTEX7	0x601C	7	28	4
MUTEX8	0x6020	8	32	4
MUTEX9	0x6024	9	36	4
MUTEX10	0x6028	10	40	4
MUTEX11	0x602C	11	44	4
MUTEX12	0x6030	12	48	4
MUTEX13	0x6034	13	52	4
MUTEX14	0x6038	14	56	4
MUTEX15	0x603C	15	60	4

28.18 NFC0 Peripheral

Base address: 0x6200 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
NFC0CR	0x6200	0	0	4
NFC0SR	0x6204	1	4	4
NFC0UID	0x6208	2	8	4
NFC0CFG	0x620C	3	12	4
NFC0TIM	0x6210	4	16	4
NFC0RXST	0x6214	5	20	4
NFC0IDX	0x6218	6	24	4
NFC0DATA	0x621C	7	28	4
NFC0TXCTL	0x6220	8	32	4
NFC0DBG	0x6224	9	36	4

28.19 GPIO4 Peripheral

Base address: 0x6300 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
P4IN	0x6300	0	0	1
P4OUT	0x6304	1	4	1
P4OUTS	0x6308	2	8	1
P4OUTC	0x630C	3	12	1
P4OUTT	0x6310	4	16	4
P4DIR	0x6314	5	20	1
P4IF	0x6318	6	24	4
P4IES	0x631C	7	28	1
P4IE	0x6320	8	32	1

P4SEL	0x6324	9	36	1
P4REN	0x6328	10	40	1
P4AFS	0x632C	11	44	4
P4TASK	0x6330	12	48	4

28.20 GPI05 Peripheral

Base address: 0x6400 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
P5IN	0x6400	0	0	1
P5OUT	0x6404	1	4	1
P5OUTS	0x6408	2	8	1
P5OUTC	0x640C	3	12	1
P5OUTT	0x6410	4	16	4
P5DIR	0x6414	5	20	1
P5IF	0x6418	6	24	4
P5IES	0x641C	7	28	1
P5IE	0x6420	8	32	1
P5SEL	0x6424	9	36	1
P5REN	0x6428	10	40	1
P5AFS	0x642C	11	44	4
P5TASK	0x6430	12	48	4

28.21 IRQROUTER Peripheral

Base address: 0x7000 (shared window, arbitrated across all harts)

Register	Address	Slot	Offset (bytes)	Size (bytes)
H0ENL	0x7000	0	0	4
H0ENM	0x7004	1	4	4
H0ENU	0x7008	2	8	4
H0ENX	0x700C	3	12	4
H1ENL	0x7010	4	16	4
H1ENM	0x7014	5	20	4
H1ENU	0x7018	6	24	4
H1ENX	0x701C	7	28	4
H2ENL	0x7020	8	32	4
H2ENM	0x7024	9	36	4
H2ENU	0x7028	10	40	4
H2ENX	0x702C	11	44	4
H3ENL	0x7030	12	48	4
H3ENM	0x7034	13	52	4
H3ENU	0x7038	14	56	4
H3ENX	0x703C	15	60	4
H4ENL	0x7040	16	64	4
H4ENM	0x7044	17	68	4
H4ENU	0x7048	18	72	4

H4ENX	0x704C	19	76	4
CLAIM	0x7800	512	2048	4
PENDL	0x7810	516	2064	4
PENDM	0x7814	517	2068	4
PENDU	0x7818	518	2072	4
PENDX	0x781C	519	2076	4
INSVCL	0x7820	520	2080	4
INSVCM	0x7824	521	2084	4
INSVCU	0x7828	522	2088	4
INSVCX	0x782C	523	2092	4

29 Errata

There are no known errata for the Castalia MCU at this time. This section will list hardware issues, their impact, and software workarounds as they are discovered.